Mark Richards

CAPCOSTS

A HANDBOOK FOR ESTIMATING MINING AND MINERAL PROCESSING EQUIPMENT COSTS AND CAPITAL EXPENDITURES AND AIDING MINERAL PROJECT EVALUATIONS

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PREFACE

From 1995 to 1997, major mining and mineral processing equipment prices were gathered under the general auspices of the CAMIRO-MPD (Metallurgical Processing Division of the Canadian Mining Industry Research Organization formerly known as MITEC, the Mining Industry Technology Council of Canada), the Canadian Mineral Processors Division of CIM, the Metal Mining Division of CIM and CANMET to revise and update CIM Special Volume 25 published in 1982. CIM Volume 25 is itself a revision of CIM Special Volumes 13 (1972) and 18 (1978) and is based on a course developed in 1978 by Professor Mular, with the title *Mining and Mineral Processing Equipment Costs and Preliminary Capital Cost Estimations.*

This newest update is effectively a handbook for (1) estimating costs of mining/mineral processing equipment, for (2) estimating capital expenditures and for (3) aiding mineral project evaluations. The handbook is entitled **CAPCOSTS**. It incorporates additional equipment and sections dealing with mineral economics and project evaluation techniques.

All data in **CAPCOSTS** have been analyzed by statistical methods unless stated otherwise. **It must be emphasized that the costs herein are not exact costs; they cannot be considered as quotes from any single manufacturer/supplier.**

Sample calculations are shown at the beginning of each section relevant to major equipment costing and, where appropriate, an explanation of data presentation is provided. A section contains various rules of thumb which may be useful for rough estimates when other data are not available.

Capital expenditures are estimated by means of either ratio methods or updated cost/capacity methods developed by O'Hara and others. O'Hara's technique to estimate total product costs is likewise employed in updated format.

Mineral project evaluation techniques are reviewed and corresponding applications are presented in a realistic manner, while distinct features of ore deposit evaluation methodology are reexamined.

A **CAPCOST** computer program can be purchased as a separate item. Users of this manual will find that the computer program reduces estimation time significantly.

ACKNOWLEDGEMENTS

We wish to express our appreciation to the major equipment firms, suppliers, engineering consulting firms and others who have contributed to this handbook by either supplying or verifying cost information.

The compilation, analysis and drafting of all data was performed by the following individuals under general supervision:

Mr. Haytham Hodaly, Technical Assistant
Mr. Paul Bialikiewicz, Technical Assistant
Mr. Kenneth Strobbe, Technical Assistant

They deserve our thanks on behalf of sponsoring organizations.

This handbook was sponsored by CAMIRO (the Canadian Mining Industry Research Organisation formerly known as MITEC), the Canadian Mineral Processors Division of CIM, the Metal Mining Division of CIM and CANMET. CAMIRO disbursed funds collected from these organizations and from the following individual sponsors: Cambior Inc., Centre de Recherche Minerales (CRM) and Cominco Ltd. Special thanks are due to Dr. Bryn Harris, who initiated interest and financial support for **CAPCOSTS** through MITEC, Mr. Ray MacDonald of CANMET, Secretary of CMP, who encouraged support by CANMET, Mr. Mike Mular, Chairman of CMP, who spearheaded support by the Canadian Mineral Processors, and Mr. Rick Zimmer, Chairman of the Metal Mining Division, who maintained corresponding interest in support from Metal Mining.

We are especially appreciative of the technical assistance provided by the chairman of CAMIRO-MPD, Mr. Art Winckers of Teck Corporation, Vancouver, British Columbia. His critical comments were most relevant to the completion of this handbook.

Continued interest, assistance and encouragement from members and affiliates of the mining and mineral processing industry was most welcome. In particular, we wish to thank Mr. John Scott (Fluor Daniel Wright) and Mr. Stu Jones (Svedala Industries) for their technical assistance and advice.

Compilation of cost data was performed while at the Department of Mining and Mineral Process Engineering, University of British Columbia, Vancouver, B. C. The assistance of Mr. Gordy Lagore and Mrs. Marina Lee, Departmental Secretaries, was greatly appreciated.

ABSTRACT

This handbook contains data in the form of graphs, tables and equations for the rapid estimation of the price of an item of major equipment used in the mining and mineral processing industry. Data collected from various sources are fitted by means of non-linear estimation to the equation, Price = aX^b, where X is a suitable parameter, e.g., motor horsepower, and a and b are appropriate constants. Equipment cost data are employed for various purposes, such as for the preliminary estimate of the fixed capital cost of a mineral processing plant. Examples are given to illustrate the calculation of key parameters that determine cost. To update a cost item, the M&S(Mine/Mill) index is employed.

Capital cost estimation procedures, based originally on the work by O'Hara (CIM Bulletin, February, 1980) and by Balfour and Pappuciyan (Annual Meeting, Canadian Mineral Processors, January, 1972, Ottawa, Canada), are presented in the form of tables. The former has been employed for open pit mines, underground mines and processing plants. The latter has been used for green field (grass roots) and battery limit (i.e., a crushing plant) processing plants, where an estimate of major equipment costs is necessary.

A preliminary total capital cost estimate is useful in several ways. First, process engineers are able to determine rapidly whether sufficient funds are available for a proposed processing/mining method; second, where total product costs (i.e., operating costs) of alternatives are similar, a preliminary total capital cost estimate aids in final selection; third, total capital costs are utilized to establish economic criteria, such as cash flows and sensitivity data, that are so necessary for overall project evaluations. Recent methodology developed by Camm (US Bureau of Mines) for prefeasibility evaluations based on quickie cost estimates is summarized for situations where specific design parameters may not be available.

Mine Mill total product cost estimation, patterned after O'Hara, is incorporated and a brief section on revenue estimation is offered.

Capital and Operating cost estimation for small deposits (CANMET SP 86-11E) and small placer mines (USBM IC 9170) is discussed.

Mineral project evaluation techniques are described and corresponding realistic applications are provided. Important features of ore deposit evaluation methodology are reviewed.

Cost information was employed to construct a windows-based computer program, called **CAPCOST**, that is menu driven. It performs otherwise tedious calculations, while the HELP menu serves as a prompt for novices. A copy of the program can be purchased separately from the **CAPCOSTS** manual. The software package comes with security protection required to run the software.

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INTRODUCTION

This is a handbook for (1) estimating costs of major mining and mineral processing equipment, for (2) estimating capital expenditures and for (3) evaluating mineral projects. The handbook title is **CAPCOSTS**, which is an update of CIM Special Volume 25 published in 1982 and entitled ***Mining and Mineral Processing Equipment Costs and Preliminary Capital Cost Estimations***. Volume 25 is itself a revision of CIM Special Volumes 13 (1972) and 18 (1978).

Cost Estimation Texts Available

There are a variety of cost estimation sources. Certified cost engineers, who are members of the American Association of Cost Engineers, have ready access to the Cost Engineers Notebook. For our purposes, several text-like sources relevant to the mining industry are:

(1) Balfour, R. J. and T. L. Papucciyan, "Capital Cost Estimating For Mineral Process Plants", **Proceedings of the 4th Annual Meeting of the Canadian Mineral Processors**, CIM, Ottawa, Ontario, 1972.

(2) Guthrie, Kenneth M., **Process Plant Estimating, Evaluaton and Control,** Craftsman Book Company of America, Solona Beach, CA., 1974, ISBN 0-910460-5-1.

(3) STRAAM Engineers, Inc., **Capital and Operating Cost Estimating System Handbook: Mining and Beneficiation**, US Bureau of Mines, OFR 10-78, 1979.

(4) Hoskins, J. R. and W. Green (Eds), **Mineral Industry Costs,** Northwest Mining Association, Spokane, WA, 1982, 248 pages, ISBN 0-931986-02-6.

(5) Woods, Donald R., **Cost Estimation For The Process Industries**, McMaster University Bookstore, McMaster University, Hamilton, Ontario, 1983.

(6) J. S. Redpath Limited, **Estimating Preproduction and Operating Costs of Small Underground Deposits**, CANMET **SP 86-11E**, Ottawa, Canada, 1986.

(7) ----------, **Bureau of Mines Cost Estimating System Handbook: Part 1. Surface and Underground Mining, IC 9142; Part 2. Mineral Processing, IC 9143;** United States Department of the Interior, US Bureau of Mines, Washington, D. C., 1987.

(8) Stebbins, Scott A., **Cost Estimation Handbook for Small Placer Mines, IC 9170**, United States Department of the Interior, US Bureau of Mines, Washington, D. C., 1987.

(9) Camm, Thomas W., **Simplified Cost Models For Prefeasibility Mineral Evaluations**, Bureau of Mines, **IC 9298**, Department of the Interior, Washington, D. C., 1991.

(10) Ruhmer, W. T., **Handbook On The estimation of Metallurgical Process Costs, 2nd Edition**, MINTEK Publication 14, Randburg, South Africa, 1991.

(11) Noakes, Michael and Terry Lanz, Eds., **Cost Estimation Handbook For the Australian Mining Industry**, Monograph 20, Aus. IMM, Parkville, Victoria, Australia, 1993

(12) ---------, **Mine and Mill Equipment Costs - An Estimator's Guide**, Western Mine Engineering, Inc., Spokane, Washington, 1994.

Some additional texts, journals and papers relevant to cost estimation is given in the section dealing with **Additional Sources of Cost Information** on page 305.

CAPCOSTS Handbook

CAPCOSTS differs from CIM Volume 25 and from the texts/booklets listed on page 1. Additional mining and mineral processing equipment and sections that deal with mineral economics and project evaluation techniques were added. Moreover, the handbook is intended for use in design courses at the university level and for individuals and groups who do not specialize in cost engineering. **No attempt was made to incorporate *detailed* equipment sizing/selection, because textbooks are available for this purpose.** Those who have not yet selected and/or sized their major equipment should refer to the following texts:

(a) Mular, Andrew L. and Bhappu, Roshan B., Eds., **Mineral Processing Plant Design**, SME-AIME, Littleton, Colorado, 1980, 946 pages.

(b) Mular, Andrew L. and Jergensen, Gerald V. II, Eds., **Design and Installation of Comminution Circuits**, SME-AIME, Littleton, Colorado, 1982, 1022 pages.

(c) Mular, Andrew L. and Anderson, Mark A., Eds., **Design and Installation of Concentration and Dewatering Circuits**, SME-AIME, Littleton, Colorado, 1986, 842 pages.

(d) Hartman, Howard L., Senior Ed., **SME Mining Engineers Handbook,** SME-AIME, Littleton, Colorado, 1992, Vol. 1 and 2.

(e) Weiss, Norman L., Ed., **SME Mineral Processing Handbook**, SME-AIME, Littleton, Colorado, 1985, Vol. 1 and 2.

MAJOR EQUIPMENT COST ESTIMATION

Estimating the cost of major equipment is unnecessary, when a firm price has been established---perhaps by calling the manufacturer/supplier for a firm quote. However, there are important reasons for estimating prices. This section deals with such methodology.

Usefulness

Typically, we wish to obtain with minimal effort an estimate of the cost, which has a known accuracy, of major equipment when (1) a procedure is being used for capital cost estimation that requires estimates of major equipment costs, (2) the cost of a standard item is being compared with an item of similar function but of different design, (3) an existing circuit is being expanded, so that additional equipment must be purchased, (4) a new plant is being designed and several alternative circuits, which provide similar grades/recoveries, must be compared, (5) existing equipment is worn out and must be replaced and (6) you are registered in a mining and

mineral processing plant design course that requires capital and operating cost estimates for reports. Since time may be an important factor, "quickie" costs with some degree of reliability are needed.

Suppliers

Chances are high that most of us have had contact with at least a few of the major suppliers of equipment in this handbook. From time to time, we must seek out suppliers of unfamiliar equipment. Fortunately, in North America, mining and mineral processing equipment suppliers and manufacturers advertise in journals such as: **CIM Bulletin**, **Mining Engineering**, **Engineering and Mining Journal**, **Canadian Mining Journal** and **Chemical Engineering**. The latter three will print equipment catalogs each year that have proven useful.

Most libraries house **Frasers Canadian (or USA) Trade Directory**, which is an extensive compilation of virtually all manufacturers/suppliers. Alternatively, the **Thomas Register** (Thomas Publishing Company, Rexdale, Ontario) reportedly locates the most qualified suppliers in all of North America with ease.

Perhaps the best source for our purposes is the **Canadian Mining Journal's Mining Sourcebook** published each year. The Sourcebook contains an up-to-date Buyers' Guide in two parts: the first is a Products and Services Listing and the second is a List of Suppliers of corresponding products/services of interest. Lists are updated each year and reflect name changes due to acquisitions and expansions. The Sourcebook can be found in most libraries and most mining companies purchase it every year.

Importance of Specifications

The cost of an item depends on (i) the basic item and its mass and/or volume, (ii) the complexity of its design, (iii) the materials of construction, (iv) the choice of accessories and (v) the nature and complexity of the drive train and motor. Specifications must be clearly stated. Thus a lubrication system, if not carefully specified, may be different from previous ones which were more reliable and less noisy. The supplier might choose to supply something less costly to permit of a lower bid.

Unless specifications are clearly stated, suppliers/manufacturers are faced with providing quality components subject to, in most cases, bidding competition. It is of interest that a buyer need not send out for bids. This choice may depend on factors other than price. For example, availability may become the important criterion. Suppose that the following is true:

Supplier A bids $2,586,000	with delivery in 3 months
Supplier B bids $2,311,000	with delivery in 6 months
Supplier C bids $2,050,000	with delivery in 12 months

Each supplier will satisfy specifications, but the buyer might go with supplier B since his delivery time is twice as fast as that of the lower bidder. In choosing a supplier, the buyer might first ask for the delivery date, perhaps because he has construction deadlines to meet in order to satisfy financial agreements concerning startup dates.

In most situations, buyers will purchase that which their experience has shown to be reliable. When choices are possible (generally the case) cost, availability, size-to-capacity and other factors become of prime importance.

Quickie Equipment Costs

Several ways to obtain quickie equipment costs are: (1) phone suppliers, (2) use a cost index with file data, (3) use a cost index in combination with the 0.7 rule and (4) use a cost index in combination with cost/ parameter relations developed from old and/or new cost data.

(1) Phone Suppliers

This is an obvious way to obtain the approximate cost of an item of equipment. However, there is a financial penalty for both the potential buyer (often he only needs a cost and never buys) and the supplier in that phone calls are necessary and some manhours are consumed. Most likely, the price cannot be provided immediately, so that a delay is involved. Clearly, individuals, both buyers and suppliers, may invest a substantial time to cost estimations. Rapid techniques are certainly called for, unless the project is at the bidding stage.

(2) Use Cost Index With Minimal File Data - Cost Indexes

It is possible that an item of equipment identical to one of interest was purchased several years ago. Because there is a record of the transaction, the specifications and the cost are in your files. The current cost can be estimated by means of a **cost index**. Thus:

$$\frac{CostNow}{CostThen} = \frac{IndexNow}{IndexThen}$$

There are a large number of indexes available. The American Association of Cost Engineers (AACE) Notebook lists 13 building cost indexes, 9 general construction/equipment indexes, 7 plant construction/equipment indexes and 7 miscellaneous ones. Obvious questions are: What is a cost index? Where does it come from? How do you use it?

A cost index is a ratio of costs at a particular time to costs at a specified base year. Where the price of an item at some time in the past is known, the current price can be estimated in several ways by application of a cost index. Such methodology has been in use since the early 1900's.

Cost indexes are based on mean costs over a period of time. They may have an accuracy of the order of ± 10 percent and have been employed with this accuracy when the time lapse is less than about six years. Accuracy decreases after about 6 years. In consequence, cost indexes should be restricted to order-of-magnitude and factored estimate costing methods (see section on Capital Cost Estimates).

Of the various cost indexes available, the following are commonly employed in the process industries: (1) Engineering News-Record construction index (ENR), (2) Marshall and Swift cost index (M&S), (3) Chemical Engineering plant construction cost index (CE) and (4) Nelson

Refinery construction cost index (NR). Each index is based on certain specific information as summarized in subsequent paragraphs (see **Cost Engineering**, October, 1968).

(i) Engineering News-Record Construction Cost Index (ENR)

The ENR index is based on the costs of labour and building materials in the following proportions: 25 cwt of structural steel, 6 bbl of Portland cement, 1.088 mfbm of 2x4 s4s lumber, and 200 hours of common labour. The ENR base year is 1913; i.e., ENR = 100.

(ii) Marshall and Swift Cost Index (M&S)

The M&S index has several values. The value most often used and referred to is the all-industry equipment index. This index is the average of the indices calculated for 47 different industries. Other M&S index values are calculated according to the type of industry, one of these being an M&S(Mine/Mill) index for the mining and milling industry. This latter index is employed in this revised handbook.

M&S indexes are based on equipment appraisals, modifying factors and judgments concerning current economic conditions (such as inflation). They are compiled quarterly by Marshall and Swift, Los Angeles, California, U.S.A. The M&S index base year is 1926.

(iii) Chemical Engineering Plant Construction Cost Index (CE)

The CE index is based on equipment, erection and installation, material, labour, engineering and supervision costs in the following percentages: 37% fabricated equipment, 14% process equipment, 20% pipes, valves and fittings, 7% process instruments and controls, 7% pumps and compressors, 5% electrical equipment., 10% structural supports, insulation and paint, 22% erection and installation labour, 7% buildings, materials and labour, 10% engineering and supervision manpower. The base year is 1957-1959.

(iv) Nelson Refinery Construction Cost Index (NR)

The NR index is employed primarily for estimations in the petroleum industry. The total index is based on material and labour in the following percentages: 24% iron and steel, 8% building material, 8% miscellaneous equipment, 30% common labour, 30% skilled labour. The NR base year is 1946.

Table 1 below compares cost index ratios (1996 value over 1986 value) for the ENR, the M&S (All-Ind), the M&S (Mine/Mill) and the CE Cost Indexes.

Table 1: Comparisons In Cost Index Ratios Over 10 Years

Index In Table 1	Ratio 1996 Index To 1986 Index
ENR	1.31
M&S(Mine/Mill)	1.30
M&S(All-Industry)	1.30
CE	1.20

Table 1, page 5, shows that the largest difference in index ratio values occurs between the ENR index and the CE index, a difference of about 9 %. Consequently, it may be argued that it does not matter which of the four indices is used, because the difference between them is within the accuracy of factored and order-of-magnitude cost estimation methods (see section on capital cost estimates). If the accuracy of these two methods is improved in a particular industry, the selection of an index will be important. The M&S index for mining and milling, denoted by M&S (Mine/Mill) or M&S(M/M), has been chosen for this handbook. Current values of cost indexes and other economic indicators are published in various journals such as **Chemical Engineering**, **Oil and Gas Journal** and **Engineering News-Record Magazine**.

Table 3, page 7, lists the ENR, M&S(All-Ind.), M&S(Mine/Mill) and CE cost index values from 1950 to 1996. A review of some index values for 1997 is shown in Table 2 below.

Table 2: Some Cost Index Values For The Year 1997

Month	ENR Index	M&S(All-Ind.)	M&S(Mine/Mill)	CE Index
Jan	5765	-	-	383
Feb	5769	1050	1082	384
Mar	5759	-	-	385
Apr	5799	-	-	385
May	5837	1056	1087	385
June	5860	-	-	386
July	5863	-	-	387
Aug	5854	1060	1092	388
Sep	5851	-	-	---
Oct	5847	-	-	---
Nov	----	----	----	---
Dec	----	-	-	---

In the event that a current M&S (Mine/Mill) cost index value cannot be obtained the data in Table 2 or 3 may be graphed and extrapolated (with caution). A rough rule of thumb (see **Rules of Thumb,** page 301 and Figures 218 and 219) is that the price of an item increased on the average by about 2.32 percent per year from 1989 to 1996. Thus, the index increased from 911 in 1989 to 1072 in 1996. This should be compared with an increase on the average of about 4.25 percent per year between the years 1987 to 1989 and an average of about 1.3% from 1982 to 1987!

Example of Use of Cost Indexes

A ball mill was purchased for $800,000 in June of 1989. What is the approximate cost of this unit today?

Letting "today" be May, 1997, then any current cost index may be chosen for the problem, although the M&S(Mine/Mill) index is preferred. From Table 2, the index "today" is 1087.

Table 3: Cost Index Values From 1950 to 1996

YEAR	ENR 1913-100	M&S/All-Ind 1926-100	M&S/M&M 1926-100	CE 1957/9-100
1950	510	168	171	74
1951	543	180	181	80
1952	569	181	182	81
1953	600	183	184	85
1954	628	185	187	86
1955	660	191	193	88
1956	692	209	216	94
1967	724	225	228	99
1958	759	229	234	100
1959	797	235	237	102
1960	824	238	241	102
1961	847	237	239	102
1962	872	239	240	102
1963	901	239	240	102
1964	936	242	243	103
1965	971	245	245	104
1966	1019	253	253	107
1967	1070	263	264	110
1968	1154	273	273	115
1969	1270	285	286	119
1970	1380	303	303	126
1971	1571	321	321	132
1972	1726	332	332	137
1973	1897	344	343	144
1974	2019	398	394	165
1975	2209	444	451	182
1976	2400	472	483	192
1977	2577	505	521	204
1978	2776	545	565	219
1979	3003	599	619	239
1980	3237	660	684	261
1981	3533	721	750	297
1982	3816	746	784	314
1983	4066	761	799	317
1984	4148	780	814	323
1985	4182	790	823	325
1986	4287	798	827	318
1987	4401	814	837	324
1988	4519	852	870	343
1989	4606	895	911	355
1990	4732	915	940	358
1991	4835	931	959	361
1992	4985	943	976	358
1993	5210	964	999	359
1994	5408	993	1028	368
1995	5471	1028	1058	381
1996	5618	1039	1072	382

May is near the middle of the year, so the index for 1989 will probably be representative. From Table 3, the M&S (Mine/Mill) for 1989 is 911. Using the formula as shown under Item (2) on page 4, then:

$$CostNow = [CostThen]\left[\frac{IndexNow}{IndexThen}\right]$$

$$= [\$800,000]\left[\frac{1087}{911}\right]$$

$$= \$954,555 \quad \text{to the nearest whole number.}$$

If the ball mill had been purchased in December of 1989, a more representative index would be the average for the years 1989 and 1990. Clearly, the application of a cost index is an averaging procedure. If an index for each item of major equipment was estimated in some way and compared yearly, significant differences would be apparent between items and from year to year. Accuracy is lost over the years for many reasons. For example, significant equipment changes and modifications may take place because of the tendency to build equipment with large capacity to size ratios. Costs will change accordingly.

(3) Use Cost Index With Cost Vs Parameter to 0.6 or 0.7 Power

Average costs of major equipment have been observed to follow, roughly, an expression written as:

$$Cost = a[Parameter]^{0.7}$$

where the exponent 0.7 has been replaced by anything from 0.6 to 2/3 to 0.7. The choice of exponent is dependent upon the particular industry and upon the degree of conservatism exhibited by the estimator. The parameter can be mass, volume, capacity, dimensions, area, power draw and any other, including any combinations (such as capacity times dimension) that work. Usually, mass or size or capacity are first guesses (see **Rules of Thumb** on page 301 of this handbook).

When a parameter that determines cost is known, the above expression is more useful in the following form:

$$\frac{(Cost)_2}{(Cost)_1} = \left[\frac{(Parameter)_2}{(Parameter)_1}\right]^{0.7}$$

where $(Cost)_2$ is the known cost at $(Parameter)_2$ and $(Cost)_1$ is the known cost at $(Parameter)_1$. The expression is often referred to as the 6/10 or the 7/10 rule by estimators.

For example, suppose it is believed that the cost of a crusher is sensitive to its capacity. If the cost of a crusher of capacity 400 stph is estimated as $265,000, what is the cost of a similar crusher with capacity 500 stph? From the above:

$$(Cost)_2 = (265,000)\left[\frac{500}{400}\right]^{0.7} = 309,801$$

rounded to the nearest whole number. If the exponent is 0.6, this becomes 302,965.

The exponent is an average and was judged years ago by estimators to be better than an educated guess.

(4) Use Cost Index With Cost Vs Parameter Relation Found From Data

When cost data (either from old files or from new quotes or other sources) are available for specific items of equipment it is possible to find an equipment parameter to which the cost is sensitive. Data must be placed onto a common index basis, so that the value of the index associated with each cost must be known. This cost is then converted to a common index using the expression on page 4. For example, suppose that file data show that in June of 1991, when the M&S (Mine/Mill) index = 959, the cost of a ball mill of standard specification is $800,000. To place this onto a common index basis of 1400, determine:

$$(Cost)_{1400} = (1400)\left[\frac{800,000}{959}\right] = 1,167,883 \text{ to the nearest whole number.}$$

Now, with each cost on a common basis, graphs of $(Cost)_{1400}$ versus equipment parameters can be plotted on log-log paper in search of straight lines. Care must be exercised to employ costs of an equipment item of comparable specifications. Thus, if costs are available for 8 different ball mills, 6 of which have single pinion drive trains and 2 have dual pinion drives, the latter two must be analyzed separately for obvious reasons. Specifications are important to the accuracy of an estimate!

When data appear to fall onto a straight line, a non-linear least squares method can be employed to fit the equation:

$$(Cost)_{1400} = a[Parameter]^b$$

where a is a constant that depends on a variety of factors and b is a constant that varies with, among other things, basic equipment type, structural features, design, and efficiency. Coefficient a can be viewed as the "intercept" of a log-log line, while coefficient b is the slope.

If the cost versus parameter data are not linear on log-log paper, alternative equations can be employed, although this additional complexity can be avoided by establishing ranges over which the data appear to give straight lines. The above equation is then fitted to each range. Of course, a and b will likely vary from range to range.

After $(Cost)_{1400}$ is estimated from the cost vs parameter graph, it must then be converted to the current cost index. Suppose $(Cost)_{1400}$ is found to be $1,170,000. What is the cost of this item when the M&S (Mine/Mill) index is 1075?

Using the expression on page 4:

$$(Cost)_{1075} = (1,170,000)\left[\frac{1075}{1400}\right] = \$898,393 \text{ to the nearest whole number.}$$

The choice of 1400 as a common base index is arbitrary.

Major Equipment Costs in CAPCOSTS

Costs of major equipment in this revised handbook are **purchased equipment costs** and **are based on the price of the equipment in US dollars**. In general, the cost in terms of Canadian currency would be: $ Canadian = (exchange ratio) times ($ US), where the exchange ratio is the ratio of the Canadian dollar to the US dollar. Exchange ratios can be obtained from various sources such as banks and newspapers. For the year 1996, Revenue Canada sets the average exchange ratio at $ Canadian = 1.3636 ($ US). However, the current exchange ratio should be applied to the estimated purchased cost in $ US to determine the current cost in $ Canadian.

Unless specifically mentioned, import duties are not included. Because import duties may contribute substantially to the total cost of an item of equipment, some rules of thumb concerning import duties are provided in the section dealing with **Rules of Thumb** (page 301).

The Marshall and Swift Cost Index (Mining & Milling) is employed for updating equipment costs reported herein. Costs are adjusted to a common base value of 1400. Index values are found in the journal, **Chemical Engineering**, McGraw-Hill, 1221 Avenue of the Americas, New York, NY, 10020. Table 3, page 7, of this handbook shows them from 1950 to 1996.

Most items of equipment in this handbook are manufactured in the U.S. and may be distributed through Canadian subsidiaries. In Canada, to account for currency fluctuations, multiply each US base cost, $(Cost)_{1400}$, by the exchange ratio (current Canadian to U.S. exchange rate) to determine a Canadian base cost if desired. **It is best to estimate $(Cost)_{now}$ in Canadian dollars as follows: First estimate $(Cost)_{now}$ in US dollars from graphs herein using the index ratio, $\frac{IndexNow}{1400}$, and then convert to Canadian dollars via the current exchange ratio.**

Unless stated otherwise, to obtain an estimate of the delivered cost of an item of equipment, multiply by the factor 1.03 which represents an overall average. The installed cost, on the average, of equipment employed for solids-fluids processes is either 1.39 times the delivered cost or 1.43 times the purchased cost.

It cannot be over-emphasized that the purchased equipment costs given in this handbook are approximations. Estimated prices of equipment must not be considered as quotes from any one source.

Each graph of $(Cost)_{1400}$ versus Cost Parameter has associated with it the ranges and units for the Cost Parameter as listed at the top of each graph. Where necessary, specifications are shown for clarity. Most of the curves involve two or more suppliers. Sometimes more than one parameter range has been fitted for a given item of equipment.

The equation:

$$(Cost)_{1400} = a[Parameter]^b$$

is used throughout, where $(Cost)_{1400}$ is the base cost in US dollars. Non-Linear least squares estimates of a and b for equipment items are shown with the corresponding graphs. Also, these coefficients have been incorporated into a computer program called **CAPCOST**.

INTRODUCTION TO FIXED/WORKING CAPITAL COST ESTIMATES

Capital cost estimation is important to many facets of mining and mineral process engineering, where decisions must be supported by financial analyses. The ***total capital investment*** in a mine or a concentrator (mill) consists of a ***fixed capital*** portion and a ***working capital*** portion.

Fixed and Working Capital Versus Total Production Cost

Fixed capital is the total amount of money needed to purchase the necessary equipment, buildings and auxiliaries such as site preparation, preproduction development and utilities. Working capital represents cash that must be available to begin the operation. Of major importance is the fact that the total capital investment is a sum that is separate from the ***total production cost*** (total operating cost) paid out per unit time from gross income per unit time. Total product costs include ***operating costs*** such as direct production costs, fixed charges and plant overhead, and ***general expenses*** such as administrative costs, distribution and marketing costs, research and development costs and gross earnings expenses.

Purpose of Estimates

Preliminary capital cost estimates are useful to engineers who are involved in selection and design. Since most companies have a limit to available capital funds and/or lines of credit, an immediate assessment of initial cash requirements is provided. In situations where capital funds must be borrowed, the less borrowed the better. In effect, a preliminary estimate may serve as the basis for a "go or no-go" decision. In situations where the total product costs of alternative, but technically feasible, processes are similar, the one with smallest capital expenditure is selected. Here the preliminary estimate serves as a way to assess processing alternatives. Obviously, depending upon their degree of accuracy, capital cost estimates will serve many other purposes. These include participation in feasibility studies, plant expansion, and presentation of bids.

Types of Fixed Capital Cost Estimates

Five major **Types** of **fixed capital cost estimates** have been proposed by the American Association of Cost Engineers. These are:

(1) Order-of-Magnitude estimate based on previous cost data and minimal knowledge. It is a ratio estimate with confidence limits that exceed ±30 percent.

(2) Factored or ratio estimate based on major equipment costs with a probable accuracy within ±30 percent.

(3) Budget authorization (preliminary) estimate based on sufficient data to permit fund procurement and budgeting. The probable accuracy is within ±20 percent.

(4) Definitive (project control) estimate based on almost complete data (some specifications missing and drawings not complete). Probable accuracy is within ±10 percent.

(5) Detailed (contractors) estimate based on complete engineering drawings, specifications and site surveys. The probable accuracy is ±5 percent.

A **Type (1)** estimate does not have the flexibility of the **Type (2)** factored estimate; the latter allows personal judgments to be made. A **Type (3)** estimate becomes more time consuming and expensive compared to **Types (1) and (2)**. **Types (4) and (5)** involve substantial time and money. Their extra accuracy is usually not justified when the preliminary feasibility of a project is still under evaluaton.

In more recent years, the **American National Standards Institute** has grouped estimates into three classes (ANSI Standard Z94.2). **Class I** is an ***order-of-magnitude estimate*** with an accuracy of +50 to -30 percent; **Class II** is a ***preliminary (budget) estimate*** with an accuracy of +30 to -15 percent; **Class III** is a ***definitive estimate*** with an accuracy of +15 to -5 percent. Accuracy can be increased as more and more information on the basic mining and processing method, capacity, equipment specifications, flowsheets, civil/structural/electrical/mechanical drawings, arrangement drawings, piping and instrumentation diagrams, site layouts, and the like is acquired. Figure 1, after Mackellar, at the bottom of this page (SME preprint number 75-B-23) summarizes the accuracy of various classes. Accuracy is well within the range of ANSI specifications. In this handbook, estimates are expected to have the accuracy of either Class I or Class II. These serve as a check on Class III estimates and may lead to further investigation of design and layout decisions.

In general, the cost of preparing an estimate will increase with the desire to increase its accuracy. Estimating costs can be as high as 2 percent of the total project cost!

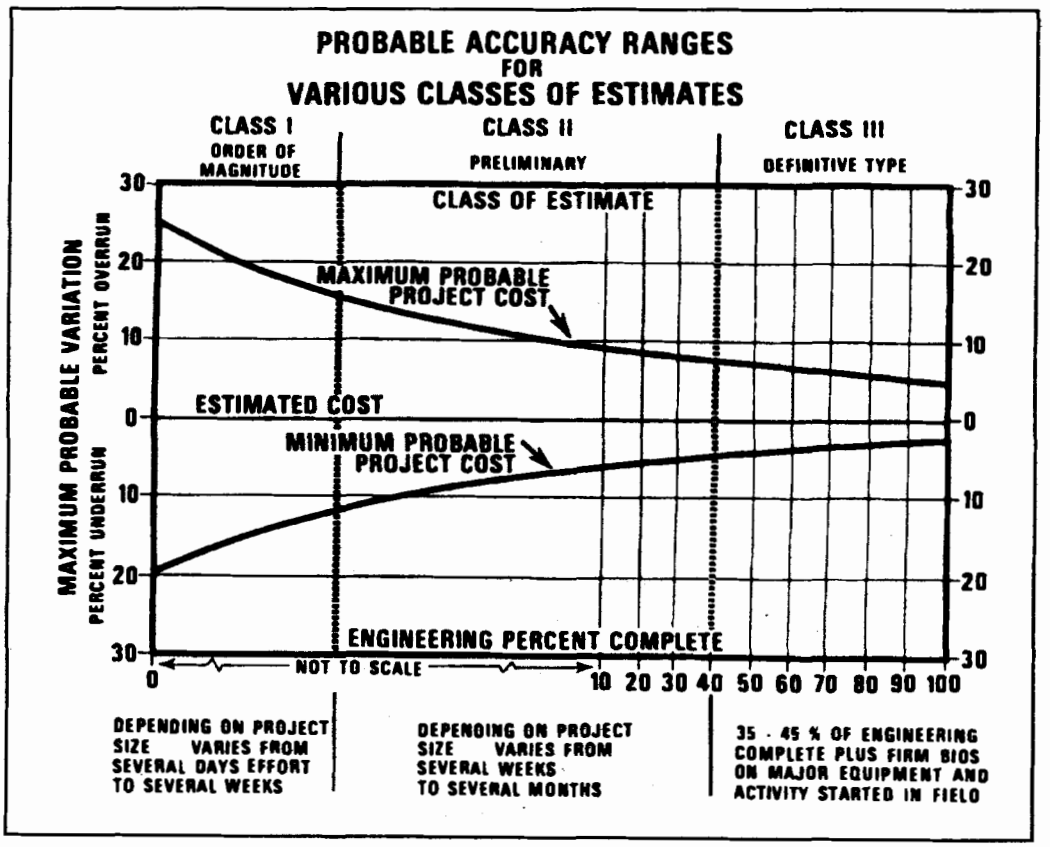


Figure 1: Probable Accuracy Ranges For Various Estimates

Types of Working Capital Cost Estimates

Working capital cost may be estimated from the following, provided the corresponding information is available:

(1) Raw materials inventory (1 month supply at cost)
(2) Materials-in-process inventory (1 month supply at cost)
(3) Product inventory (1 month at manufactured cost)
(4) Accounts receivable (1 month at selling price)
(5) Available cash (to meet expenses of wages, raw material, utilities, supplies for I month at manufactured cost)

(6) **Working capital** = (1)+(2)+(3)+(4)+(5)

The **Canadian Mining Journal's Mining Sourcebook**, may be helpful for completing the table above.

O'Hara (see "Quick Guides to the Evaluation of Orebodies", **CIM Bulletin**, February, 1980) recommends that **working capital should be equivalent to 4 months of estimated operating costs** on a full production basis.

An alternative to the above methods is to estimate **working capital cost** as a percentage of annual estimated sales revenue. Of the order **of 30 percent of annual sales** has been recommended in the chemical industry.

Another alternative has been to estimate working capital required as a percentage of the fixed capital investment. Anywhere from **10 to 20 percent of fixed capital will be necessary, with 15% being reasonable**.

Invariably, working capital requirements are estimated on the assumption that startup will be relatively trouble free. Feldman (**Chemical Engineering**, November 3, 1969) stresses that **startup costs**, which include operating costs, depend on fixed capital expenditure, newness of the process and the technology to the company, newness of the equipment type (i.e., is it the first 38 ft diameter ball mill?), labour quantity and quality and the extent of interplant dependency. To estimate startup costs:

Startup Cost = A(0.1 + B + C + D + NE)

where A = fixed capital cost
B = factor for newness of process and technology
C = factor for newness of type of equipment
D = factor for quantity and quality of labour available
N = number of plants or sections that make up the process chain
E = factor for interplant dependency

Note: Startup Cost = 10% of A when B, C, D and NE are zero.

For some chemical plants, Feldman suggests:

B(process/tech.)	C(equipment type)	D(labour supply)	E(plant dep.)
.05 radically new	.07 radically new	.04 very short	.04 very dependent
.02 relatively new	.04 very new	.02 short	.02 moderate
-.02 old	.02 relatively new	-.01 surplus	-.02 independent
	-.03 old		

To estimate startup time, a similar expression is employed with A equal to construction time. Following the sequence at the top of this page, B is .15 or .05 or -.01; C is .15 or .08 or .05 or -.01; D is .15 or .05 or -.01; E is .25 or I or -.02. Unfortunately, the data necessary to determine factors specific for mining and mineral processing startup are not readily available. Such factors as winter versus summer startup can be introduced if necessary.

Useful Prior Information

Regardless of the method being employed to estimate capital expenditures, prior technical information will be necessary. Specific details on the mining method (s), the basic flowsheet for milling, material/energy balances, major equipment necessary, approximate equipment sizes/capacities, infrastructure envisaged and other details may be necessary.

Appropriate information for a given capital cost estimation method used in this handbook is listed with the section that describes the method.

CAPITAL COST ESTIMATIONS FOR MINING

Capital cost estimations in mining usually involve cost versus parameter relations, where the parameter is, say, the capacity. A number of methods, where capacity parameters are coupled with other important relationships, are summarized in this section.

Using The 0.6 Rule For Mines and Mills/Example

O'Hara (see **CIM Bulletin**, February, 1980) has employed the 6/10 rule to estimate the combined cost of a mine/mill project. His data were converted to US dollars and then adjusted from a base year M&S(Mine/Mill) index of 565 for 1978 to a new base of 1400. Thus the correction ratio, 1400/565, applied as a multiplier results in expressions involving T_m , milling capacity (tons/day):

$$\text{Capital Cost} = 869{,}239 T_m^{0.6} \quad \text{\$ US (open pit/mill project)}$$

$$\text{Capital Cost} = 1{,}738{,}478 T_m^{0.6} \quad \text{\$ US (underground/mill project)}$$

The above expressions serve as very crude guides only. The costs of many projects differ widely from the average values predicted by the equations. The point to retain is that the 0.6 rule for estimating the cost of a mine/mill, as noted by O'Hara, should be used only as a very rough estimate. Predictions will improve by quantifying the effects of specific local conditions

such as high stripping ratios, vein thickness underground, complexity of mineral processing, degree of isolation and many other factors.

Example Calculation

Estimate the capital cost of an open pit/mill complex as of May, 1997. The design capacity of the mill is 20,000 TPD. From Table 2, page 6, the M&S(Mine/Mill) index is 1087. Thus:

$$\text{Capital Cost} = \left[\frac{1087}{1400}\right](869,239)[20,000]^{0.6} = \text{US\$ } 256,955,968$$

Fixed Capital Cost Estimation For Open Pit Mines/Example (O'Hara Method)

O'Hara stresses that a more accurate estimate of mine capital costs is obtained by judging the influence of specific conditions unique to a given mine. For open pit mines, capital costs were distributed as follows:

(1) Site Preparation; Plant and pit roads.
(2) Pre-Production Stripping Costs.
(3) Open pit equipment - Sizing and Costs.
(4) Open Pit Maintenance Facilities.
(5) Electric Power Supply/Distribution and Water Supply part of Processing Plant; General Plant Services, Access Road to Main Thoroughfare, Townsite and Housing estimated as Infrastructure.
(6) Feasibility Studies; Design Engineering; Planning
(7) Project Supervision; Contract Management; Expediting & General Construction Facilities including Camp Costs.
(8) Administration; Accounting; Legal; Pre-Production Employment of key operating staff.

Cost-Parameter expressions were developed for items (1) to (4); item (5) is estimated as part of processing or as part of infrastructure costs; cost items (6) to (8) may be expressed as percentages of the others. Table 4 summarizes the approach. Note that the pre-production stripping ratio, the number of trucks required and the number of shovels necessary must be estimated. Stripping ratios and tonnages of overburden to remove can be approximated from analyses of drill core available from exploration of the orebody.

To estimate the shovel size, S, in cubic yards, use $S = .13\ T^{0.4}$, where T = ton/day of ore+waste to be mined, and round off S to the nearest standard size. Then calculate N_s, the number of shovels to employ, from $N_s = .007\ (1/S)T^{0.8}$ where N_s is rounded to the next whole number.

The size of a truck required may be estimated from $t = 8\ S^{1.1}$ with t expressed in tons of capacity per truck. The number of trucks is found from $N_t = .2\ (1/t)T^{0.8}$ with t rounded to the nearest standard truck size and N_t to the next whole number.

Example Calculation

Assume that an open pit mine is located in rugged terrain with heavy tree growth. Soil will be stripped at a rate of 25,000 tons per day for 60 working days, whereas rock overburden will be stripped at a rate of 12,500 tons per day for 300 working days. The amount of ore plus waste

TABLE 4: Summary of Open Pit Mine Capital Cost Estimation*

Cost Item	Cost Parameter	Graph Range	Cost Equation	Comments
1) Site preparation; Plant/Pit Roads.	T = ton/day of ore+waste	10^3 to 10^5	$C_{11} = 4346\ T^{0.5}$ $C_{12} = 10866\ T^{0.5}$	flat terrain; lightly treed rugged terrain; heavily treed
2) Preproduction Stripping.	T_0= tons of overburden	10^4 to 10^7	$C_{21} = 1739 T_0^{0.5}$	for soil overburden
			$C_{22} = 18471 T_0^{0.5}$	for rock overburden
3) Open pit Equipment.	S = shovel size, cu.yd	4 to 15	$C_{31} = 499813\ N_s\ S^{0.73}$	shovel fleet; N_s = no. shovels
	t = truck size, tons	35 to 150	$C_{32} = 19558\ N_t\ t^{0.85}$	truck fleet; N_t= no. trucks
			$C_{33} = 1134575\ N_s\ S^{0.73}\ T^{-0.2}$	drilling equipment
4) Maintenance Facilities.	T = ton/day of ore+waste	10^3 to 10^5	$C_4 = 325964\ T^{0.3}$	includes maintenance equipment

5) Electric Power Supply/Distribution and Water Supply part of Processing Plant; General Plant Services, Access Road to Townsite and MainThoroughfare and Housing estimated as infrastructure.

6)Feasibility, Engineering, Planning.	4 to 6% of $(C_{11} + C_{12} + C_{21} + C_{22})$ plus 6 to 8% of $(C_{31} + C_{32} + C_{33} + C_4)$
7)Supervision, Management,Camp Construction Facilities.	8 to 10% of $(C_{11} + C_{12} + C_{21} + C_{22} + C_{31} + C_{32} + C_{33} + C_4)$
8)Admin, Acct, Key Staff, Legal.	4 to 7% of $(C_{11} + C_{12} + C_{21} + C_{22} + C_{31} + C_{32} + C_{33} + C_4)$

*Adapted from O'Hara (CIM Bulletin, Feb, 1980)

to be mined per day for normal operation is 25,000, where mill design capacity is 10,000 tons per day (i.e., a stripping ratio of 2.5). Estimate the capital cost of the mine from Table 4 when the M&S (Mine/Mill) index is at 1060.

From Table 4:

(1) $C_{11} = 0$
$C_{12} = (1060/1400)(10866)(25000)^{0.5}$ = $1,300,821

(2) $C_{21} = (1060/1400)(1739)(25000 \times 60)^{0.5}$ $1,612,587
$C_{22} = (1060/1400)(18471)(12500 \times 300)^{0.5}$ $27,082,196

$S = .13(25000)^{0.4} = 7.5$ (use 8 cu. yd. shovels)
$N_s = .007(1/8)(25000)^{0.8} = 2.9$ (use 3 shovels)
$t = 8(8)^{1.1} = 78.8$ ton (use 85 ton trucks)
$N_t = 0.2(1/85)(25000)^{0.8} = 7.8$ (use 8 trucks)

(3) $C_{31} = (1060/1400)(499813)(3)(8)^{0.73}$ = $5,180,388
$C_{32} = (1060/1400)(19558)(8)(85)^{0.85}$ = $5,171,285
$C_{33} = (1060/1400)(1134575)(3)(8)^{0.73}(25000)^{-0.2}$ = $1,551,672

(4) $C_4 = (1060/1400)(325964)(25000)^{0.3}$ = $5,149,078

Items **(1) + (2) + (3) + (4)** = $47,048,027

(6) .04**((1) + (2))** plus .06**((3) + (4))** $2,222,970

(7) .08**((1) + (2) + (3) + (4))** = $3,763,842

(8) .05**((1) + (2) + (3) + (4))** = $2,352,402

Fixed Capital Cost = **((1) + (2) + (3) + (4) + (6) + (7) + (8)) = $US 55,387,241**

Fixed Capital Cost Estimation For Underground Mines/Example (O'Hara Method)

For underground mines, O'Hara itemizes fixed capital costs that may be divided into the following categories:

(1) Shaft Sinking
(2) Mine Development (ore reserve tonnage equal to 2000 times daily mill tonnage)
(3) Hoisting Plant
(4) Compressor Plant
(5) Underground Mining Equipment
(6) Underground Mine Maintenance Facilities
(7) Electric Power Supply/Distribution and Water Supply part of Processing Plant; General Plant Services, Access Road to Main Thoroughfare, Townsite and Housing estimated as Infrastructure.
(8) Feasibility, Design Engineering, Planning

(9) Project Supervision, Management, Expediting and General Construction Facilities including Camp Costs
(10) Administration, Accounting, Legal, Key Operating Staff Employment

Cost versus Parameter equations were developed for items (1) to (6); item (7) is estimated as part of the processing or as infrastructure; items (8) to (10) are estimated as percentages of others. Table 5, page 19, is a summary of the estimation procedure where cost equations have been updated to an M&S (Mine/Mill) index of 1400.

The rectangular shaft area, A in sq. ft., is estimated from $A = 17\ T^{0.33}$ where T is the daily tonnage of ore hoisted. For circular shafts, $D_s = 5.2\ T^{0.15}$, where D_s is shaft diameter in ft.

Hoist drum diameter is calculated from: $D = (40\ T+100\ h^{0.5}T^{0.6}+ h^{0.3}T^{1.2})^{0.357}$, where T is ton/day hoisted and h is hoisting distance in ft. The height of the headframe complex, L in ft, is found from the fitted expression: $L = .25\ D + 5.5\ (D/100)^3 + 6.3\ T^{0.33}$. Hoisting speed, S in ft/min, is estimated from $S = 1.6\ h^{0.5}T^{0.4}$. Once S is known, the hoist motor horsepower, H, is determined from $H = .5\ S(D/100)^{2.4}$. Approximately the weight of structural steel in a headframe, lbs, where hoisting ropes are 1/80 of the hoist drum diameter, will be $.12\ h^3\ (D/100)^2$. To estimate the air consumed underground, the equation $Q = 200T^{0.46}$ is employed, where Q is cu ft/min and T is ton/day mined.

Example Calculation

Estimate the fixed capital cost of an underground mine that will mine 5000 ton/day. Up to 6000 ton/day may be hoisted, using double drum hoists (one for ore and the other for cage service), in a circular shaft of 2000 ft maximum depth. Average stope widths are estimated at 30 ft. Note that circular shafts are for weak and incompetent rock. Assume that the M&S(Mine/Mill) index is 1060.

Find: $D_s = 5.2(6000)^{0.15} = 19.1$ ft (say 19 ft)

(1) $C_{11} = 0$
$C_{12} = (1060/1400)(173848(19)^{0.5}+ 496(2000)(19)^{0.7}) =$ \$6,473,277

(2) $C_2 = (1060/1400)(86924)(5000)(30)^{-0.8} =$ \$21,656,613

Find: $D = (40(6000) +100(2000)^{0.5}(6000)^{0.6} + (2000)^{0.3}(6000)^{1.2})^{0.357}$
$= 156$ inch drum

Find: $L = .25(156) + 5.5(156/100)^3 + 6.3(6000)^{0.33}$
$= 171$ ft of headframe

Find: $S = 1.6(2000)^{0.5}\ (6000)^{0.4} = 2322$ fpm

Find: $H = .5(2322)(156/100)^{2.4} = 3375$ HP

(3) $C_{31} = (1060/1400)(740)(3375)^{0.2}\ (156)^{1.4} =$ \$3,345,328

TABLE 5: Summary of Underground Mine Capital Cost Estimation*

Cost Item	Cost Parameter	Graph Range	Cost Equation	Comments
1) Shaft sinking.	A = cross sect. area, sq. ft.	125 - 320	$C_{11} = 130387\ A^{0.25} + 184\ FA^{0.45}$	rectangular; sets blocked to rock; F = ft. of shaft sunk.
	D = diameter,ft.	13 - 20	$C_{12} = 173848 D_s^{0.5} + 496 F D_s^{0.7}$	circular, concreted shafts
2) Mine development.	T = ton/day, mined	500 - 7000	$C_2 = 86924\ TW^{-0.8}$	W = avg. width stope, ft. development reserves = 2000T
3) Hoist plant.	D = hoist drum diameter, inch	72 - 168	$C_{31} = 740\ H^{0.2} D^{1.4}$	hoist equipment; H = motor HP
			$C_{32} = 88\ D^{1.8}$	hoist installation cost
			$C_{33} = .204\ D^{3.2}$	hoist room cost
			$C_{34} = 1.033\ L^{1.8} D^{1.2}$	headframe complex; L= ft of height
4) Mine compressor plant.	Q = cfm of air	2000 - 12000	$C_{41} = 805\ Q^{0.8}$	compressor equipment cost
			$C_{42} = 381\ Q^{0.7}$	compressor installation cost
5) Underground mining equip	T = ton/day mined	500 - 7000	$C_5 = 36943\ W^{-0.3} T^{0.8}$ $C_5 \approx 651937^{0.6}$(large stopes)	includes installation cost W = stope width, ft (10 to 50)
6) Underground maintenance facilit.	T = ton/day mined	500 - 7000	$C_6 = 30424\ T^{0.5}$	

7) Electric Power Supply/Distribution and Water Supply part of Processing Plant; General Plant Services, Access Road to Main Thoroughfare, Townsite and Housing estimated as infrastructure.

8) Feasibility, design — 4 to 6% of $(C_{11} + C_{12} + C_2)$ plus 6 to 8% of $(C_{31} + C_{32} + C_{33} + C_{34} + C_{41} + C_{42} + C_5 + C_6)$

9) Supervision, camp — 8 to 10% of $(C_{11} + C_{12} + C_2 + C_{31} + C_{32} + C_{33} + C_{34} + C_{41} + C_{42} + C_5 + C_6)$

10) Admin., Acct. legal, key staff — 4 to 7% of $(C_{11} + C_{12} + C_2 + C_{31} + C_{32} + C_{33} + C_{34} + C_{41} + C_{42} + C_5 + C_6)$

*After O'Hara(CIM Bulletin, Feb, 1980)

C_{32} = (1060/1400)(88)(156)$^{1.8}$ = $590,588
C_{33} = (1060/1400)(.204)(156)$^{3.2}$ = $1,609,929
C_{34} = (1060/1400)(1.033)(171)$^{1.8}$(156)$^{1.2}$ = $3,390,942

(4) C_{41} = (1060/1400)(805)(10000)$^{0.8}$ = $965,993
C_{42} = (1060/1400)(381)(10000)$^{0.7}$ = $182,013

(5) C_5 = (1060/1400)(36943)(30)$^{-0.3}$(5000)$^{0.8}$ = $9,178,033

(6) C_6 = (1060/1400)(30424)(5000)$^{0.5}$ = $1,628,843

(1)+(2) = 28,129,890 and **(3)+(4)+(5)+(6)** = 20,891,669

(8) .06(28129890)+.08(20891669) = $3,359,127

(9) .1(28129890 +20891669) = $4,902,156

(10) .07(28129890 +20891669) = $3,431,509

Fixed Capital Cost **=(1) + (2) + (3) + (4) + (5) + (6) + (8) + (9) + (10)= $US 60,714,351**

Small Underground Mines and Processing Plants: CANMET SP 86-11E

CANMET sponsored J. S. Redpath Limited, North Bay, Ontario, to prepare a manual for estimating costs of mining and processing small underground deposits. Both capital and operating costs were estimated in some detail for mining, although associated processing costs are very general (a cost-capacity curve for the concentrator and another for the tailings disposal area). Methodology can be useful. A sensible calculation procedure, which uses a series of blank forms to be filled out via computational procedures presented in the body of the text, is recommended. Calculation forms are:

Form 1 Basic Information
Form 2 Operating Costs
- (A) Operating Costs
- (B) Manpower Schedule

Form 3 Capital Costs
- (A) Preproduction Capital Costs
- (B) Ongoing Capital Costs

Form 4 Regional Cost Factors
Form 5 Mineral Deposit Value
- (A) Summary Of Geological Tonnes & Grade
- (B) Geological Tonnes & Grade - Reserve By X-Section
- (C) Summary Of In-Situ Tonnes & Grade
- (D) In-Situ Tonnes & Grade - Reserve By X-Section
- (E) Mineable Tonnes & Grade To Mill
- (F) Mineral Deposit Value

Form 6 Preliminary Cash Flow Summary

Graphs and Tables are employed for estimation purposes. Procedures are relevant to North America and must be updated via cost indexes from about 1985.

Small Placer Mines and Processing Plants: USBM IC 9170

Stebbins has written the **Cost Estimation Handbook for Small Placer Mines**. It was published as IC 9170 by the US Bureau of MInes in 1987. It relies on prior estimation strategies developed, initially, by STRAAM Engineers, and published in updated form as a **Bureau of Mines Cost Estimating System Handbook: Part 1. Surface and Underground Mining, IC 9142; Part 2. Mineral Processing, IC 9143**.

The handbook has filled a gap, in that small placer mines are relatively unique. Mining and processing costs therein are representative of operations in the Western US and Alaska. The latter is typical of Canadian placer operations in the Klondike and Yukon Territories. To estimate capital costs, fitted equations (each of which may involve one or more multiplying factors) are provided for the following:

Capital Costs Equations Supplied For Placer Mining:

Exploration:
- Panning
- Churn drilling
- Bucket drilling
- Trenching
- General reconnaissance
- Camp costs
- Seismic surveying
- Rotary drilling
- Helicopter rental

Development:
- Access roads
- Clearing

Preproduction overburden removal:
- Bulldozers
- Draglines
- Front-end loaders
- Rear-dump trucks
- Scrapers

Mine equipment:
- Backhoes
- Bulldozers
- Draglines
- Front-end loaders
- Rear-dump trucks
- Scrapers

Processing equipment:
- Conveyors
- Feed hoppers
- Jig concentrators
- Sluices
- Spiral concentrators
- Table concentrators
- Trommels
- Vibrating screens

Supplemental:
- Buildings
- Employee housing
- Generators
- Pumps
- Settling ponds

In addition, fitted equations are supplied in the Stebbins handbook, along with corresponding multiplying factors, for each of the following to estimate operating costs:

Operating Cost Equations Supplied For Placer Mining:

Overburden Removal:
- Bulldozers
- Draglines
- Front-end loaders
- Rear-dump trucks
- Scrapers

Mining:
- Backhoes
- Bulldozers
- Draglines
- Front-end loaders
- Rear-dump trucks
- Scrapers

Tailings Removal:
- Bulldozers
- Draglines
- Front-end loaders
- Rear-dump trucks
- Scrapers
- Trommels
- Vibrating screens

Processing:
- Conveyors
- Feed hoppers
- Jig concentrators
- Sluices
- Spiral concentrators
- Table concentrators

Supplemental:
- Employee housing
- Generators
- Lost time and general services
- Pumps

All capital and operating cost equations were calculated in January, 1985 $US. Methodology to update costs is described in the manual. Site adjustment factors are used and labor rates are relative to a base of $15.69 per hour for mining and $15.60 per hour for milling.

CAPITAL COST ESTIMATIONS FOR MINERAL PROCESSING

The O'Hara method and the popular ratio and factored estimation techniques to determine capital expenditures for processing plants are described with corresponding examples. Infrastructure costs are estimated in a separate section.

Capital Cost Estimation For Processing Plants/Example (O'Hara Method)

O'Hara has itemized mineral processing plant fixed capital costs which may be divided into the following categories:

(1) Plant-Site Clearing and Mass Excavation
(2) Concrete Foundations and Detailed Excavations
(3) Crushing Plant, Coarse Ore Storage, Conveyors
(4) Concentrator Building
(5) Grinding Section, Fine Ore Storage

(6) Flotation and/or Processing Section
(7) Thickening and Filtering Section
(8) Concentrate Storage and Loading
(9) Tailings Storage
(10) Electric Power Supply and Distribution (Mine/Mill)
(11) Water Supply (Mine/Mill)
(12) General Plant Services, Access Road to Main Thoroughfare, Townsite and Housing estimated as Infrastructure.
(13) Feasibility Studies, Design Engineering, Technical Planning
(14) Project Supervision, Contract Management, Expediting and General Construction Facilities including Camp Costs
(15) Administration, Accounting, Legal, Pre-Production Employment of Key Operating Staff

Cost-Parameter equations were developed for all items other than items (11) to (13) which are calculated as percentages of the others.

Table 6, page 24, is a summary of the estimation procedure. To use Table 6, values of cost equation factors must be employed. These are presented below.

Factor	Value	Application
F_s = site factor	1.0	flat sites; less than 10 ft of overburden
	1.5	moderate slopes; some blasting required
	2.5	steep slopes; extensive blasting required
F_c = rock factor	1.0	solid rock for foundation support
	1.8	gravel/sand as support
	3.5	moist soil as support; piled foundations
F_w = climate factor	1.0	mild climate
	1.8	cold climate
	2.5	severe climate
F_g = grind factor	1.0	soft ores; 55% -200 mesh; work index under 12
	1.5	medium ores; 70% -200 mesh; work index = 15
	1.8	hard ores; 80% -200 mesh; work index = 17
F_p = process factor	1.0	Au ores; cyanidation
	1.2	flotation; coarse low grade Cu ores
	1.6	flotation; hi grade Cu/Zn ores
	2.0	selective flotation; complex base metal ores
	3.0	complex Au ores; float, roast, cyanide
	5.0	gravity concentration
F_t = process factor	1.0	low grade Cu ores
	1.6	hi grade Cu/Zn ores
	2.0	complex Pb/Zn/Ag or Cu/Zn/Pb ores
	3.0	cyanided Au ores

TABLE 6: Summary of Mineral Process Plant Capital Cost Estimation*

Cost Item	Parameter	Graph Range	Cost Equation	Comment
1) Clear/excav.	T=capacity, tpd	500-7000	$C_1 = 86924\ F_s\ T^{0.3}$	F_s = site factor
2) Foundation	T=capacity, tpd	500-7000	$C_2 = 43463\ F_c\ T^{0.5}$	F_c = rock factor
3) Crush/conv.	T=capacity, tpd	500-7000	$C_3 = 97790\ T^{0.5}$	
4) Mill bldg.	T=capacity, tpd	500-7000	$C_4 = 65193\ F_w\ T^{0.5}$	F_w = climate factor
5) Grind/storage	T=capacity, tpd	500-7000	$C_5 = 17386\ F_g\ T^{0.7}$	F_g = grind factor
6) Flotation/etc.	T=capacity, tpd	500-7000	$C_6 = 5433\ F_p\ T^{0.7}$	F_p = processing factor
7) Thicken/filt.	T=capacity, tpd	500-7000	$C_7 = 10866\ F_t\ T^{0.5}$	F_t = thickening factor
8) Con. storage	T_c=con., tpd	20-500	$C_8 = 8693\ T_c^{0.8}$	
9) Tail pond	T=capacity, tpd	500-700	$C_9 = 6520\ T^{0.5}$	Dam; flat terrain
10) Power, lines	P=peak load kw	2000-30000	$C_{101} = 99963\ P^{0.6}$	coal-fired generator
			$C_{102} = 9780\ P^{0.6}$	diesel generator
	M=miles of lines =0 if paid by utility		$C_{103} = 761P^{0.8} + 13038M$	utility substation
			$C_{104} = 1305\ P^{0.8}$	low volt lines
11) Water	Q = water IGPM L = miles pipe	500-6500	$C_{111} = 761\ LQ^{0.9}$	pipe costs
			$C_{112} = 4999\ Q^{0.6}$	fresh water pumps
			$C_{113} = 6520\ Q^{0.6}$	reclaim water pumps

12) Plant services, access roads, townsite, housing estimated as part of infrastructure. Item 10) and Item 11) include mine requirements as well.

13) Feasibility, plan,design [4 to 6% of ((1)+(2))] + [6 to 8% of (sum of items (3) to (11))]

14) Supervise/camp 8 to 10% of sum of items (1) to (11)

15) Admin, staff 4 to 7% of sum of items (1) to (11)

*Adapted from O'Hara (CIM Bulletin, 1980)

Water requirements for Table 6 calculations are determined from the table below:

Q, IGPM	Calculated From:	Application
Fresh water	$Q = 12\ T^{0.6}$	plentiful supply; 1 mile away
Fresh water	$Q = 2.5\ T^{0.6}$	scarce supply; open pit, hi tonnage
Reclaim water	$Q = .026\ T^{1.2}$	reclaim when fresh supply scarce

Pipe diameters, D_p in inches, are estimated from $D_p = .15\ Q^{0.6}$ for either fresh or reclaim water.

To estimate the total number of employees, N, refer to the section on Mine-Mill Total Product Cost Estimation (O'Hara Method)-----see Table 13, page 39.

Peak power loads, KW, are found from the following: for open pit/mill complexes, $P = 136\ T^{0.5}$, while for underground/mill complexes, $P = 27\ T^{0.7}$ with T in mill capacity, ton/day.

Concentrate tonnage is calculated as the mill capacity, ton/day, multiplied by recovery and the ratio of feed grade over concentrate grade.

Example Calculation

Estimate the fixed capital cost (at an M&S(Mine/Mill) index of 1060) of a mill, associated with an underground mine, that will treat 5000 ton/day of a 2% Cu ore (Bond work index = 15 kwhr/ton) and that will produce a concentrate averaging 30% Cu with 92% recovery. A blast hole mining method is contemplated with stopes averaging 30 ft width. An access road will be 10 miles in length; two 40-ft bridges will be necessary. Fresh water is scarce at a distance of 3 miles; reclaim water must be pumped from the tailing pond located 1 mile from the mill. The mill is to be sited on solid rock at moderate slope in a cold climate. Power is obtainable from a utility company and the company agrees to pay for high voltage lines to the substation to be located nearby. A total of 392 employees are required with a subsidized family townsite.

Referring to Table 6, page 24, and to the tables at the bottom of page 23 and the top of page 25:

(1) $C_1 = (1060/1400)(86924)(1.5)(5000)^{0.3} =$ $1,270,866

(2) $C_2 = (1060/1400)(43463)(1)(5000)^{0.5} =$ $2,326,926

(3) $C_3 = (1060/1400)(97790)(5000)^{0.5} =$ $5,235,489

(4) $C_4 = (1060/1400)(65193)(1.8)(5000)^{0.5} =$ $6,282,555

(5) $C_5 = (1060/1400)(17386)(1.5)(5000)^{0.7} =$ $7,669,160

(6) $C_6 = (1060/1400)(5433)(1.6)(5000)^{0.7} =$ $2,556,328

(7) $C_7 = (1060/1400)(10866)(1.6)(5000)^{0.5} =$ $ 930,792

T_c = .92(5000)C2/30) = 307 ton/day of concentrates

(8) $C_8 = (1060/1400)(8693)(307)^{0.8}$ = $ 642,774

P = $27(.5000)^{0.7}$ = 10487 KW peak load (underground/mill complex)

(9)) $C_9 = (1060/1400)(6520)(5000)^{0.5}$ = $ 349,068

(10) $C_{101} = 0$
$C_{102} = 0$
$C_{103} = (1060/1400)(761(10487)^{0.8} + 0)$ = $ 948,601
$C_{104} = (1060/1400)(1305)(10487)^{0.8}$ = $1,626,707

$D_p = .15(849)^{0.6}$ = 8.6 (use 9 inch fresh water line of 3 mile length)
$D_p = .15(714)^{0.6}$ = 7.7 (use 8 inch reclaim water line, 1 mile length)

$Q = 2.5(5000)^{0.6}$ = 415 IGPM fresh water, scarce supply
$Q = .026(5000)^{1.2}$ = 714 IGPM reclaim water

(11) $C_{111} = (1060/1400)(761(4)(849)^{0.9}$ = $ 996,872
$C_{112} = (1060/1400)(4999)(415)^{0.6}$ = $ 140,893
$C_{113} = (1060/1400)6520)(714)^{0.6}$ = $ 254,475

(1) + (2) = $3,597,792
(3)+(4)+(5)+(6)+(7)+(8)+(9)+(10)+(11) = $27,634,314

(13) .05(3,597,792) + .07(27,634,314) = $2,114,292
(14) .09(3,597,792 + 27,634,314) = $2,810,890
(15) .06(3,597,792 + 27,634,314) = $1,873,927

Fixed Capital Costs = Sum of all items = US $38,031,215

Capital Cost Estimation For Processing Plants Via Cost Ratio Methods

Balfour and Papucciyan (**CMP Proceedings**, 1972, Ottawa) describe four ways to estimate fixed capital costs. These are (1) Six-Tenths Rule, (2) Plant Cost Ratio Method, (3) Equipment Cost Ratio Method and (4) Plant Component Cost Ratio Method. To use methods (2), (3) and (4) the following information should be available: a rough flowsheet showing major items of equipment and their corresponding sizes along with sufficient information to estimate plant size and complexity. Material balances and energy balances will be of value.

Methods (3) and (4) are critically dependent upon cost information files and records of actual costs. Class II accuracy is obtainable, provided factors employed are accurate.

(1) Six-Tenths Rule

To use the six-tenths rule, the fixed capital cost of a plant of known capacity must be available.

In such cases:

$$\frac{(Cost)_2}{(Cost)_1} = \left[\frac{(Capacity)_2}{(Capacity)_1}\right]^{0.6}$$

For example, let a 10,000 TPD concentrator cost $\$60 \times 10^6$ to build in 1989, when the M&S(Mine/Mill) index is 911 (see Table 3, page 7). Suppose that the cost of a similar 20,000 TPD plant is desired at an M&S(Mine/Mill) Index of 1060.

Using the 0.6 rule:

$$(Cost)_2 = (60 \times 10^6)\left[\frac{20000}{10000}\right]^{0.6}$$

= \$90.94 x 10^6 at an M&S (Mine/Mill) index of 911.

To update the cost, the current cost index of 1060 must be employed. The cost index expression shown on page 4, can be applied such that:

$$(Cost)_{now} = (90.94 \times 10^6)\left[\frac{1060}{911}\right]$$

= \$105.82 x 10^6 .

The exponent, 0.6, is an average and in fact depends on the type of plant. Some estimators prefer to use a 0.7 rule as described in a previous section. Factors such as type of site, economic conditions prevalent, geographic location and regional productivity are responsible for substantial variation.

(2) Plant Cost Ratio Method

This method requires an estimate of the cost of major items of **delivered** major process equipment. If this cost is equal to N (often inflated to account for auxiliary items) then:

Cost of solid process plant	= 3.10 (N)
Cost of solid-fluid process plant	= 3.63 (N)
Cost of fluid process plant	= 4.74 (N)

Multipliers above (Lang factors) can be broken down into parts as follows:

Multiplier	F/I	P	C/B	OH
3.1	1.43	1.10	1.50	1.31
3.63	1.43	1.25	1.50	1.35
4.74	1.43	1.64	1.50	1.35

where F/I represents foundations, supports, chutes, installation; P represents piping costs; C/B represents construction, engineering, building; OH represents overhead. The procedure assumes that reasonably accurate estimates of equipment costs are available and nothing is "abnormal".

Delivered costs have been estimated as 1.03 times purchased costs, while **installed** costs are sometimes estimated as **Q times delivered** costs. The value Q is 1.45 for solids plants; 1.39 for solids-fluids plants; 1.47 for fluids plants (note: average is 1.43). **An example** of the plant cost ratio method: The current cost of **delivered** equipment items for a 20,000 tpd primary crushing station happens to be $\$1.4 \times 10^6$, so that the cost of the installation is estimated as:

$$\text{Cost} = 3.10\ (1.4 \times 10^6) = \mathbf{\$4.34 \times 10^6}$$

(3) Equipment Cost Ratio Method

Compared with estimating method **(2)**, page 27, more accuracy results from multiplying categories of equipment of similar nature by corresponding ratio factors and then calculating the resulting sum. Let the cost of the ith major process unit be C_i . Then a "plant" cost associated with item i is equal to $F_i\ C_i$ where F_i is an appropriate factor. The cost of the total plant is then:

$$\text{Plant Cost} = \sum_{i=1}^{n} F_i C_i$$

for a plant containing n items of major equipment and key auxiliaries.

Table 7 below shows typical F_i values (see Balfour and Papucciyan). Such tables may be expanded after several projects have been analyzed.

Table 7: Equipment Cost Ratios

Equipment Category	Factor, F_i
Bucket elevator	2.0
Mixer	2.0
Furnace	2.1
Drum dryer	2.2
Kiln	2.2
Conveyor	2.3
Compressor	2.3
Electrostatic precipitator	2.5
Blower, fan	2.5
Refrigeration unit	2.5
Boiler	2.8
Mills	3.0
Vacuum rotary dryer	3.2
Dry dust collector	3.5
Storage tank	3.5
Crusher	3.5
Process tank	4.1
Instrumentation	4.1
Heat exchanger	4.8
Wet dust collector	6.0
Pump	6.8
Electric motor	8.5

(4) Plant Component Cost Ratio Method

This method provides considerable flexibility and involves a breakdown of fixed capital costs into plant components whose costs are a ratio of major equipment costs. The ratios are sometimes referred to as factors (i.e., factored estimates).

Balfour and Papucciyan provide case examples of the plant component cost ratio method as shown below in Table 8 for various types of processing plants. It is of interest to note from the table that the fixed capital cost of an asbestos plant, a zinc refinery and a copper refinery respectively is 2.43, 2.28 and 3.24 times the corresponding major equipment costs.

A generalized version of Table 8 is given in Table 9, page 30, which shows ranges for cost ratios. This table is typical of most plant component cost ratio methods.

Table 10, page 31, is an example of the plant component cost ratio method suggested for process plants and used with success in chemical engineering.

Table 8: Typical Plant Component Cost Ratios For Processing Plants

Plant Component	Asbestos Plant Cost relative to equipment	Zinc Refinery Cost relative to equipment	Copper Refinery Cost relative to equipment
Equipment	1.00	1.00	1.00
Equipment installation	0.22	0.21	0.19
Process Piping	0.05	0.03	0.17
Electrical	0.20	0.10	0.32
Instrumentation	0.06	0.03	0.11
Process buildings	0.37	0.34	0.62
Auxiliary buildings	0.07	0.10	0.09
Plant services	0.05	0.09	0.15
Site improvements	0.02	0.07	0.13
Field indirects	0.13	0.07	0.09
Project management	0.26	0.24	0.37
Total:	2.43	2.28	3.24

Table 9, page 30, can be extremely flexible. Such components as geographic location (desert versus far North), type of industry (Cu versus Au versus Cu-Ni), type of labour pool and other components are readily incorporated as desired, when cost files of the correct type can be evaluated.

TABLE 9: Plant Component Cost Ratio Method (patterned after Balfour and Papucciyan)

1. Delivered equipment costs from references and on current cost index basis.
   (if delivery costs unavailable use 1.03 times purchased equipment costs).....$000,000

2. Equipment installation.
   (0.17 to 0.25 times Item 1) $000,000

3. Piping, material and labour, excluding service piping.
   (0.07 to 0.25 times Item 1) $000,000

4. Electrical, material and labour, excluding building lighting.
   (0.13 to 0.25 times Item 1) $000,000

5. Instrumentation.
   (0.03 to 0.12 times Item 1) $000,000

6. Process buildings, including mechanical services and lighting.
   (0.33 to 0.50 times Item 1) $000,000

7. Auxiliary buildings, including mechanical services and lighting.
   (0.07 to 0.15 times Item 1) $000,000

8. Plant services such as fresh water systems, sewers, compressed air, etc.
   (0.07 to 0.15 times Item 1) $000,000

9. Site improvements such as fences, roads, railroads, etc.
   (0.03 to 0.18 times Item 1) $000,000

10. Field expenses related to construction management.
    (0.10 to 0.12 times Item 1) $000,000

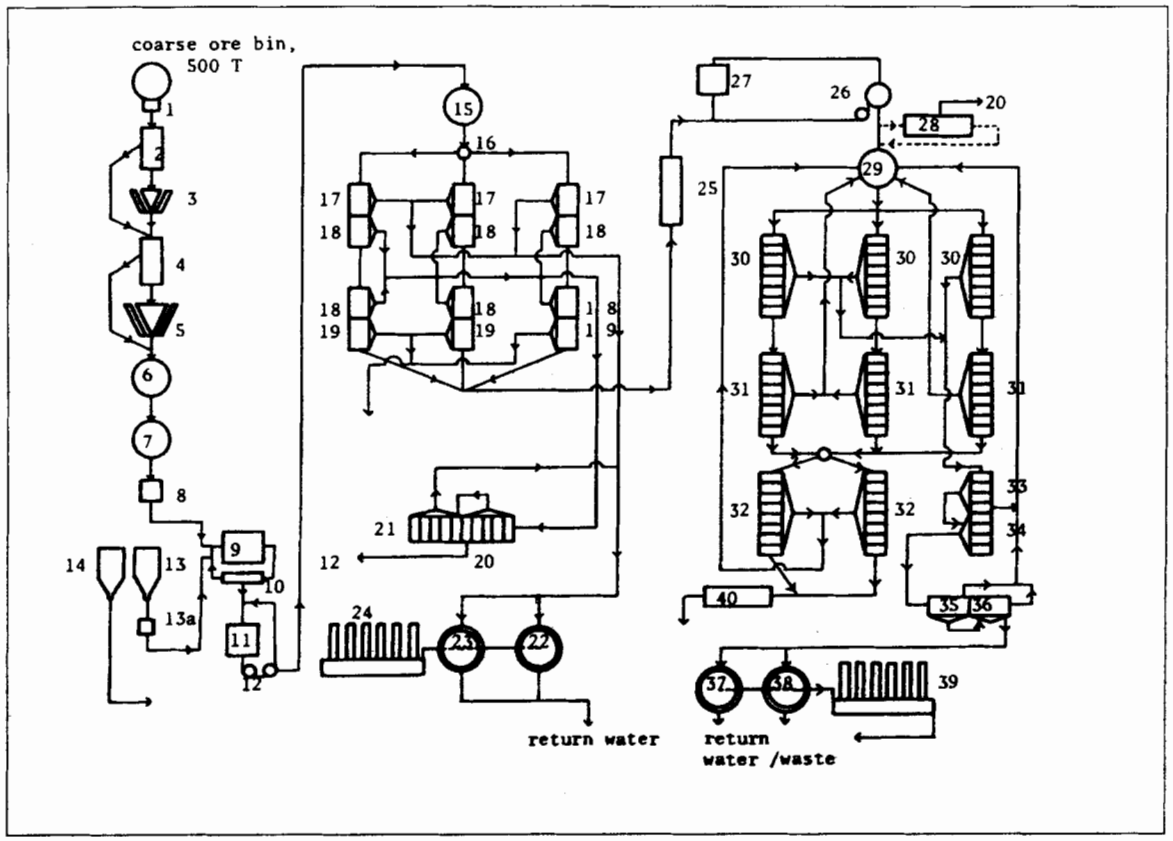
11. Project management including engineering and construction.
    (0.30 to 0.33 times Item 1) $000,000

12. Fixed capital costs = 1+2+3+4+5+6+7+8+9+10+11 .....$000,000

Example of Ratio Estimation For Mineral Processing

Consider the flowsheet shown in Figure 2, page 32, where each process unit has been numbered. This flowsheet is typical of the type found in the literature.

A superficial glance indicates that adequate information is available for estimation purposes, while close examination will show that in many cases important details are missing. For example, in Item 41, an equipment parameter such as pump capacity is not specified. In Item 13, the material of construction has been omitted. It is important to specify those equipment parameters, such as capacity and materials of construction, which have been related to the cost of the process unit. Before attempting a fixed capital cost estimate using ratio methods,



1. 1 - 36"x60" vibrating feeder
2. 1 - 4'x8; single rod deck screen
3. 1 - 4-1/4' standard cone crusher
4. 1 - 5'x8' single rod deck screen
5. 1 - 4' short head cone crusher
6. 1 - 32'x32' open steel ore bin
7. 1 - 32'x32' open steel ore bin
8. 1 - weightometer
9. 1 - 9'xl2' rod mill
10. 1 - 54" classifier
11. 1 - 7'x11' ball mill
12. 2 - 20-B cyclones
13. 1 - 20'xl8' elevated soda ash storage tank & pneumatic conveyor
14. 1 - bulk line storage
15. 1 -18'x8' super conditioner
16. 1 - 5'x3-way distributor
17. 12 - primary #24 roughers
18. 24 - secondary #24 roughers
19. 12 - tertiary #24 scavenger
20. 5 - primary #24 cleaners
21. 5 - secondary #24 cleaners
22. 1 - 30'xl0' thickener
23. 1 - 30'xl0' thickener
24. 1 - 9'x6' disc filter
25. 1 - 4'x30' blanket table
26. 2 - 20-B cyclones
27. 1 - 9'xl0' ball mill
28. 6 - #24 scavengers
29. 1 - 8'x8' super conditioner
30. 24 - primary #24 roughers
31. 24 - secondary #24 roughers
32. 16 - scavengers
33. 4 - primary #24 cleaners
34. 4 - secondary, #24 cleaners
35. 4 - #24 tertiary cleaners
36. 4 - #24 final cleaners
37. 1 - 30'x10' thickener
38. 1 - 30'xl0' thickener
39. 1 - 9'x6' disc filter
40. 1 - 4'x30' blanket table
41. 1 - tailings disposal pump

Figure 2:
Flow Sheet For Estimating Mill Costs.

Table 10: Factored Capital Cost Estimate Guide

From **Chemical Engineering Plant Design**, Vilbrandt, Frank C. and Dryden, Charles E., McGraw Hill, N.Y., 1959, by permission of publisher

1. Purchased equipment costs from references and on current index basis		$000,000
2 .Installed equipment costs		
a. From references; current index basis		$000,000
b. Item 1 multiplied by 1.43		$000,000
3. Process piping		$000,000
Type plant:	Percent of Item 2:	
Solid	7-10	
Solid-Fluid	10-30	
Fluid	30-60	
4. Instrumentation		$000,000
Amount of automatic control:	Percent of Item 2:	
None	2-5	
Some	5-10	
Extensive	10-15	
5. Buildings and site development		$000,000
Type plant:	Percent of Item 2:	
Outdoor	5-20	
Outdoor-Indoor	20-60	
Indoor	60-100	
6. Auxiliaries (e.g., electric power)		$000,000
Extent:	Percent of Item 2:	
Existing	0	
Minor additions	0-5	
Major additions	5-25	
New facilities	25-100	
7. Outside lines		$000,000
Average length:	Percent of Item 2:	
Short	0-5	
Intermediate	5-15	
Long	15-25	
8. Total physical plant costs = Sum of Items 2+3+4+5+6+7		$000,000
9. Engineering and construction		$000,000
Complexity:	Percent of Item 8:	
Simple	20-35	
Difficult	35-60	
10. Contingencies		$000,000
Type process:	Percent of Item 8:	
Firm	10-20	
Subject to change	20-30	
Speculative	30-50	
Average	30	
11 .Size factor		$000,000
Size plant:	Percent of Item 8:	
Large commercial	0-5	
Small commercial	5-15	
Pilot plant	15-35	
12. **Fixed capital costs** = Sum of Items 8+9+10+11		$000,000

the reader should become familiar with the equipment cost equations in this handbook. This will ensure that the correct equipment parameter is involved in specifications.

The table below follows Table 10 format and shows that purchased equipment costs based on an M&S(Mine/Mill) index of 1060 amount to about $2,912,210 for a 1200 TPD facility. The fixed capital cost is estimated as $9,678,122 and the total capital investment is $10,839,497. Note that working capital is taken as12 percent of the fixed capital cost.

Total Capital Investment For a 1200 TPD Plant

1. Purchased equipment costs (M&S = 1060).....$2,912,210
   - Crushing: $541,678
   - Grinding: $881,350
   - Copper Flotation: $525,596
   - Zinc Flotation: $963,586

2. Installed equipment costs (1.43 times Item 1).....$4,164,460

3. Process piping (10 percent of Item 2).....$416,416

4. Instrumentation (3 percent of Item 2) $124,934

5. Buildings and site development (35 percent of Item 2).....$1,457,561
   - ..... Mill Building
   - ..... Crusher Building
   - ..... Assay, Machine Shop
   - ..... Administration Building

6. Auxiliaries (10 percent of Item 2).....$416,416
   - ..... Water Supply
   - Diesel Standby Power
   - ..... Tailings Disposal

7. Outside lines (8 percent of Item 2).....$333,157

8. Total physical plant costs (2+3+4+5+6+7).....$6,912,944

9. Engineering and construction (25 percent of Item 8).....$1,728,236

10. Contingencies (10 percent of Item 8).....$691,295

11. Size factor (5 percent of Item 8).....$345,647

12. Fixed Capital Costs (8+9+10+11).....**US $9,678,122**

13. **Working Capital** (12 percent of Item 12).....**US $1,161,375**

14. **Total Capital Investment** (12 + 13).....**US $10,839,497**

INFRASTRUCTURE COSTS FOR A MINE/MILL COMPLEX/EXAMPLE (O'HARA)

Originally, infrastructure costs were estimated jointly for a mine/mill complex by O'Hara. These costs are largely attributable to the mine, if it is underground, and not to the mill. Some flexibility to assign separately a *proportion* of infrastructure costs to the mill and the rest to the mine is obtained by calculating the joint costs and apportioning them according to experience.

Infrastructure costs are determined herein as follows:

(1) General Plant Services
(2) Access Road to Main Thoroughfare
(3) Townsite and Housing
(4) Feasibility, Planning and Design
(5) Supervision, Camp
(6) Administration, Accounting, Legal, Key Staff

Items (1) and (3) depend upon an estimate of the number of employees at the mine/mill complex; item (2) depends on estimates of the lengths of roads and bridges. Items (4) to (6) reflect a portion of these costs that are associated with infrastructure and previously estimated as part of the processing plant.

Table 11 below summarizes infrastructure capital cost estimation, which uses equations developed by O'Hara.

Table 11: Summary of Infrastructure Capital Cost Estimation

Cost Item	Parameter	Cost Equation	Comment
1) Plant services	N= no. employees	$C_{12} = 17386\ N^{0.8}$	
2) Access road	R=miles of road b=bridge length, ft	$C_{131} = 651930\ R$ $C_{132} = 283\ b^{1.5}$	road bridge
3) Townsite, housing	N=no. employees	$C_{141} = 119520\ N$ $C_{142} = 43463\ N$	family townsite bunkhouse camp
4) Feasibility, plan,design	[4 to 6% of item (2)] + [6 to 8% of (items (1) + (3))]		
5) Supervision/camp	8 to 10% of (items (1) + (2) + (3))		
6) Admin, staff	4 to 7% of (items (1) + (2) + (3))		

If the number of employees is not known with any reliability, it may be estimated from Table 13, page 39, in the section dealing with Mine-Mill Total Product Cost Estimation (O'Hara Method).

Example Calculation

Calculate the infrastructure associated with the underground mine and mill complex described in the example problem on page 25, where a total of 392 employees are required with a subsidized family townsite. The M&S (Mine/Mill) index is 1060.

From Table 11:

(1) C_{12} = (1060/1400)(17386)(392)$^{0.8}$ = $1,563,171

(2) C_{131} = (1060/1400)(651930)(10) = $4,936,042

(3) C_{132} = (1060/1400)(283)(80)$^{1.5}$ = $ 202,498

(4) C_4 = .06 (4,936,042) + .08 (37,036,707) = $3,259,105

(5) C_5 = .1 (41,972,849) = $4,197,285

(6) C_6 = .06 (41,972,849) = $2,518,371

Infrastructure Capital Cost = sum of items (1) to (6) = US $16,676,472

It should be noted that, because the bulk of employees at the underground/mill complex will be underground miners, some 75% of this cost is associated with the mining operation.

SIMPLIFIED CAPITAL/OPERATING COST ESTIMATION MODELS: USBM IC 9298

Camm (USBM Information Circular 9298) has presented quickie estimates of the cost to develop mineral deposits such as gold in the southwest. The approach permits the novice to estimate capital and operating costs with minimum effort. Using a combination of empirical mine dilution and recovery factors and Hugh Taylors rule, capacity (short tons/day) was calculated from:

$$C = \frac{T}{350L} = \frac{T^{0.75}}{70}$$ for 350 operating days per year (7 day/ week) or

$$C = \frac{T}{260L} = \frac{T^{0.75}}{52}$$ for 260 operating days per year (5 day/ week).

In the above, T is the total tonnage, in short tons, to be mined, L is the mine life in years and C is capacity in short tons/day. To estimate L use $L = 0.2T^{0.25}$ which is known as Taylors rule. If R_d is the deposit reserve (short tons) and M_r is the recovery factor for the mining method, then T = R_d M_r (1 + D_m) where D_m is the mine dilution factor. Both M_r and D_m are expressed as fractions (not percents) and should be chosen from the averages reported for Canadian mines in Canada.

Costs are based on average 1989 US dollars. A number of building cost indexes, as well as the M&S (Mine/Mill) index, are used for cost prediction. US labour costs, including mining labour, are employed and infrastructure such as power and access roads is estimated separately. It should be noted that the Camm methodology is adaptable to most deposits. Cost-Capacity equations must be updated before using (such as by means of the M&S(M/M) Index).

Table 12 below summarizes many of the total cost equations developed by Camm. High correlation coefficients were observed for each. Each cost model is associated with a particular set of conditions as reported in IC 9298. Operating costs are in US$ per short ton.

Table 12: Total Cost Equations Reported In USBM IC 9298

Open pit mine models:	Capacity Range in X, stpd	Capital cost, US$	Operating cost, US$/st
Small Open Pit	$1x10^3$ to $2x10^4$	$160,000(X)^{0.515}$	$71.0(X)^{-0.414}$
Large Open Pit	$2x10^4$ to $2x10^5$	$2,670(X)^{0.917}$	$5.14(X)^{-0.148}$
Underground mine models:			
Block Caving	$4x10^3$ to $4x10^5$	$64,800(X)^{0.759}$	$48.4(X)^{-0.217}$
Cut-and-Fill	$2x10^2$ to $8x10^3$	$1,250,000(X)^{0.461}$	$279(X)^{-0.294}$
Room-and-Pillar	$5x10^2$ to $4x10^4$	$97,600(X)^{0.644}$	$35.5(X)^{-0.171}$
Shrinkage Stope	$1x10^2$ to $6x10^3$	$179,000(X)^{0.620}$	$74.9(X)^{-0.160}$
Sublevel Longhole	$4x10^2$ to $1x10^4$	$115,000(X)^{0.552}$	$41.9(X)^{-0.181}$
Vertical Crater Retreat	$4x10^2$ to $9x10^3$	$45,200(X)^{0.747}$	$51.0(X)^{-0.206}$
Mill models:			
Autoclave-CIL-EW	$5x10^2$ to $1x10^4$	$96,500(X)^{0.770}$	$78.1(X)^{-0.196}$
CIL-EW	$5x10^2$ to $1x10^4$	$50,000(X)^{0.745}$	$84.2(X)^{-0.281}$
CIP-EW	$1x10^3$ to $2x10^4$	$372,000(X)^{0.540}$	$105(X)^{-0.303}$
CCD-MC	$5x10^2$ to $2x10^4$	$414,000(X)^{0.584}$	$128(X)^{-0.300}$
Float-Roast-Leach	$1x10^3$ to $2x10^4$	$481,000(X)^{0.552}$	$101(X)^{-0.246}$
Flotation, One Product	$5x10^2$ to $4x10^4$	$92,600(X)^{0.667}$	$121(X)^{-0.335}$
Flotation, Two Product	$5x10^2$ to $4x10^4$	$82,500(X)^{0.702}$	$149(X)^{-0.356}$
Flotation, Three Product	$5x10^2$ to $4x10^4$	$83,600(X)^{0.708}$	$153(X)^{-0.344}$
Gravity	$1x10^2$ to $1x10^3$	$35,300(X)^{0.529}$	$67.8(X)^{-0.364}$
Heap Leach	$1x10^3$ to $2x10^4$	$296,500(X)^{0.512}$	$31.5(X)^{-0.223}$
Solvent Extraction-EW	$6x10^3$ to $7x10^4$	$14,600(X)^{0.596}$	$3.00(X)^{-0.145}$

The parameter X reported in Table 12 is the capacity in short tons per day as calculated from the expressions at the bottom of page 35. Flowsheets and/or mining methods that are associated with each of the models above are listed in **IC 9298** and summarized below.

Open Pit Mine Models

These include an assumption on pit geometry (inverted lamp shade) and use of basic equipment such as diesel haul trucks. Costs are distributed to labor, equipment, steel, fuel,

lube, explosives, tires, construction material, taxes, and haulage distance factor adjustments. Initial equipments costs account for a large portion of capital costs.

Underground Mine Models

Each model includes the cost of development, daily production, mine plant construction and operation and each is based on adit entry (a depth factor is used to correct for shaft entry). The mining method determines the cost equation to use, where costs are broken into categories similar to open pit mine models. The **Block Caving** mine model involves a high development cost and is appropriate for large massive deposits. Stope dimensions of 200' length 70' width 300' height are assumed. The **Cut-and-Fill** mine model is assumed to be vertical with hydraulic sandfill for mined stopes. The method is used when selectivity is required with good recovery and adaptability. Walls can be weak. The **Room-and-Pillar** mine model is appropriate when ore bodies are horizontal (or nearly so), where rock bolts, in addition to pillars, will support openings. Ore recovery is about 85% with little dilution. The **Shrinkage Stope** mine model applies for vertical ore bodies and moderate to weak walls. Veins must be steep with a regular dip. Recovery is assumed to be 90% with 10% dilution. The **Sublevel Longhole** mine model is used for ore bodies that are vertical or steeply dipping. There must be strong hanging walls and foot walls. For weak to moderate wallrock, the **Vertical Crater Retreat** mine model is preferred.

Mill models

Each model involves a simplified flowsheet. A tailings pond is necessary for each, where its cost is basically the cost of preparing the pond site and then building an impoundment dam around it. The **autoclave-CIL-EW** mill model is for sulfide (refractory) ores and includes crushing, a primary ball mill grinding circuit, thickening, an autoclave circuit, CIL tanks, carbon stripping tanks, electrowinning, refining/casting and tailings discharge. The **CIL-EW** mill model is for oxide deposits and incorporates crushing, grinding in a SAG mill-ball mill circuit, CIL tanks, thickening, carbon stripping tanks, electrowinning with steel wool cathodes, refining/casting and tailings discharge. The **CIP-EW** mill model is for oxide gold deposits. It incorporates crushing, a rod mill-ball mill grinding circuit, thickening, leach tanks, CIP tanks, thickening, carbon columns, carbon stripping tanks, electrowinning, refining/casting and tailings discharge. The **CCD-Merrill Crowe** process mill model for gold ores involves crushing, a primary ball mill grinding circuit, leach tanks, CCD circuit, thickening, precipitation, refining/casting and tailings discharge. The **Float-Roast-Leach** mill model is for gold deposits that require fluid bed roasting (carbonaceous or sulfide ores). Involved is crushing, a SAG mill-ball mill grinding circuit closed with mineral jigs, a flotation circuit closed with a regrind mill, thickening, disk filtration of concentrate, fluid bed roasting, carbon columns, electrowinning, refining/casting and tailings discharge. The **Flotation, One Product**, mill model produces a single Au concentrate via flotation and incorporates crushing, a rod mill-ball mill circuit, flotation, thickening, disk filtration of concentrate and tailings discharge. The **Flotation, Two Product**, mill model is for deposits from which two concentrates are produced (lead and zinc concentrates for example). Involved is crushing, a rod mill-ball mill grinding circuit, a differential flotation circuit that produces two concentrates and a final tailings, thickening and disk filtration. The **Flotation, Three Product**, mill model incorporates unit operations similar to the two product plant but expands the flotation-dewatering section to include a third concentrate stream (e.g., a Pb con, a Zn con and a Cu con). The **Gravity** mill model is intended for deposits

amenable to gravity separations (coarse Au or scheelite ore). Unit operations are crushing, coarse jigs, rod mill grinding, spiral classification and gravity tables, flotation, thickening, disk filtration and final tailings. A jig/flotation concentrate is produced for shipment to a refinery. The **Heap Leach** model is designed for operations that involve heap leaching of low grade Au and/or oxide Ag ore. Involved is crushing, transport to leach pads (cyanide leach circuit), carbon columns, carbon stripping tanks, electrowinning (steel wool cathodes), refining and casting. The **Solvent Extraction-Electrowinning** mill model is used to produce a copper cathode product such as for leach solutions from copper waste dumps or insitu low grade deposits. Involved are heap leach pads, two extraction stages and one stripping stage. Barren solution flows from the 2nd stage to leach pads; pregnant organic is sent to stripping by electrolyte from the EW circuit. Barren organic goes to the 2nd extraction stage; electrolyte loaded with copper is electrolyzed to produced high purity cathodes.

The potential user of the Camm costing procedure should read IC 9298 carefully. The M&S(mine/mill) index may be used to update approximately each equation.

MINE-MILL TOTAL PRODUCTION COST ESTIMATION

Broadly, total production costs may be divided into two main categories, namely, manufacturing or **operating costs** and **general expenses**. Operating costs include **direct production costs** (i.e., raw materials, operating labour and supervision, power and utilities, maintenance and repairs, operating supplies, laboratory costs, etc.); **fixed costs** (i.e., depreciation, property taxes, insurance and rent); **plant overhead costs** (i.e., medical, safety, fringe benefits, recreation, townsite subsidies, storage, etc.). On the other hand, general expenses consist of **administrative charges** (i.e., executive salaries, clerical wages, legal and consulting costs, office maintenance and communications) and **distribution-marketing expenses** (i.e., shipping, advertising, technical sales, salesmen, etc.). However, there are many alternative ways to categorize total production costs.

O'Hara Method/Example

O'Hara (**CIM Bulletin**, February, 1980) chooses to divide costs into two categories, namely, **Operating Costs and Supplies** and **Administration and General Services.** This approach is intended specifically for operating mines and mills in the mining industry. Consequently, it is employed herein with minor modifications.

Table 13, page 39, summarizes the procedure employed to estimate personnel requirements.

Underground personnel are distributed percentage-wise in accord with the mining method employed. For estimation purposes, the table below may be utilized.

	% Distribution, Underground Labour			
	Shrinkage	Cut&Fill	Blasthole	Room&Pillar
Development:	9	12	17	14
Stoping:	31	31	23	45
Mine Services:	29	20	22	12
Maintenance:	18	23	23	13
Mine Staff:	13	14	15	16

Table 13: Estimating Personnel Requirements*

I. Operating Labour: $N_0 = N_1+N_2+N_3$

1) Open Pit Mines: $N_1 = N_{11} +N_{12} +N_{13} +N_{14} +N_{15} +N_{16}$

Drill/ Blast	Load	Haulage/ Roads	Maint-enance	Mine Staff	Misc.
$N_{11} = .075(T_p)^{0.5}$	$N_{12} = .11(T_p)^{0.5}$	$N_{13} = .035(T_p)^{0.7}$	$N_{14} = .21(T_p)^{0.5}$	$N_{15} = .108(T_p)^{0.5}$	$N_{16} = .07(T_p)^{0.5}$

T_p = TPD of ore+waste

2) Underground Mines: N_2

Shrinkage	Cut & Fill	Blasthole	Room&Pillar
$N_2 = 4.5T^{0.7}W^{-0.5}$	$5.46T^{0.7}W^{-0.5}$	$19.64T^{0.5}W^{-0.5}$	$2.16T^{0.7}W^{-0.5}$

T = TPD mined W = stope width, ft

3) Processing Plant: N_3

Au Ores	Simple Base Metal Ores	Complex Base Metal Ores
$N_3 = 1.14T^{0.5}$	$1.06T^{0.5}$	$1.10T^{0.5}$

T = TPD milled

II. General Plant Service Employees: $N_g = N_6+N_7+N_8$

6) Electrical Services: N_6
- $N_6 = .03N_0$ Substations
- $N_6 = .05N_0$ Diesel Generating Plant
- $N_6 = .08N_0$ Coal-Fired Generating Plant

7) Surface Plant and Road Maintenance Services: N_7
- $N_7 = .04N_0$

8) Townsite Employees: N_8
- $N_8 = 0$ Existing Town
- $N_8 = .03N_0$ Bunkhouse Camp
- $N_8 = .05N_0$ Subsidized Family Townsite

III. Administration Employees: $N_a = N_{10}$

10) General Administration: N_{10}
- $N_{10} = .07N_0$

IV. Total Employees: $N = N_0 +N_g +N_a$ *based on O'Hara (CIM Bulletin, 1980)

Table 14: Mine-Mill Total Production Cost Estimation*

I. Operating Costs; Supplies; Power: $(C_1+C_2+C_3+C_4+C_5)$ $/day

1) Open Pit Mines: $C_1 = (C_{11} + C_{12})$ $/day

Labour: $C_{11} = 155.59(T_p)^{0.5} + 9.540(T_p)^{0.7}$

Supplies: $C_{12} = 29.12(T_p)^{0.5} + 2.695(T_p)^{0.7} + 1.955(T_p)^{0.8}$

2) Underground Mines: $C_2 = (C_{21} + C_{22})$ $/day

	Blasthole	Cut & Fill	Shrinkage	Room & Pillar
Labour: C_{21} =	$4936.7T^{0.5}W^{-0.5}$	$1369.8T^{0.7}W^{-0.5}$	$1129.2T^{0.7}W^{-0.5}$	$546.0T^{0.7}W^{-0.5}$
Supplies: C_{22} =	$57.58T^{0.8}W^{-0.2}$	$28.7T^{0.9}W^{-0.2}$	$46.72T\ W^{-0.2}$	$30.54T^{0.9}W^{-0.2}$

3) Processing Plant: $C_3 = (C_{31} + C_{32})$ $/day

	Simple Base Metal Ores	Complex Base Metal Ores	Gold Ores
Labour: C_{31} =	$212.5T^{0.5}$	$247.2T^{0.5}$	$257.7T^{0.5}$
Supplies: C_{32} =	$40.8T^{0.7}$	$46.72T^{0.7}$	$32T^{0.7}$

4) Power: C_4 = $/day

	Open Pit Mine/Mill	Underground Mine/Mill
Diesel: C_4 =	$222.7T^{0.5}$	$44.1T^{0.7}$
Utility: C_4 =	$83.4T^{0.5}$	$16.5T^{0.7}$
Coal: C_4 =	$156.6T^{0.5}$	------

5) Supplies-General Plant Services: $C_5 = 13.05T^{0.5}$ $/day

II. Wages/General Plant Services: $(C_6 + C_7 + C_8 + C_9)$ $/day

6) Electrical Services(35% fringe benefits): $C_6 = 1.35(180.7N_6)$

7) Surface Plant Services(35% fringe benefits): $C_7 = 1.35(140.8N_7)$

8) Townsite Employees(35% fringe benefits): $C_8 = 1.35(132.8N_8)$

9) Townsite Operating Cost:

	Existing	Bunkhouse	Subsidized Town
C_9 =	0	$34.53N_0$	$13.28N_0$

III. Wages/Administration Expenses: $(C_{10} + C_{13})$ $/day

10) General Administration Wages: $C_{10} = 1.35(15.82N_0)$
(35% fringe benefits)

13) General Expenses: $C_{13} = 10.63N_0$

IV. Total Costs = $(C_1 + C_2 + C_3 + C_4 + C_5 + C_6 + C_7 + C_8 + C_9 + C_{10} + C_{13})$ $/day

T p = TPD,ore+waste; T= TPD mined or milled; W= stope width, ft

*adapted from O'Hara (CIM Bulletin, 1980)

Table 14, page 40, summarizes the mine-mill total production cost estimation procedure. Operating costs and supplies have been related to capacities, while administration and general services are left as functions of the number, N_0, of operating personnel. Each wage dependent expression **in Canadian dollars**, namely C_{11}, C_{21}, C_{31}, C_6, C_7, C_8, C_9, C_{10} and C_{13}, employed in the table has been updated from Special Volume 25 by the ratio (1000/600). The value 1000 is an easily-remembered common wage base in Canadian dollars. **To calculate wage-dependent expressions in US dollars**, multiply each by the ratio ((current average Canadian weekly earnings in mining times current US to Canadian exchange ratio)/1000).

In Canada, the average weekly earnings statistic in mining is obtainable from **STATISTICS CANADA** (see phone book under Federal Government), Catalog #72002. In the US, the Bureau of Labor Statistics (BLS) should be contacted for current rates.

Equipment-Tool dependent expressions in Table 14, page 40, have been updated from Special Volume 25 to a base M&S(Mine/Mill) index of 1400 by means of the ratio .877(1400/800), where 800 is the base index employed in Special Volume 25 and .877 is the 1978 average US to Canadian currency exchange rate. To update equipment-tool dependent equations in Table 14, multiply by the ratio ((current M&S (Mine/Mill) index)/1400). These items are: C_{12}, C_{22}, C_{32} C_4 and C_5 and **are in US dollars**.

Example Calculation

Estimate the mine-mill total production cost when a simple base metal ore is to be mined at the rate of 7000 TPD (5-day work week) and milled at the rate of 5000 TPD (7-day work week). Primarily blasthole mining methods will be employed with average stope widths of 40 ft. Diesel generated power will be necessary. A family townsite must be subsidized for employees.

Assume that the current average weekly earnings in the mining industry is at $1045, while the current M&S(Mine/Mill) index is 1060. Assume an exchange ratio of .7334 (US to CDN ratio).

Referring to Table 13, page 39, first estimate the number of personnel required for the operation as follows:

I. Operating Labour, N_0

$N_1 = 0$	
$N_2 = 19.64(7000)^{0.5}\ (40)^{-0.5} = 259.8$	260 persons
$N_3 = 1.06(5000)^{0.5} = 77.8$	78 persons
$N_0 = 260 + 78 =$	338 persons

II. General Plant Service Employees, N_g

$N_6 = .05(338) = 16.9$	17 persons
$N_7 = .04(338) = 13.5$	14 persons
$N_8 = .05(338) = 16.9$	17 persons
$N_9 = 17 + 14 + 17 =$	48 persons

III. Administration Employees, N_a

$N_a = N_{10} = .07(338) = 23.7$	24 persons

IV. Total Employees = 338 + 48 + 24 — 410 persons

The above results indicate that approximately 410 employees will be required. The distribution of 260 underground personnel may be estimated from the table at the bottom of page 38. For job categories of other personnel, the **Canadian Mining Journal, Mining Sourcebook**, may be helpful.

Once employee requirements are estimated, Table 14, page 40, may be utilized to estimate daily total product cost. Remember to update each equation in the table by its appropriate ratio. In this example, wage ratio = (.7334)(1045/1000)= .7664 for US dollars and index ratio = (1060/1400) = 0.7571.

Again, referring to Table 14 and in US dollars per day:

I. Operating Costs; Supplies; Power

1) $C_1 = 0$
2) $C_2 = C_{21} + C_{22}$
   $C_{21} = (.7664)(4936.7)(7000)^{0.5}(40)^{-0.5}$ = \$50051
   $C_{22} = (.7571)(57.58)(7000)^{0.8}(40)^{-0.2}$ = \$24837
   $C_2 = 50051 + 24837$ = \$74888
3) $C_3 = C_{31} + C_{32}$
   $C_{31} = (.7664)(212.5)(5000)^{0.5}$ = \$11516
   $C_{32} = (.7571)(40.8)(5000)^{0.7}$ = \$11998
   $C_3 = 11516 + 11998$ = \$23514
4) $C_4 = (.7571)(44.1)(5000)^{0.7}$ = \$12968
5) $C_5 = (.7571)(13.05)(5000)^{0.5}$ = \$ 699

Total, Item I = 74888 +23514 + 12968 + 699 = \$112069

II. Wages/General Plant Services

6) $C_6 = (.7664)(1.35)(180.7 \times 17)$ = \$ 3178
7) $C_7 = (.7664)(1.35)(140.8 \times 14)$ = \$ 2040
8) $C_8 = (.7664)(1.35)(132.8 \times 17)$ = \$ 2336
9) $C_9 = (.7664)(1368 \times 338)$ = \$ 3440

Total, Item II = 3178 + 2040 + 2336 + 3440 = \$ 10994

III. Wages/Administration Expenses

10) $C_{10} = (.7664)(1.35)(15.82 \times 338)$ = \$ 5532
13) $C_{13} = (.7664)(10.63 \times 338)$ = \$ 2754

Total, Item III = 5532 + 2754 = \$ 8286

IV. **Total Estimated Cost** = I + II + III **= US \$131,349 /day**

To calculate the total production cost on a yearly basis, remember that some cost items are associated with a 5-day week; others with a 7-day week. Thus:

C_{21} = 50051 x 260 days =	\$13,013,260
C_{22} = 24837 x 260 days =	\$6,457,620
C_{31} = 11516 x 365 days =	\$4,203,340
C_{32} = 11998 x 365 days =	\$4,379,270
C_4 = 12968 x 365 days =	\$4,733,320
C_5 = 699 x 260 days =	\$181,740
C_6 = 3178 x 365 days =	\$1,159,970

C_7 = 2040 x 260 days	=	$530,400
C_8 = 2336 x 260 days	=	$607,360
C_9 = 3440 x 365 days	=	$1,255,600
C_{10} = 5532 x 365 days	=	$2,019,180
C_{13} = 2754 x 365 days	=	$1,005,210

Total Estimated Cost per year = US $39,546,270 per year

The above suggests that the estimated yearly cost is equivalent to about $131,035 per day multiplied by 301.8 operating days equivalent.

Dilution and Recovery In Revenue Estimaton

The estimation of revenue has a direct bearing on an assessment of project success and is covered in subsequent sections of this manual. A brief summary of the work reported by O'Hara to assess dilution and recovery is provided for completeness.

The estimation of revenues is dependent on recoverable ore reserves, on the amount of waste rock in mill feed, on the mill recovery and on the nature of the smelter contract negotiated by the mine/mill and purchaser of concentrates. For reasonably regular ore reserve blocks, at least 95% of the ore reserve should be classed as recoverable.

The percentage dilution (i.e., the percent of waste rock in the ore mined) depends on the mining method. Likewise, for underground mines, the stope width and the dip of the orebody are factors. Their influence on dilution may be estimated from the table below.

	Blasthole Stopes	Shrinkage Stopes	Cut&Fill Stopes	Room&Pillar Stopes
%Dilution:	$100/(W^{0.5}Sin(A))$	$60/(W^{0.5}Sin(A))$	$45/(W^{0.5}Sin(A))$	$70/(W^{0.5}Sin(A))$

where W = stope width, ft, and A = dip of orebody, degrees of angle.

To estimate mill recovery, Table 15, page 44, is useful. Normally, recoveries are estimated from actual laboratory/pilot plant studies conducted by process engineers on drill core or exploration adit samples. Referring to Table 15, note that each equation expresses recovery as a function of the average feed assay for various base metal and gold ores, along with a few miscellaneous ores.

Smelter contracts vary considerably and will involve *penalties* imposed by the smelter for detrimental impurities in concentrates, *payments* by the smelter (on an agreed basis) for valuable metals present, *treatment charges* paid to the smelter by the mine/mill and *shipping charges* paid by the mine/mill. O'Hara suggests that shipping charges can be estimated from:

$$F = 0.17(M_t)^{0.9} + 0.26(M_r)^{0.7} + 0.8D_0$$

where F = cost in $/ton of concentrate, M_t = miles trucked by road, M_r = miles transported by rail and D_0 = days of loading, ocean travel, and unloading when shipped by 15,000 ton freighters. The shipping cost expression is based on average 1978 costs.

Table 15: Recovery of Various Ores Milled By Flotation (head grades in %)

		Recovery Formulas:	Typical Conc. Grade:
Base metal ores:			
Cu	in straight chalcopyrite ores	$R = 100\% (1- 0.07\ Cu^{-0.8})$	28.5% Cu
	in oxidized copper ores (sulphides)	$R_{CuS} = 100\% (1- 0.08\ CuS^{-0.8})$	Variable
	in oxidized copper ores (oxides)	$R_{CuO} = 100\% (1- 0.40\ CuO^{-0.3})$	Variable
	in copper-zinc ores	$R = 100\% (1- 0.16\ Cu^{-0.8})$	25.5% Cu
	in lead-zinc-copper ores	$R = 100\% (1- 0.22\ Cu^{-0.8})$	22.0% Cu
MoS_2	in straight molybdenite ores	$R = 100\% (1- 0.04\ MoS_2^{-0.8})$	88.0% MoS_2
	in copper-moly ores	$R = 100\% (1- 0.06\ MoS_2^{-0.8})$	Variable
Zn	in straight sphalerite ores	$R = 100\% (1- 0.25\ Zn^{-0.6})$	56% Zn
	in lead-zinc ores	$R = 100\% (1- 0.32\ Zn^{-0.6})$	53% Zn
	in copper-zinc ores	$R = 100\% (1- 0.45\ Zn^{-0.6})$	52% Zn
	in copper-lead-zinc ores	$R = 100\% (1- 0.55\ Zn^{-0.6})$	50% Zn
Pb	in straight galena ores	$R = 100\% (1- 0.13\ Pb^{-0.8})$	60% Pb
	in iead-zinc ores	$R = 100\% (1- 0.18\ Pb^{-0.8})$	53% Pb
	in copper-lead-zinc ores	$R = 100\% (1- 0.28\ Pb^{-0.8})$	45% Pb
Miscellaneous Ores:			
Tungsten ores (gravity separation)		$R = 100\% (1 - 0.33\ WO_3^{-0.5})$	75% $W0_3$
Nickel in nickel-copper ores		$R = 100\% (1- 0.20\ Ni^{-0.6})$	10% Ni
Uranium ores (flotation-leaching)		$R = 100\% (1- 0.16\ U_30_8^{-0.8})$	77% U_30_8
Iron ores (gravity-magnetic)		$R = 100\% (1- 1.5\ Fe^{-0.6})$	65% Fe
Precious Metal Ores and Precious Metals in Base Metal Ores: (head grades in oz per ton)			**Process**:
Gold in siliceous ores		$R = 100\% (1- 0.013\ Au^{-0.8})$	Cyanidation only
	in pyrite ores	$R = 100\% (1- 0.03\ Au^{-0.8})$	Flot./Roast./Cyan.
	in base metal ores	$R = 100\% (1- 0.3\ Au^{-0.8})$	Flotation
Silver in straight silver ores		$R = 100\% (1- 0.22\ Ag^{-0.6})$	Flot./Grav. Separation
	in base metal ores (<1.0 oz/t)	$R = 100\% (1- 0.40\ Ag^{-0.6})$	Flotation

MINERAL PROJECT EVALUATION TECHNIQUES

The objective of any financial analysis is to evaluate the fundamental financial and economic characteristics of a project. In mining these characteristics are based on the geological and engineering parameters of the project, including the environment, and on the anticipated economic, social, and political aspects that can be quantified.

The evaluation of mineral projects concerns the mineral supply process. The process of attaining economic production of minerals consist of a multi-stage series of activities by which minerals are converted from undefined geological resources to marketable commodities. Mineral projects are evaluated and subsequent decisions are made in light of their estimated economic characteristics. The economics of mineral supply is comprised of costs, risks and returns and expressed by the net return.

The starting point for any economic study to support an investment decision is the gathering together and development of relevant experience and information. Economic evaluation techniques are then applied to reduce these future estimates to a few indicators of the economic attractiveness of alternative investment decision measures. These measures are intended to portray the quantitative economic consequences of the investment in terms of expected value, sensitivity and risk.

Economic evaluation techniques rely on the concepts of cash flow and the time value of money. These combine in various ways to assess discounted cash flow criteria including net present value, rate of return, equivalent annual value and present value ratio. An exception is the payback period criterion, a commonly used economic indicator, based only on the estimation of future cash flows and not on the time value of money. To refine the evaluation, the impact of taxation may be introduced, along with the effect of inflation, to convert before-tax cash flow estimates to an after-tax basis.

Single point estimates of expected future conditions are combined to assess the expected values of the discounted cash flow indicators. Sensitivity analysis may be used to examine the sensitivity of the discounted cash flow measures to realistic variations of input variables away from their expected values. Risk analysis translates perceived uncertainties concerning the input variables into probability distributions of possible values for the discounted cash flow indicators.

The expected value, sensitivity and risk analyses, together with an appreciation of factors not quantifiable , are the ultimate criteria to compare before the investment decision is made.

Concept of Cash Flow

A cash flow analysis is based on provisional financial statements for a proposed mining project. These statements establish the benefits and costs of the project during the planned production period. The cash flow is the difference between the anticipated annual benefits (i.e., revenues) and the anticipated annual costs incurred by the project over a given period. The benefits are the earnings of the mine and the costs are the capital expenditures and annual expenses (such as wages, raw materials, operating costs, maintenance costs and income taxes).

The cash flow distribution describes a particular situation by its stream of cash inflows and outflows over time. Cash inflows and outflows occur intermittently, at irregular intervals. To facilitate economic analysis, all inflows and outflows occurring during a particular time period, typically one year, are summed up and assumed to have occurred at the end of that time period. With the end-of-period convention, a cash flow distribution is represented by a positive or a negative amount of money occurring at equally spaced points in time.

Figure 3 below is a representative cash flow diagram. Each of these arrows represents the algebraic total of in- and out-of-pocket operations. Three phases can easily be identified on the diagram. The first is the exploration phase where money is spent in relatively small increments but over a long period of time. The second phase is development of the mineral property where large capital cost expenses are required. Finally, the third phase is production where a stream of revenue occurs over time to compensate and reward the two investment phases.

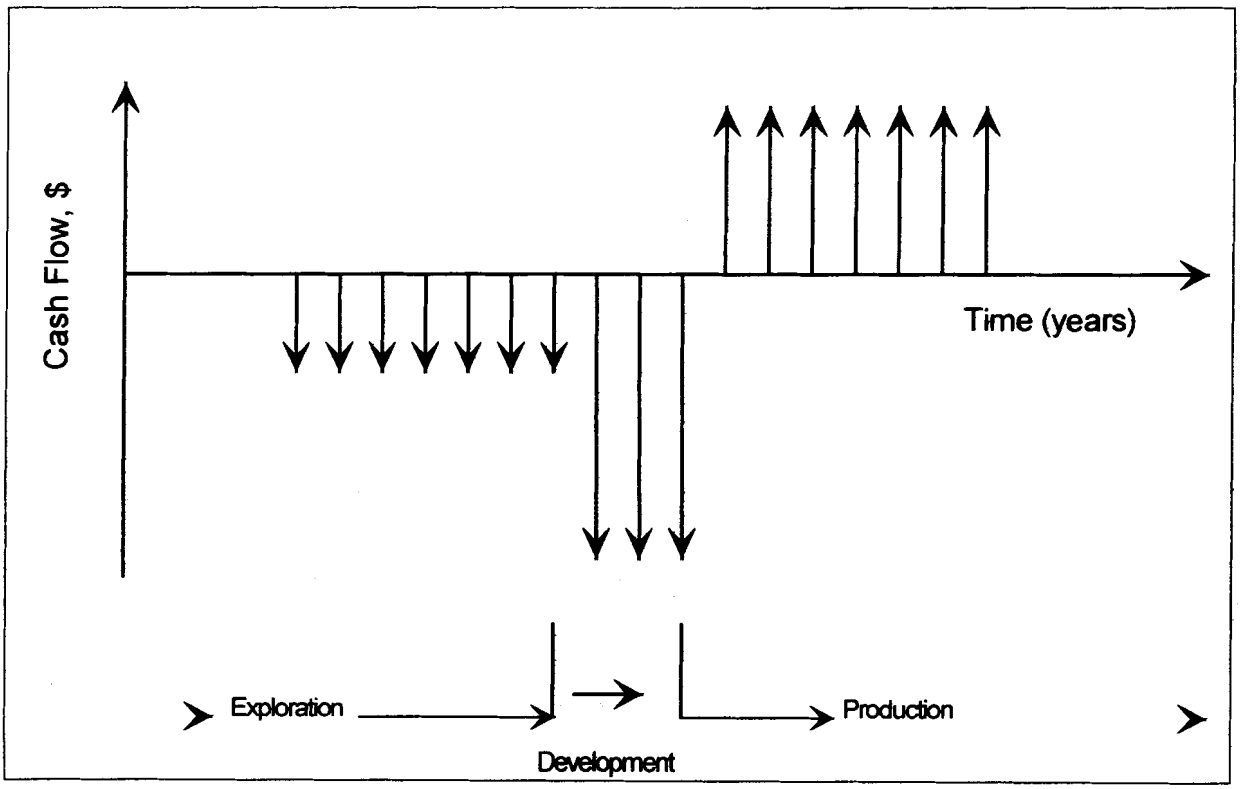


Figure 3: Cash Flow Diagram Showing Three Identifiable Phases.

The simplified calculation of a **cash flow is** as follows:

	Operating Revenue
less	**Operating Expenses**
less	**Taxes**
less	**Capital Expenditure**
=	**Cash Flow**

Capital cost allowance is used to calculate taxable income but does not appear above because it is not a cash operation even though it as an indirect effect through taxes. If capital cost allowance is subtracted from cash flow the result would be profit. Cash flow itself is not a measure of profitability and may merely reflect a high depreciation rate. However, cash flow is used to calculate profitability. Profit is an accounting concept while cash flow will represent how much cash surplus a project will generate in relation to how much cash outlay it required.

Payback Criteria

One of the primary concerns of most business people is whether money invested in a project can be recovered and when. The payback method screens projects on the basis of how long it takes for net receipts to equal investment outlays.

A common standard used in determining whether to pursue a project is that no project may be considered unless its payback period is shorter than some specified period of time. If the payback period is within the acceptable range, a formal project evaluation may begin. One consequence of insisting that each proposed investment have a short payback period is that investors can assure themselves of being restored to their initial position within a short span of time. By restoring their initial position, investors can take advantage of other investment possibilities that may arise. If the venture has some uncertainty, payback time gives the length of time initial investment is at risk.

The payback period is determined by adding the expected proceeds for each year until the sum is equal to or greater than zero. The cumulative cash flow equals zero at the point that cash inflows exactly match or payback the cash outflows. Table 17 shown below illustrates the payback time calculations.

Table 17: Projected After-Tax Cash Flows And Their Cumulative Sum

Period	Cash flow	Cumulative Cash flow
0	-2000	-2000
1	+ 500	-1500
2	+ 500	-1000
3	+ 700	- 300
4	+1000	+ 700
5	+1000	+1700
6	+ 500	+2200

If linearity is assumed in the cash flow distribution of Table 17, the payback period becomes 3.3 years.

There are several advantages to payback time. It is a simple, easily understood method and is commonly used in the business world. The method is particularly adaptable to simple problems. Initial project screening by the payback method reduces the information search by

focusing on that time at which the firm expects to recover the initial investment. It may also eliminate some alternatives, thus reducing overall analytical efforts by the firm.

The principal objection to the payback method is its failure to measure profitability. Simply measuring how long it will take to recover initial investment outlays contributes little to gauging the earning power of a project. Payback period analyses ignore differences in the timing of cash flows and hence fail to recognize time value of money.

Time Value of Money

Monetary values at different points in time may seem superficially equivalent. This is not because they have the same purchasing power, but because they have the same earning power. Equivalence is the basis of comparison between alternatives. Money has a time value because a dollar received today is worth more than a dollar received tomorrow: at the very least, today's dollar can earn the going rate of interest in a savings account, and more generally it can grow in value through other types of investment.

Money is a productive resource so money lenders require compensation such as interest. Discounting is the application of interest rate over time, so that it is important to distinguish between simple, compound, nominal and effective interest rate.

Simple interest considers the interest to be earned on only the principal amount during each interest period. In other words, under simple interest, the interest earned during each interest period does not earn additional interest in the remaining periods, even though it is not withdrawn. Compound interest is when the interest for each period is based on the total amount owed at the end of the previous period. This total amount includes the original principal plus the accumulated interest that has been left in the account.

The nominal interest rate refers to the sum over one year of the interest rates used for each compounding period. Suppose there are m = 4 compounding periods in a year and that for each period a 3% rate of return is obtained. In this case, the nominal interest, i, is said to be 12%. However, the effective interest rate, EIR, which is calculated as if the compounding were done once a year instead of m times a year, is obtained from the following formula:

$$EIR = \left[1 + \frac{i}{m}\right]^m - 1 = \left[1 + \frac{0.12}{4}\right]^4 - 1 = 0.1255$$

where i = the nominal rate expressed as a fraction, m = the number of compounding periods and EIR = the resulting effective interest rate. For the example of a 12% nominal interest compounded quarterly , the EIR is 12.55%.

In project evaluation, it is usual to use a one year compounding period for interest or rate of discount. This rate of discount can be said to be the cost of capital associated with raising funds to invest in the project. It is the percentage rate of return that the project must generate to compensate those outside investors who supplied funds. It is referred to as the weighted average cost of capital and is the rental cost of money. From the lenders point of view, the discount rate is the interest rate that will compensate him for the loss of opportunity and for taking the risk of not being repaid.

Organizations invest in those activities that have expected revenues in excess of costs, but it would be incorrect to directly compare the cost incurred early in the life of a project with revenues received only later. Cash flows must be adjusted for the time value of money.

Time Value Factors

Cash flow equivalence for amounts at different points in time can be converted by interest formulas. These are based on the following components:

F =	future worth
P =	present worth
A =	stream of equal end-of-period values
i =	effective interest rate per period
n =	number of interest periods

The **compound amount factor**, (F/P, i, n), will give a future value equivalent of a given present value, where / means "given". The expression is read: "future worth given a present worth with i rate and n periods". The **present worth factor** (P/F, i, n) gives a present worth equivalent of a given future value. The **sinking fund factor** (A/F, i, n) yields a uniform series of end-of-period payments equivalent to a given future value, while the **series compound amount factor** (F/A, i, n) yields the future value equivalent of a uniform series of end-of-period payments. The **capital recovery factor** (A/P, i, n) yields a uniform series of end-of-period payments equivalent to a given present value. The **cumulative present value factor** (P/A, i, n) yields the present worth equivalent of a uniform series of end-of-period payments.

These six factors are traditionally found in tables for a spectrum of n and i values. Today, any spreadsheet software will transform cash flows at will. The relationships are relatively simple and can be condensed to the following algebric formulas:

$$F = P[1+i]^n$$

$$F = A\left[\frac{(1+i)^n - 1}{i}\right]$$

$$A = P\left[\frac{i}{1-(1+i)^{-n}}\right]$$

Special cases of the above relations such as gradient series and geometric series can also be mentioned. The emergence of modern computing tools eliminates the need to match each of the cases with a specific formula.

Discounted Cash Flow (DCF) Criteria

The two main criterion used to evaluate the cash flows of projects are the net present value (NPV) and the rate of return (also referred to as internal rate of return or DCF rate of return).

The net present value is the amount of money at time zero that is equivalent to a sequence of cash flows discounted at a specified interest rate. Here, i, the discount rate, is considered to be either the cost of capital or an opportunity cost. If money is borrowed, return on investment

must be at least equal to the cost of investment funds. If money is invested in a project, it cannot be invested elsewhere. This lost return is an opportunity cost. Investment as well as cost of capital must be recovered from project benefits; thus a project is acceptable if the NPV is greater than zero. A positive NPV implies that the project will yield a return in excess of the rate used in the calculation procedure.

There may be several discount rates when alternative scenarios are analyzed, so that the NPV should be graphed against the cost of capital. The graph will illustrate the size of the returns expected for a present value profile.

When projects have different initial investments or different lifetimes, the use of NPV to rank alternative projects is limited, because NPV does not take such factors into account. When comparing projects with different capital costs, the *present value ratio* (the NPV divided by the present worth of the investment) is a more valid ranking tool that measures the efficiency of each investment. As for projects with different lifetimes, the equivalent annual value, which is a stream of equal end-of-period values, can be used for comparisons.

The rate of return is defined as that interest rate which equals the sum of the present value of the cash inflows and those of the outflows for a project. It is the discount rate that will make the NPV equal to zero. The calculation procedure involves a trial-and-error solution. The return on investment is analogous to the interest rate received from a bank account, where the investment is recovered as well. The project is acceptable if the rate of return is greater than the cost of capital or a pre-defined hurdle rate. The incremental rate of return is often viewed as a risk premium. An implicit assumption is that cash flows generated by the project are reinvested at the rate of return. This is not necessarily a valid assumption; especially when the project has a high rate of return. The value of the project may be overstated in relation to some alternative investment if the funds released or generated by the project are reinvested at some rate less than the calculated rate of return.

Neither the rate of return nor the NPV, take into account factors such as different investment levels or lifetimes (lives) for project ranking. Multiple rates of return are possible for particular cash flow distributions. There can be as many roots as there are sign changes in the CF distribution. Whether all roots are real depends on the magnitude of the cash flows.

A particular problem is created when NPV and rate of return produce contradictory results. This arises because of the different time patterns embodied in the cash flow distributions and the implicit assumptions about reinvestment rates for the two criteria. Of course, the use of incremental or marginal analysis techniques will give more coherent results in the case of mutually exclusive projects. However, a more simple and conservative way is to favor NPV over rate of return, because the reinvestment rate remains the same for any proposal.

Other criteria can be used, but either these are not widely accepted or they are for very specific applications. The discounted payback period is similar to the payback period described earlier but uses discounted investment and discounted revenues. The benefit-cost ratio, mainly used for public investment, is the ratio of the present worth of benefits and the present worth of costs. The investment would be acceptable only when the ratio is larger than one.

Because all the criteria are based on the principle of discounted cash flow, the present is always favored in relation to the future via a direct relation to the rate being used.

Sensitivity Analysis

A sensitivity analysis examines the effects of variations in project parameters on investment decision criteria. The profitability of a mining operation is directly influenced by variations in cost, revenue and recovery parameters and by other economic and financial variables.

A base case is established using the most likely values for each variable in the NPV analysis. Then a range of possible NPV's is calculated by testing the extent to which individual variables influence the economic attractiveness of an investment. Usually, a series of calculations is made using various values of the variable under consideration while holding all other variables constant. This is the simplest form of sensitivity analysis, examining the effect of one variable at a time. By varying a particular parameter over a range of values such as 10%, 20% and 30% of its most likely estimate, a range of profitability values is obtained for the project. By analyzing all parameters in this way, the sensitivity analysis enables a decision-maker to define the parameters to which decision criteria are most sensitive. These strategic parameters can be given more attention at the estimation stage and given careful scrutiny at the subsequent risk analysis stage.

The results of a sensitivity analysis are often presented in graphical form, by means of spider diagrams as shown in Figure 4. Relative changes in particular factors are plotted against relative changes in the profitability of a project. The slope of each of the curves expresses the relative sensitivity to the factor such as the project life.

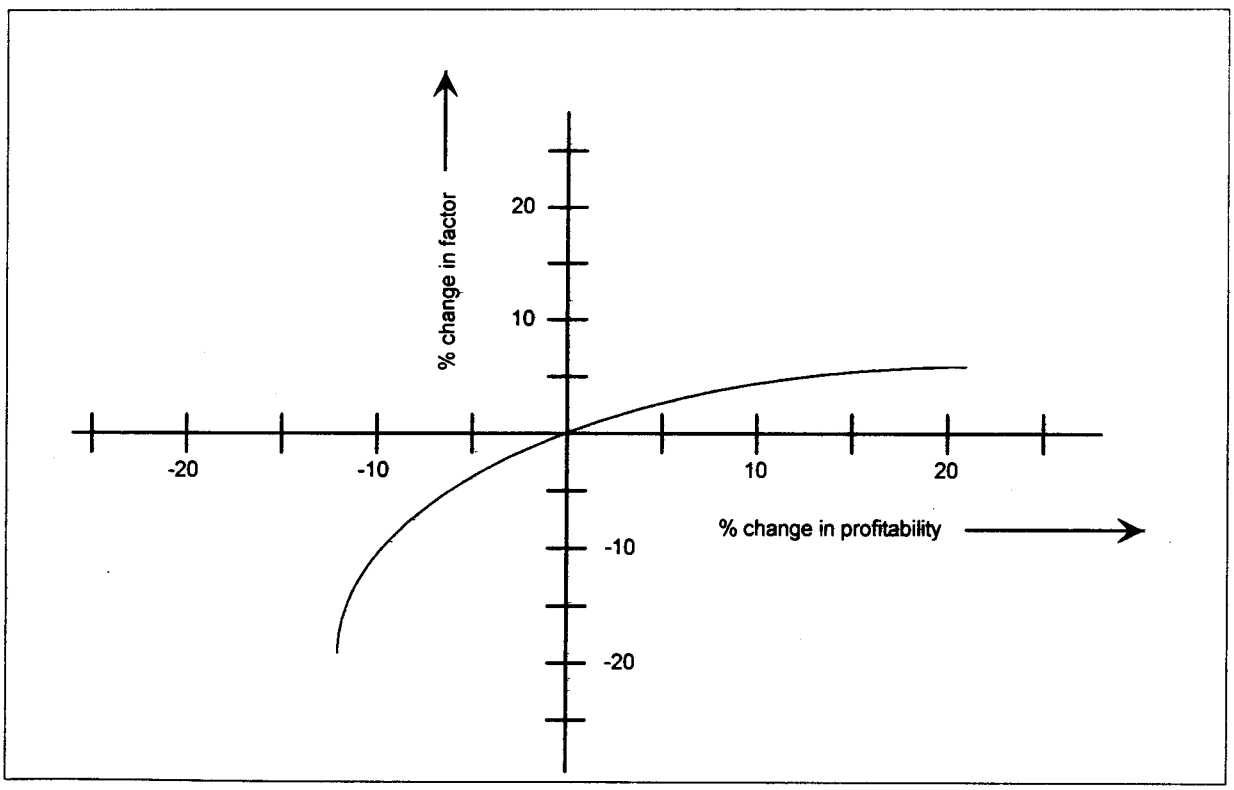


Figure 4: Spider Diagrams Associated With Sensitivity Analyses

The validity of the results depends on the appropriateness of the parameters fed into the equations. Three families of parameters are used, namely, cost, revenue and recovery. Cost per ton mined or milled, operating cost and capital cost are parameters that can be varied. Since one individual cost can be lowered to overall productivity expenses, cost per unit of metal could be more representative. Revenue would comprise price of metal on the market and exchange rate. Recovery is a catch all term here including production level, grade and tonnage, construction delays, start-up time required and rated capacity. These last parameters influence profitability. So does proper timing during the cash flow period to create best results.

Sensitivity is used to answer "what if" questions. This would include best case and worse case scenarios, where many parameters are varied simultaneously.

Risk Analysis

The discounted cash flow criterion allows ranking of investment alternatives by comparing benefits with costs after adjusting for the time value of money. However, it is risky to assume that future benefits and costs are known with certainty.

Risk is a measure of the degree of variability of possible future revenues and costs. Risk analysis is where probabilities are attributed to the results of a sensitivity analysis carried out on influential project parameters.

In the mineral development framework, risk factors can be grouped into three categories, economic, technical and political.

Economic risk is determined by the economic system as a whole. It is caused by the variability in mineral prices, in mineral demand and in foreign-exchange rates. Price risk, demand risk and foreign-exchange risk can be partially countered by resorting to the future markets.

Technical risk comprises reserve risk (quantity and quality of reserves), completion risk (delays or overruns) and production risk (capacity). These risks are partially under the control of the organization active in mineral development.

Political risk is the result of actions by government. Actions include changes in regulations concerning environment, taxation, ownership and currency convertibility.

Risk evaluation attempts to establish the probability of variations in project profitability resulting from uncertainties in the project. Evaluating the cash flow of an investment opportunity will produce the most likely estimate of the net present value, which is the mean of the NPV distribution. The expected NPV (or the average of all possible NPV's) requires the calculation of cash flow for each of all the possible outcomes.

A simple way to deal with risk is to raise the discount rate, that is to use a risk-adjusted discount rate. The rationale is that income from successful risky projects must be large enough to make up for losses incurred from unsuccessful risky projects. Raising the discount rate reduces the NPV if a project starts first with negative and then with subsequently positive cash flows. A correction factor could be employed, but this introduces a bias of arbitrary effect and magnitude.

A more scientific approach is to use either combinatorial analysis or a simulation technique. When uncertainties are formulated with discrete distributions, a combinatorial analysis involves the determination of profitability associated with all of the possible combinations of project parameters together. Under the assumption of no interdependence, the probability of each combination is the product of the probabilities of occurrence of each individual parameter considered in the combination. The result of this form of analysis is a discrete distribution of probability values, reflecting the variability associated with the investment project.

Simulation techniques, such as Monte Carlo simulations, constitute an alternative when analytical techniques are not feasible because of the number of alternatives, or because of the lack of specific values or probability distributions for some factors. The unknown distribution of profitability values is sampled by simulating random values of the input parameters. By repeating this process a large number of times, the unknown distribution is approximated by a histogram. Expected value and risk criteria are derived from this set of values.

The Influence of Taxation And Inflation On Evaluation

Taxation is an added cost in project evaluation. Since it affects the cash flow distribution associated with a project, it should always be considered in the analysis. Because taxation does not affect capital costs and operating costs in the same way, the best alternative on a before-tax basis may not be the best alternative on an after-tax basis.

All operating expenses are deductible from revenues for the purpose of determining taxable income. The after-tax reduction in cash flow resulting from taxation is proportional to the effective tax rate. It is different for capital costs because generally they are not fully deductible from revenue in the year they are incurred. Rather, they are written off over time with some form of capital cost allowance. These tax shields reduce the taxable income during future time periods.

Because of the complexity of taxation, a complete year by year cash flow distribution should be transformed to an after-tax basis. Of course, if only an isolated cost must be analyzed, the use of tax factors can be contemplated. These relate tax rates, depreciation and discounting to give an after tax cost.

Mining project taxation is complicated by the presence, along with usual municipal and business profit taxes, of a specific mining tax that varies widely from province to province and from state to state. Rates vary, but more importantly, so do the rules of application and their form.

Inflation is a general decrease in the purchasing power of money over time. It must be accounted for in project evaluation to avoid distortion of investment criteria. Inflation coupled with taxation and debt financing has significant effects on project profitability. Also investment criteria must be based on monetary values of equal earning power and purchasing power.

Capital cost allowance is not indexed and is based on the price of an asset at the time of purchase. However, revenues and operating costs inflate with time, while the capital cost allowance does not. Thus, the relative tax shield created by the capital cost allowance decreases with time. The tax payments increase in current dollar terms as well as in constant

dollar terms. The overall effect is the reduction of the real after-tax profitability of the project. The effect is the inverse for project financing. Interest and principal are repaid in current dollars, not in an index form. These payments become relatively less important with respect to current dollar operating profits, thereby increasing the real profitability of the project. In an inflationary environment, taxation reduces the real profitability of a project while debt has the opposite effect. The overall effect of both items taken together depends on many factors, including the depreciation method, the debt-equity ratio and the inflation rate.

To be consistent, after tax cash flows should be converted into constant dollars before applying any investment criteria. The monetary unit must have a constant value from year to year to permit comparison. Choosing a deflating rate can be complicated if high escalation is present. The different components of a project rarely change with the same magnitude. If costs and revenues are affected differently, the impact is magnified in the cash flow.

The inclusion of taxation and inflation effects in a cash flow analysis will bring the results closer to reality and avoid the use of ad hoc corrections or adjustments.

Example Calculations

A project with an expected life of 5 years requires an initial investment at time zero of $240,000. Annual cash flow will most likely be $67,000 and the equipment used in the project is expected to yield a salvage value of $70 000.

(1) Define the cash flow, (2) calculate the payback period, (3) calculate the net present value at a discount rate of 10% and the present value ratio, (4) calculate the equivalent annual value, (5) calculate the rate of return and (6) Perform a sensitivity analysis on the effect of varying annual cash flow and project life.

(1) Define Cash Flow

The following table can be constructed from the above information.

Time, years	Cash Flow	Cumulative Cash Flow
0	-$240,000	-$240,000
1	$ 67,000	-$173,000
2	$ 67,000	-$106,000
3	$ 67,000	-$ 39,000
4	$ 67,000	+$ 28,000
5	$137,000	+$165,000

(2) Calculate Payback Period

The cash flow becomes negative at time t = 0. Thereafter, $67,000 accrues each year for 5 years. At the end of the 5th year, the equipment is sold so the cash flow becomes 70,000 + 67,000 = $137,000. The cumulative cash flow becomes zero somewhere between the 3rd and 4th year. If the positive cash flow is linear during year 4, then 39,000 becomes zero at 39,000/67,000 or 0.58 years. Hence, the payback period is 3.58 years.

(3) Calculate Present Value Ratio and Net Present Value At a 10% Discount Rate

The net present value, NPV, is the algebraic sum of the cash flows discounted to time zero.

$$NPV = -240,000 + \frac{67,000}{(1+0.1)^1} + \frac{67.000}{(1+0.1)^2} + \frac{67,000}{(1+0.1)^3} + \frac{67,000}{(1+0.1)^4} + \frac{67,000}{(1+0.1)^5} = 57,447$$

The present value ratio is NPV divided by the worth of the investment ($240,000) at time zero, which is:

$$\text{Present value ratio} = \frac{57,447}{240,000} = 0.24$$

(4) Calculate Equivalent Annual Value

The equivalent annual value is the five equal payments that would be equivalent to a present value of $57,447. Referring to the formula on page 49, we have:

$$A = P\left[\frac{i}{1-(1+i)^{-n}}\right]$$

$$A = 57,447\left[\frac{0.1}{1-(1+0.1)^{-5}}\right] = \$15,154$$

(5) Calculate Rate of Return

The rate of return is the rate that brings the NPV to zero. In the expression,

$$NPV = -240,000 + \frac{67,000}{(1+0.1)^1} + \frac{67.000}{(1+0.1)^2} + \frac{67,000}{(1+0.1)^3} + \frac{67,000}{(1+0.1)^4} + \frac{67,000}{(1+0.1)^5} = 57,447$$

we must set NPV to zero and let the value i (0.1 in the example) be the variable whose unique value will cause NPV to become zero. Rewriting the above, we have:

$$NPV = -240,000 + \frac{67,000}{(1+i)^1} + \frac{67.000}{(1+i)^2} + \frac{67,000}{(1+i)^3} + \frac{67,000}{(1+i)^4} + \frac{67,000}{(1+i)^5} = 0$$

The above is quasi-parabolic so it will have a single minimum. The value of i is found by any of a number of techniques. The easiest are direct search methods---in this case involving a unidimensional search in the i direction. An iterative method shows that i = 0.18. Thus 18% is the rate of return on the investment.

(6) Sensitivity Analysis: Effect Of Annual Cash Flow and Project Life

The table below shows the sensitivity of Annual Cash Flow (ACF), $, and Project Life (PL), years, on the Project Rate of Return (PROR), %, and the Cash Rate of Return (CROR), %.

Change In parameter	Annual ACF	Cash CROR	Flow %change	PL	Project PROR	Life % change
-40%	40200	3.6	-80.2	3	5.6	-68.8
-20%	53600	11.0	-39.0	4	13.4	-25.5
0	67000	18.0	0	5	18.0	0
20%	80400	24.8	37.9	6	20.9	16.3
40%	93800.	31.5	74.8	7	22.9	27.1

ORE DEPOSIT AND REVENUE EVALUATION

Ore Deposit Evaluation Methodology

Methodology to evaluate ore deposits must result in a clear-as-possible statement of metal in place. There is a need: **for** geological work and know-how to outline a sufficiently continuous minable orebody, **for** estimation of quantity and grade of the minable ore, and **for** presentation of the information to interested parties via an ore classification system that is precise and unambiguous. Thus, to arrive at the size and grade of (1) a recently-discovered body of mineralization, (2) a potential ore resource, (3) an orebody proposed for mining or (4) the ore reserves in a mine, fundamental physical, chemical, geological and geometrical analyses/studies are necessary. Methodology is grouped by Owens and Armstrong (see "Ore Reserves---The Four C's", 97th AGM, CIM, Volume 87, No. 979, 1995) under the headings: **C**haracter (type or nature) of mineralization, **C**ontinuity of mineralization types in the orebody, **C**alculation methodology and **C**lassification of the mineral estimate---the four **C's** of ore reserve evaluation.

A good understanding with a clear representation of ore body composition, shape, size, variability and limit is needed at the evaluation and development stage. Such characterization requires a high quality geological database. The key elements are location data, lithological data, structural data, alteration data and mineralization data. Because samples represent an infinitesimal fraction of the rock mass, the estimation of mineral content should come after a critical review of sampling is complete. Random and systematic errors introduced during the collection, preparation, analysis, and evaluation of samples must be recognized and accounted for. Samples may be from trenches, shafts, or underground workings, as well as from drill cuttings and drill core. Whether a sample is representative of a given mass, will depend upon factors such as homogeneity, size and spatial distribution of mineral grains and the specific gravity difference between mineralization and gangue.

The preparation of samples for assay depends on sample size, physical properties, and on the analytical method to be used. Generally preparation involves crushing and grinding of the sample and homogenization followed by splitting to an assayable size. Error can be introduced during sample preparation by factors such as particle shape, hardness, specific gravity, friability, malleability or moisture. The desired end result of sample preparation is a rock powder that contains the elements to be analyzed in the same concentrations and proportions

as in the original sample received. Sample preparation procedures are bias prone and should be controlled carefully. Implementation of statistical quality control (SQC) at the analytical stage is mandatory.

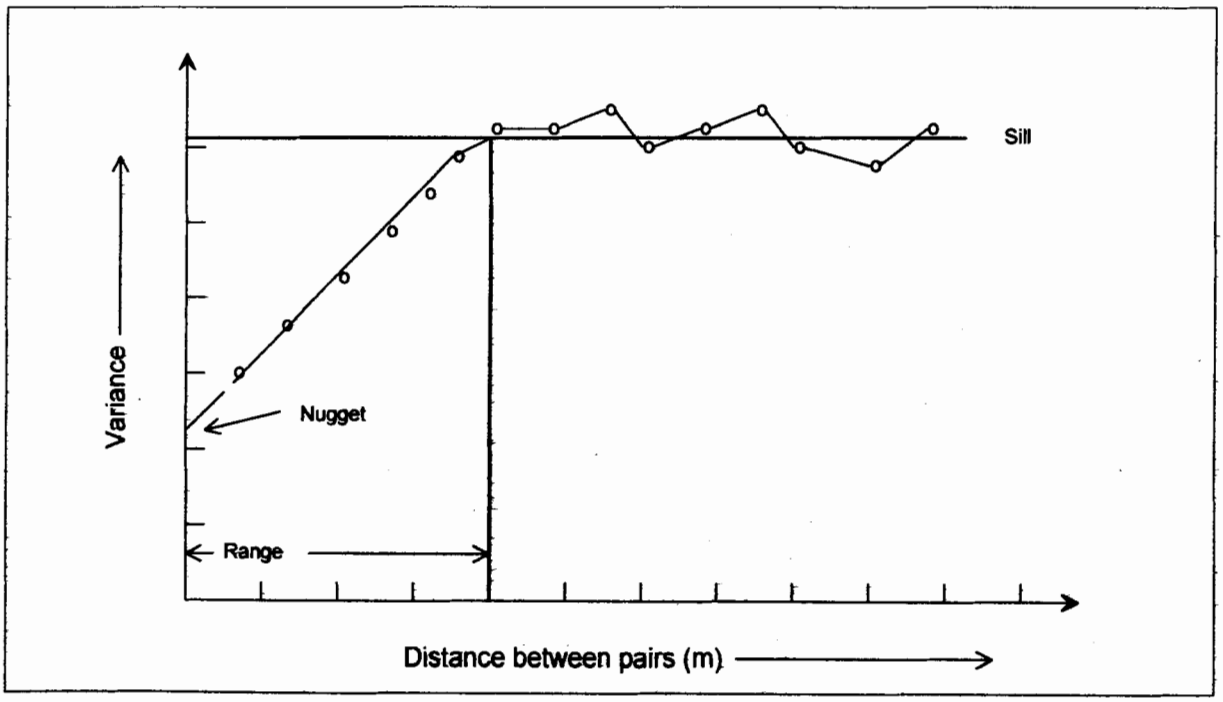
Two classes of assay methods are available, namely, semi-quantitative and quantitative. Generally, samples are taken during exploration drilling. Split portions will be sent for quantitative analysis. Wet assay and fire assay procedures are most common and results are employed for ore reserve estimation.

Resource estimation requires proper interpretation (a) of geological data to develop an orebody model which defines the various geological domains and (b) of the results of the geostatistical analysis of assay data. Among other things, geological domains can represent rock types, alteration types, mineralogy, and structural features such as major faults. Geostatistical analysis includes the preparation of histograms, cumulative frequency plots and variograms.

Histograms and cumulative frequency plots are used to distinguish distinct grade populations within geological domains. In general, the spatial variance of grade is evaluated by means of experimental variographs to determine the area of influence for a set of data points. Geological domains with distinct sample populations are evaluated separately.

A variogram is a graph of the variance between measured values (sample assays for example) as a function of the distance between these measured values. A typical variogram has a shape similar to that shown in Figure 5. The most important features of Figure 5 are the *nugget*, the *range* and the *sill*. The nugget effect represents random and short distance variability such as sampling error, assay error and erratic mineralization. Many gold deposits display non-uniform gold distributions, because the coarse grains cause a "nugget effect". Low nugget values are indicative of a finely and evenly distributed mineralization. In general, there

Figure 5: Typical Variogram



is a positive correlation between ore grades up to the distance indicated by the range, which can be seen as the range of influence for the data points in a particular direction. Accordingly, the sampling density should be within the range and a sampling spacing of about 2/3 of the range is adopted accordingly. The sill value is equal to the sample variance. The slope and shape of the variogram may vary in different directions, displaying directional anisotropy in the mineralization.

Methods for resource estimation include polygonal and block modeling procedures. The polygonal methods involve assigning grade, bulk density and area of influence to a polygon in a certain plane. They have the advantage of simplicity and ease of implementation and they provide an inexpensive preliminary assessment. However, subjectivity is involved in assigning the area of influence.

In block modeling, the deposit is divided into equal-sized blocks. Inverse-Distance weighting and kriging methods are used to assign grade and density to each block. Inverse distance weighting depends on assigning a weight proportional for each sample to an inverse power of the distance from the location of the estimate to the sample. On the other hand, kriging is a method developed to provide the best linear unbiased estimate for grade based on a least squares minimization of the error of an estimation. This method has a theoretical base to resolve complex situations. Block size is related to the mining method and selectivity required for grade control; height of a bench is related to geologic controls, at least 1/2 the size of the smallest geologic feature, and to drill hole spacing, 1/2 to 1/4 of average spacing. These are conflicting requirements and final size will be a compromise.

In ore resource evaluation mining and mineral beneficiation dilution and losses will be apportioned accordingly. This will result in an ore reserve estimate that can be used for estimation of revenue. Dilution is waste mined with ore; losses are ore left in the ground as a result of partial recovery.

Estimation of Revenue

The Net Smelter Return, $, provides a convenient way to express revenue in dollars per tonne. NSR are the proceeds from the sale of mineral products after deducting off-site processing and distribution costs. The following formulas contain most of the components of smelter contracts:

$$\$ = \left[\frac{G_c - U_d}{100}\right]\frac{P_r}{100}[P - R_c] - [T_c + a_t(P - P_0)] + C_r - P_n - T_r$$

or

$$\$ = \left[\frac{G_c - U_d}{100}\right]\frac{P_r}{100}[P + a_a(P - P_0) - R_c] - T_c + C_r - P_n - T_r$$

where a_t is called an escalator and is used when treatment charges are adjusted. When accountable metal units are being adjusted, the base price participation factor, a_a, is used.

$ = Value per tonne of concentrate

G_c = grade of product in concentrate (%)

U_d = unit deduction (%)

P_r = proportion of concentrate paid for (%)

P = price settlement ($/t of metal)

R_c = refining charge ($/t of metal)

T_c = treatment charge ($/t of concentrate)

a_t = escalator (fraction) applied to treatment charges (a = 0)

a_a = price participation (fraction) applied to accountable metal units (a_t = 0)

P_0 = contract base price ($/t of metal)

C_r = credit for by product ($/t of concentrate)

P_n = penalty ($/t of concentrate)

T_r = transport cost ($/t of concentrate)

The treatment charge or accountable meal units can be subject to adjustment by a fraction of the difference between contract price and settlement price. The fraction will be different depending on whether the difference is positive or negative. Additional clauses to the formula could be weighing, sampling, moisture determination and arbitration in case of disputes.

Credit for byproduct elements such as gold is calculated from:

$$[G_b - U_d]P_r[P_b - R_{cb}]$$

where G_b = grade of byproduct in concentrate (g/t); U_d = unit deduction (g/t); P_r = proportion of concentrate paid for (%); P_b = price settlement ($/g of byproduct); R_{cb} = refining charge ($/g of byproduct). Typically, concentrate penalties are calculated from:

$$P_n = [A_a - A_m]P_p$$

where A_a = content of penalty element in conc, %; A_m = minimum content of penalty element below which there is no penalty, %; P_p = penalty in $/tonne conc/% penalty element.

Example Calculation

Calculate the annual revenue from a Cu/Au deposit with a 500,000 tpy capacity and a mill feed grade of 2% Cu, 1.5 g/t Au and 0.02% As. Recovery is, respectively, 90, 80 and 60%. Copper, the main product, assays 22% in the concentrate under the following conditions: unit deduction = 1%, refining charge = $150/t, treatment charge = $75/t, proportion of concentrate paid = 99%, contract base price = $2200/t, settlement price = $2640/t, escalator is 0.1 when difference, P - P_0, is positive and .05 when negative. A_m = 0.1%; P_p = $10/t per percent of As. The credit for Au involves a unit deduction of 1 g/t at a proportion of concentrate paid for of 95%, a refining charge of $0.18/g of Au and a price settlement of $15/g of Au. Transport charge is $108/t of concentrate. If f is feed grade with the same units as G_c, then recovery is determined from:

Recovery = $\left[\frac{tpyConc}{tpyFeed}\right]\left(\frac{G_c}{f}\right)$ expressed as a fraction.

Concentrate grades of elements can be found from:

Concentrate Grade = $Recovery\left[\frac{tpyFeed}{tpyConc}\right]f = G_c$

Hence the tpyConc = $\frac{Recovery(f)(tpyFeed)}{G_c} = \frac{.90x2x500000}{22} = 40909$ tonne

For gold, $G_b = .80\left[\frac{500000}{40909}\right]1.5 = 14.56 g/t$

For Arsenic, $G_b = .60\left[\frac{500000}{40909}\right].02 = 0.1466\%$

Credit for Au = $[14.56 - 1]\left(\frac{95}{100}\right)[15 - 0.18] = \$190.91$ per tonne of concentrate

Penalty for As = (0.1466 - 0.1)($10) = (.0466)($10) = $0.47 per tonne of concentrate

In this example an escalator is used , although price participation is the norm for Cu. Thus:

$\$ = \left[\frac{22-1}{100}\right]\left(\frac{99}{100}\right)[2640 - 150] - [75 + 0.1(2640 - 2200)] + 190.91 - 0.47 - 108$

= 517.67 - 119 + 190.91 - 0.47 - 108

= $481.11/tonne of Concentrate

Since there are 40909 tonnes of concentrate, then the annual revenue must be:

= $481.11(40909) = $19.682x10^6.

MINERAL PROJECT APPLICATIONS

Equipment replacement

The comparison between existing equipment with a short useful life and new equipment with a relatively longer useful life calls for the use of an equivalent annual cost method. The equivalent annual cost is the stream of equal end-of-period values that represent preferably the after-tax costs associated with the economic life of the equipment. Thus, the equipment with the lowest equivalent annual cost is chosen. The use of this method implies that services are required indefinitely, the new equipment can be replaced by identical equipment, and the existing equipment will eventually be replaced by the new equipment.

For a valid comparison, both existing and new equipment must be evaluated under their best economic conditions, i.e. under their lowest cost life. For the new equipment, this life period corresponds to its economic life, i.e., the time period over which the equivalent annual cost of

ownership is minimized. This cost of ownership of an asset consists of the initial capital investment, operating and maintenance costs which increase as the asset ages and deteriorates, and a decreasing realizable salvage value as the asset ages.

Thus the cost of ownership of existing equipment over one or more years consists of increasing operating and maintenance costs as the asset deteriorates, decreasing salvage values as the asset ages, and possibility of an overhaul that would reduce operating and maintenance costs for a few future years. When the overhaul alternative is not considered, the minimum equivalent annual cost is achieved for a period of one more year and higher costs are associated with longer periods of time. Although an overhaul may require a significant capital investment it would typically decrease operating and maintenance costs over future years. After overhauls, the economic life must be determined by the same procedure as that used for the new equipment.

The economics of replacement should be periodically reviewed in the light of technological improvements and unforeseen increases in costs. The method described assumes that the existing and the new equipment will produce equivalent benefits. If this is not so, after-tax benefits should be added to the cost components.

Investments in Mining/Further Reading

Investments in mining require that a detailed cash flow analysis be applied with provision for taxation and inflation. This is illustrated via a comprehensive example, where a mining project has an expected capital cost of 102 million dollars completed by time zero and a stream of revenue of 50 millions for 6 years with operating costs of 20 millions. General inflation is 5 % and tax is 40 %. We want to calculate the net present value and the rate of return. Assume straight line depreciation.

The following table presents the results.

Cash Flow Over Project Life

Time: yr	0	1	2	3	4	5	6
Revenue		52.50	55.13	57.80	60.78	63.81	67.00
Operating cost		21.00	22.05	23.15	24.31	25.53	26.80
Operating		31.50	33.08	34.73	36.47	38.28	40.20
Capital cost allowance		17.00	17.00	17.00	17.00	17.00	17.00
Taxable revenue		14.50	16.08	17.73	19.47	21.28	23.20
Taxes at 40%		5.80	6.43	7.09	7.78	8.51	9.28
Profit		8.70	9.65	10.64	11.69	12.77	13.92
Capital cost	102						
Cash flow (current $)	102	25.7	26.65	27.64	28.69	29.77	30.92
Cash flow(constant $)	102	24.481	24.17	23.87	23.60	23.33	23.07

NPV @ 10% $1.79 million
ROR 10. 6 %

Revenue and cost are inflated because any taxes and debt are paid in present dollars. Notice that capital cost allowance however is not inflatable. This means that the tax shield given by depreciation is relatively reduced with time. The resulting cash flow, in current dollars is deflated so that project evaluation criteria can be calculated with a constant dollar cash flow. The net present value is calculated with a 10% discount rate. This rate is theoretically the

riskless rate, such as the Treasury Bond rate plus a risk premium. The lower limit of the discount rate is the cost of capital. No project should be considered if it can not exceed the cost of the capital needed. Companies will use a higher rate, called the hurdle rate, to rank investment opportunities.

Readers are referred to the following texts which have proven useful for mineral project evaluation techniques:

Gocht, W. R., Zantop, H. and Eggert, R. G. (1988) **International Mineral Economics**, Springer-Verlag.

Gentry, D. W. and O'Neil, T. J., (1984) **Mine Investment Analysis**, SME-AIME, Littleton,Colorado.

Vallee, M. (1992), **Guide to the Evaluation of Gold Deposits**, CIM Special Volume 45, Montreal, Quebec.

MAJOR MINING EQUIPMENT COSTS

Major mining equipment items are placed in the following categories: **Drilling**; **Excavating and Loading**; **Haulage**; **Ventilation and Cooling**; **Power Generation**. Categories are presented in this order.

For each category, examples of equipment cost estimations that utilize corresponding graphs herein, are incorporated for clarity.

Equipment costs are given in the form of logarithmic scale graphs. The cost in US dollars at an M&S(Mine/Mill)$_{1400}$ index of 1400 is plotted against a suitable parameter (s). In some cases, additional graphs are provided to relate important parameters such as capacity and/or horsepower to cost parameters.

Once the cost parameter is known, the equipment cost is estimated in US dollars. The current exchange rate (Canadian to US) is employed to convert cost to Canadian dollars.

I. Drilling Equipment

Drilling equipment has been subdivided further as **Borers**, **Drills**, **Stopers** and **Rigs**.

(1) Example Estimates - - - - - - - - **Page 64**

(2) Tunnel Borer - - - - - - - - - **Page 65**

(3) Raise Borers, AC and DC - - - - - - - **Page 66**

(4) Drill Pipe - - - - - - - - - - **Page 67**

(5) Jackleg Drills (hand-operated) - - - - - - - **Page 68**

(5) Stopers (hand-operated) - - - - - - - **Page 69**

(7) Production Drill Rigs, Percussive Crawler-Mounted - - - **Page 70**

(8) Production Drill Rigs, RotaryTruck-Mounted - - - - **Page 71**

(9) Production Drill Rigs, Underground Jumbo - - - - - **Page 72**

EXAMPLE ESTIMATES

Drilling equipment prices have been related to equipment parameters such as size, mass and force.

Example 1:

What is the price of a 15 foot tunnel borer in Canadian dollars when the M&S (Mine/Mill) Index is 1090?

Figure 6, page 65, shows that the price is about US$6,900,000 at an M&S/(M/M) Index of 1400. This price is first converted via the Index ratio as $69000000\left[\frac{1090}{1400}\right]$ = US$5,370,000. With an exchange ratio of 1.4, this becomes CAD$7,518,000 at an Index of 1090.

Example 2:

What is the cost of a 10 foot variable speed DC drive raise boring machine with 100 feet of drill pipe at an M&S(M/M) Index of 1400 in Canadian dollars?

Figure 7, page 66, shows that a DC drive raise borer should be about US$1,800,000. On page 67, Figure 8 shows that at a cutter diameter of 10 foot, 100 feet of drill pipe is around US$141,000. Total cost is then US$1,941,000 at an M&S(M/M) Index of 1400. At an exchange ratio of 1.4, this becomes CAD$2,717,400.

Example 3:

What is the price of a production truck-mounted rotary drill rig with a bit pulldown force of 100,000 lb?

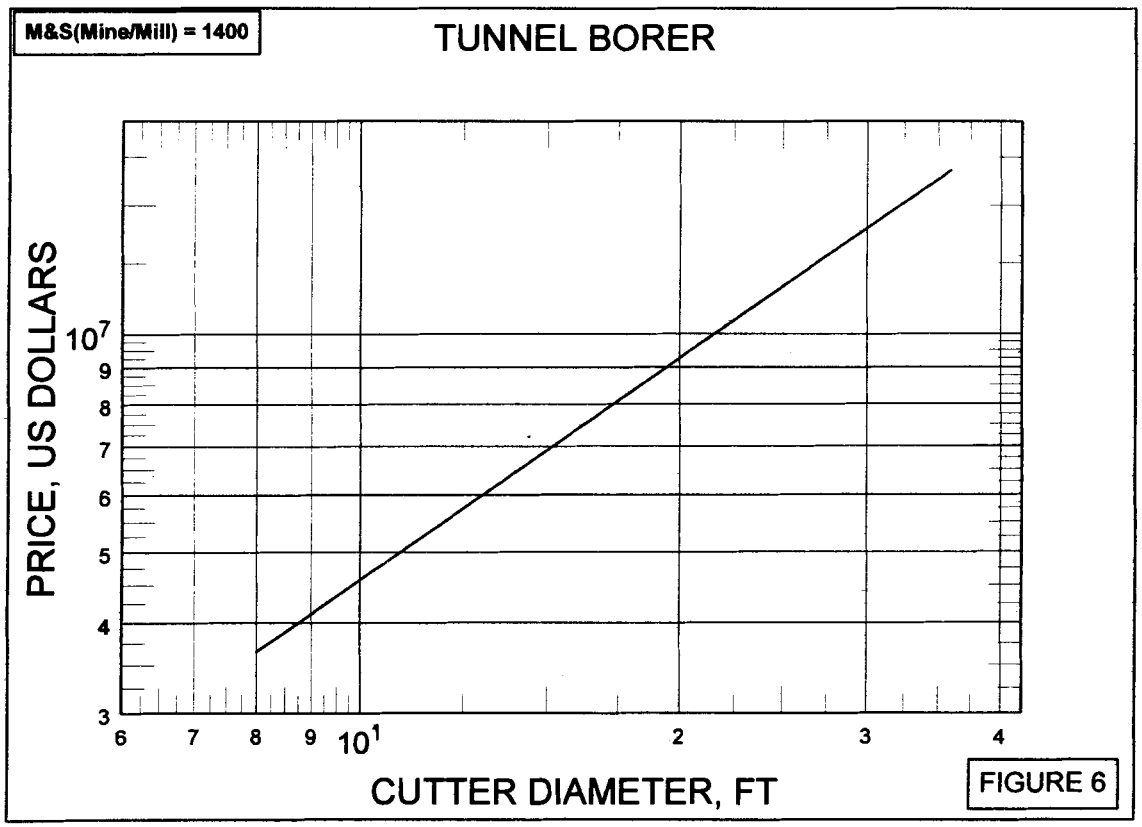
Referring to Figure 12, page 71, the price is estimated as US$960,000 at an M&S(M/M) Index of 1400. If the Index today is 1090, then this price becomes $\left[\frac{1090}{1400}\right]960000$ = US$747429. This is CAD$1,046,400 at the exchange ratio of 1.4.

TUNNEL BORER

Price = aX^b, US dollars

where **X is the cutter diameter in feet**. Price includes basic machine; excludes special materials transport equipment such as belts, trucks and tunnel lining machinery.

Range in X, ft.: 8 to 36 a = 440300 b = 1.0126



RAISE BORERS, AC and DC

Price = aX^b, US dollars

where **X is the cutter diameter in feet**. Price includes basic machine; excludes crawler, starter pipe, cutter, pitot bit and drive. It should be noted that these machines are sized by their rated cutter diameters. However, cutters of up to 40% larger than the rated size have been successfully used on these machines.

AC Range in X, ft:	4 to 15	a = 98630	b = 1.262
DC Range in X, ft:	5 to 15	a = 186600	b = 0.9896

NOTE: An *hydraulic* unit is of 5 ft cutter diameter and has an average cost of US$867958 at the M&S(M/M) Index of 1400.

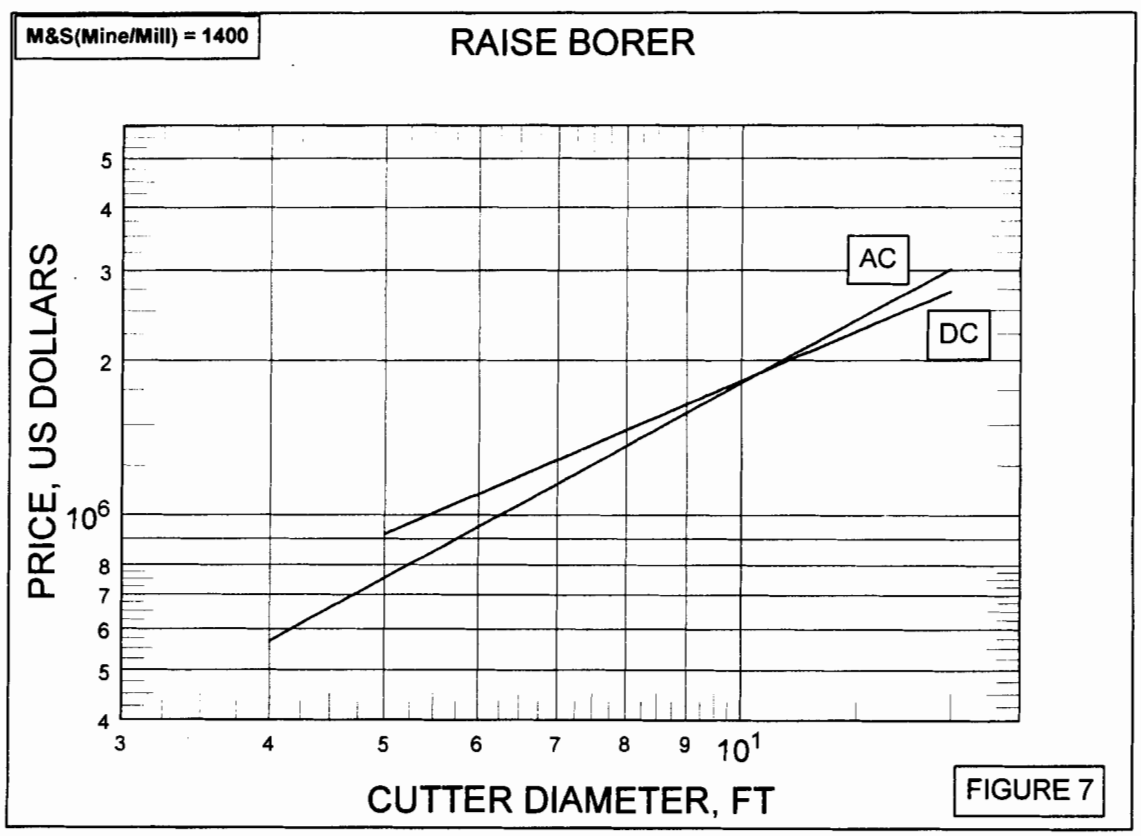


FIGURE 7

DRILL PIPE
(For Raise Borer)

Price = aX^b, US dollars

where **X is the raise borer cutter diameter in feet**. Price is for 100 feet of drill pipe and includes guides.

Range in X, ft: 4 to 16 a = 14460 b = 0.9895

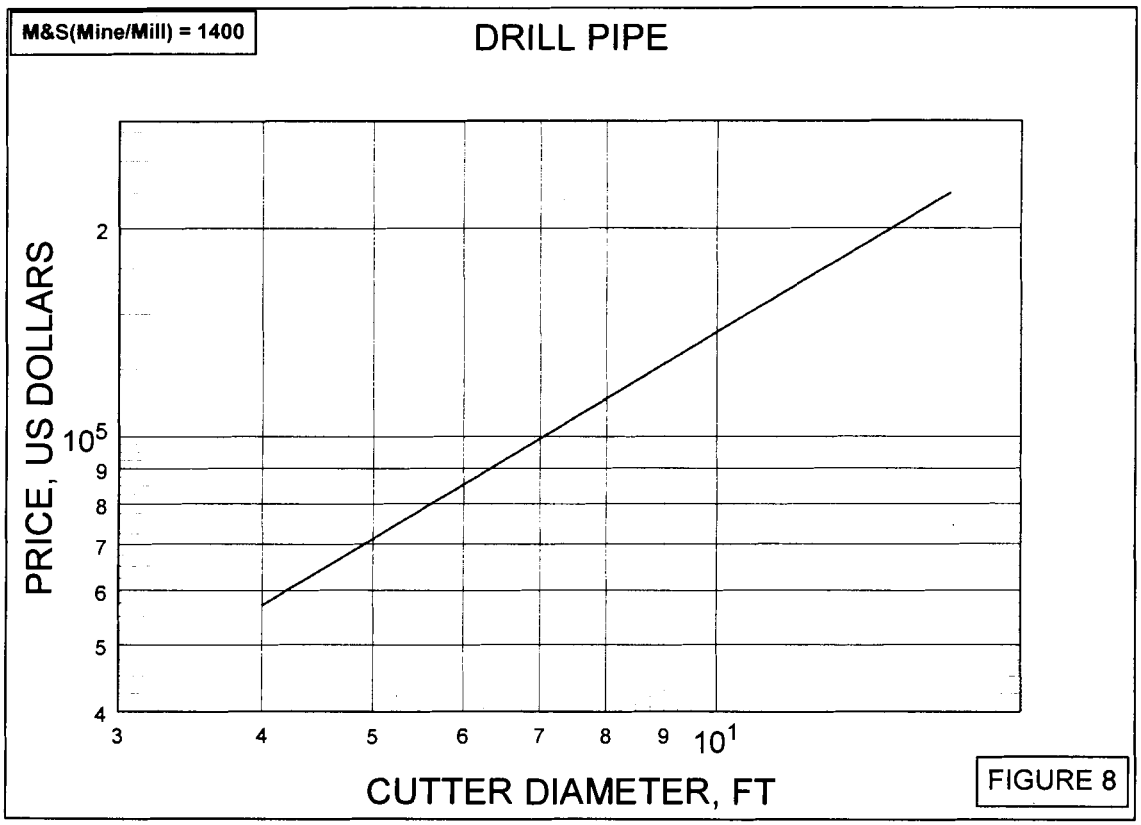


FIGURE 8

JACKLEG DRILLS
(Hand Operated)

Price = aX^b, US dollars

where **X is drill weight in lbs**. Price covers light, medium and heavy machines; includes drill and jackleg; excludes drill steel oiler and hoses.

Range in X, lb: 48 to 77 a = 191.1 b = 0.8269

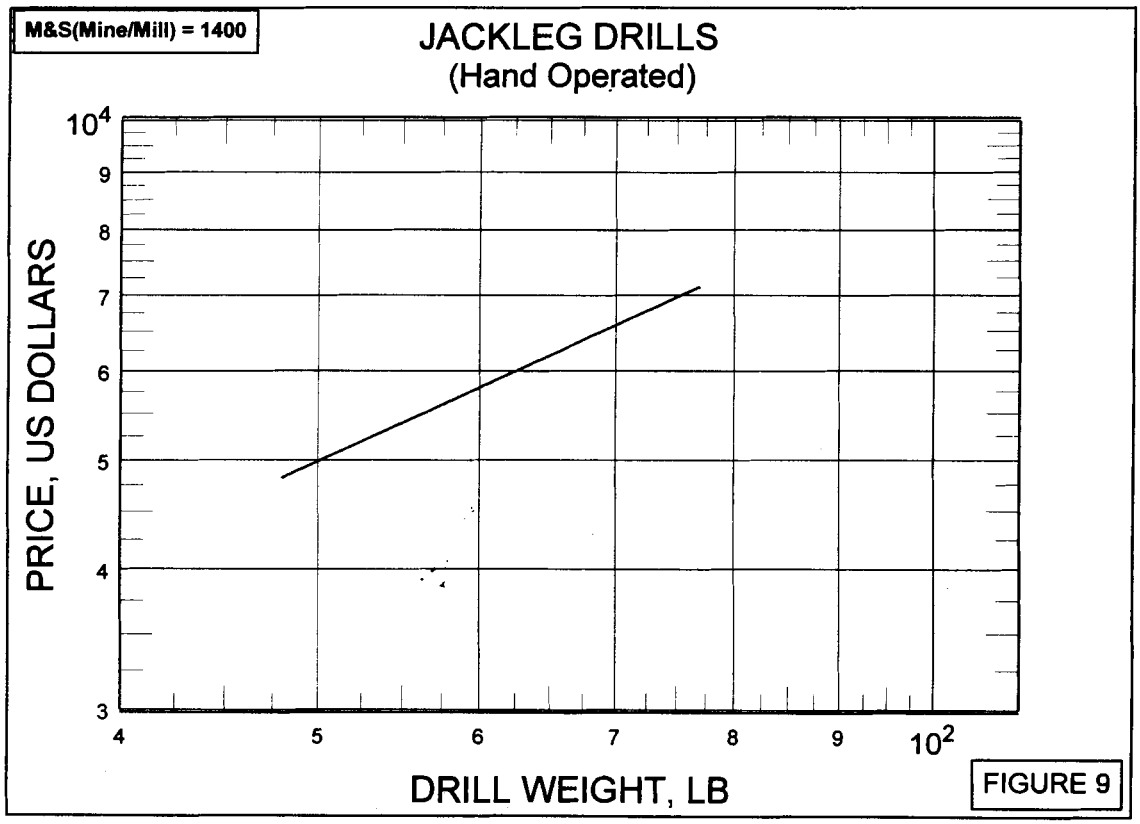


FIGURE 9

STOPERS
(Hand Operated)

Price = aX^b, US dollars

where **X is the number of stopers**. Price covers 86 lb machine; includes leg; excludes drill steel, oiler and hoses.

Range in X: 1 to 10 units a = 5298 b = 1.000

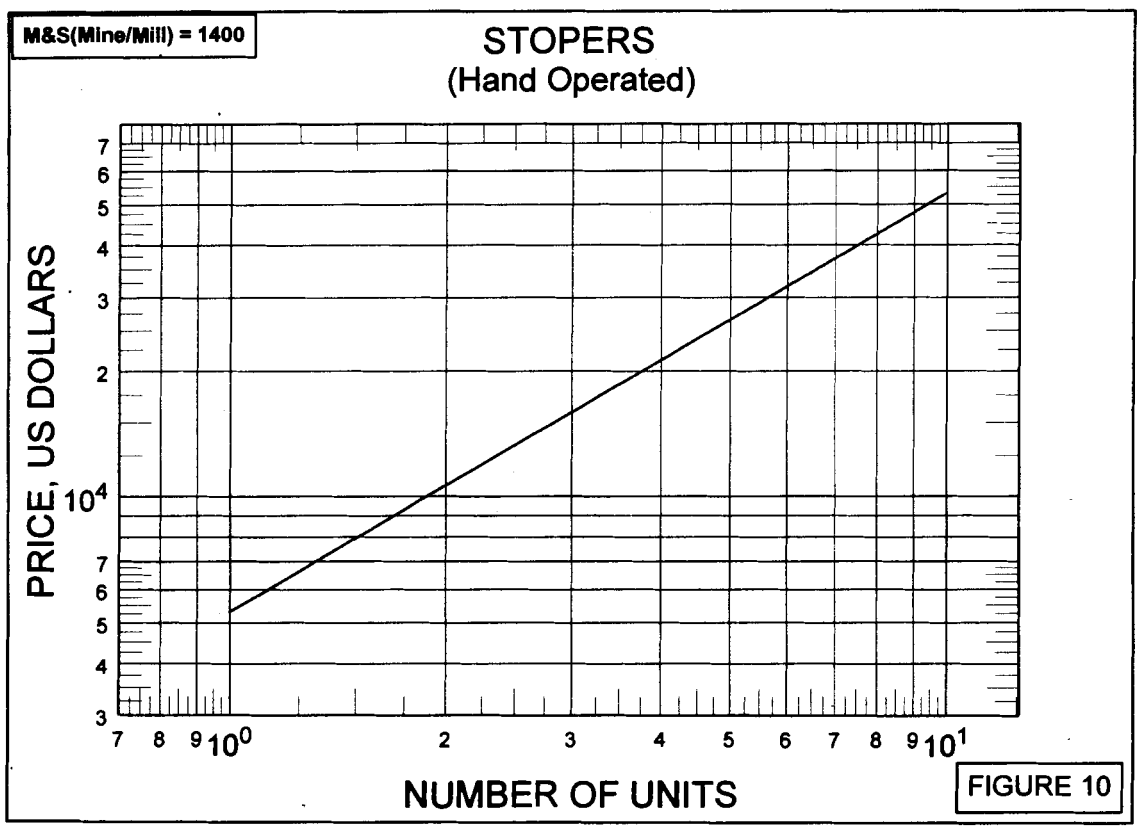


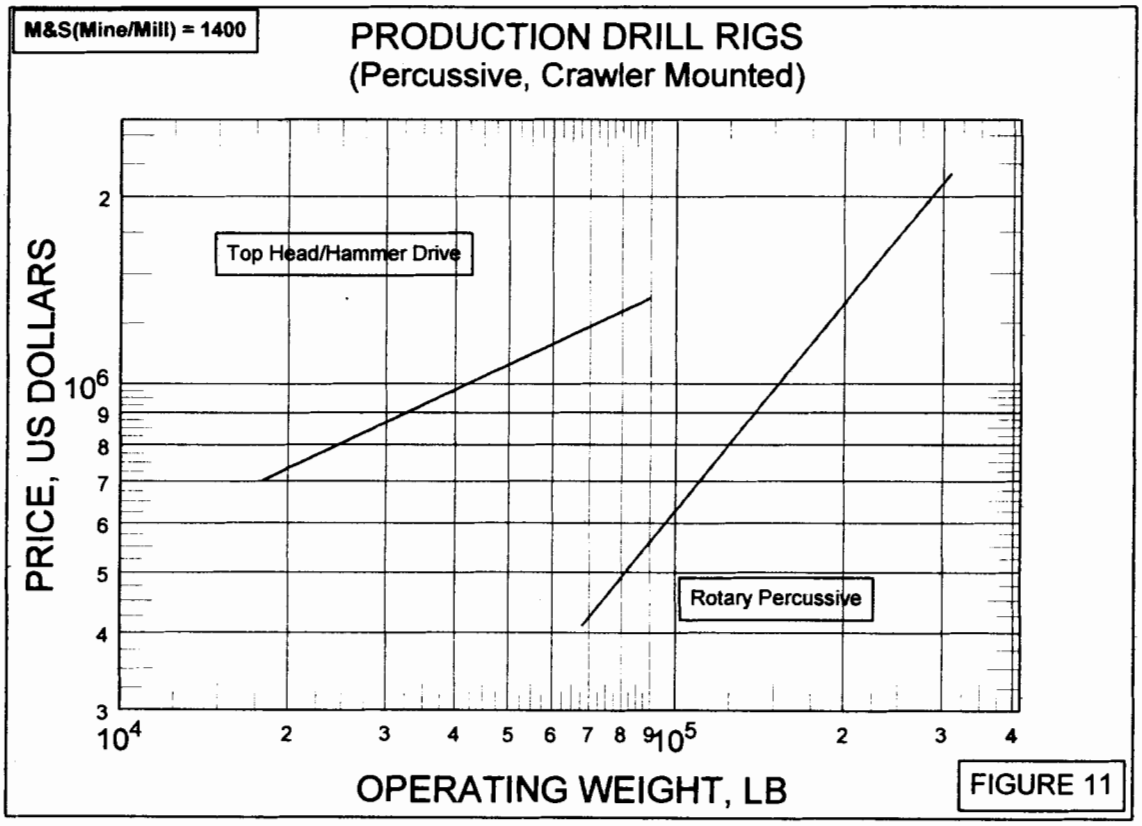
FIGURE 10

PRODUCTION DRILL RIGS
(Percussive, Crawler Mounted)

Price = aX^b, US dollars

where **X is operating weight in pounds (lb)**.

Top Head/Hammer Drive Range in X, lb: 18000 to 90000	a = 11740	b = 0.4172
Rotary Percussive Range in X, lb: 68000 to 310000	a = 1.988	b = 1.1000



PRODUCTION DRILL RIGS
(Rotary, Truck Mounted)

Price = aX^b, US dollars

where **X is the bit pulldown force in pounds (lb)**.

Range in X, lb: 30000 to 126000 a = 409.1 b = 0.6739

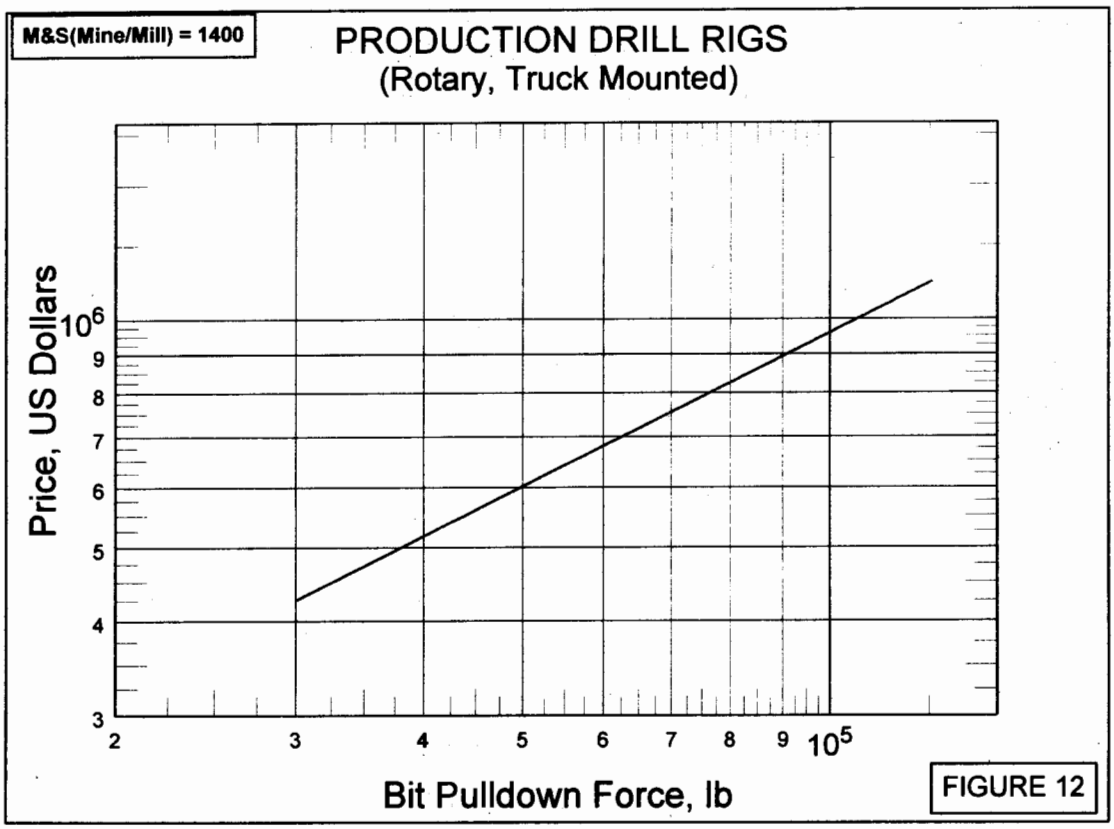


FIGURE 12

PRODUCTION DRILL RIGS
(Underground Jumbos)

Price = aX^b, US dollars

where **X is operating weight in kilograms**. Note: 0.454 kg = 1 lb.

Range in X, kg:			
	8500 to 19200	a = 2674	b = 0.569
	19200 to 22000	a = 7.537E-9	b = 3.265

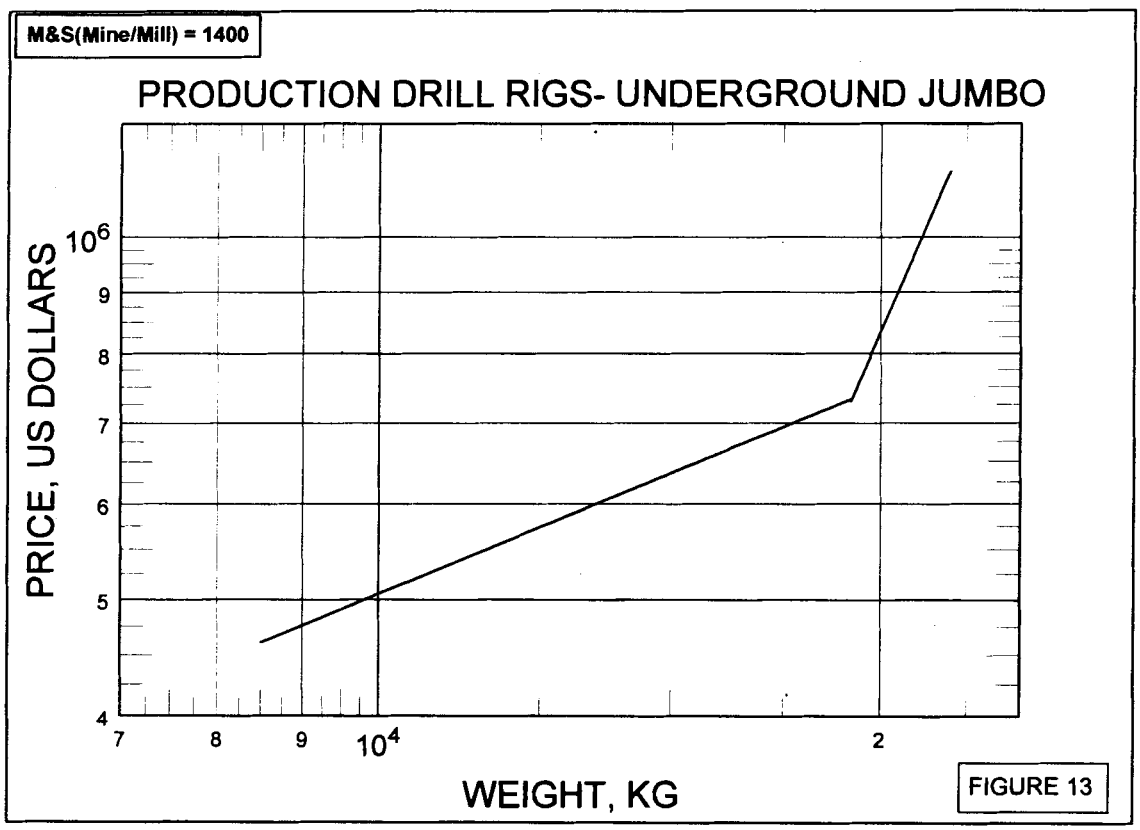


FIGURE 13

II. Excavating and Loading Equipment

Excavating and Loading equipment has been further categorized as **Excavators**, **Loaders**, **Movers**, **Shovels** and **Miners**.

(1) Example Estimates - - - - - - - - Page 74

(2) Walking Draglines - - - - - - - - Page 75

(3) Bucket Wheel Excavators - - - - - - - Page 76

(4) Dozers (Track/Wheel) - - - - - - - - Page 78

(5) Graders - - - - - - - - - - Page 79

(6) Front End Loaders - - - - - - - - Page 80

(7) Scoop Trams - - - - - - - - - Page 82

(8) Cable Shovels - - - - - - - - - Page 83

(9) Scrapers - - - - - - - - - - Page 84

(10) Hydraulic Shovels - - - - - - - - Page 86

(11) Continuous Miners - - - - - - - - Page 87

(12) Continuous Miners, Boom Type - - - - - - Page 88

(13) Longwall Miners - - - - - - - - Page 90

EXAMPLE ESTIMATES

Prices of excavating and loading equipment have been related to power, capacity and size parameters.

Example 1:

What is the price of a 25 cubic yard, 230 foot boom walking dragline at an M&S(M/M) Index of 1400?

Figure 14, page 75, shows that for (boom length)^2(bucket size) = (230/3)^2(25) = 146944.44, the price is approximately US$13,400,000. This covers the basic machine and includes the bucket.

Example 2:

4000 tons per hour of material is to be excavated by means of a bucket wheel excavator. What is the cost of a suitable machine today?

From Figure 16, page 77, a bucket wheel excavator with a nominal capacity of 4000 tph has a diameter of 24 feet. The price of a 24 foot diameter unit, from Figure 15, page 76, is about US$7,120,000 at an M&S(M/M) Index of 1400. If today's index is 1090, then this becomes approximately:

$$\left[\frac{1090}{1400}\right]7120000 = \text{US\$5,540,000.}$$

Example 3:

A wheel mounted front end loader with a flywheel horsepower of 400, is observed by technical personnel to have desirable operating characteristics during a mine site visit. What is the nominal capacity in (yds)^3 of this unit and what is its cost in current dollars?

Figure 20, page 81, shows that the nominal capacity of a 400 flywheel horsepower, wheel mounted unit is about 8 (yds)^3. From Figure 19, page 80, the price at an M&S(M/M) Index of 1400 is approximately US$670,000. If today's index is 1090, then this becomes approximately:

$$\left[\frac{1090}{1400}\right]670000 = \text{US\$522,000.}$$

Example 4:

A long wall miner, operating with a face conveyor, requires a minimum height of 50 inches of operating height. 900 tons of roof support is judged necessary. What are the costs associated with this equipment at an M&S(M/M) Index of 1400?

Figure 29, page 90, shows that the price of the miner is about US$1,300,000. Add US$149,400 for 900 tons of roof support; add US$1,800,000 for a face conveyor.

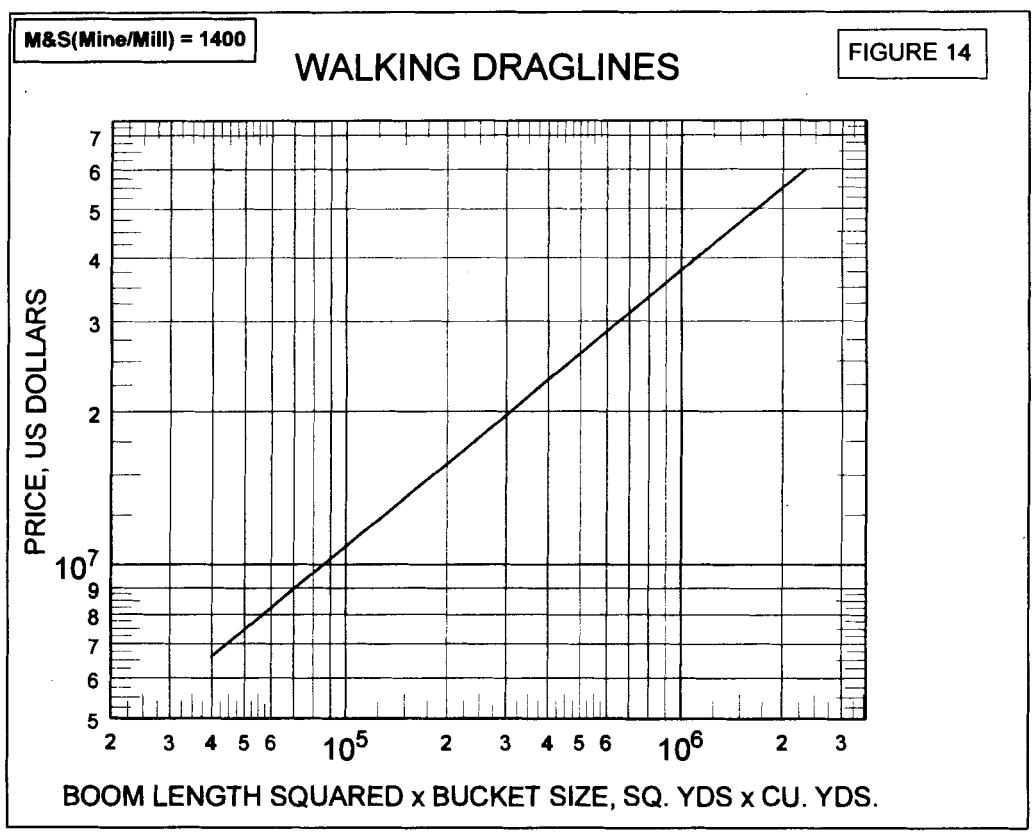
WALKING DRAGLINES

Price = aX^b, US dollars

where **X is the boom length squared (square yards) times the bucket size (cubic yards).** Price includes basic machine and bucket.

Range in X, $(yds)^5$: 39822 to 2352000 a = 21290 b = 0.5414

Boom Length, yds:	With Bucket size, $(yds)^3$:
71.67	14
73.33	22
76.67	25
88.33	25
81.00	35
91.67	45
106.67	45
110.00	60



BUCKET WHEEL EXCAVATORS

Price = aX^b, US dollars

where **X is the bucket wheel diameter in feet**. Price covers the standard machine.

Range in X, ft:	3.5 to 18.15	a = 262000	b = 0.8757
Range in X, ft:	18.15 to 37.7	a = 985.5	b = 2.796

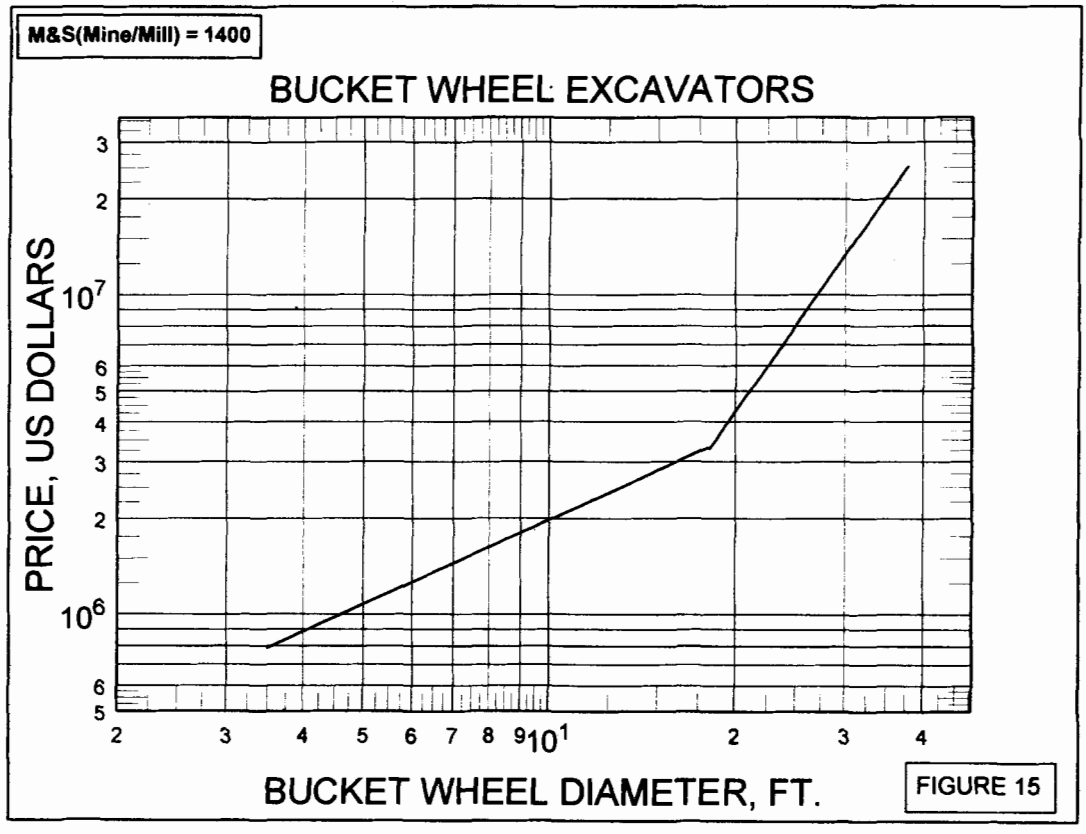


FIGURE 15

BUCKET WHEEL EXCAVATORS (CONTINUED)

Figure 15 below shows how bucket wheel diameter in feet varies with a nominal capacity in tons per hour.

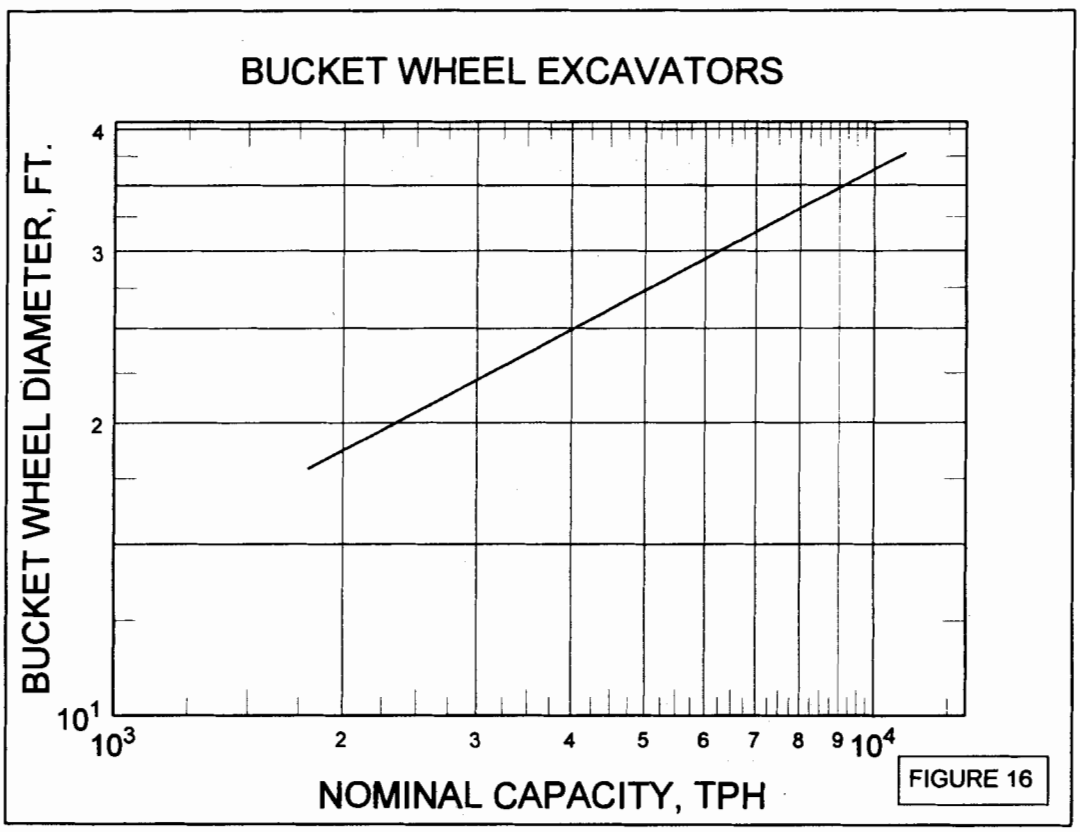


FIGURE 16

DOZERS (TRACK/WHEEL)

Price = aX^b, US dollars

where **X is flywheel horsepower**. Price includes basic machine fitted for mining applications.

Track-Type Range in X, HP:	70 to 770	a = 1287	b = 1.067
Wheel-Type Range in X, HP:	210 to 690	a = 1050	b = 1.08

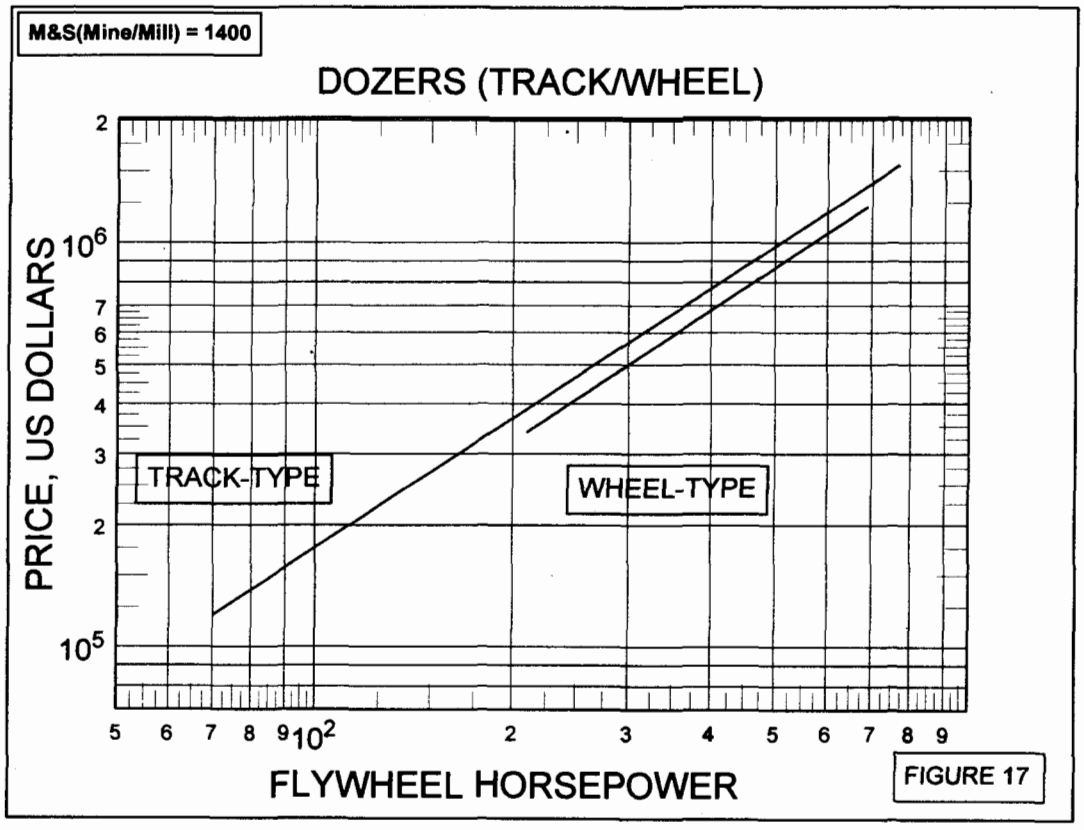


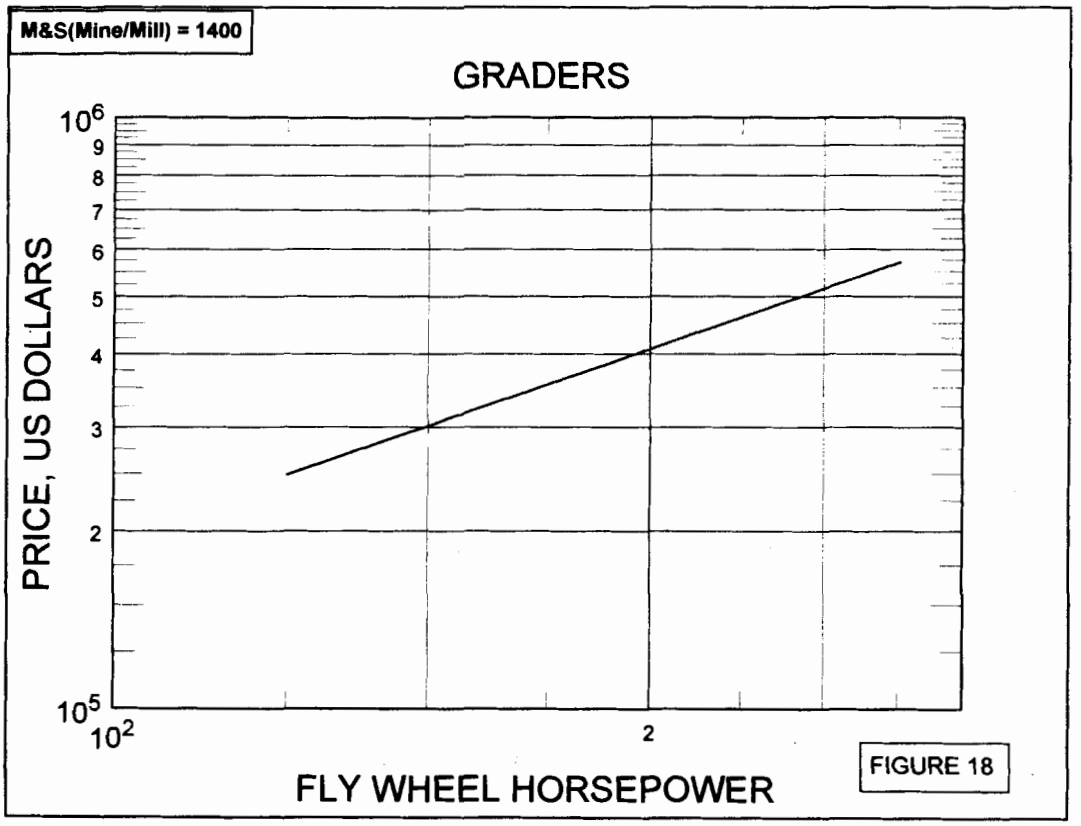
FIGURE 17

GRADERS

Price = aX^b, US dollars

where **X is flywheel horsepower**. Price covers basic machine and includes blade.

Range in X, HP: 125 to 276 a = 1566 b = 1.049



FRONT END LOADERS
(Track/Wheel)

Price = aX^b, US dollars

where **X is flywheel horsepower**. Price includes tires or tracks and bucket.

Track Range in X, HP:	65 to 210	a = 1123	b = 1.122
Wheel Range in X, HP:	78 to 1600	a = 569.2	b = 1.181

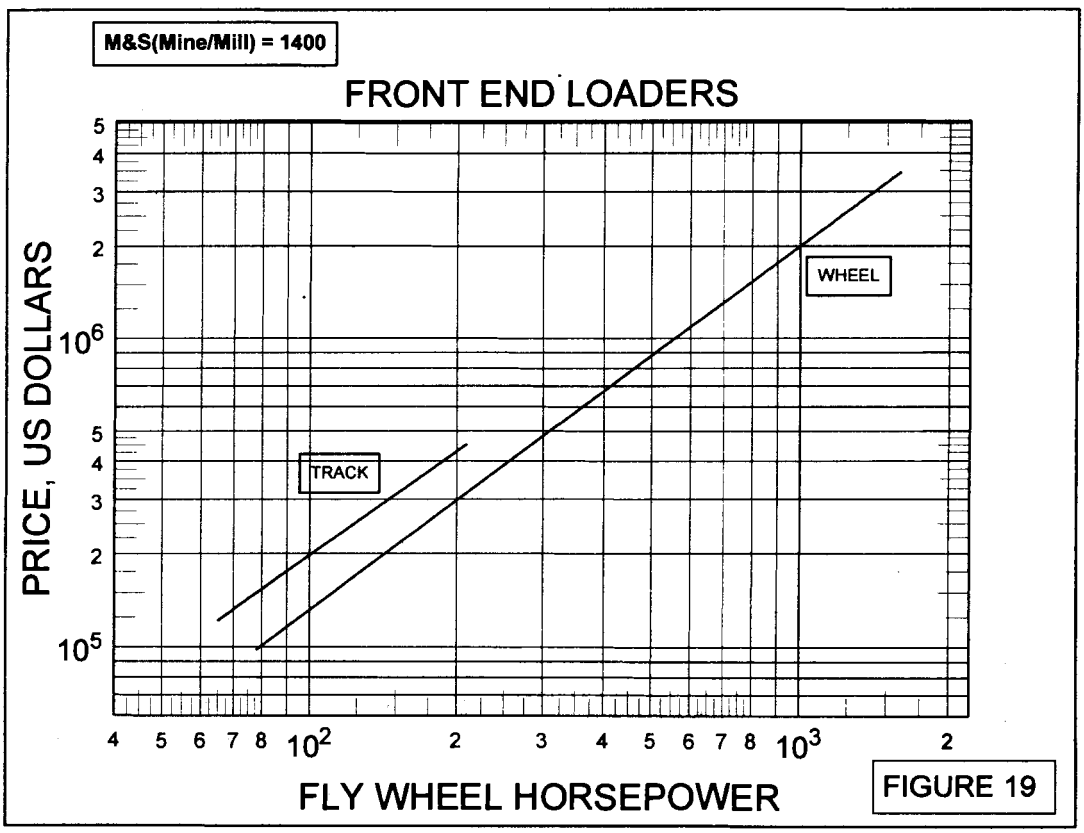


FIGURE 19

FRONT END LOADERS (CONTINUED)

Figure 19 below shows how flywheel horsepower varies with the nominal capacity of front end loaders of the track and wheel types.

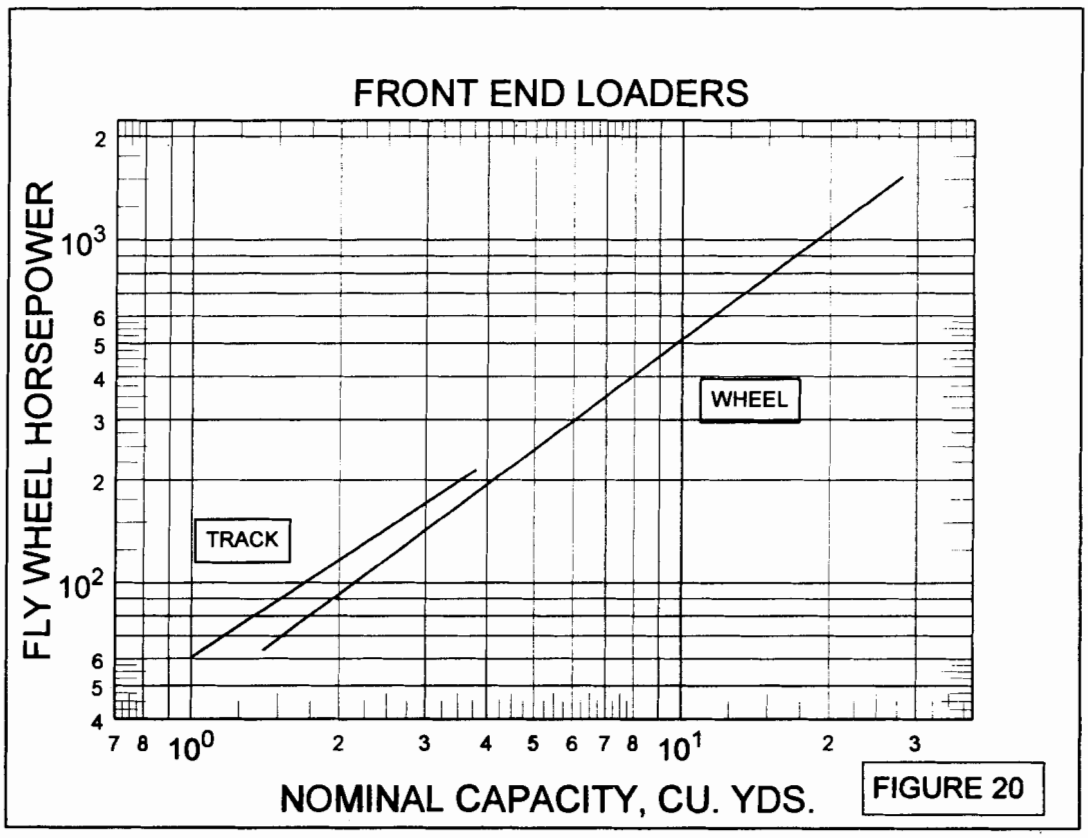


FIGURE 20

SCOOP TRAMS

Price = aX^b, US dollars

where **X is capacity in pounds**.

Coal Range in X, lb: 5000 to 15000	a = 73.18	b = 0.861
Hard Rock Range in X, lb: 5000 to 39000	a = 70.59	b = 0.841

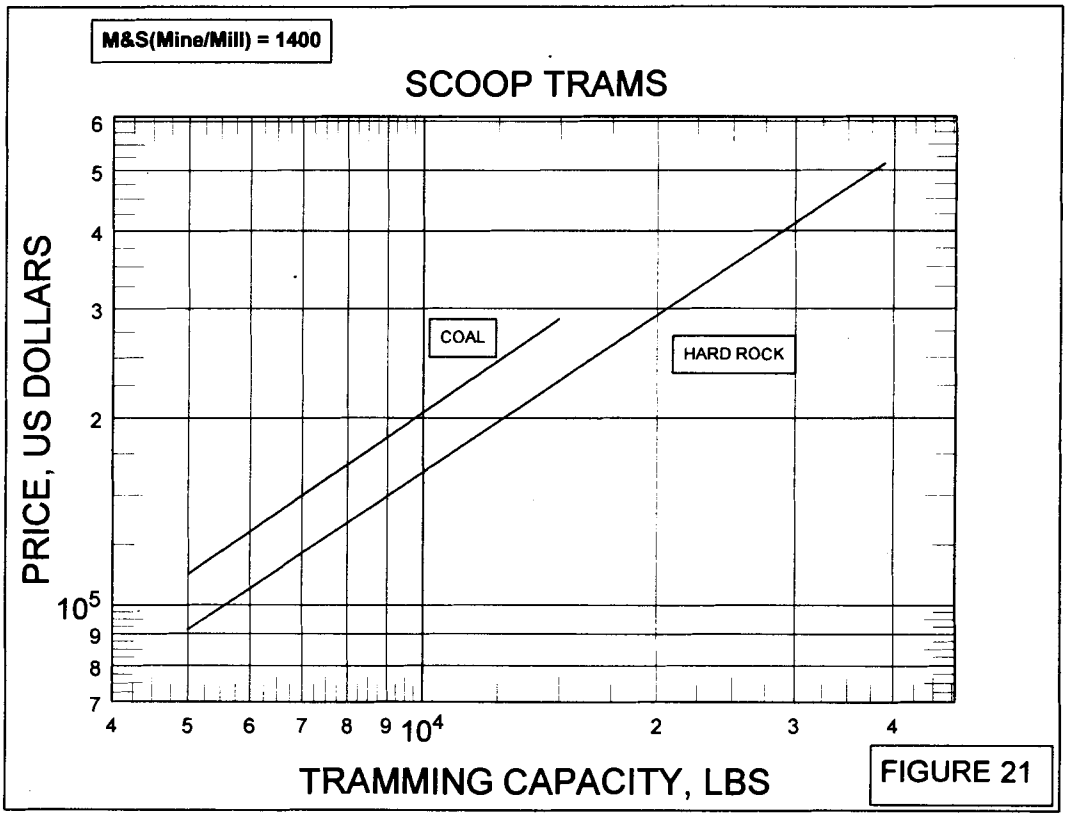


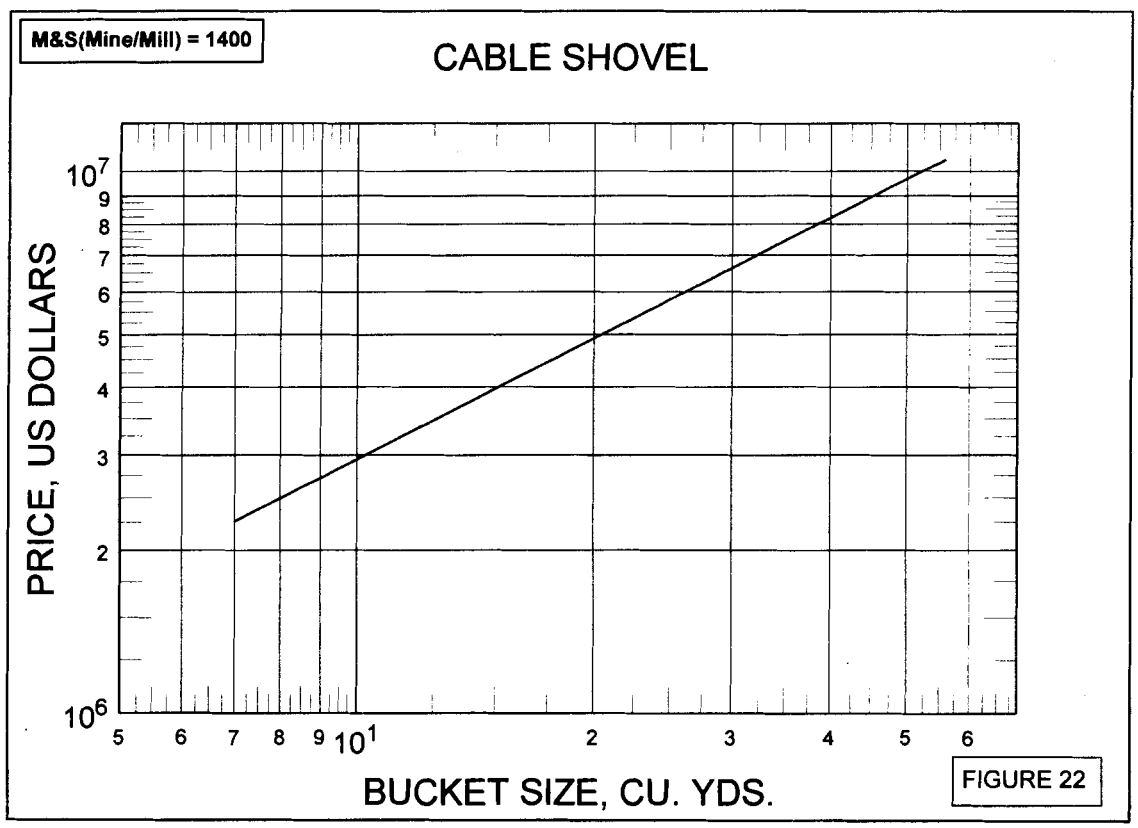
FIGURE 21

CABLE SHOVELS

Price = aX^b, US dollars

where **X is bucket size in cubic yards**. Price includes motor with motor horsepower increasing with increasing bucket size. LIkewise, as bucket size increases, boom length varies from 35 to 70 ft, while boom power varies from 250 kW to 1350 kw.

Range in X, yd^3: 7 to 56 a = 535300 b = 0.7395



SCRAPERS

Price = aX^b, US dollars

where **X is flywheel horsepower**. Price refers to standard machine used for stripping overburden.

Standard Range in X, HP: 330 to 550	a = 438.3	b = 1.236
Tandem Push-Pull Range in X, HP: 290 to 950	a = 11280	b = 0.7068
Elevating Range in X, HP: 175 to 475	a = 638	b = 1.181

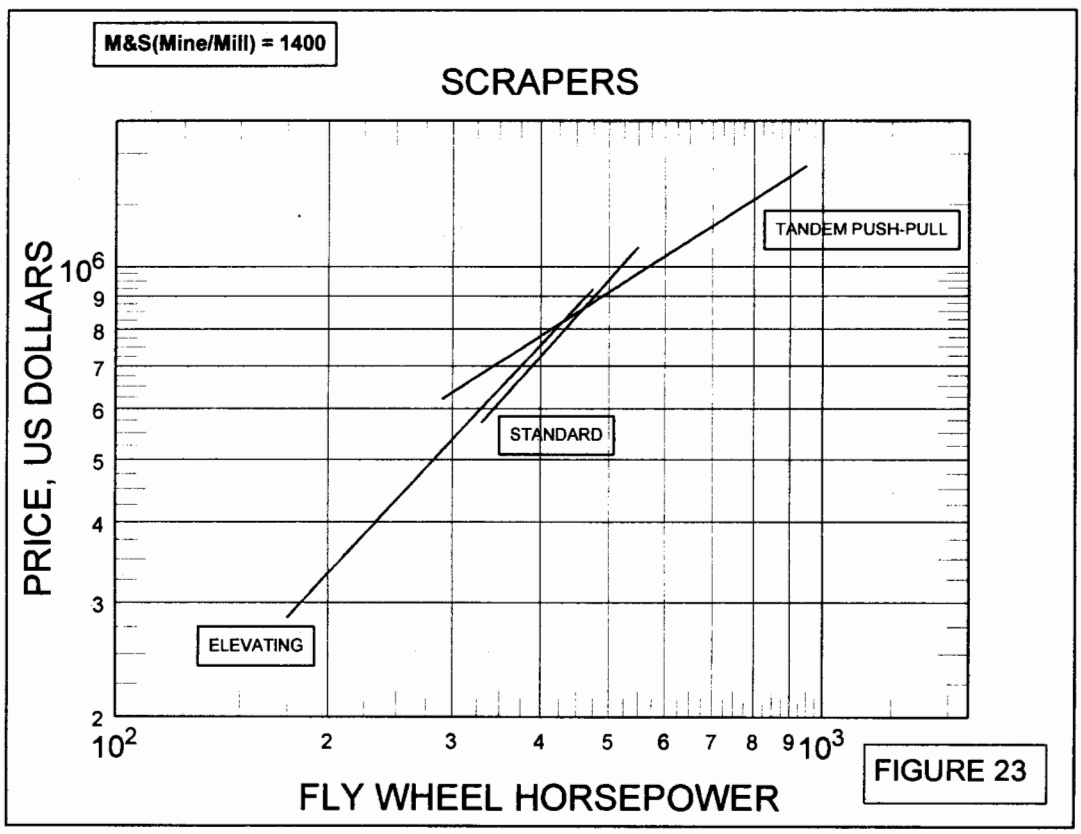


FIGURE 23

SCRAPERS (CONTINUED)

Figure 22 below shows how the flywheel horsepower of scrapers will vary with nominal capacity in cubic yards.

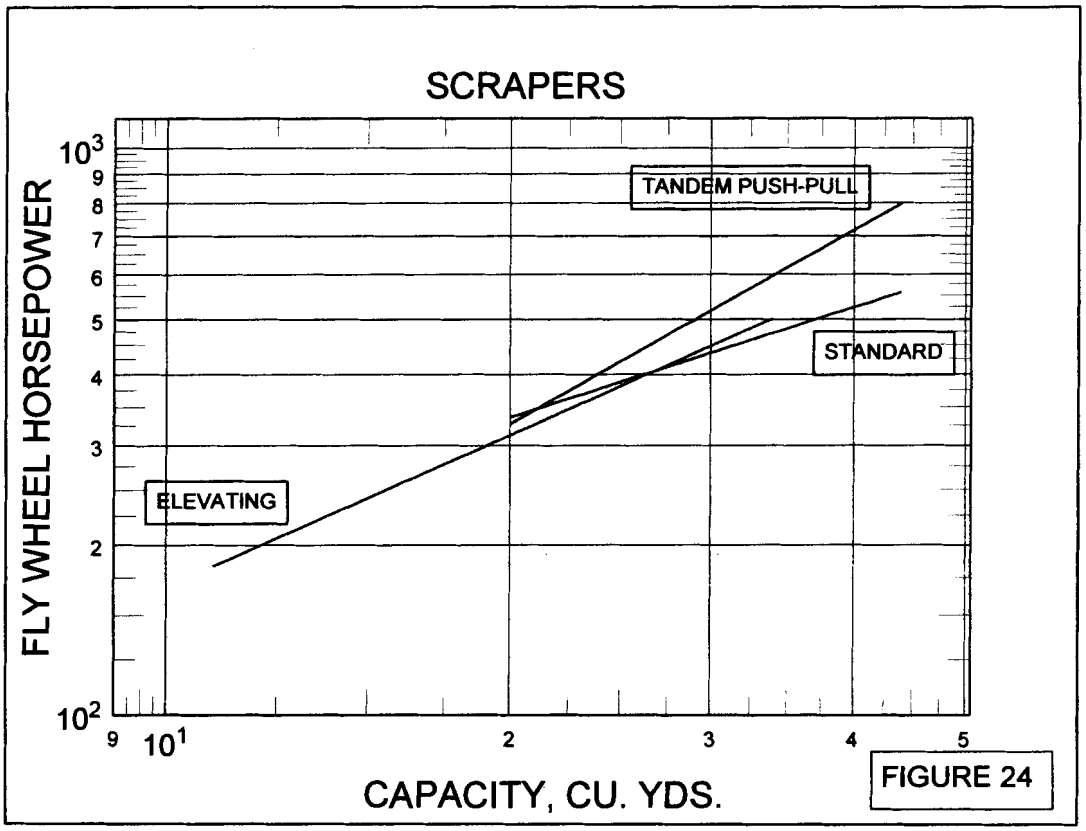


FIGURE 24

HYDRAULIC SHOVELS

Price = aX^b, US dollars

where **X is bucket size in cubic yards**. Price includes basic machine and bucket; add an average of 3% of the shovel price for options.

Range in X, $(yds)^3$: 3 to 35 a = 125000 b = 1.128

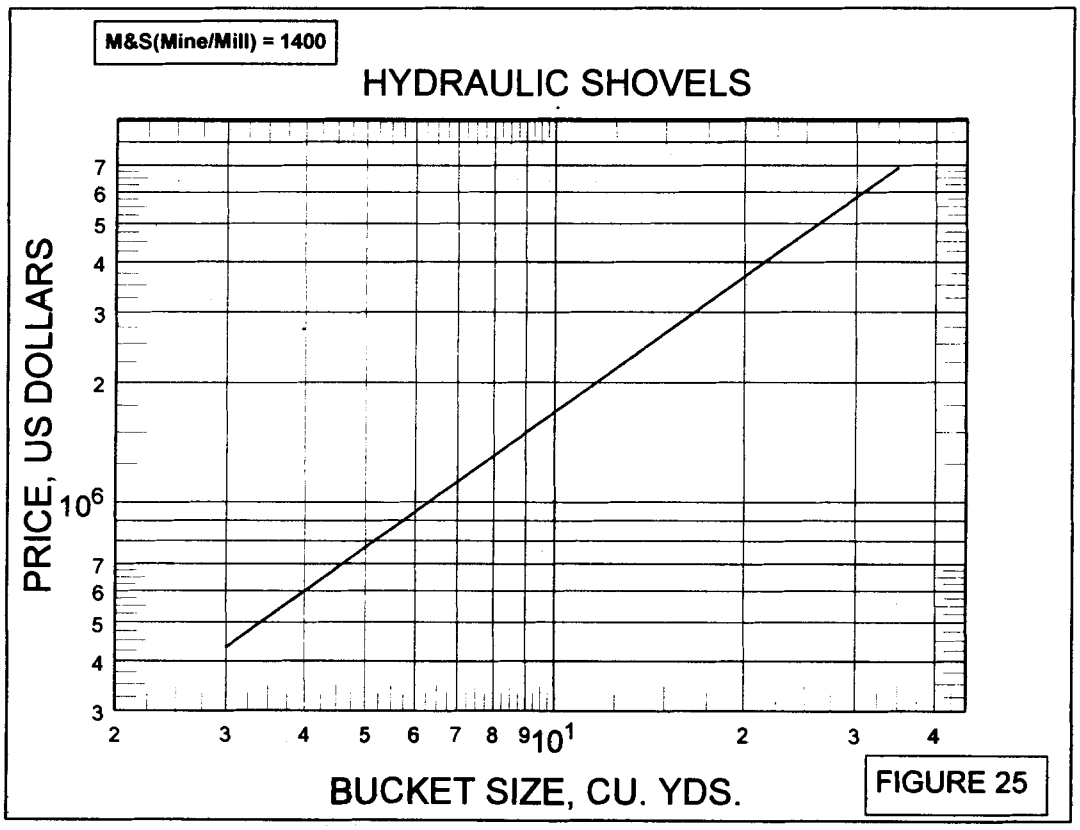


FIGURE 25

CONTINUOUS MINERS

Price = aX^b, US dollars

where **X is the average capacity in tons per hour**. Price includes hard coal cutting head and covers standard machine.

Range in X, tph: 300 to 960 a = 98550 b = 0.3342

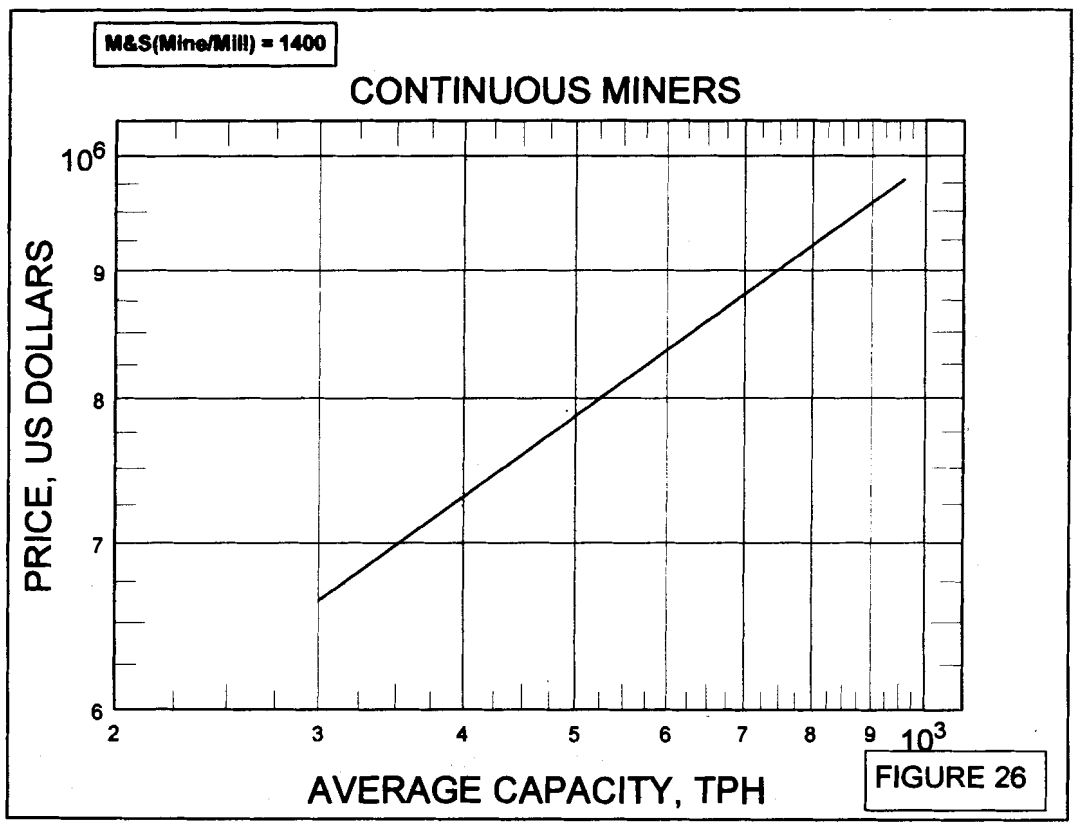


FIGURE 26

CONTINUOUS MINERS, BOOM TYPE (ROAD HEADER)

Price = aX^b, US dollars

where **X is horsepower**. Price covers standard machine.

Range in X, HP: 210 to 1073 a = 54160 b = 0.5449

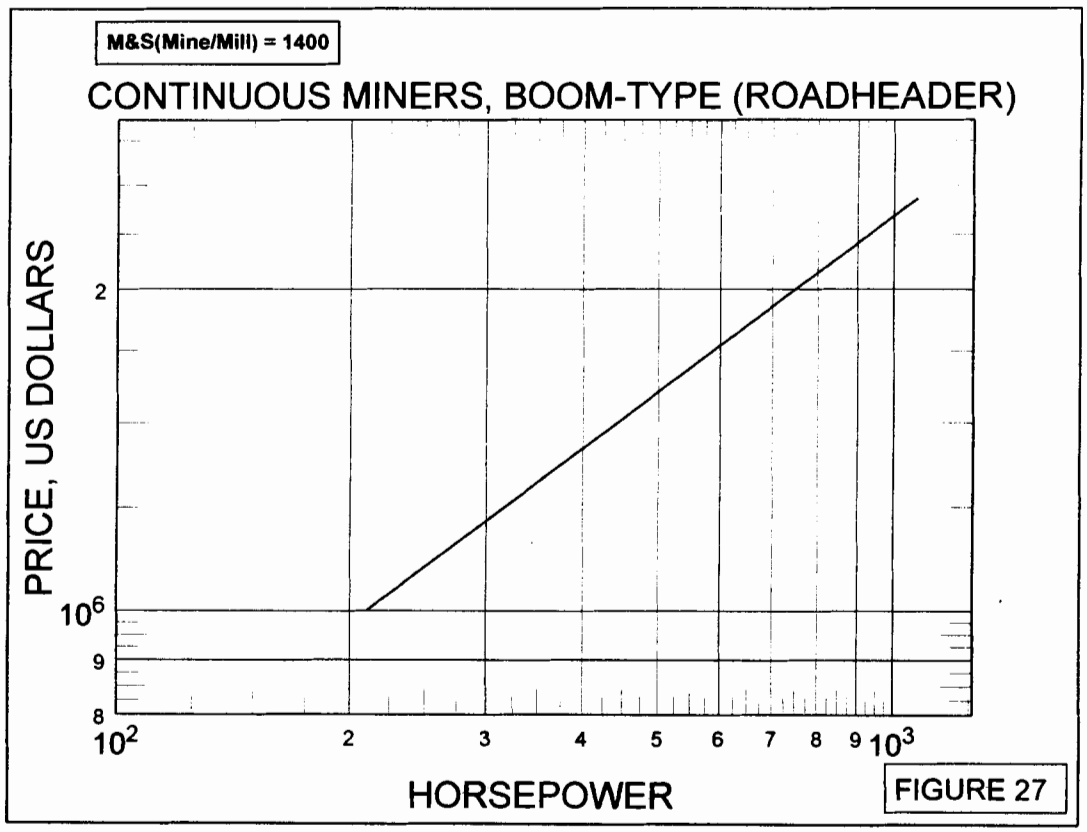


FIGURE 27

CONTINUOUS MINERS, BOOM-TYPE (ROADHEADERS), CONTINUED

Figure 26 below shows how total horsepower for Boom-Type Roadheaders depends upon the height times width of cut $(ft)^2$.

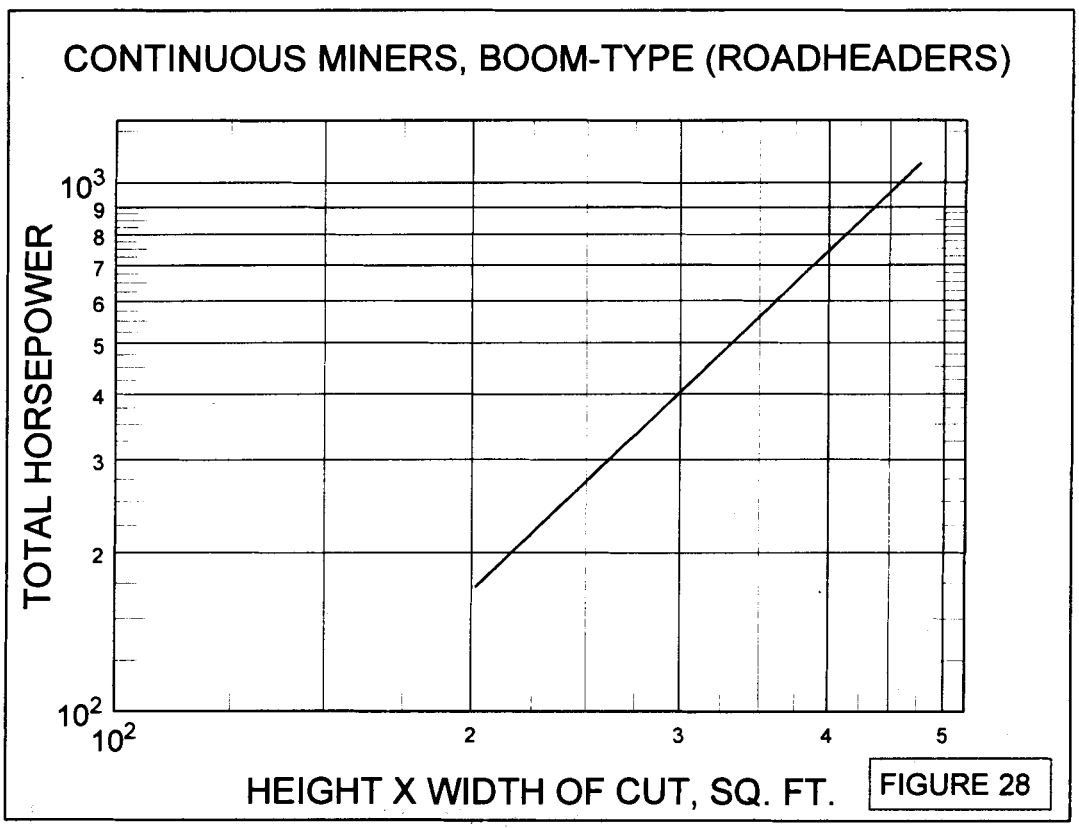


FIGURE 28

LONG WALL MINERS

Price = aX^b, US dollars

where **X is the minimum roof to floor operating height in inches**. Price covers standard machine.

Range in X, inches: 36 to 66 a = 13980 b = 1.162

At an M&S(M/M) Index of 1400, add US$83,000 for each 500 tons of roof support; add US$1,840,000 for a face conveyor.

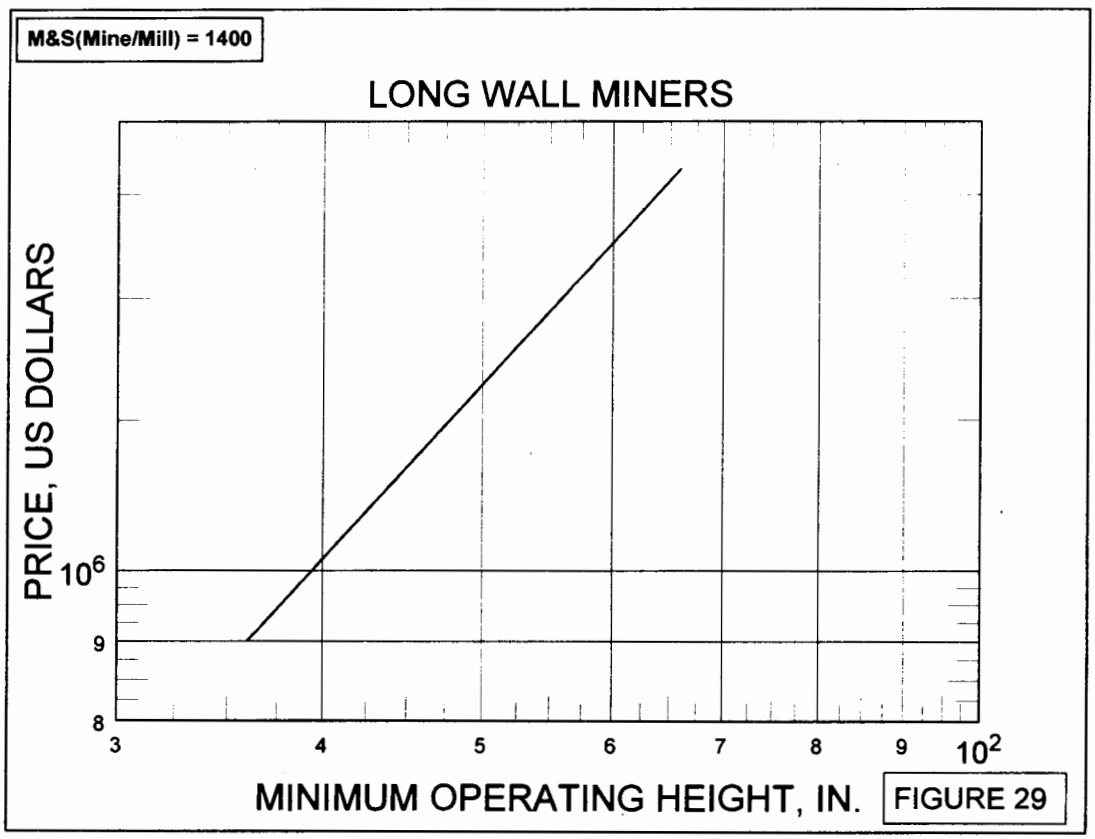


FIGURE 29

III. Haulage Equipment

Haulage equipment has been further identifiied as **Trucks**, **Trains**, **Dumpers**, **Hoists**, **Muckers** and **Slushers**.

(1) Example Estimates - - - - - - - Page 92

(2) Surface Haulage Trucks - - - - - - Page 93

(3) Truck Tires - - - - - - - - Page 94

(4) Locomotive Rails - - - - - - - Page 95

(5) Underground Haulage Vehicles (Trucks) - - - - Page 96

(6) Mine Locomotives - - - - - - - Page 97

(7) Car Dumpers - - - - - - - - Page 98

(8) Mine Hoists - - - - - - - - Page 99

(9) Mucking Machines (Rubber Tired) - - - - - Page 100

(10) Mucking Machines (Rail Mounted) - - - - - Page 101

(11) Slushers - - - - - - - - Page 102

EXAMPLE ESTIMATES

In this section, haulage equipment prices are related to parameters such as size, mass and capacities and load sizes.

Example 1:

Estimate the cost of a 240 ton surface haulage truck that has a mechanical drive.

From Figure 30, page 93, the price is approximately US$2,500,000 at an M&S(M/M) Index of 1400.

Example 2:

Some 1200 yards of steel that weighs 100 lb per yard is required. What is the price at an M&S(M/M) Index of 1400?

The number of yards in 13 yard (39 foot) lengths is 1200/13 or approximately 93. This number will weigh (93)(13)(100/2000) = 60.45 tons. From Figure 32, page 95, the cost is US$47,500.

Example 3:

Estimate the price of a 10 foot diameter friction mine hoist.

Figure 36, page 99, shows that at an M&S(M/M) Index of 1400, the price (excluding motor, starter and rope) will be approximately US$2,500,000.

Note that the selection of rope and motor horsepower depends on speed, depth, load and acceleration. The table on page 99 can be useful. Likewise, Table 5, page 19, of CAPCOSTS is useful for selection purposes.

Example 4:

Compare the estimated price of a 20 kw-ton electric slusher with one that is air operated.

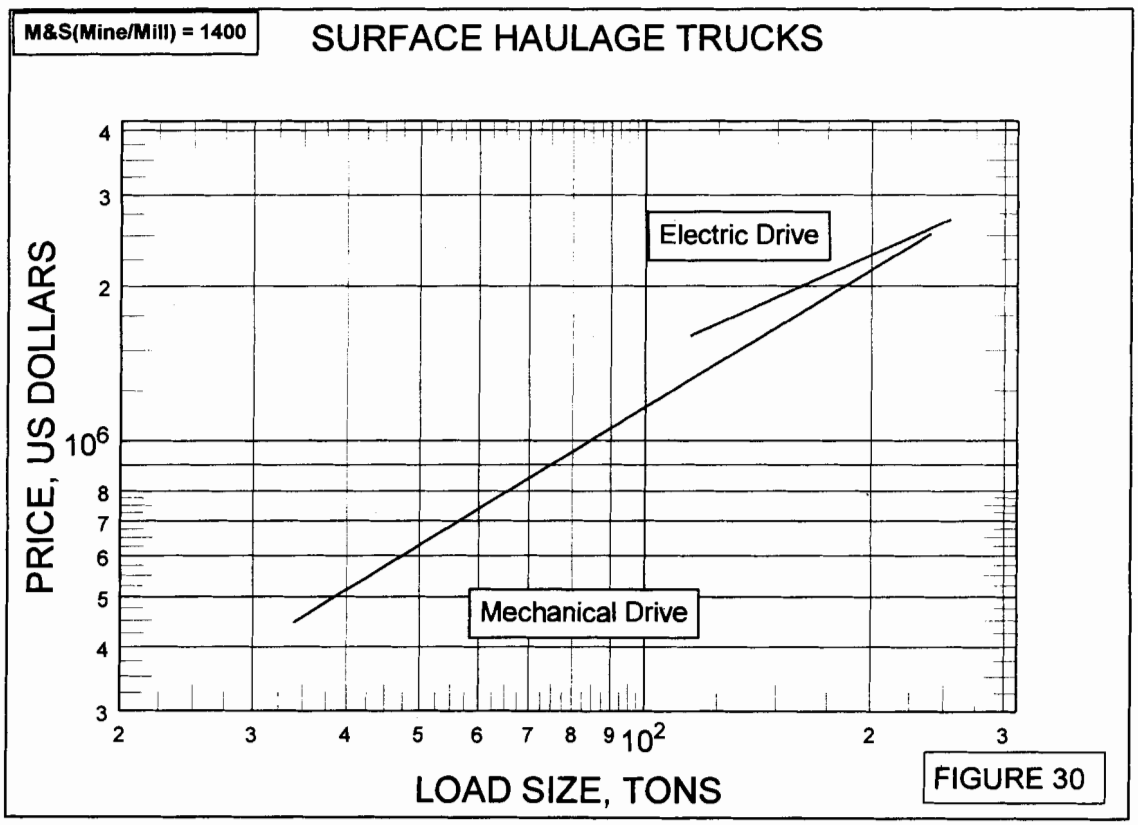
Figure 39, page 102, shows that each one will cost around US$12,000 at an M&S(M/M) Index of 1400.

SURFACE HAULAGE TRUCKS

Price = aX^b, US dollars

where **X is the weight of the rated load in tons**. Price covers complete standard machine.

Electric Drive Range in X, tons:	115 to 255	a = 72710	b = 0.6513
Mechanical Drive Range in X, tons:	34 to 240	a = 19570	b = 0.8862



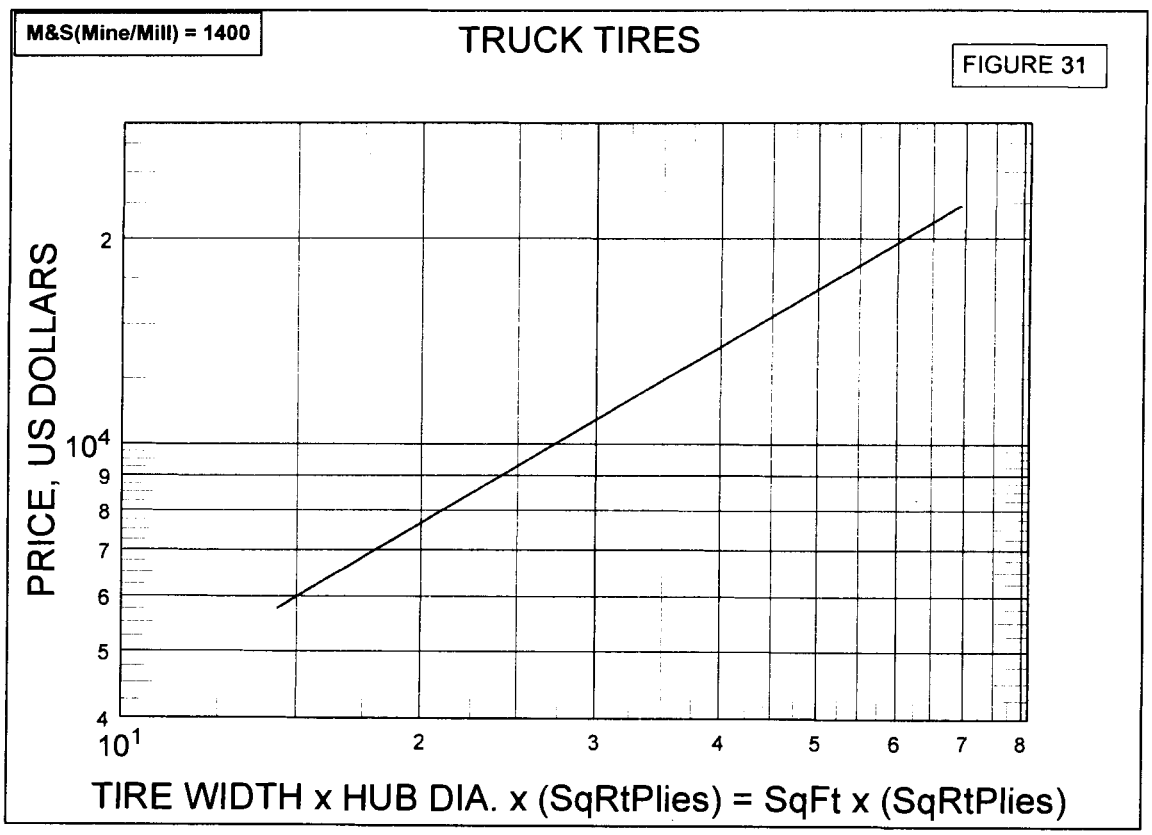
TRUCK TIRES

Price = aX^b, US dollars

where **X is the tire width (ft) times hub diameter (ft) times the square root of the number of plies (plies)$^{0.5}$.** Price includes tubes if applicable.

Range in X, $(ft)^2(plies)^{0.5}$: 14 to 69 a = 582.2 b = 0.8599

Note: number of plies ranges from 36 to 60



LOCOMOTIVE RAILS

Price = aX^b, US dollars

where **X is the net weight of rail in tons** and rail weighs 100 pounds per yard. Price covers straight rail in 13 yard (39 foot) lengths.

Range in X, tons: 1 to 100 a = 785.8 b = 1.00

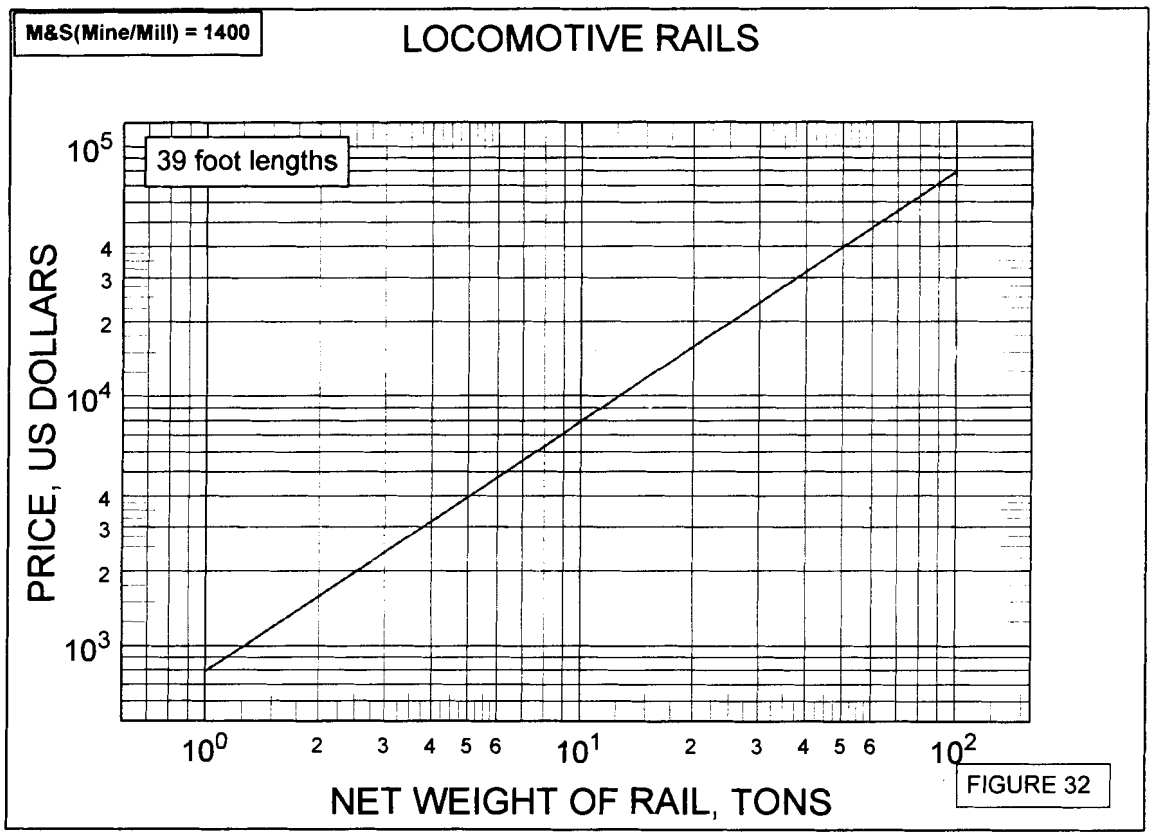


FIGURE 32

UNDERGROUND HAULAGE VEHICLES
(Trucks)

Price = aX^b, US dollars

where **X is the rated capacity in tons**. Price covers standard machine.

Range in X, tons: 5 to 55 a = 33050 b = 0.7841

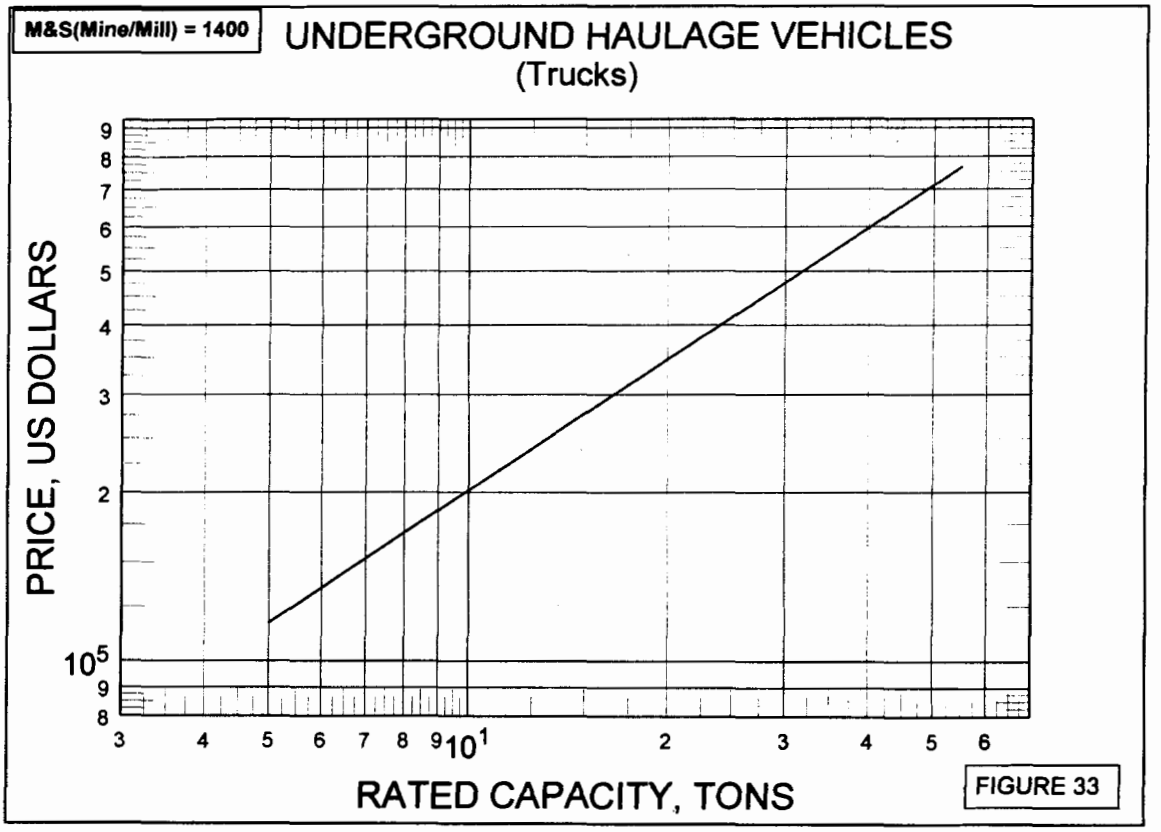


FIGURE 33

MINE LOCOMOTIVES

Price = aX^b, US dollars

where **X is locomotive weight in tons**. Price covers battery, trolley and diesel locomotives

Trolley Range in X, tons: 2 to 30	a = 39060	b = 0.7305
Battery Range in X, tons: 2 to 25	a = 38280	b = 0.6048
Deisel Range in X, tons: 5 to 40	a = 46800	b = 0.4726

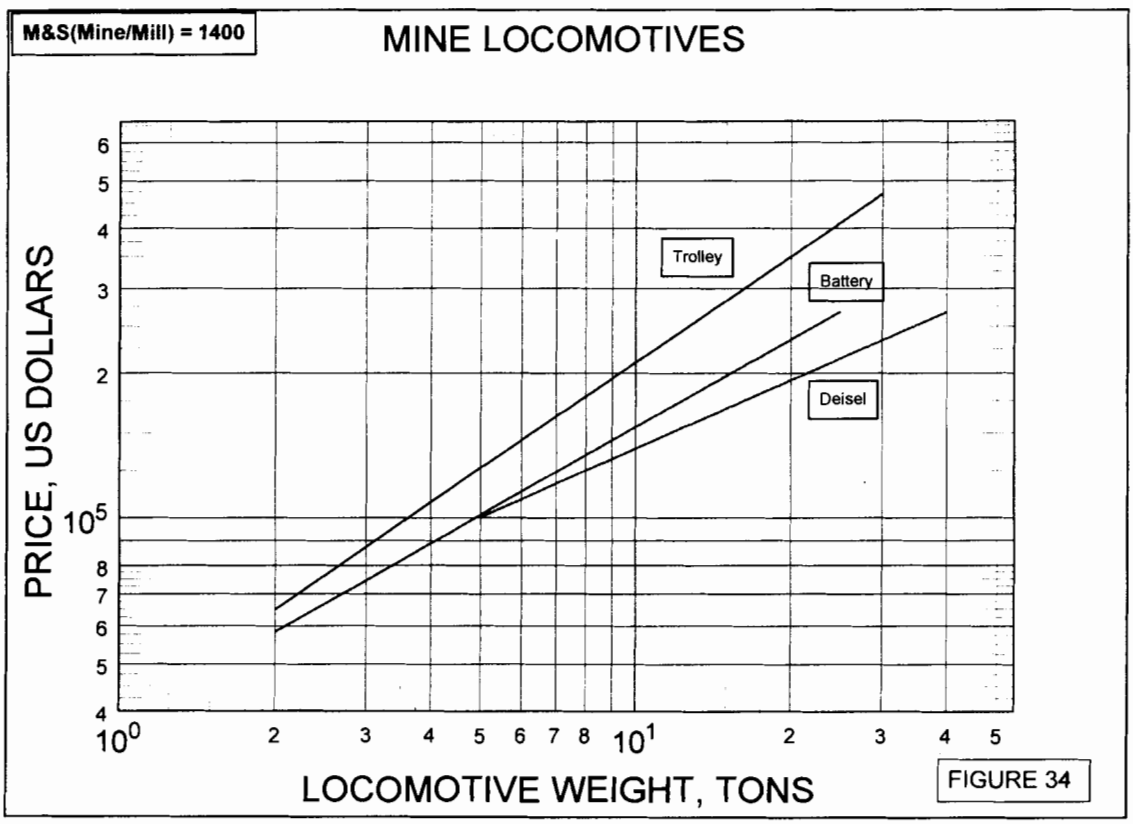


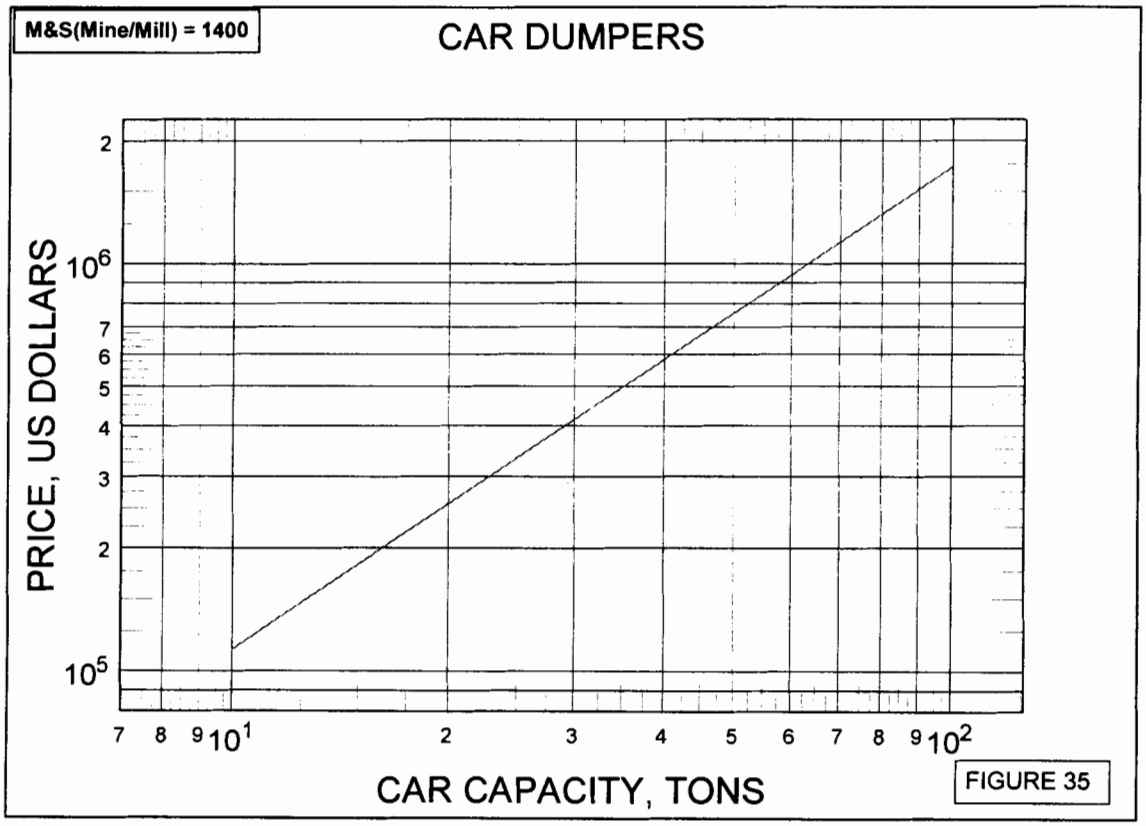
FIGURE 34

CAR DUMPERS

Price = aX^b, US dollars

where **X is the rated capacity of car dumped in tons**. Price covers rotary car dumper and hydro-mechanical side tippler.

Range in X, tons: 10 to 100 a = 7432.55 b = 1.1815



MINE HOISTS

Price = aX^b, US dollars

where **X is the drum diameter in feet**. Price excludes motor, starter and rope. Motor horsepower depends on speed, depth, load and acceleration. Table 5, page 19, can be helpful.

Friction Hoist Range in X, ft: 5 to 15 a = 30949 b = 1.9063

Double Drum Hoist Range in X, ft: 5 to 7.24 a = 112741 b = 1.4024
7.24 to 15 a = 41208 b = 1.9109

Diameter, ft	Nominal HP
5	400
6	500
7	600
8	1000
9	2000
10	4000
11	5000
12	8000
13	8000
14	9000
15	10000

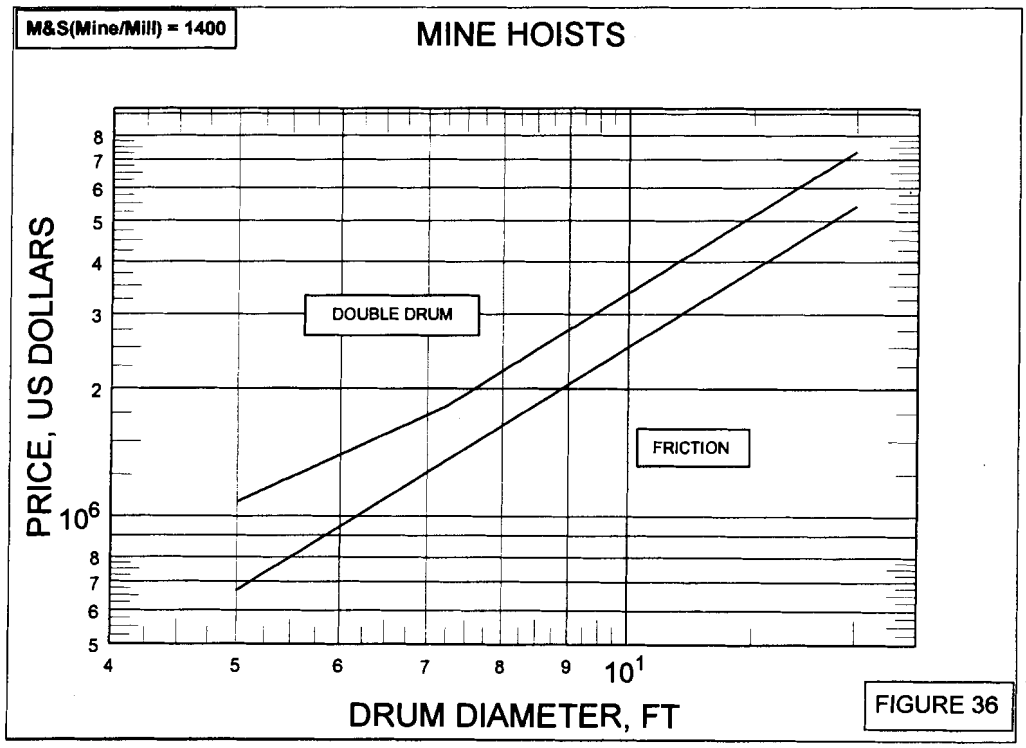


FIGURE 36

MUCKING MACHINES
(Rubber Tires)

Price = aX^b, US dollars

where **X is the bucket capacity in cubic feet**.

Range in X, ft^3: 10.59 to 21.19 a = 9605 b = 0.950

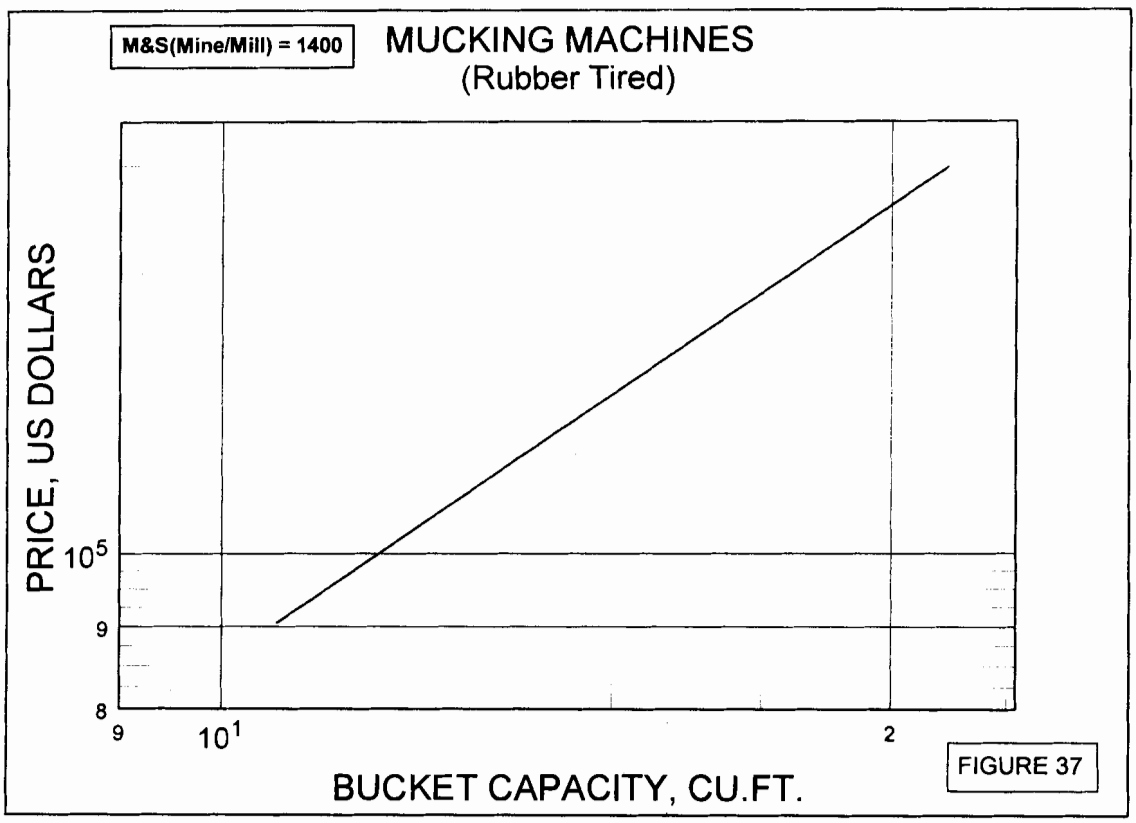


FIGURE 37

MUCKING MACHINES
(Rail Mounted)

Price = aX^b, US dollars

where **X is bucket capacity in cubic feet**. Price is for the standard machine.

Range in X, ft^3: 4.5 to 21 a = 8718 b = 0.845

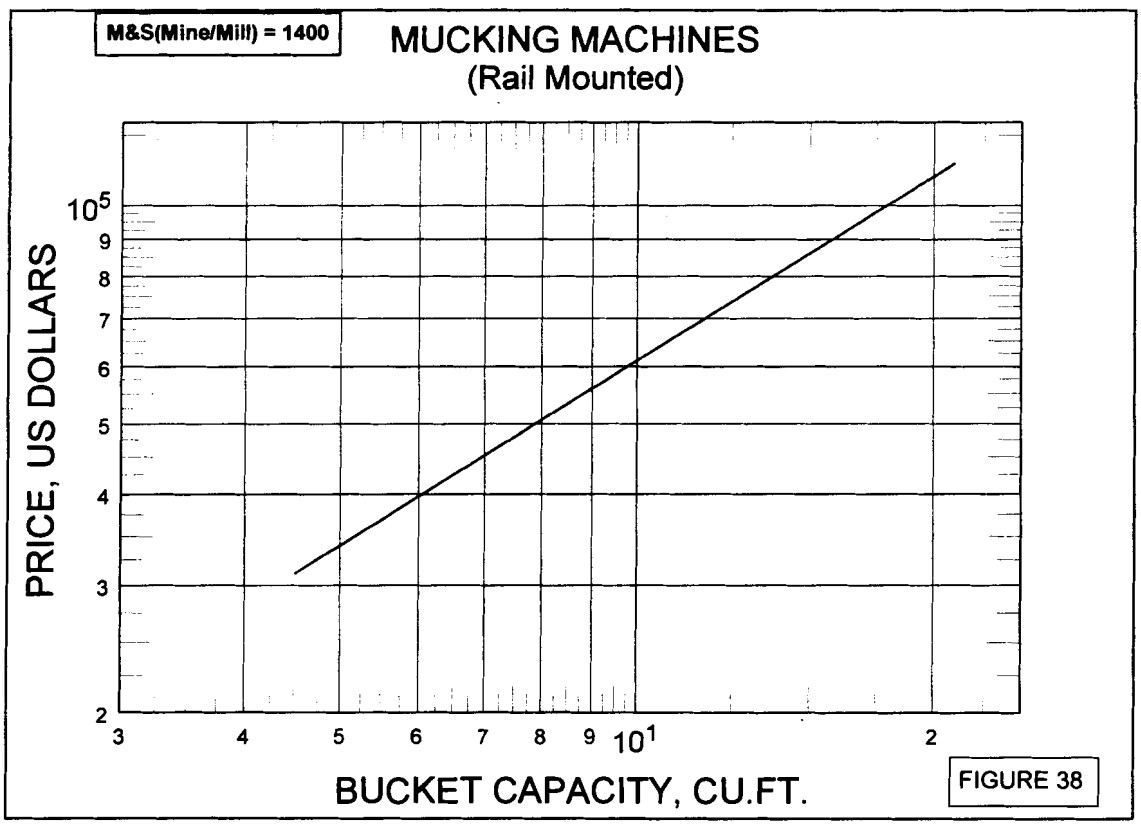


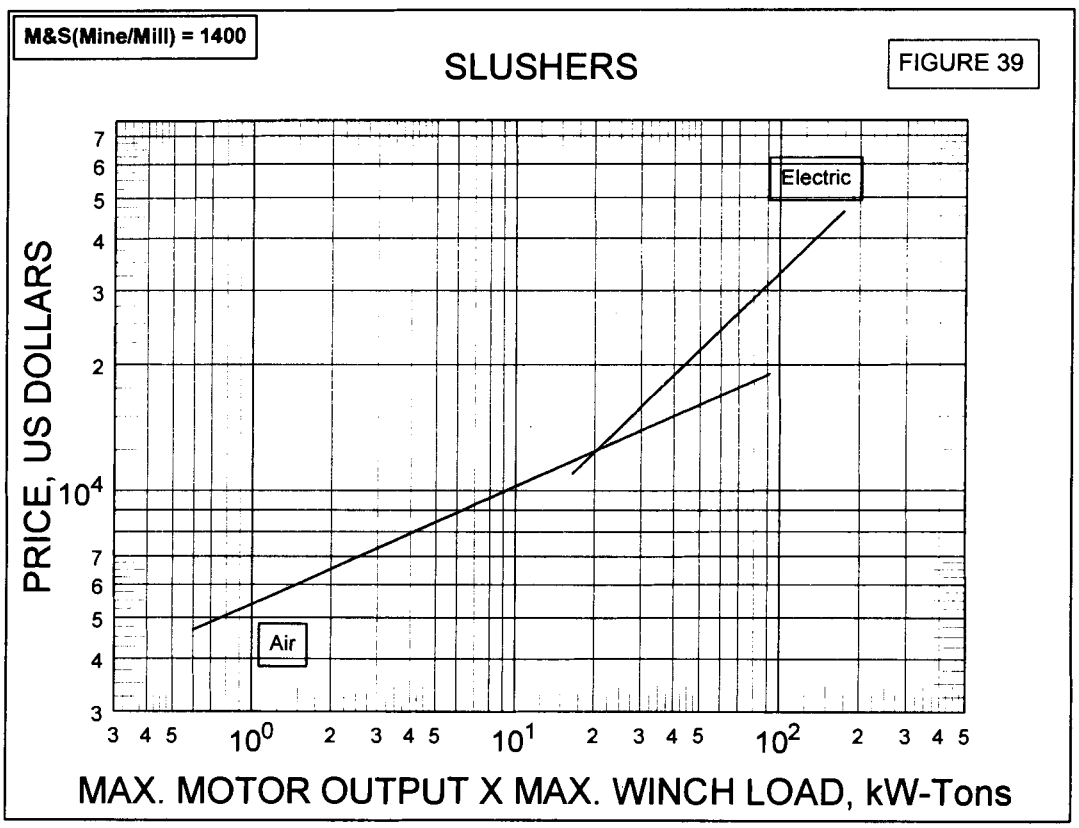
FIGURE 38

SLUSHERS
(Electric and Air)

Price = aX^b, US dollars

where **X is the maximum motor output in kilowatts times the maximum winch load in tons.** Price excludes slusher blade and rope.

Electric Range in X, kw-tons:	16 to 175	a = 1967	b = 0.6114
Air Range in X, kw-tons:	0.6 to 93	a = 5374	b = 0.2783



IV. Ventilation And Cooling Equipment

Sub-Categories in the class are, broadly, **Compressors**, **Cooling Towers** and **Fans**.

(1) Example Estimates - - - - - - - **Page 104**

(2) Air Compressors - - - - - - - **Page 105**

(3) Cooling Towers - - - - - - - - - **Page 107**

(4) Axial Flow Fans (Booster Fan) - - - - - **Page 108**

(5) Centrifugal Fans, (Main Fan) - - - - - - **Page 109**

EXAMPLE ESTIMATES

In this section ventilation and cooling equipment items have been priced as a function of parameters such as capacity, size and distance.

Example 1:

A portable air compressor of capacity 1000 cfm is required. How much will it cost and what size motor is necessary?

From Figure 40, page 105, the price is about US$90,000 at an M&S(M/M) Index of 1400.

From Figure 41, page 106, the nominal horsepower drawn by the unit will be close to 225.

Example 2:

A cooling tower of 1 million BTU per hour capacity is required. How much is the unit expected to cost?

Since 1 ice ton is 11940 BTU/hr, then the cooling capacity is $\frac{10^6}{11940} = 83.75$ ice tons. Figure 42, page 107, shows that the price will be around US$9,200 at an M&S(M/M) Index of 1400.

Example 3:

An axial flow booster fan of capacity 10000 cfm of air is to be purchased. What is the approximate cost excluding air filters and heat exchangers?

The table on page 108 shows that a 24 inch diameter unit has the desired capacity.

Figure 43 on page 108 shows that the cost is approximately US$2700 at an M&S(M/M) Index of 1400.

AIR COMPRESSORS

Price = aX^b, US dollars

where **X is the capacity in cubic feet per minute at 102 PSIG**.

Portable and Stationary Piston; Range in X, cfm: 90 to 3500 a = 384.4 b = 0.8062

Stationary Screw; Range in X, cfm: 120 to 1200 a = 2720 b = 0.4615

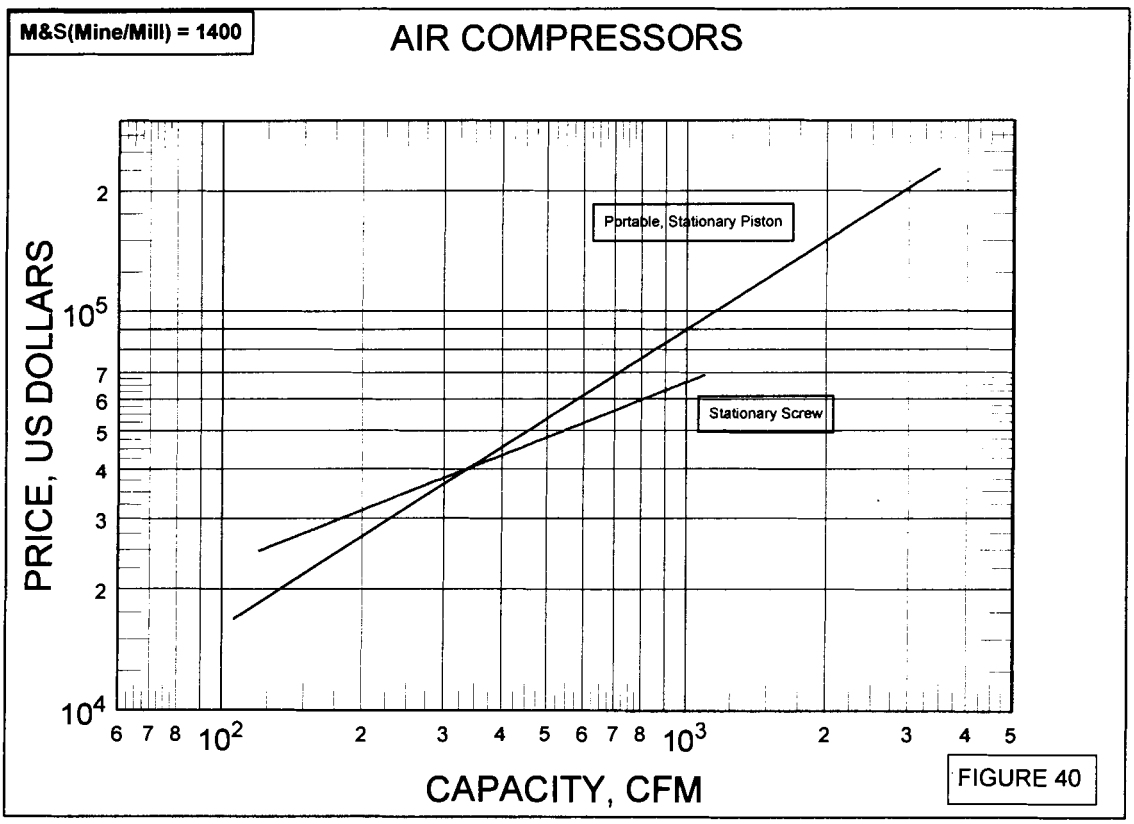


FIGURE 40

AIR COMPRESSORS (CONTINUED)

Figure 39, shown below, relates the capacity of air compressors to their nominal horsepower.

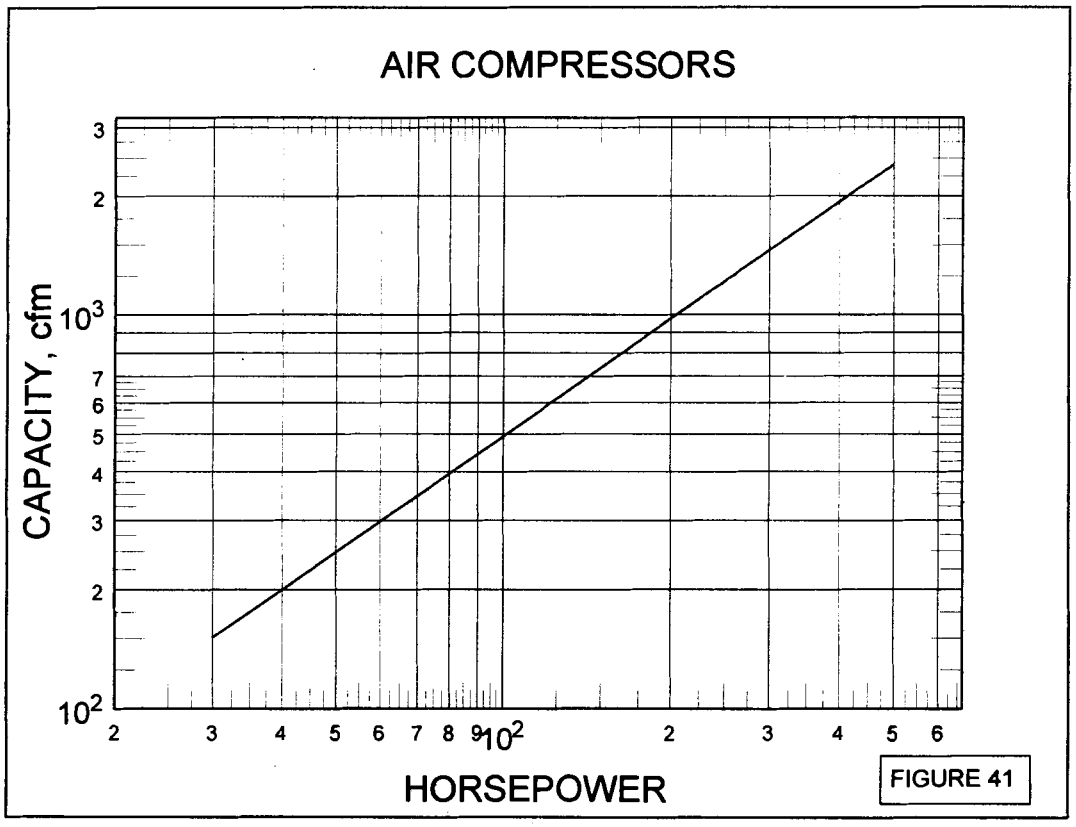


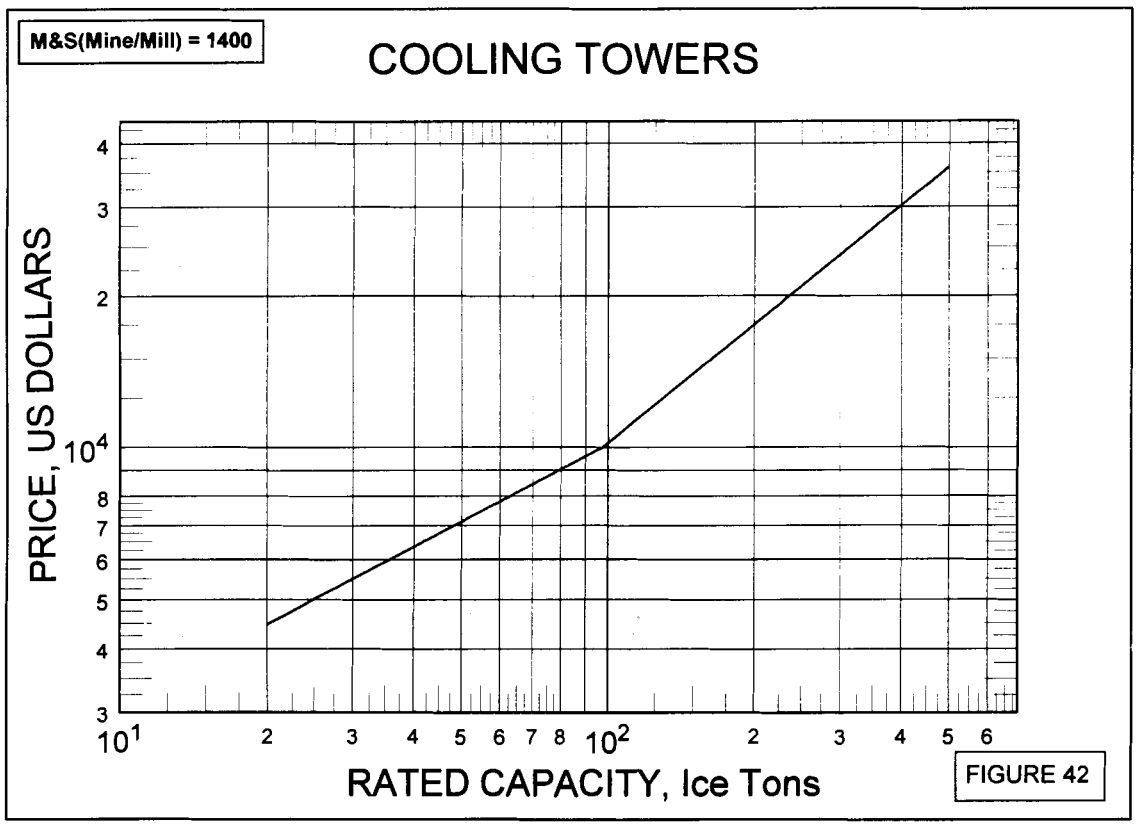
FIGURE 41

COOLING TOWERS

Price = aX^b, US dollars

where **X is capacity in ice tons**. One ice ton is equivalent to 11940 BTU/hour.

Range in X, ice tons:			
20 to 97.38	a = 976.8	b = 0.5071
97.38 to 500	a = 275.8	b = 0.7833



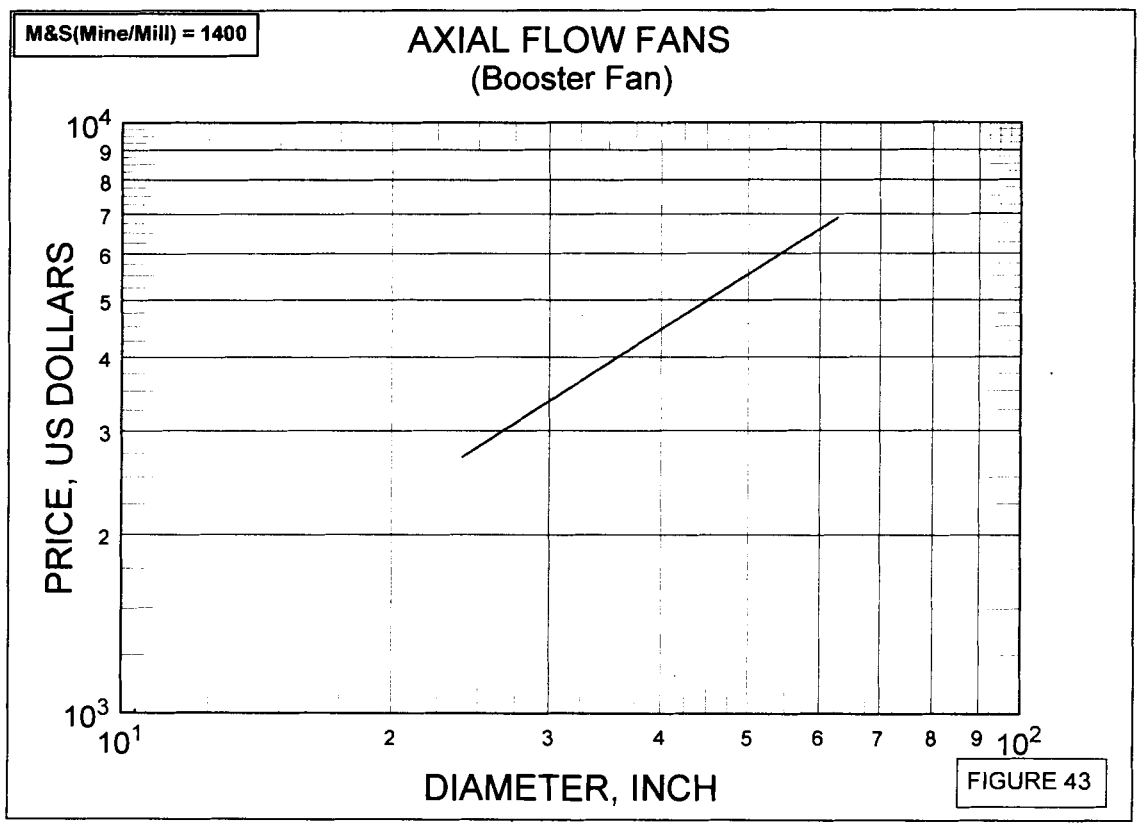
AXIAL FLOW FANS
(Booster Fan)

Price = aX^b, US dollars

where **X is fan diameter in inches**. Price excludes air filters and heat exchangers; includes motors.

Range in X, inch: 24 to 63 a = 123.4 b = 0.9715

Diameter, inch	Capacity, cfm	Disch. Pressure, inch WG
12	900	0.2
24	10000	0.8
30	20000	1.5
48	40000	1.6
63	100000	3.0

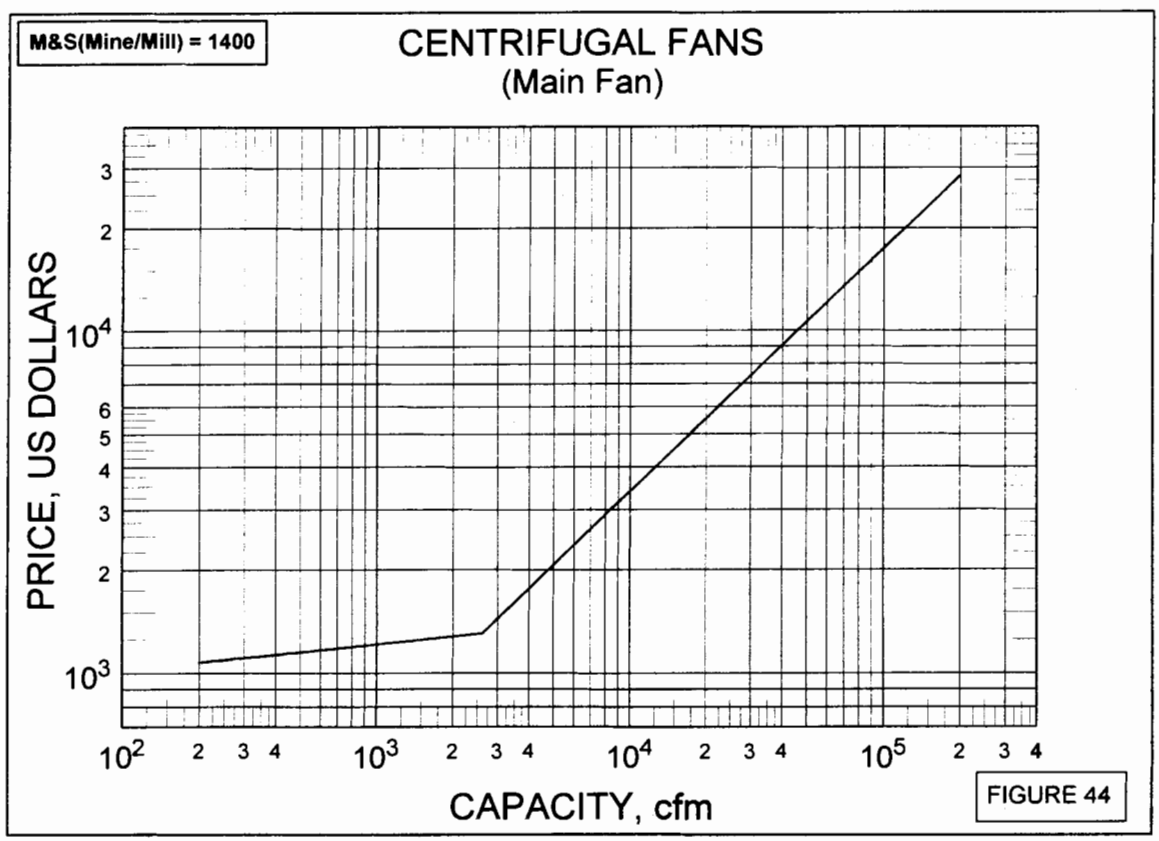


CENTRIFUGAL FANS
(Main Fan)

Price = aX^b, US dollars

where **X is capacity in cubic feet per minute**. Price excludes air filters and heat exchangers; includes motor and drive.

Range in X, ft^3:			
200 to 2608.73	a = 724	b = 0.0743
2608.73 to 200000	a = 4.837	b = 0.7110



V. Power Generation Units

Generation of electricity is lumped into the category **Power Generation**.

(1) Example Estimates - - - - - - - Page 111

(2) Coal-Fired Generator Plants - - - - - - Page 112

(3) Diesel Powered Generator Plants - - - - - Page 113

(4) Utility Substation - - - - - - - Page 114

(4) Transmission Lines (69kV&139kV) - - - - - Page 115

EXAMPLE ESTIMATES

In this section, power generating systems have been costed on the basis of peak load in kilowatts.

Example 1:

Compare the cost of a utility substation with that of a coal-fired generating plant at a peak load of 10000 kw.

From Figures 45 and 47, pages 112 and 114 respectively, prices at an M&S(M/M) Index of 1400 are:

Utility Substation:	US$1,160,000
Coal-Fired Generator Plant:	US$24,000,000

A substation is by far the cheapest. However, the cost of power distribution lines from the substation to the mine/plant must be considered.

Example 2:

Estimate the cost of a 138 kV transmission line that will carry power a distance of 100 km.

From Figure 48, page 115, the cost is estimated, at an M&S(M/M) Index of 1400, to be US$18,000,000.

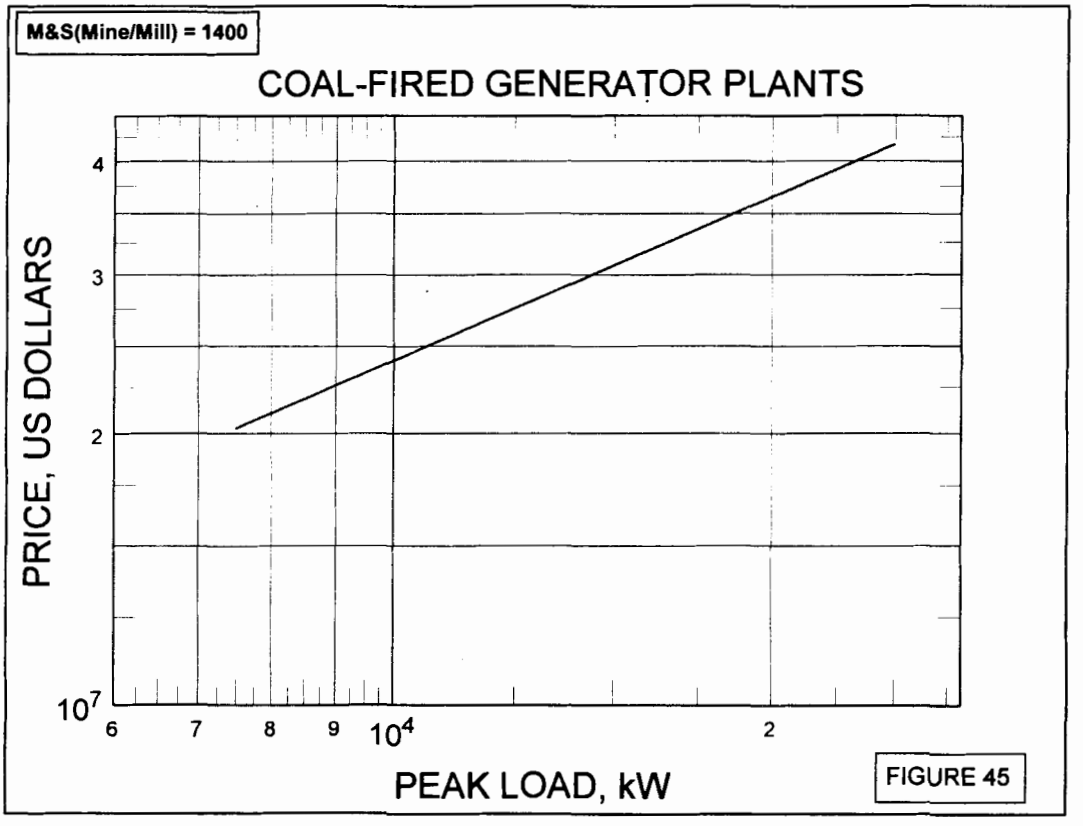
COAL-FIRED GENERATOR PLANTS

(Update from CIM Vol. 25)

Price = aX^b, US dollars

where **X is Peak Load, in kilowatts**.

Range in X, kW: 7500 to 25000 a = 95780 b = 0.600



DIESEL-POWERED GENERATOR PLANTS

Price = aX^b, US dollars

where **X is Peak Load, in kilowatts**.

Range in X, kW:			
	30 to 350	a = 3978	b = 0.5167
	400 to 930	a = 7.997	b = 1.579

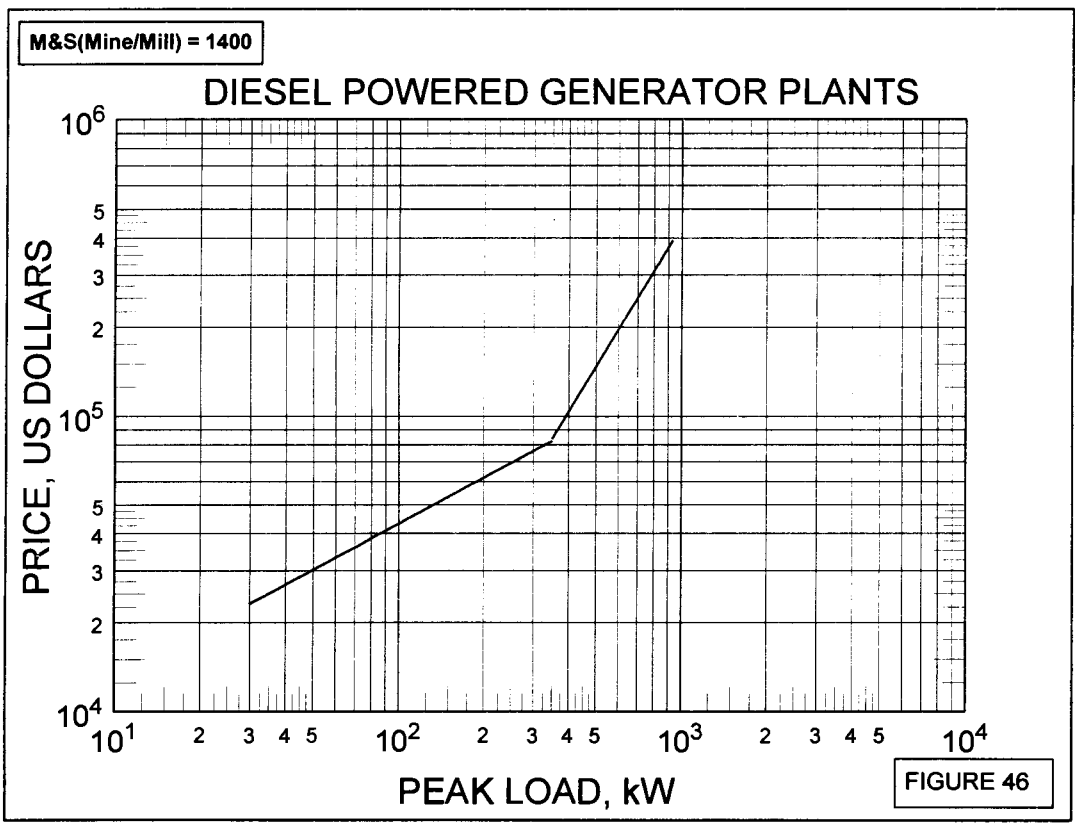


FIGURE 46

UTILITY SUBSTATION
(Update from CIM Vol. 25)

Price = aX^b, US dollars

where **X is Peak Load, in kilowatts**.

Range in X, kW: 2000 to 30000 a = 729.4 b = 0.800

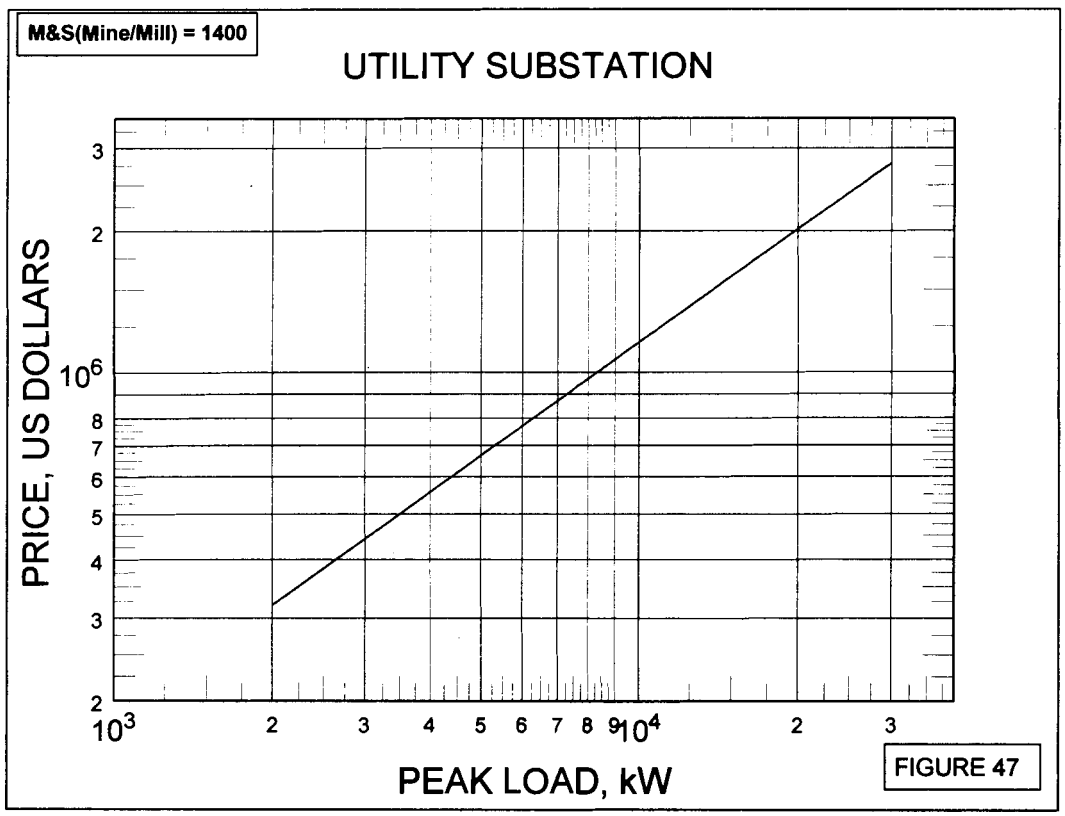


FIGURE 47

TRANSMISSION LINES
(Includes R/W and Clearing Costs)

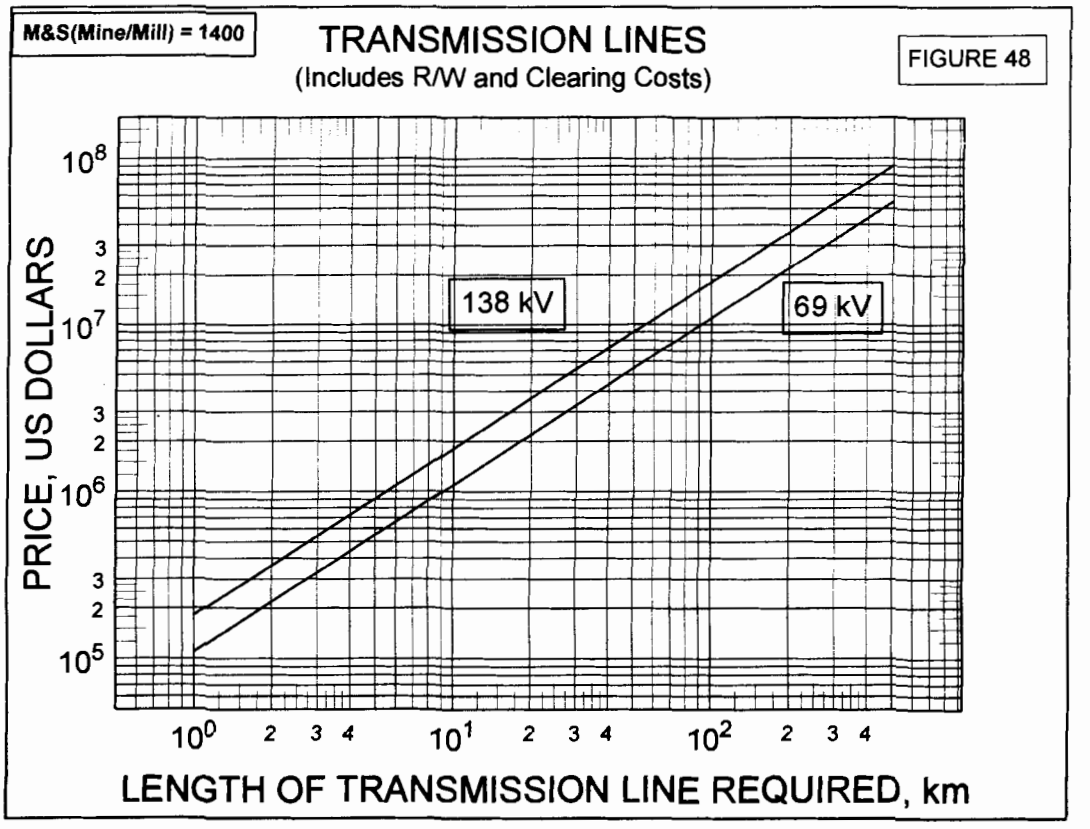
Price = aX^b, US dollars

where **X is length of transmission line required in kilometers**. For 69 kV lines, price is based on use of a 267 kcmil ACSR 'Partridge' conductor which can supply a 30 MVA load. In the case of 138 kV lines, price is based on use of single woodpile construction and the use of a 267 kcmil ACSR 'Partridge" conductor.

69 kV Line Range in X, km: 1 to 500 a = 108700 b = 1.00

138 kV Line Range in X, km: 1 to 500 a = 179600 b = 1.00

Note: 1 km = 0.62137 miles



MAJOR MINERAL PROCESSING EQUIPMENT COSTS

Major mineral processing equipment items have been placed in the following categories: **Comminution**; **Sizing and Classification**; **Fluid Flow, Materials Handling, Storage and Preparation**; **Dust Collection**; **Solid-Solid Separation**; **Solid-Liquid Separation**; **Mixing and Leaching**; **Liquid-Liquid Separation**.

For most categories and/or sub-categories, examples of cost estimations that utilize appropriate graphs are incorporated herein.

Equipment costs are given in the form of logarithmic scale graphs. The cost in US dollars at an $M\&S(Mine/Mill)_{1400}$ index of 1400 is plotted against a suitable parameter (s). In many cases, additional graphs/tables are provided to relate important parameters such as capacity and/or horsepower to cost parameters.

Once the cost parameter is known, the equipment price is estimated in US dollars. The current exchange rate (Canadian to US) is employed to convert to Canadian dollars.

VI. Comminution Equipment

Comminution equipment has been placed in two sub-categories, namely, **Crushers/Pulverizers** and **Grinding Mills**.

(1) Crushers/Pulverizers - - - - - - - - Page 116

(a) Example Estimates - - - - - - - Page 117

(b) Jaw Crushers - - - - - - - - Page 119

(c) Gyratory Crushers - - - - - - - Page 121

(d) Cone Crushers - - - - - - - - Page 123

(e) Roll Crushers - - - - - - - - Page 125

(f) High Pressure Grinding Rolls - - - - - - Page 127

(g) Impact Crushers - - - - - - - Page 128

(h) Hammer Mills - - - - - - - - Page 130

(i) Pulverizers - - - - - - - - Page 131

EXAMPLE ESTIMATES

The price of crushing equipment has been related to the dimension or dimensions most commonly used when referring to a particular crusher. Graphs which relate size to horsepower and size to capacity often is included so that, given a size, a horsepower or a capacity requirement, a price can be found. In most cases, tables are provided to indicate some of the standard sizes of corresponding equipment.

Capacities of the various crushing units have been based upon medium-hard material weighing approximately 100 lb per cubic foot.

Example 1:

How much would a 60-inch by 48-inch (60 x 48) jaw crusher cost, where the 60 inches refers to the long dimension (length) and the 48 inches refers to the gape or width?

In Figure 49, page 119, the price has been related to the area of the feed opening. The table above Figure 49 shows that the feed area for a 60 x 48 jaw crusher is equal to 2880 $in.^2$. The cost of this unit is thus approximately US$600,000 at an M&S(M/M) Index of 1400.

Note that Figures 50 and 51 relate a nominal horsepower and a nominal capacity respectively to area of feed opening. Such information is approximate only.

Example 2:

The price of an open circuit standard cone crusher that will handle 250 short tons per hour is required.

Figure 55, page 123, shows that the price has been related to the diameter of the crusher at the discharge point. From Figure 57, page 124, a four ft standard cone crusher can handle from 200 to 350 stph, depending upon crusher set and chamber selected. Ore hardness and feed size are important as well.

The price of a 4' unit, from Figure 55, is approximately US$280,000 at an M&S(M/M) Index of 1400.

Example 3:

What is the price of a 54 inch (roll diameter) x by 30-inch (length of roll face) double roll crusher and what is its approximate capacity and horsepower?

The unit parameter = $(54)^2(30)$ = 87480 $in.^3$. From Figure 58, page 125, the price is around US$225,000 at an M&S(M/M) Index of 1400. On page 126, Figure 59 shows that about 250 horsepower is required at 30 inches, while Figure 60 indicates a capacity of approximately 1000 stph.

EXAMPLE ESTIMATES (CONTINUED)

Example 4:

An impact crusher is being considered for limestone reduction, where the capacity desired is 900 stph. Estimate the horsepower and price of the crusher.

Figure 63, page 129 shows that a crusher of feed opening 4500 $in.^2$ should be considered.

From Figure 64, the estimated horsepower is 550. Figure 62, page 128 shows that the price is approximately US$295,000 at an M&S(M/M) Index of 1400.

Example 5:

A primary gyratory crusher is being consider for an open pit operation. The capacity is nominally 3500 stph. Estimate the cost of a crusher. What will be the approximate horsepower of the motor?

Figure 53, page 122, shows that at about 3500 stph, the feed opening (gape) by mantle diameter is approximately 6500 $in.^2$. The table on page 121 shows that a 60 x 109 (6540 $in.^2$) primary crusher should be a good choice.

Figure 52, page 121, shows that the price at an M&S(M/M) Index of 1400 is around US$2.7 million. Note that the cost includes drive and lubrication system but not the motor.

Figure 54, page 122, shows that at 6540 $in.^2$, a nominal horsepower is about 950.

Example 6:

A hammer mill is required to treat 150 tph of approximately -1 inch material. Estimate the cost of the unit and the approximate horsepower required.

The table on page 130 suggests that a suitable hammer mill will be one of rotor diameter 36 inches and rotor length 60 inches. The horsepower will be approximately 175 (see table) at a feed opening of 18 inch by 60 inch. Since rotor diameter by length is 2160, then from Figure 65, the price is estimated as US$150,000 at an M&S(M/M) Index of 1400.

Example 7:

A 78-inch diameter high pressure grinding rolls of 31.5-inch length is being considered for a diamond processing circuit. What is the estimated cost of this unit?

Figure 61, page 127 shows that the cost is approximately US$5,200,000 at an M&S(M/M) Index of 1400.

JAW CRUSHERS

Price = aX^b, US dollars

where **X is the area of the feed opening in square inches**. Price includes drive and lube system; excludes motor.

Range of X, $in.^2$:		a =	b =
	36 to 550	11840	0.3367
	550 to 2880	110.3	1.079

Feed Opening, in. x in.	Area of Feed Opening, in. x in.	Range of Open Side Set, inches
48 x 36	1728	5 to 8
48 x 42	2016	5 to 10
60 x 48	2880	6 to 10
84 x 60	5040	6 to 12

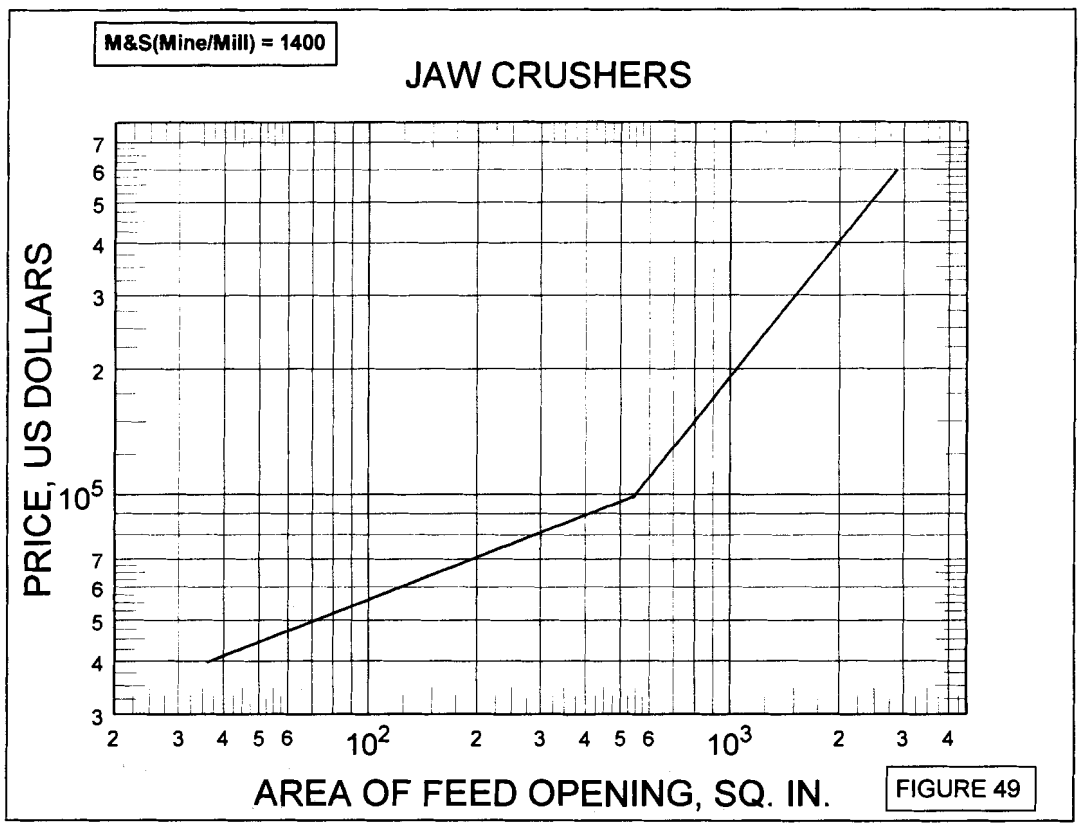


FIGURE 49

JAW CRUSHERS (CONTINUED)

Both nominal horsepower and capacity (min/max) are expressed versus area of feed opening.

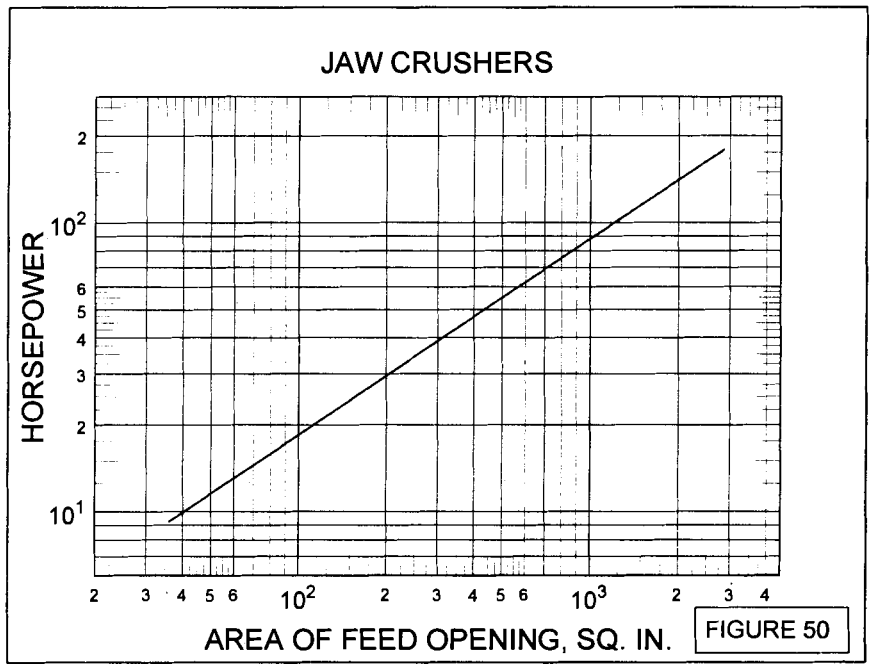


FIGURE 50

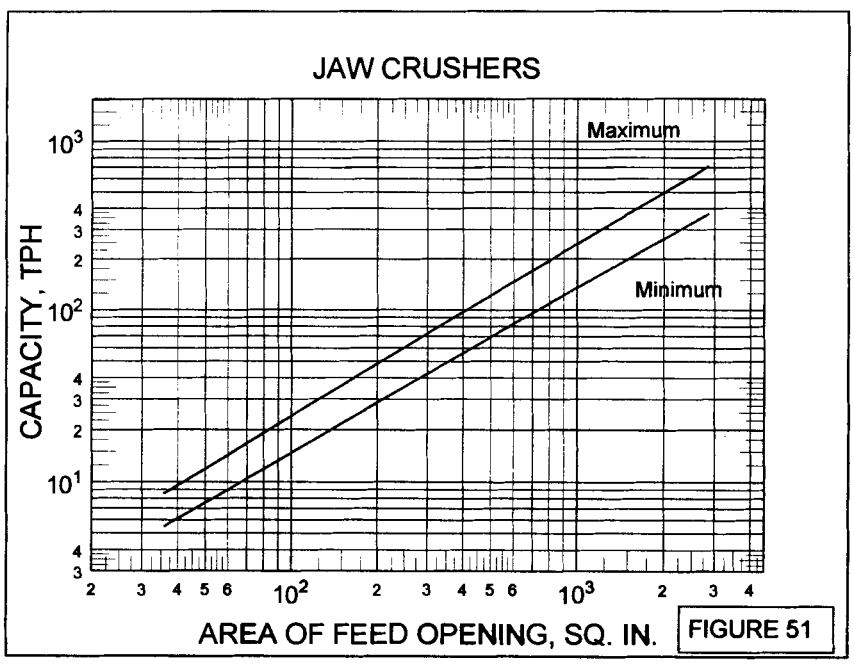


FIGURE 51

GYRATORY CRUSHERS

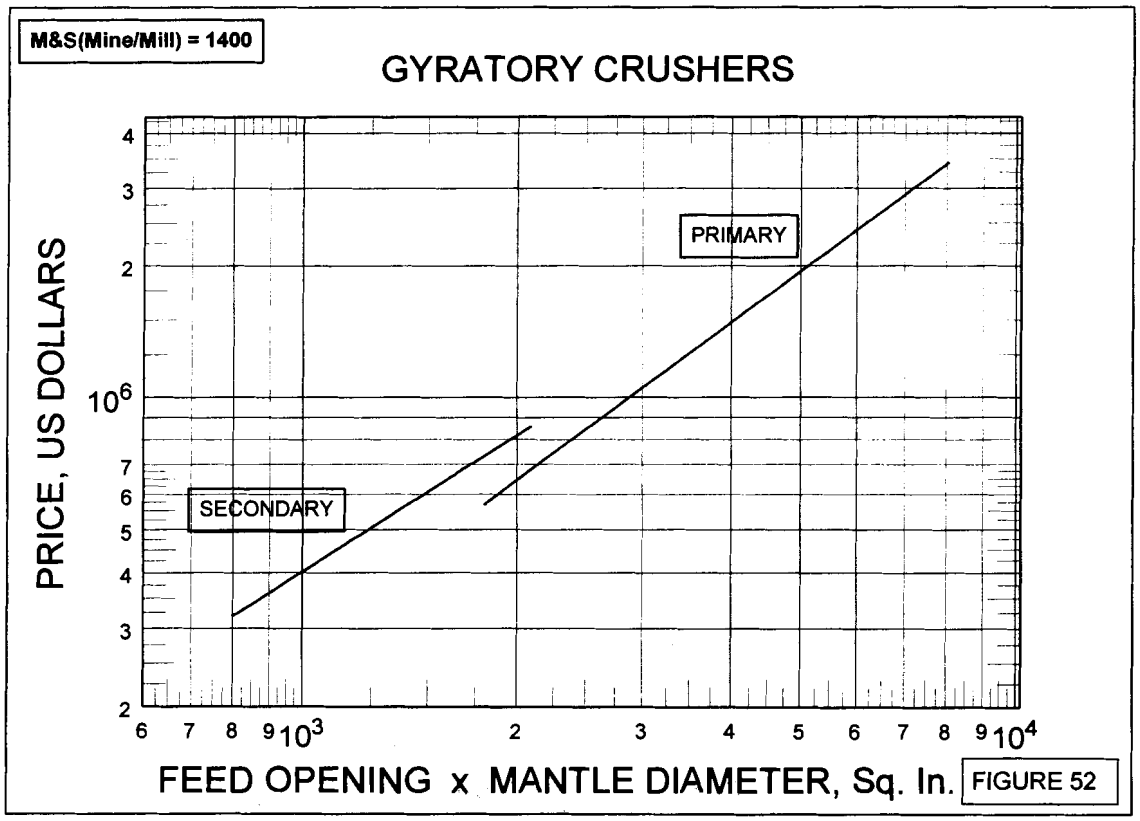
Price = aX^b, US dollars

where **X is the feed opening (gape = G) by mantle diameter (D) in square inches**. Cost includes drive and lubrication system; excludes motor.

Primary Crusher Range, $in.^2$: 1780 to 8200 a = 71.25 b = 1.199

Secondary Crusher Range, $in.^2$: 800 to 2100 a = 336.9 b = 1.025

Type	G by D, in. x in.	G x D, sq. in.	Range, Discharge Opening (set), in.
Primary	30 x 55	1650	2.5 to 5.5
	42 x 65	2730	4.5 to 7
	54 x 74	3996	5.5 to 8
	60 x 89	5340	6 to 9
	60 x 109	6540	8.5 to 12
Secondary	16 x 50	800	1.5 to 3.5
	24 x 60	1440	1.5 to 4.5
	30 x 70	2100	2 to 5.5



GYRATORY CRUSHERS (CONTINUED)

Figures 53 and 54 show nominal capacity and horsepower versus gape x mantle diameter, $in.^2$.

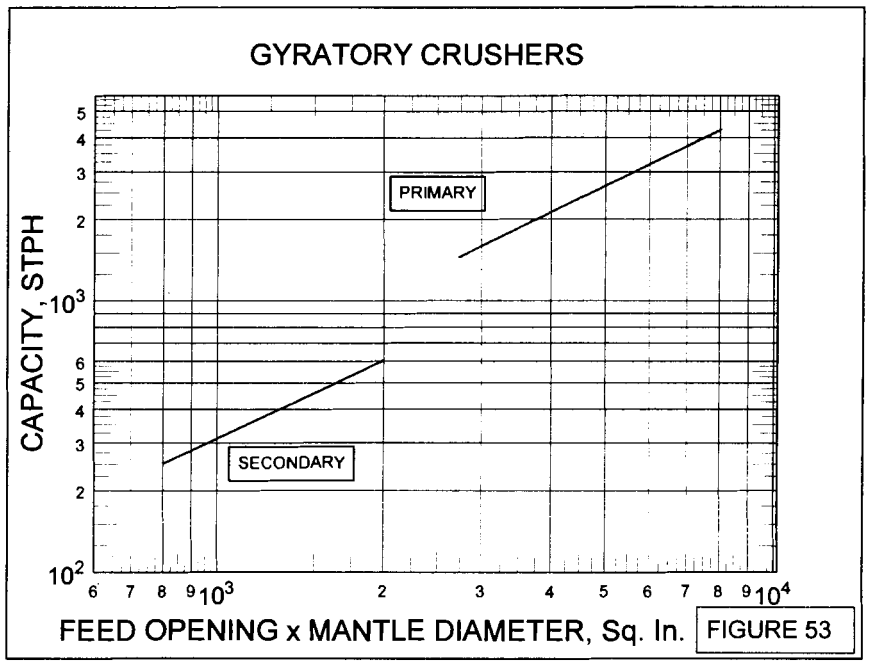
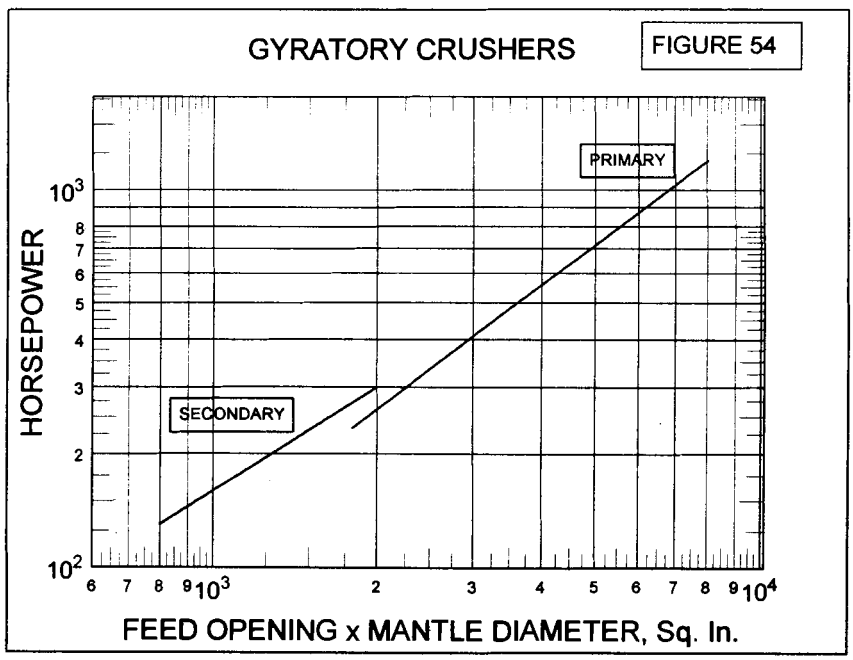


FIGURE 53



CONE CRUSHERS

Price = aX^b, US dollars

where **X is the diameter of the mantle at the discharge annulus in feet.** Cost includes drive and lubrication system; excludes motor. Usual diameters are 2, 3, 3.75, 4, 4.25, 5, 5.5, and 7.

Standard Cone Range in X, ft.: 3 to 7 a = 25070 b = 1.756

Shorthead Cone Range in X, ft: 3 to 7 a = 30010 b = 1.700

Crusher Diameter at Annulus, ft	Range of Discharge Sizes (closed sets), inch
3	1.25 to 1.5
4	0.375 to 1.5
4.25	0.5 to 2
5	0.625 to 2
5.5	0.625 to 2.5
7	0.75 to 2.5

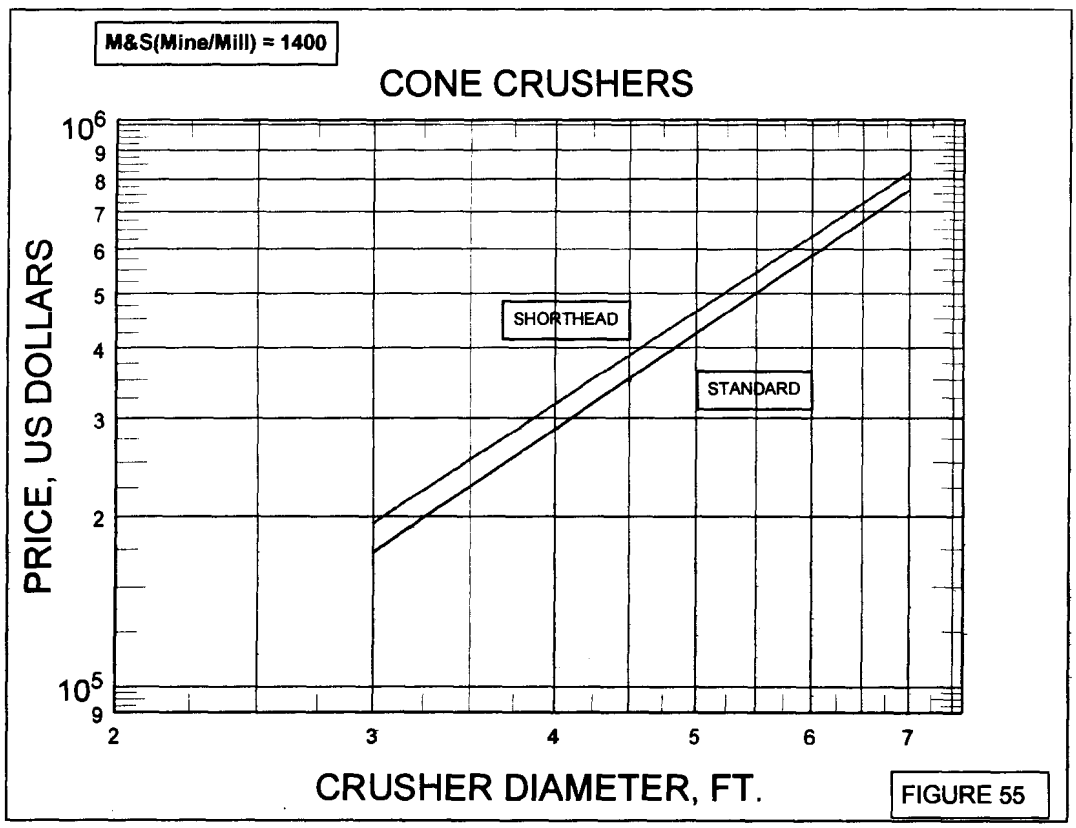


FIGURE 55

CONE CRUSHERS (CONTINUED)

Figures 56 and 57 show horsepower and open circuit capacity (min/max) versus diameter, ft.

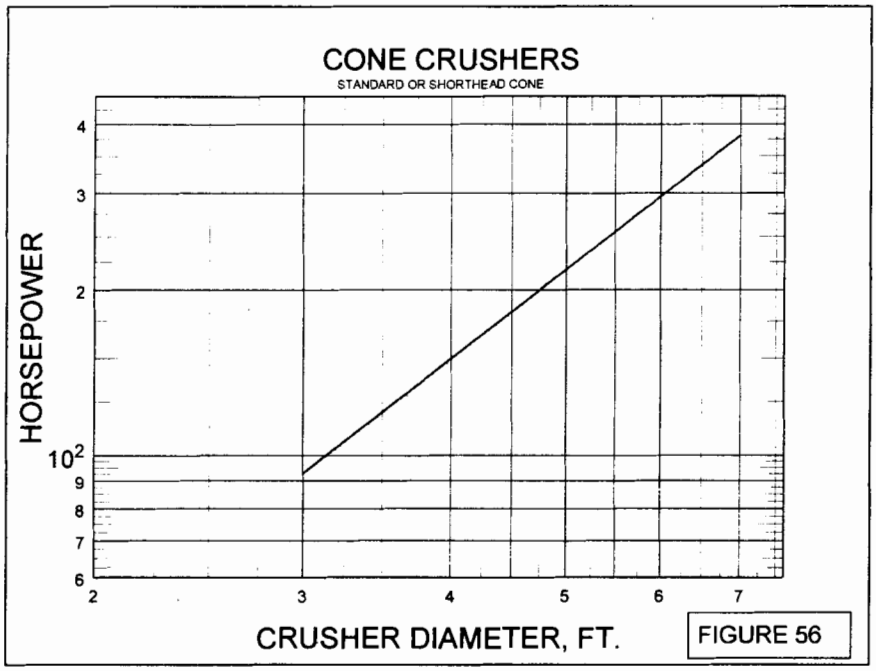


FIGURE 56

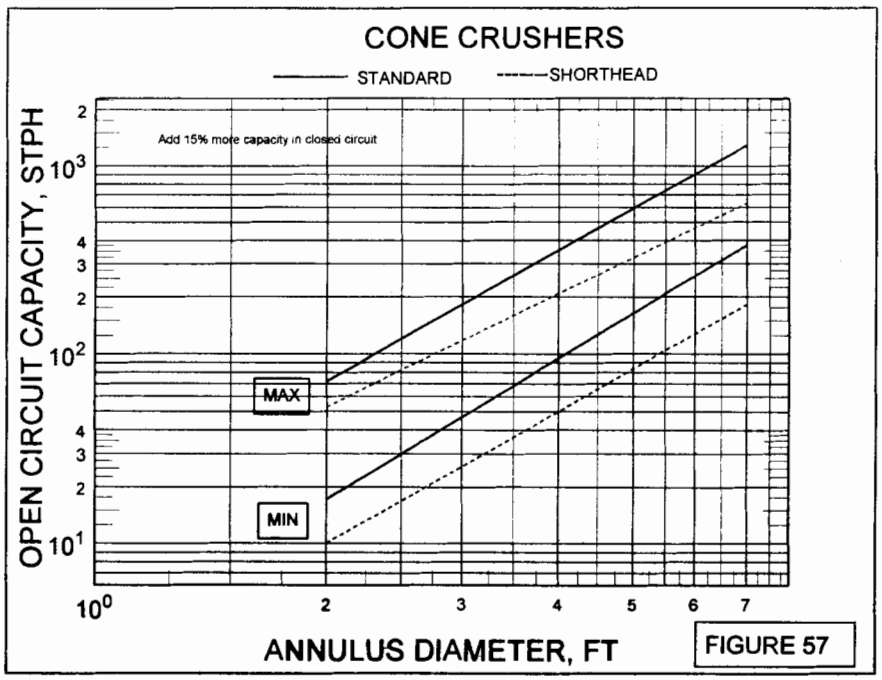


FIGURE 57

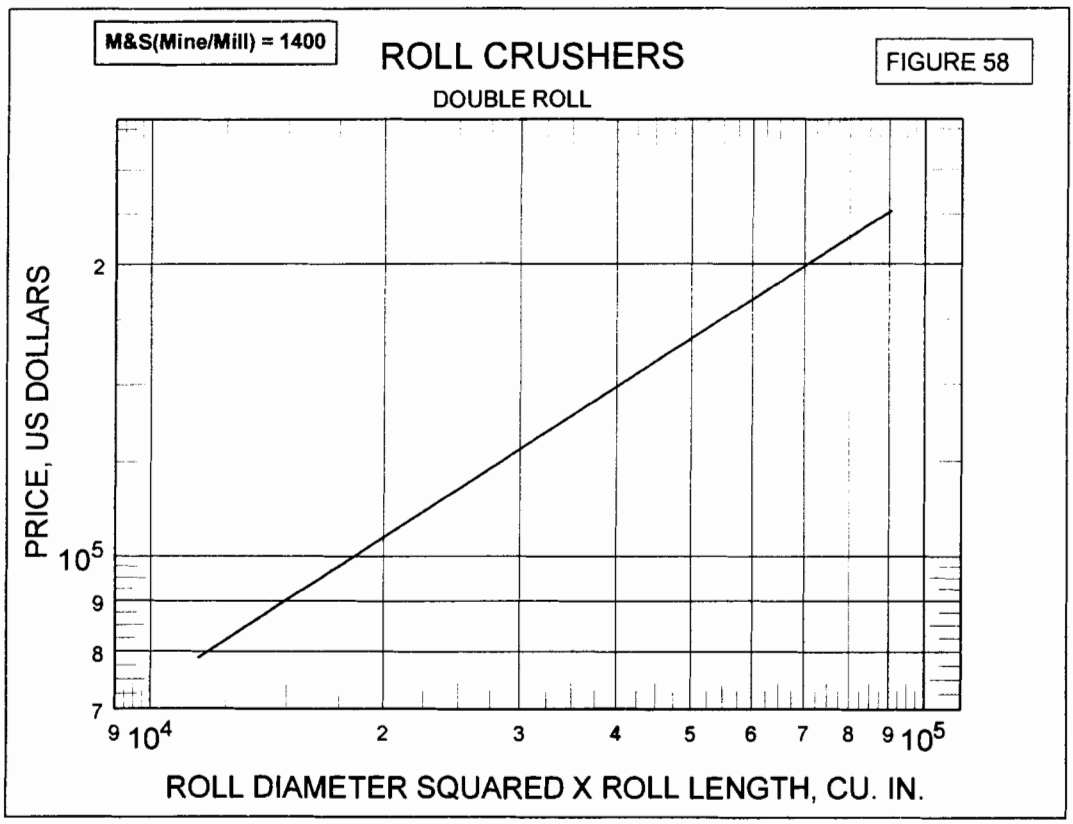
ROLL CRUSHERS
(Double Roll)

Price = aX^b, US dollars

where **X is the roll diameter squared times length in cubic inches.** Price includes drive, feed box and lubrication system; excludes motor.

Range in X, $in.^3$: 11520 to 90750 a = 650 b = 0.5129

Duty	Typical Diameter, ft	Product Size Range, in.
Heavy	2	2 to 6
	2.5	3 to 8
	3	4 to 12
Light	2	0.75 to 6
	2.5	0.75 to 8
	3	0.75 to 10



ROLL CRUSHERS (Double Roll)

Figures 59 and 60 show how horsepower and capacity vary with length of roll in inches.

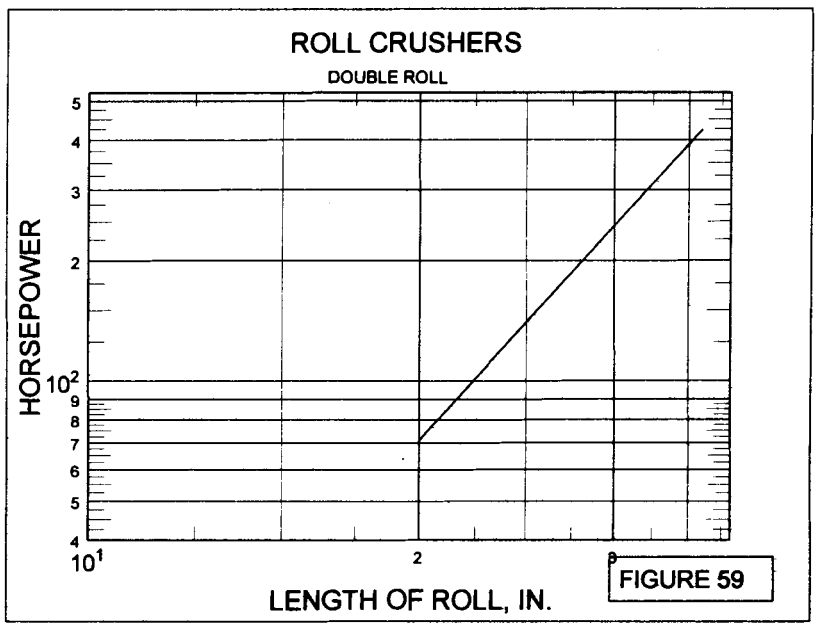


FIGURE 59

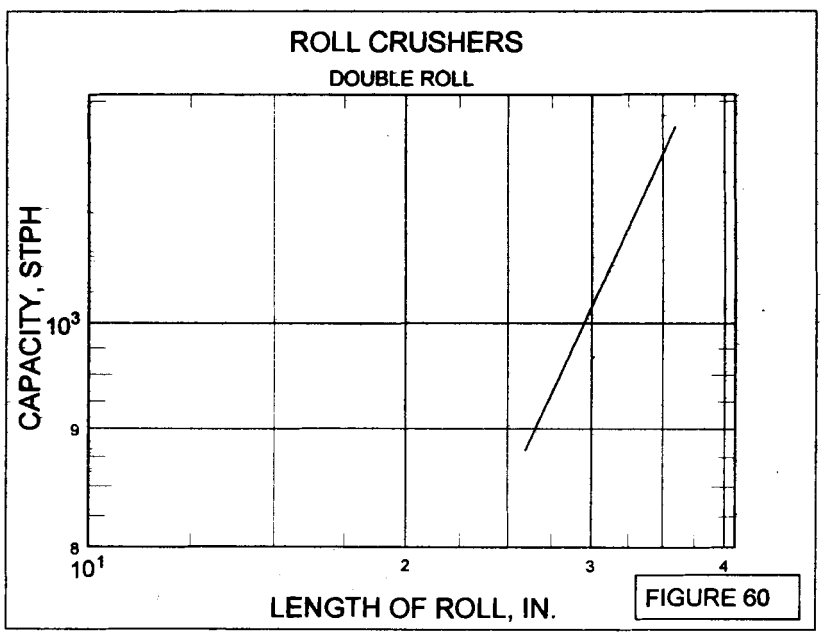


FIGURE 60

HIGH PRESSURE GRINDING ROLLS

Price = aX^b, US dollars

where **X is roll diameter in inches**. Price includes standard motors and surfaces, feed chute, lubrication system, hydraulic system, drive train, auxiliary drive, special tools and field instrumentation.

Range in X, in.:			
15.75 to 60	a = 2478	b = 1.716
60 to 94.49	a = 189.1	b = 2.346

Max. Diameter, in.	Length, in.	Capacity, stph	Horsepower	Speed, rpm
41.73	15.75	72	402	17-27.6
54.33	19.69	121	751	20-25.7
68.90	23.62	199	1475	16.5-23
77.95	31.50	331	2213	13-21.1
86.61	39.37	485	2548	11-20
96.85	39.37	573	3353	11-19.1

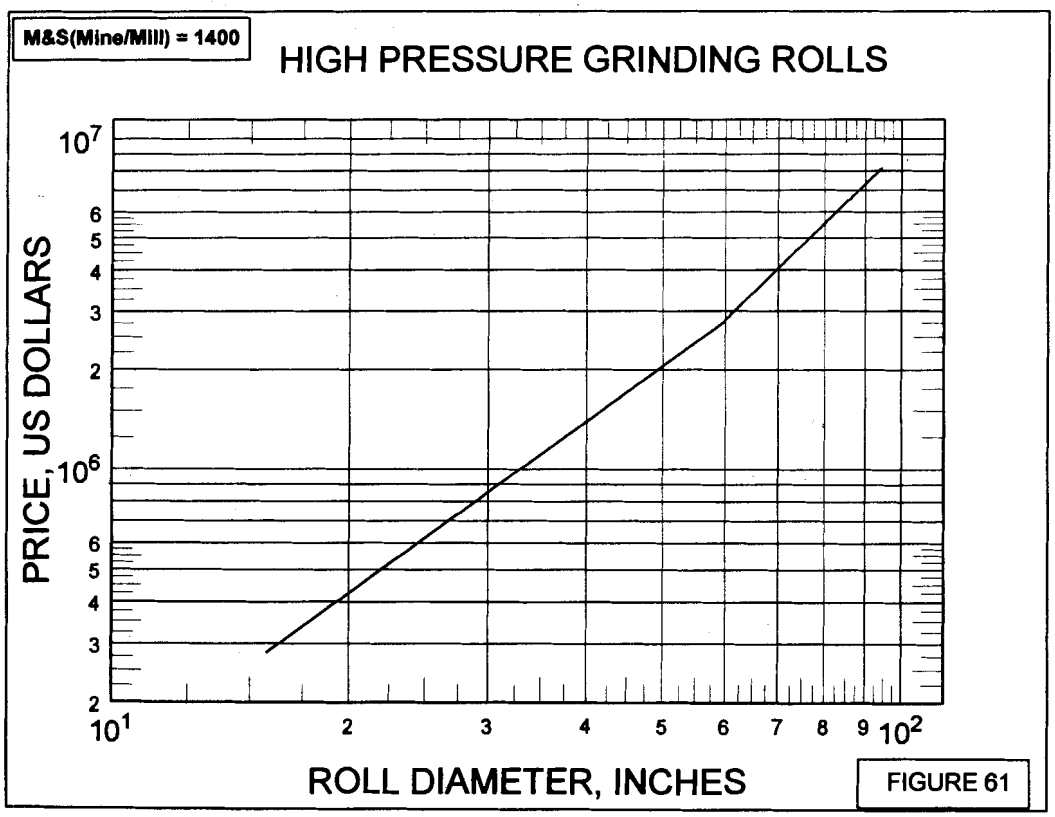


FIGURE 61

IMPACT CRUSHERS

Price = aX^b, US dollars

where **X is is the area of the feed opening, square inches**. Price includes drive, feed box, lubrication system; excludes motor.

Range in X, $in.^2$:			
1190 to 4600	a = 103.2	b = 0.9457
4600 to 7030	a = 0.3554	b = 1.6180

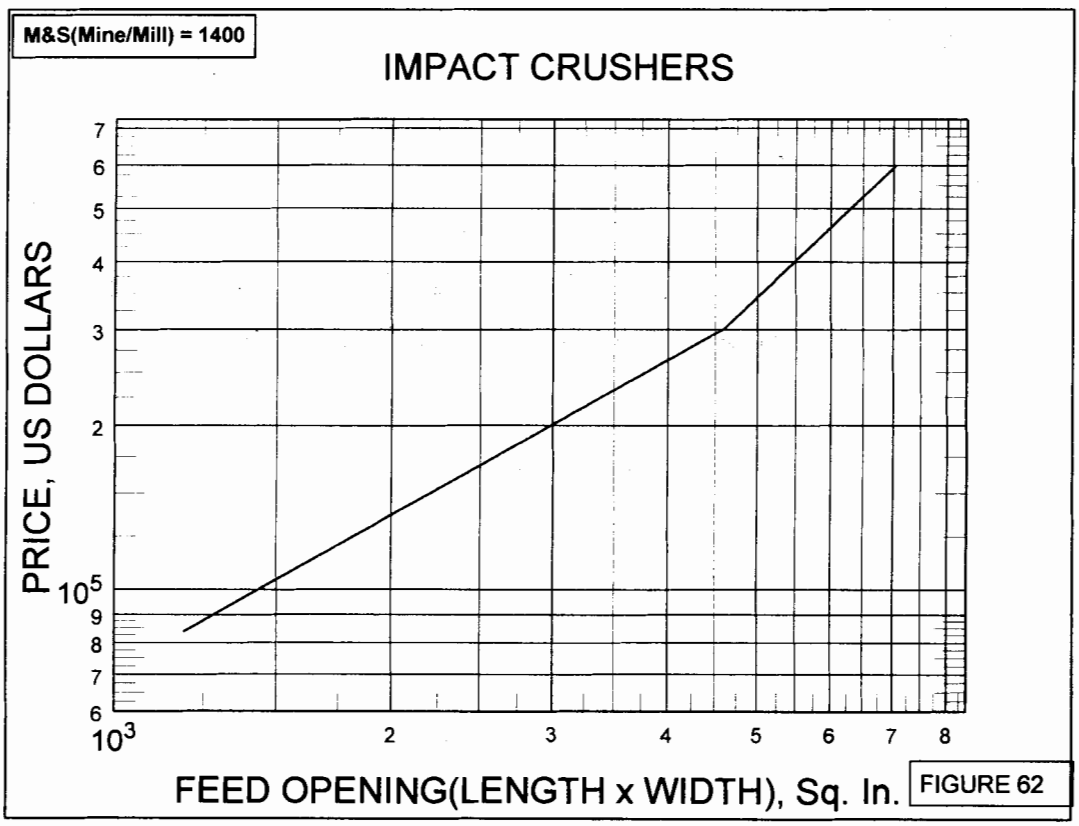
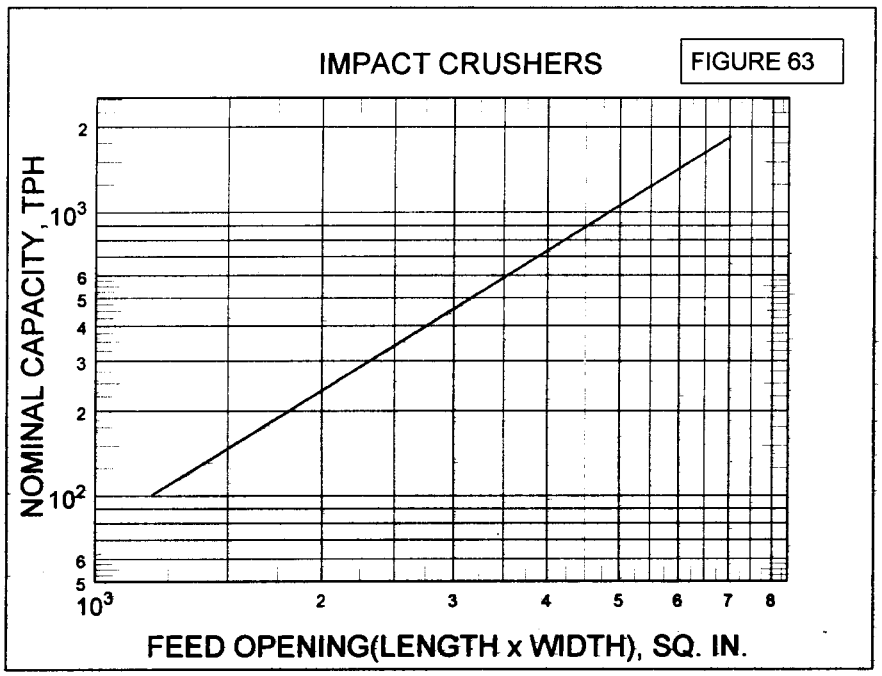
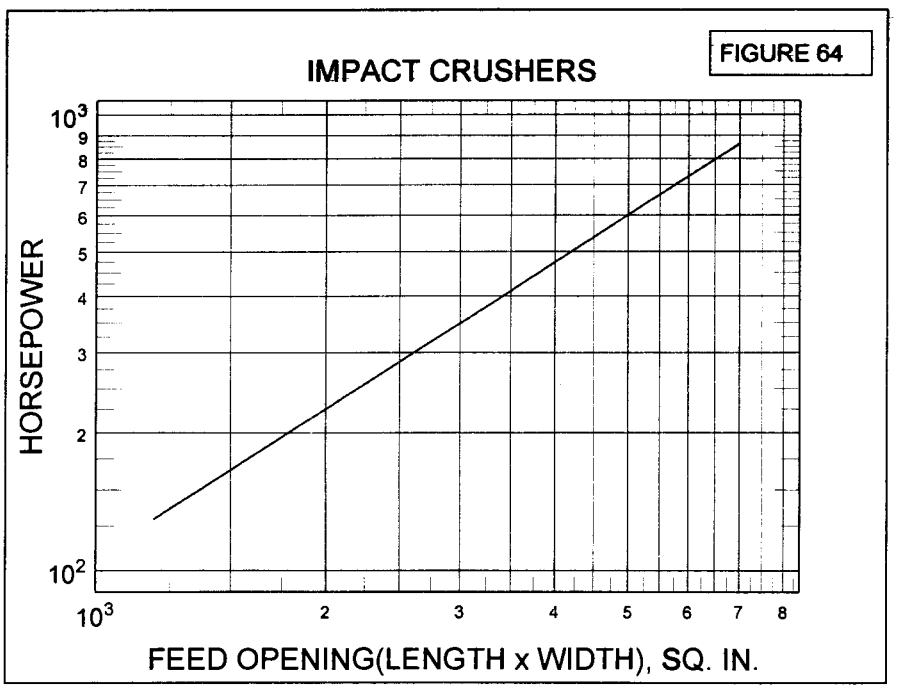


FIGURE 62

IMPACT CRUSHERS (CONTINUED)

Figures 63 and 64 show how nominal capacity and horsepower vary with feed opening.





HAMMER MILLS

Price = aX^b, US dollars

where **X is rotor diameter times rotor length, square inches**. Price includes drive, feed box and lubrication system; excludes motor.

Range in X, $in.^2$:	24 to 1716	a = 976	b = 0.6307
	1716 to 7680	a = 3.00	b = 1.407
Non-Reversible Range in X, $in.^2$:	500 to 6000	a = 311.9	b = 0.9126

Dia. x Length in. x in.	Horsepower Range	Feed Opening in. x in.	Capacity, stph, at Discharge Size .125 inch	1 inch
20 x 12	12 to 20	11 x 11	2 to 3	6 to 8
24 x 20	30 to 40	13 x 20	6 to 8	18 to 20
36 x 36	75 to 100	18 x 36	20 to 25	75 to 90
36 x 60	150 to 175	18 x 60	45 to 50	140 to 160
42 x 66	175 to 250	22 x 66	75 to 85	175 to 220

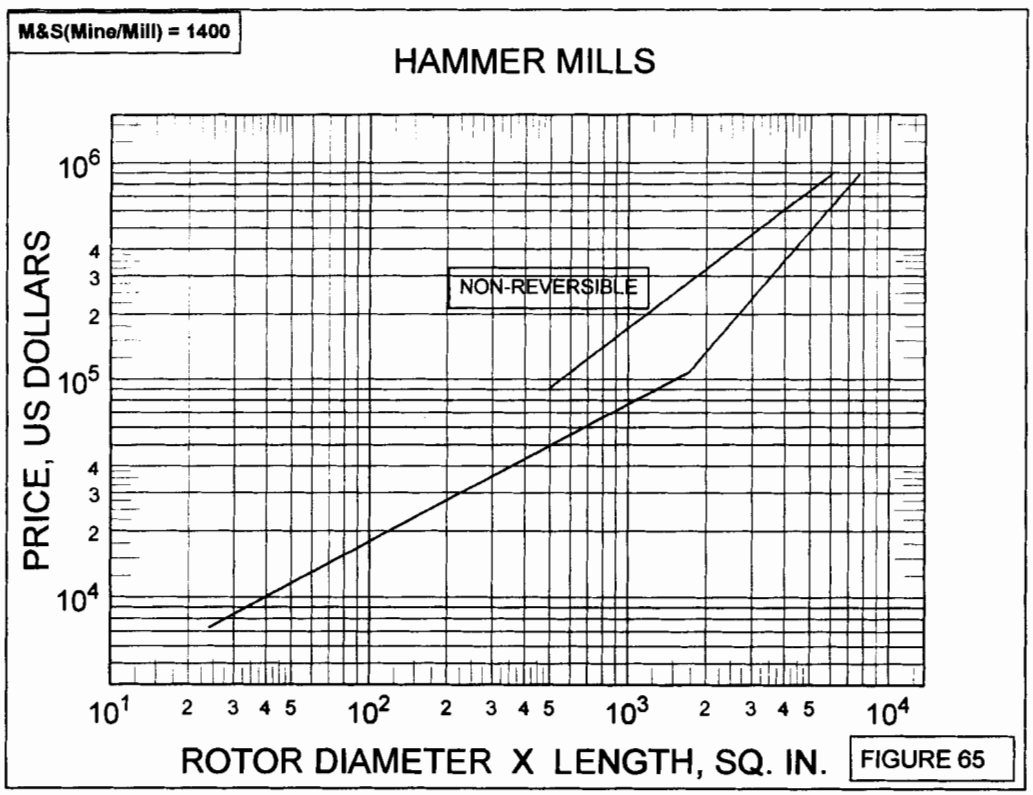


FIGURE 65

PULVERIZERS

Price = aX^b, US dollars

where **X is required horsepower**.

Light Duty Range in X, hp:	5 to 150	a = 5370	b = 0.445
Heavy Duty Range in X, hp:	10 to 62.3	a = 13800	b = 0.3698
	62.3 to 200	a = 2181	b = 0.8162

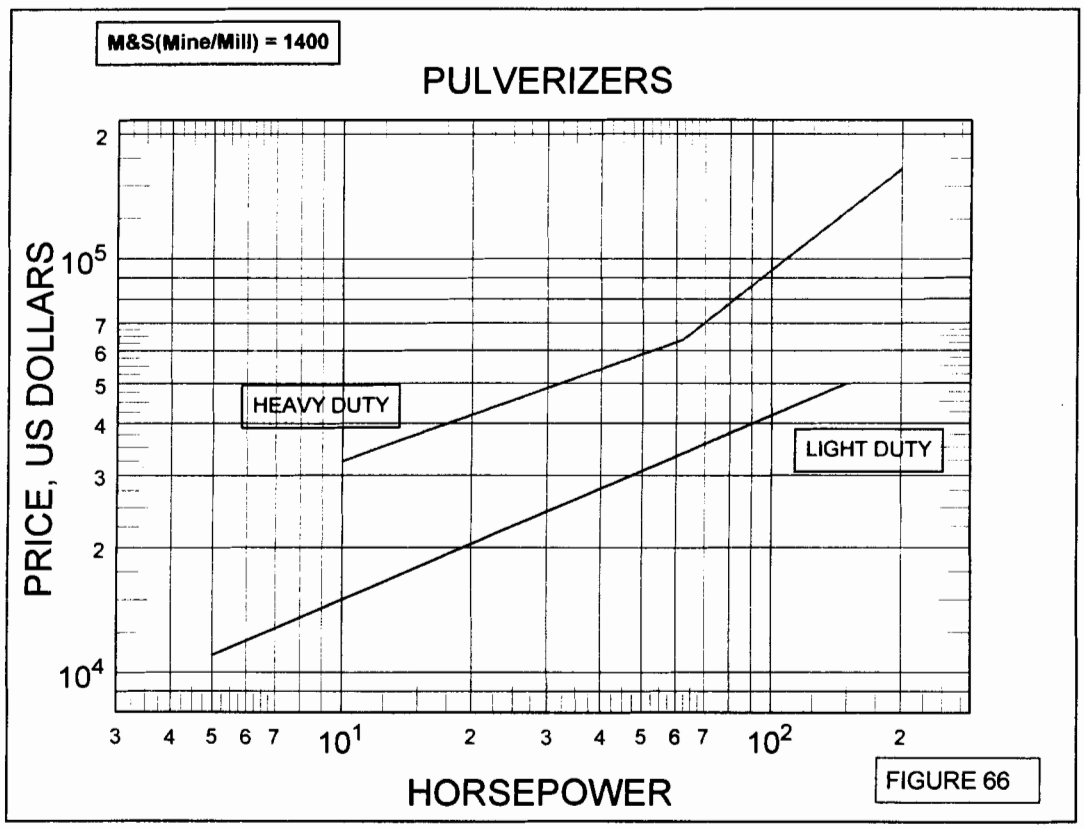


FIGURE 66

VI. Comminution Equipment (Continued)

(2) Grinding Mills - - - - - - - - Page 132

(a) Example Estimates - - - - - - - Page 133

(b) SAG Mills - - - - - - - - Page 135 ✓

(c) Ball Mills - - - - - - - - Page 137 ✓

(d) Pebble Mills - - - - - - - - Page 139

(e) Rod Mills - - - - - - - - Page 141

(f) Vertically Stirred Ball Mills - - - - - - Page 143 ✓

(g) Ring-Roller Mills - - - - - - - Page 145

(h) Rotary Breaker (Bradford) - - - - - - Page 146

EXAMPLE ESTIMATES

Grinding mill prices have been related, by means of equations and corresponding graphs, to either the nominal size or the horsepower of the mills. For most types of grinding mills a table or graph has been included, which shows a relation between mill size and horsepower. Thus, if a horsepower (or size) is known, an appropriate size (or horsepower) can be estimated such that a corresponding price can be determined.

Approximate capacity versus horsepower curves or tables in this section are limited, because these depend upon many factors including the ore hardness and final grind size. Example 1, below, illustrates the use of Bond's equation for estimating required horsepower per ton. This expression should be used with caution without further information and experience.

For convenience, Table 16, page 147, gives the equivalence between mesh sizes and screen aperture sizes in either inches or micrometers (microns). Both the US and Canadian sieve series are compared with the Tyler equivalent.

Example 1:

Estimate the power required to grind 100 stph (dry) in a ball mill and classifier closed circuit that operates at a circulating load ratio of 2.5. The feed to the circuit is 80% passing size F_{80} in micrometers, while the product leaving the circuit is 80% passing size P_{80} in micrometers. The Bond Work Index, W_I, is found to be 14.7.

Bond's equation to estimate power per stph is:

$$W = 10W_I\left[\frac{1}{\sqrt{P_{80}}} - \frac{1}{\sqrt{F_{80}}}\right]$$

where W = kilowatt hours per short ton required to grind from size F_{80} to P_{80}
W_i = the Bond work index

Suppose the feed to the circuit is 80% passing 1/4 inch, while the final product is 80% passing 65 mesh Tyler. Table 16, page 147, shows that 65 mesh corresponds to 212 μm, while 1/4 inch is equal to 6350 μm. Consequently:

$$W = 10(14.7)\left[\frac{1}{\sqrt{212}} - \frac{1}{\sqrt{6350}}\right] \text{ Kw Hr/Ton}$$

W = 8.25 Kw Hr/Ton

= 1.34(8.25) = 11.06 Hp Hr/Ton

Various correction factors are added to the above for several reasons (See Rowland, C. A., "Selection of Rod Mills, Ball Mills, Pebble Mills and Regrind Mills", Chapter 23 in Design and Installation of Comminution Circuits, Eds. Andrew L. Mular and Gerald V. Jergensen, II, SME-AIME, Littleton, Colorado, 1982). As a crude approximation, add 10 percent for losses in

EXAMPLE ESTIMATES (CONTINUED)

drive efficiency. Mills of larger diameter can be more efficient so that a diameter correction factor must be added (this can represent a reduction in required power of up to 17% for a 20 ft diameter mill). Ignoring corrections for diameter and factors other than drive efficiency, the power required becomes: $W = 11.06 + .1(11.06) = 12.17$ Hp Hr/Ton. To treat 100 stph, then the required power is: 100(12.17) = 1217 horsepower, say, a 1250 hp motor.

The computation roughly applies to rod, ball and pebble mills.

Example 2:

Scale up from grinding tests in a calibrated 6' diameter SAG mill established that a 30' diameter by 12' length SAG mill should have a capacity of 1000 stph to reach a desired final size P_{80}. What is the estimated price of a 30' x 12' SAG mill and what is the nominal horsepower?

Page 135 shows that the price of a SAG mill is determined from: Price $= aD^b\left[\frac{L}{D}\right]$, where D is mill diameter, ft, L is mill length, ft, and a and b are constants. If a and b are known, then D = 30' and L = 12' are substituted into the equation to obtain Price.

The graph in Figure 67, page 135, provides the price at the $\frac{D}{L}$ ratio of 2.65 ($\frac{L}{D} = .3775$). To **convert the price in the graph** to an alternative ratio, **you must multiply the price from the graph by** $2.65\left[\frac{L}{D}\right]$. From the graph at D = 30', the unconverted price is US$4,400,000 (where the diameter to length ratio = 2.65 and M&S(M/M) Index = 1400). In our case, the diameter to length ratio is 2.5. Hence the price becomes $4400000(2.65)\left[\frac{12}{30}\right] = 4660000$ US$.

Figure 68, page 136, shows that the **nominal horsepower per unit length** of a SAG mill at a diameter of 30 ft is about 530. Since the length is 12 ft, then the estimated horsepower becomes 530(12) = 6360. At a ten percent drive loss, this becomes about 7000 hp.

Example 3:

The horsepower required for a particular capacity and grind was 650, as found from the Bond equation. What size ball mill (for closed circuit grinding) is necessary and how much will it cost complete with liners and ball charge?

Referring to Figure 70, page 138, a 10 ft diameter (inside liners) mill has a horsepower per foot requirement of about 65. Using a length to diameter ratio of 1 means that a 10' by 10' ball mill would be a suitable size. The cost of the mill is obtained directly from the graph in Figure 69, page 137, as approximately US$540,000 at an M&S(M/M) Index of 1400. Note that at L/D ratios other than one, the cost obtained from the graph must be multiplied by L/D.

Assuming the charge to occupy 45% of the mill volume, then the cubic feet of loose balls required is $0.45(\pi\ (5)^2(10)) = 353.4\ ft^3$. The tons of balls required is then $353.4(280\frac{lbs}{ft^3})(1/2000)$ = 49.5 tons. At $600 per ton, the charge will cost about US$29,700 at an M&S(M/M) Index of 1400. Total cost becomes approximately US$570,000.

SEMIAUTOGENOUS MILLS

$$\text{Price} = aD^b\left[\frac{L}{D}\right], \text{ US dollars}$$

where **D is mill diameter in ft. of a mill with a length to diameter ratio of** $\left[\frac{L}{D}\right]$. Price includes liners, backing, lubrication system, twin drive train; excludes motor and steel balls. At an M&S(M/M) Index of 1400, ball charge (excluding shipping, handling, packaging) will cost US$625 per ton, while a liner-handling crane (hydraulic drive with boom) and motor will cost approximately US$190,000. Ball charge bulk density is about 280 lb/ft^3. Usual ball charge is 10 to 18% of internal mill volume.

Range in D, ft: 16 to 38 a = 8202 b = 2.134

To use the graph below to estimate prices when $\frac{L}{D}$ is other than 0.3775, multiply the price taken from the graph by $2.65\left[\frac{L}{D}\right]$, where $\left[\frac{L}{D}\right]$ is the new ratio.

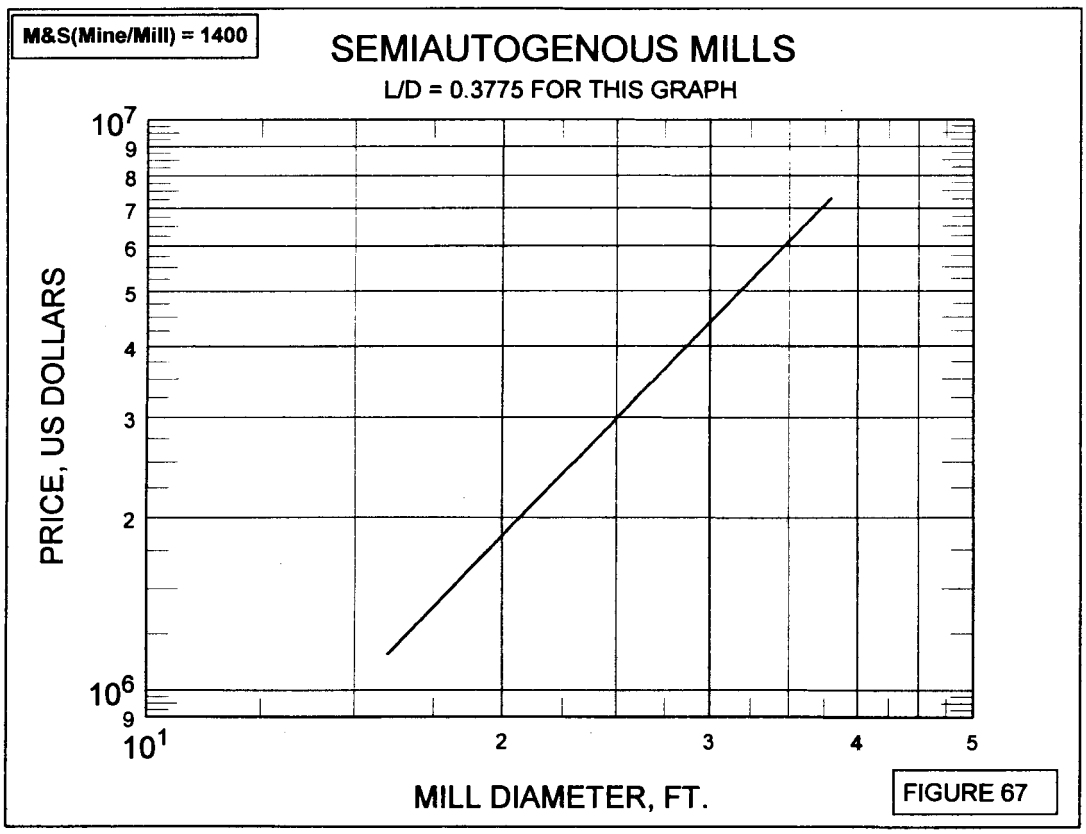
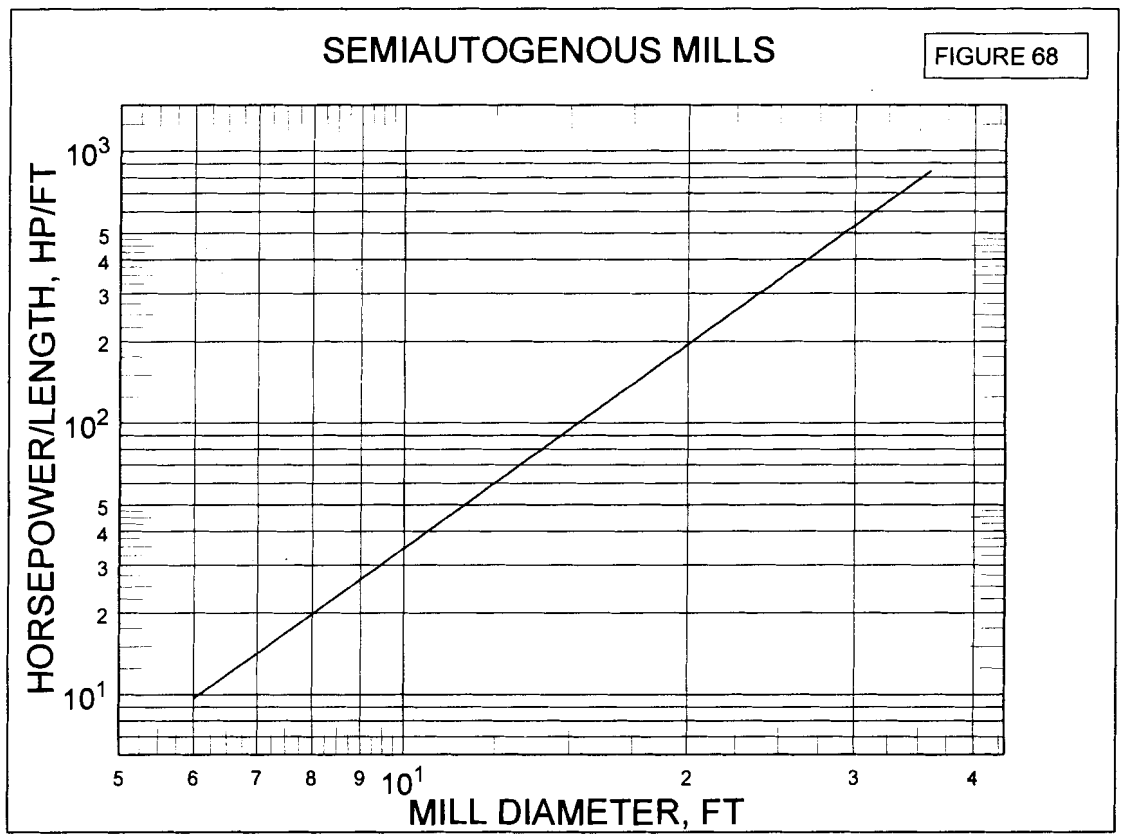


FIGURE 67

SEMIAUTOGENOUS MILLS (CONTINUED)

Figure 68 below shows how a nominal horsepower per unit of effective grinding length will vary with mill diameter.



The capacity of a full scale operating SAG mill will vary widely with a number of factors, such as feed size distribution, ore type (related to geological characteristics that influence hardness and toughness), steel load, mill speed, discharge grate opening and so on. The capacity of a 34' x 16' SAG mill at a high tonnage open pit mine is known to vary from 300 to 2400 tonnes per hour, depending on the rock type, ore body faulting and degree of hydrothermal alteration. SAG mills are fed at a rate that maintains the mill holdup at a constant weight.

A rough relationship (a variation of the Bond equation) between capacity and horsepower can be employed as follows:

$$\frac{horsepower}{stph} = 13.4 W_{sag}\left[\frac{1}{\sqrt{P_{80}}} - \frac{1}{\sqrt{F_{80}}}\right]$$

where P_{80} and F_{80} are 80% passing sizes in product and feed and W_{sag} is the "autogenous" work index which can be as high as 30 and more.

BALL MILLS

$$\text{Price} = aD^b\left[\frac{L}{D}\right], \text{ US dollars}$$

where **D is the mill diameter in ft. of a mill with length to diameter ratio of** $\left[\frac{L}{D}\right]$. Price includes liners, rubber backing, ring gear/single pinion/clutch assembly, lubrication system; excludes motor, starter and ball charge. At an M&S(M/M) Index of 1400, ball charge (excluding shipping, handling, packaging) for primary mills will cost US$600 per ton and for regrind mills will cost US$675 per ton. Ball charge bulk density is about 280 lb/ft^3. Usual ball charge is 40 to 50% of internal mill volume.

Range in D, ft.:			
	5 to 17.8	a = 14150	b = 1.578
	17.8 to 24	a = 14310	b = 4.778

Note: Cost of liners and backing is approximately 12% of total mill cost.

To use the graph below to estimate prices, multiply price taken from the graph by $\left[\frac{L}{D}\right]$, where $\left[\frac{L}{D}\right]$ is the length to diameter ratio of the mill being costed.

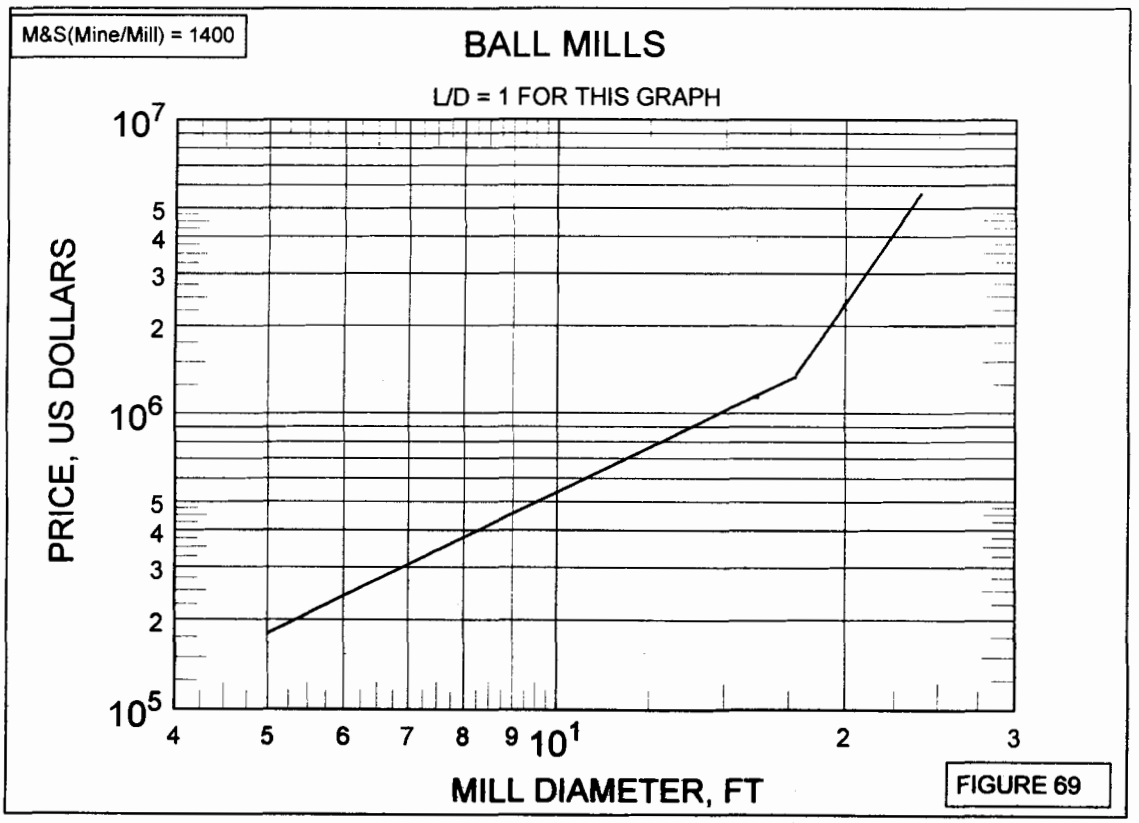


FIGURE 69

BALL MILLS (CONTINUED)

Figures 70 shows horsepower/ft of mill length vs mill diameter (inside liners) based on75% critical speed and ball charge of 45%. Figure 71 shows a nominal capacity, stph, vs horsepower for 5/8" feed being ground to 80% passing 48 mesh as calculated from the Bond equation for medium hard ore.

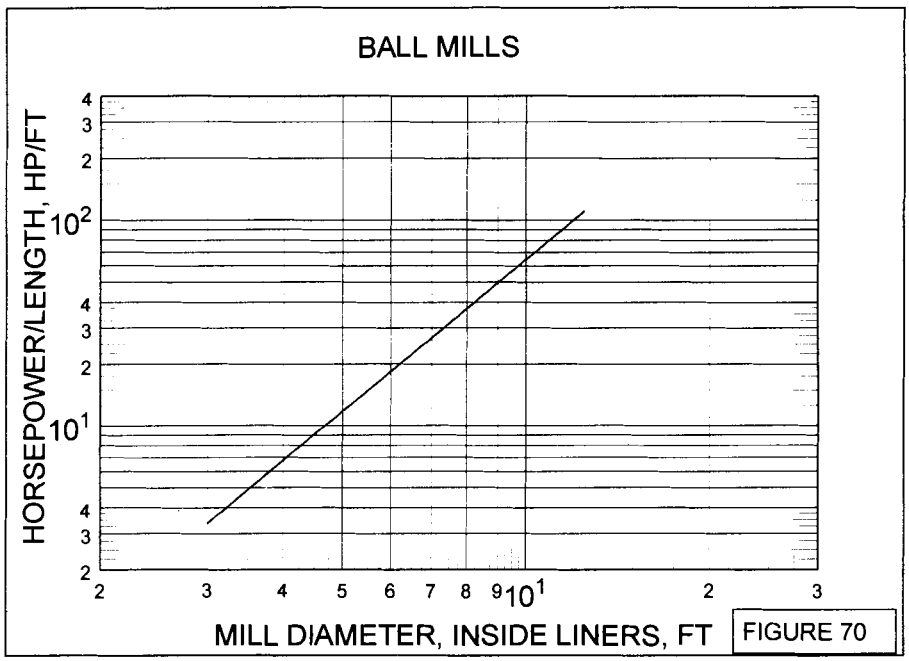


FIGURE 70

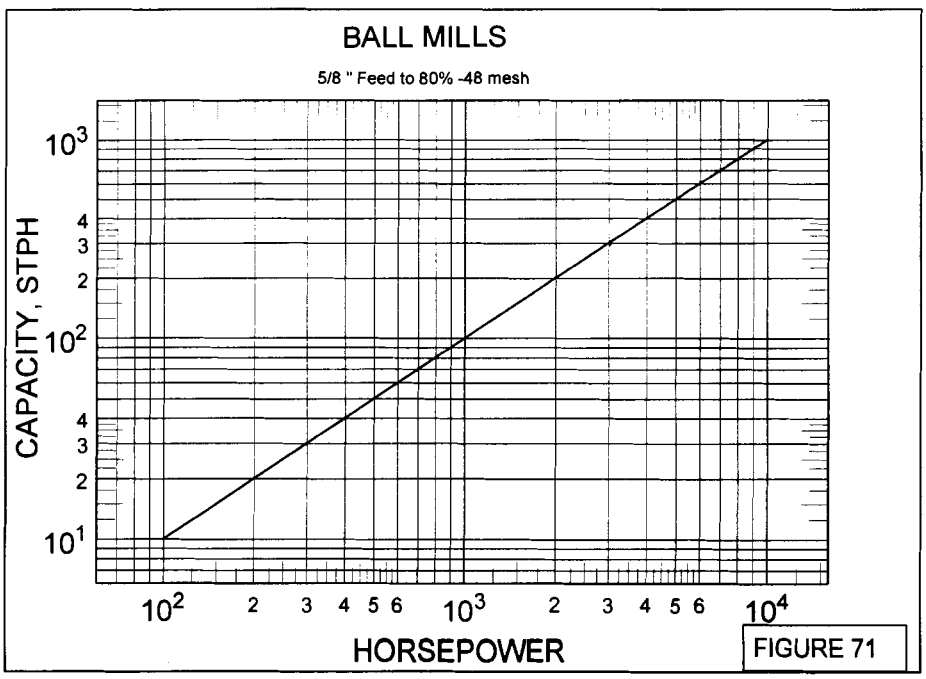


FIGURE 71

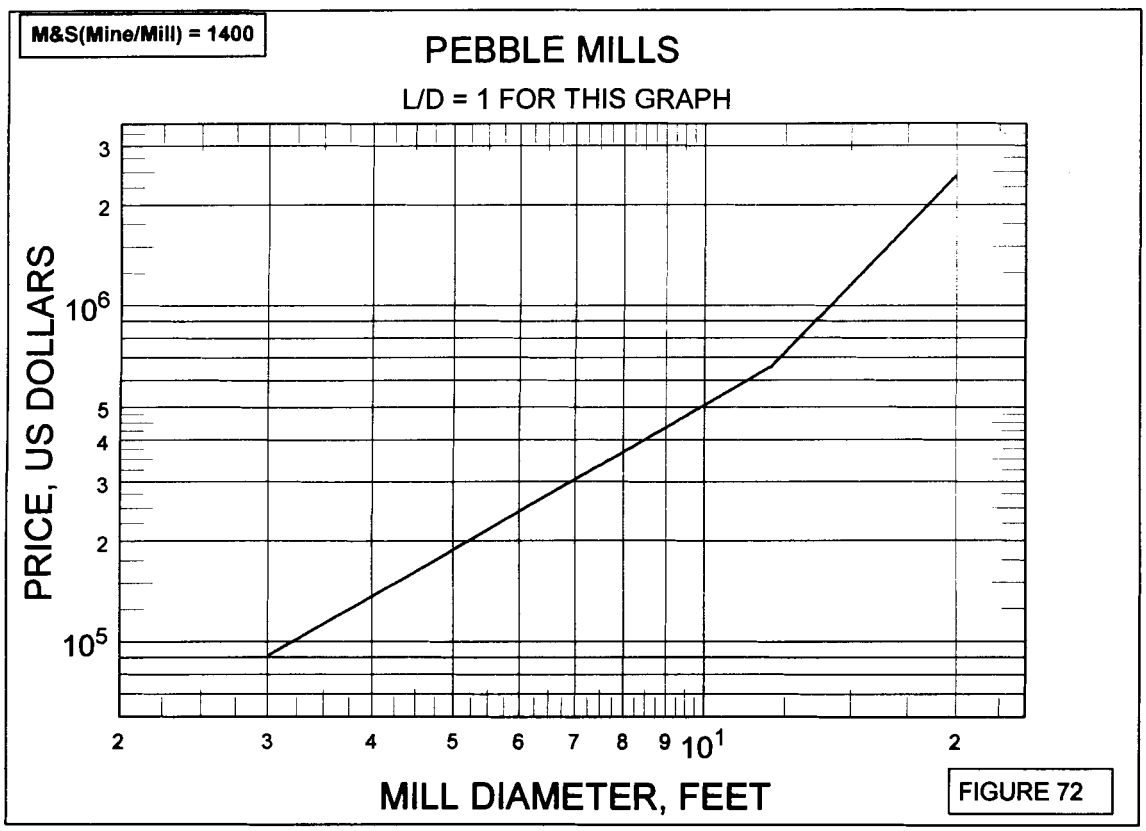
PEBBLE MILLS

$$\text{Price} = aD^b\left[\frac{L}{D}\right], \text{ US Dollars}$$

where **D is the mill diameter in ft. of a mill with length to diameter ratio of** $\left[\frac{L}{D}\right]$. Price includes liners, rubber backing, gear/pinion/clutch assembly and lube system; excludes motor/starter.

Range in D, ft:			
3 to 12	a = 18840	b = 1.428
12 to 20	a = 1061	b = 2.583

To use the graph below to estimate prices, multiply price taken from the graph by $\left[\frac{L}{D}\right]$, where $\left[\frac{L}{D}\right]$ is the length to diameter ratio of the mill being costed.



PEBBLE MILLS (CONTINUED)

Figure 73 shows horsepower per mill length (Hp/ft) versus mill diameter (ft) based on a pebble charge of 35% of mill volume, pebbles of specific gravity 2,65, 75% critical speed and 75% feed solids. Figure 74 shows capacity (stph) versus horsepower for fine grinding of ore of specific gravity 2.65.

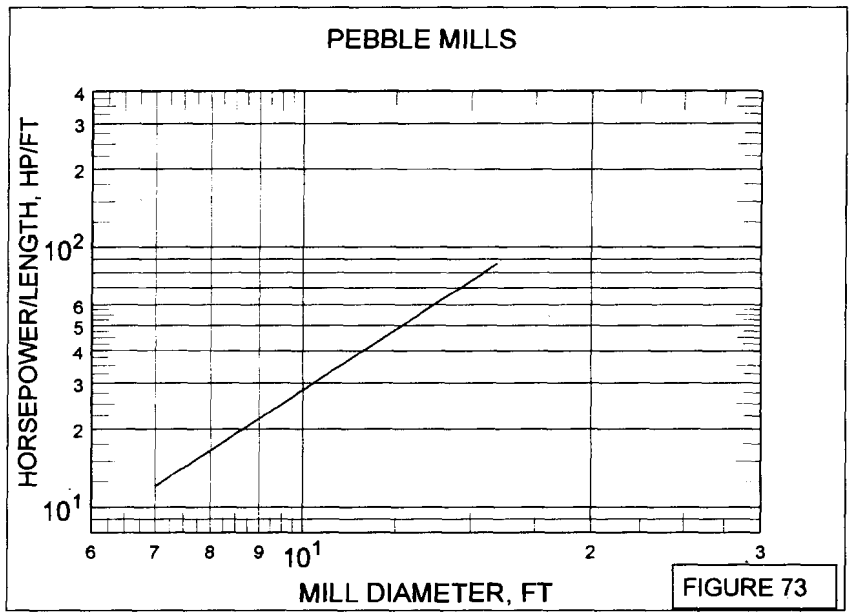


FIGURE 73

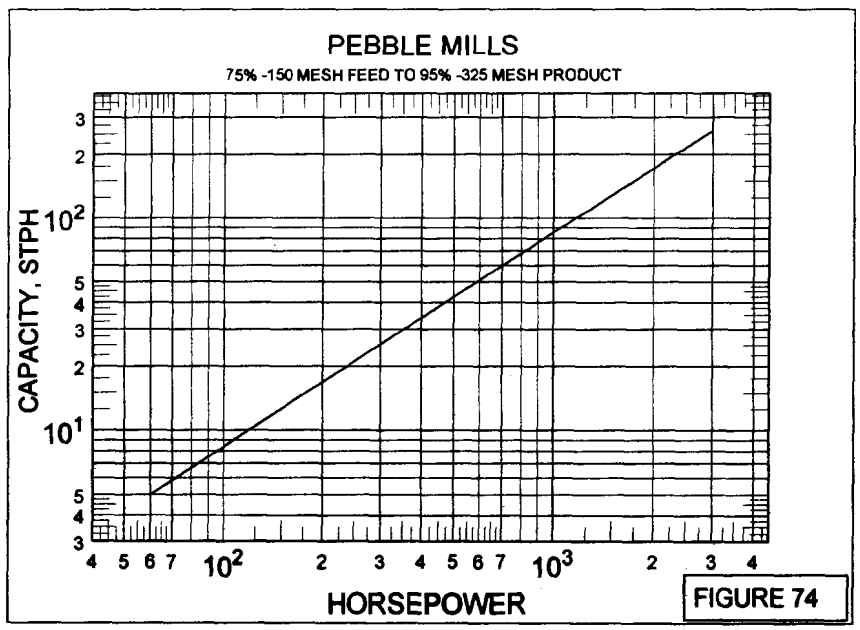


FIGURE 74

ROD MILLS

$$\text{Price} = aD^b\left[\frac{L}{D}\right], \text{ US Dollars}$$

where **D is mill diameter in ft. of a mill with a length to diameter ratio of** $\left[\frac{L}{D}\right]$. Price includes liners, rubber backing, gear/pinion/clutch assembly, lubrication system; excludes motor/starter and rod charge. At an M&S(M/M) Index of 1400: a rod charger (pneumatic/hydraulic) with motor is approximately US$110,000, while steel rods will cost approximately US$550 per short ton. Charge weighs about 350 lb/ft^3. Usual charge is 35 to 45% of internal mill volume.

Range in D, ft: 5 to 14 a = 12440 b = 1.658

Note: Cost of liners and backing is approximately 15% of the total mill cost.

If you wish to use the graph below for a length to diameter ratio other than 1.25, **multiply the price taken from the graph by** $0.8\left[\frac{L}{D}\right]$, where $\left[\frac{L}{D}\right]$ is the new length to diameter ratio.

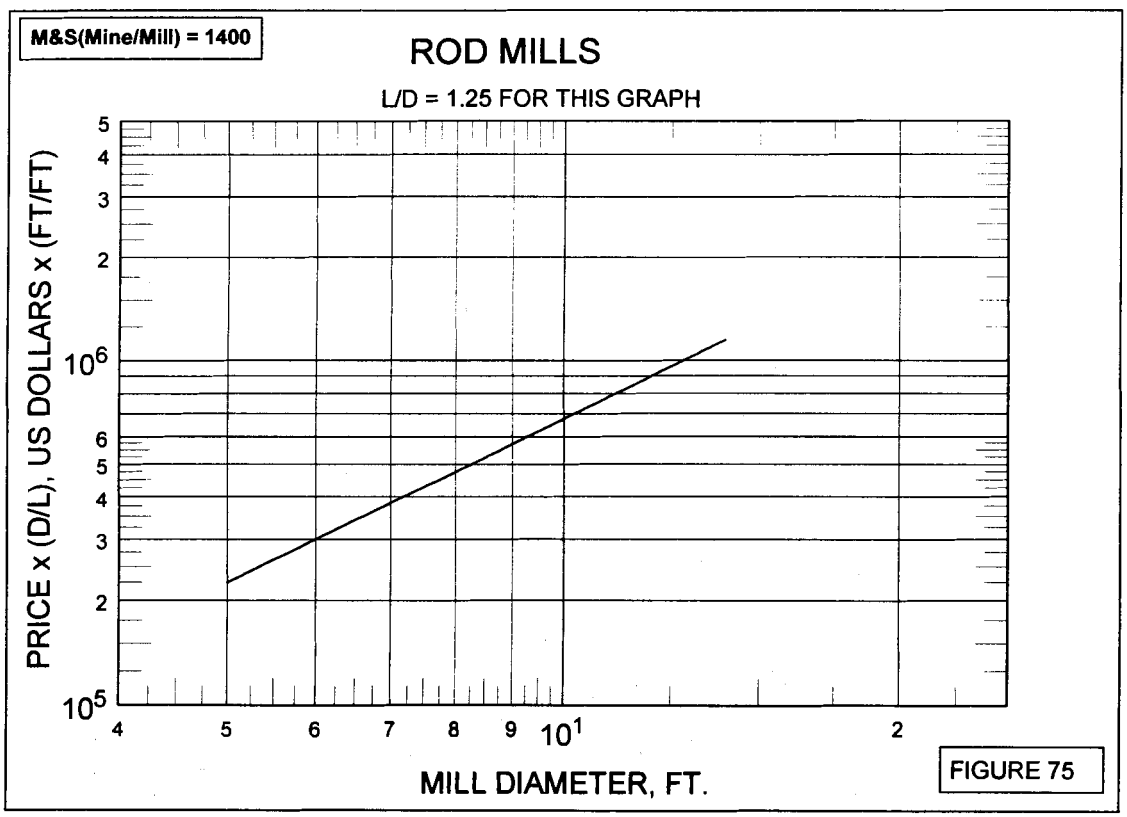
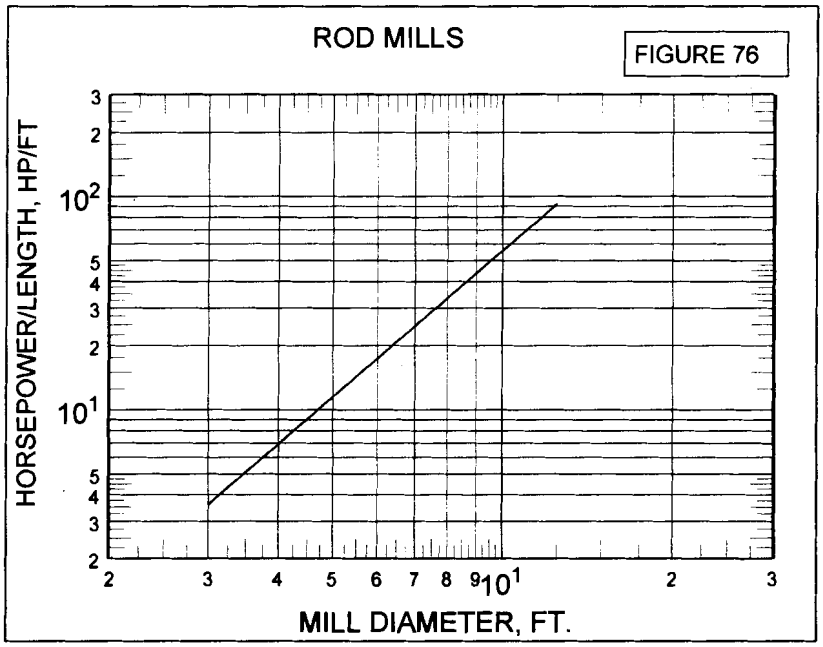
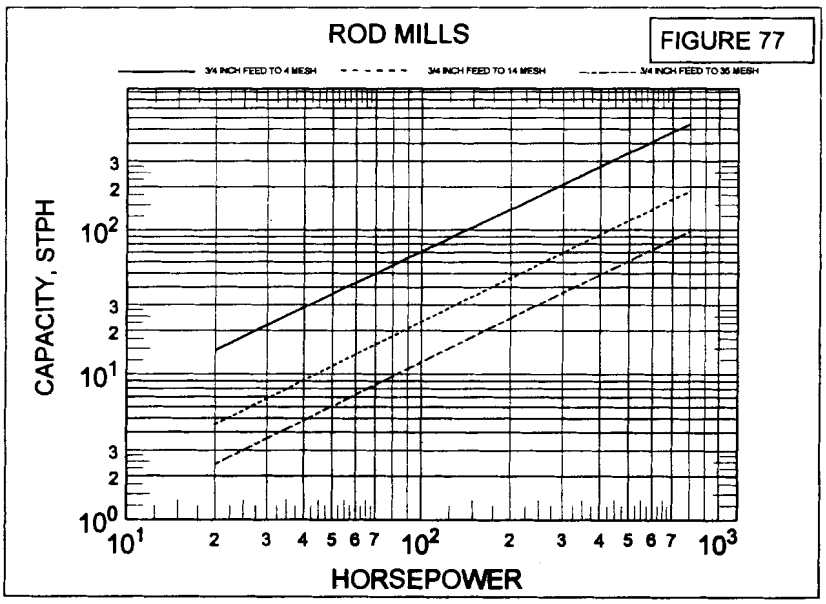


FIGURE 75

ROD MILLS (CONTINUED)

Figure 76 relates horsepower per unit length to mill diameter (inside length) based on a mill speed of 75% critical speed and rod charge of 45% mill volume. For mills greater than 7 ft diameter, subtract 0.5 ft from the shell diameter to estimate inside liner diameter. Figure 77 plots capacity versus horsepower for F_{80} = 3/4" and different P_{80} values with medium hard ore.





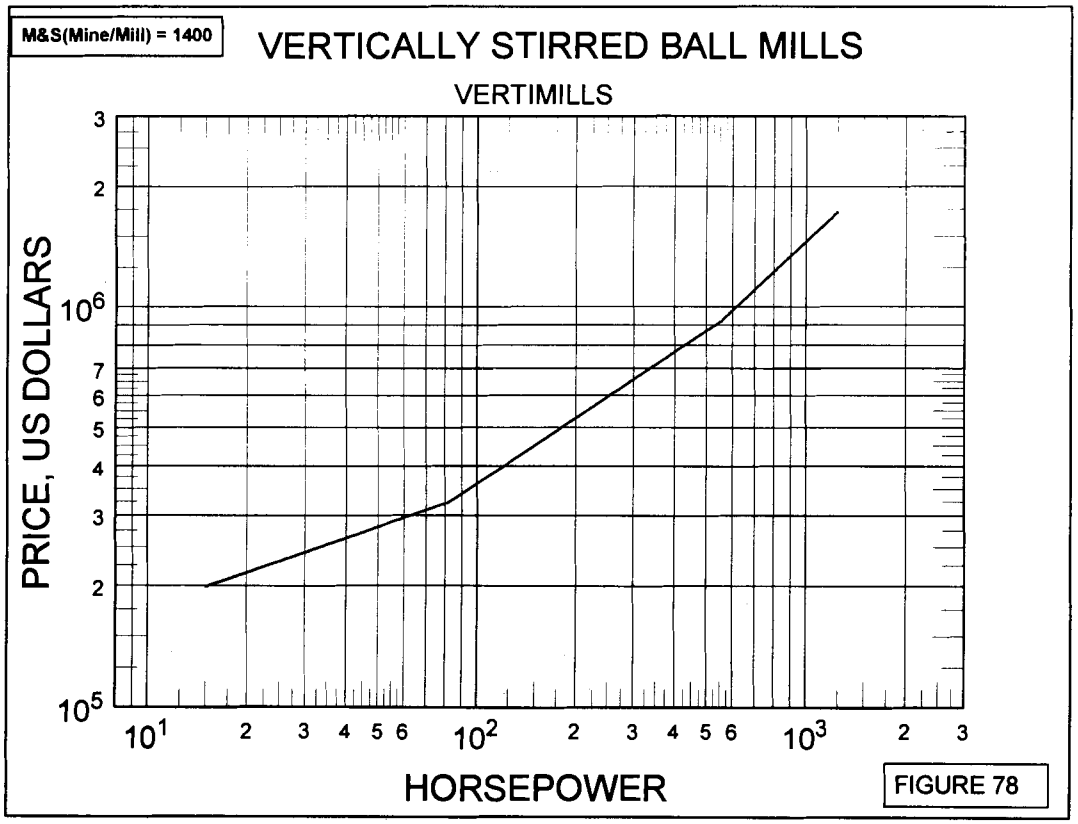
VERTICALLY STIRRED BALL MILLS
(Vertimills)

Price = aX^b , US Dollars

where **X is the horsepower drawn by the mill**. Price includes magnetic body liners, alloy steel agitator liners, speed reducer drive and motor. Steel balls and motor starter are excluded. At an M&S(M/M) Index of 1400, steel balls will cost approximately US$675 per short ton. Charge weighs about 280 lb/ft^3.

Range in X, hp:			
15 to 81	a = 91190	b = 0.2867
81to 530	a = 29130	b = 0.5458
530 to 1250	a = 6887	b = 0.7744

Typical U by V ratios are: 11.88' x 17.17', 9.33' x 17.33', 7.25' x 16.67', 5.27' x 14', 3.42' x 14', where V is height at the operating level and U is tank diameter.



VERTICALLY STIRRED BALL MILLS (CONTINUED)

In Figure 79A, capacity is related to horsepower via the Bond equation using a medium hard ore, where work inputs are scaled by vertimill factors. The upper curve is for coarser grinds; the lower one is for finer grinds. Figure 79B relates horsepower to the ball load.

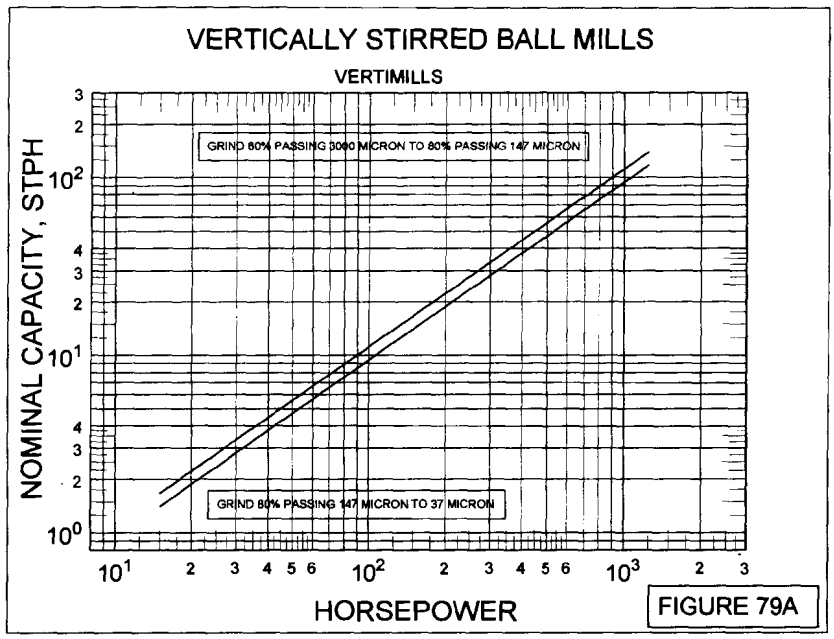


FIGURE 79A

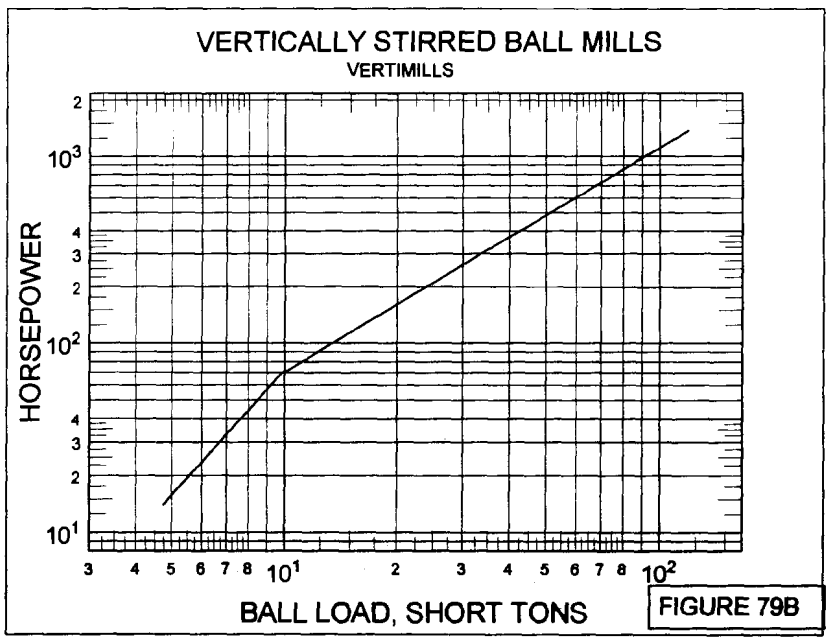


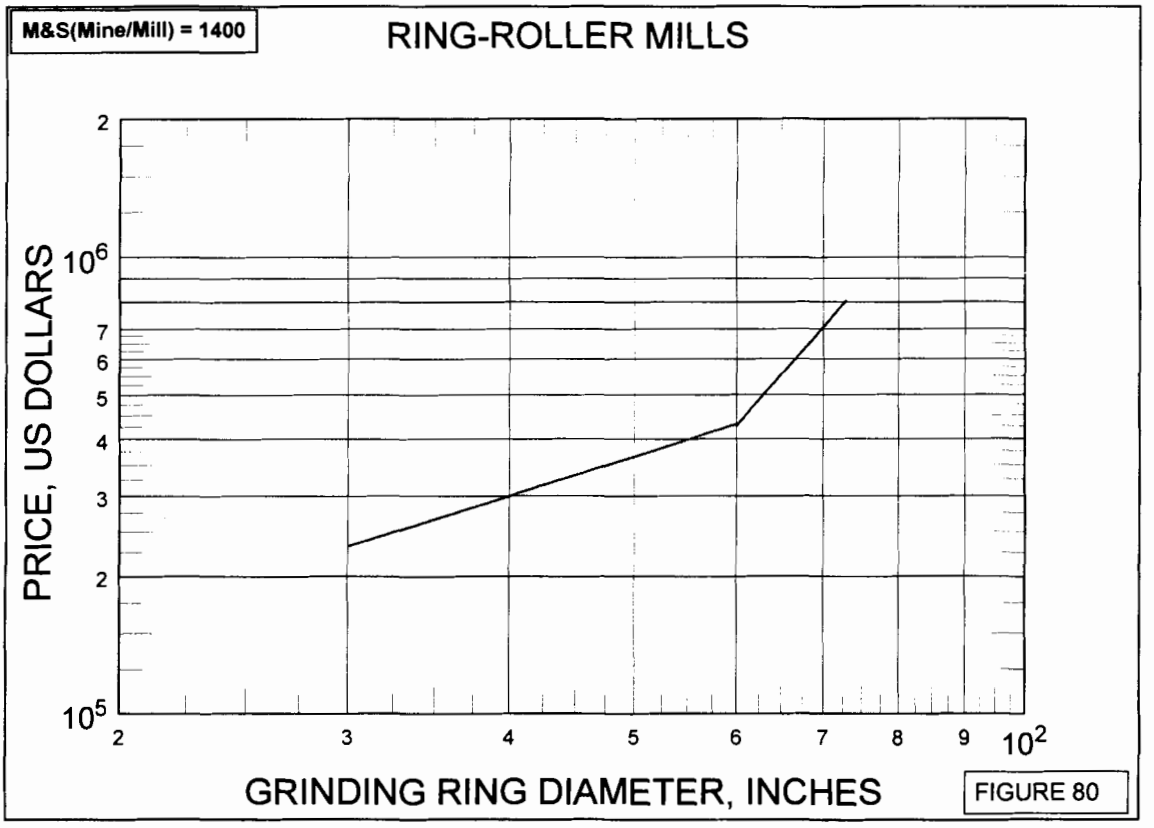
FIGURE 79B

RING-ROLLER MILLS

Price = aX^b, US Dollars

where **X is grinding ring diameter in inches**. These mills are used in the cement and ceramics industries. Price includes built in air classifier; excludes motor.

Range in X, inches:			
	30 to 60	a = 11130	b = 0.8924
	60 to 73	a = 0.6867	b = 3.2570



ROTARY BREAKER (BRADFORD)

Price = aX^b , US Dollars

where **X, in ft^3, is diameter squared (ft^2) times length (ft)**. Cost includes drive; excludes motor and starter.

Range of X, cu. ft.: 1296 to 5488 a = 2182 b = 0.6118

Diameter x Length ft. x ft.	HP Range	Capacity Range STPH	Coal Hardness
6 x 8	10	75 - 150	soft
7 x 14	15 - 20	125 - 250	soft
9 x 17	40 - 50	275 - 450	medium
10.5 x 19	60 - 75	500 - 750	medium/hard
12 x 22	100 - 150	1000 - 1500	medium/hard

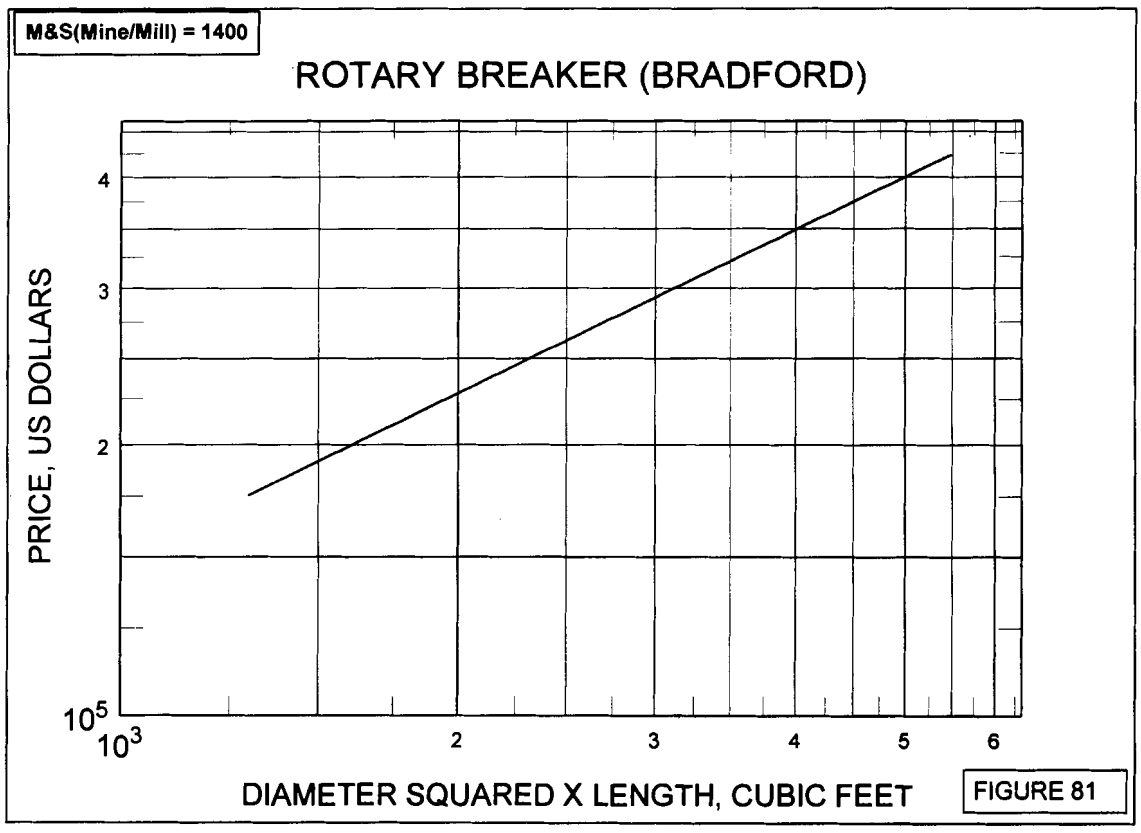


TABLE 16: CONVERSION OF SIEVE SIZES

Comparison Table of U.S.A., Tyler, Canadian, British, French and German Standard Sieve Series

U.S.A. (1)		TYLER (2)	CANADIAN (3)		BRITISH (4)		FRENCH (5)		GERMAN (6)
*Standard	Alternate	Mesh Designation	Standard	Alternate	Nominal Aperture	Nominal Mesh No.	Opg. M.M.	No.	Opg.
125 mm	5″		125 mm	5″					
106 mm	4.24″		106 mm	4.24″					
100 mm	4″		100 mm	4″					
90 mm	3½″		90 mm	3½″					
75 mm	3″		75 mm	3″					
63 mm	2½″		63 mm	2½″					
53 mm	2.12″		53 mm	2.12″					
50 mm	2″		50 mm	2″					
45 mm	1¾″		45 mm	1¾″					
37.5 mm	1½″		37.5 mm	1½″					
31.5 mm	1¼″		31.5 mm	1¼″					
26.5 mm	1.06″	1.05″	26.5 mm	1.06″					
25.0 mm	1″		25.0 mm	1″					25.0 mm
22.4 mm	⅞″	.883″	22.4 mm	⅞″					
19.0 mm	¾″	.742″	19.0 mm	¾″					20.0 mm
									18.0 mm
16.0 mm	⅝″	.624″	16.0 mm	⅝″					16.0 mm
13.2 mm	.530″	.525″	13.2 mm	.530″					
12.5 mm	½″		12.5 mm	½″					12.5 mm
11.2 mm	7/16″	.441″	11.2 mm	7/16″					
									10.0 mm
9.5 mm	⅜″	.371″	9.5 mm	⅜″					
8.0 mm	5/16″	2½	8.0 mm	5/16″					8.0 mm
6.7 mm	.265″	3	6.7 mm	.265″					
6.3 mm	¼″		6.3 mm	¼″					6.3 mm
5.6 mm	No. 3½	3½	5.6 mm	No. 3½					
							5.000	38	5.0 mm
4.75 mm	4	4	4.75 mm	4					
4.00 mm	5	5	4.00 mm	5			4.000	37	4.0 mm
3.35 mm	6	6	3.35 mm	6	3.35 mm	5			
							3.150	36	3.15 mm
2.80 mm	7	7	2.80 mm	7	2.80 mm	6			
2.36 mm	8	8	2.36 mm	8	2.40 mm	7	2.500	35	2.5 mm
2.00 mm	10	9	2.00 mm	10	2.00 mm	8	2.000	34	2.0 mm
1.70 mm	12	10	1.70 mm	12	1.68 mm	10	1.600	33	1.6 mm
1.40 mm	14	12	1.40 mm	14	1.40 mm	12			
							1.250	32	1.25 mm
1.18 mm	16	14	1.18 mm	16	1.20 mm	14			
1.00 mm	18	16	1.00 mm	18	1.00 mm	16	1.000	31	1.0 mm
850 µm	20	20	850 µm	20	850 µm	18			
							.800	30	800 µm
710 µm	25	24	710 µm	25	710 µm	22			
							.630	29	630 µm
600 µm	30	28	600 µm	30	600 µm	25			
500 µm	35	32	500 µm	35	500 µm	30	.500	28	500 µm
425 µm	40	35	425 µm	40	420 µm	36			
							.400	27	400 µm
355 µm	45	42	355 µm	45	355 µm	44			
							.315	26	315 µm
300 µm	50	48	300 µm	50	300 µm	52			
250 µm	60	60	250 µm	60	250 µm	60	.250	25	250 µm
212 µm	70	65	212 µm	70	210 µm	72			
							.200	24	200 µm
180 µm	80	80	180 µm	80	180 µm	85			
							.160	23	160 µm
150 µm	100	100	150 µm	100	150 µm	100			
125 µm	120	115	125 µm	120	125 µm	120	.125	22	125 µm
106 µm	140	150	106 µm	140	105 µm	150			
							.100	21	100 µm
90 µm	170	170	90 µm	170	90 µm	170			90 µm
							.080	20	80 µm
75 µm	200	200	75 µm	200	75 µm	200			
									71 µm
63 µm	230	250	63 µm	230	63 µm	240	.063	19	63 µm
									56 µm
53 µm	270	270	53 µm	270	53 µm	300			
							.050	18	50 µm
45 µm	325	325	45 µm	325	45 µm	350			45 µm
							.040	17	40 µm
38 µm	400	400	38 µm	400					

(1) *U.S.A. Sieve Series - ASTM Specification E-11-70*
(2) *Tyler Standard Screen Scale Sieve Series.*
(3) *Canadian Standard Sieve Series 8-GP-1d.*
(4) *British Standards Institution, London BS-410-62.*
(5) *French Standard Specifications, AFNOR X-11-501.*
(6) *German Standard Specification DIN 4188.*

**These sieves correspond to those recommended by ISO (International Standards Organisation) as an International Standard and this designation should be used when reporting sieve analysis intended for international publication.*

VII. Sizing and Classification Equipment

Sizing and Classification equipment has been placed in the sub-category, **Classifiers and Screens**.

(1) Classifiers and Screens - - - - - - - Page 148

(a) Example Estimates - - - - - - - Page 149

(b) Air Classifiers/Cyclones - - - - - - Page 152

(c) Hydrocyclones - - - - - - - - - Page 154

(d) Rake Classifiers - - - - - - - - - Page 156

(e) Spiral Classifiers - - - - - - - Page 157

(f) Vibrating Grizzlies - - - - - - - Page 158

(g) DSM Screens - - - - - - - - - Page 159

(h) Rotary Trommel Screens - - - - - - Page 160

(i) Vibrating Screens, Inclined, Woven Wire Deck - - - Page 161

(j) Vibrating Screens, Inclined, Polyurethane Deck - - - Page 162

(k) Stationary Screens - - - - - - - Page 163

(l) Screen Decks - - - - - - - - - Page 164

(m) Banana Screens - - - - - - - Page 165

EXAMPLE ESTIMATES

In this section, costs have been related generally to size. In most cases, size-to-capacity relationships in the form of tables or graphs are included. For screens, an example of capacity estimation is provided below in *Example 3.*

Example 1:

A 20-inch hydrocyclone lined with urethane/ceramic is required. How much will it cost?

From Figure 85, page 154, a 20-inch unit will cost approximately US$7,700 at an M&S(M/M) Index of 1400. If the current index is 1100, then the cost is closer to $7700\left[\frac{1100}{1400}\right]$ = US$6060.

Example 2:

An air classifier is to be used for recycling oversized material back to a dry grinding mill. The feed to the unit is 60 tph. How much will the classifier cost and what is the estimated horsepower and estimated maximum air flow rate?

From Figure 84, page 153, a 60 tph unit will have a diameter of about 18 ft. The price, from Figure 82, page 152, is about US$225,000 at an M&S(M/M) Index of 1400. Figure 84, page 153, shows that the horsepower of an 18 ft diameter air classifier is nominally close to 250. Figure 83, page 153, shows that at a horsepower of 250, the maximum air flow is approximately 45,000 cfm.

Example 3:

Estimate the area of a screen to be installed in secondary crushing circuit. The area can be approximated from the following:

$$\text{Area} = S = \frac{T}{CAB} = \text{square feet}$$

where T is the tons of feed of bulk density, $d_b = 100 \frac{lb}{ft^3}$ fed to the screen. C is the capacity in tons per square foot per hour passed by a screen for material of bulk density, $d_b = 100$. If the density is other than 100, vary C in direct proportion. A is a factor correcting for the percent of oversize in the feed, while B is a factor correcting for percent of feed to the screen that passes holes one half the size of the openings in the screen. There are other correction factors, but C, A, and B are major ones. These are shown in the table at the top of page 150, where feed is dry, screen openings are square and **d_b = 100**. The **screen analysis of the feed is**:

% Passing:	100	91.5	83	75	66	58	49	39	28	22	15	8	4
Size, inch:	5	4.5	4	3.5	3	2.5	2	1.5	1	0.75	0.5	0.25	0.125

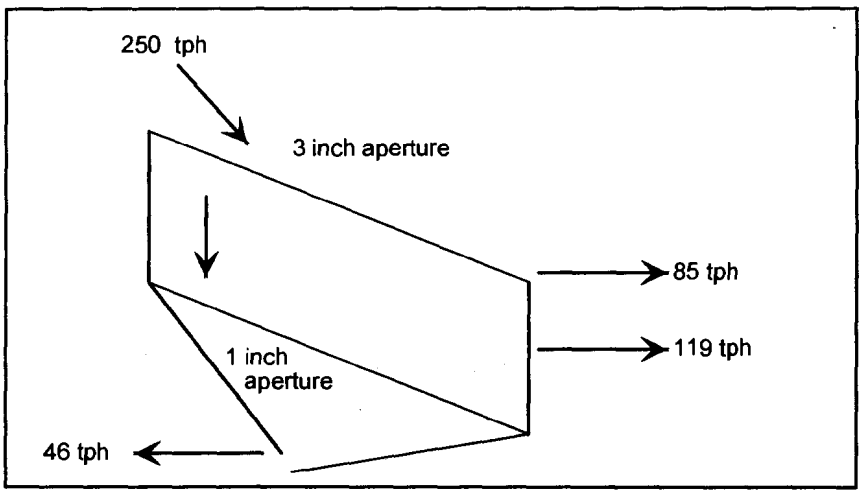
e.g., the above states that 66% of the dry feed will pass a 3-inch square opening. Suppose that the dry **feed rate** to a double deck screen (the screen opening is known for each deck) is 250 stph. We have the screen analysis so that a mass balance is obtainable. The balance is shown in the diagram on page 150.

EXAMPLE ESTIMATES (CONTINUED)

Table of Factors for Screening Calculations

Aperture X, inch	C, ton/ft^2/hr	% Oversize in Feed	Factor A	% feed passing 1/2 of size X	Factor B
0.028	0.400	10.000	0.940	10.000	0.700
0.033	0.500	20.000	0.970	20.000	0.800
0.046	0.600	30.000	1.030	30.000	0.900
0.065	0.700	40.000	1.090	40.000	1.000
0.093	0.850	50.000	1.180	50.000	1.200
0.131	1.100	60.000	1.320	60.000	1.400
0.185	1.500	70.000	1.550	70.000	1.600
0.250	2.000	80.000	2.000	80.000	1.800
0.500	3.100	90.000	3.600	90.000	2.000
0.750	3.700				
1.000	4.350				
2.000	5.700				
3.000	7.250				

Mass Balance Around Double Deck Screen



Top Deck Area Calculation:

3-inch square openings, so C = 7.25 ton/ft^2/hr from above table.
% oversize in feed = 34% (from 100 - 66% passing 3-inch)
Factor A = 1.05 from above table (from 1.03 +((34 - 30)/10) x (1.09 - 1.03))
% passing 1/2 of 3 = 1.5 inch is 39%
Factor B = 0.99 from above table (from 0.9 + ((39 - 30)/10) x (1 - 0.9))

Area required = 250/(7.25 x 1.05 x 0.99) = **33.2 ft^2**

EXAMPLE ESTIMATES (CONTINUED)

Bottom Deck Area Calculation:

1-inch square openings, so C = 4.35 ton.ft^2/hr from table, page 150
% oversize in feed = 57.6% (from (100% - 100(28/66)% passing 1-inch)
Factor A = 1.29 from table on page 150 (from 1.18 +((57.6 - 50)/10) x (1.32 - 1.18))
% passing 1/2 of 1 = 0.5 inch is about 23% (from 100(15/66))
Factor B = 0.84 from table on page 150 (from 0.8 + ((23 - 20)/10) x (0.9 - 0.8))

Area required = 165/(4.34 x 1.29 x 0.84) = **35.0 ft^2**

The deck requiring the greatest area determines the size of the screen. In this case, it is the bottom deck at 35 ft^2. Ideally, the length should be about 3 times the width, W, or 3W^2 = 35. W is thus around 3.42. Actual screen areas of top decks are roughly 90%. For a bottom deck it is roughly 75%. This is due to design features which blank off part of the screen areas. Hence, take 3.42 as 4, **so that a 4' by 12' screen should suffice**.

Example 4:

One hundred tons per hour of an ore with a specific gravity of 2.75 is to be classified in a simplex spiral classifier. What size is required and what is the estimated price?

From the table on page 157, a spiral classifier of tank length 23 1/4 feet and spiral diameter 48 inches is selected.

From Figure 88, page 157, a 48-inch spiral simplex classifier should cost approximately US$50,000 at an M&S (M/M) Index of 1400.

Example 5:

Estimate the cost of of a DSM screen of 5' 3" radius and 4' wide.

From Figure 90, page 159, the screen price is approximately US$7,700 at an M&S (M/M) Index of 1400.

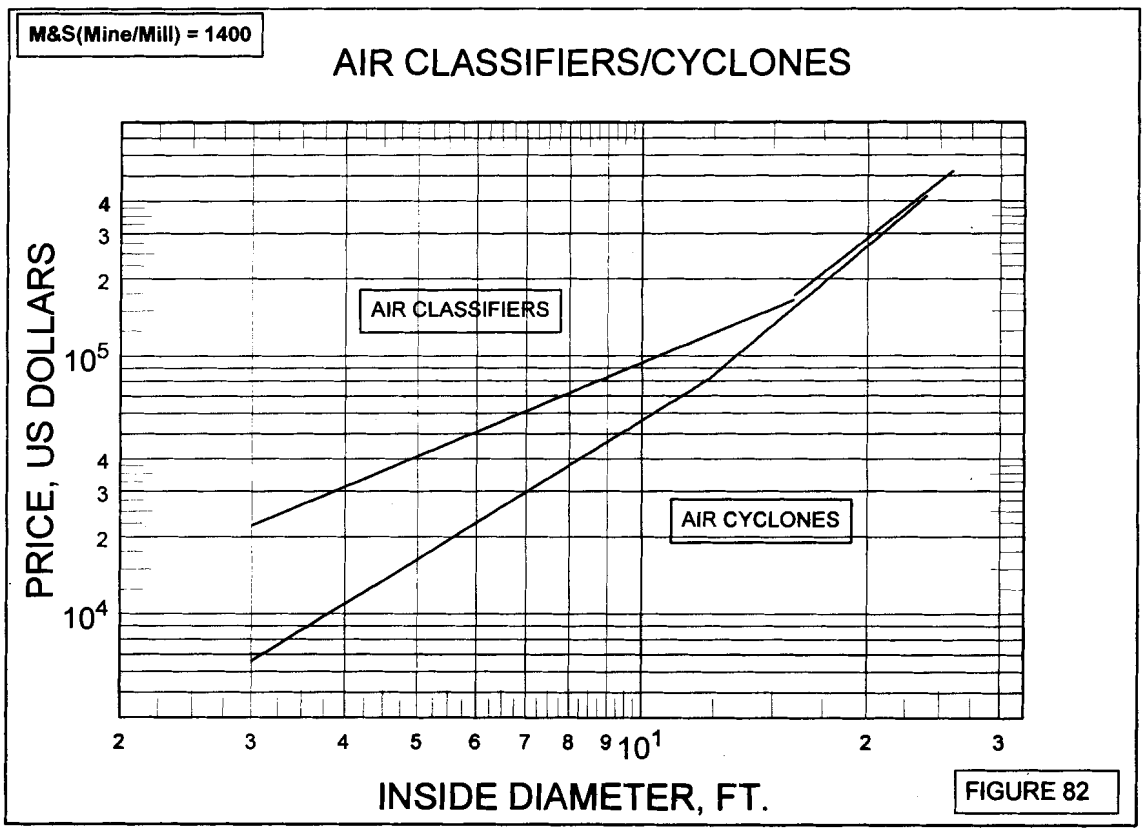
Note that this includes the feed box with standard design and reversible screen at 60 degrees but excludes the rapping or vibrating mechanism.

AIR CLASSIFIERS/CYCLONES

Price = aX^b, US Dollars

where **X is classifier/cyclone diameter in ft**. Price includes standard attachments; excludes motor and ducting. Blower is excluded for air cyclones. Capacity based on silica particles, -200 microns.

Air Classifier Range, ft:	3 to 16	a = 5862	b = 1.203
	16 to 26	a = 304.4	b = 2.285
Air Cyclone Range, ft:	3 to 12.3	a = 929.1	b = 1.779
	12.3 to 24	a = 171.5	b = 2.453



AIR CLASSIFIERS/CYCLONES (CONTINUED)

Figure 83 shows how blower capacity depends upon horsepower, while Figure 84 shows capacity or horsepower versus diameter.

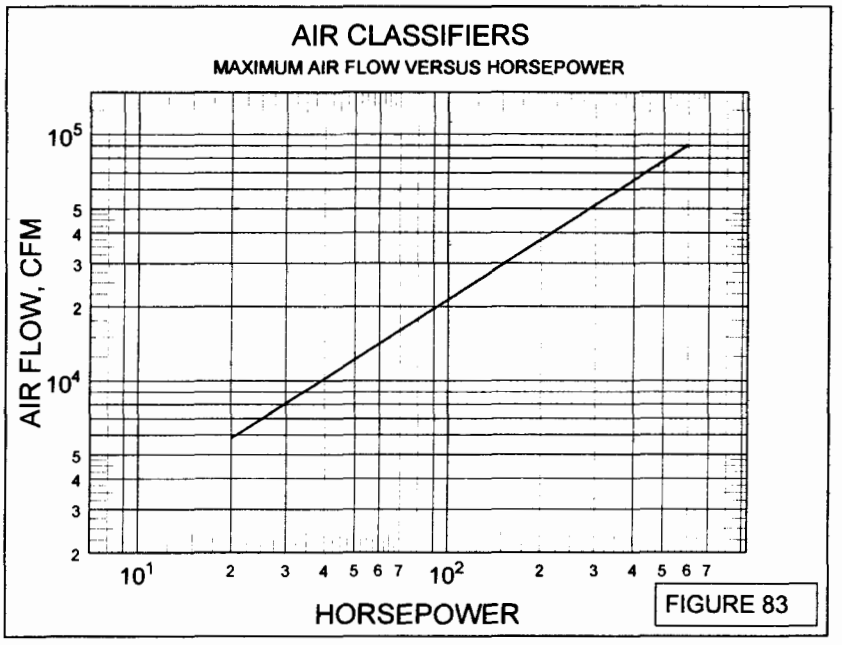


FIGURE 83

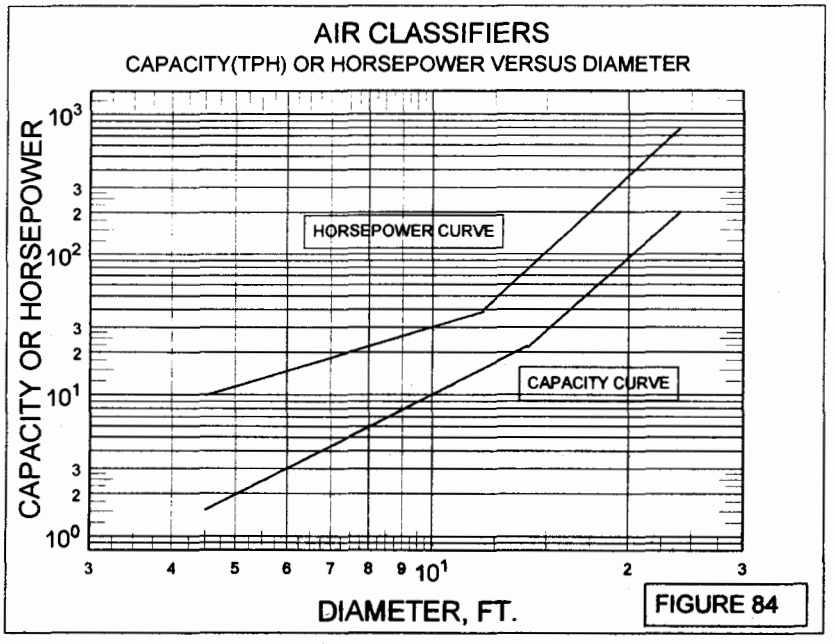


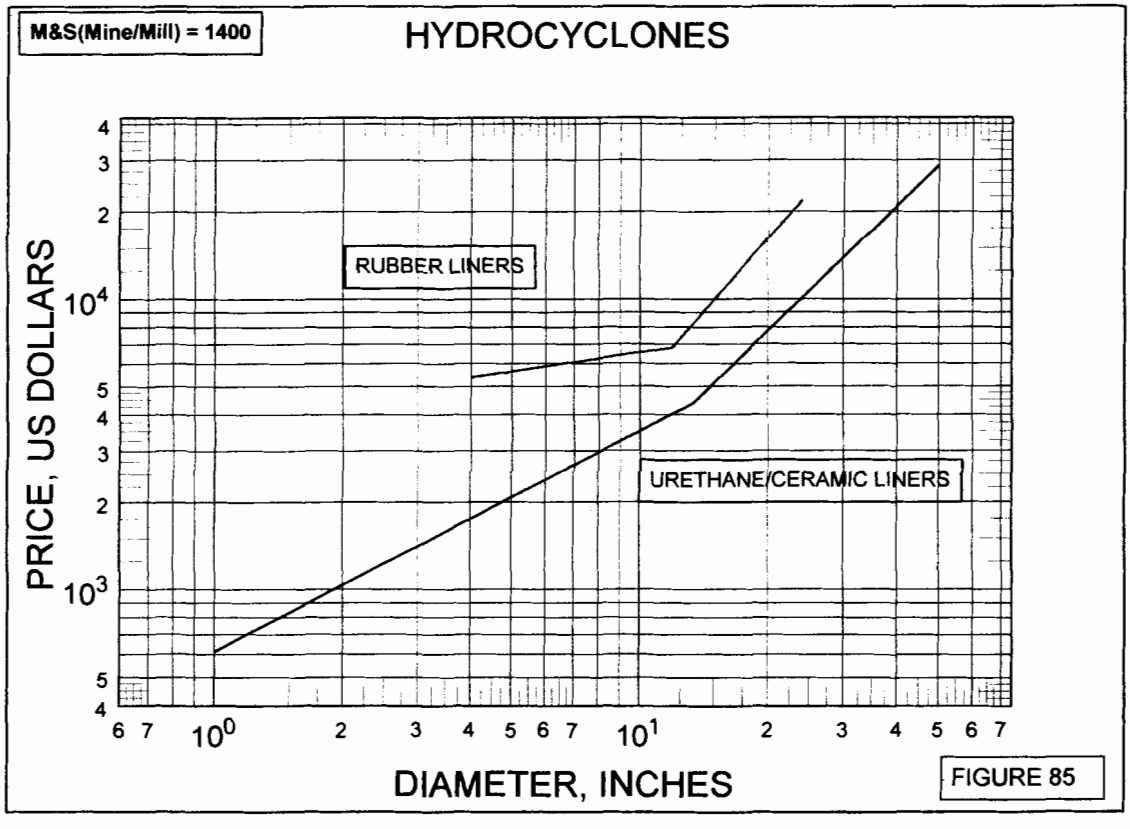
FIGURE 84

HYDROCYCLONES

Price = aX^b, US Dollars

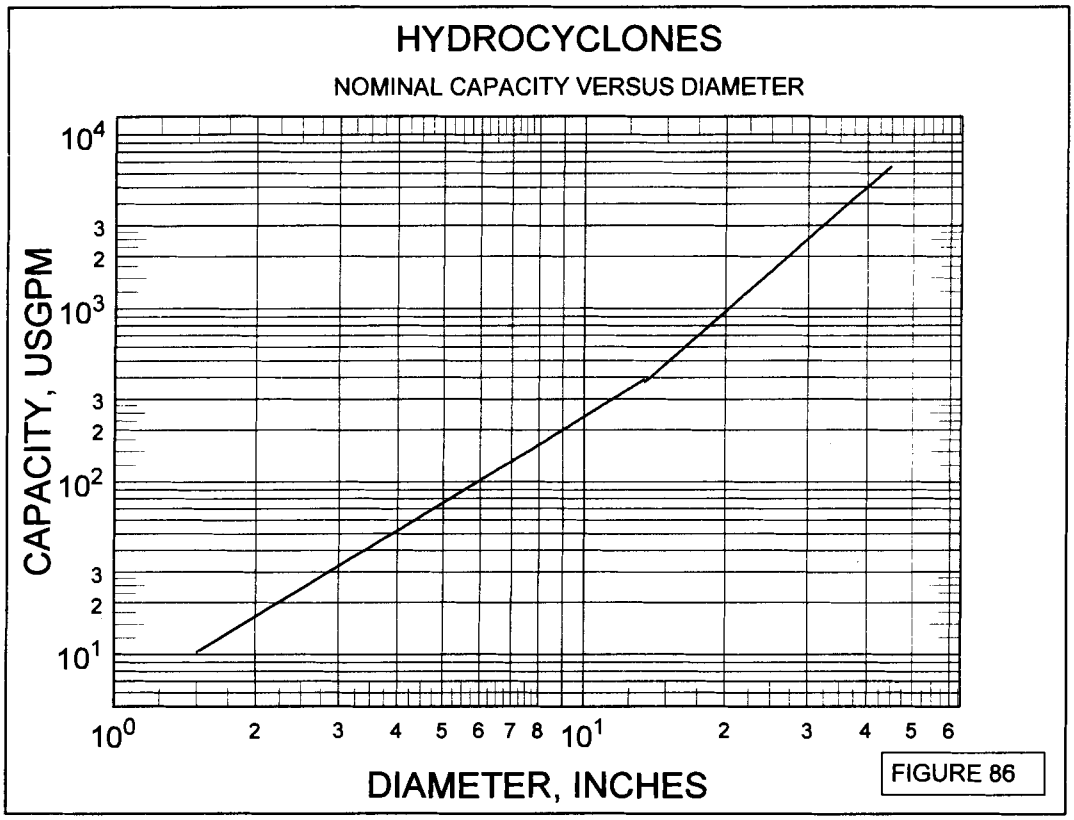
where **X is hydrocyclone diameter in inches.** Cast iron/steel construction. Cost includes fittings and liners.

Urethane/Ceramic Liners, Range in X, inches:	1 to 13.46	a = 609.5	b = 0.7582
	13.46 to 50	a = 106.3	b = 1.430
Rubber Liners, Range in X, inches:	4 to 12	a = 3969	b = 0.2152
	12 to 24	a = 103.5	b = 1.684



HYDROCYCLONES (CONTINUED)

Figure 86 shows the variation of nominal capacity versus hydrocyclone diameter. Capacities are approximate and depend upon geometric variables such as diameter, inlet pressure and characteristics of feed slurries.



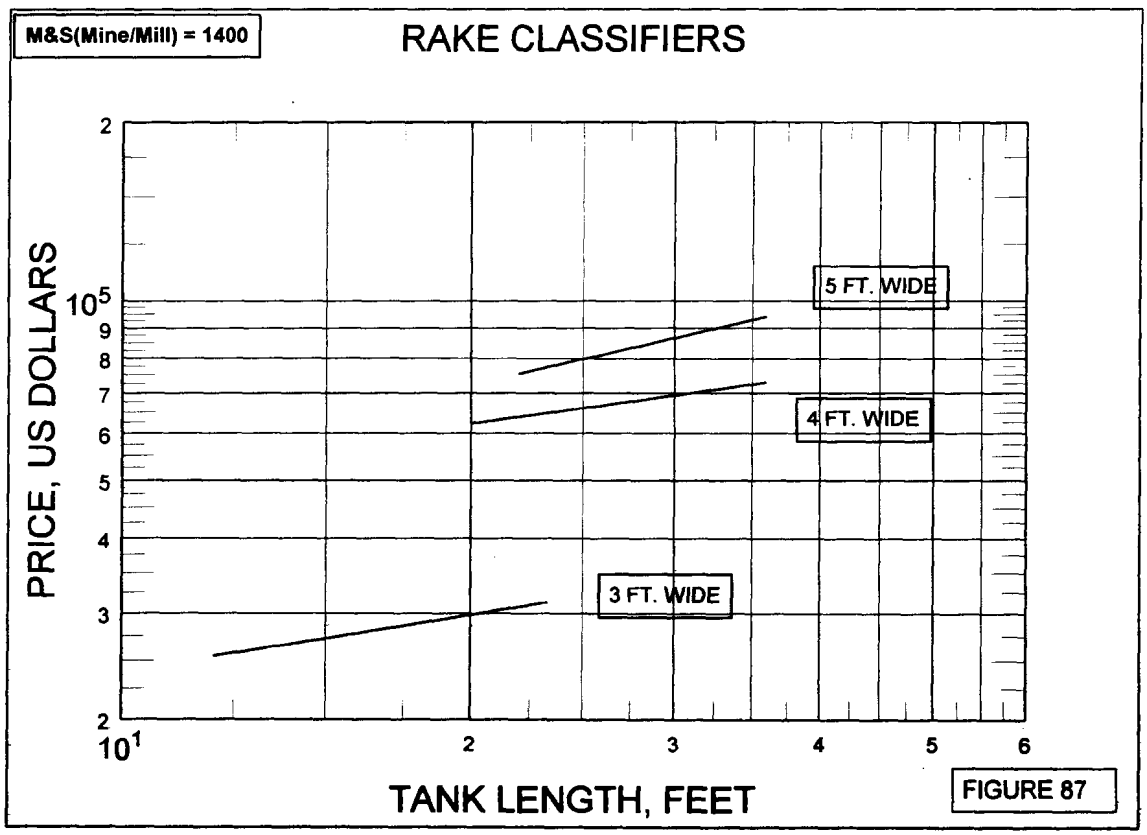
RAKE CLASSIFIERS

Price = aX^b, US Dollars

where **X is length of tank in feet.**

3' Width, Range in X, ft:	12 to 23.33	a = 11770	b = 0.3098
4' Width, Range in X, ft:	20 to 36	a = 27620	b = 0.2703
5' Width, Range in X, ft:	22 to 36	a = 18690	b = 0.4503

Tank Width, ft.	Overflow Capacity, tph
3	3.75 to 75
4	5 to 100
5	6.25 to 125



SPIRAL CLASSIFIERS

Price = aX^b, US Dollars

where **X is spiral diameter in inches**. Cost includes motor.

Simplex, Range in X, in.:	24 to 49.82	a = 783.7	b = 1.057
	49.82 to 84	a = 35.93	b = 1.872
Duplex, Range in X, in.:	36 to 66	a = 62.4	b = 1.893

X, inches	Length, ft.	Raking Capacity, tph	Horsepower
24	13	16 to 20	3
36	19.25	33 to 65	5
48	23.25	54 to 120	7.5
54	26.25	74 to 147	10
60	27.75	75 to 175	10 to 15
66	29.75	100 to 205	10 to 15
72	32	125 to 220	15 to 20
78	35.33	160 to 300	15 to 20

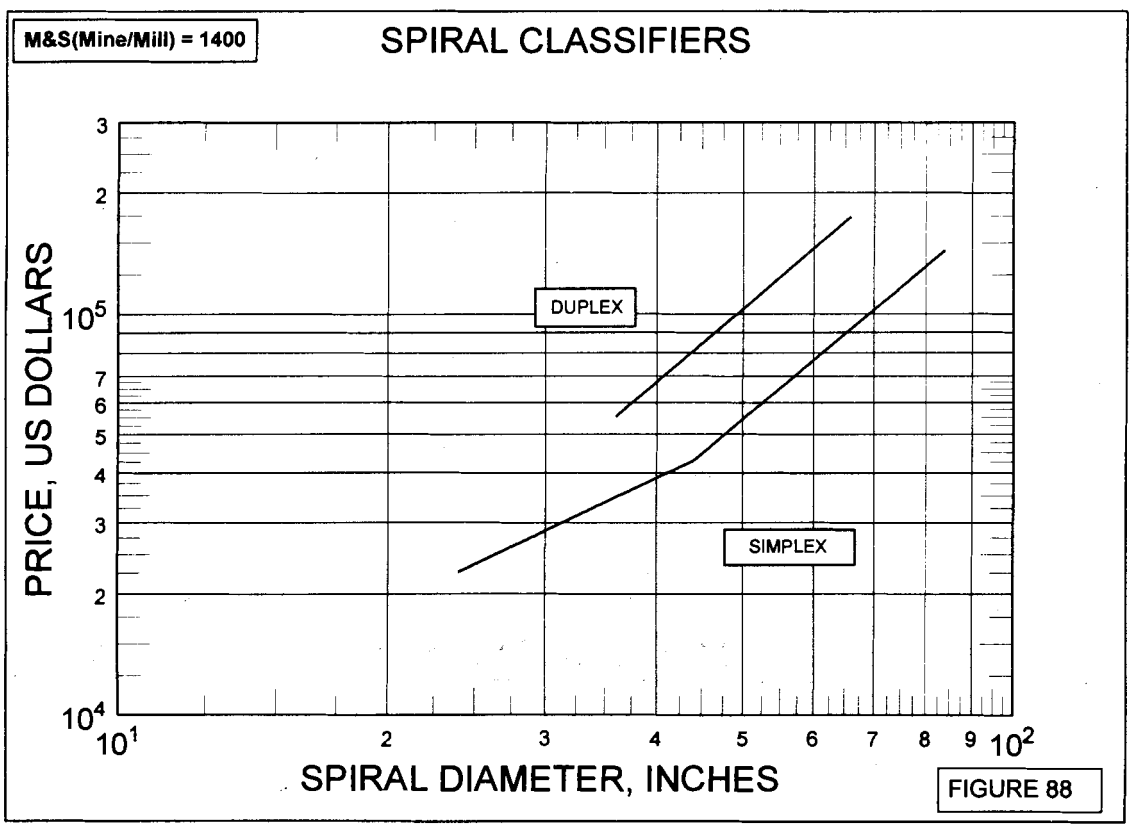


FIGURE 88

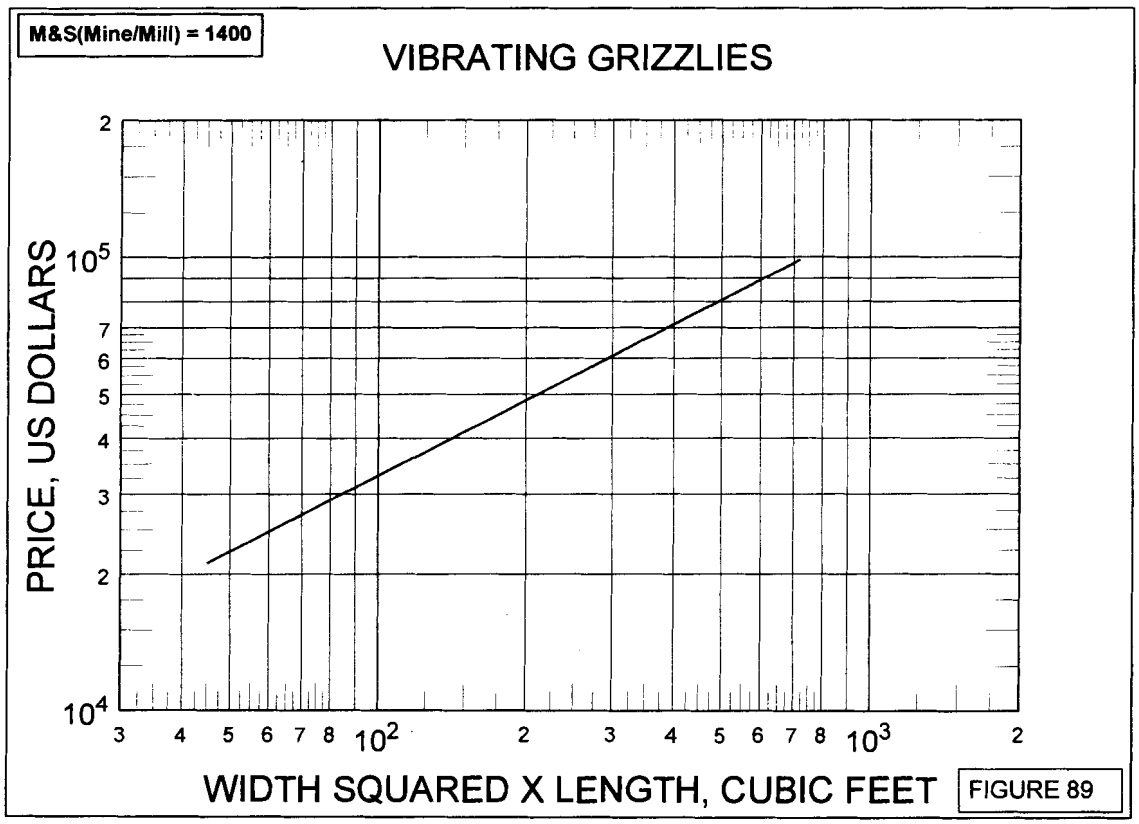
VIBRATING GRIZZLIES

Price = aX^b, US Dollars

where **X, in ft^3, is width squared times length**. Price includes drive and feed box; excludes motor.

Range in X, ft^3: 45 to 725 a = 2543 b = 0.5557

Typical Width, in.	Typical Length, in.
24	36
36	60
42	120

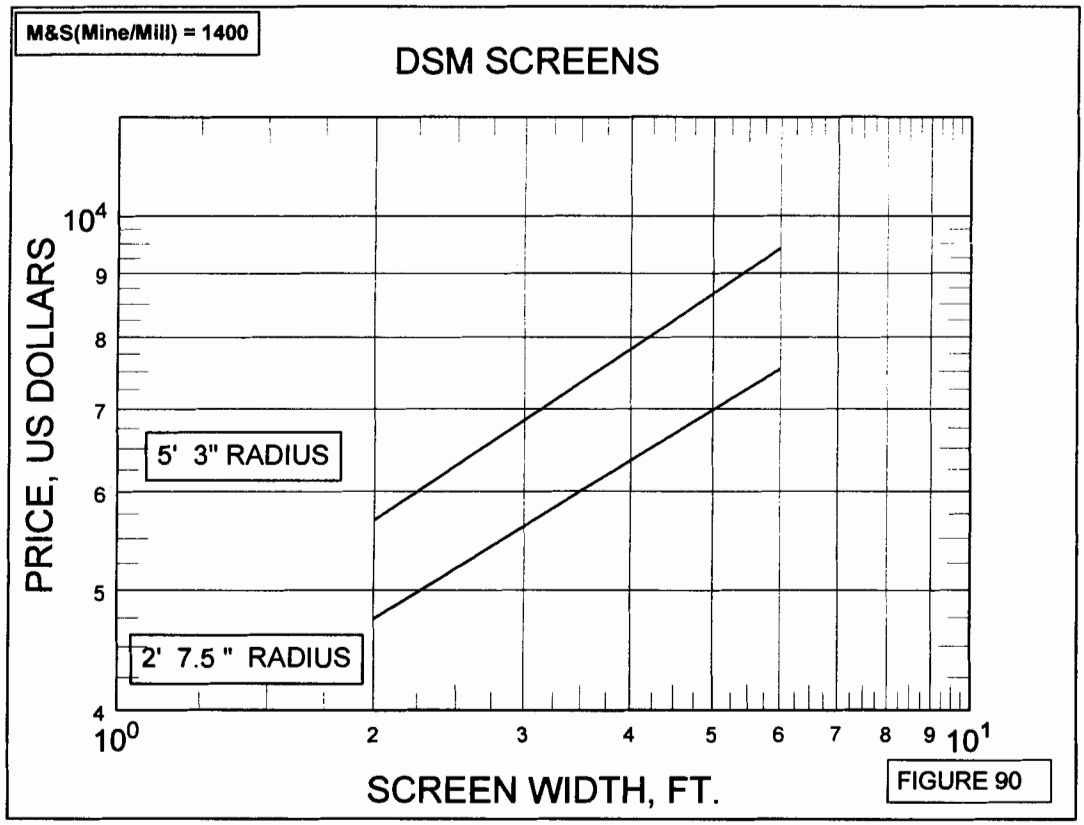


DSM SCREENS

Price = aX^b, US Dollars

where **X is screen width in feet**. Cost includes feed box with standard design for coal; excludes rapping or vibration mechanism. Estimation covers reversible screens at 60 degrees.

5' 3" Radius, Range in X, ft.: 2 to 6	a = 4133	b = 0.4595
2' 7.5" Radius, Range in X, ft.: 2 to 6	a = 3535	b = 0.4224

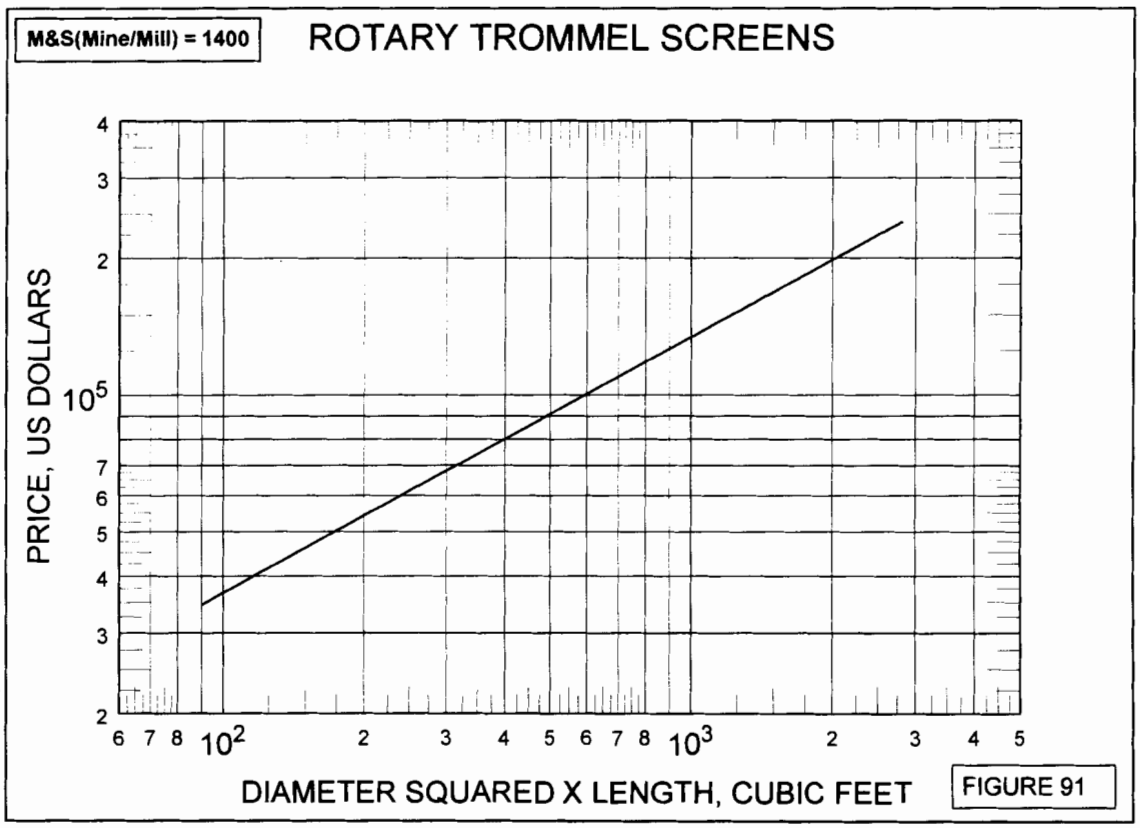


ROTARY TROMMEL SCREENS

Price = aX^b, US Dollars

where **X, in ft^3, is screen diameter squared times length**. Cost includes drive and feed box; excludes motor and starter.

Range in X, ft^3: 90 to 2800 a = 2743 b = 0.5627



INCLINED VIBRATING SCREENS
(Woven Wire Deck)

Price = aX^b, US Dollars

where **X, in ft^3, is screen width squared times length**. Cost includes drive and feed box; excludes screen cloth, motor and starter.

Single Deck, Range in X, ft^3:	125 to 1250	a = 1041	b = 0.5877
Double Deck, Range in X, ft^3:	180 to 1600	a = 2033	b = 0.5224
Triple Deck, Range in X, ft^3:	125 to 1600	a = 3757	b = 0.4237

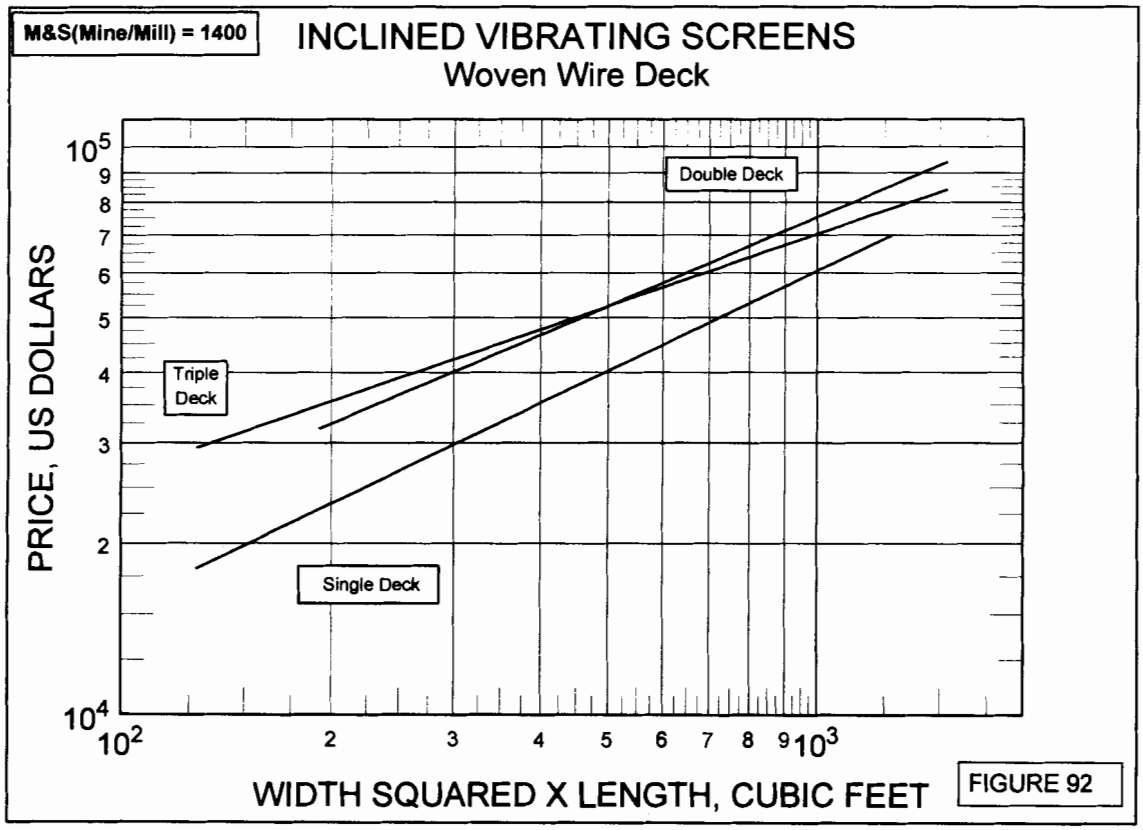


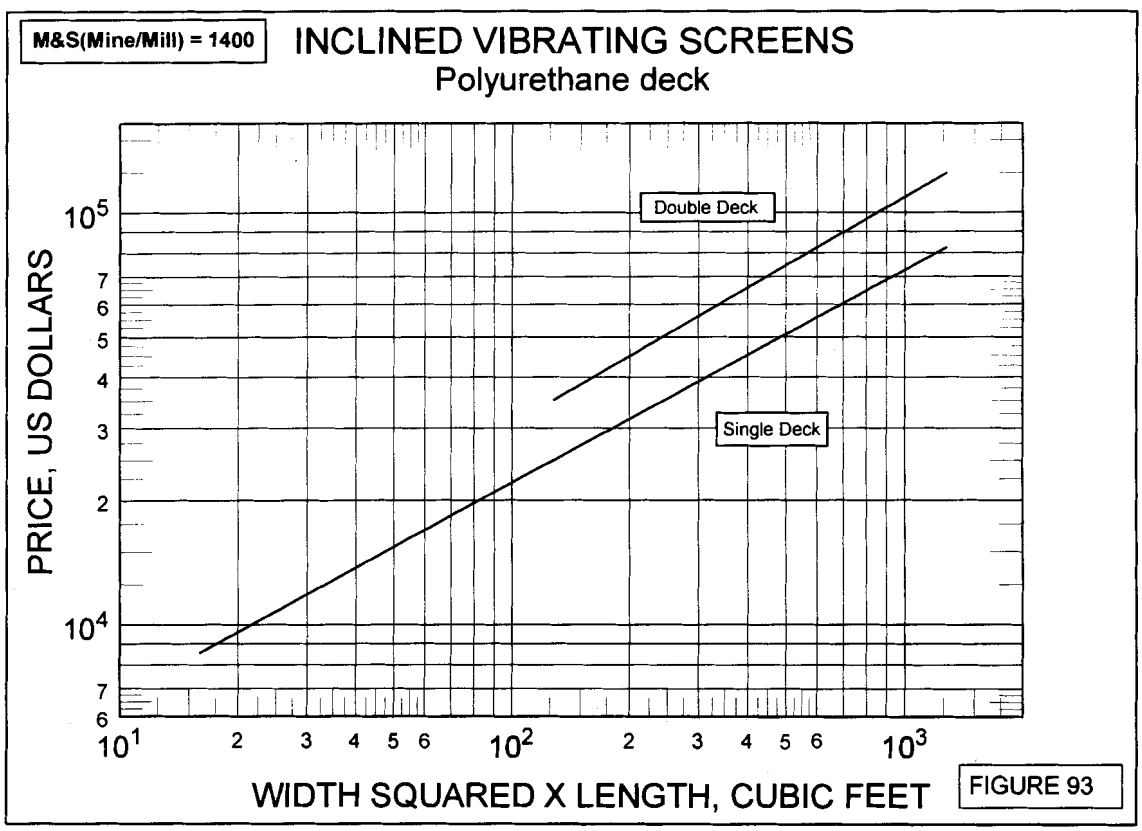
FIGURE 92

INCLINED VIBRATING SCREENS
(Polyurethane Deck)

Price = aX^b, US Dollars

where **X, in ft^3, is screen width squared times length**. Cost includes drive and feed box; excludes screen cloth, motor and starter.

Single Deck, Range in X, ft^3:	16 to 1250	a = 2033	b = 0.5172
Double Deck, Range in X, ft^3:	125 to 1250	a = 2413	b = 0.5514

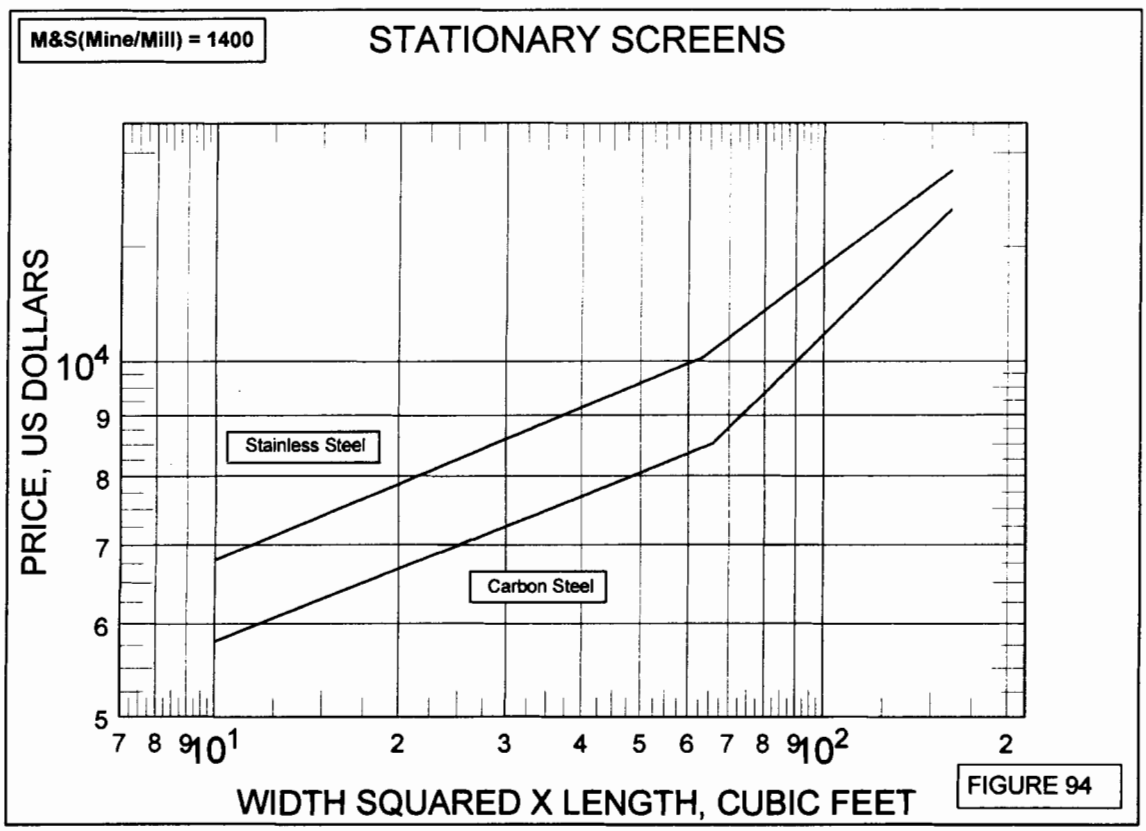


STATIONARY SCREENS

Price = aX^b, US Dollars

where **X, in ft^3, is screen width squared times length**. Cost includes standard accessories and screen cloth

Carbon Steel, Range in X, ft^3:	10 to 66	a = 3626	b = 0.2035
	66 to 1665	a = 1014	b = 0.5079
Stainless Steel, Range in X, ft^3:	10 to 62	a = 4151	b = 0.2135
	62 to 1665	a = 2021	b = 0.3870



ESTIMATED SCREEN DECK PRICES

(Expressed in US dollars per square foot)

Deck Type	Cost, US$ per Sq. Ft.
Rotary Twin Trommel (urethane)	204.60
Flat Deck Screens:	
Modular (urethane or rubber)	82.80
hooked (urethane)	77.90
hooked (rubber)	68.20
Wire Mesh (10 mesh to 4 inch):	
heavy duty	18.10
medium duty	13.80
light duty	10.90
Profile Rod or Wedge Wire,	
Standard, 28 mesh to 1/4 inch	69.60
Polyurethane, Standard,	
10 mesh to 1 inch	77.90
Performated Plate, Standard:	
10 mesh to 3/4 inch	29.00
1 inch to 4 inch	32.50

Prices based on 4 foot by 8 foot sheet of screen decking. Cost includes standard attachments. Actual prices vary with thickness (or gauge), percent open area and duty.

Prices at M&S(M/M) Index of 1400.

BANANA SCREENS

Price = aX^b, US Dollars

where **X is screen area in square feet**. Price includes standard drive, Devcon coated cleat supports, huck bolts, feed box/discharge box with AR liners, rubber lined cross members and drip angles. Motors and screen decking not included. To estimate screen deck prices: Polyurethane is US$103 per sq ft; Rubber is US$109 per sq ft; Wire is US$26 per sq ft at an M&S(M/M) Index of 1400.

Double Deck; Range in X, ft^2:	120 to 200	a = 719	b = 1.048
Single Deck; Range in X, ft^2:	120 to 160	a = 8044	b = 0.4907
	160 to 200	a = 305.9	b = 1.136

Typical Sizes, Width, ft. x Length, ft.: 6 x 20; 8 x 20; 10 x 20.

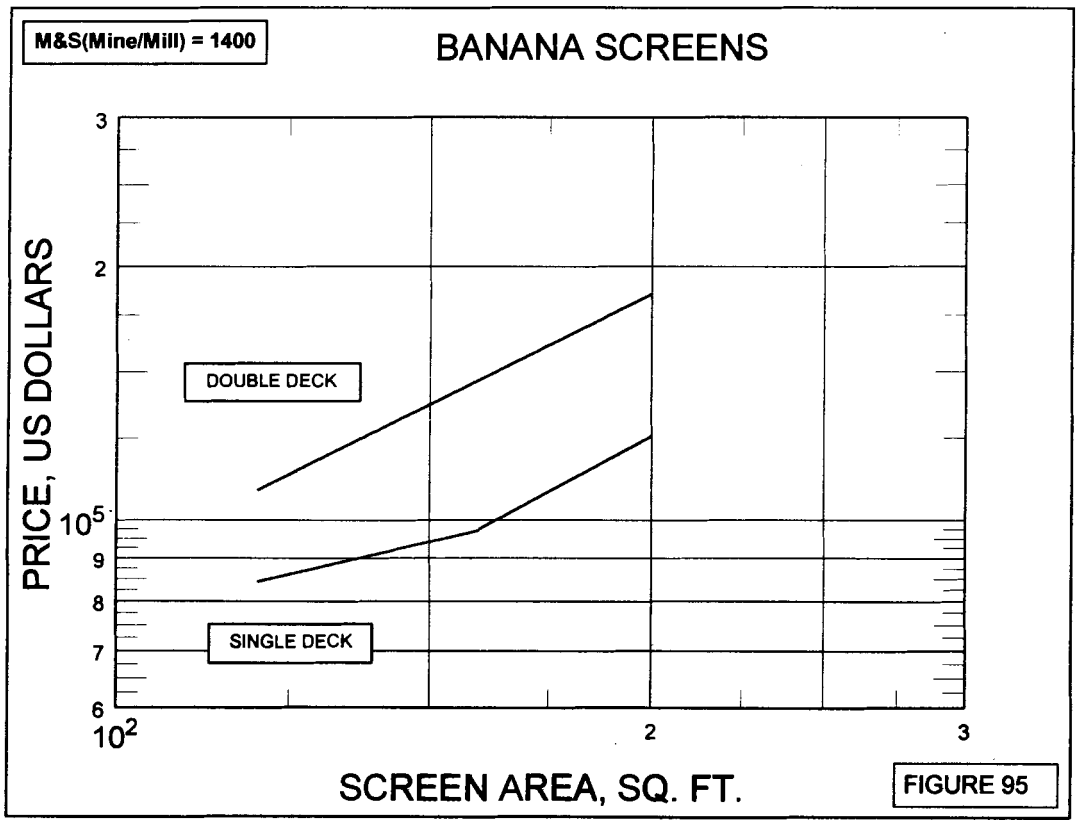


FIGURE 95

VIII. Storage, Handling, Motors and Pumping Equipment

Storage, Handling, Motors and Pumps have been placed in the following sub-categories: **Ore Storage**, **Conveyors (Conveying)**, **Feeders**, **Motors**, **Heating Units**, **Blowers** and **Pumps**.

(1) Example Estimates - - - - - - - Page 168

(2) Ore Storage

(a) Bins (Hoppers) - - - - - - - Page 170
(b) Boom Stacker-Reclaimer - - - - - Page 171

(3) Conveyors, Conveying

(a) Rectangular Suspension Magnets - - - - Page 172
(b) Belt Conveyors - - - - - - - Page 173
(c) Pneumatic Conveyors - - - - - - Page 175
(d) Screw Conveyors - - - - - - Page 176
(e) Conveyor Trippers - - - - - - Page 177
(f) Vibrating Conveyors - - - - - - Page 178
(g) Overhead Cranes - - - - - - Page 180
(h) Bucket Elevators - - - - - - Page 181

(4) Feeders

(a) Ore
(i) Apron Feeders - - - - - - Page 182
(ii) Grizzly Feeders - - - - - Page 183
(iii) Vibrating Feeders - - - - - Page 184
(b) Reagents
(i) Liquid Feeder, 4-10 cup - - - - Page 185
(ii) Liquid Feeder, 20 cup - - - - - Page 186
(iii) Liquid Feeder, Hi Capacity - - - - Page 187

(5) Motors

(a) Motor Starters - - - - - - - Page 188
(b) Voltage Rectifiers - - - - - - Page 189
(c) DC Motors - - - - - - - Page 190
(d) Drip Proof Motors (575V) - - - - - Page 191
(e) Drip Proof Motors (2300 V) - - - - - Page 192
(f) Drip Proof Motors (4000V) - - - - - Page 193
(g) Explosion Proof Motors (575V) - - - - Page 194
(h) TEFC Motors (575V) - - - - - - Page 195
(i) TEFC Motors (2300V) - - - - - - Page 196
(j) TEFC Motors (4000V) - - - - - - Page 197
(k) Synchronous Motors - - - - - - Page 198

VIII. Storage, Handling, Motors and Pumps (Continued)

(6) Heating Units

(a) Fire Tube Boilers - - - - - - Page 199
(b) Shell / Tube Heat Exchangers - - - - Page 200

(7) Blowers

(a) Axial Flow - - - - - - - Page 201
(b) Centrifugal Fan - - - - - - - Page 202

(8) Pumps

(a) Multistage, Centrifugal - - - - - - Page 203
(b) Centrifugal, Stainless Steel - - - - - Page 204
(c) Vertical Centrifugal, Sump - - - - - Page 205
(d) Proportioning Diaphragm - - - - - Page 206
(e) Gear Pumps (Oil) - - - - - - Page 207
(f) Proportioning Meter - - - - - - Page 208
(g) Air Diaphragm, Reciprocating - - - - - Page 209
(h) Piston Diaphragm - - - - - - Page 210
(i) Reciprocating Piston - - - - - - Page 211
(j) Single Stage Centrifugal, Open Impeller - - - Page 212
(k) Single Stage Centrifugal, Closed Impeller - - - Page 213
(l) Vacuum, Water Seal - - - - - - Page 214
(m) Vacuum, Dry Seal - - - - - - Page 215

EXAMPLE ESTIMATES

In this section, the various process units have been related by means of graphs, equations and tables to size parameters or horsepower or capacity. In most cases, capacity/size/horsepower correlations are included.

Example 1:

A blower (centrifugal fan type) is needed for six flotation cells, each cell of 400 ft^3 capacity. What capacity blower is required and how much will it cost?

The maximum discharge need not exceed a pressure of 35-inch water gauge. Assuming 1.5 cubic feet per minute of blower capacity is required for each cubic foot of flotation cell capacity, the blower should have a capacity of 600 cfm/cell. Hence 6 x 600 = 3600 cfm are required.

From Figure 128, page 202, the cost of a blower of this capacity is about US$4800 at an M&S(M/M) Index of 1400.

Example 2:
A bucket elevator is needed to transport 120 tons per hour of a material of density 75 lb/ft^3 a distance of 80 feet. What size bucket is needed and what will the elevator cost?

The capacity to be moved in cubic feet per hour is equal to 120 x 2000/75 = 3200 cubic feet per hour.

From the table shown above Figure 107, page 181, the size of bucket required is 18 x 8 inches. The cost of this elevator is approximately US$46500 at an M&S(M/M) Index of 1400.

Example 3:

A liquid feeder for a 10 percent solution of ethyl xanthate is required for a flotation circuit. A 20 cup size made of 316 stainless steel is judged necessary.

The amount of reagent to be added is 2000 cc/min. What is the estimated price of the feeder?

From the table on page 186 above Figure 112, sizes 1 to 4 are appropriate. Figure 112 suggests that size one is the choice, since it has a lower cost of about US$1900 at an M&S(M/M) Index of 1400.

Example 4:

A 240 rpm, 5000 HP synchronous motor is required for a 16 1/2 ft by 28 ft ball mill. What is the estimated price of the motor?

EXAMPLE ESTIMATES (CONTINUED)

Figure 124, page 198, shows that the price of the motor is approximately US$740000 at an M&S(M/M) Index of 1400. Remember to convert from an index of 1400 to the current index. Thus, if the current value is 1100, then the estimated price becomes approximately:

$$\left[\frac{1100}{1400}\right]740000 = \text{US\$ } 581{,}400.$$

Example 5:

A horizontal SRL (soft rubber lined) closed impeller slurry pump is required to transport 7000 USGPM of slurry in a grinding circuit. Estimate the price of the pump.

Figure 139, page 213, shows that the estimated cost is about US$63000 at an M&S(M/M) Index of 1400.

Example 6:

A 3000 ton per hour boom stacker-reclaimer is necessary to generate a live storage pile. Located underneath the pile in a reclaim tunnel is a feeder that will discharge to conveyor belts that transfer ore feed to SAG mills. What is the estimated cost of the boom stacker-reclaimer?

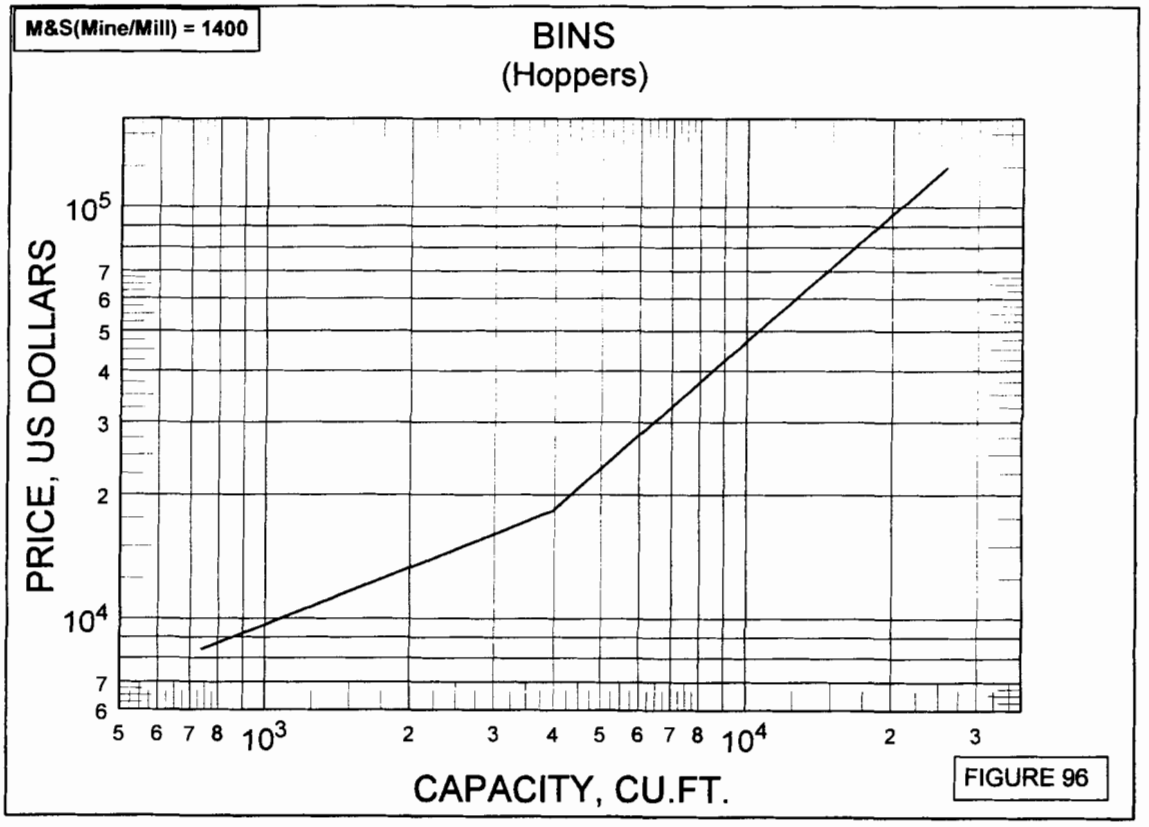
Figure 97, page 171, shows that the price is about US$4500000 at an M&S(M/M) Index of 1400. Note that a yard conveyor is not included in the price.

BINS (HOPPERS)

Price = aX^b, US Dollars

where **X is bin capacity in cubic feet**. For material weighing 80 to 100 pounds per cubic foot; either 45 or 60 degree hopper; steel.

Range in X, ft^3:	740 to 3960.7	a = 392.8	b = 0.4631
	3960.7 to 25912	a = 3.743	b = 1.025

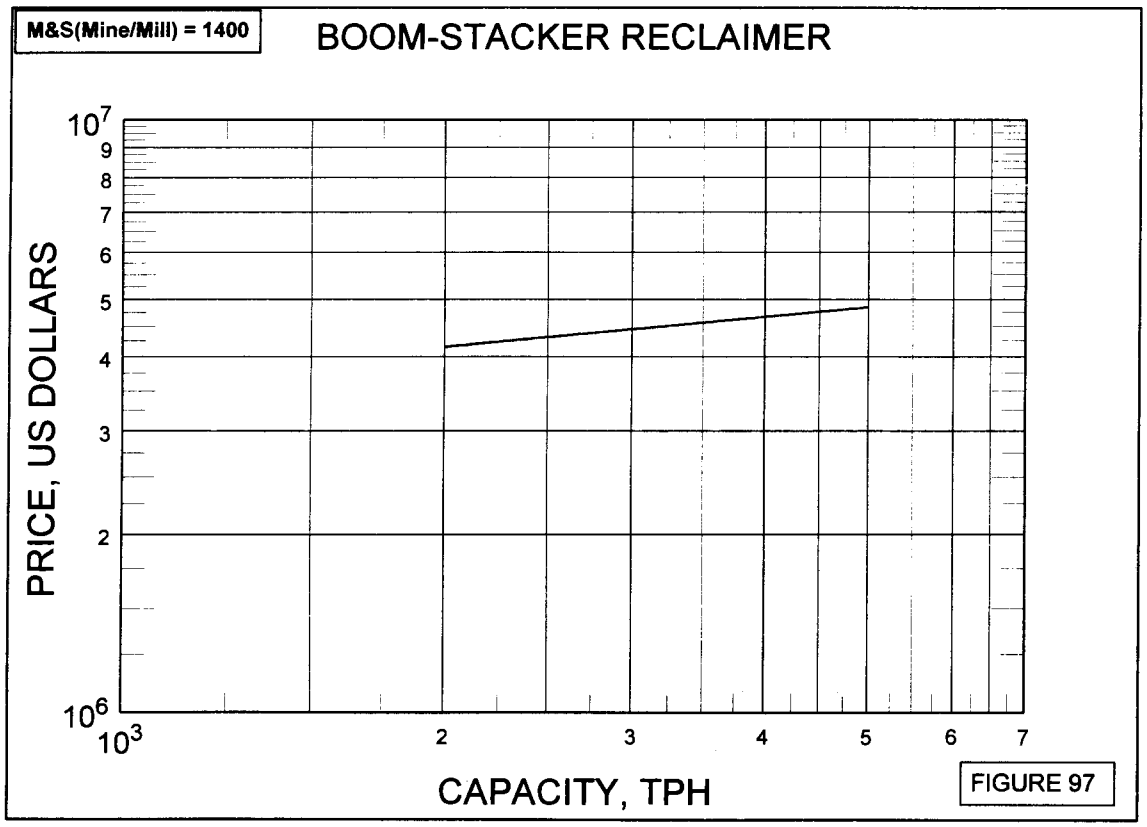


BOOM STACKER-RECLAIMER

Price = aX^b, US Dollars

where **X is capacity in tons per hour**. Reclaimer capacity about 50% less than stacker capacity. Price includes idlers, head-end drive, head frame, pulleys, gravity take-up on tail pulleys, belting and ground mounted steel structure. Estimation covers conveyors carrying material weighing 100 lb per cu. ft. with surcharge angle of 20 degrees at belt speed of 500 fpm. Yard conveyor not included in price.

Range in X, tph: 2000 to 5000 a = 1140000 b = 0.1699



RECTANGULAR SUSPENSION MAGNETS

Price = aX^b, US Dollars

where **X is the product of belt width in feet times the suspension height in cubic inches**. Electromagnets.

Range in X, ft x $in.^3$:	104.96 to 2842.6	a = 1793	b = 0.249
	2842.6 to 419840	a = 251.6	b = 0.4959

Suspension Height, inch	Belt Width Range, ft.
4	1.64 to 3.9
6	1.64 to 3.9
8	2.00 to 3.9
10	2.00 to 4.6
12	3.00 to 4.9
14	3.40 to 5.4
16	3.40 to 5.9
20	4.60 to 6.56
32	5.90 to 6.56
40	5.90 to 6.56

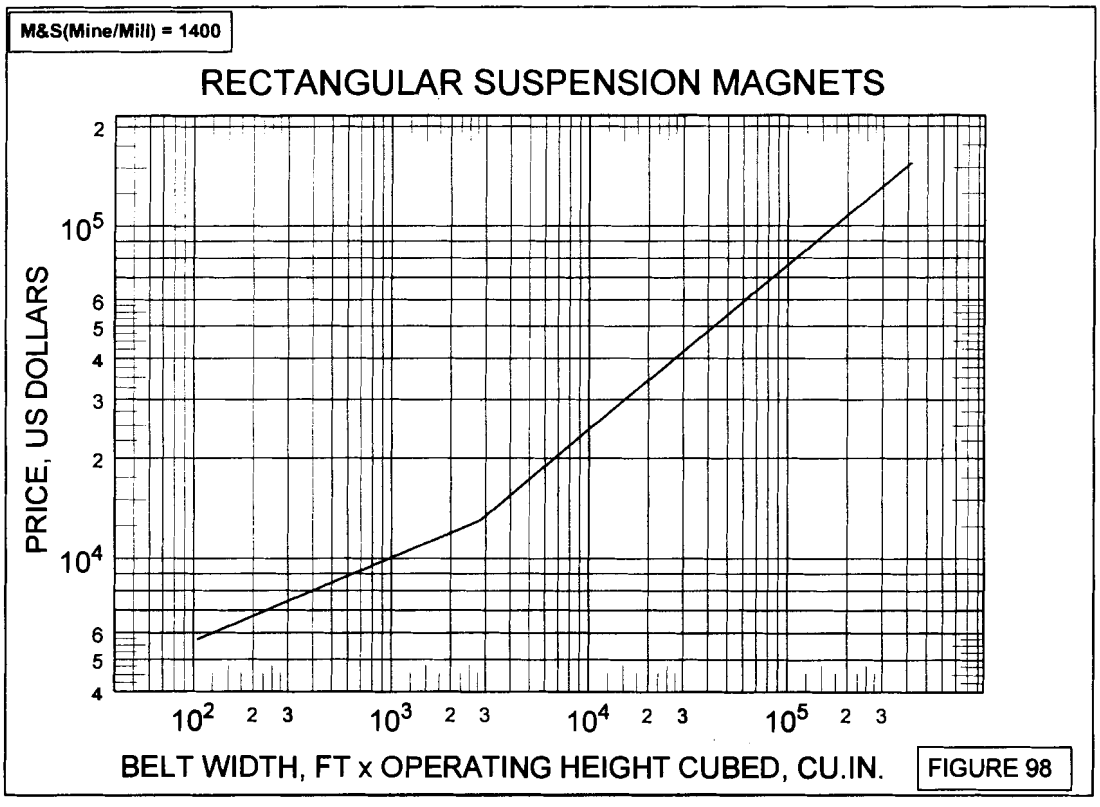


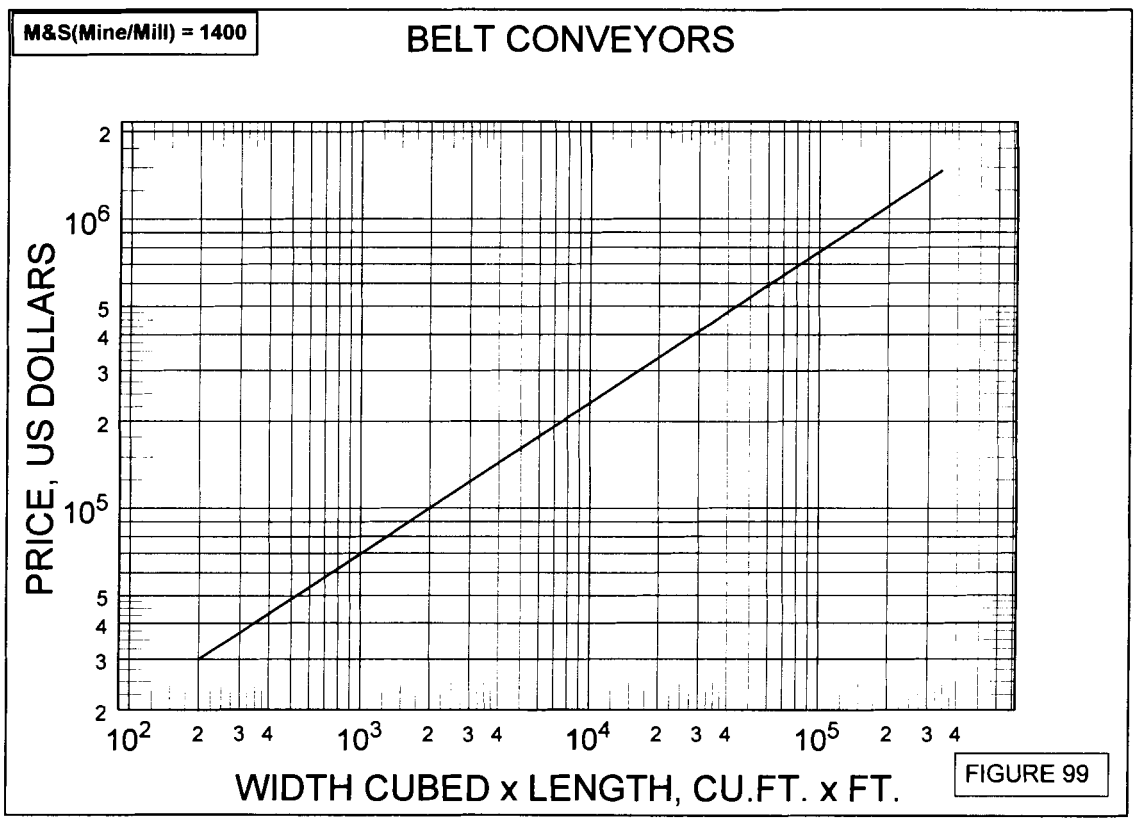
FIGURE 98

BELT CONVEYORS

Price = aX^b, US Dollars

where **X is belt width cubed times belt length (cubic feet times feet)**. Price includes idlers, head-end drive, head frame, pulleys, gravity take-up on tail pulleys, belting and ground mounted steel structure. Conveyor carries material weighing 100 lb per cu ft at a surcharge angle of 20 degrees at a belt speed of 500 cfm.

Range in X, ft^3xft: 200 to 343000 a = 1875 b = 0.5225



BELT CONVEYORS (CONTINUED)

Figure 100 below shows how nominal capacity of material of bulk density 100 lb/ft^3 varies with belt width at 500 feet per minute (fpm).

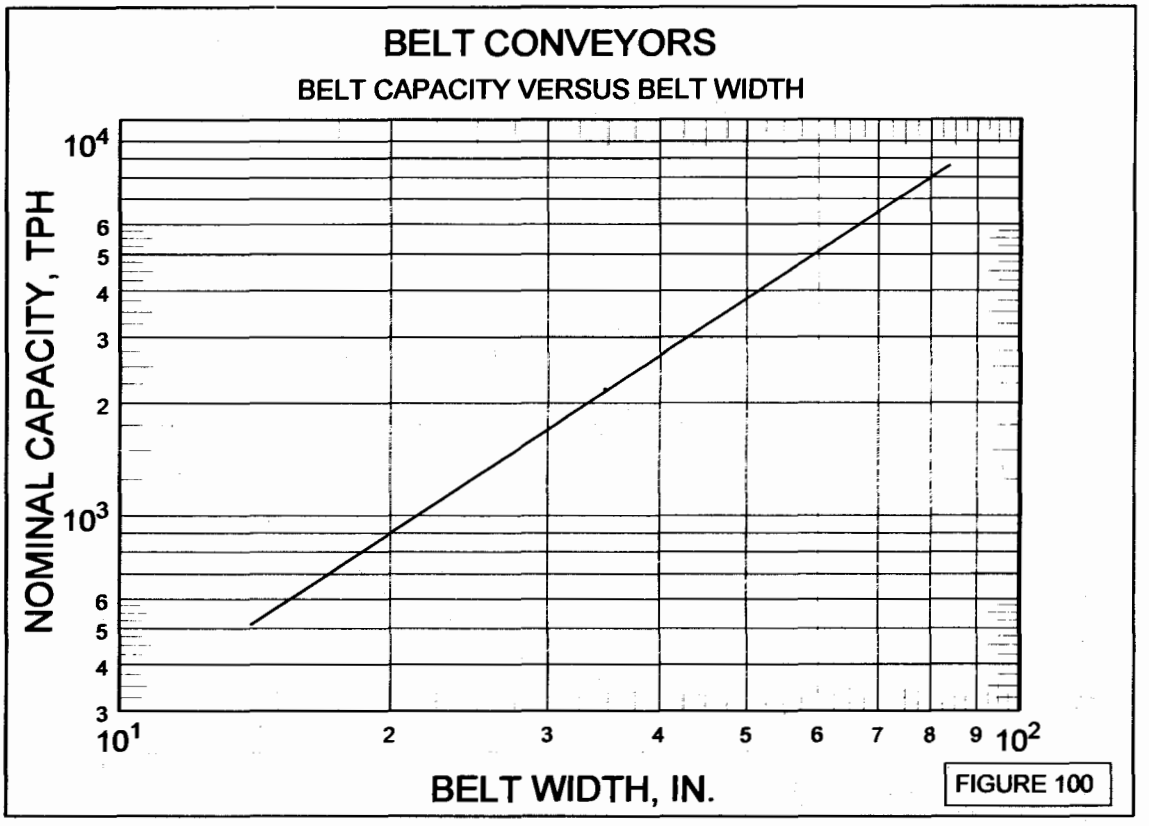


FIGURE 100

PNEUMATIC CONVEYORS

Price = aX^b, US Dollars

where **X is diameter squared times length, cubic feet**. Costs are for ranges of continuous vacuum, vacuum/pressure and pressure systems. Systems are custom designed with a wide variety of piping layouts, motors, blowers, control systems, diverter and feed valves and receivers.

Stainless Steel, Range in X, ft^3:	6.25 to 62.22	a = 69910	b = 0.1549
	62.22 to 444.4	a = 25280	b = 0.4012
Carbon Steel, Range in X, ft^3:	6.25 to 72.66	a = 31850	b = 0.1506
	72.66 to 444.4	a = 12400	b = 0.3708

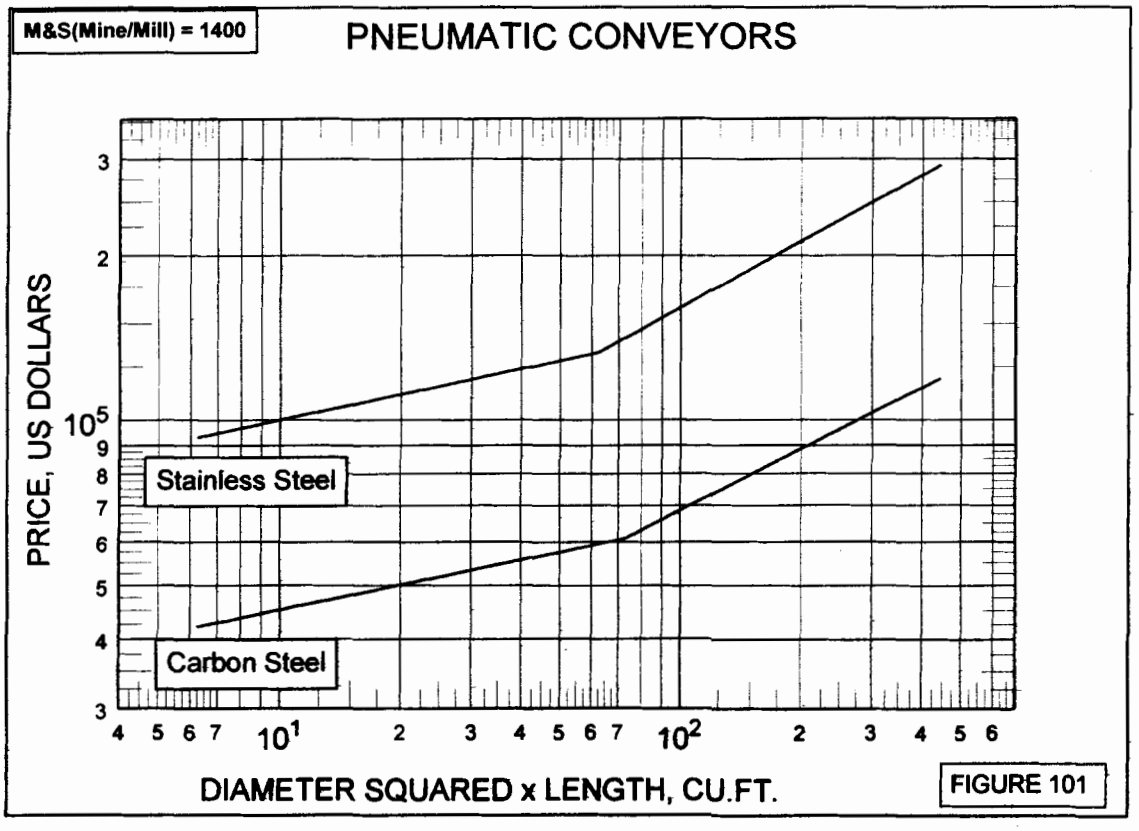


FIGURE 101

SCREW CONVEYORS

Price = aX^b, US Dollars

where **X is screw diameter in ft. times length in ft.**

Range in X, ft^2:		a =	b =
	7.5 to 43.85	474.4	0.8091
	43.85 to 1333.33	148.2	1.117

Typical Diameter, ft.	Typical Length Range, ft.
0.75	10 to 40
1.00	12 to 48
1.33	12 to 48

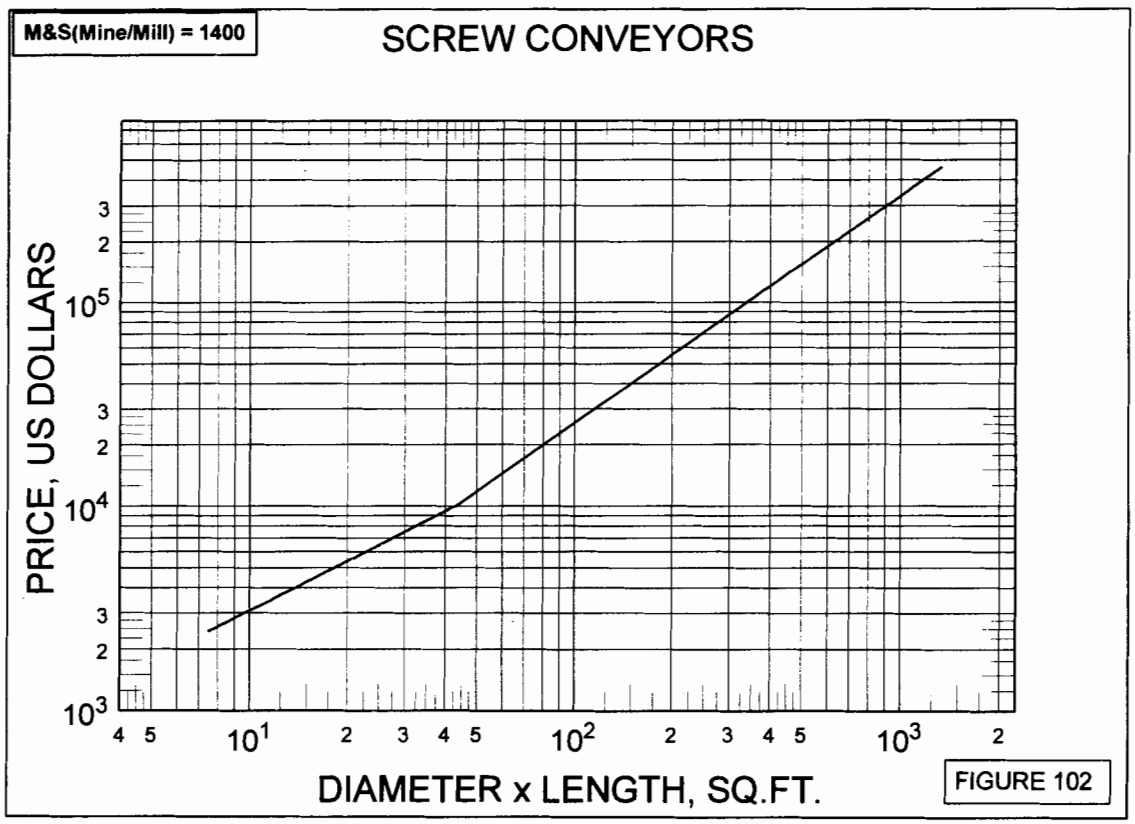


FIGURE 102

CONVEYOR TRIPPERS

Price = aX^b, US Dollars

where **X is belt size, inches**. Manually operated; 3-way discharge.

Range in X, in.:			
30 to 35.63	a = 594.9	b = 1.255
35.63 to 60	a = 5005	b = 0.6591

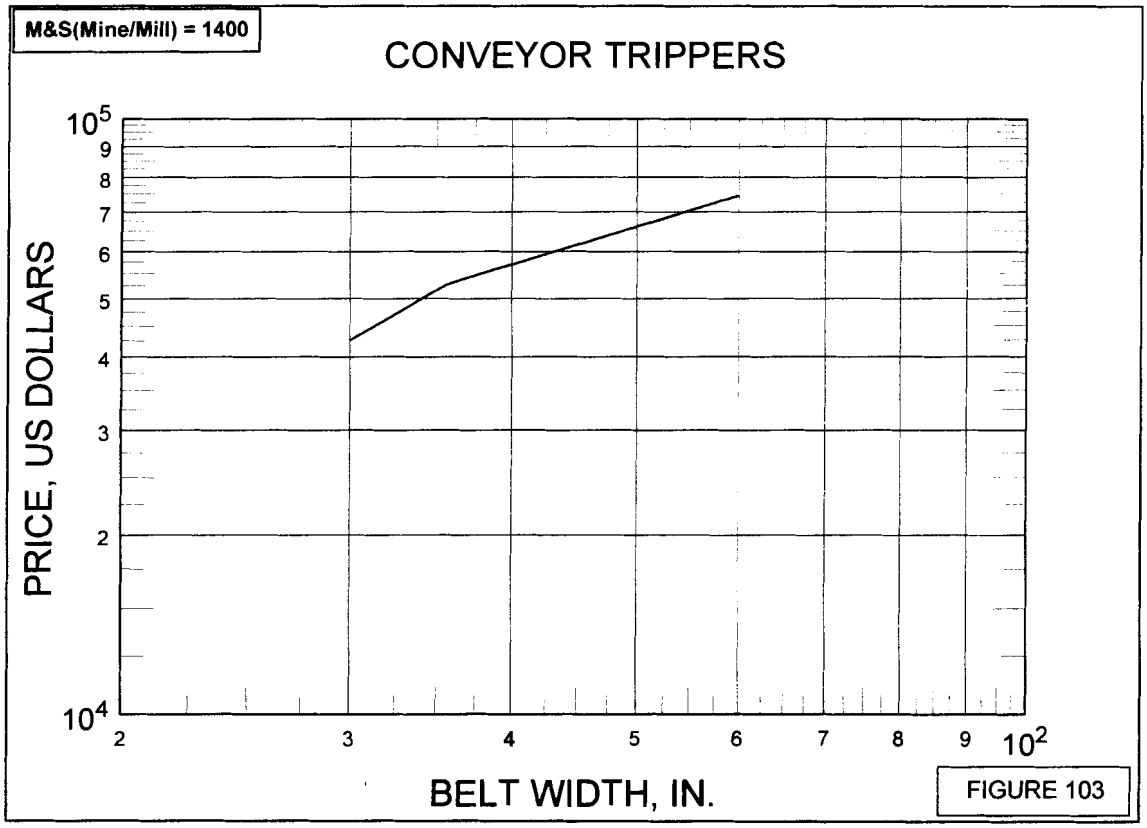


FIGURE 103

VIBRATING CONVEYORS

Price = aX^b, US Dollars

where **X is length times width in square feet**. Heavy duty units; cost includes drive; excludes motor. For dry solids of bulk density 100 lb/ft^3.

Range in X, ft^2: 10 to 850 a = 654.8 b = 1.00

Width, ft	Length, ft	HP
1	20	1
1	50	5
1.5	20	1.5
1.5	50	5
2	20	1.5
2	50	5
3	20	2
3	50	7.5

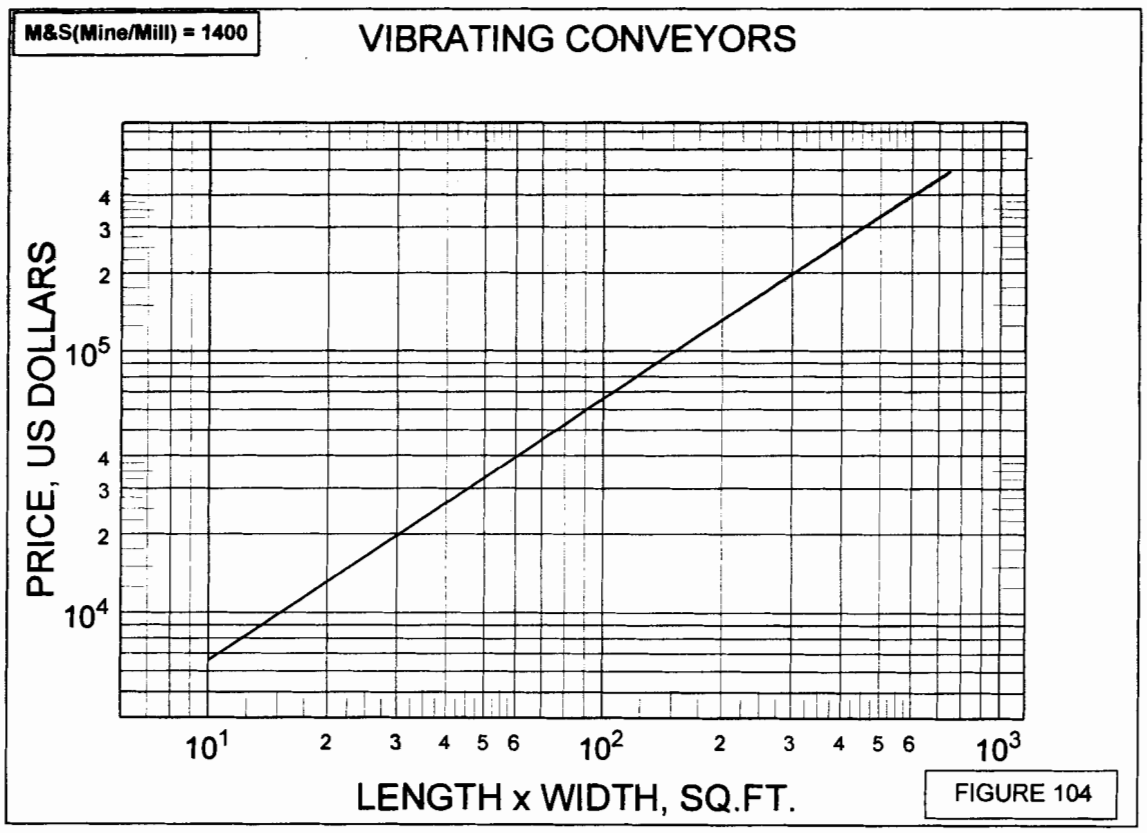


FIGURE 104

VIBRATING CONVEYORS (CONTINUED)

Figure 105 below shows how the nominal capacity of vibrating conveyors will vary with length times width.

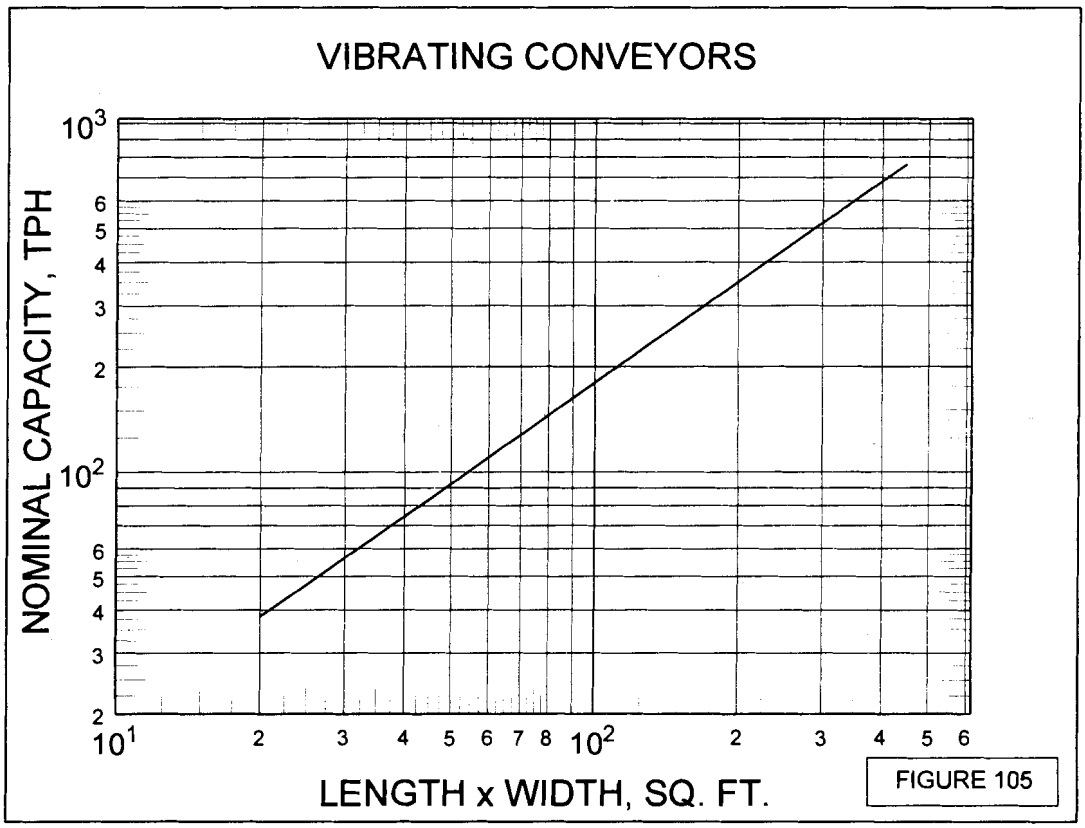


FIGURE 105

OVERHEAD CRANES

Price = aX^b, US Dollars

where **X is lift capacity in tons**. Price includes motors, required span type, cables, hooks, electrical controls; excludes tracks and crane support structures.

50 ft Span; Range in X, tons:	5 to 25	a = 31580	b = 0.4098
100 ft Span; Range in X, tons:	5 to 50	a = 70870	b = 0.3669
150 ft Span: Range in X, tons:	5 to 50	a = 240100	b = 0.1504

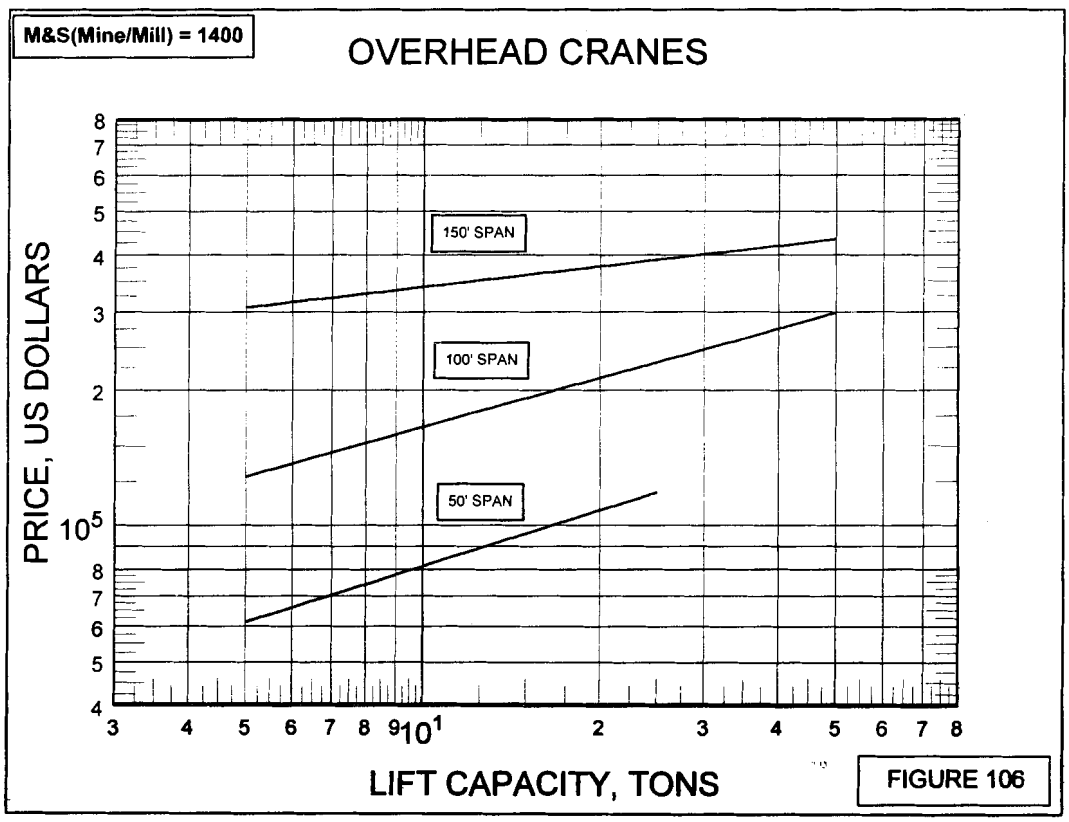


FIGURE 106

BUCKET ELEVATORS

Price = aX^b, US Dollars

where **X is bucket size in square inches**. Price includes drive, standard hood section and head section.

80 ft Centres; Range in X, sq. in.: 24 to 144 a = 5138 b = 0.4432

40 ft Centres; Range in X, sq. in.: 24 to 144 a = 2977 b = 0.469

Bucket Size, in. x in.	Capacity,cu. ft. /hr	Capacity, tph (density 100 lb/cu. ft.
6 x 4	300	15
8 x 5	600	30
10 x 6	1000	50
12 x 7	1500	75
14 x 7	1800	90
16 x 8	2400	120
18 x 8	3300	165

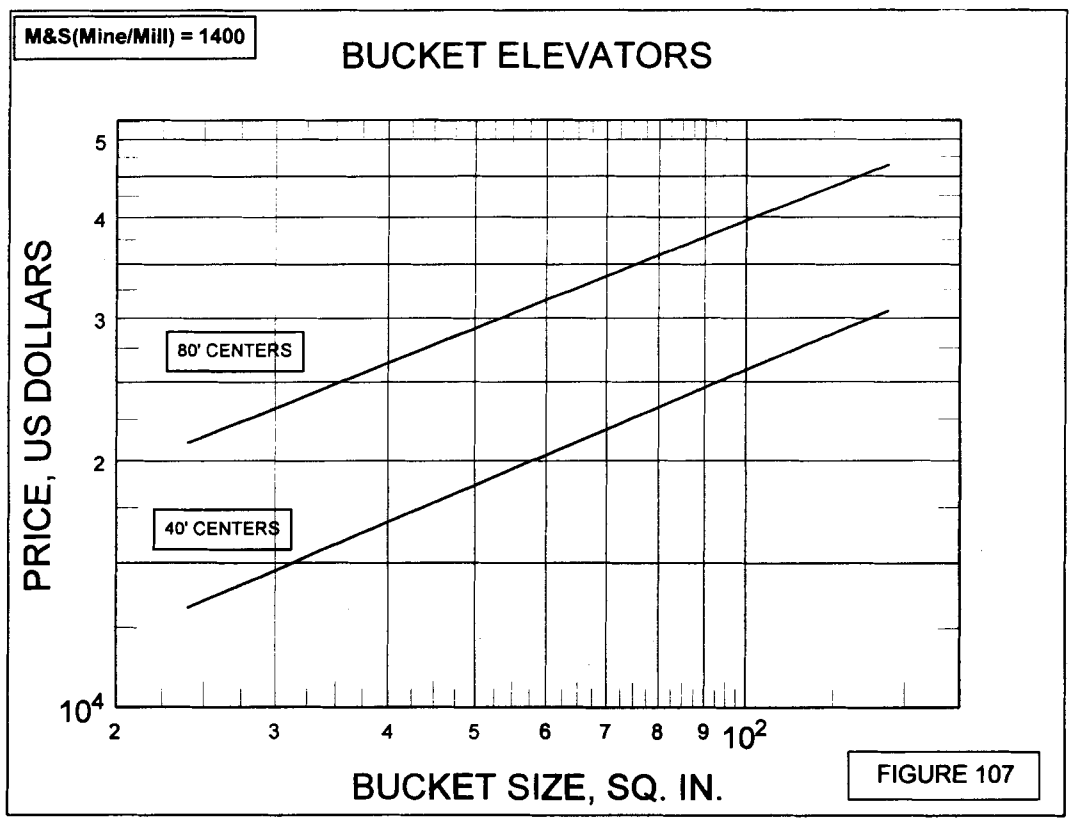


FIGURE 107

APRON FEEDERS

Price = aX^b, US Dollars

where **X is in cubic feet (width squared times length)**.

Range in X, ft^3: 32 to 1200 a = 4809 b = 0.5732

Width, in.	Length, ft.	Capacity, tph at 25 fpm
24	6 to 15	150
30	6 to 18	234
36	9 to 21	338
42	9 to 21	459
48	12 to 27	600
54	12 to 27	759
60	15 to 30	957
72	15 to 30	1350
84	18 to 30	1838

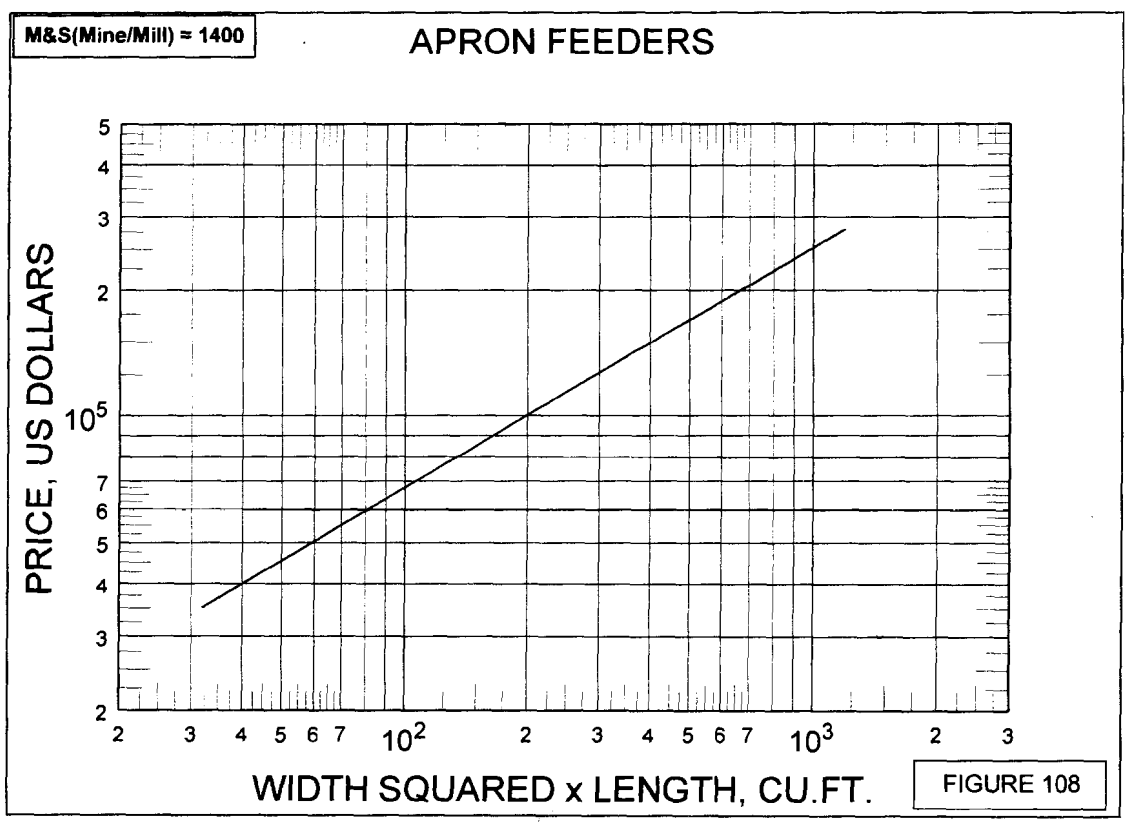


FIGURE 108

GRIZZLY FEEDERS

Price = aX^b, US Dollars

where **X is in square feet (width times length)**. Price includes feed box.

Range in X, ft^2: 3.5 to 190 a = 3352 b = 0.645

width x Length in. x in.	Magnet Size Watts
12 x 42	250
14 x 42	250
18 x 42	525
24 x 42	525
24 x 54	1000
30 x 60	1750
30 x 72	1750
36 x 72	2300
54 x 72	4600

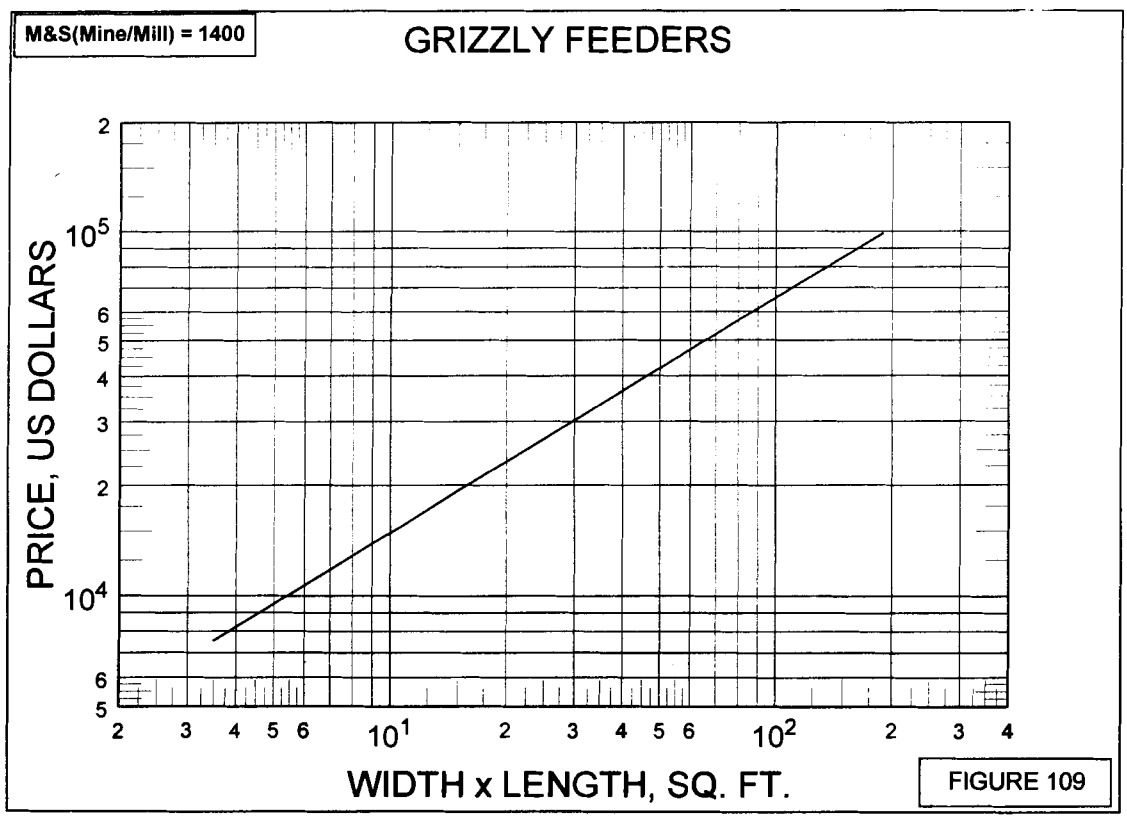


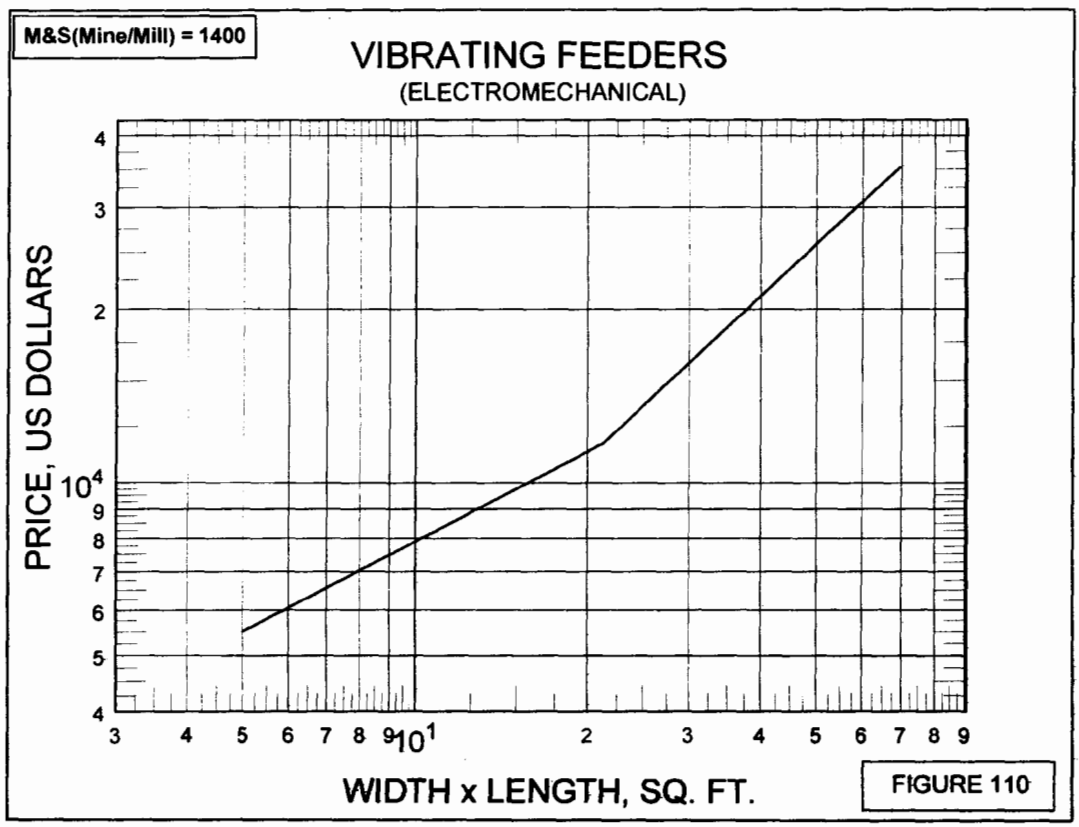
FIGURE 109

VIBRATING FEEDERS
(Electromechanical)

Price = aX^b, US Dollars

where **X is in square feet (length times width)**. Price includes drive, motor and starter.

Range in X, ft^2:			
5 to 21	a = 2378	b = 0.5203
21 to 70	a = 674.8	b = 0.9321



LIQUID FEEDER
(4 to 10 Cup)

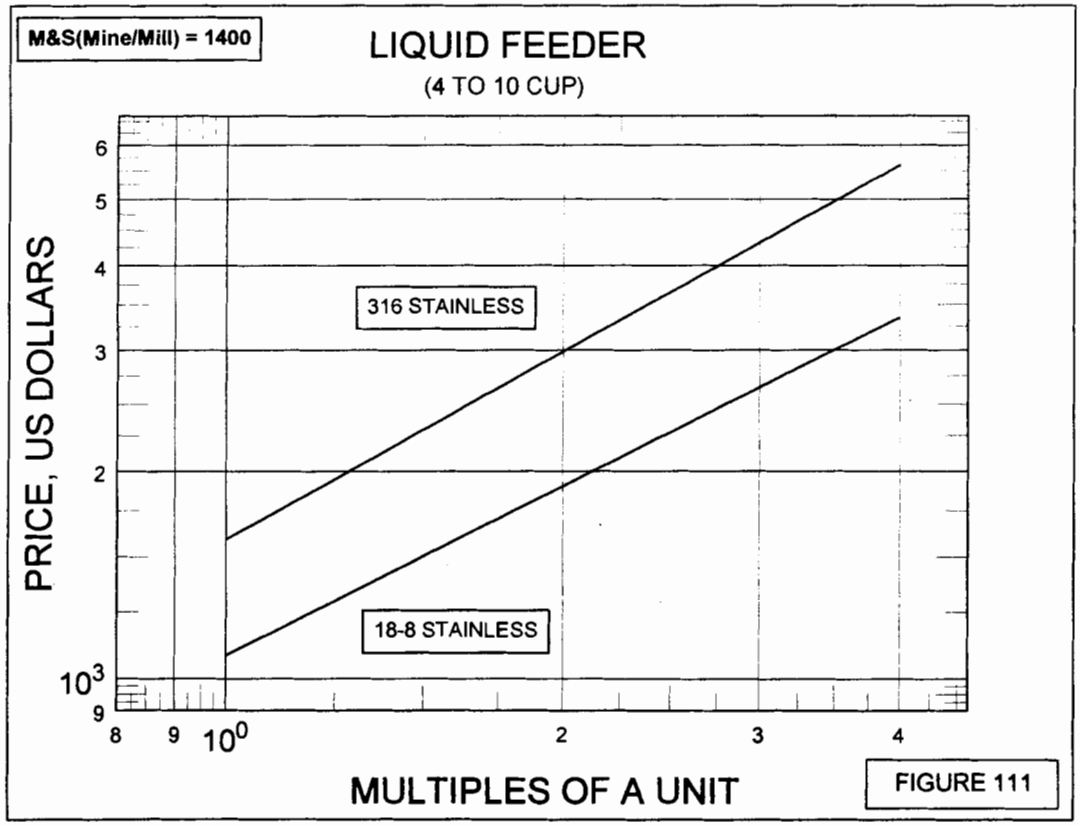
Price = aX^b, US Dollars

where **X is a multiple of a single unit (range 1 to 4)**.

18-8 Stainless; Range in X: 1 to 4 a = 1078 b = 0.8169

316 Stainless; Range in X: 1 to 4 a = 1585 b = 0.911

Unit	Capacity Range, cc/min
1	40 to 1000
2	80 to 2000
3	120 to 3000
4	160 to 4000



LIQUID FEEDER
(20 Cup)

Price = aX^b, US Dollars

where **X is a multiple of a single unit (range 1 to 4).**

18-8 Stainless; Range in X: 1 to 4 a = 1118 b = 0.82

316 Stainless; Range in X: 1 to 4 a = 1910 b = 0.91

Unit	Capacity Range, cc/min
1	200 to 2000
2	400 to 4000
3	600 to 6000
4	800 to 8000

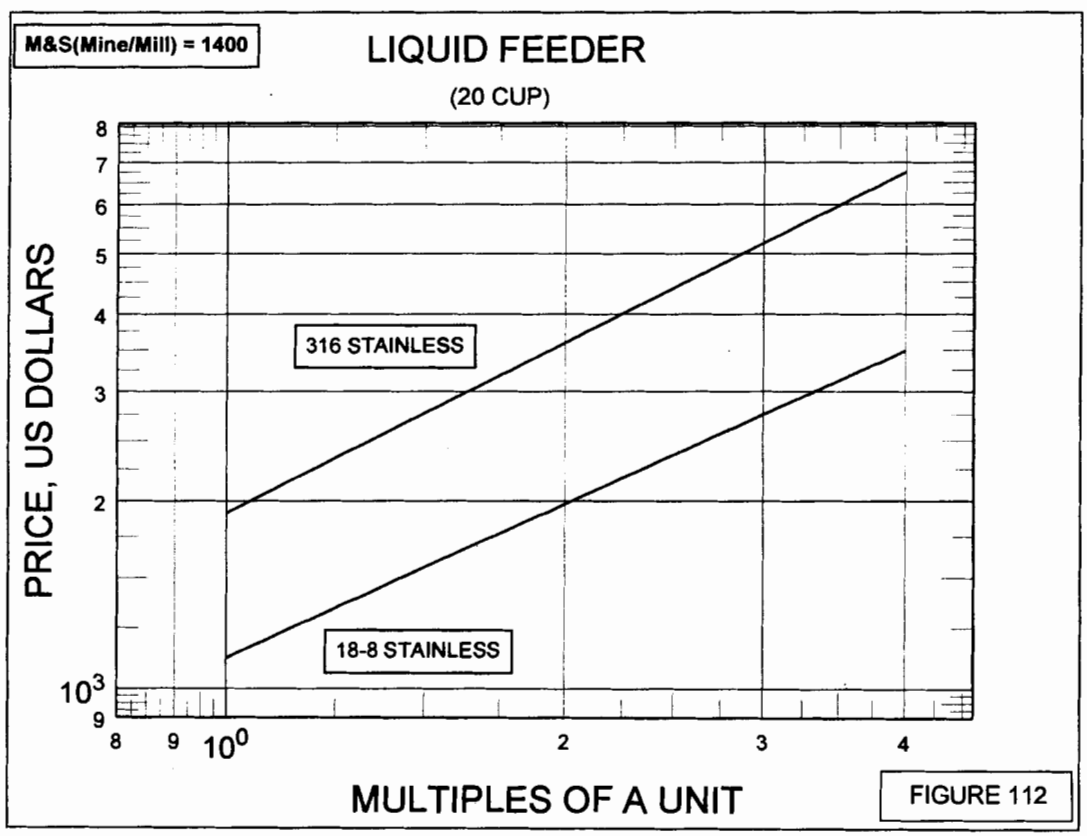


FIGURE 112

LIQUID FEEDER
(High Capacity)

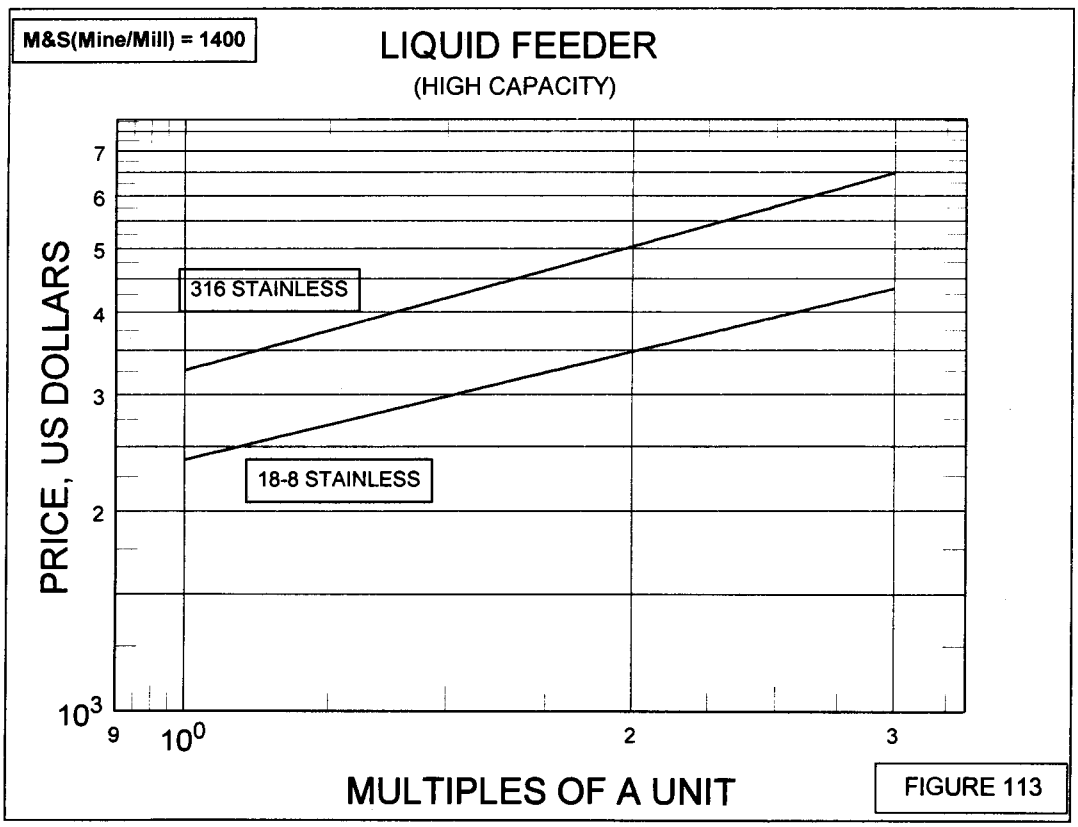
Price = aX^b, US Dollars

where **X is a multiple of a single unit (range 1 to 3)**.

18-8 Stainless; Range in X: 1 to 3 a = 2384 b = 0.5449

316 Stainless; Range in X: 1 to 3 a = 3259 b = 0.6249

Unit	Capacity Range, cc/min
1	1000 to 8000
2	2000 to 16000
3	3000 to 24000

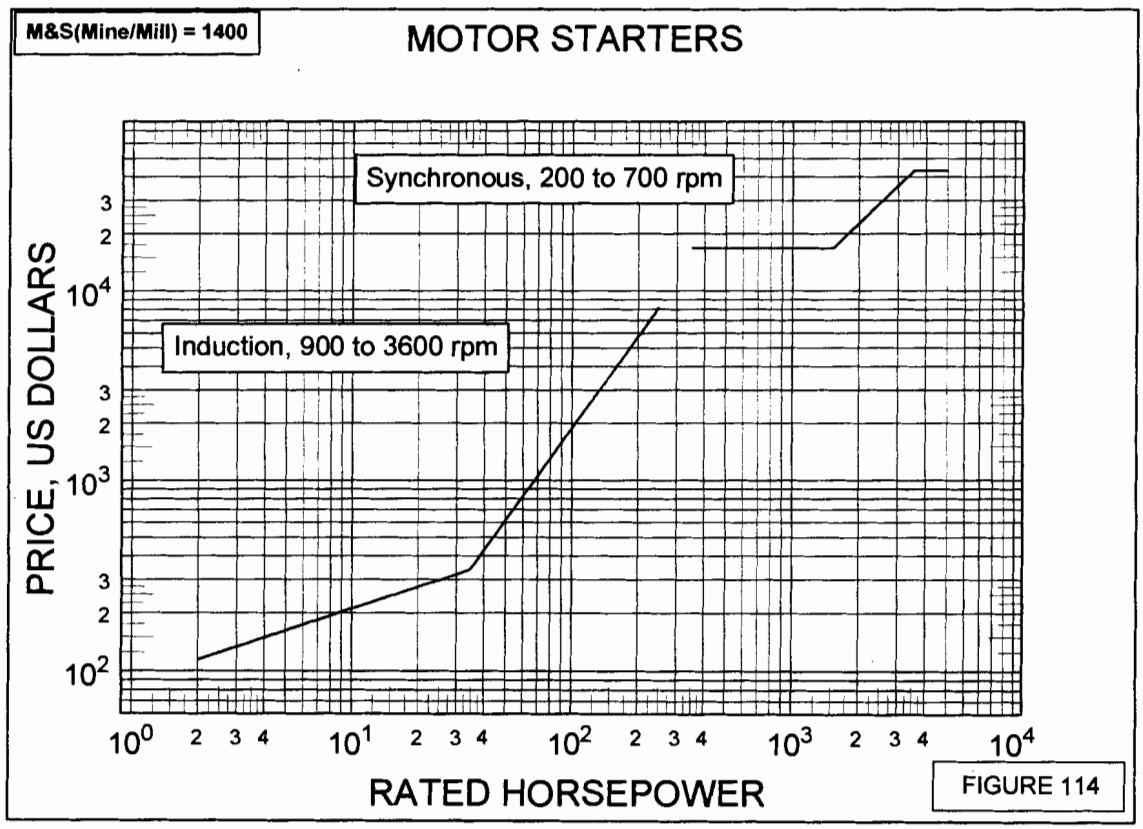


MOTOR STARTERS

Price = aX^b, US Dollars

where **X is rated horsepower**. For induction motor starters: motor voltage 575V, magnetic, full voltage starting, single speed, non-reversing, block type overload, EEMAC 1 enclosure, non-combination type. For synchronous motor starters: voltage @ 4160V, breaker type, full voltage, power factor 1, for brush type motors, EEMAC 1 enclosure, 3 phase, 60 HZ, non-reversing, fuse starter.

Synchronous; Range in X, hp:	350 to 1539	a = 16745	b = 0
	1539 to 3551	a = 4.291	b = 1.127
	3551 to 5000	a = 42952	b = 0
Induction; Range in X, hp:	2 to 34.5	a = 87.74	b = 0.3774
	34.5 to 250	a = 1.118	b = 1.61



VOLTAGE RECTIFIERS

Price = aX^b, US Dollars

where **X is rated horsepower**. For DC motors; price includes isolation transformer.

Range in X, hp: 250 to 4000 a = 205.8 b = 0.9856

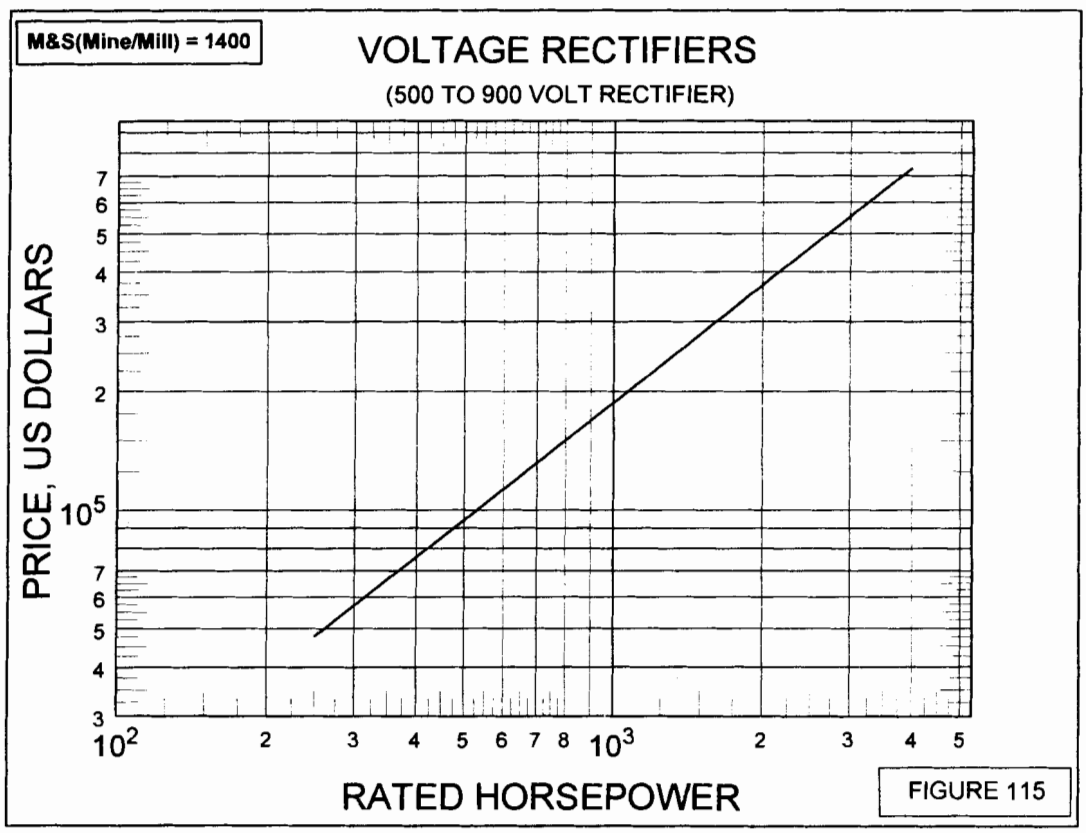


FIGURE 115

DC MOTORS

Price = aX^b, US Dollars

where **X is motor horsepower**. Price excludes rectifier.

1800 rpm, 240V; Range in X, hp: 1 to 250	a = 557.2	b = 0.8635
180 rpm, 750V; Range in X, hp: 1000 to 6000	a = 379.4	b = 0.8982
80 rpm, 750V; Range in X, hp: 1000 to 6000	a = 2423	b = 0.7301

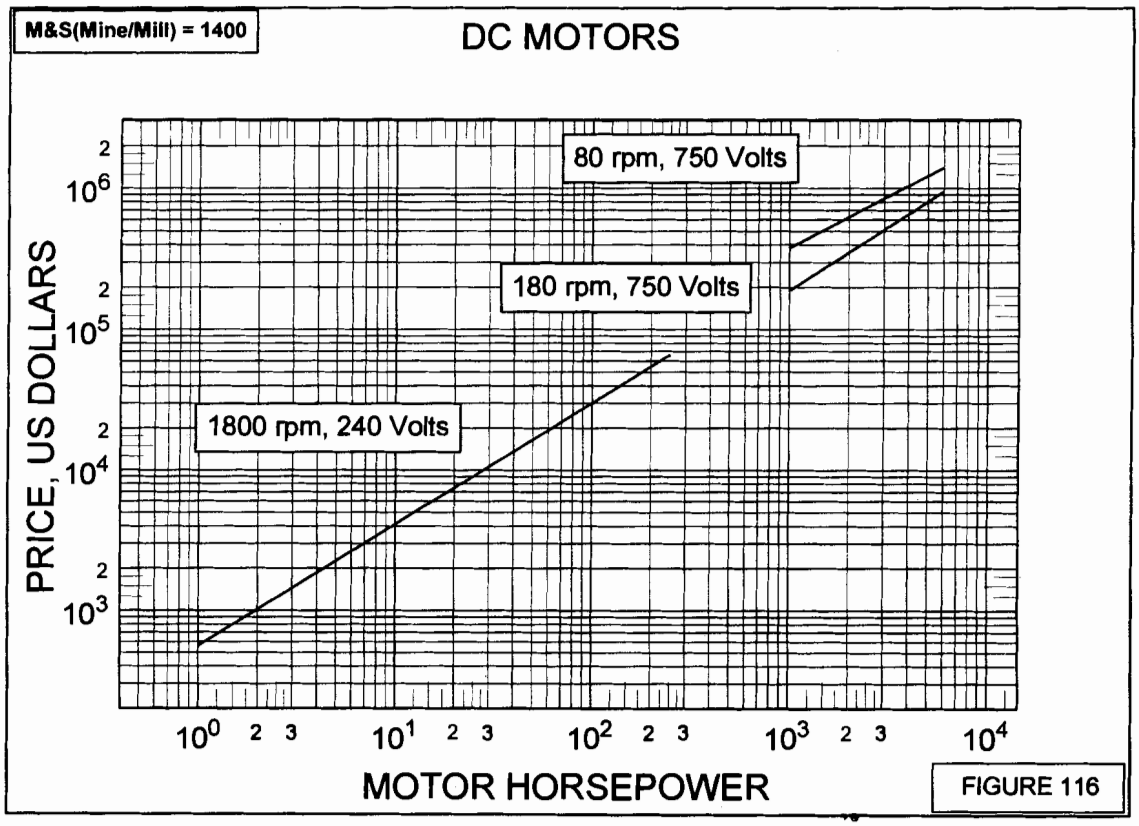


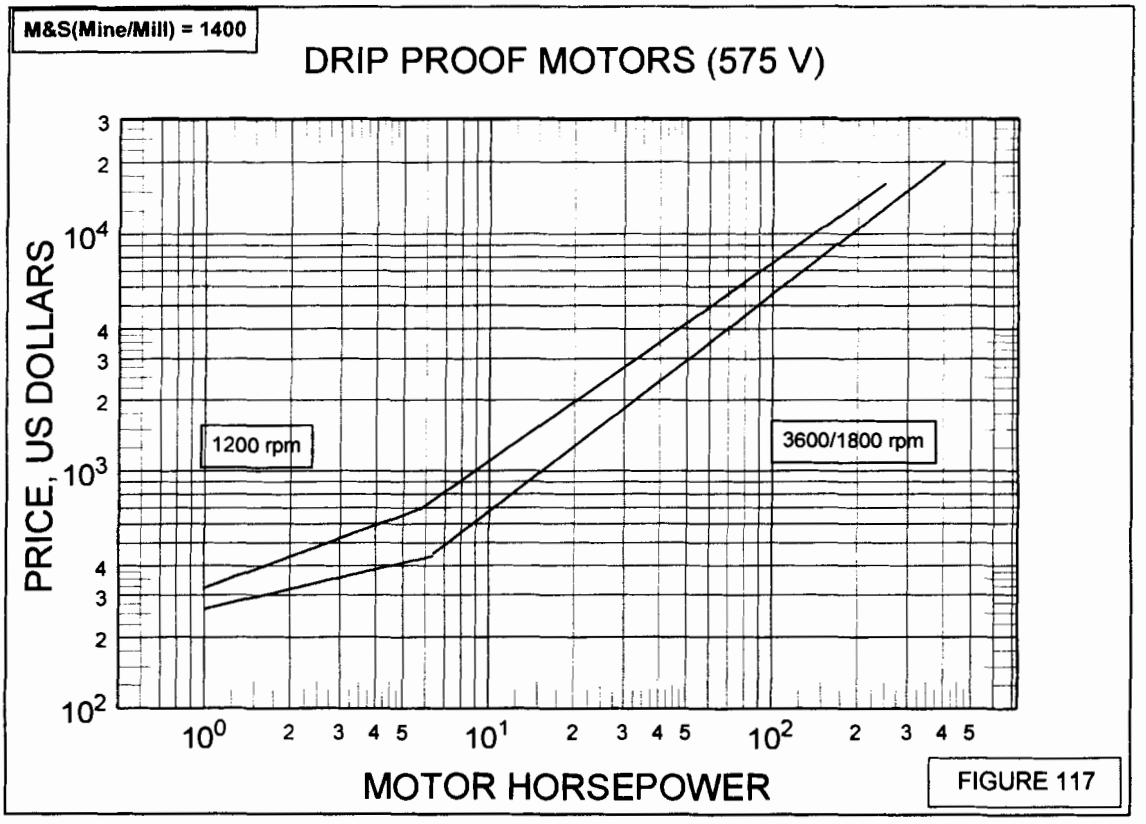
FIGURE 116

DRIP PROOF MOTORS (575 V)

Price = aX^b, US Dollars

where **X is motor horsepower**.

3600 and 1800 rpm; Range in X, hp:	1 to ~~64~~ 6.4	a = 261.3	b = 0.2769
	6.4 ~~64~~ to 400	a = 81.44	b = 0.9164
1200 rpm; Range in X, hp:	1 to 5.75	a = 319.8	b = 0.4409
	5.75 to 250	a = 159.8	b = 0.836



DRIP PROOF MOTORS (2300 V)

Price = aX^b, US Dollars

where **X is motor horsepower**.

3600 rpm; Range in X, hp:	250 to 1000	a = 742.1	b = 0.6514
1800 rpm; Range in X, hp:	250 to 1000	a = 339.4	b = 0.7754
1200 rpm; Range in X, hp:	250 to 1000	a = 655.7	b = 0.6972
900 rpm; Range in X, hp:	250 to 1000	a = 663.7	b = 0.7259

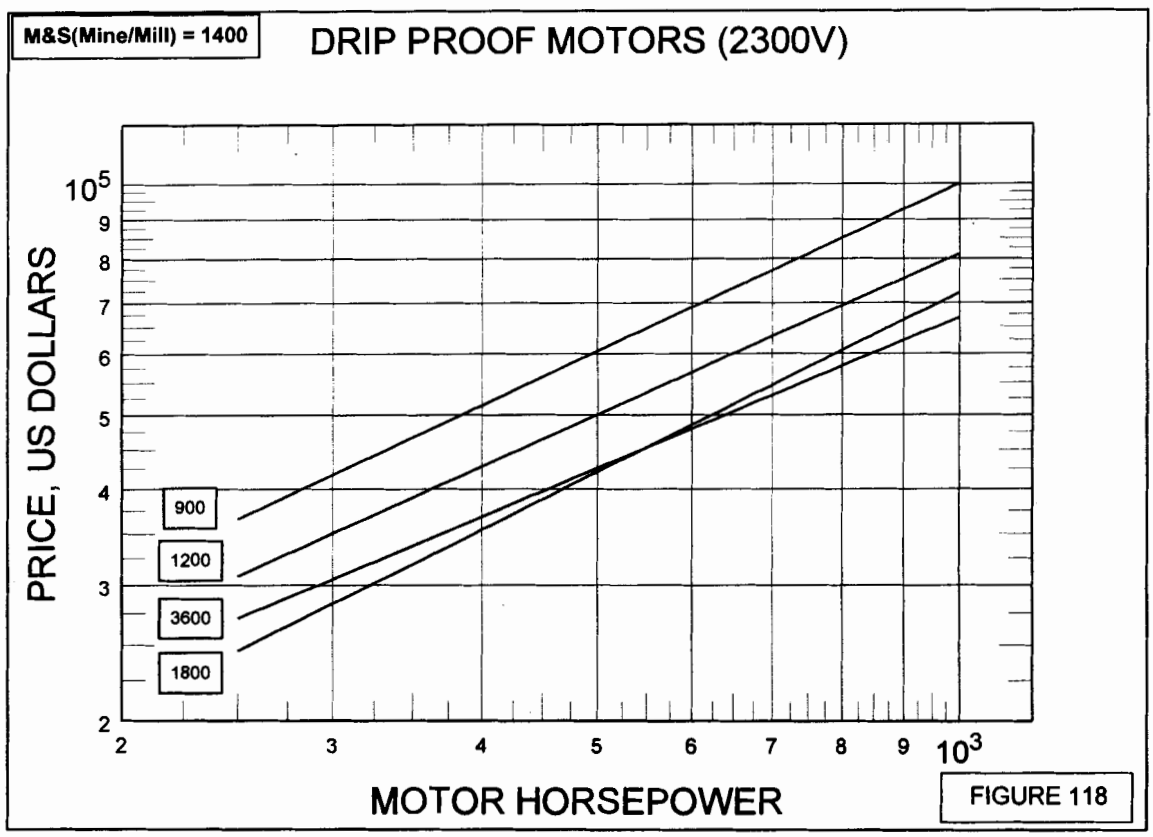


FIGURE 118

DRIP PROOF MOTORS (4000 V)

Price = aX^b, US Dollars

where **X is motor horsepower**.

3600 rpm; Range in X, hp:	250 to 1000	a = 742.1	b = 0.6514
1800 rpm; Range in X, hp:	250 to 1000	a = 339.4	b = 0.7754
1200 rpm; Range in X, hp:	250 to 1000	a = 655.7	b = 0.6972
900 rpm; Range in X, hp:	250 to 1000	a = 663.7	b = 0.7259

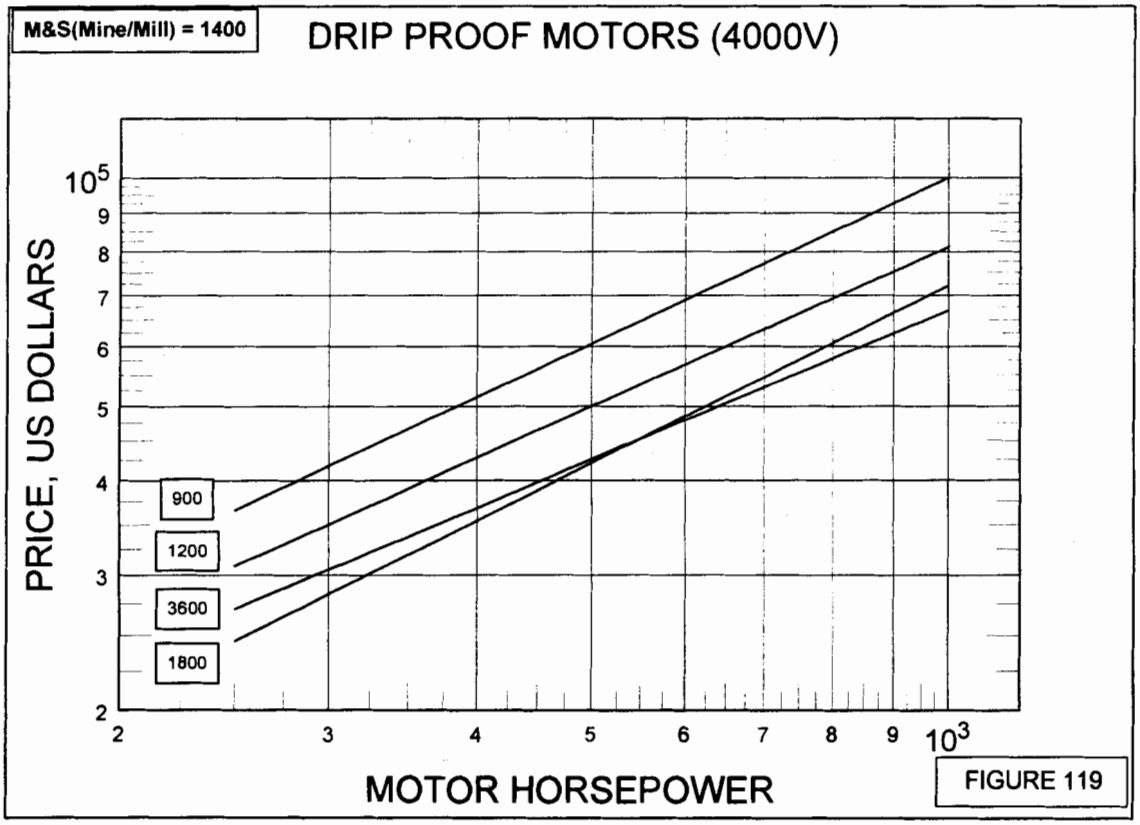


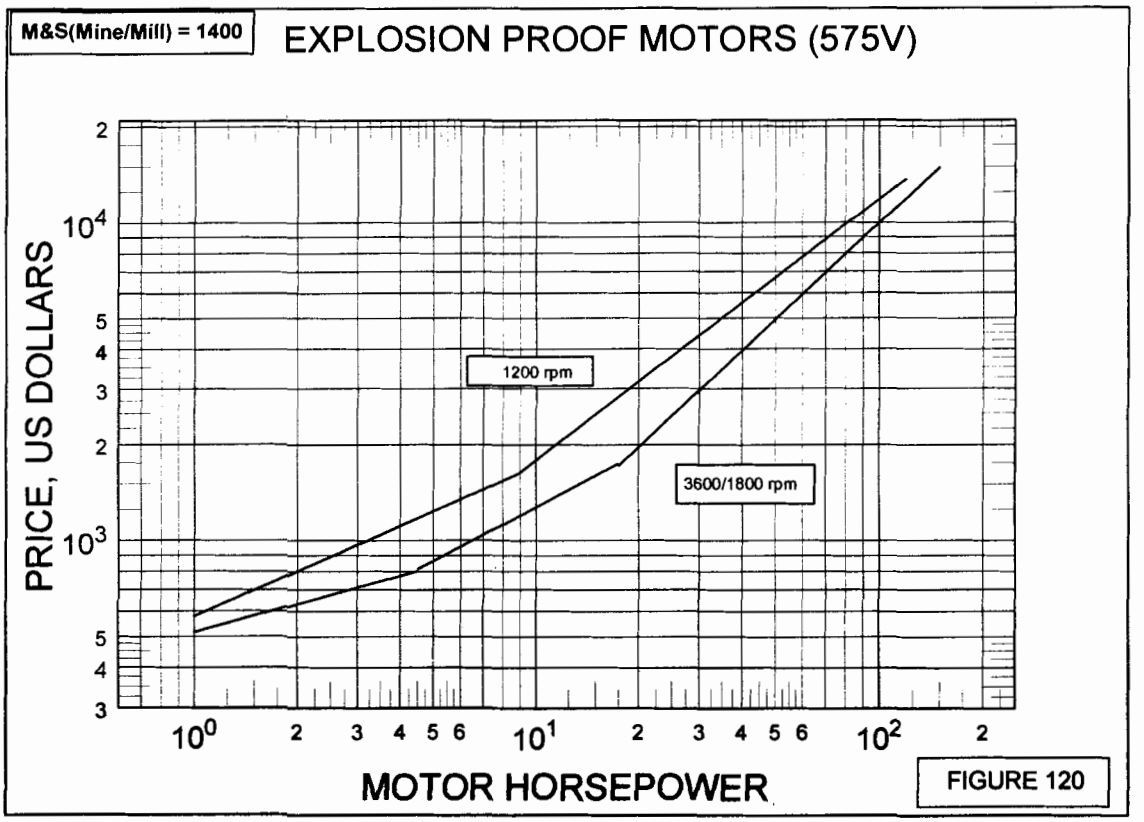
FIGURE 119

EXPLOSION PROOF MOTORS (575 V)

Price = aX^b, US Dollars

where **X is motor horsepower**.

3600 and 1800 rpm; Range in X, hp:	1 to 4.5	a = 513.5	b = 0.2938
	4.5 to 17.5	a = 347.3	b = 0.563
	17.5 to 150	a = 95.84	b = 1.007
1200 rpm; Range in X, hp:	1 to ~~90~~ 9.0	a = 574.7	b = 0.4738
	~~90~~ 9.0 to 120	a = 268.8	b = 0.8201



TEFC MOTORS (575 V)

Price = aX^b, US Dollars

where **X is motor horsepower**.

3600 and 1800 rpm; Range in X, hp:	1 to 12	a = 274.8	b = 0.4198
	12 to 200	a = 56.55	b = 1.067
1200 rpm; Range in X, hp:	1 to 9.5	a = 327.5	b = 0.5667
	9.5 to 200	a = 162.1	b = 0.8921
900 rpm; Range in X, hp:	1 to 45	a = 386.6	b = 0.7243
	45 to 200	a = 99.29	b = 1.084

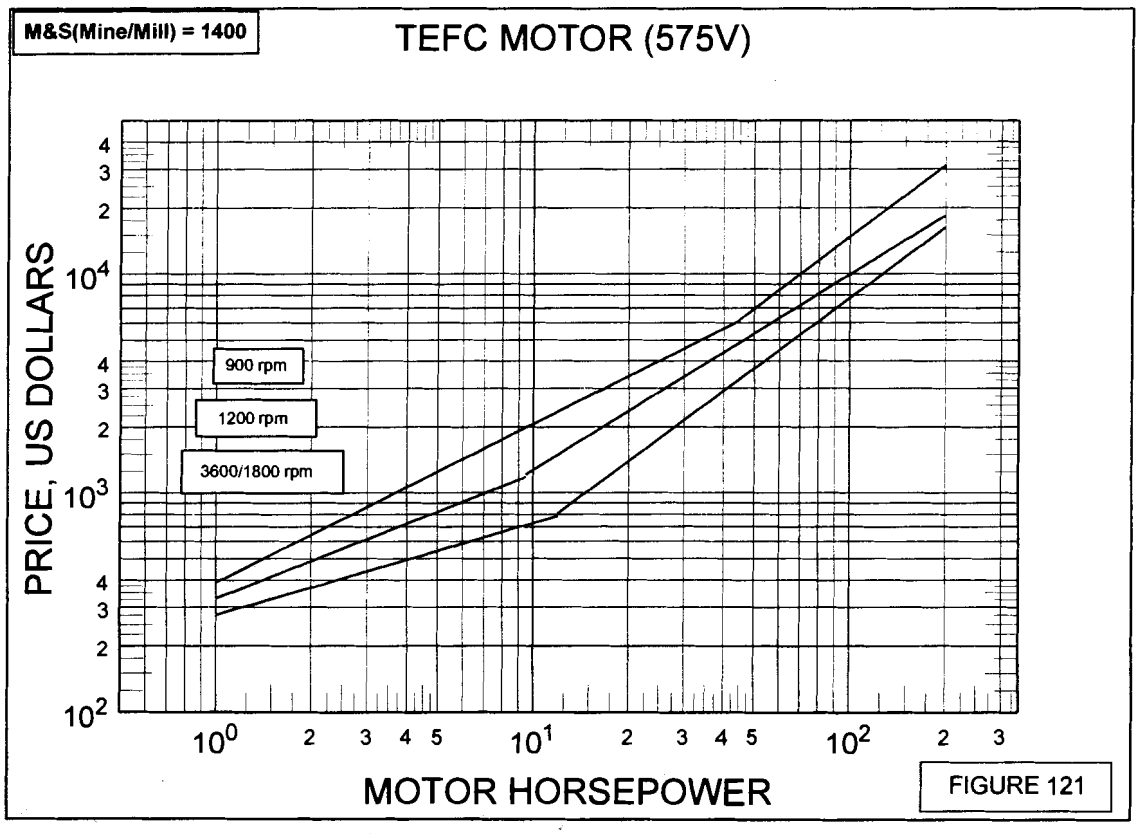


FIGURE 121

TEFC MOTORS (2300 V)

Price = aX^b, US Dollars

where **X is motor horsepower**.

3600 and 1800 rpm; Range in X, hp:	250 to 500	a = 221.4	b = 0.9353
	500 to 1000	a = 4021	b = 0.4683
1200 rpm; Range in X, hp:	250 to 480	a = 439.3	b = 0.8443
	480 to 1000	a = 7476	b = 0.3859
900 rpm; Range in X, hp:	250 to 460	a = 819.4	b = 0.7773
	460 to 1000	a = 15440	b = 0.2992

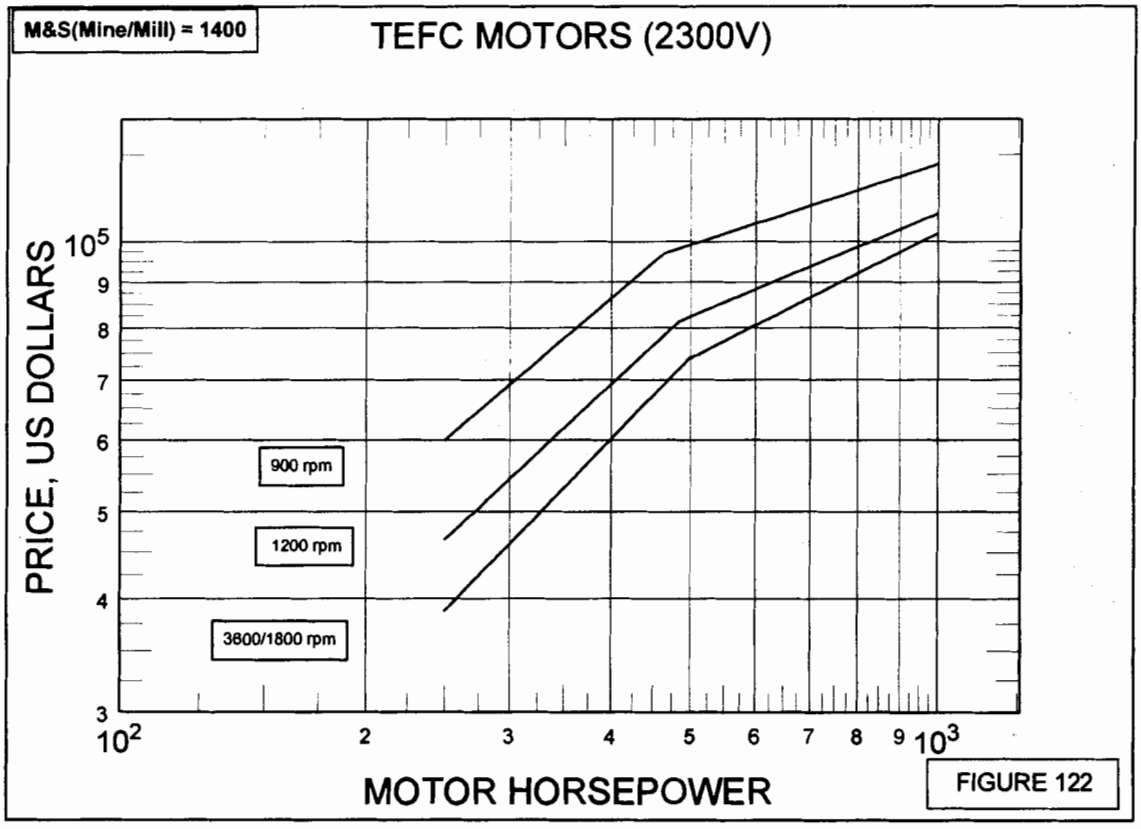


FIGURE 122

TEFC MOTORS (4000 V)

Price = aX^b, US Dollars

where **X is motor horsepower**.

3600 and 1800 rpm; Range in X, hp:	250 to 500	a = 221.4	b = 0.9353
	500 to 1000	a = 4021	b = 0.4683
1200 rpm; Range in X, hp:	250 to 480	a = 439.3	b = 0.8443
	480 to 1000	a = 7476	b = 0.3859
900 rpm; Range in X, hp:	250 to 460	a = 819.4	b = 0.7773
	460 to 1000	a = 15440	b = 0.2992

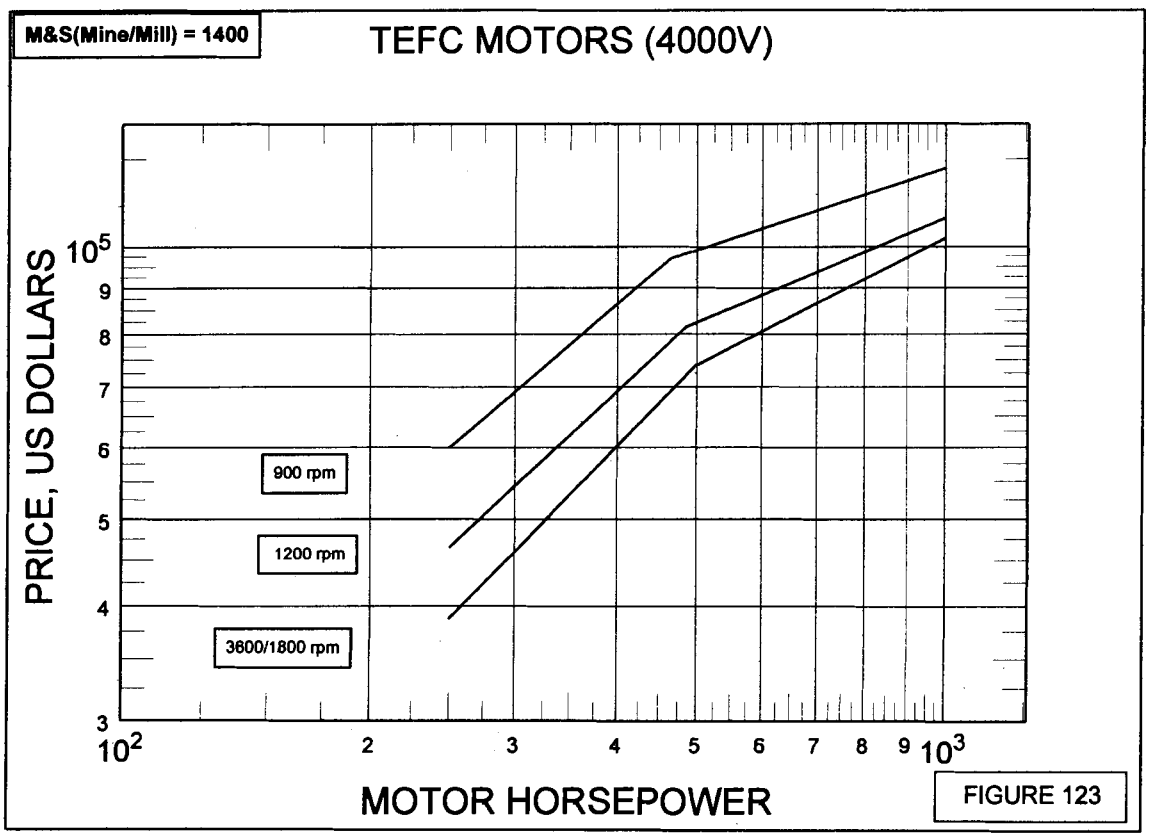


FIGURE 123

SYNCHRONOUS MOTORS

Price = aX^b, US Dollars

where **X is motor horsepower**.

700 rpm; Range in X, hp:	2500 to 5000	a = 1130	b = 0.7586
240 rpm; Range in X, hp:	1500 to 5500	a = 66260	b = 0.2829
	5500 to 6500	a = 191.9	b = 0.9611

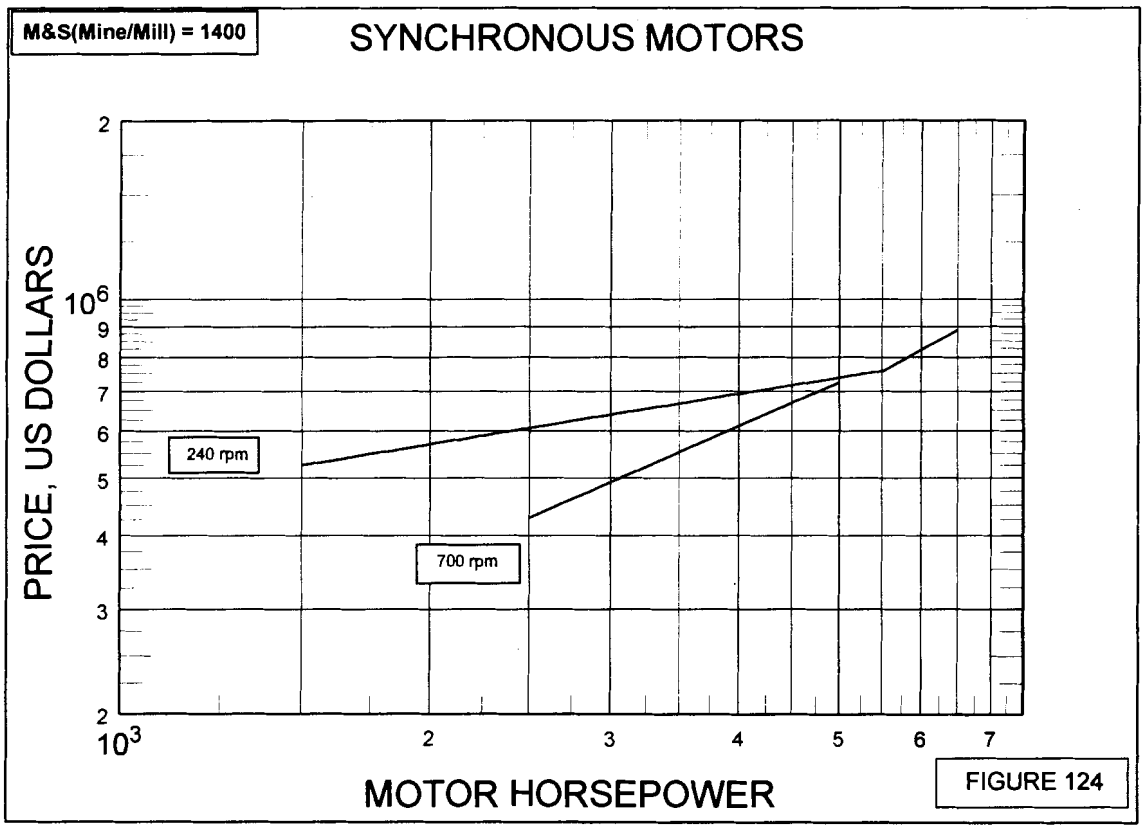


FIGURE 124

FIRE TUBE BOILER

Price = aX^b, US Dollars

where **X is boiler horsepower**. Price covers boiler operating at 150 PSI steam pressure.

Range in X, hp: 10 to 700 a = 1685 b = 0.6505

Horsepower	Evaporation Rate, lb/hr
25	850
50	1700
100	3400
150	5000
200	6800
250	8500
300	10200
400	13800
600	20700

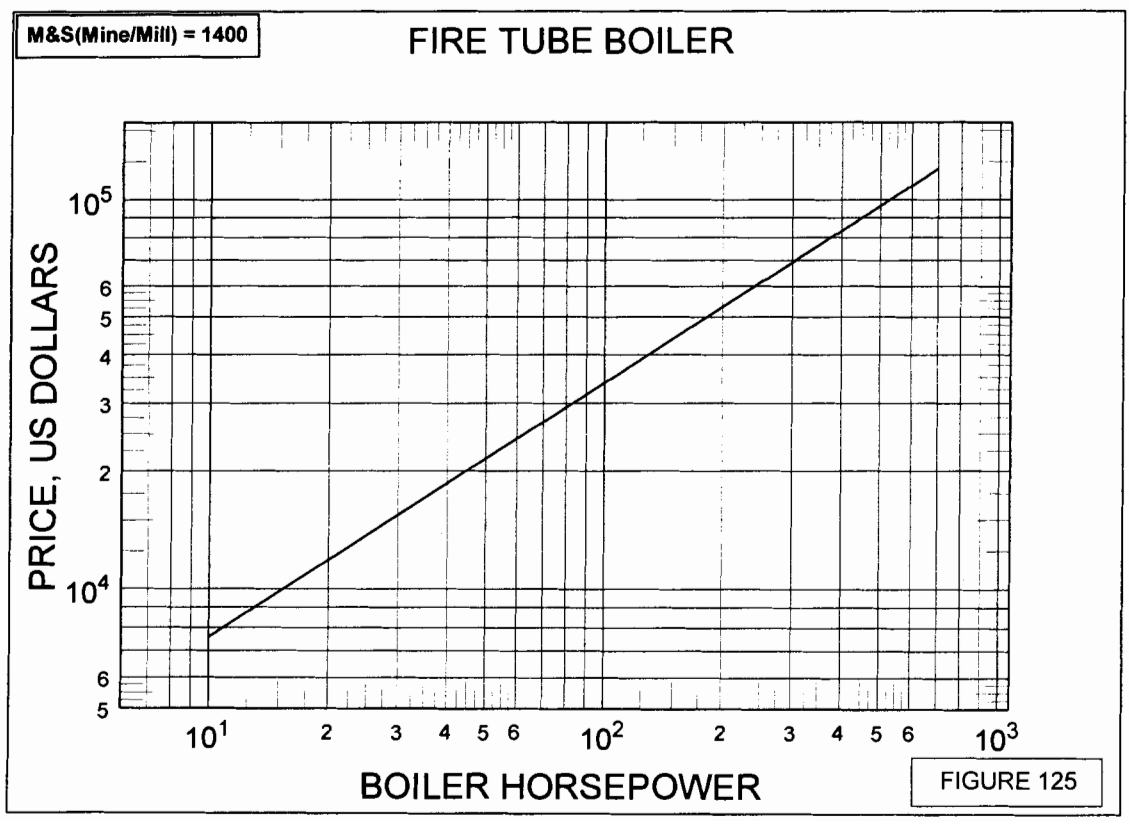


FIGURE 125

SHELL AND TUBE HEAT EXCHANGERS

Price = aX^b, US Dollars

where **X is heat exchange area, square feet.**

Tantalum tubes; Range in X, ft^2:	2.36 to 9.52	a = 3949	b = 0.5505
	9.52 to 223	a = 1529	b = 0.9714
Copper tubes; Range in X, ft^2:	500 to 7500	a = 339.4	b = 0.7354
Teflon tubes; Range in X, ft^2:	101.8 to 538	a = 1661	b = 0.4394
Titanium tubes; Range in X, ft^2:	3.1 to 12.16	a = 1947	b = 0.2583
	12.16 to 918	a = 1097	b = 0.4879

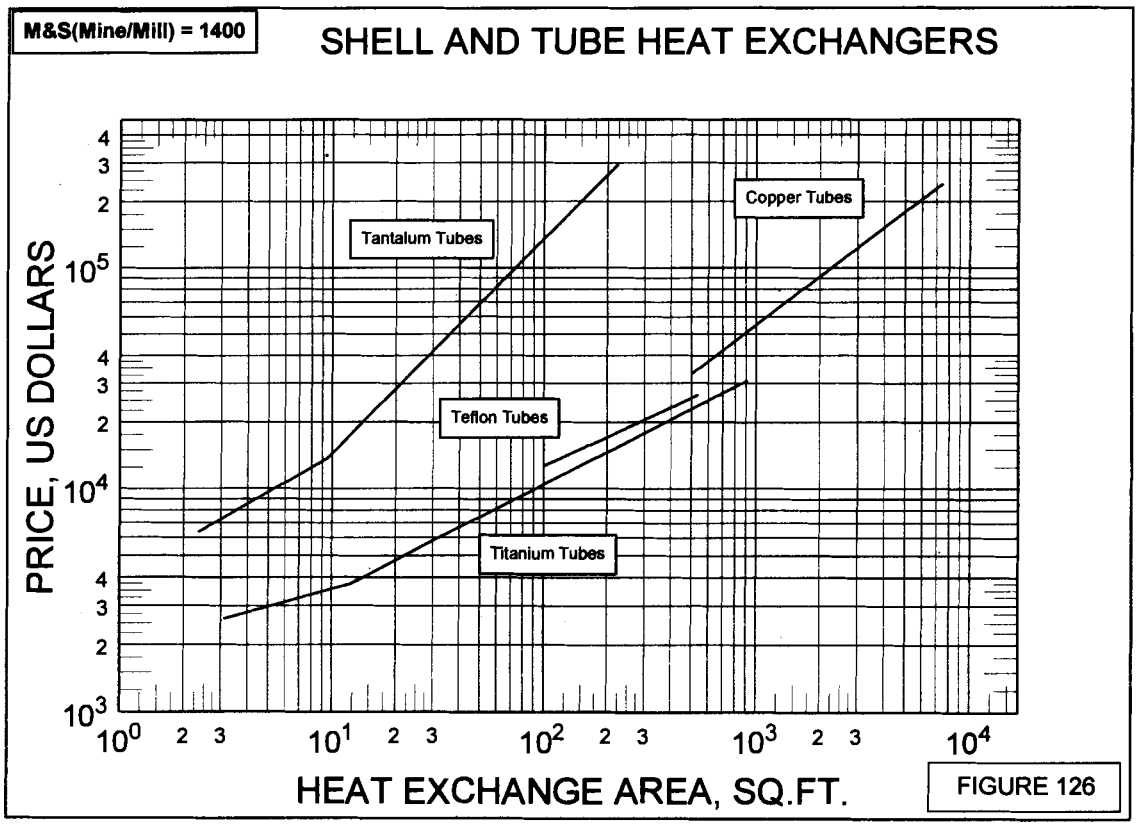


FIGURE 126

AXIAL FLOW BLOWERS

Price = aX^b, US Dollars

where **X is capacity in cubic feet per minute**. Discharge pressure is 1 atmosphere; vacuum pressure is 1/2 atmosphere. Cost includes motor; excludes mounts and air filter.

Range in X, cfm:	100 to 716.88	a = 208.1	b = 0.4271
	716.88 to 2550	a = 4.818	b = 0.9998

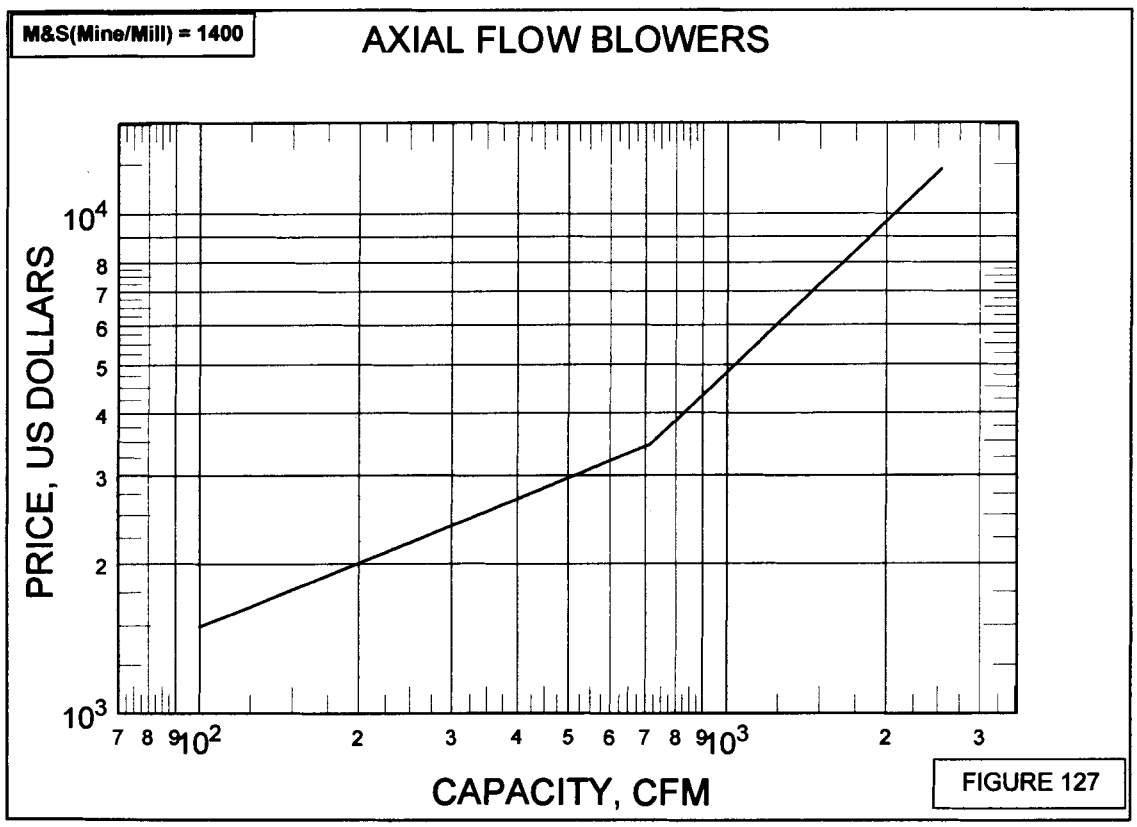


FIGURE 127

CENTRIFUGAL FAN BLOWERS

Price = aX^b, US Dollars

where **X is capacity, cubic feet per minute**. Price includes motor; excludes mounts and air filter.

Disch. Pressure 35" WG; Range in X, cfm:	100 to 3600	a = 1257	b = 0.1636
Disch. Pressure 15" WG; Range in X, cfm:	170 to 4000	a = 235.5	b = 0.3419

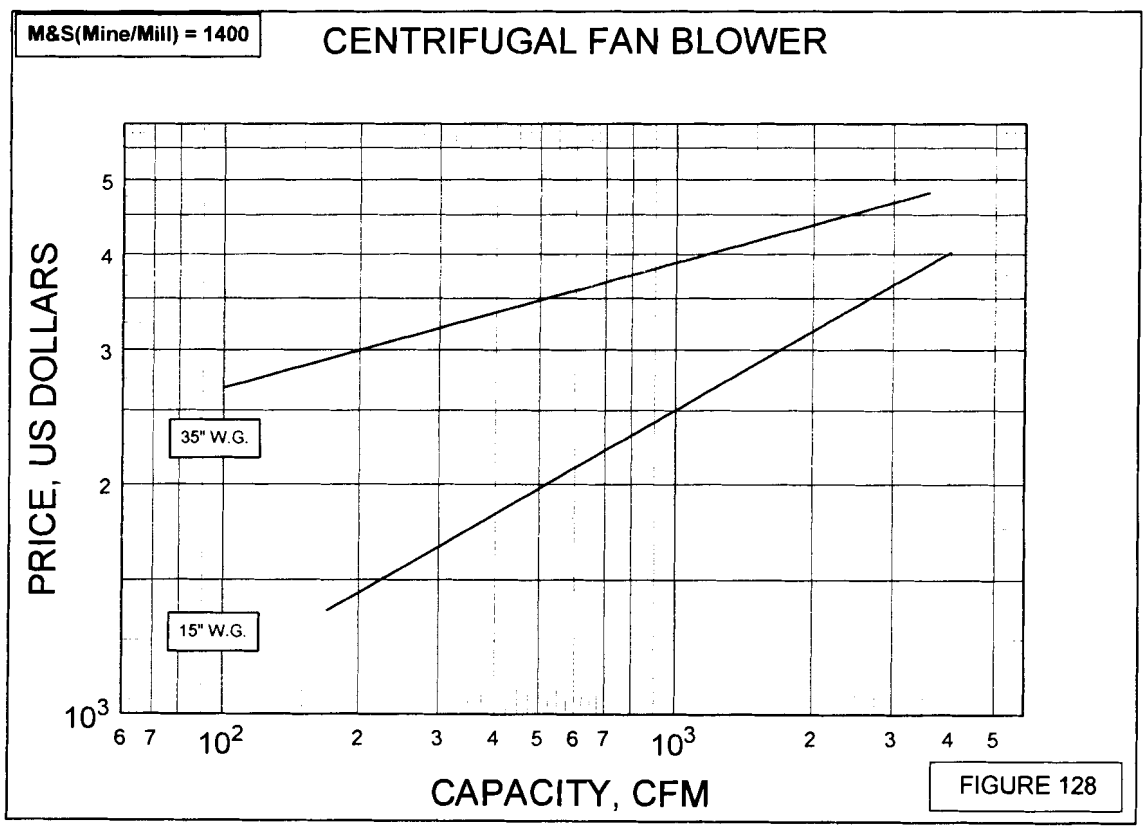


FIGURE 128

MULTI-STAGE CENTRIFUGAL PUMPS

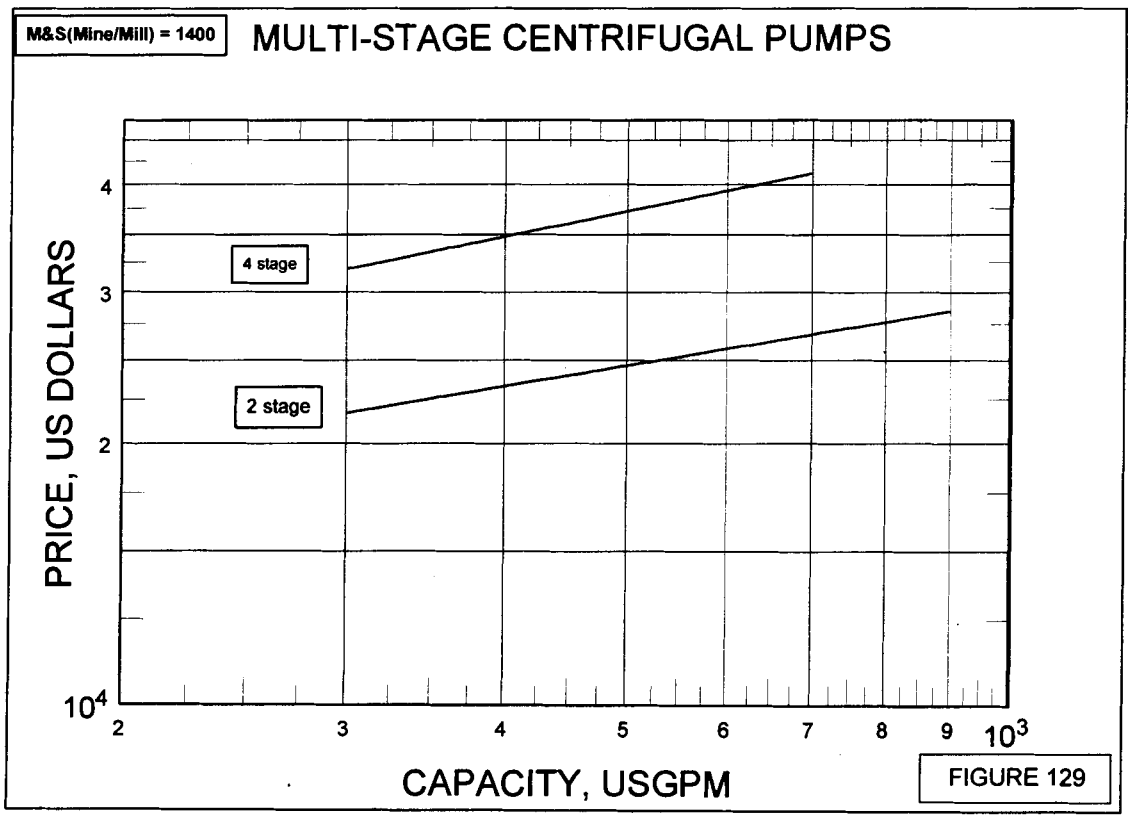
Price = aX^b, US Dollars

where **X is capacity in US gallons per minute**. Iron casing and bronze impeller, water only, capacity at 400 ft head and speed near maximum efficiency. Price includes drive; excludes motor.

4-Stage; Range in X, USGPM: 300 to 700 a = 5699 b = 0.3018

2-Stage; Range in X, USGPM: 300 to 900 a = 5288 b = 0.2474

Capacity, USGPM	HP (2-Stage)	HP (4-Stage)
300	50	100
500	75	125
700	110	175
900	150	200



CENTRIFUGAL PUMPS
(Stainless Steel)

Price = aX^b, US Dollars

where **X is capacity in US gallons per minute**, at near maximum efficiency. Price includes pump, base and coupling; excludes motor. Heads from 7 to 100 ft. of water.

Stainless Steel; Range in X, USGPM:	50 to 995	a = 4241	b = 0.1613
	995 to 12000	a = 221.9	b = 0.5887
Titanium; Range in X, USGPM:	100 to 500	a = 3654.4	b = 0.1787

Capacity, USGPM	Nominal HP, clear water
100	10
300	20
600	32
1000	50
1600	53
3000	95
4250	141
7000	216
9000	291
12000	357

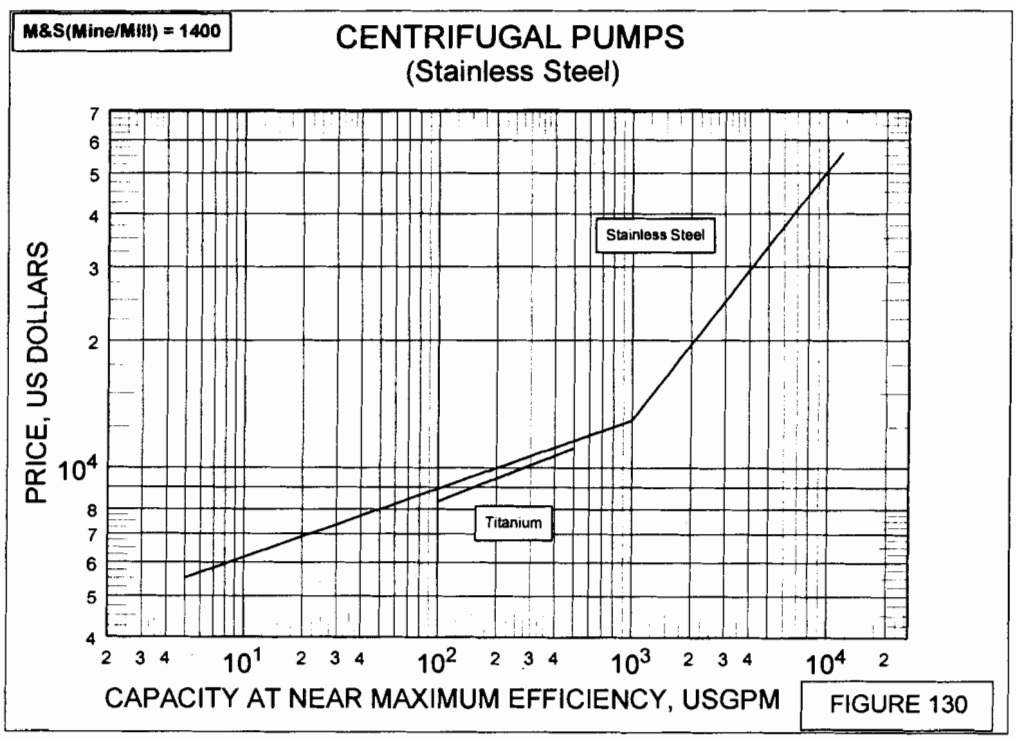


FIGURE 130

VERTICAL CENTRIFUGAL PUMPS
(Sump Pumps)

Price = aX^b, US Dollars

where **X is the capacity in US gallons per minute**. Price includes drive and excludes motor. Pumps have closed impellers unless noted otherwise. Head of 100 ft at near maximum efficiency.

Rubber, open impeller; Range in X, USGPM: 80 to 3750 a = 565.3 b = 0.3982

Hi-Chrome; Range in X, USGPM: 80 to 3750 a = 519.9 b = 0.4376

Rubber lined; Range in X, USGPM: 160 to 2775 a = 2459 b = 0.3257

Capacity, USGPM	HP, clear water
200	11
500	21
700	28
2000	39
3000	102

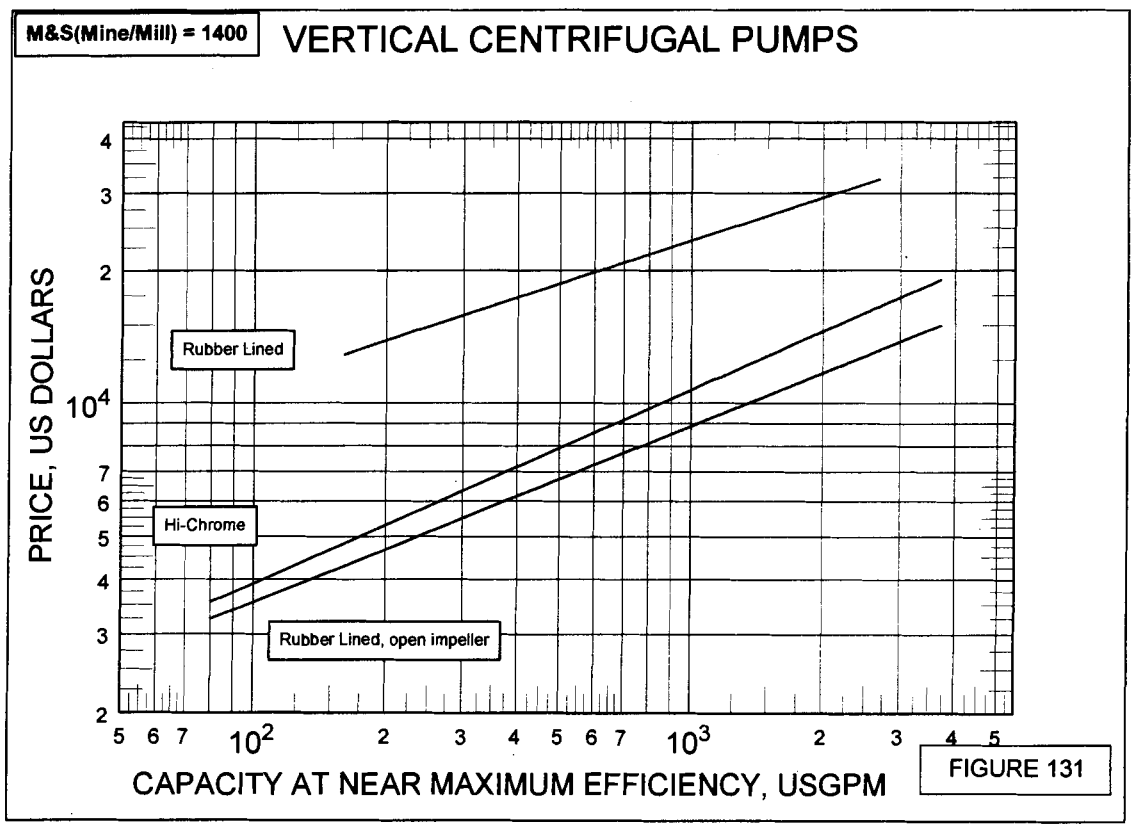


FIGURE 131

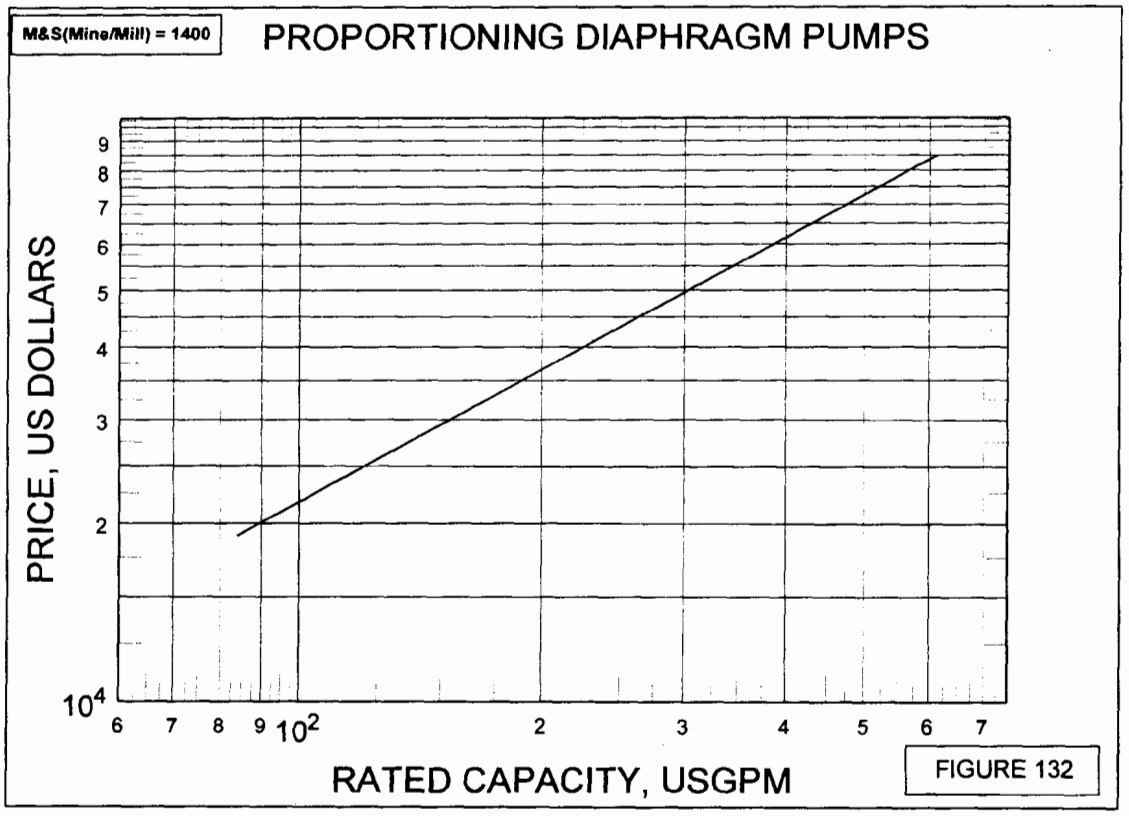
PROPORTIONING DIAPHRAGM PUMPS

Price = aX^b, US Dollars

where **X is the rated capacity in US gallons per minute for slurry at 50% solids**. Price includes drive; excludes motor.

Range in X, USGPM: 84 to 614 a = 682.8 b = 0.7509

Capacity, USGPM	HP (50% solids)
84	1.5
130	2
242	5
334	5
614	7.5



GEAR PUMPS (OIL)

Price = aX^b, US Dollars

where **X is the capacity in US gallons per minute at 200 PSI of pressure**. Price includes drive; excludes motor.

Range in X, USGPM:	1 to 9.87	a = 1325	b = 0.1345
	9.87 to 50	a = 798	b = 0.3559

Capacity, USGPM	HP, nominal
1	0.5
5	1
10	2
20	5
50	10

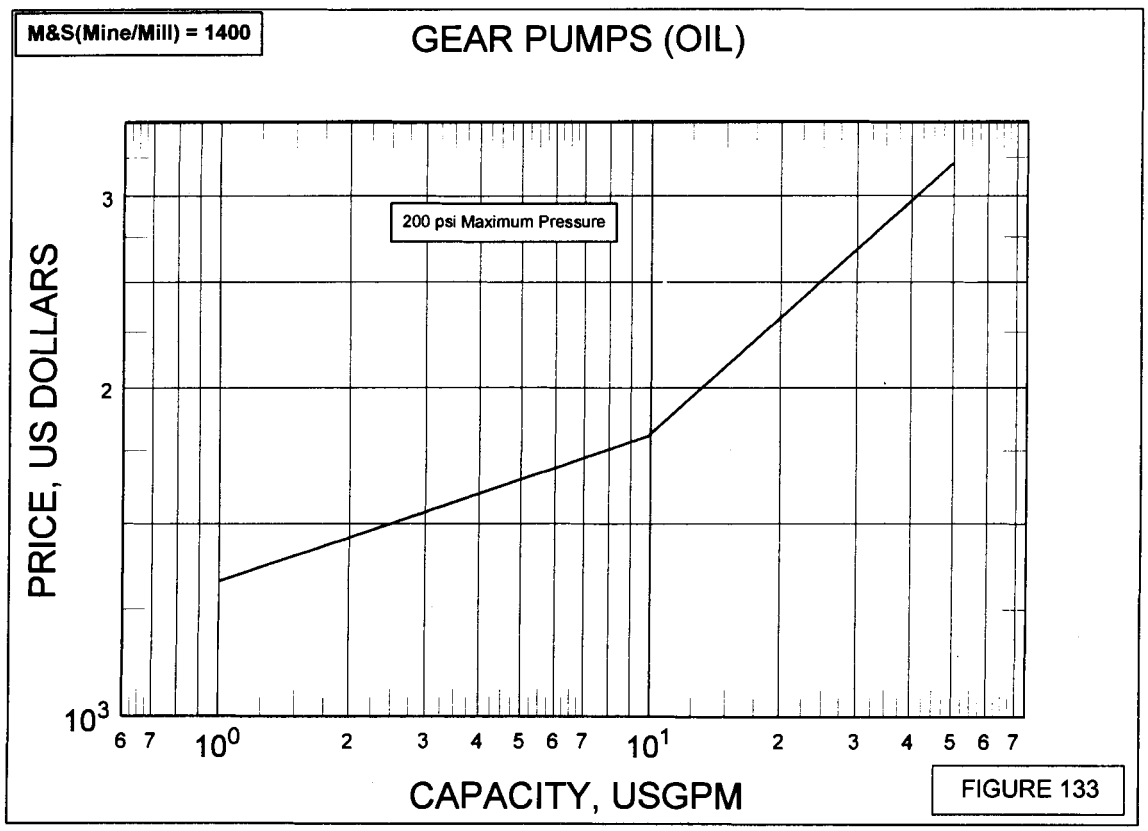


FIGURE 133

PROPORTIONING METER PUMPS
(Progressive Cavity Type)

Price = aX^b, US Dollars

where **X is the rated capacity in US gallons per minute**. Price includes drive and excludes motor.

Range in X, USGPM:			
.03 to 1.98	a = 2668	b = 0.0398
1.98 to 10	a = 2168	b = 0.3434

Capacity, USGPM	HP, clear water
.03	.25
.30	.75
2	.75
10	1.5

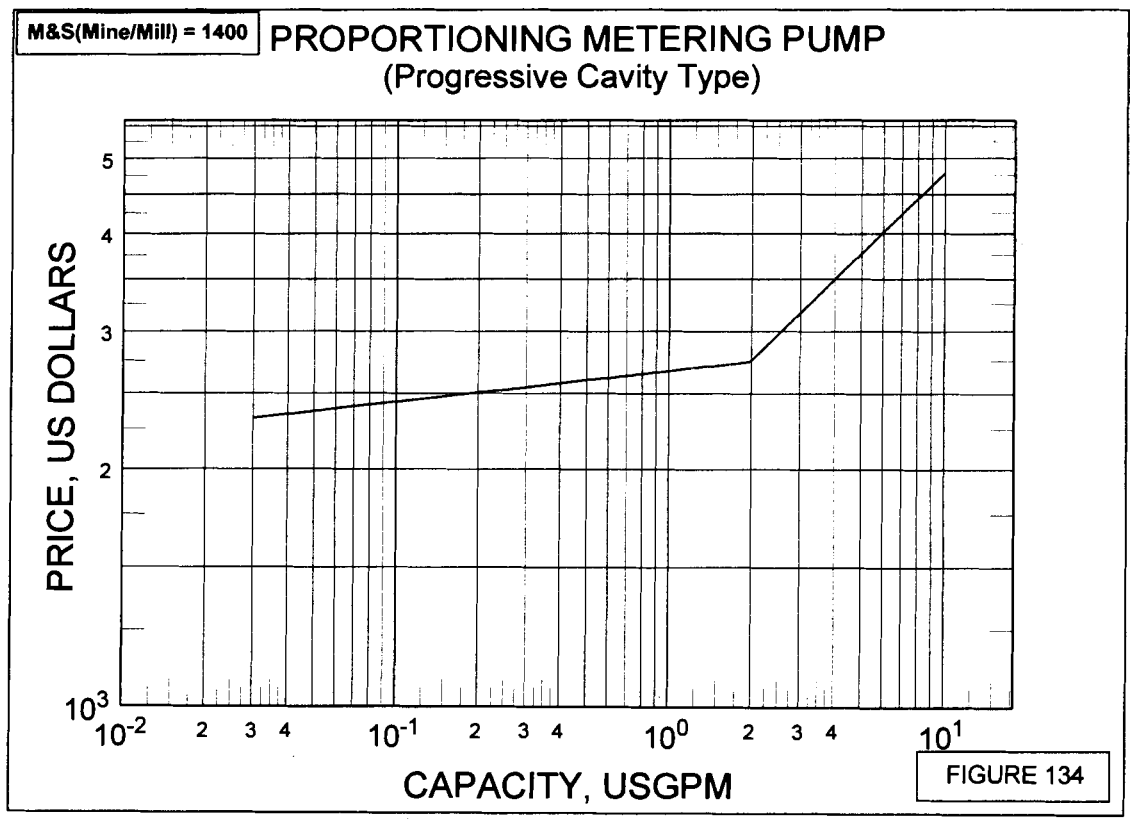


FIGURE 134

AIR DIAPHRAGM PUMP
(Reciprocating, Slurry or Water)

Price = aX^b, US Dollars

where **X is the maximum capacity in US gallons per minute**. Cost excludes base and air couplings and is based on pump with neoprene diaphragms and balls.

Aluminum Casing; Range in X, USGPM:	30 to 240	a = 217.1	b = 0.3809
316 SS Casing; Range in X, USGPM:	30 to 240	a = 150	b = 0.6664
Cast Iron Casing; Range in X, USGPM:	70 to 240	a = 94.37	b = 0.6076

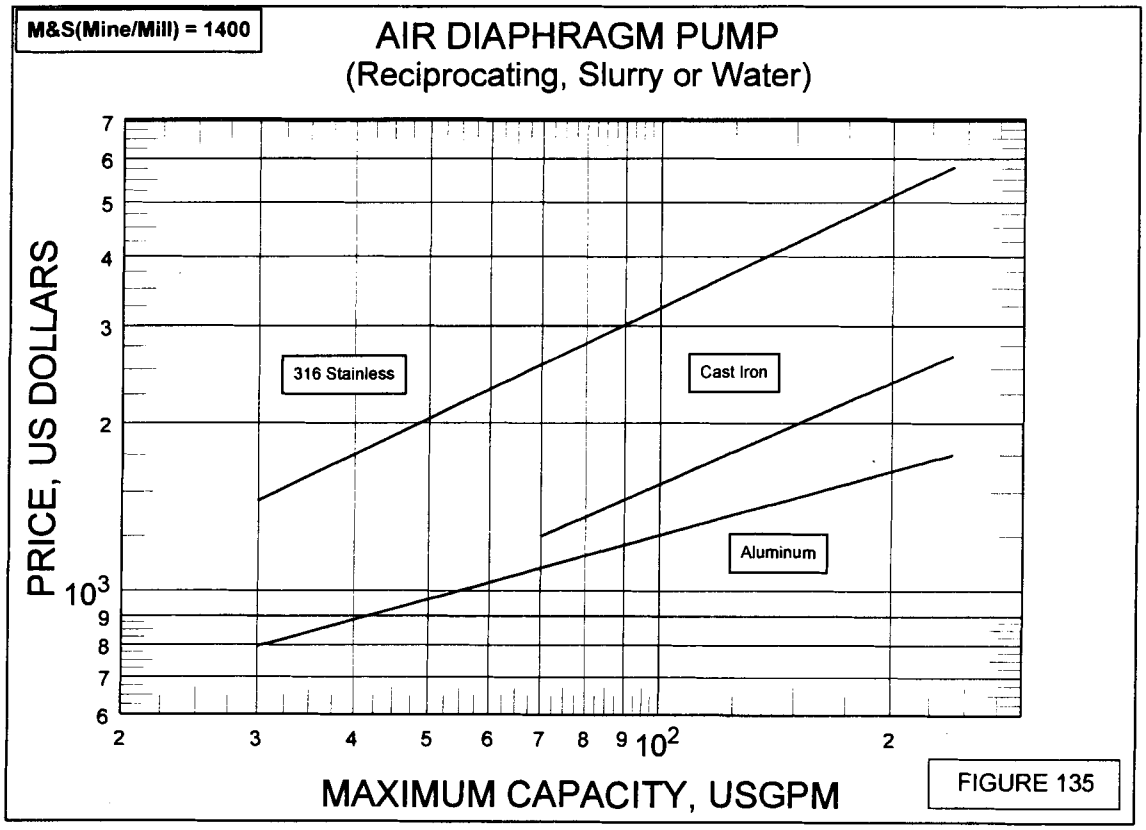


FIGURE 135

PISTON DIAPHRAGM PUMPS
(Thickener Pump)

Price = aX^b, US Dollars

where **X is the maximum capacity in US gallons per minute**. Cost excludes motor and drive.

Simplex; Range in X, USGPM:	6 to 46.23	a = 6774	b = 0.053
	46.23 to 330	a = 414.1	b = 0.782
Duplex; Range in X, USGPM:	12 to 88.46	a = 8492	b = 0.065
	88.46 to 660	a = 383.5	b = 0.756

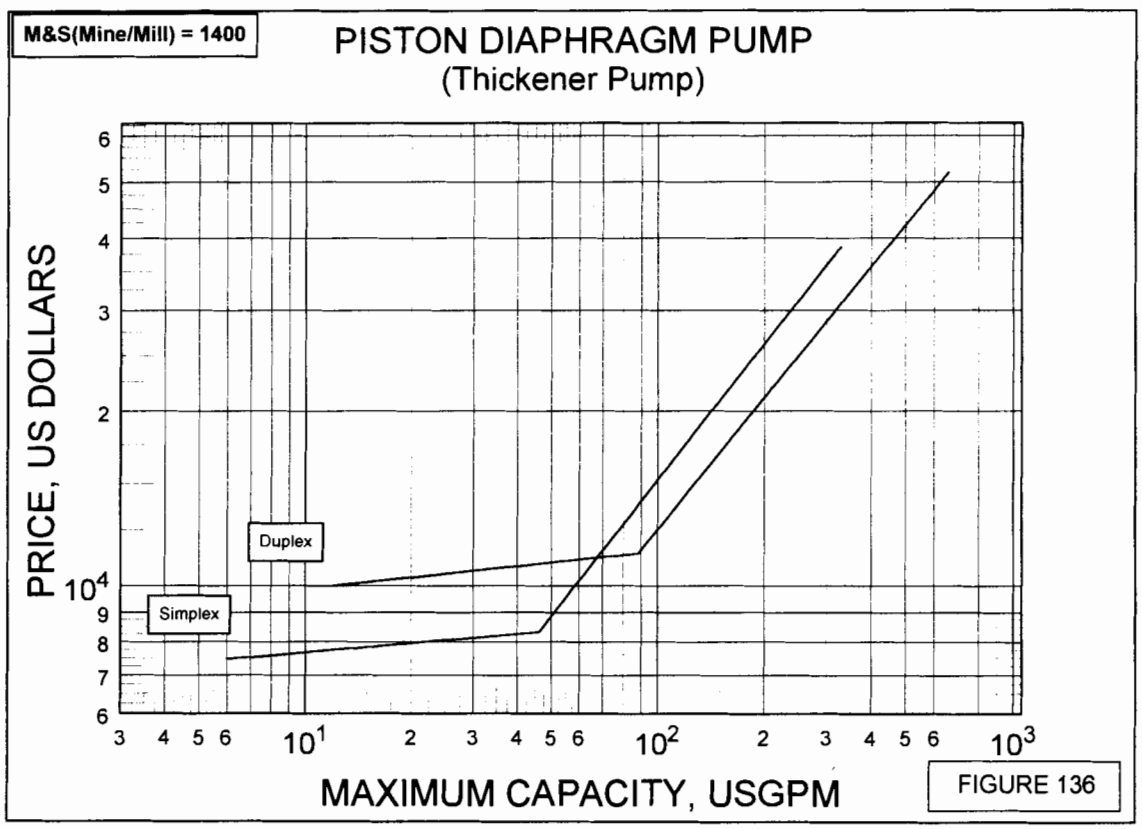


FIGURE 136

RECIPROCATING PISTON POSITIVE DISPLACEMENT PUMP

Price = aX^b, US Dollars

where **X is the rated capacity in US gallons per minute at 100 rpm**. Stroke rate at maximum efficiency for slurries. Price excludes motor.

Range in X, USGPM: 40 to 420 a = 14260 b = 0.5314

Capacity, USGPM	HP, clear water
40	80
60	150
170	350
250	600
420	1000

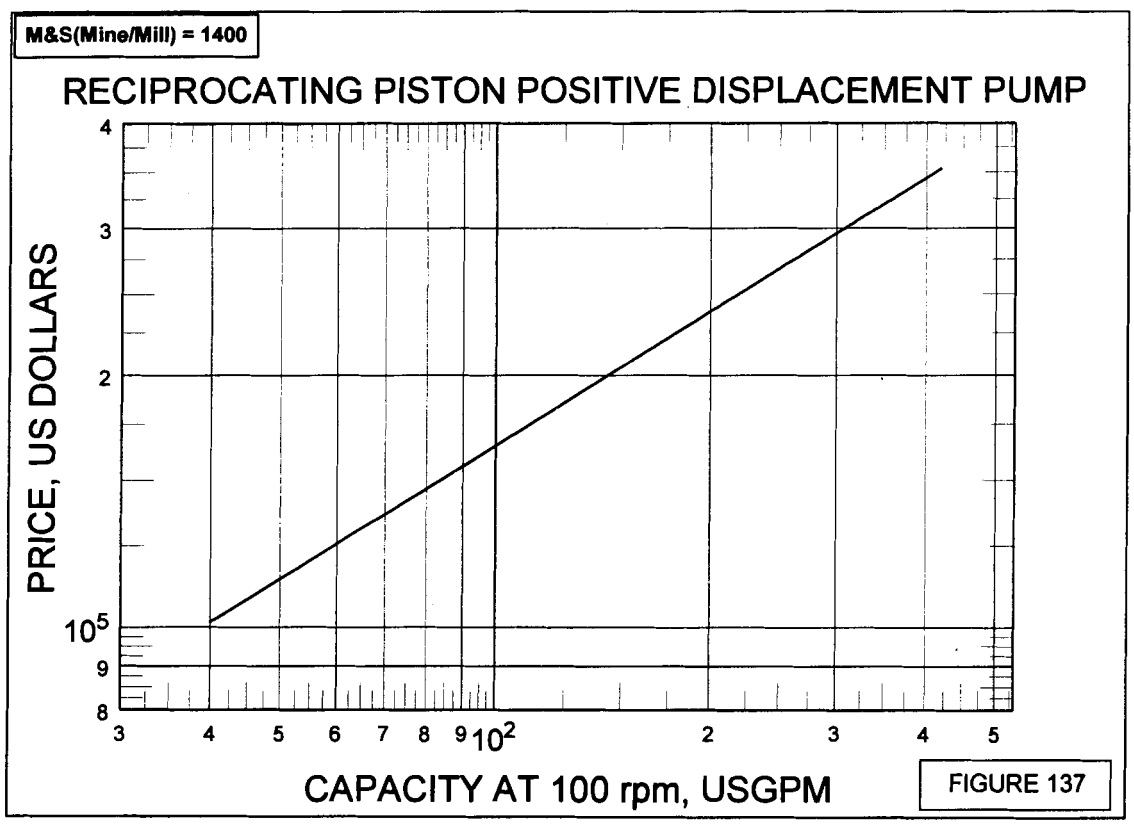


FIGURE 137

SINGLE STAGE CENTRIFUGAL PUMPS
(Open Impeller)

Price = aX^b, US Dollars

where **X is the maximum capacity in US gallons per minute**. Price excludes motor and drive.

Water, Cast Iron; Range in X, USGPM:	100 to 3500	a = 976.3	b = 0.4619
Slurry, Rubber Lined; Range in X, USGPM:	70 to 475	a = 6112	b = 0.1078
	475 to 12000	a = 272.3	b = 0.6141
Slurry; Hi-Cr Lined; Range in X, USGPM:	40 to 2500	a = 1635	b = 0.2183

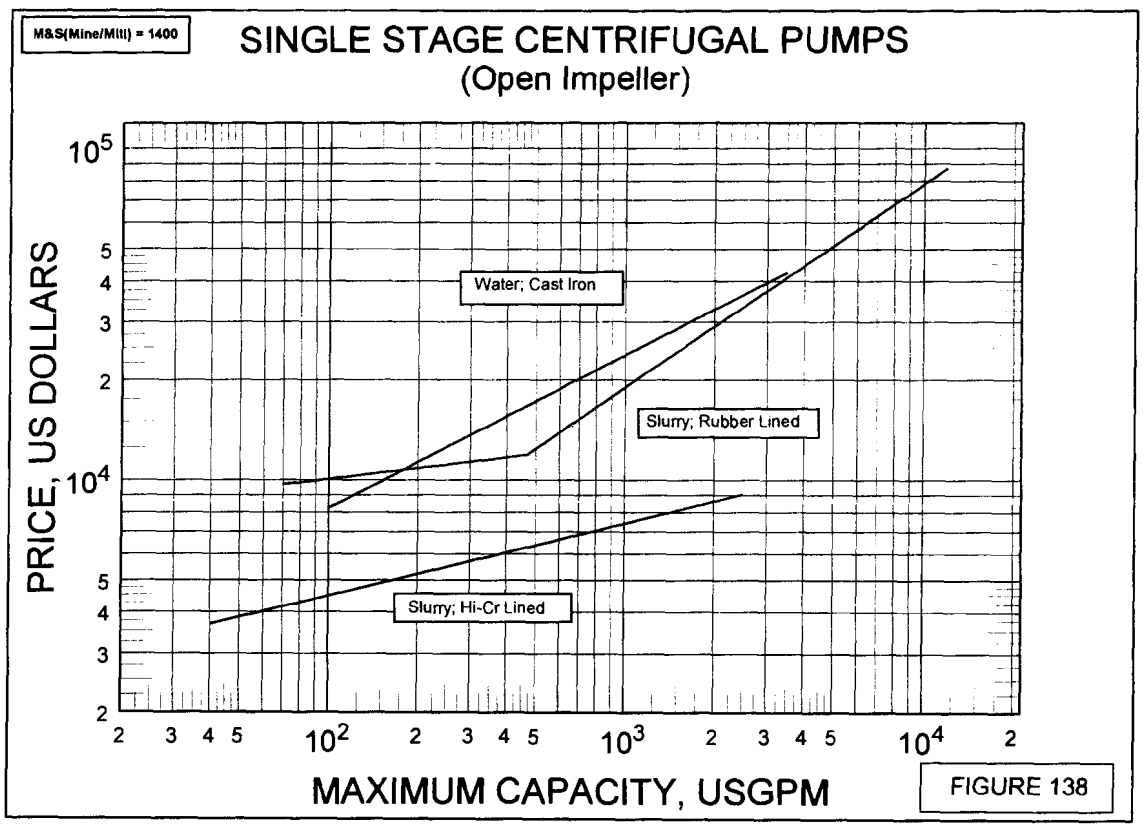


FIGURE 138

SINGLE STAGE CENTRIFUGAL PUMP
(Closed Impeller)

Price = aX^b, US Dollars

where **X is the maximum capacity in US gallons per minute for a pump operating at 100 ft of head near optimum efficiency**. Price excludes motor and drive.

Slurry, rubber lined; Range in X, USGPM:	70 to 463	a = 6112	b = 0.1078
	463 to 12500	a = 272.3	b = 0.6141
HC-250; Range in X, USGPM:	500 to 22500	a = 759.5	b = 0.5262

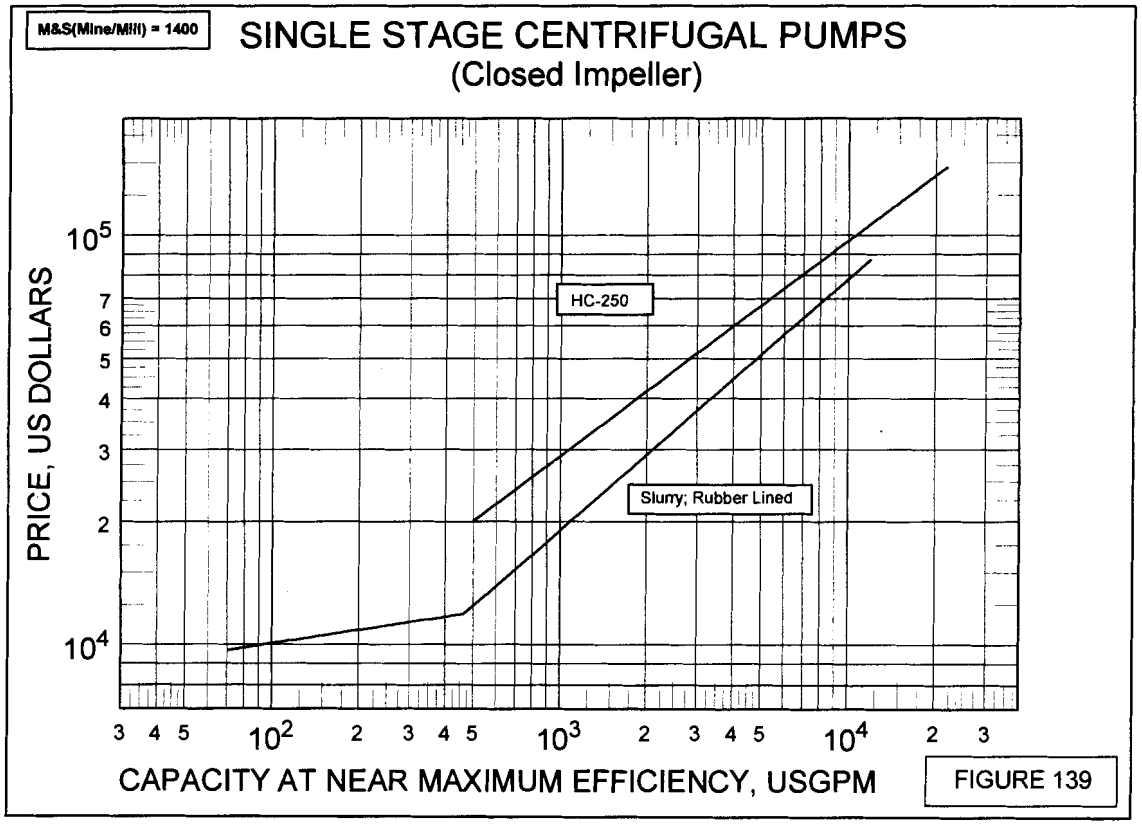


FIGURE 139

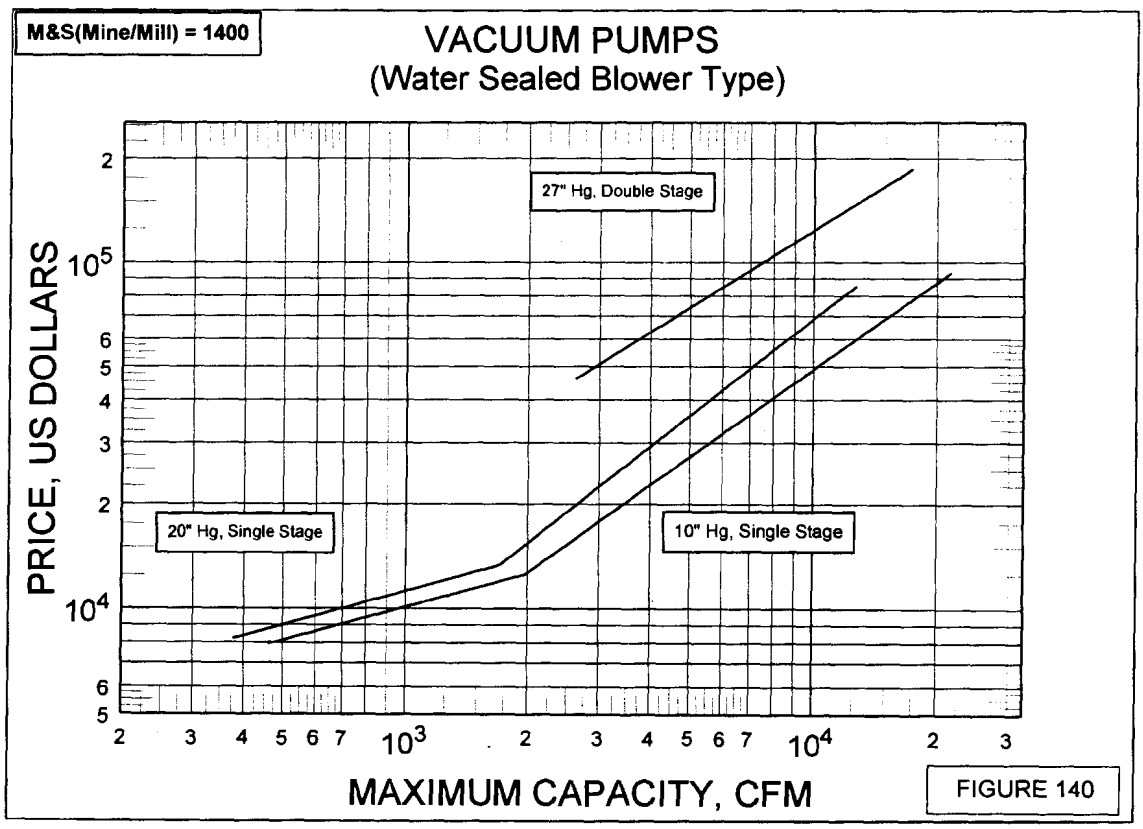
VACUUM PUMPS
(Water Sealed Blower Type)

Price = aX^b, US Dollars

where **X is the maximum capacity in cubic feet per minute**. Capacity based on 68 °F inlet temperature. Price excludes motor, base and noise suppressors.

Single Stage 10" Vac.; Range in X, cfm:	459 to 1977	a = 1135	b = 0.3162
	1977 to 21650	a = 21.5	b = 0.8388
Single Stage 20" Vac.; Range in X, cfm:	378 to 1716	a = 1209	b = 0.3223
	1716 to 12750	a = 13.5	b = 0.9258
Double Stage 27" Vac.; Range in X, cfm:	2620 to 17450	a = 138.6	b = 0.7371

Type	Brake HP Range
10"	17 to 485
20"	28 to 597
27"	108 to 680



VACUUM PUMPS
(Dry Sealed Blower Type)

Price = aX^b, US Dollars

where **X is capacity in cubic feet per minute**. Price excludes motor, base and noise suppressors.

15 in. Hg Vac.; Range in X, cfm:	239 to 510.5	a = 4692	b = 0.1146
	510.5 to 2565	a = 114.1	b = 0.7107
22 in. Hg Vac.; Range in X, cfm:	142 to 311.3	a = 4942	b = 0.1161
	311.3 to 2045	a = 302.4	b = 0.6028

Type, in.	Brake HP Range
15 in.	13.4 to 108.3
22 in.	19 to 157

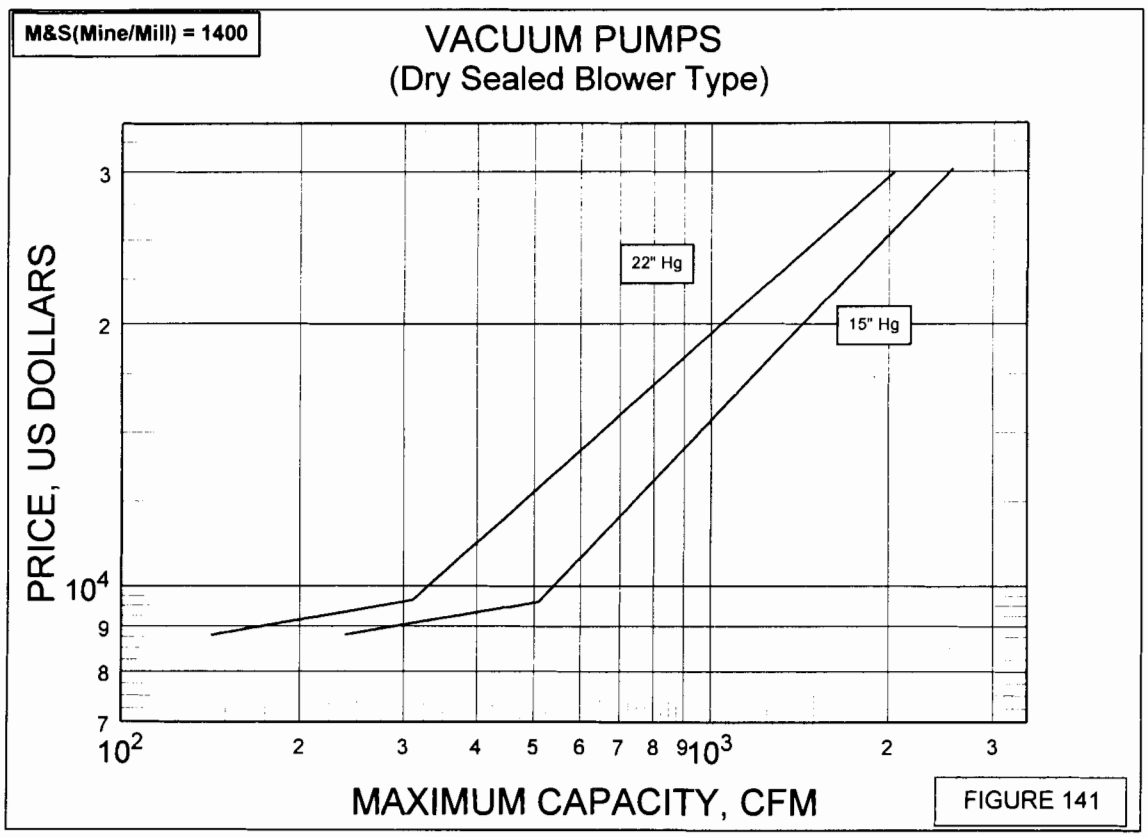


FIGURE 141

IX. Dust Collection Equipment

Dust collection equipment has been identified broadly as **dust collectors**, with differing principles of particle separation involved.

(1) Example Estimates - - - - - - - Page 217

(2) Dust Scrubbers, Wet (Dynamic, Venturi, Cyclone) - - - Page 218

(3) Dust Scrubbers, Wet (Centrifugal Collector Types) - - - Page 219

(4) Dry Cyclone Dust Collector - - - - - - Page 220

(5) Electrostatic Precipitators - - - - - Page 222

(6) Bag House Filters - - - - - - - Page 223

EXAMPLE ESTIMATES

Cost of dust collectors has been related to either capacity of a size parameter. A comparison of dust collector efficiencies is shown in the table below:

Cyclones	70 to 95 percent
Bag Filters	95 to 99.9 percent
Electrostatic Precipitators	80 to 99.5 percnt
Wet Scrubbers	90 to 99 percent

Example 1:

Compare the cost of several types of dust collectors at an M&S(M/M) Index of 1400. The capacity required is 10,000 cfm (cubic feet per minute).

From Figure 143, page 219, the cost of a wet centrifugal dust scrubber of the irrigated type is about US$30,000. Others of this class are of the same order of cost. A dry cyclone unit, from Figure 145, page 221, would have a diameter of about 4 feet at 10000 cfm. The cost from Figure 144, page 220, would be approximately US$11,000. On the other hand, a high voltage electrostatic precipitator without precleaners, from Figure 146, page 222, would cost (by extrapolation) around US$250,000.

From the above prices, it is easy to see which item of equipment is the least expensive. Apart from cost, determining factors in the selection are: (a) the necessary efficiency of the collector equipment, (b) the effect of water in or on any exit gases and (c) the allowable pressure drops.

DUST SCRUBBERS
(Wet Dynamic, Venturi, Cyclone)

Price = aX^b, US Dollars

where **X is capacity in cubic feet per minute**. Price excludes blowers, motors and ducting where applicable.

Wet dynamic; Range in X, cfm:	2000 to 35000	a = 4.44	b = 1.004
Venturi; Range in X, cfm:	5000 to 32943	a = 847.9	b = 0.3684
	32943 to 100000	a = 4.215	b = 0.8783
Cyclone; Range in X, cfm:	250 to 3468.6	a = 253.4	b = 0.3488
	3468.6 to 100000	a = 2.985	b = 0.8936

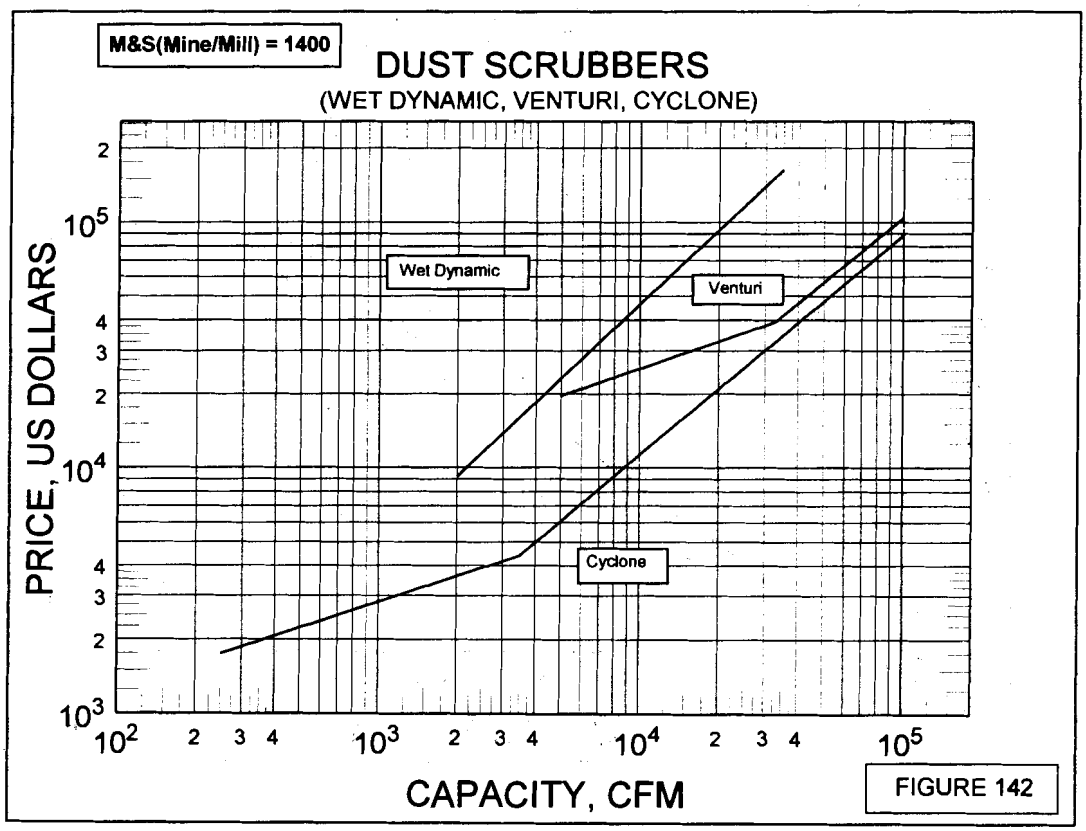


FIGURE 142

DUST SCRUBBERS
(Wet Centrifugal Type)

Price = aX^b, US Dollars

where **X is capacity in cubic feet per minute**. Price excludes blowers, motors and ducting.

Irrigated; Range in X, cfm: 2000 to 35000	a = 6.35	b = 0.9161
Kinetic; Range in X, cfm: 2000 to 35000	a = 10.62	b = 0.8553
Spray; Range in X, cfm: 2000 to 35000	a = 5.869	b = 0.9005

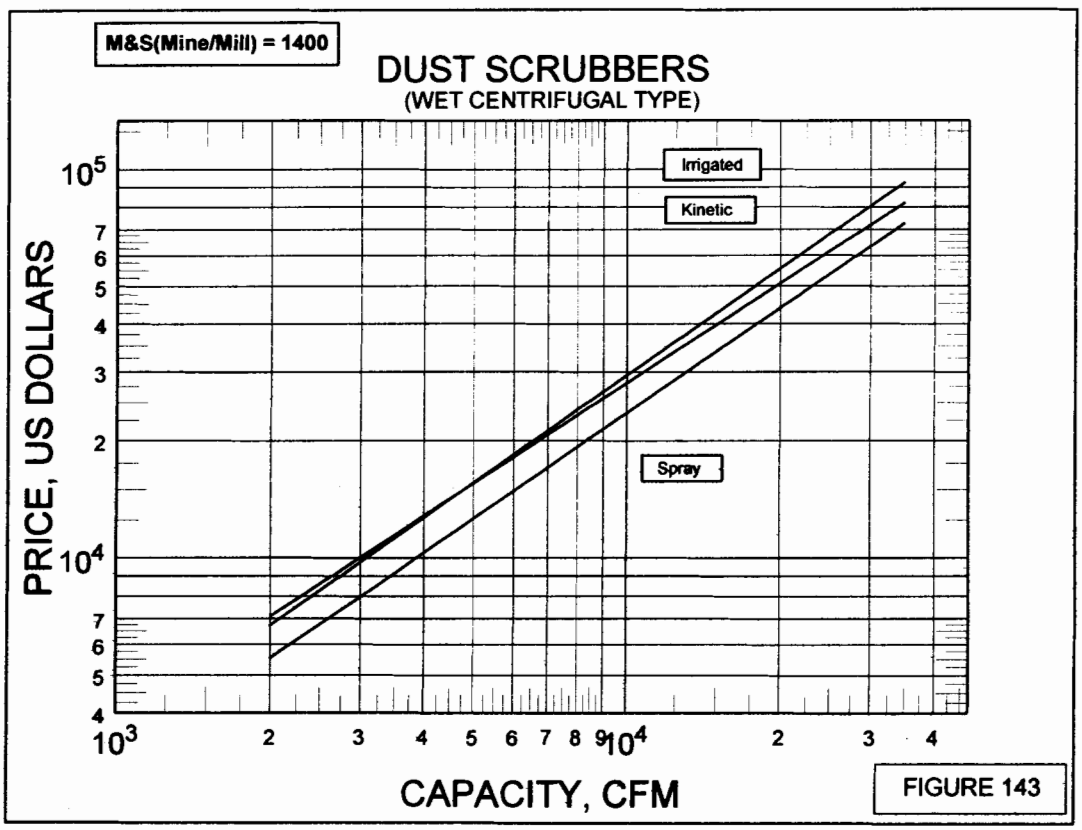


FIGURE 143

DRY CYCLONE DUST COLLECTOR

Price = aX^b, US Dollars

where **X is the cyclone diameter in feet**. Price excludes blowers, motors and ducting.

Range in X, ft:			
3 to 8.8	a = 3869	b = 0.7541
8.88 to 18	a = 1511	b = 1.184

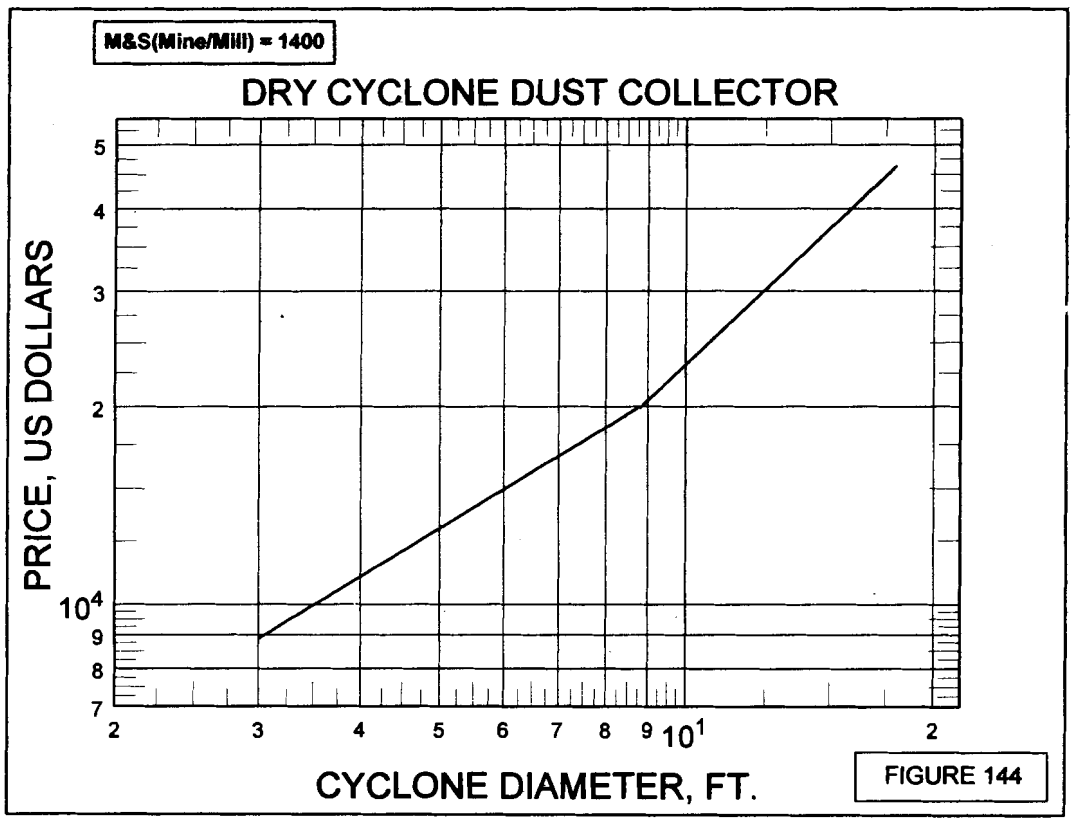


FIGURE 144

DRY CYCLONE DUST COLLECTOR (CONTINUED)

Figure 145 below shows how nominal dry cyclone capacity varies with cyclone diameter.

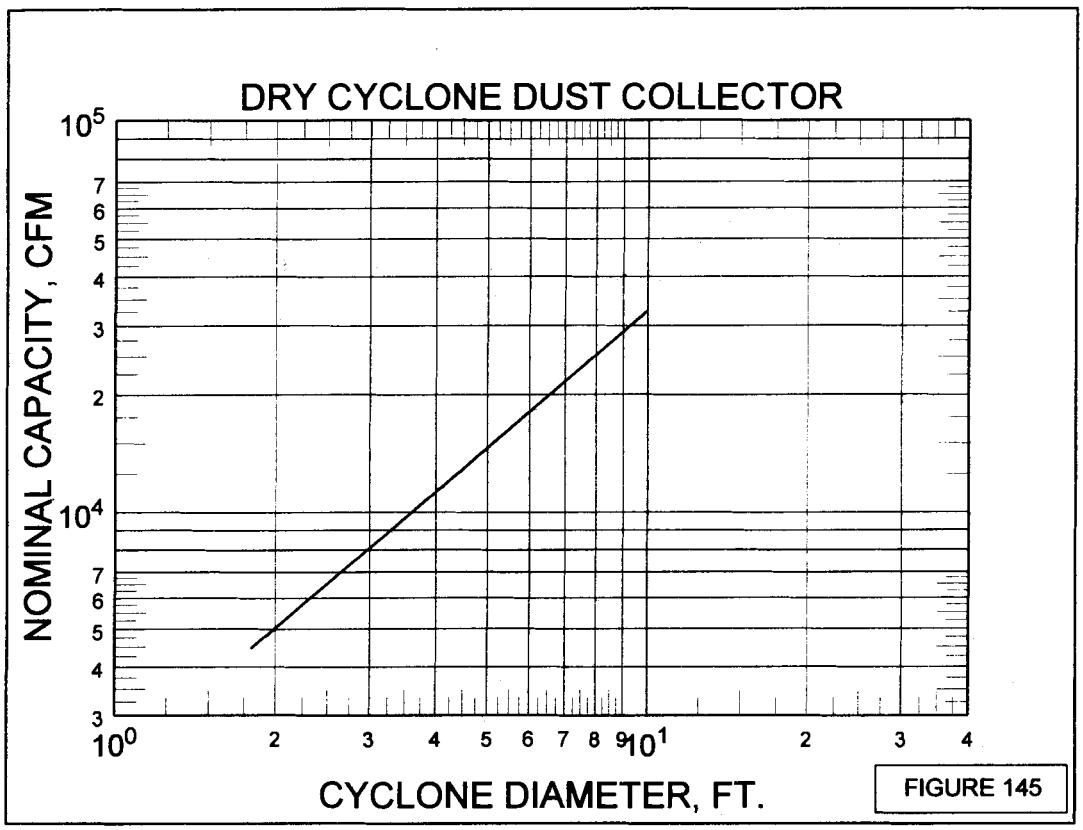


FIGURE 145

ELECTROSTATIC PRECIPITATORS

Price = aX^b, US Dollars

where **X is gas volume in cubic feet per minute**. Price excludes blowers and precleaners.

Range in X, cfm:			
12000 to 88000	a = 8807	b = 0.3623
88000 to 210000	a = 3.104	b = 1.06

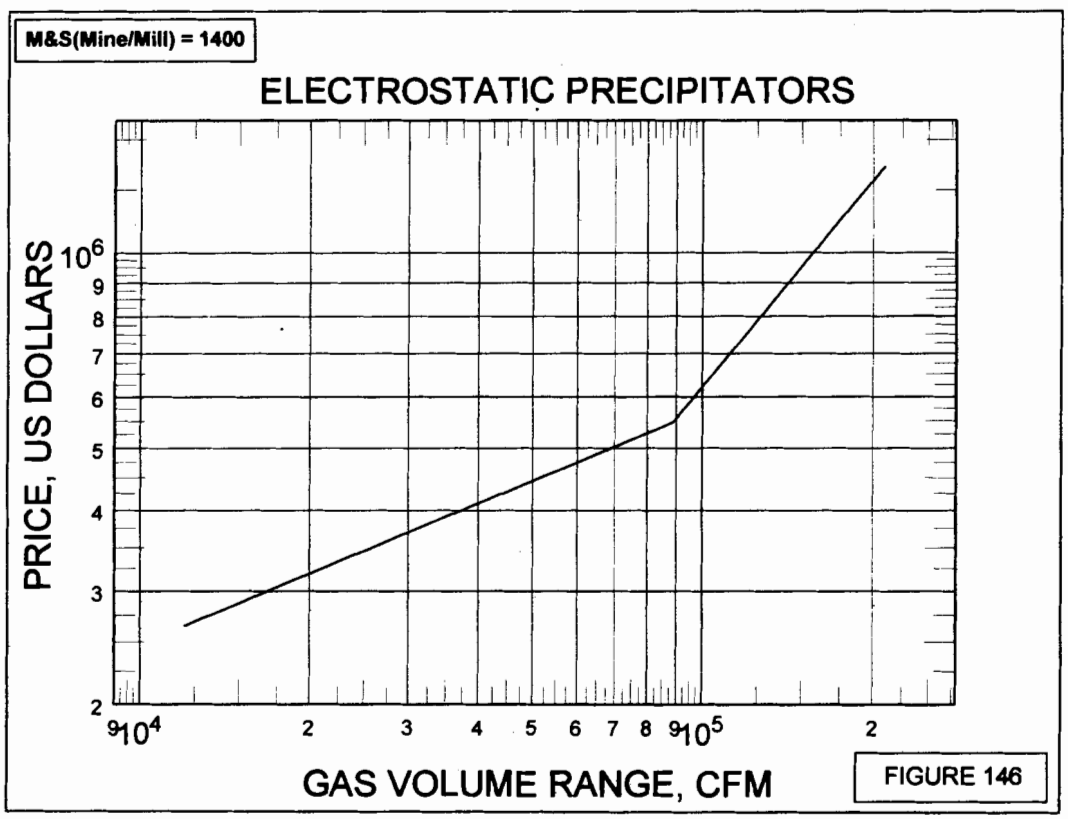


FIGURE 146

BAGHOUSE FILTERS

Price = aX^b, US Dollars

where **X is filter area in square feet**. Price includes standard attachments; excludes precleaners, blowers, motors and ducting.

Range in X, ft^2: 200 to 10000 a = 53.3 b = 0.8772

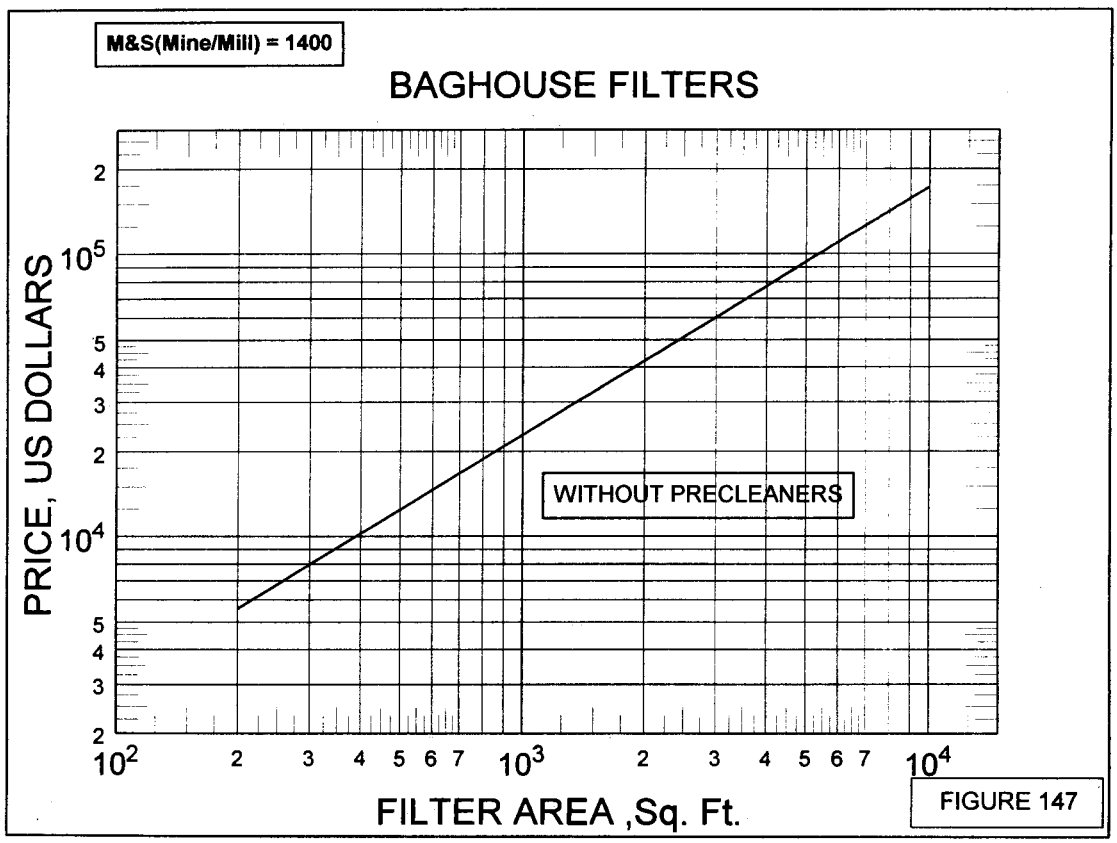


FIGURE 147

X. Solid-Solid Separation Equipment

Solid-Solid separation equipment has been placed in the following sub-categories: **Gravity/ Centrifugal Separators**, **Flotation Cells**, **Electrostatic Separators** and **Magnetic Separators**. These are broken into additional sub-sub-categories as necessary.

(1) Example Estimates - - - - - - - Page 225

(2) Gravity/Centrifugal Separators

(a) Reichert Cones - - - - - - - Page 227
(b) Humphrey Spirals - - - - - - - Page 228
(c) Shaking Tables - - - - - - - - Page 229
(d) Dense Medium Cyclones - - - - - Page 230
(e) Water Only Cyclones - - - - - - Page 232
(f) Heavy Media Separator Assemblies - - - - Page 234
(g) Baum Coal Jigs - - - - - - - - Page 235
(h) Fine Coal Jigs - - - - - - - - Page 236
(i) Circular Jigs - - - - - - - - Page 237
(j) Bendelari Jigs - - - - - - - - Page 238
(k) Mineral Jigs - - - - - - - - Page 239
(l) Centrifugal Concentrators - - - - - Page 240

(3) Flotation Cells

(a) Self-Aerating Flotation Cells - - - - - Page 241
(b) Conventional Flotation Cells - - - - - Page 242
(c) Column Cells - - - - - - - - Page 243

(4) Electrostatic Separators - - - - - - Page 244

(5) Magnetic Separators

(a) Wet Drum Magnetic Separator - - - - Page 245
(b) Wet Permanent Magnet, Alternate Pole Drum Separtor - Page 246
(c) Induced Roll Magnetic Separators - - - - Page 247
(d) Wet High Intensity Magnetic Separator - - - Page 248
(e) Permanent Guard Magnet Overband, Crossbelt - - Page 250
(f) Permanent Guard Magnet Overband, Incline - - - Page 251

EXAMPLE ESTIMATES

As previously mentioned, solid-solid separation equipment has been sub-divided into the various sub-categories. Prices of equipment items have been related to size, a capacity or a horsepower. In cases where cost has been related to size or horsepower, a table or figure showing corresponding capacities is provided.

Example 1:

A gold ore is to be treated by Reichert cones at a rate of 400 stpd. What will be the cost of the corresponding cones?

From page 227, approximately 400/88 or 5 units (rounded off) are required. Hence 5 x 88 = 440 stph of capacity will cost (from Figure 148, page 227) around US$420,000 at an M&S(M/M) Index of 1400.

Example 2:

Dense medium cyclones are required to treat 200 stph of material. What is the cost of cyclones for this purpose?

From Figure 152, page 231, a 20 inch cyclone has a nominal capacity of about 20 stph. Hence twelve (roughly 2 spares) of these in parallel should be adequate.

From Figure 151, page 230, the cost with ceramic liners is about 12 x 15,000 = US$180,000 at an M&S(M/M) Index of 1400.

Example 3:

A mineral jig will be employed to process 350 stpd of river gravel. What size jig is necessary and at what cost?

From the table on page 239, an appropriate size jig appears to be a 16 x 24 Duplex with 384 square inches of area per cell. This jig has a capacity range of 200 to 500 stpd.

From Figure 160, page 239, the price of a 500 tpd maximum capacity jig is US$23,000 at an M&S(M/M) Index of 1400.

Example 4:

Eight flotation cells, each of 400 ft^3 volume, are required for a rougher flotation section. How much will these cells cost?

From Figure 163, page 242, the cost of a single cell is US$36,000 at an M&S(M/M) Index of 1400. Hence the cost of 8 cells is 8 x 36000 or approximately US$288,000.

EXAMPLE ESTIMATES (CONTINUED)

The cost of launders will be approximately 8 x 8 x US$150 or about US$9600, where each cell is taken to have a length of about 8 ft. If paddles are necessary, then there is an additional cost of 8 x .09 x 36000 or US$26,000. Note that the motors likely needed are 60 HP per 2 cells and should be costed. If blowers are necessary, corresponding costs may be obtained in this manual. Note that about 1 to 1.5 ft^3/min of air is necessary per cubic foot of cell capacity.

Example 5:

A wet high intensity magnetic separator is required to treat 25 stph of hematite ore. What is the price of a suitable unit?

From Figure 170, page 249, the horsepower for this capacity is around 45. Figure 169, page 248, shows that the cost will be around US$520,000 at an M&S(M/M) Index of 1400.

Example 6:

It is desired to purchase an electrostatic separator with a 14-inch rotor of 18-inch length. What will be the cost and what is the capacity.

The cost parameter is 14 x 18 x 1 = 252 sq. in. From Figure 165, page 244, the cost is estimated as US$27,000 at an M&S(M/M) Index of 1400. The power supply, as noted on page 244, must be priced separately. In this case, add an additional US$14,000.

The effective rotor length is corrected for the ends, where 2 inches are ineffective at each end. Thus 18 - 4 = 14 inches of effective rotor length. Since capacity is estimated as approximately 90 lb per hour per inch of effective rotor length, then capacity is about 90 x 14 or 1260 lbs per hour.

REICHERT CONES

Price = aX^b, US Dollars

where **X is nominal capacity in short tons per hour**. Cost is for feed box, supports, rougher-cleaner-recleaner-scavenger unit per 88 tph of capacity. Capacity based on 30% feed solids of size -2 mm.

Range in X, stph: 88 to 1056 a = 959.6 b = 1.00

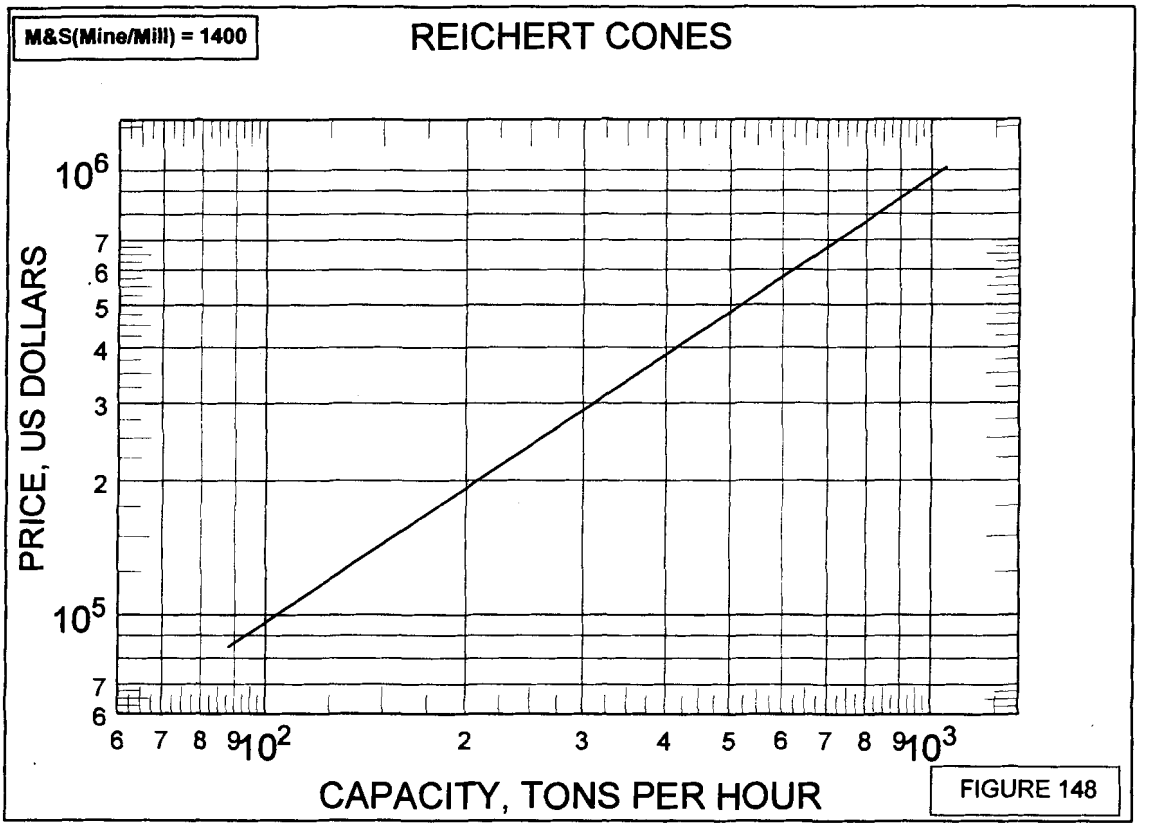


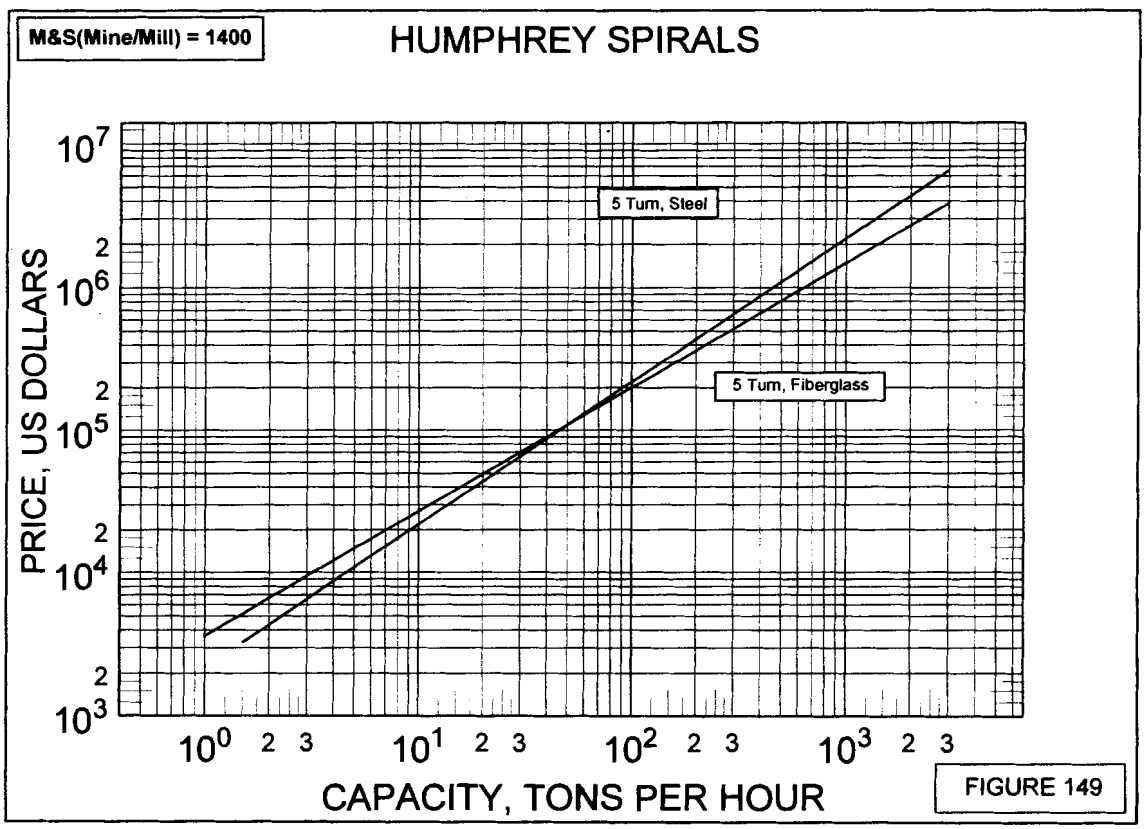
FIGURE 148

HUMPHREY SPIRALS

Price = aX^b, US Dollars

where **X is capacity in short tons per hour**. Cost includes standard unit with feed box and supports. Nominal capacity is 1.5 stph per unit.

5-turn Steel; Range in X, stph: 1.5 to 3000	a = 3600	b = 0.87
5-turn Fiberglass; Range in X, stph: 1 to 3000	a = 2165	b = 1.00



SHAKING TABLES

Price = aX^b, US Dollars

where **X is deck area in square feet**. Price includes motor, rubber top and riffles. Price refers to single deck units. *To estimate cost of double and triple deck unit, multiply single deck cost by 2.35 or 3.55 respectively.*

Rectangular Deck; Range in X, ft^2: 5 to 90 a = 7041 b = 0.3619

Diagonal Deck; Range in X, ft^2: 5 to 90 a = 7698 b = 0.2147

Deck size, ft x ft	Area, sq. ft.	Capacity, tpd	Horsepower
2 x 4.2	8.33	1 to 5	1/6
4 x 9	36	10 to 90	1
6 x 15	90	15 to 150	2

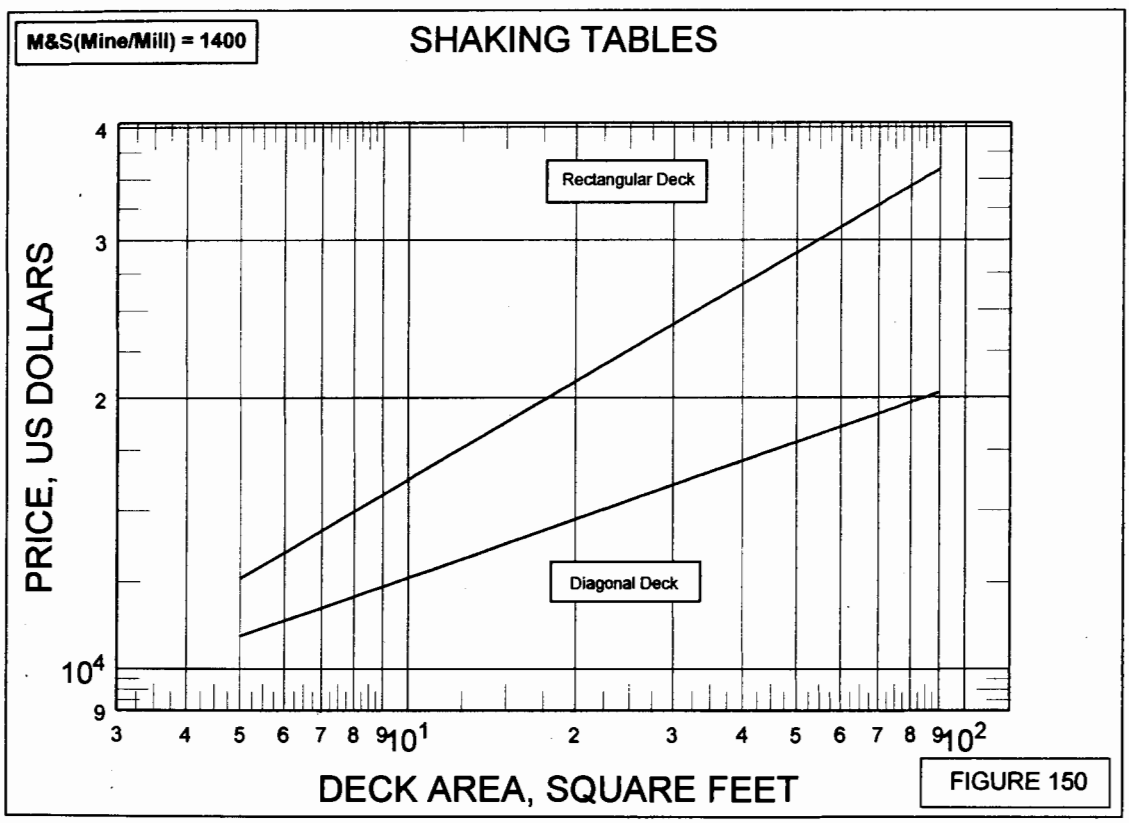


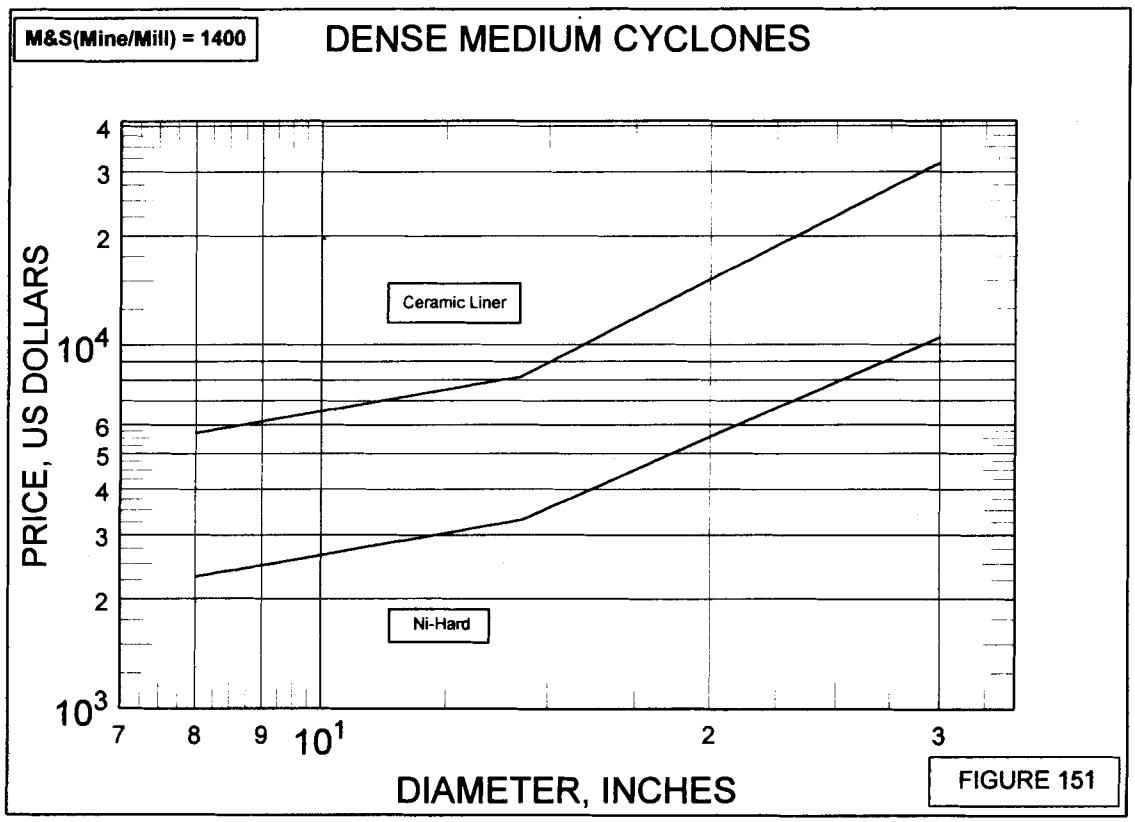
FIGURE 150

DENSE MEDIUM CYCLONES

Price = aX^b, US Dollars

where **X is cyclone diameter in inches**. Cost includes liners and fittings.

Ceramic Liner; Range in X, in.:	8 to 14.3	a = 1568	b = 0.6195
	14.3 to 30	a = 62.93	b = 1.829
Ni-Hard Liner; Range in X, in.:	8 to 14.3	a = 629.7	b = 0.6214
	14.3 to 30	a = 51.59	b = 1.561



DENSE MEDIUM CYCLONES (CONTINUED)

Figure 152 below shows the variation of nominal capacity versus dense medium cyclone diameter. The graph is intended as a rough approximation only.

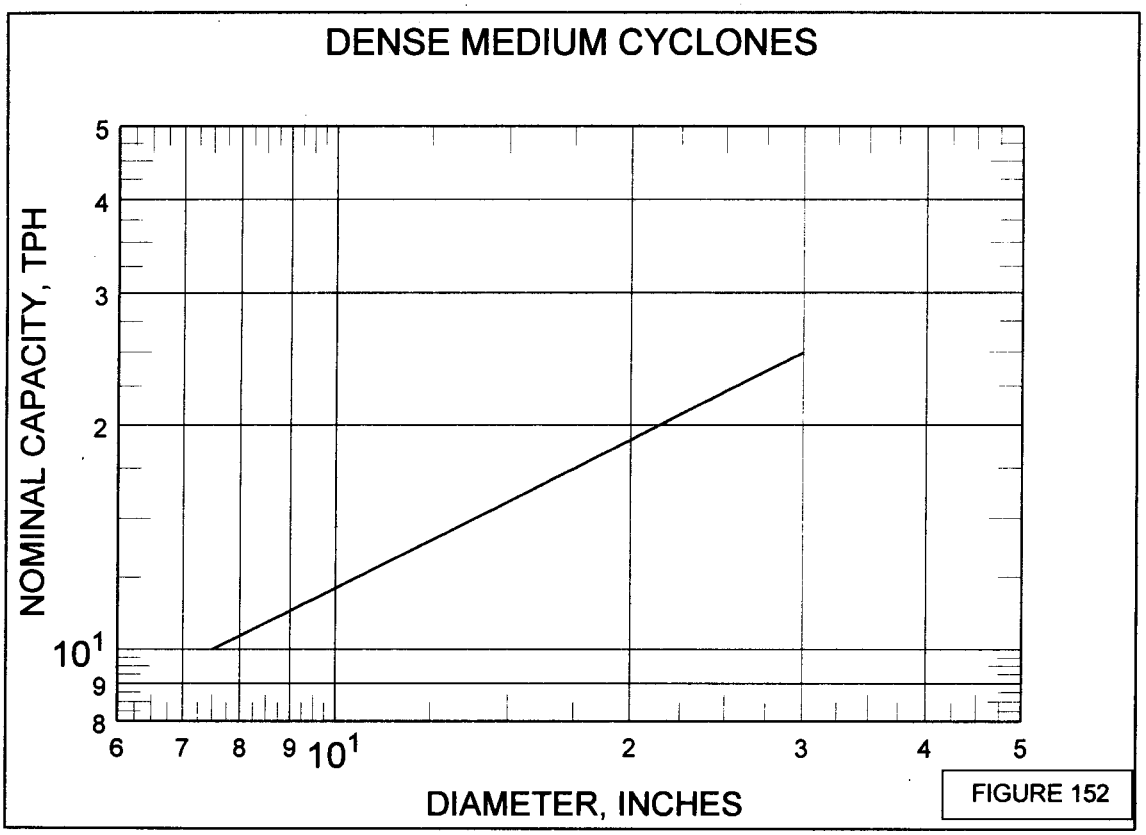


FIGURE 152

WATER ONLY CYCLONES

Price = aX^b, US Dollars

where **X is cyclone diameter in inches**. Cost includes all fittings and liners.

Iron/Steel Housing; Range in X, in.:	4 to 24	a = 1602	b = 0.7416
Polyester Housing; Range in X, in.:	5 to 28	a = 443.2	b = 1.054

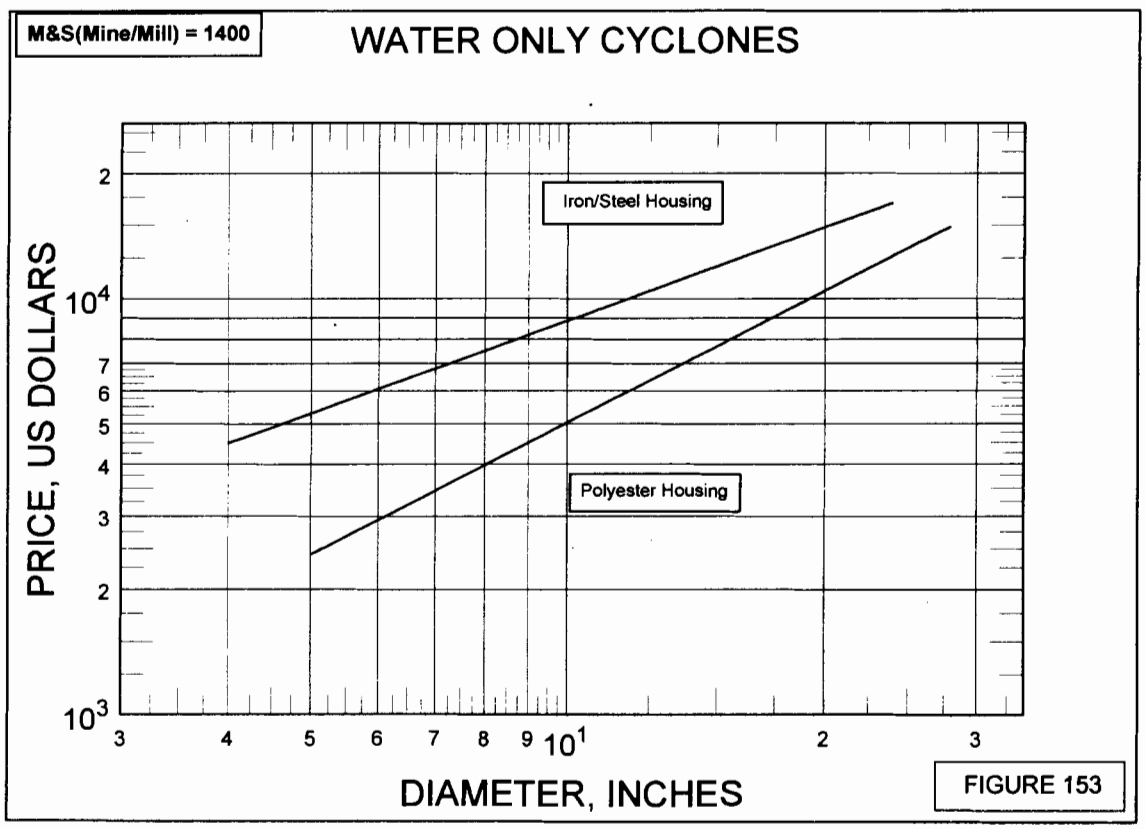


FIGURE 153

WATER ONLY CYCLONES (CONTINUED)

Figure 154 below shows nominal capacity (stph, dry) versus cyclone diameter for water only cyclones. The curve should be employed for rough estimations only.

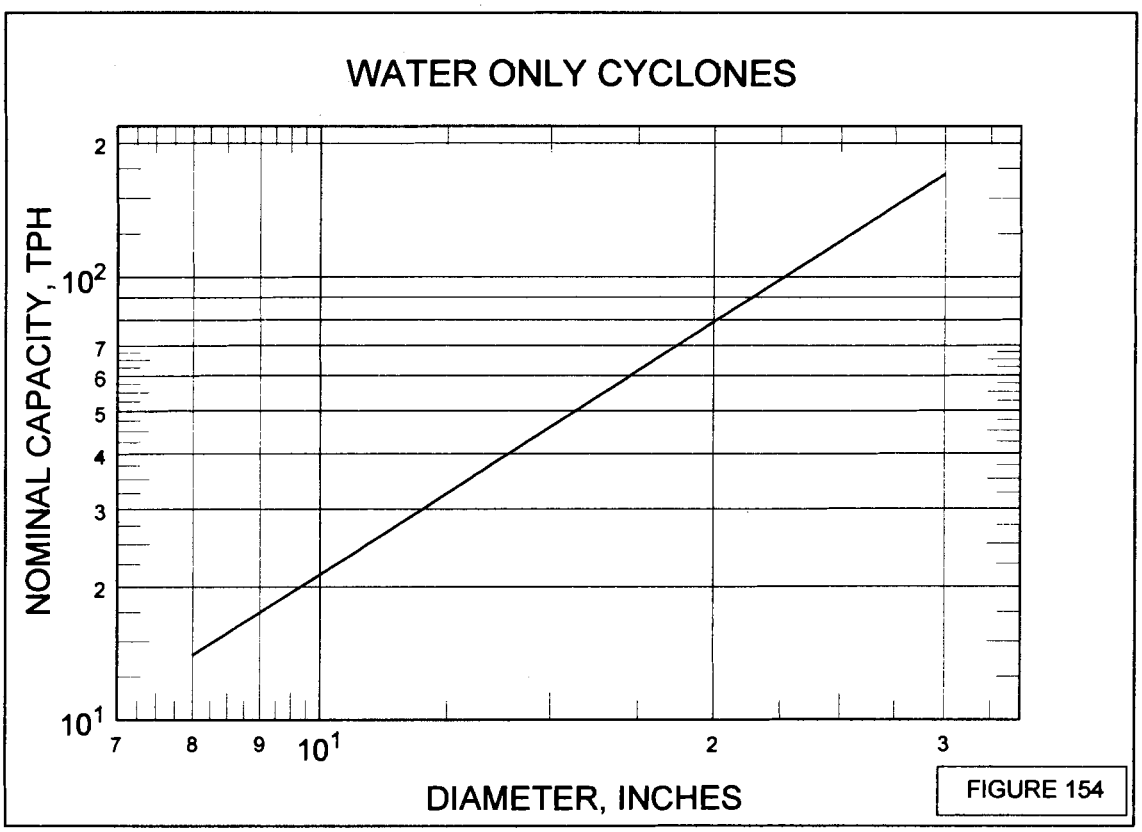


FIGURE 154

HEAVY MEDIA SEPARATOR ASSEMBLIES

Price = aX^b, US Dollars

where **X is capacity in dry short tons per hour**. Price includes heavy media drum or cone separator, screens densifier, magnetic separator, pumps, hoppers, sumps, launders and motor controls.

Range in X, stph: 15 to 20 a = 21860 b = 0.23

Drum, Dia x Length	Cone Diameter, ft	Capacity Range, stph
4 ft x 4 ft	5	5 to 20
6 ft x 6 ft	7	20 to 70
8 ft x 8 ft	10	40 to 140
10 ft x 10 ft	14	8- to 250

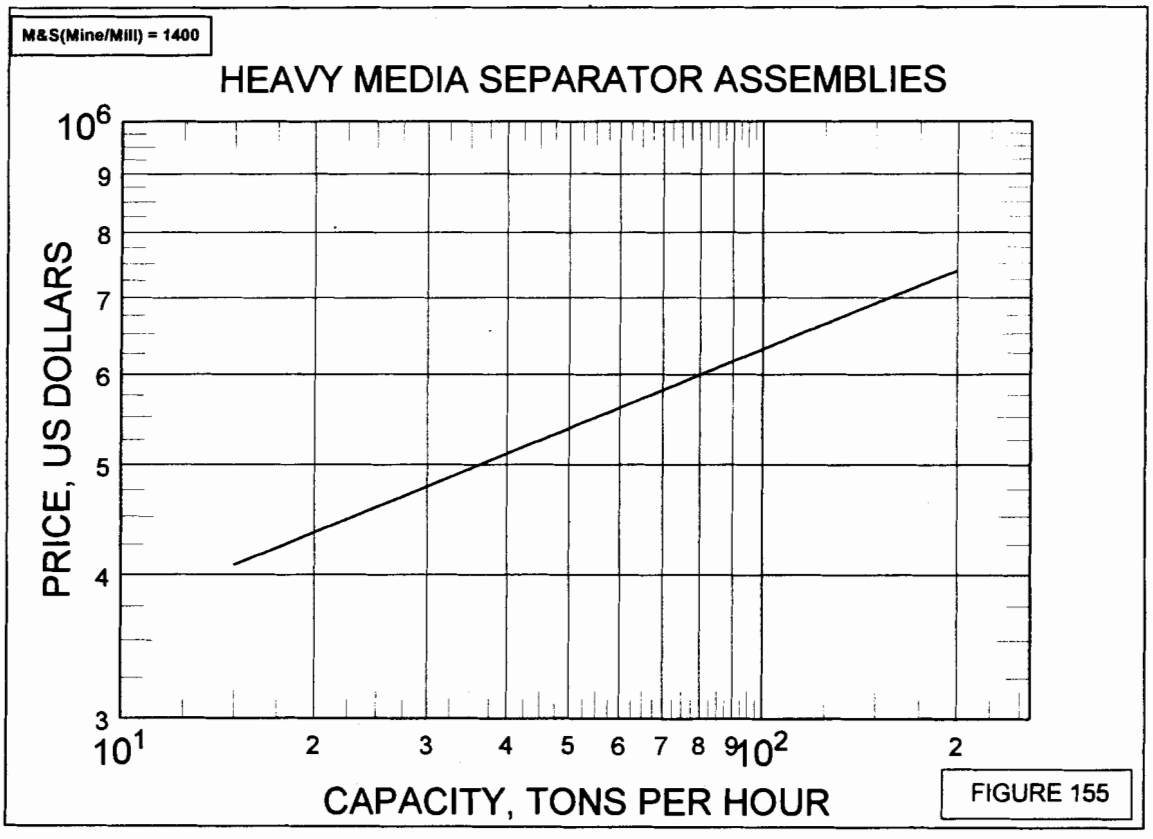


FIGURE 155

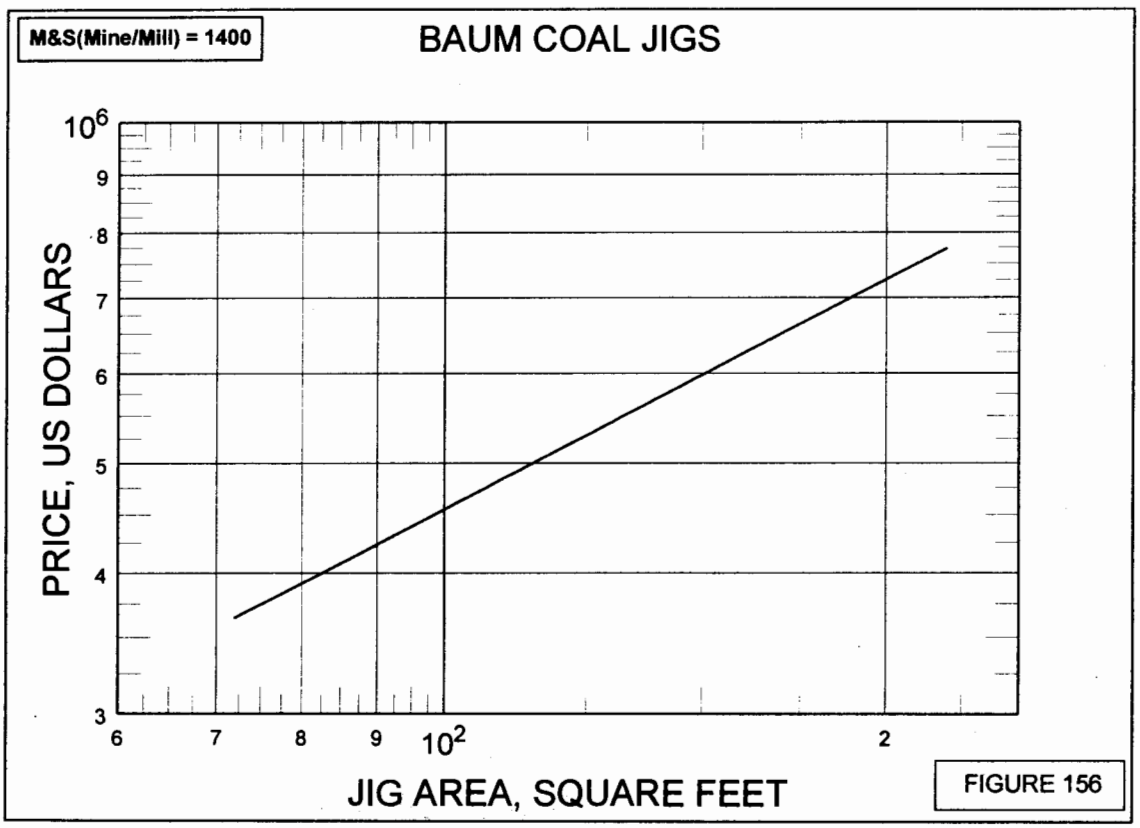
BAUM COAL JIGS

Price = aX^b, US Dollars

where **X is the jigging area in square feet**. Price includes push water box, bypass flume and blower.

Range in X, ft^2: 72 to 220 a = 20340 b = 0.6746

Size, ft	# Compartments	# Cells	Jigging Area, sq. ft.
6	2	4	72
7	2	5	105
7	3	8	168



FINE COAL JIGS

Price = aX^b, US Dollars

where **X is jig area in square feet.**

Range in X, ft^2: 178 to 267 a = 114800 b = 0.434

Typical Jig Area	Nominal Capacity, stph	Horsepower
178 ft^2	400	130
222 ft^2	500	170
267 ft^2	600	230

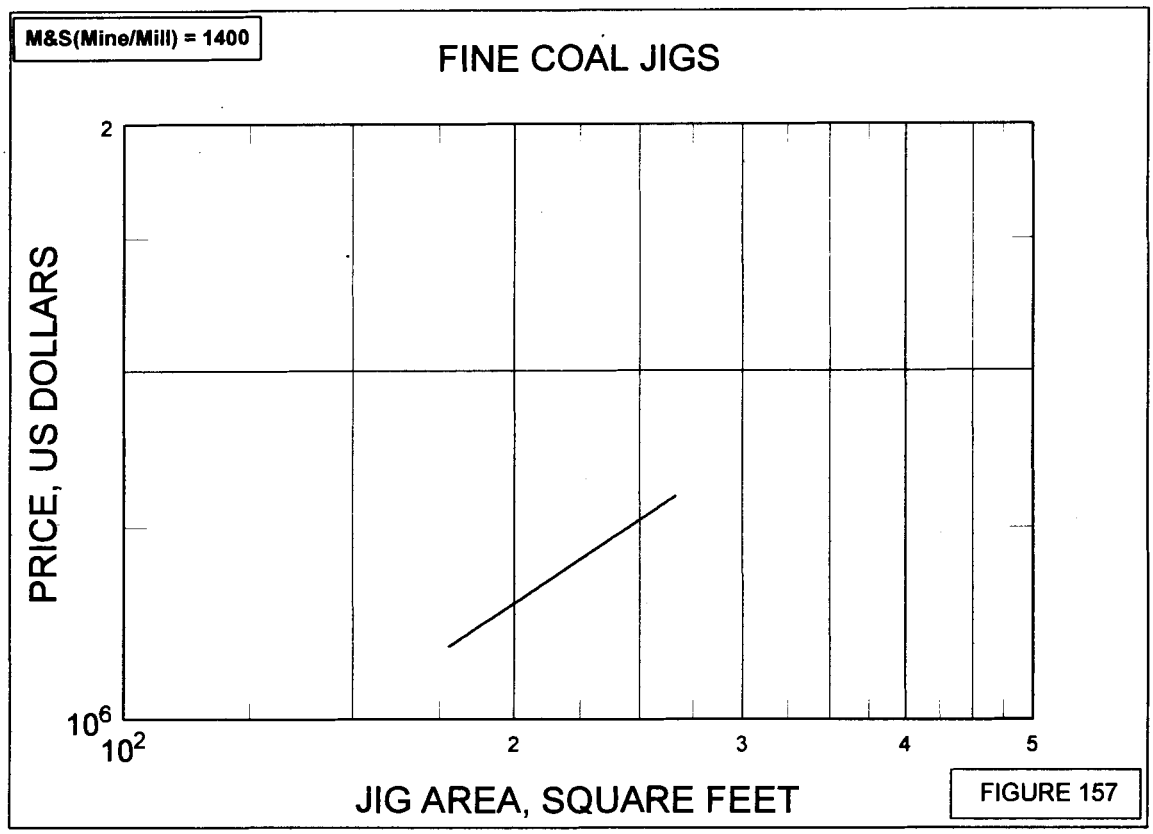


FIGURE 157

CIRCULAR JIGS
(Radially Arranged Jig Cells)

Price = aX^b, US Dollars

where **X is total jig area in square feet**. Price includes motor, concentrate/tailing chutes, reduction gearbox, skimmer with blades, screen grid, hutch, diaphragm, and walking platform. For gold and heavy metallics/non-metallics.

Range in X, ft^2: 48 to 450 a = 1153 b = 0.9477

Approx. Diameter, ft.	Cells per Jig	Horsepower	Capacity, cu. yd/hr
8	1	3	20 to 40
12	3	7.5	40 to 100
18	6	10	100 to 200
24	12	12.5	200 to 400

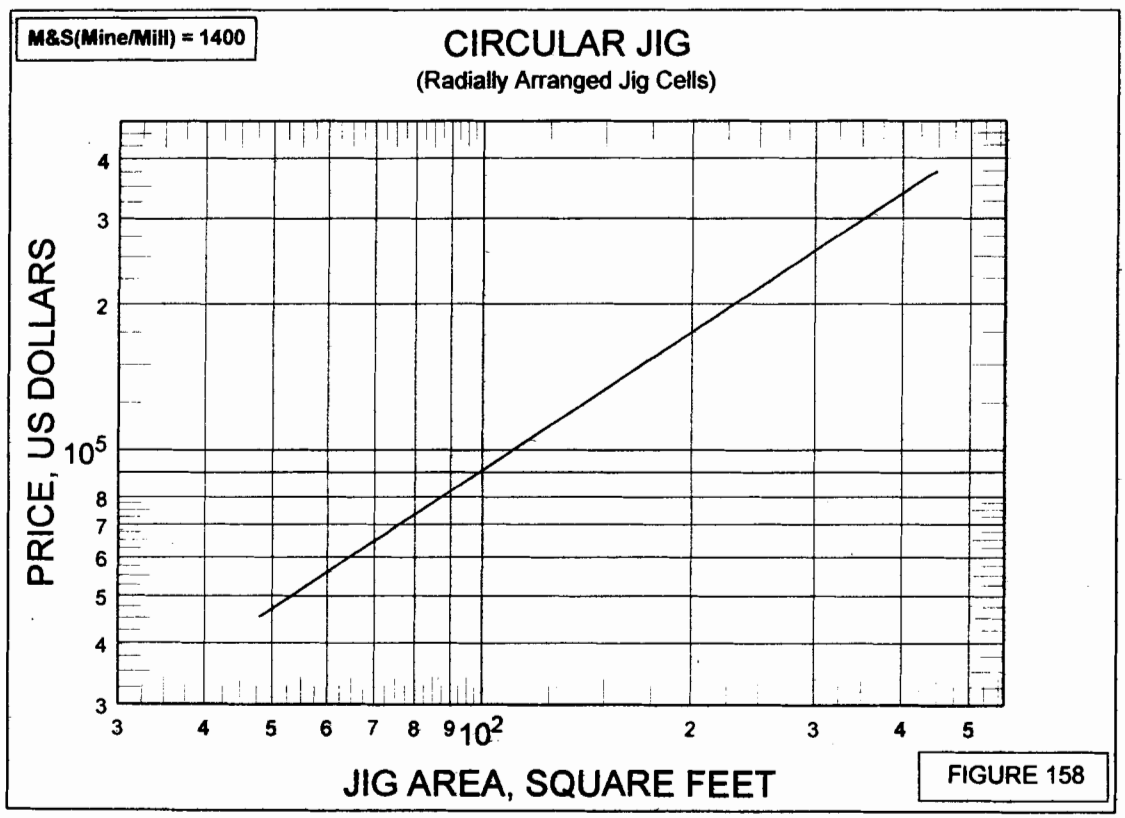


FIGURE 158

BENDELARI JIGS

Price = aX^b, US Dollars

where **X is jig area in square inches**. Price includes internal mechanical diaphragm, motor and drive system; excludes platforms, walkways and discharge launders. Add about 40% for excluded extras.

1-Cell; Range in X, in^2:	48 to 230	a = 2265	b = 0.1462
	230 to 1764	a = 293.6	b = 0.5223
2-Cell; Range in X, in^2:	144 to 1764	a = 175.8	b = 0.6767
3-Cell; Range in X, in^2:	144 to 1764	a = 232.5	b = 0.6749

Size, in. x in.	1-Cell Capacity*, stph	Horsepower
12 x 12	to 0.5	0.5
18 x 18	1.5 to 2	0.5
26 x 26	2 to 4	2
36 x 36	3 to 6	3
42 x 42	4 to 8	3

*Capacity increases roughly in proportion to the number of cells.

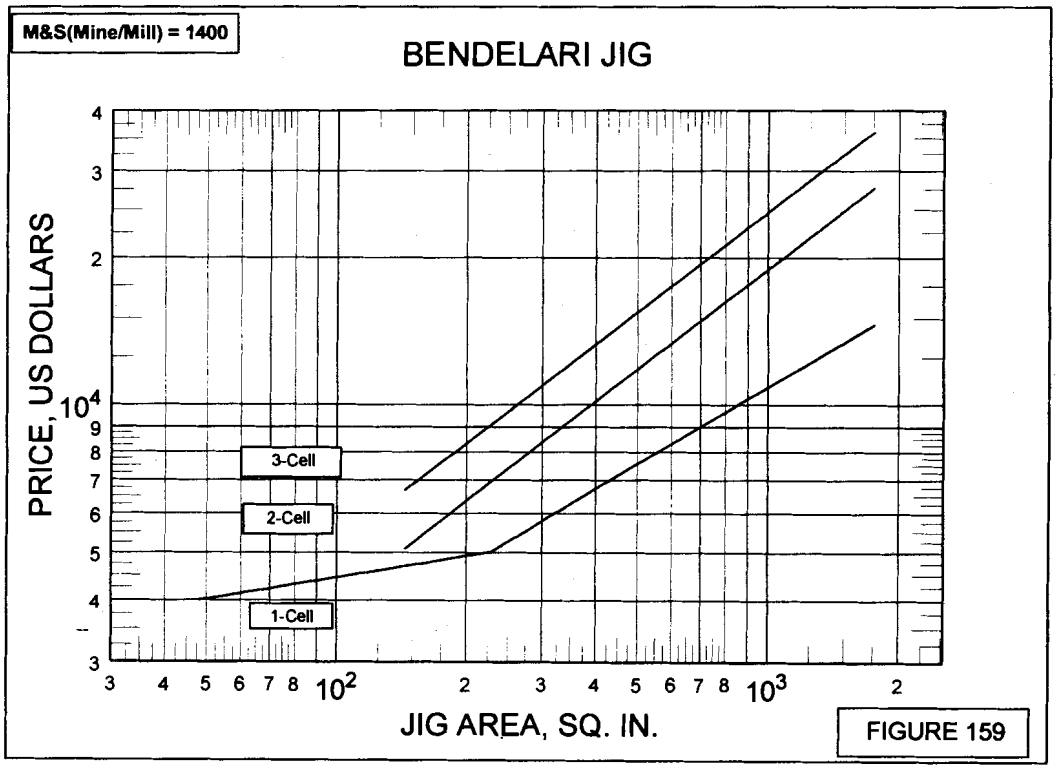


FIGURE 159

MINERAL JIGS

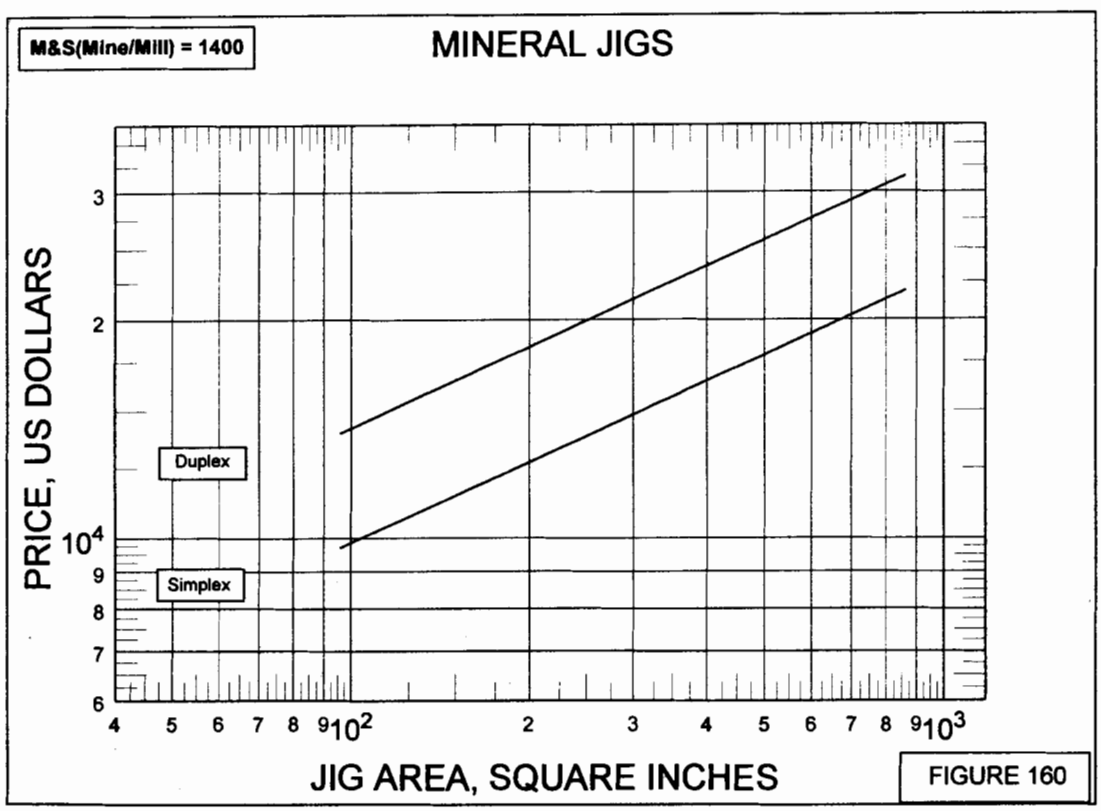
Price = aX^b, US Dollars

where **X is jigging area in square inches**. Price includes motor; excludes ragging.

Duplex; Range in X, in^2: 96 to 864 a = 2580 b = 0.3701

Simplex; Range in X, in^2: 96 to 864 a = 1791 b = 0.3701

Jig Size, in. x in.	Capacity Range, tpd
8 x 12 simplex	1 to 35
8 x 12 duplex	15 to 50
12 x 18 duplex	50 to 200
16 x 24 duplex	200 to 500
24 x 36 duplex	400 to 1000



CENTRIFUGAL CONCENTRATORS
(Knelson Separator)

Price = aX^b, US Dollars

where **X is maximum feed rate in tons per hour**.

300g's Concentrator; Range in X, tph: 0.1 to 100	a = 20740	b = 0.5458
60g's Auto Discharge; Range in X, tph: 4 to 100	a = 14450	b = 0.5846
60g's Manual Discharge; Range in X, tph: 0.1 to 40	a = 13950	b = 0.308

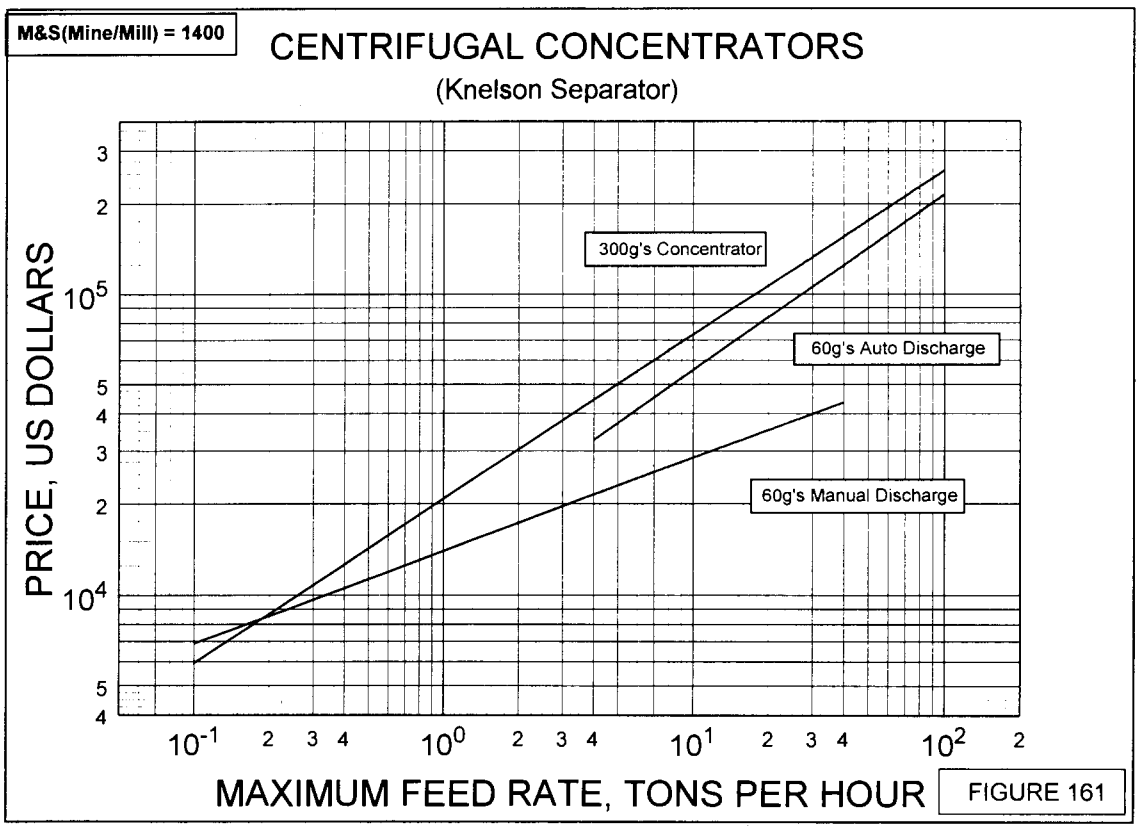


FIGURE 161

SELF AERATING FLOTATION CELLS

Price = aX^b, US Dollars

where **X is cell capacity in cubic feet**. Price is based on a 10-cell row and includes paddles with drives and feed/junction/discharge boxes; excludes motors and launders.

Range in X, ft^3:			
	11 to 685	a = 3205	b = 0.3953
	685 to 1000	a = 35.03	b = 1.087

Note: Data on page 242 can be useful to estimate horsepower and tons per day treated per cell.

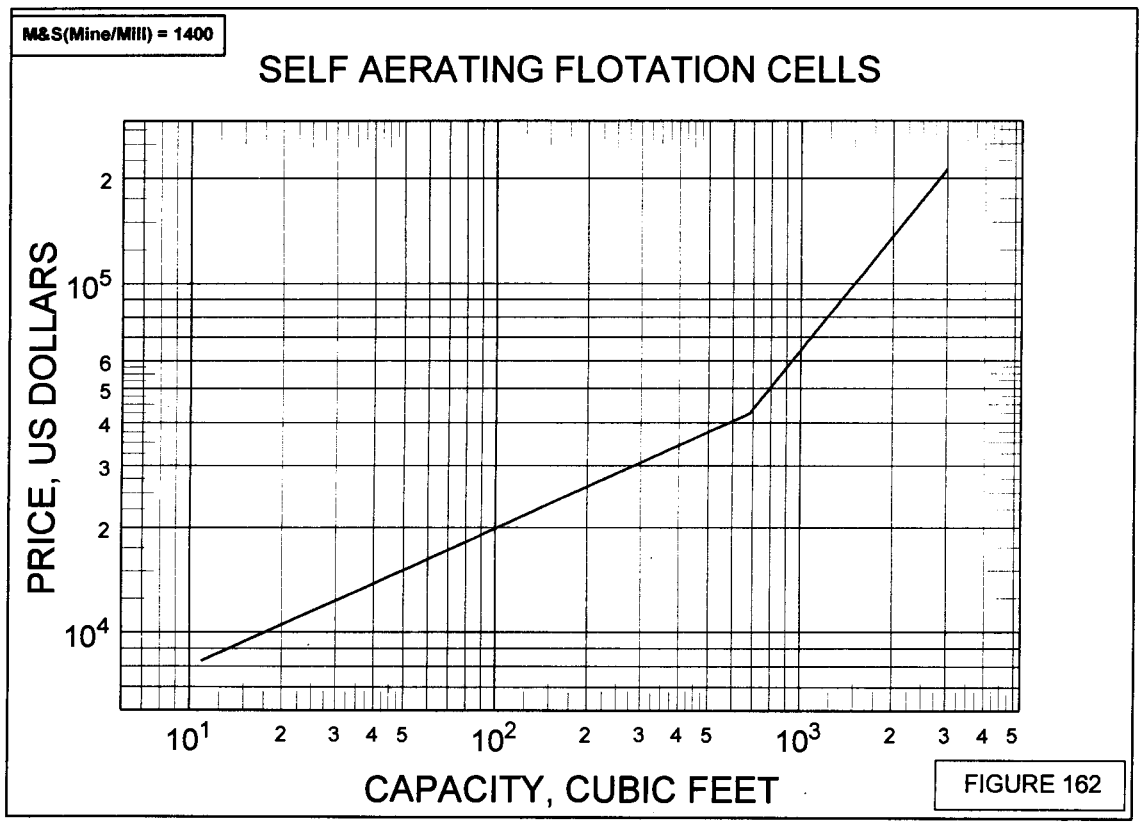


FIGURE 162

CONVENTIONAL FLOTATION CELLS

Price = aX^b, US Dollars

where **X is cell capacity in cubic feet**. Cost includes motor guard, feed/discharge boxes; excludes motors, launders and paddles. Launders are approximately US$150 per foot of cell length. Estimate paddle cost as 9% of the cost per cell. For blowers, about 1 to 1.5 cfm is required per cubic foot of cell volume.

Range in X, ft^3:			
9 to 450	a = 3223	b = 0.3999
450 to 1000	a = 264.2	b = 0.8089

Cell Volume	----------Tons per day per cell----------					
cubic feet	Cu	Fe	Zn	Pb	Coal	HP/cell
25	25-40	30-40	20-30	25-40	20-25	7.5
40	50-80	60-80	40-60	50-80	40-50	10
50	80-120	100-120	50-85	80-120	60-70	15
100	130-200	165-200	85-140	130-200	100-115	25
180	200-300	250-310	125-210	200-300	150-175	40
300	400-600	500-600	150-425	400-600	300-350	50
500	-	-	-	-	-	80
1275	-	-	-	-	-	150

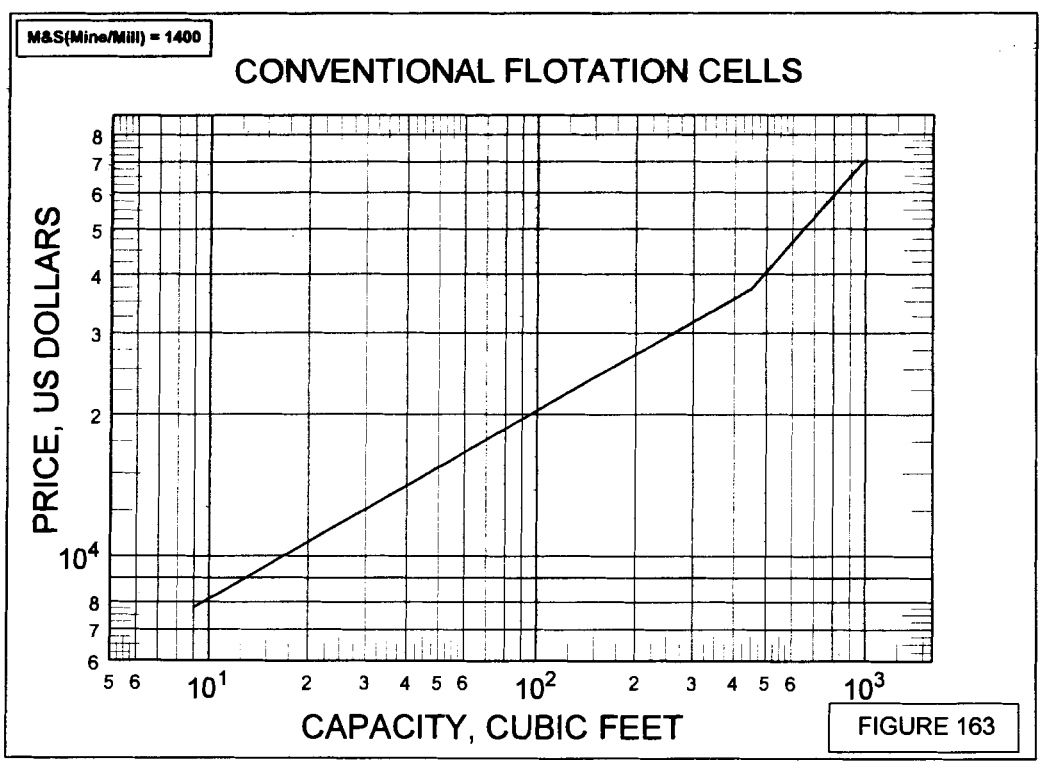


FIGURE 163

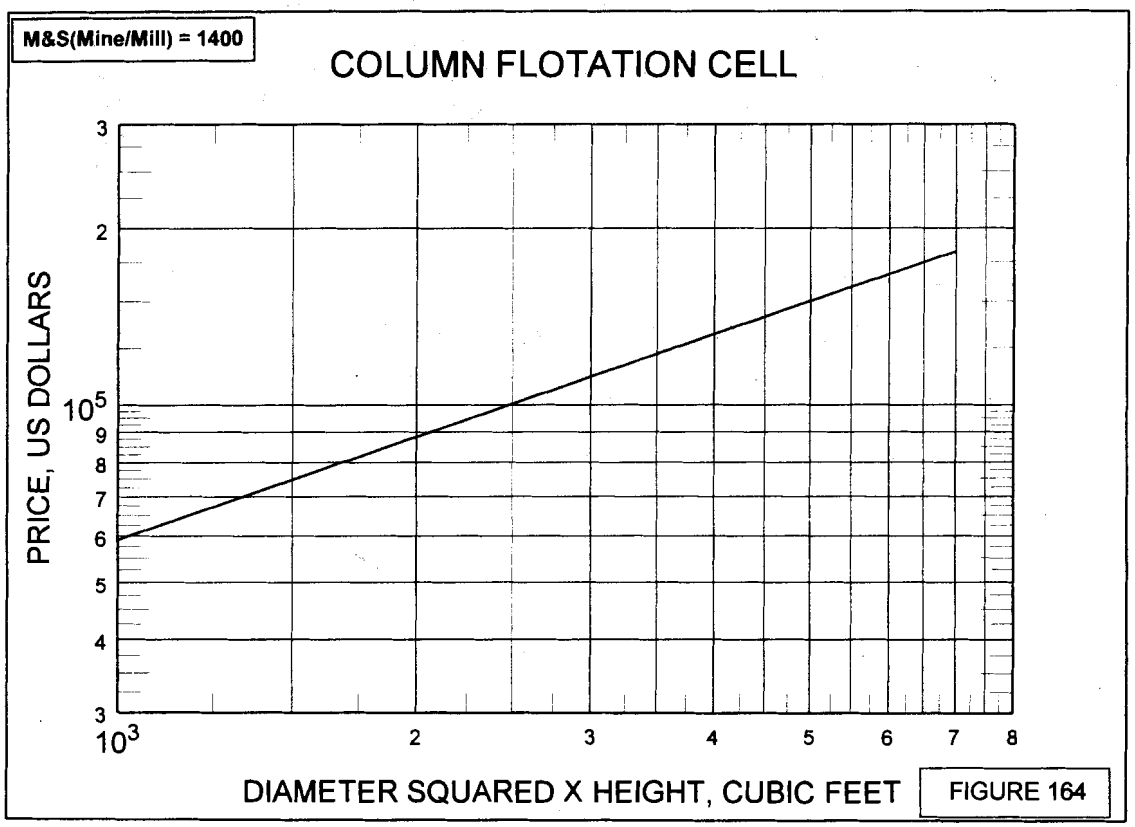
COLUMN FLOTATION CELLS

Price = aX^b, US Dollars

where **X, in cubic feet, is cell diameter squared times cell height**. Price excludes pumps, feed and discharge lines.

Range in X, ft^3: 1000 to 7000 a = 1074 b = 0.5799

Note: Cells are typically 30 to 45 ft in height; diameters vary from 1.5 to 12 ft diameter and more. Remember that the retention time of a cell is its volume divided by the volume flow rate of feed slurry.



ELECTROSTATIC SEPARATORS
(Carpco Type)

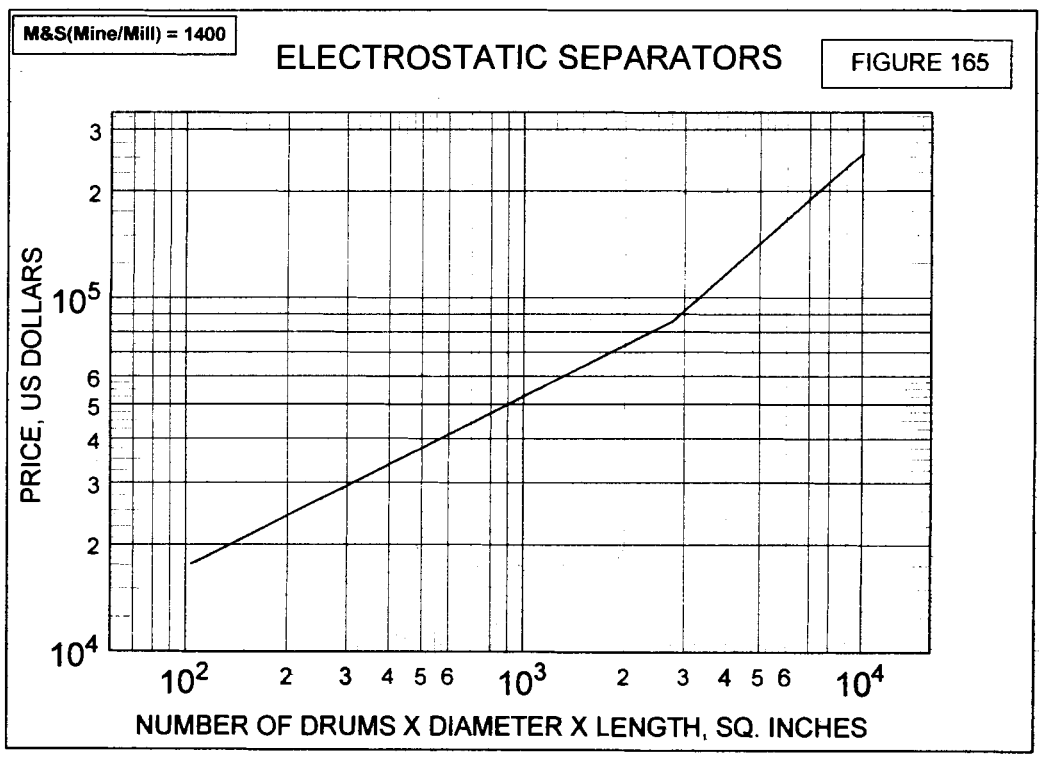
Price = aX^b, US Dollars

where **X, in square inches, is the product of drum (rotor) diameter times drum length times number of drums in the unit**. Cost includes motor; excludes power supply. To estimate the cost of a power supply (High Voltage rectifier, AC wiper and controller) add US$14,000 at an M&S(M/M) Index of 1400.

Range in X, in^2:	105 to 2768	a = 1866	b = 0.4826
	2768 to 10080	a = 101.9	b = 0.8494

Rotor Dia. inch	Length, inch	# Rotors	Horsepower
10	60	4	3
14	18	1	0.5
14	60	1	1
14	60	4	4

Note: Estimate capacity as 90 lb/hr/inch of effective rotor length. Subtract 4 inches from length to estimate effective rotor length.



WET DRUM MAGNETIC SEPARATORS

Price = aX^b, US Dollars

where **X, in square inches, is roll diameter times length**. Capacity is approximately 80 USGPM per foot of roll length.

36-inch Drum; Range in X, in^2:	2592 to 4320	a = 129.6	b = 0.6638
30-inch Drum; Range in X, in^2:	1440 to 2225	a = 4.779	b = 1.086
	2225 to 3600	a = 193.2	b = 0.6062

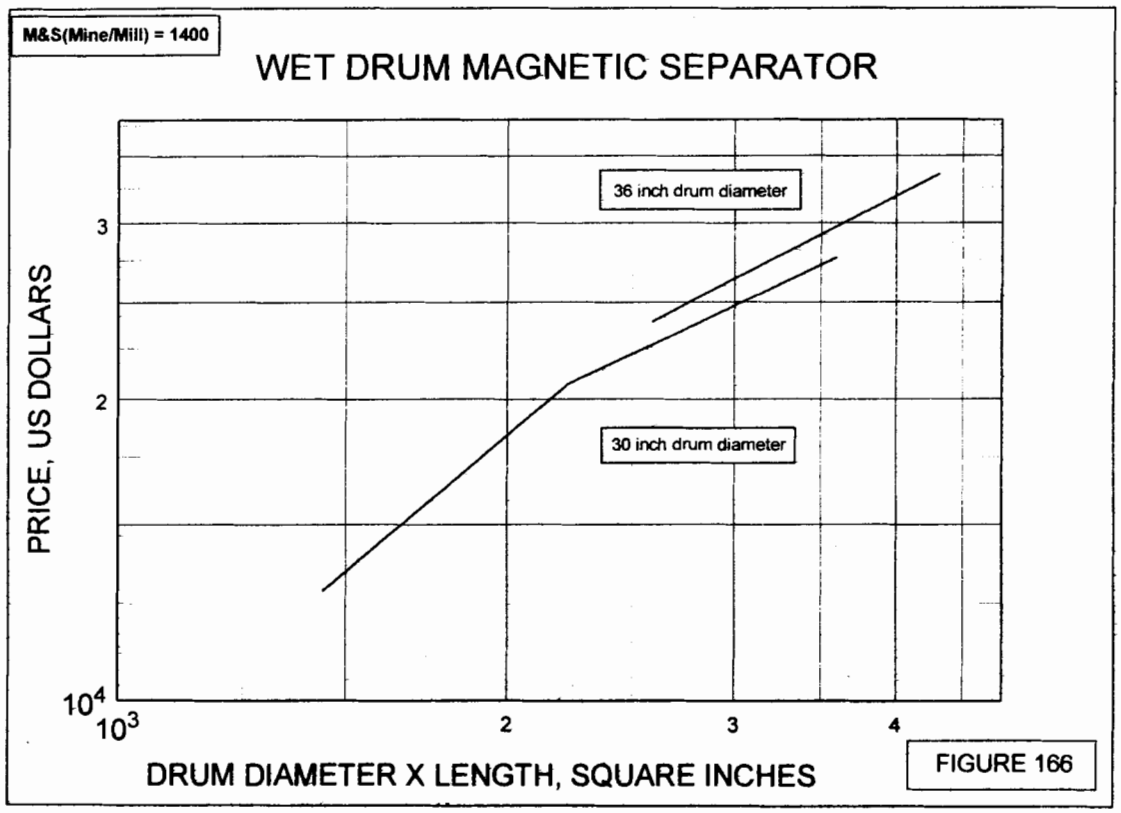


FIGURE 166

WET MAGNETIC DRUM SEPARATORS
(Permanent Magnet, Alternate Pole)

Price = aX^b, US Dollars

where **X, in cubic inches, is drum diameter squared times length**. Capacity is approximately 2 to 10 tph per foot of drum length at 1300 gauss.

Range in X, in^3: 1728 to 23328 a = 1418 b = 0.2936

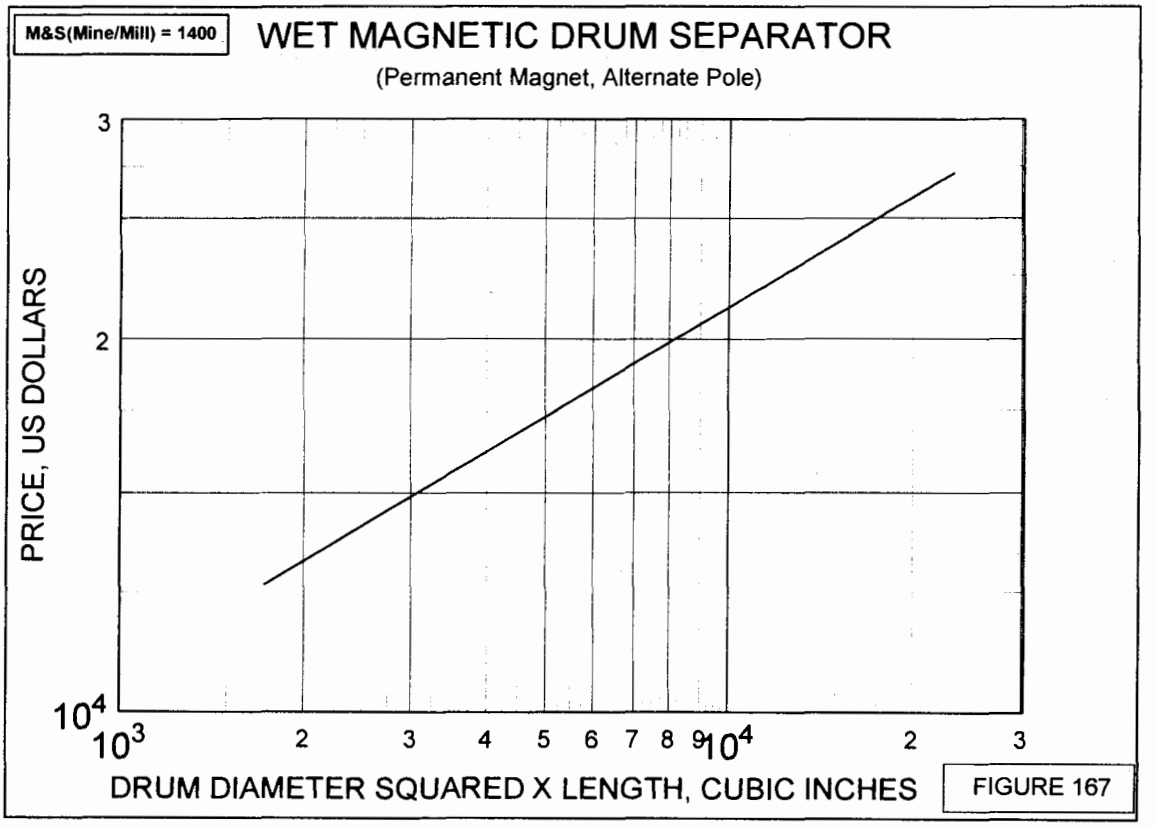


FIGURE 167

INDUCED ROLL MAGNETIC SEPARATORS

Price = aX^b, US Dollars

where **X is nominal capacity in pounds per hour**. Capacity is approximately 90 lb per hour per inch of effective rotor length; 12000 gauss field.

Range in X, lb/hr: 1600 to 25600 a = 286.9 b = 0.5841

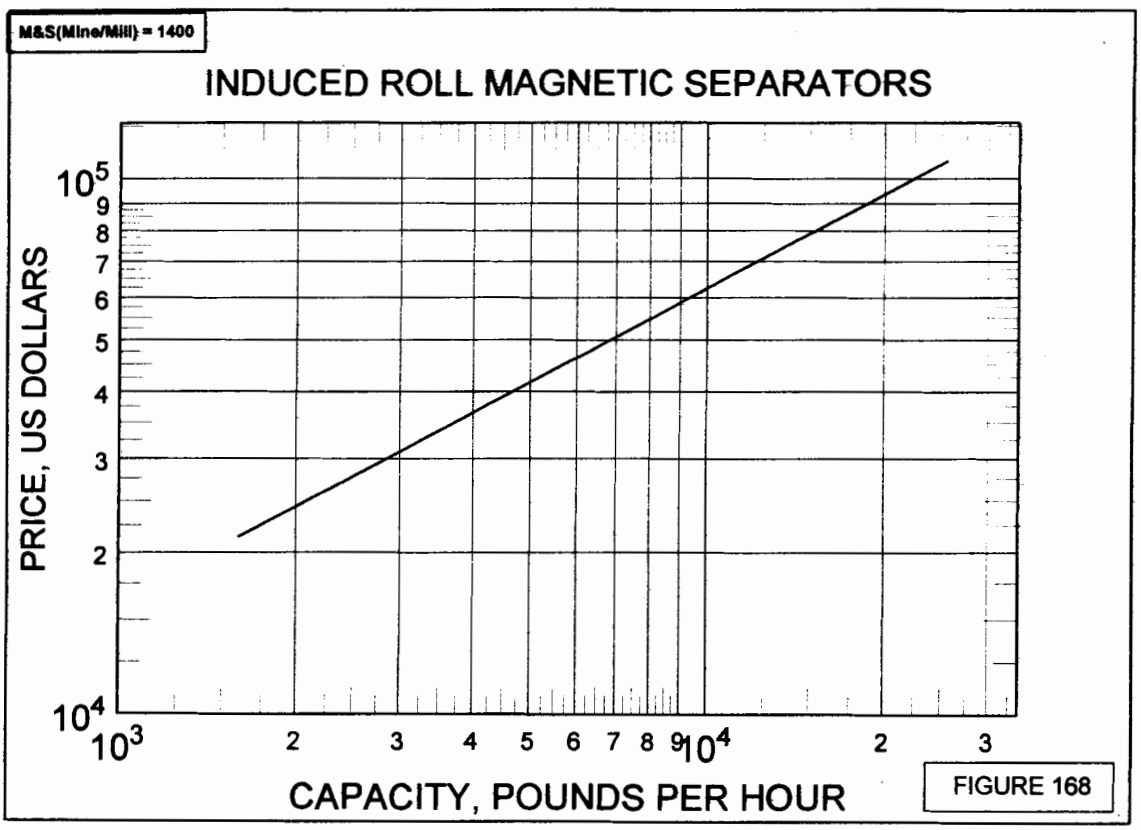


FIGURE 168

WET HIGH INTENSITY MAGNETIC SEPARATORS

Price = aX^b, US Dollars

where **X is horsepower**.

Range in X, HP: 13 to 98 a = 13900 b = 0.951

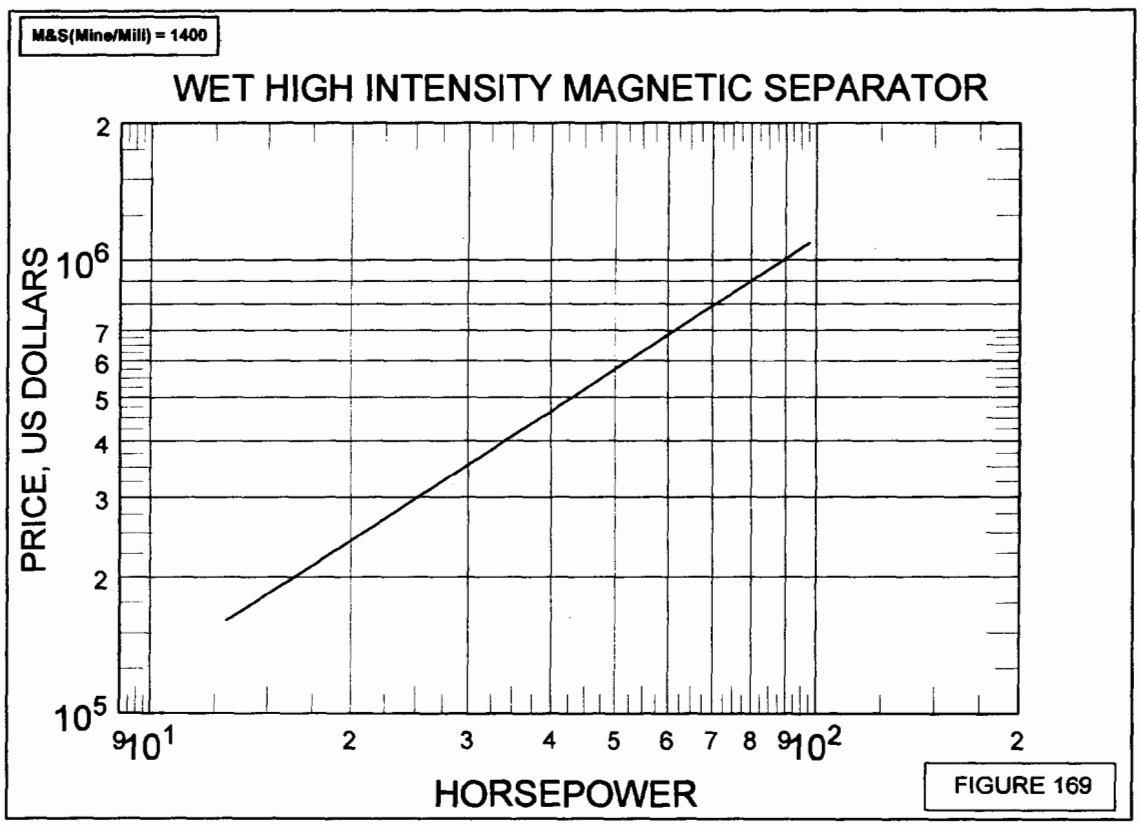
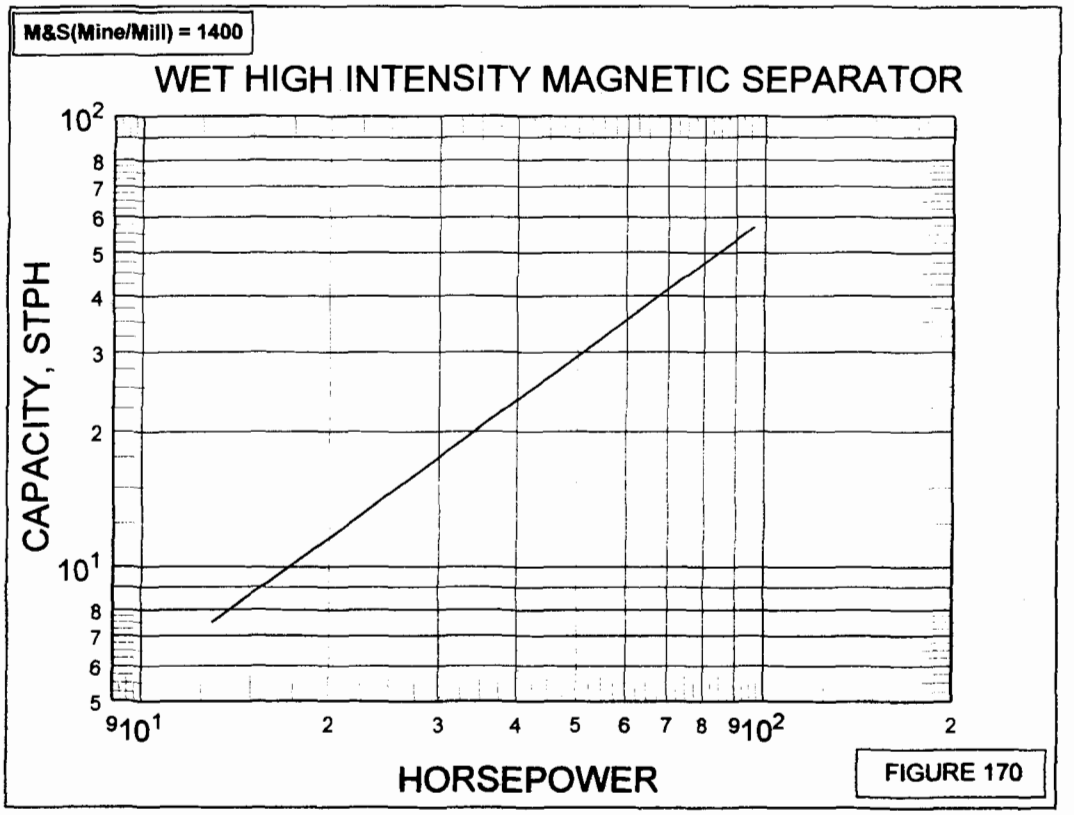


FIGURE 169

WET HIGH INTENSITY MAGNETIC SEPARATOR (CONTINUED)

Figure 170 below is plot of average capacity for many different materials versus horsepower drawn by the high intensity magnetic separator. Optimal conditions are assumed.



PERMANENT GUARD MAGNETS
(Overband Crossbelt)

Price = aX^b, US Dollars

where **X is belt width in inches**.

Range in X, inches: 20 to 42 a = 1356 b = 0.8262

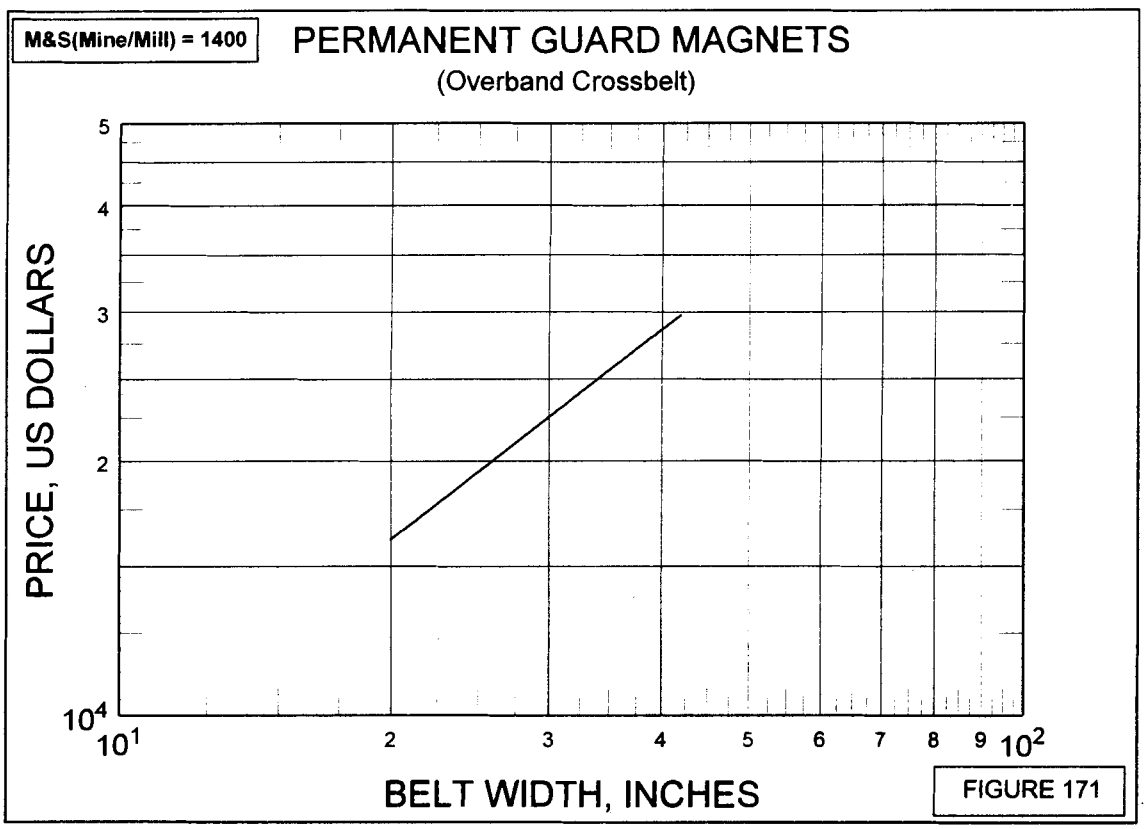


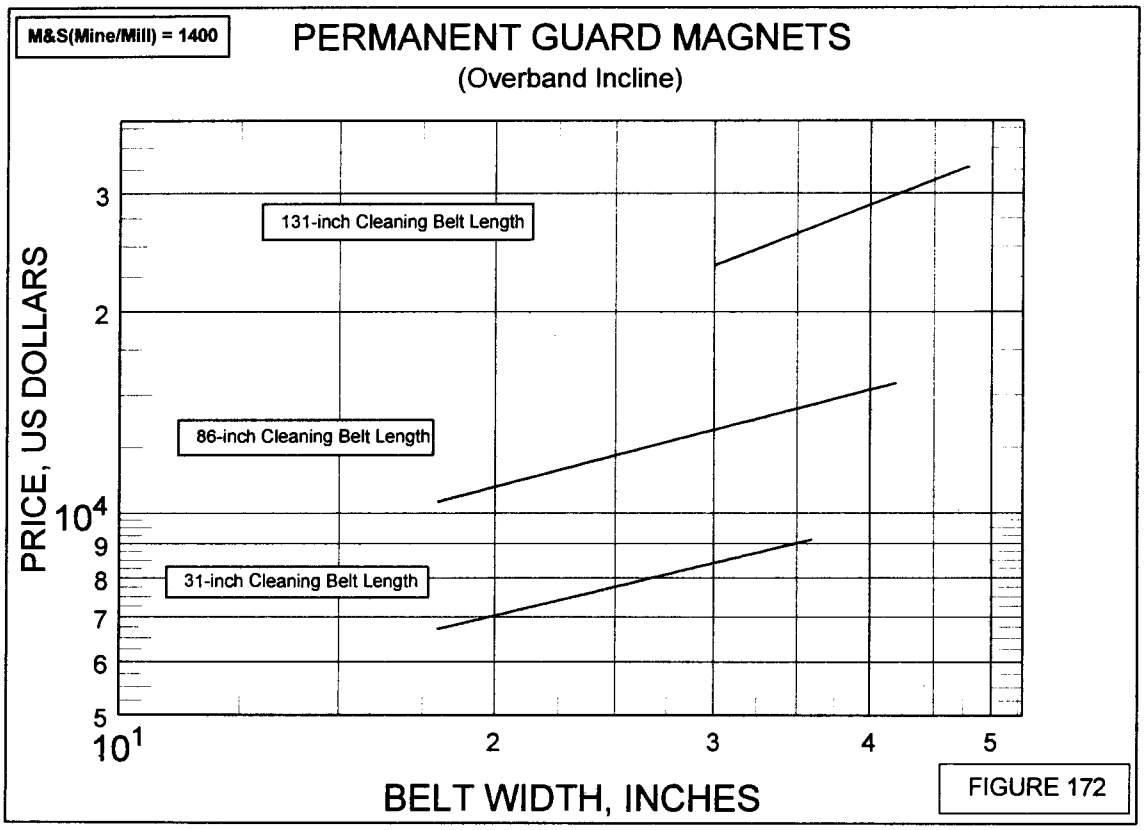
FIGURE 171

PERMANENT GUARD MAGNETS
(Overband Incline)

Price = aX^b, US Dollars

where **X is belt width in inches**. Price depends on belt cleaning length as shown below.

131-inch; Range in X, inches: 18 to 36	a = 2007	b = 0.7220
86-inch; Range in X, inches: 18 to 42	a = 2569	b = 0.4829
31-inch; Range in X, inches: 30 to 48	a = 1851	b = 0.4447



XI. Solid-Liquid Separation Equipment

Although the equipment in this section may not separate completely a solid from a liquid, the distinction between solid-solid separation and solid-liquid separation is made to permit of convenient categorization. Equipment has been placed in the following sub-categories: **Continuous Centrifuges**, **Dryers**, **Filters** and **Thickeners**. These have been broken down further as necessary.

(1) Example Estimates - - - - - - - Page 253

(2) Continuous Centrifuges
- **(a) Perforated Basket and Pusher Discharge - - - Page 255**
- **(b) Scroll Discharge, Stainless Steel - - - - Page 256**
- **(c) Scroll Discharge, Carbon Steel - - - - Page 257**
- **(d) Screen Cone - - - - - - - Page 258**

(3) Dryers
- **(a) Fluidized Bed - - - - - - - Page 259**
- **(b) Rotary Gas, carbon steel - - - - - Page 260**
- **(c) Spray - - - - - - - - Page 261**

(4) Filters
- **(a) Pressure Filters**
  - **(i) Vertical Plate and Frame - - - - Page 263**
  - **(ii) Horizontal Plate and Frame - - - - Page 264**
- **(b) Vacuum Filters**
  - **(i) Horizontal Belt - - - - - - Page 266**
  - **(ii) Rotary Disc - - - - - - Page 267**
  - **(iii) Rotary Drum - - - - - - Page 268**
  - **(iv) Rotary Tilting Pan - - - - - Page 269**

(5) Thickeners
- **(a) Conventional - - - - - - - Page 270**
- **(b) High Capacity - - - - - - - Page 271**

EXAMPLE ESTIMATES

Equipment prices in this section have been mainly related to size parameters. Relevant capacity/power data are provided in most cases to ensure of rough relationships between size, power and capacity.

Example 1:

A thickener is required to dewater a tailings stream which contains 100 tons of dry solids per day. What size of thickener is required and how much will it cost? Assume that the tailings is reject from copper beneficiation.

Normally, to specify capacity of a thickener, the Coe-Clevinger or Talmadge-Fitch-Roberts methods or variations thereof are employed, where laboratory settling tests are conducted. In this case a very simple method is used. Thus, from the table on page 270, the number of square feet of thickener needed per ton per day is 4 to 10. Choosing 8 ft^2/tpd, the total area required is then 8 x 100 = 800 ft^2.

The diameter of the thickener is then $\sqrt{\left(\frac{4}{\pi}\right)800}$ = 32 ft. From Figure 189, page 270, the cost of the 32-foot thickener including mechanism and tank walls is US$110,000 at an M&S(M/M) Index of 1400.

Example 2:

A laboratory leaf filter test showed that for a safety factor of 0.65 a filter that handles 34 lb of dry solids per square foot per hour is required. What size of disc filter is necessary to treat 100 tons of dry solids per day and what is the cost of the unit? What size drum filter might substitute and at what cost compared with a disc?

From 100(2000/24) = 8,333 lb/hr, it is found that the total surface area of filter needed is (8,333/34) = 245 square feet. From a table of standard disc filter sizes (see page 267), a 6-foot diameter disc filter with 5 discs has a nominal surface area of 250 ft^2. A 4-disc filter of the same diameter has a nominal surface area of 200 ft^2. Thus, one possible size of interest is a filter of diameter 6 feet with 5 discs.

From Figure 186, page 267, it is apparent that the cost of this disc filter is approximately US$110,000 at an M&S(M/M) Index of 1400.

An equivalent area drum filter is 8-foot in diameter and 10-foot in length. This would cost around (see Figure 187, page 268) US$150,000 at an M&S(M/M) Index of 1400.

Example 3:

A hot air rotary dryer for reducing moisture content of a concentrate must have a volume of 1000 cubic feet. What is the cost of such a unit?

EXAMPLE ESTIMATES (CONTINUED)

A rotary dryer of 1000 ft^3 volume is equivalent to a drum of diameter 8 ft and length 20 ft. Thus D^2L = 1280 ft^3. From Figure 178, page 260, the corresponding cost at 1280 square feet is roughly US$200,000 at an M&S(M/M) Index of 1400.

Example 4:

A stacked horizontal chamber pressure filter (e.g., the Larox) is being considered for the filtration of flotation concentrate. What is the estimated price of a filter that will treat 10 tons of concentrate per hour?

Figure 184, page 265, shows that treating 10 tph of typical concentrate will require about 200 ft^2 of filtration area.

From Figure 183, page 264, the category of filter with the correct range is the Larox PF. The cost is approximately US$675,000 at an M&S(M/M) Index of 1400. If the current index is 1095, then the current cost becomes:

US$528,000.

Example 5:

Estimate the price of a perforated basket centrifuge that will dewater 40 tph of calcium sulfate sludge.

The table on page 255 suggests that a 36-inch basket is required. From Figure 173, page 255, the estimated price is US$60,000 at an M&S(M/M) Index of 1400.

CONTINUOUS CENTRIFUGES
(Perforated Basket and Pusher Discharge)

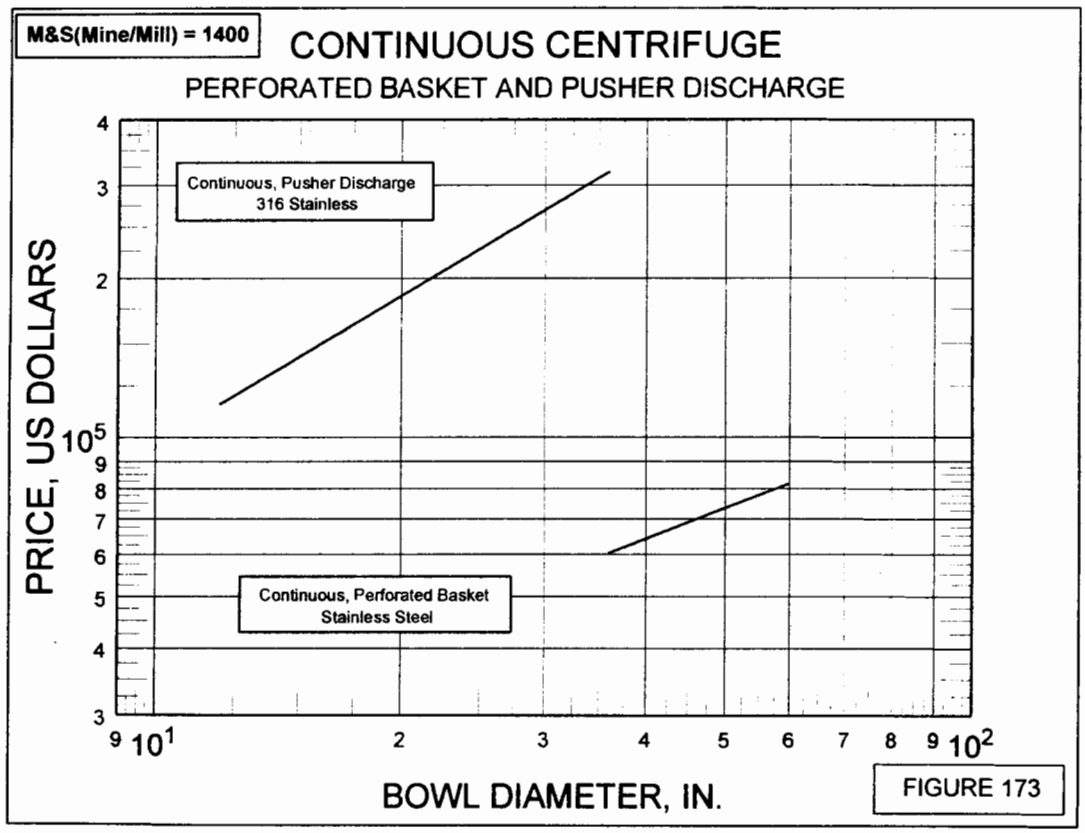
Price = aX^b, US Dollars

where **X is bowl diameter in inches**. For pusher discharge, cost includes wash, isolation system and main drive motor. For perforated basket, cost includes motor and drive.

Pusher Discharge; Range in X, in.: 12 to 36 a = 11624 b = 0.9234

Perforated Basket; Range in X, in.: 36 to 60 a = 7185 b = 0.5934

Diameter, in.	Capacity, tph	Type
20	25	pusher discharge
36	40	pusher discharge
36	35 to 80	perforated basket
48	110 to 200	perforated basket



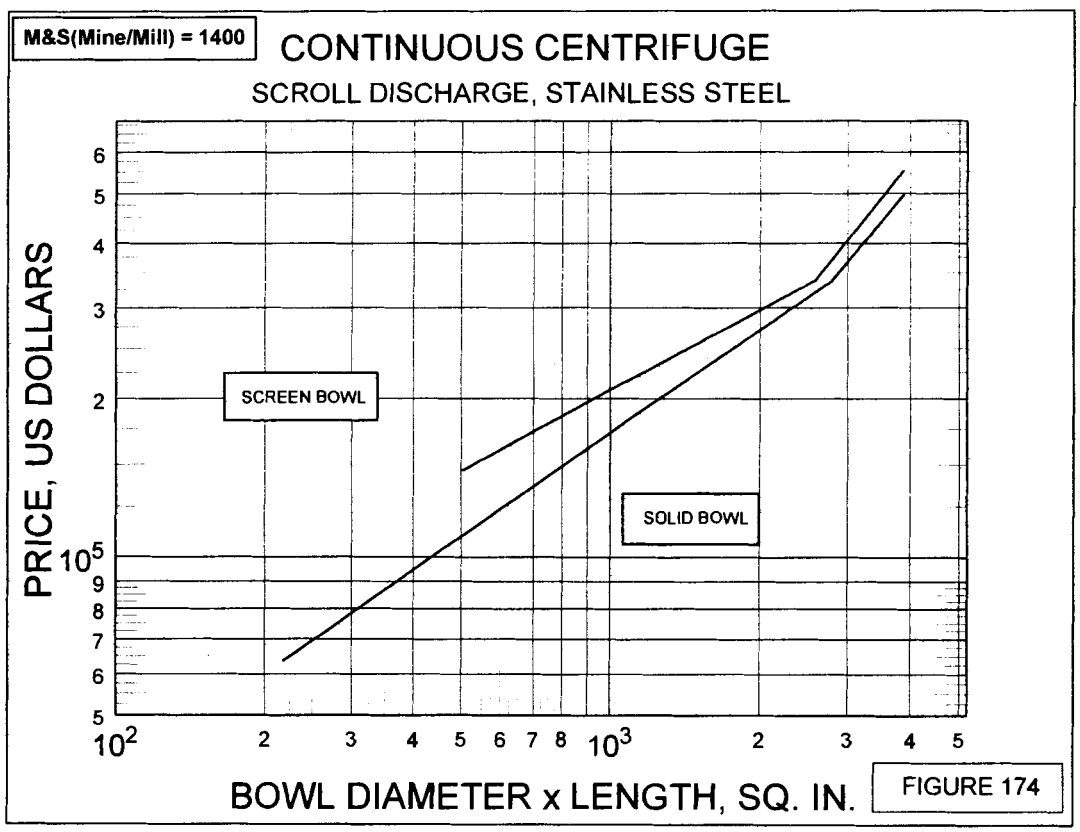
CONTINUOUS CENTRIFUGES
(Scroll Discharge, Stainless Steel)

Price = aX^b, US Dollars

where **X, in square inches, is bowl diameter times length**. Cost includes main drive motor. See page 257 for solid bowl capacities.

Screen Bowl; Range in X, in^2:	504 to 2592	a = 5993	b = 0.5131
	2592 to 3888	a = 25.71	b = 1.207
Solid Bowl; Range in X, in^2:	218 to 2799	a = 1865	b = 0.655
	2799 to 3888	a = 32.65	b = 1.165

Dia. x Length, sq. in.	Capacity, tons/hour	Type
18 x 28	3 to 5	screen
24 x 38	10 to 12	screen
40 x 60	25 to 35	screen
54 x 70	60 to 80	screen



CONTINUOUS CENTRIFUGES
(Scroll Discharge, Carbon Steel)

Price = aX^b, US Dollars

where **X, in square inches, is bowl diameter times length**. Cost includes main drive motor.

Screen Bowl; Range in X, in^2:	504 to 1837	a = 10320	b = 0.4249
	1837 to 5808	a = 1387	b = 0.692
Solid Bowl; Range in X, in^2:	200 to 5808	a = 4303	b = 0.5359

Dia. x Length, in. x in.	Capacity, tons per hr.	Type
18 x 18	2	solid bowl
24 x 60	8	solid bowl
36 x 96	20	solid bowl
44 x 132	40	solid bowl

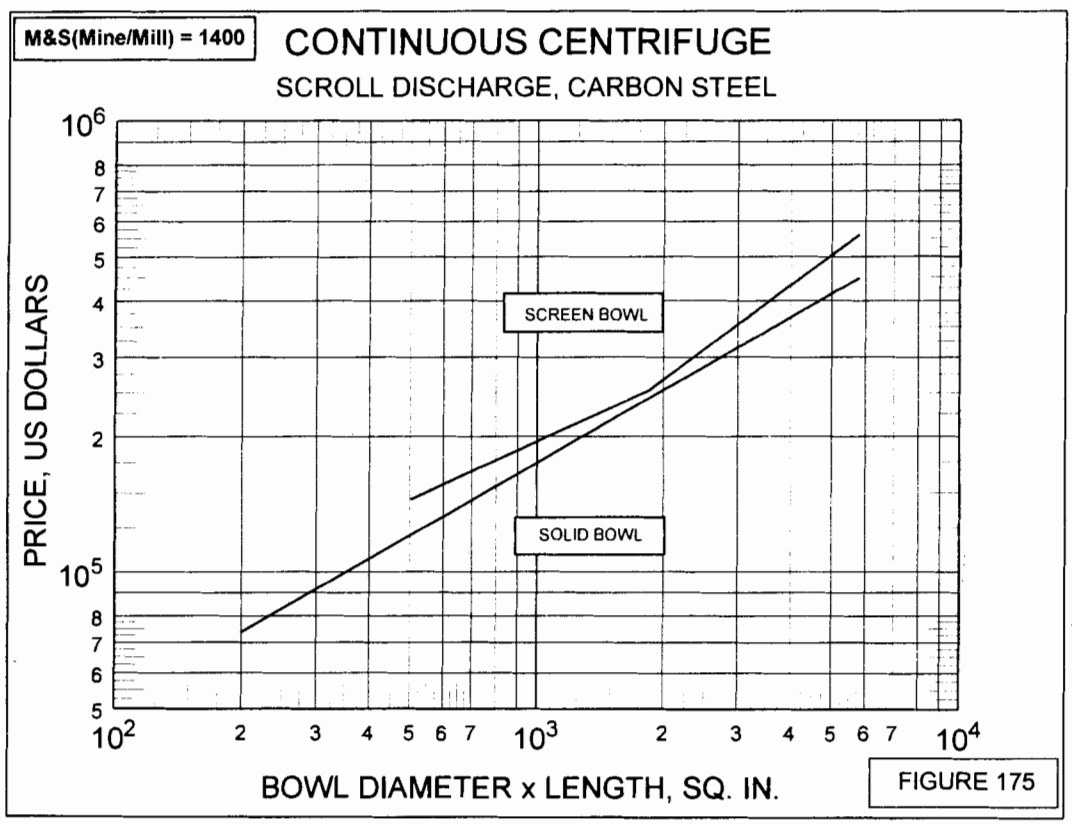


FIGURE 175

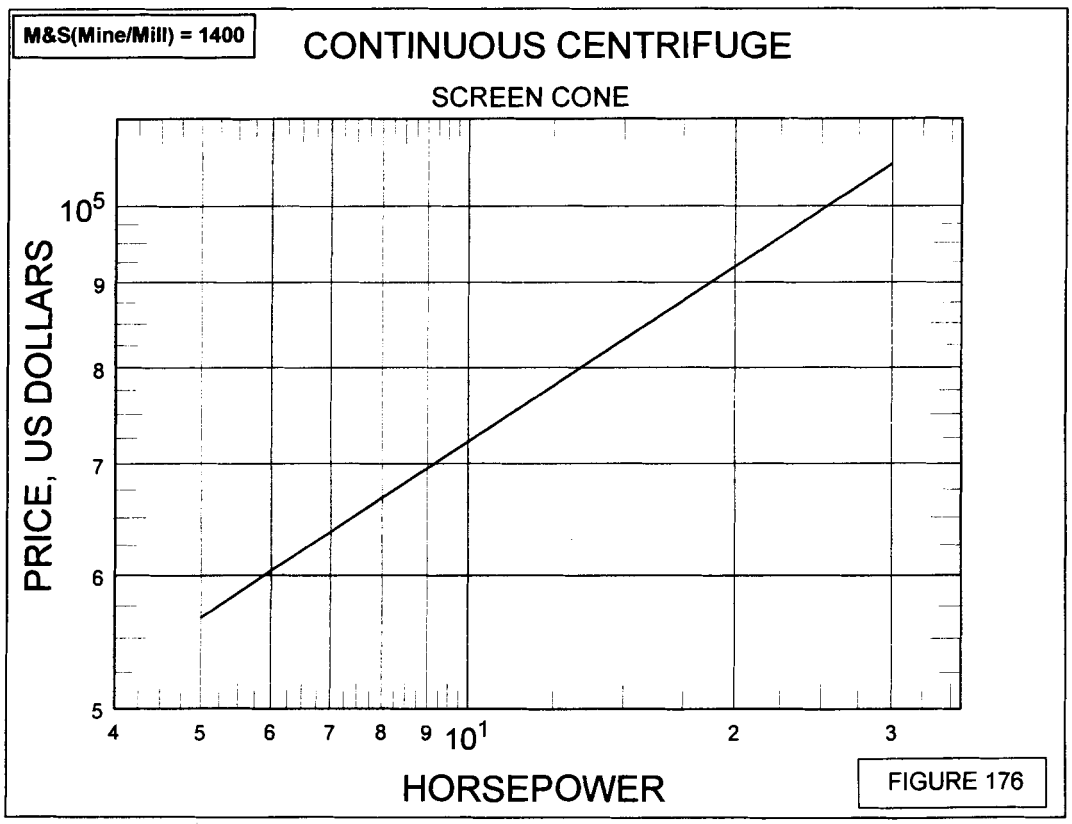
CONTINUOUS CENTRIFUGES
(Screen Cone)

Price = aX^b, US Dollars

where **X is in horsepower**. Screen cone is stainless steel.

Range in X, Hp: 5 to 30 a = 32217 b = 0.35

Screen Area, sq. in.	Capacity, US GPM	HP
145	1 to 25	5
405	15 to 150	30



FLUIDIZED BED DRYERS

Price = aX^b, US Dollars

where **X is evaporation capacity in pounds of water per hour**. Cost includes air handling system, motors and starters.

Range in X, lbs/hr:	300 to 837	a = 4180	b = .4326
	837 to 1800	a = 384.4	b = 0.7872

Evaporation Capacity, lbs/hr	Diameter, inch
300	24
700	34
950	52
1800	64

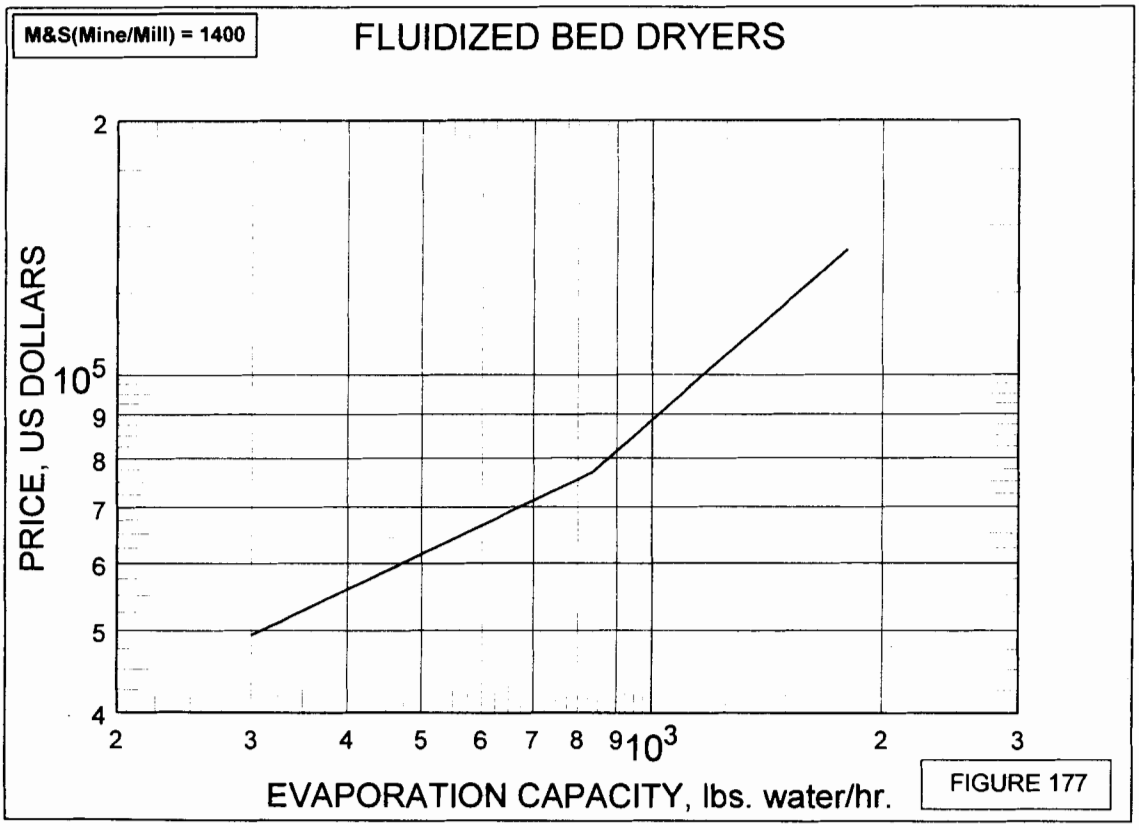


FIGURE 177

ROTARY GAS DRYERS

Price = aX^b, US Dollars

where **X, in cubic feet, is diameter squared times length**. Price excludes air handling system, motor starter and air heaters.

Range in X, ft^3: 480 to 19600 a = 8309 b = 0.4409

Dia., ft	Length, ft	HP	!	Dia., ft	Length, ft	HP
3	30	5	!	8	50	40
4	40	10	!	10	40	60
5	50	20	!	10	90	125
6	30	15	!	12	65	125
6	50	25	!	12	90	200

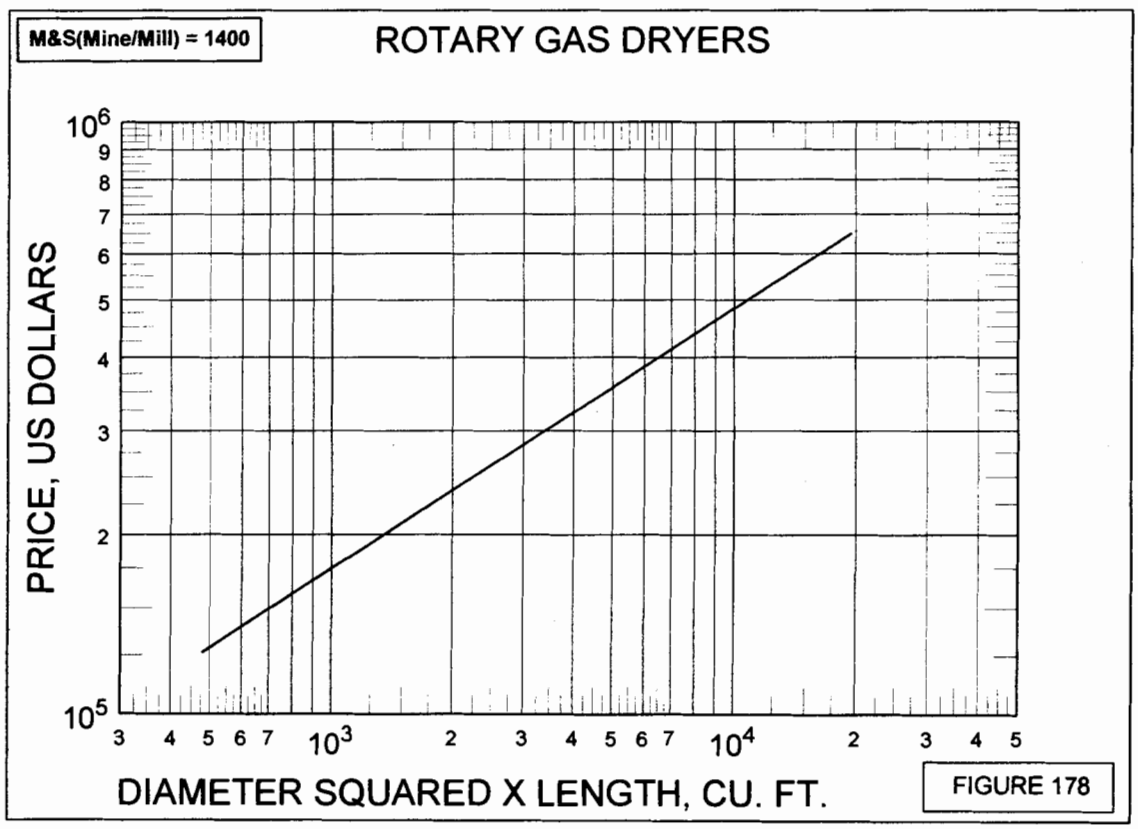


FIGURE 178

SPRAY DRYERS

Price = aX^b, US Dollars

where **X is spray diameter in feet**. Cost excludes motors.

Range in X, ft:	12.5 to 17.8	a = 4712	b = 1.557
	17.8 to 25	a = 669.2	b = 2.236

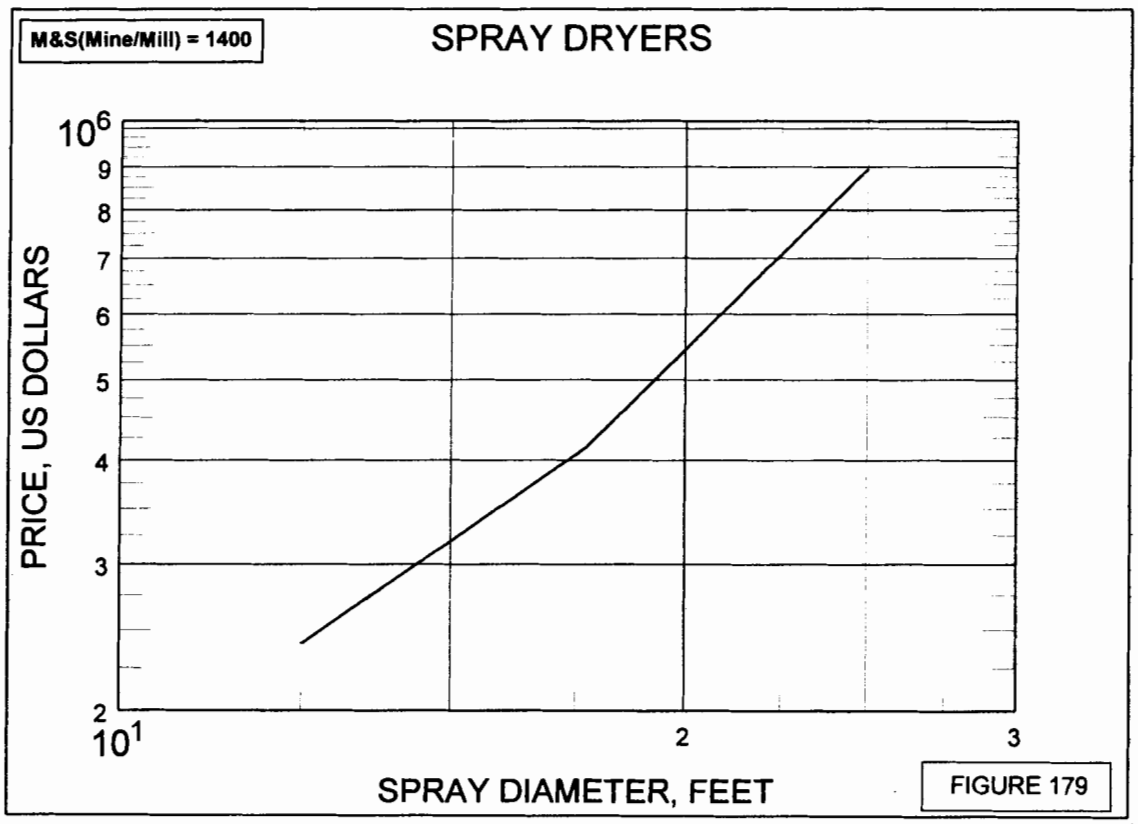


FIGURE 179

SPRAY DRYERS (CONTINUED)

Figure 180 below shows how evaporation capacity depends on spray diameter, while Figure 181 provides the horsepower versus evaporation capacity curve. **For both figures, the inlet/outlet temperatures are at 500 ^{0}F.**

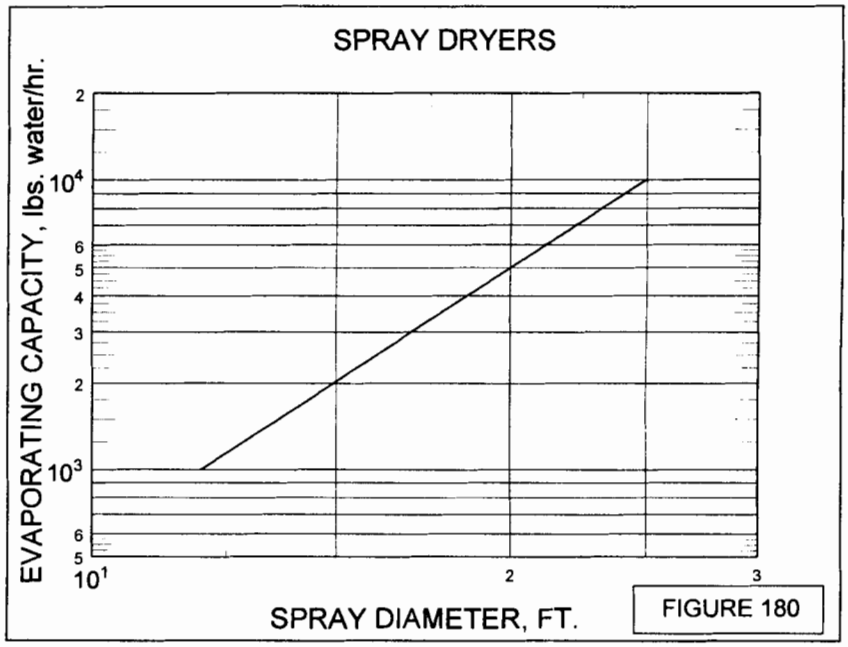


FIGURE 180

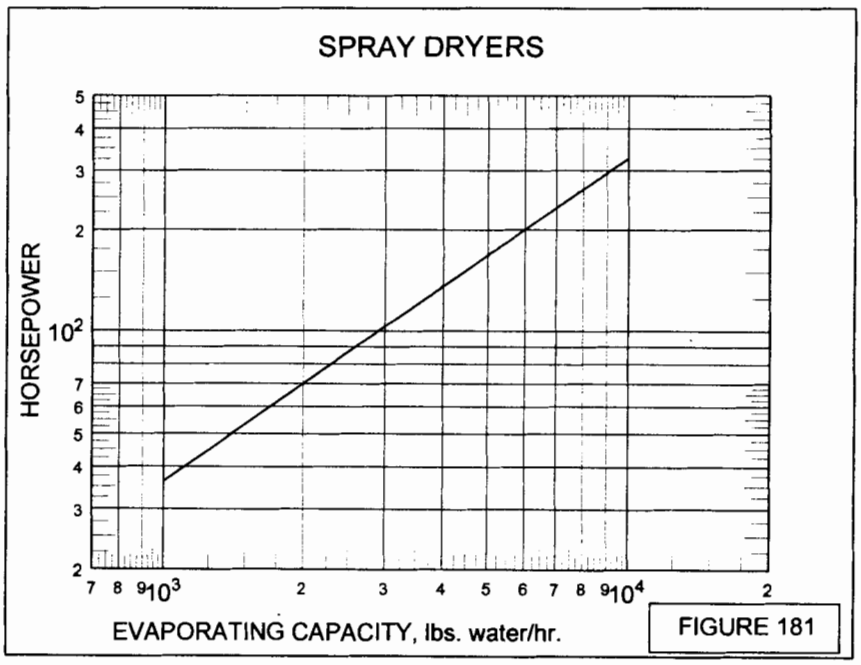


FIGURE 181

VERTICAL PLATE AND FRAME PRESSURE FILTERS

Price = aX^b, US Dollars

where **X is filter area in square feet**. Price excludes feed pumps and storage tanks.

Manual; Range in X, ft^2:	30 to 291	a = 3491	b = 0.4699
	291 to 798	a = 851.6	b = 0.7185
Automated; Range in X, ft^2:	134 to 798	a = 4085	b = 0.5409

Filtrate*, cu. ft.	Plate Size, in.x in.	Press Length, in.	# Chambers	Area, sq. ft.
1.2	18 x 18	76	9	30
5	24 x 24	96	22	133
10	30 x 30	140	26	269
20	36 x 36	171	43	534
30	48 x 48	199	34	798

*at 100 psi filtration.

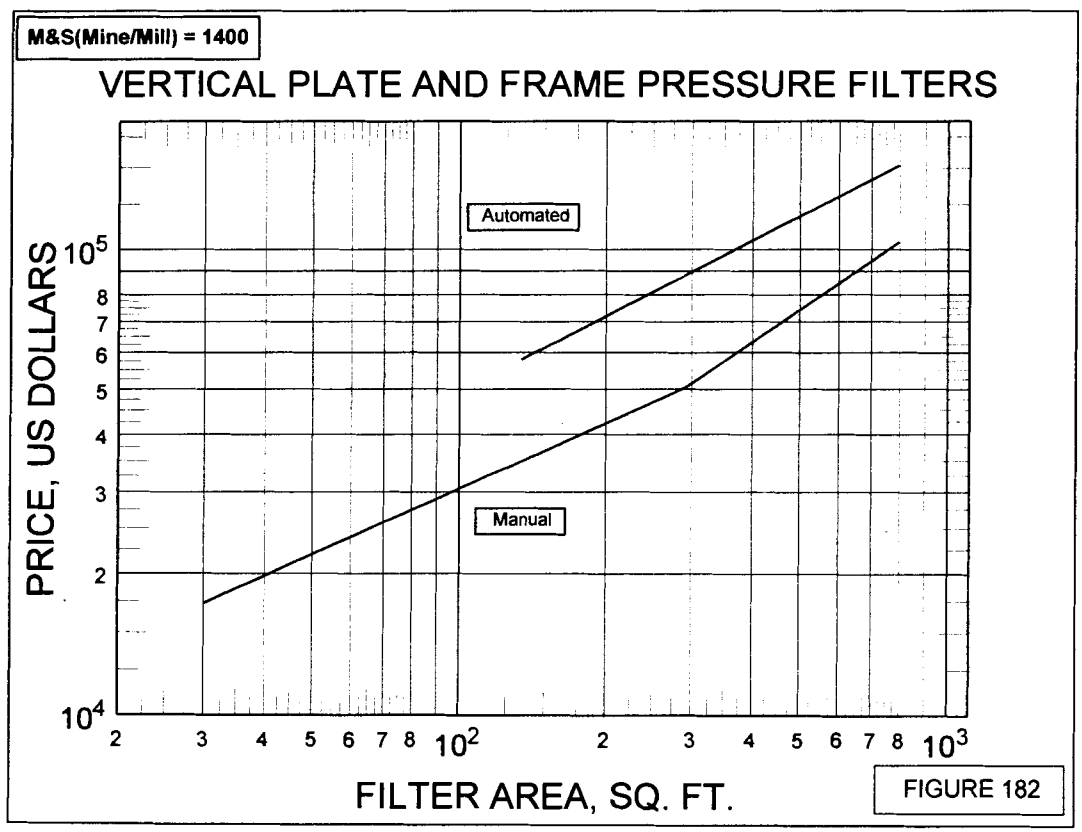


FIGURE 182

HORIZONTAL PLATE AND FRAME PRESSURE FILTER
(Larox Variety)

Price = aX^b, US Dollars

where **X is total filtration area in square feet**. Prices, as in most cases in this manual, are FOB source. *NOTE:* 1 tonne/hr = 1.1023 short tons/hr; 1 square meter = 10.7638 square feet.

MINIMAX; Range in X, ft^2: 17.22 to 67.81 a = 233400 b = 0.115

PF; Range in X, ft^2: 102.26 to 409.02 a = 55800 b = 0.473

POWER PF; Range in X, ft^2: 645.83 to 1550 a = 198500 b = 0.3217

Type:	2.5	5.0	9.5	16.0	22.0	28.0	38	60	84	120	144
Area:	27.4	54.9	101.7	169.5	237.3	305.1	406.7	645.6	903.8	1291.2	1549.4
Plate Size:	1'-8"	1'-8"	2'-11"	---same up to----------			2'-11"	4'-11"	-----same up to---		4'-11"
	2'-9"	2'-9"	5'-9"	-----same up to----------			5"-9"	13'-2"	----same up to----		13'-2"
# Plates:	6	12	6	10	14	18	24	10	14	20	24
Total* HP:	13.5	13.5	26.5	26.5	26.5	46.5	46.5	88	88	88	118

* Clamping mechanism; cloth tension device; cloth drive mechanism; water pump

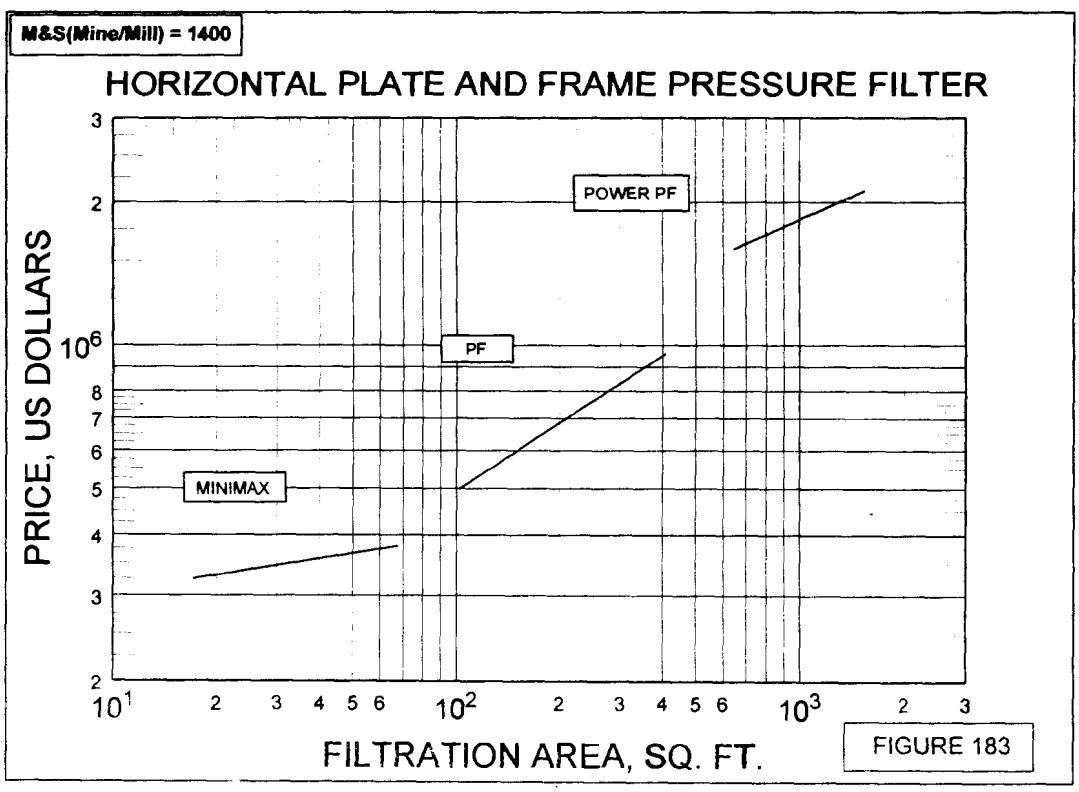


FIGURE 183

HORIZONTAL PLATE AND FRAME PRESSURE FILTER (CONTINUED)

(Larox Variety)

Figure 184 below shows the capacity in stph versus filtration area in square feet. Conditions are: chalcopyrite concentrate cake at 8% moisture; feed slurry is 65% solids; product is 80% passing 325 mesh.

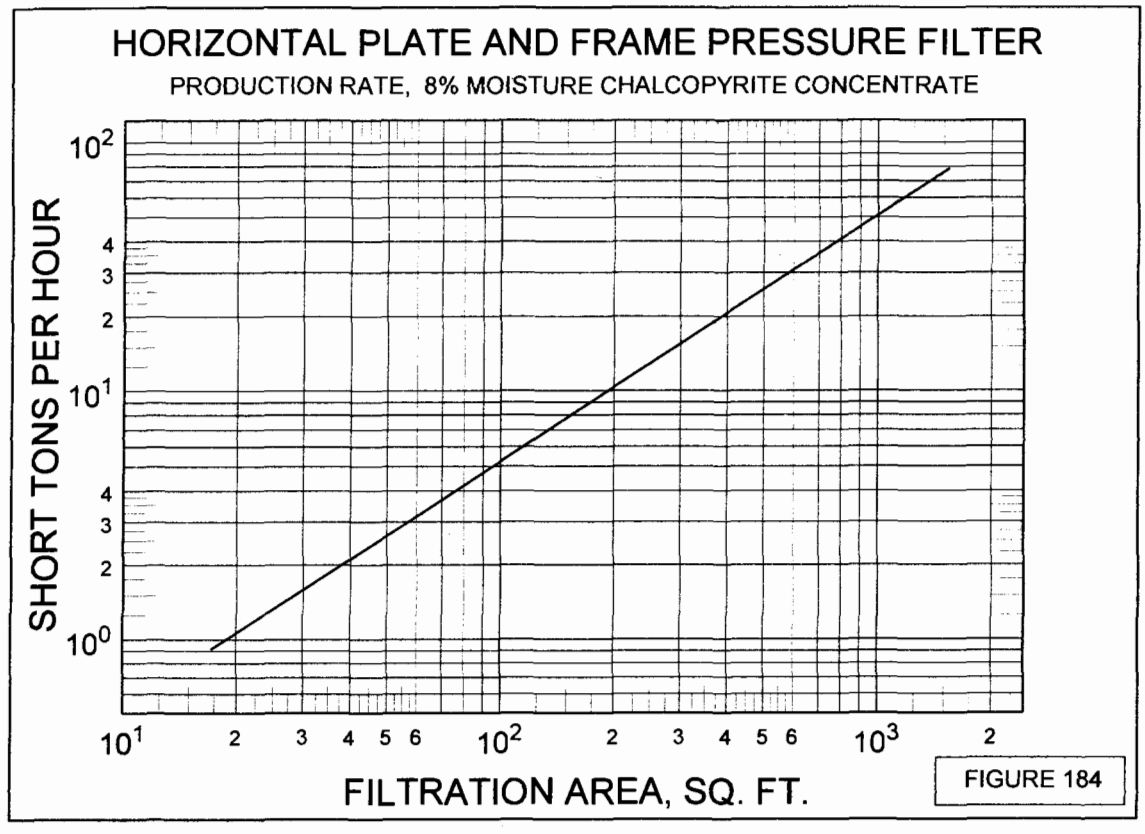


FIGURE 184

HORIZONTAL BELT VACUUM FILTERS

Price =aX^b, US Dollars

where **X, in square feet, is belt width times length**. Add 14% of cost if wetted parts are to be stainless steel or rubber coated.

Range in X, ft^2: 10 to 192 a = 22210 b = 0.4916

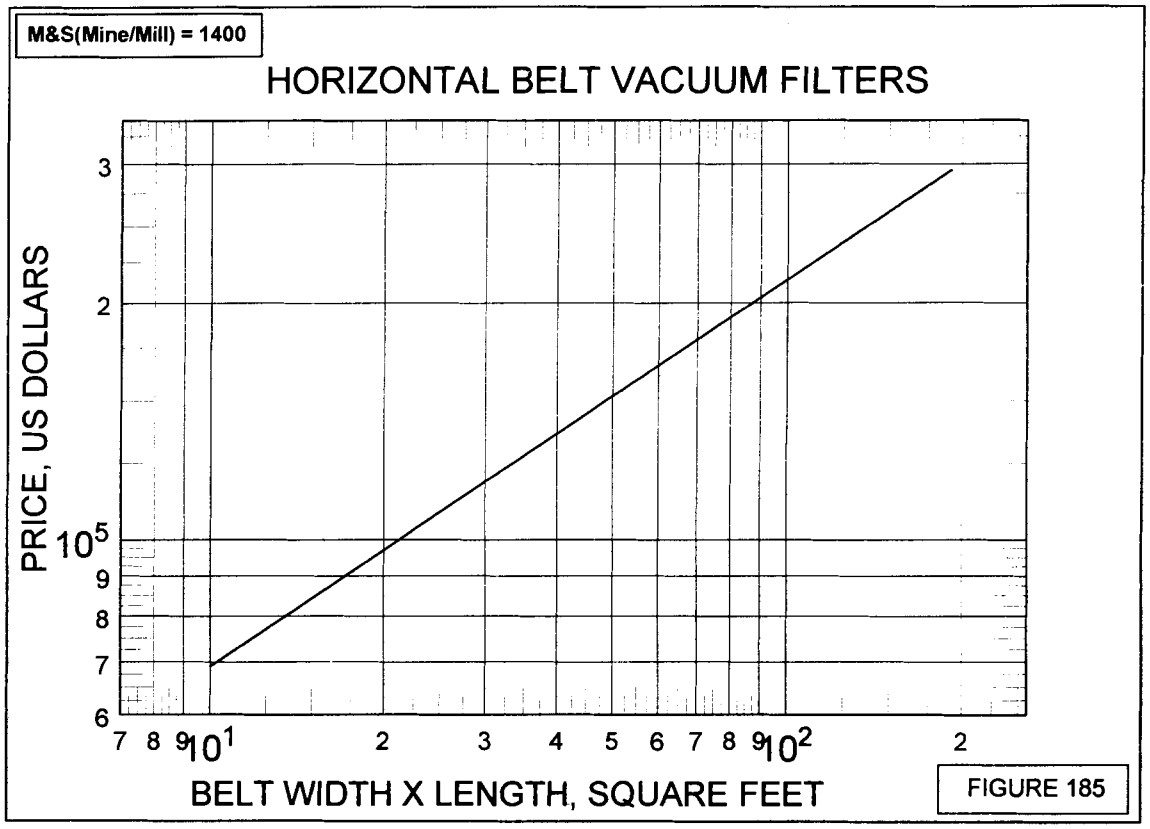


FIGURE 185

ROTARY DISC FILTERS

Price = aX^b, US Dollars

where **X is filter area in square feet**. Cost includes vacuum fittings and motor.

Range in X, ft^2: 225 to 2500 a = 3002 b = 0.6478

Disc Dia., ft	Area per Disc, sq. ft.
4	22
6	50
7.5	75
9	107
11	160
13 1/4	240

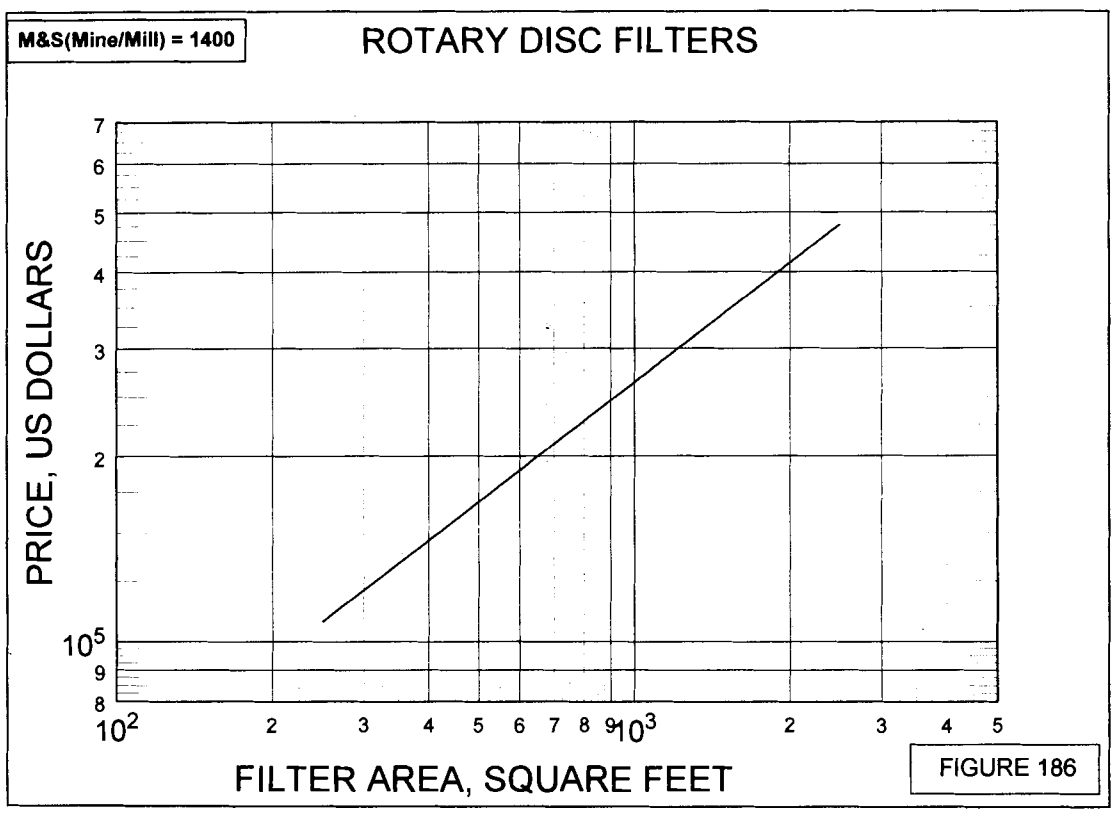


FIGURE 186

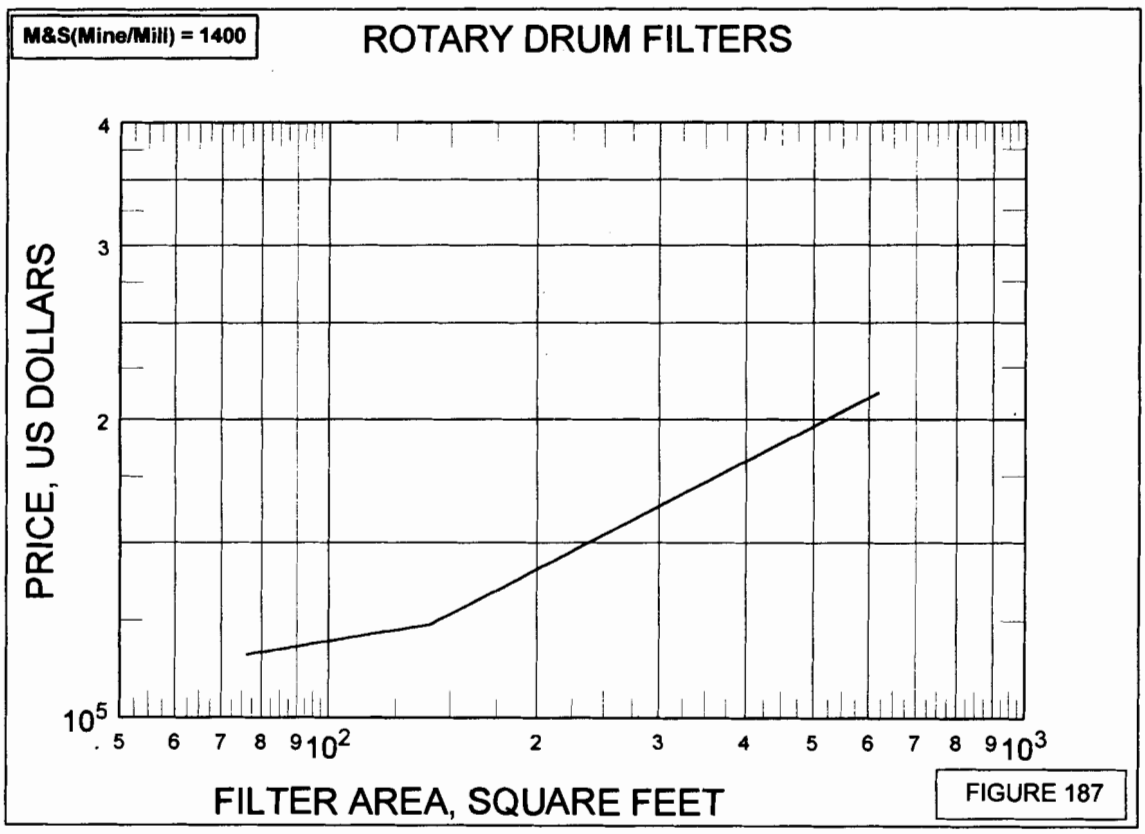
ROTARY DRUM FILTERS

Price = aX^b, US Dollars

where **X is filter area in square feet**. Price includes vacuum fittings and motor. Add 20% if wetted parts are to be stainless steel or rubber coated.

Range in X, ft^2:			
76 to 140	a = 70550	b = 0.1136
140 to 620	a = 20460	b = 0.364

Dia. x Length, sq. ft.	Area, sq. ft.
4 x 3	37.5
4 x 5	62.5
6 x 4	75
6 x 5	95
6 x 6	113
8 x 10	250

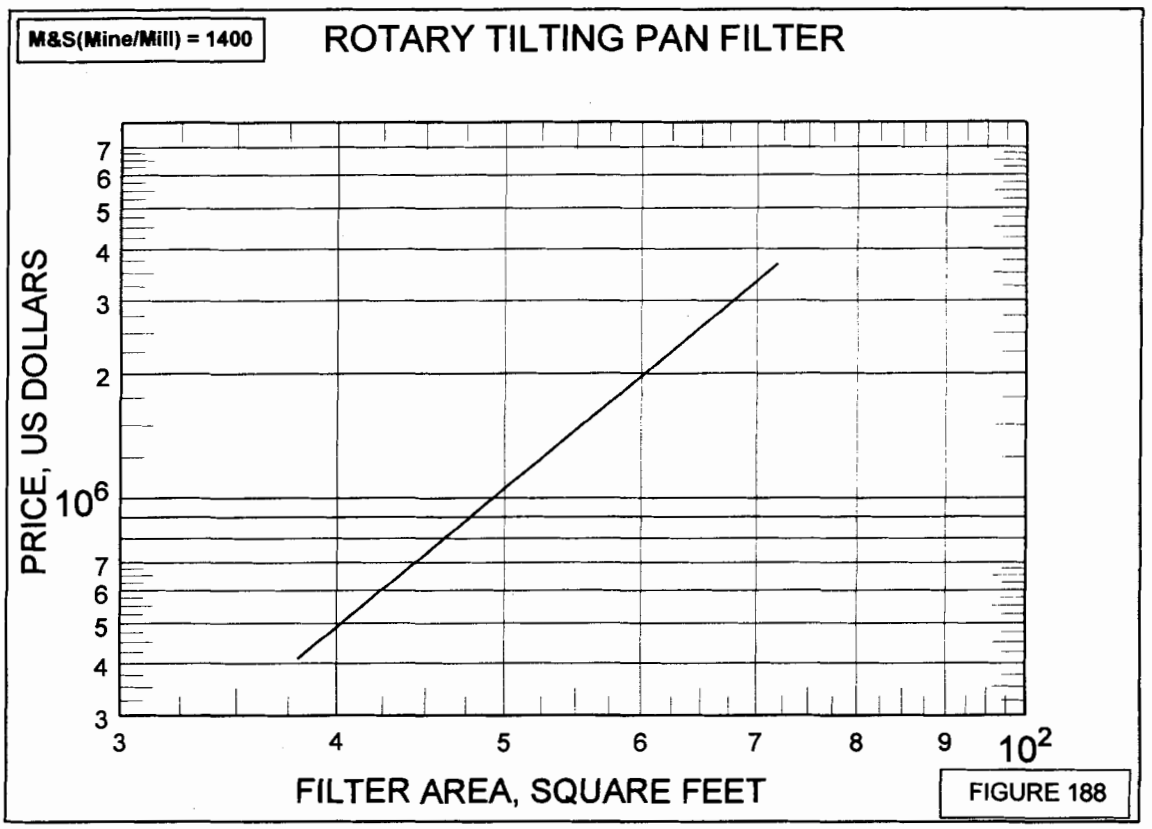


ROTARY TILTING PAN FILTERS

Price = aX^b, US Dollars

where **X is filter area in square feet**. Wetted parts of 316 stainless steel; cost includes fittings and motor.

Range in X, ft^2: 38 to 72 a = 1.562 b = 3.43



CONVENTIONAL THICKENERS

Price = aX^b, US Dollars

where **X is tank diameter in feet**. Price includes thickener drive, drive motor, lift, lift motor, controls, thickener mechanism (mild steel) with rakes, tank shell and anchor channel and prime painted; excludes erection, service, finish paint, remote controls, computers, motor starters, foundations or freight/boxing.

Range in X, ft:		a	b
	17 to 100	5159	0.8832
	100 to 260	182.6	1.607

Material	Settling Area, sq. ft./tpd	Material	Settling Area, sq. ft./tpd
Cu con	2 to 6	Ni con	5 to 20
Cu tail	4 to 10	Zn con	3 to 7
Fe con	.4 to .8	Potash Sl.	40 to 200
Fe tail	4 to 10	Mo Slimes	10 to 15
Co-Ni S	5 to 10	Mo con	10 to 15
Pb con	2 to 6	U-ppt.	50-125

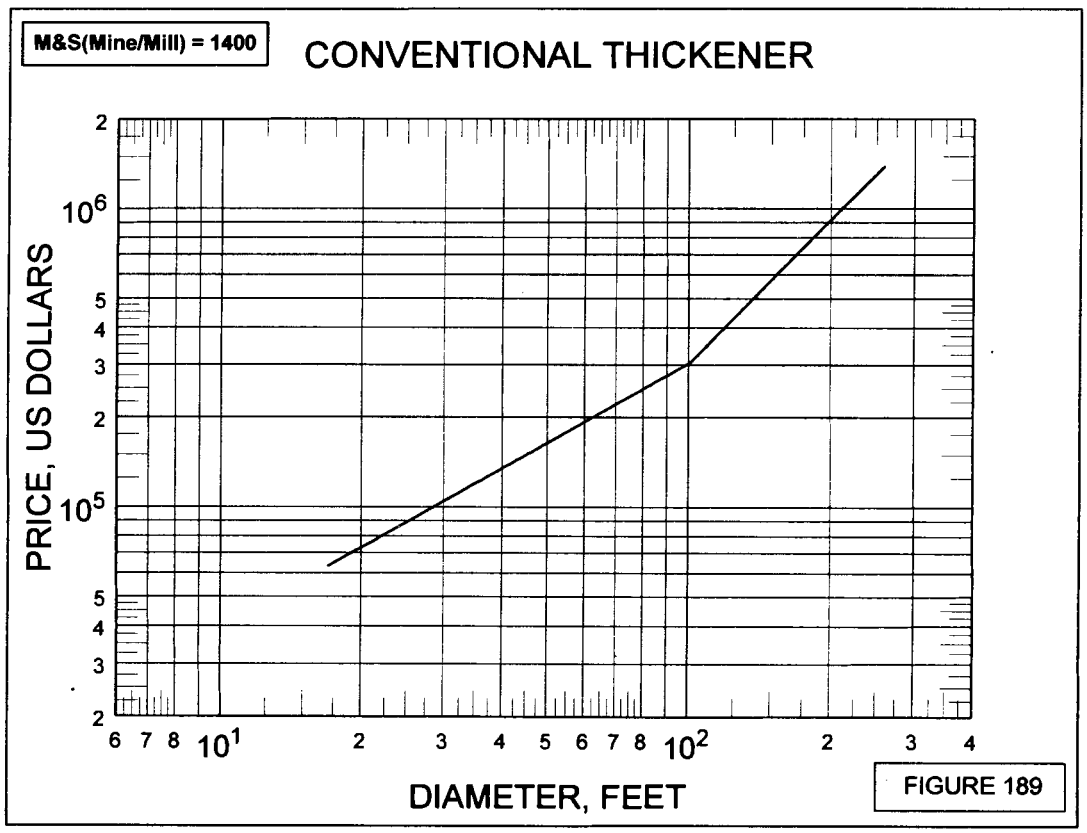


FIGURE 189

HIGH CAPACITY THICKENERS

Price = aX^b, US Dollars

where **X is thickener diameter in feet**. High capacity and E-CAP thickeners **must use flocculants**. Price includes thickener drive, drive motor, lift, lift motor, controls, thickener mechanism (mild steel) with rakes, tank shell and anchor channel and prime painted; excludes erection, service, finish paint, remote controls, computers, motor starters, foundations or freight/boxing. E-CAP thickeners/clarifiers include erection; exclude a drive or rake mechanism.

E-CAT; Range in X, ft:	10 to 36	a = 1476	b = 1.385
High Capacity; Range in X, ft:	16 to 100	a = 5809	b = 0.8786
	100 to 240	a = 133.7	b = 1.698

For typical unit areas, (settling areas in ft^2/tpd) see page 270. Typical E-CAP height to diameter ratios: (28.2/9.84), (33.8/16.4), (53.8/39.4).

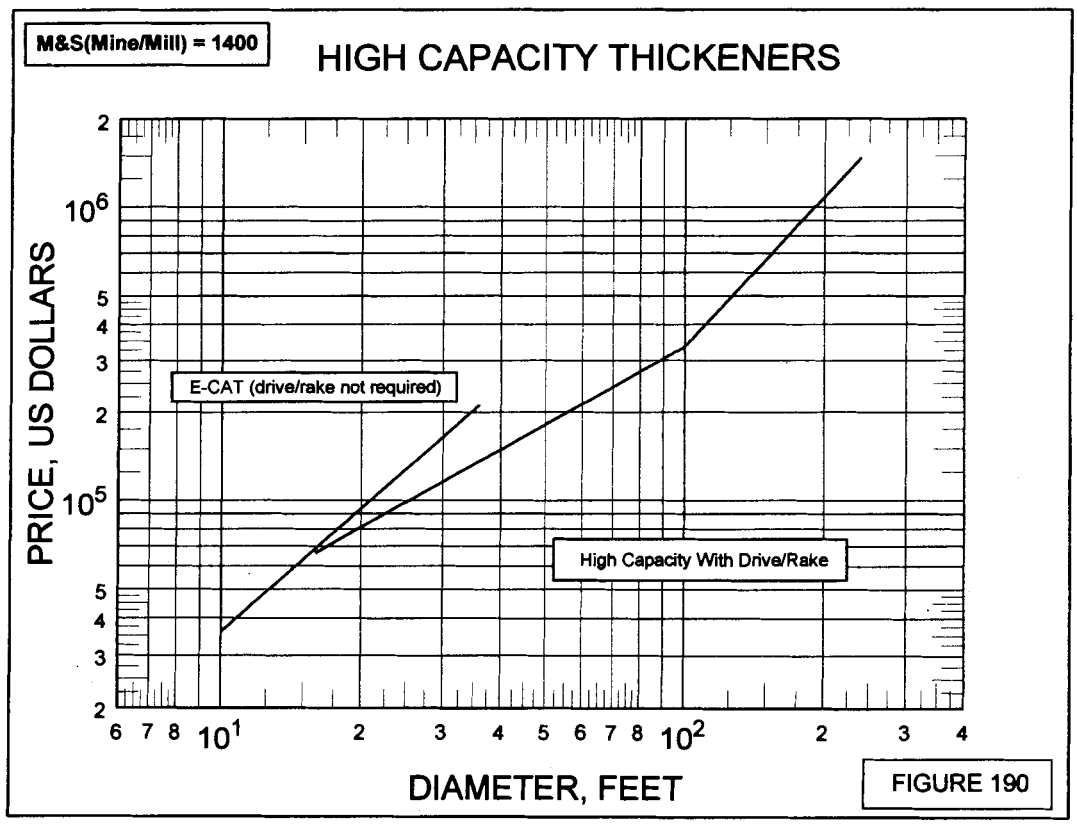


FIGURE 190

XII. Mixing, Leaching and Container Equipment

In this section, mixing, leaching and container equipment is placed in the following sub-categories: **Mixer/Blenders**, **Pelletizers** and **Containers**. These are broken down further as noted below.

(1) Example Estimates - - - - - - - Page 273

(2) Mixers/Blenders

(a) Rotary Dry Blenders - - - - - - Page 274

(b) Slurry Mixer Mechanisms - - - - - Page 275

(c) Propeller Agitator Mechanisms - - - - Page 276

(3) Pelletizers

(a) Rotary Drum - - - - - - - Page 277

(b) Rotating Disc - - - - - - - Page 279

(4) Containers

(a) Wooden Tanks, Vertical - - - - - - Page 281

(b) Epoxy Tanks, Horizontal - - - - - Page 282

(c) Polyethylene Tanks - - - - - - Page 283

(d) Fiberglass Tanks - - - - - - Page 284

(e) Vertical Fiberglass Tanks, Closed Top - - - Page 285

(f) Glass-Lined Steel Tanks - - - - - Page 286

(g) Stainless Steel Tanks, Vertical - - - - Page 287

(h) Steel Bulk Storage Tanks - - - - - Page 288

(i) Steel Pressure Tanks - - - - - - Page 289

(j) Steel Fuel Tanks - - - - - - - Page 290

EXAMPLE ESTIMATES

In this section, prices are presented as a function of size, horsepower or capacity. Where relevant, tables or some additional graphs provide information to convert from these three parameters.

Example 1:

Estimate the price of a 70 cubic foot rotary dry blender at an M&S(M/M) Index of 1095.

Figure 191, page 274, shows that the price is about US$30,000 at an M&S(M/M) Index of 1400. This is then converted as follows:

$$\left[\frac{1095}{1400}\right]30000 = \text{US\$23,500.}$$

Example 2:

What is the approximate price and size of a rotating disc pelletizer that will produce 10 stph of pellets?

Figure 197, page 279, shows that the pelletizer should cost around US$64,000 at an M&S(M/M) Index of 1400.

Figure198, page 280, shows that the approximate diameter of the disc unit is around 10 foot. Roughly disc depth is about 12% of the diameter, or 1.2 ft.

Figure 199, page 280, shows that the approximate horsepower of the unit is about 22 (say 25 to the next size motor).

Example 3:

Estimate the cost of a 650 US gallon vertical fibreglass tank, closed at the top and with a dish-shaped bottom.

Figure 204, page 285, shows that the price is about US$2,000 at an M&S(M/M) Index of 1400.

If the Index is currently at 1095, the current price becomes:

$$\left[\frac{1095}{1400}\right]2000 = \text{US\$1560.}$$

ROTARY DRY BLENDERS

Price = aX^b, US Dollars

where **X is operating volume in cubic feet**. Cost includes motor and drive.

Range in X, ft^3:			
5 to 78.5	a = 5071	b = 0.415
78.5 to 550	a = 1819	b = 0.650

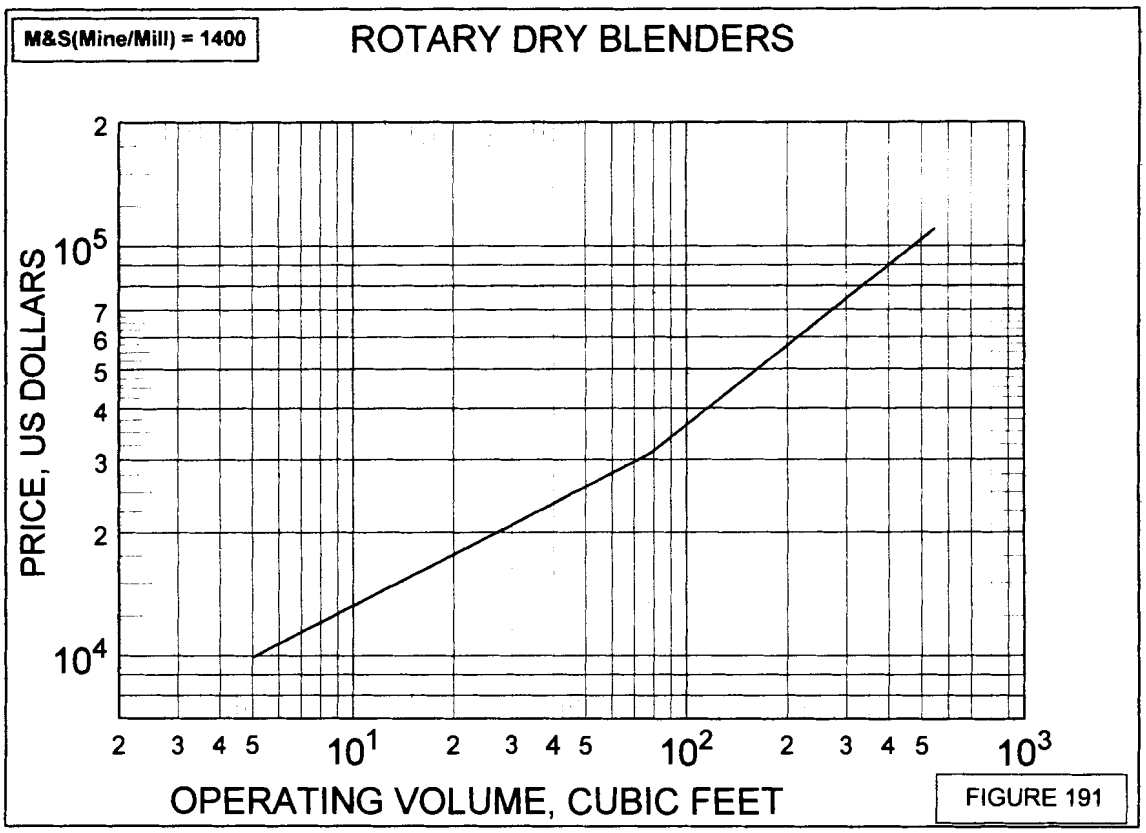


FIGURE 191

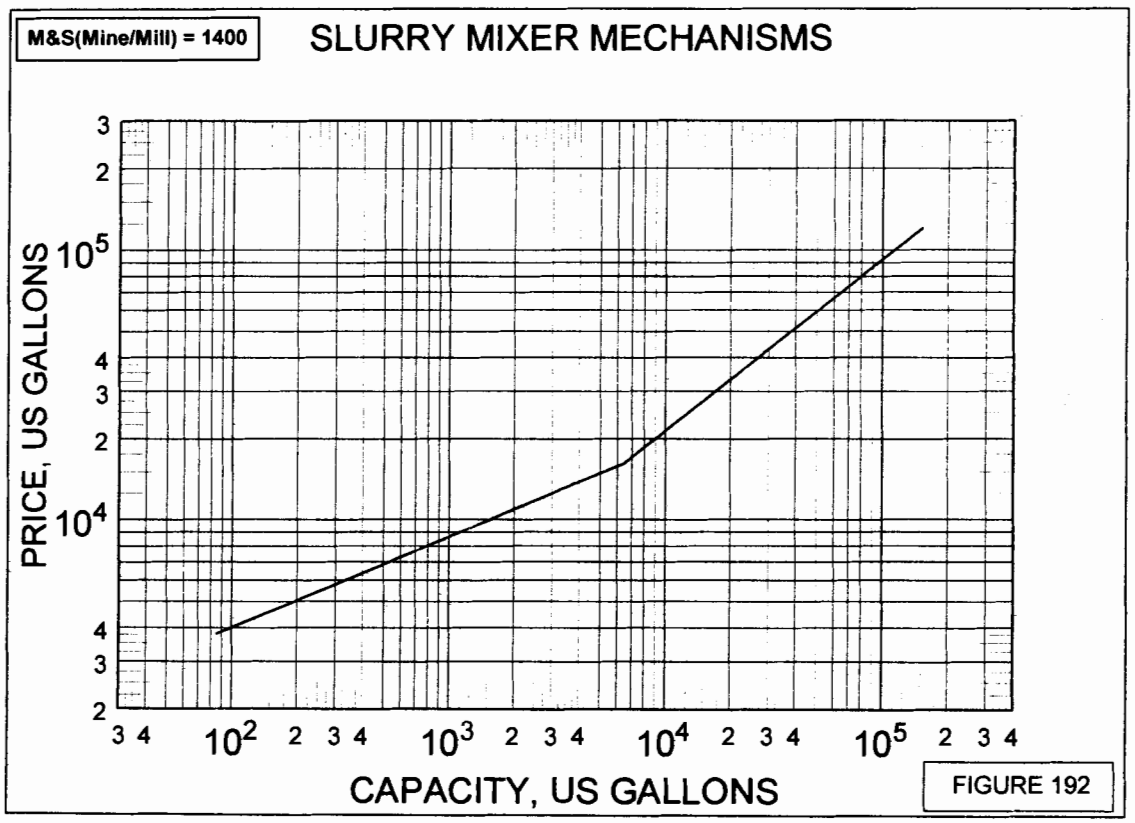
SLURRY MIXER MECHANISMS

Price = aX^b, US Dollars

where **X is volume of slurry to be mixed in US gallons**. Cost excludes tank; includes motor, drive, shaft and impeller. Slurry is at 30% solids of specific gravity 3.

Range in X, US Gal.:			
85 to 6399	a = 851.9	b = 0.3348
6399 to 154000	a = 61.6	b = 0.6345

Volume, US Gal.	HP, nominal
1600	25
2300	40
5000	75



PROPELLER AGITATOR MECHANISMS

Price = aX^b, US Dollars

where **X is, in US gallons, volume of fluid to be agitated**. Cost includes motor; excludes tank.

Range in X, US Gallons: 15 to 1200 a = 174 b = 0.4476

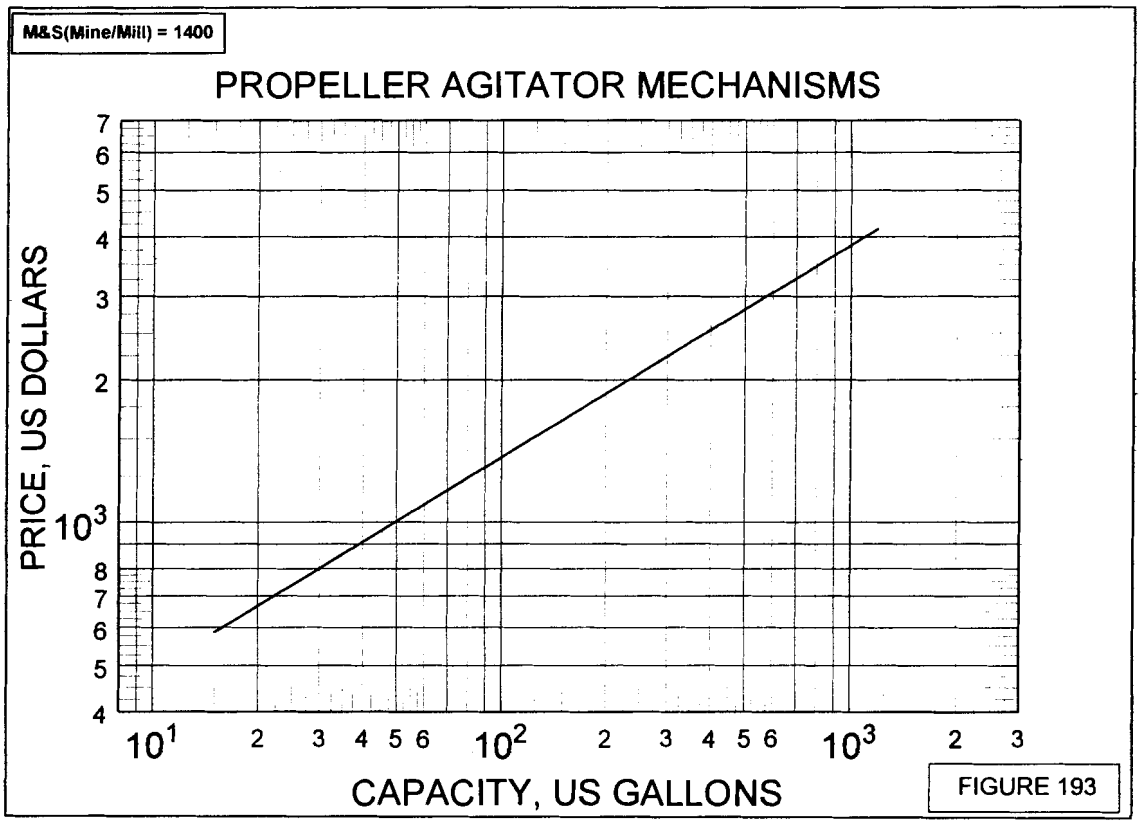


FIGURE 193

ROTARY DRUM PELLETIZER

Price = aX^b, US Dollars

where **X is drum diameter in feet**. Cost includes motor and auger feed system.

Range in X, ft:			
2 to 5.4	a = 7422	b = 1.48
5.4 to 14	a = 3030	b = 1.004

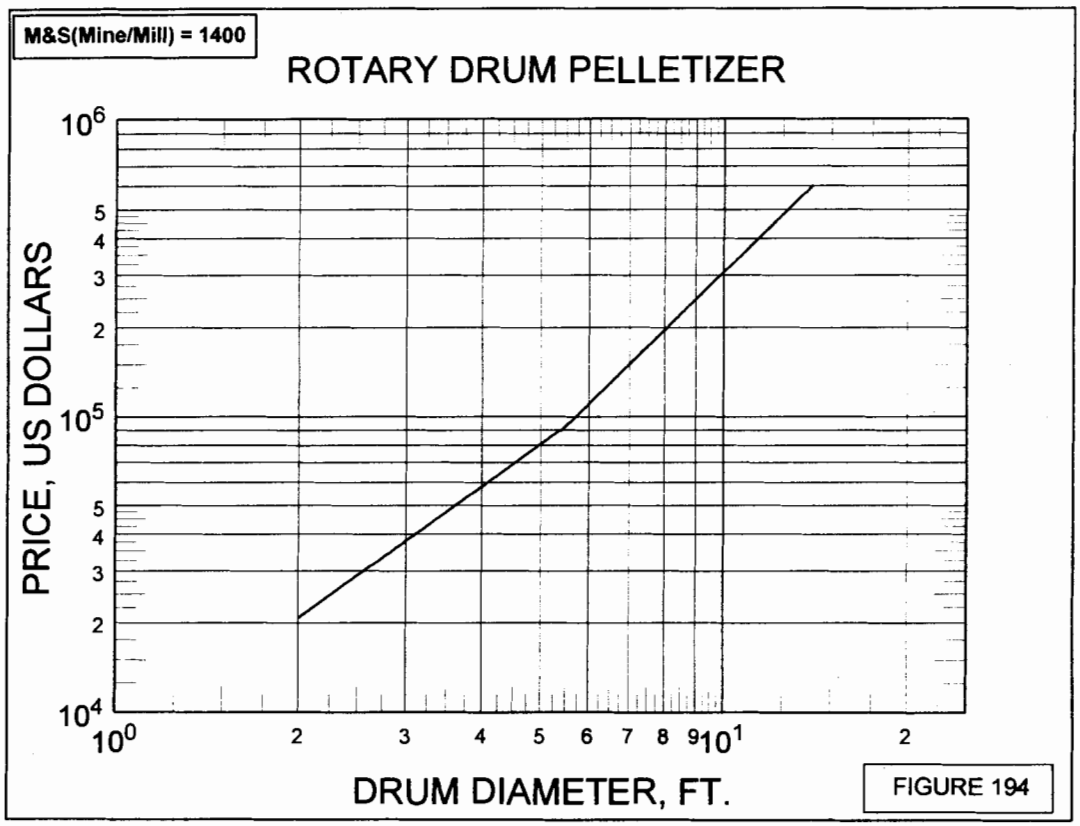


FIGURE 194

ROTARY DRUM PELLETIZER (CONTINUED)

Figure 195 below shows how nominal capacity varies with drum diameter, whereas Figure 196 shows how horsepower varies with capacity. Rough approximations depend on bulk density and feed size.

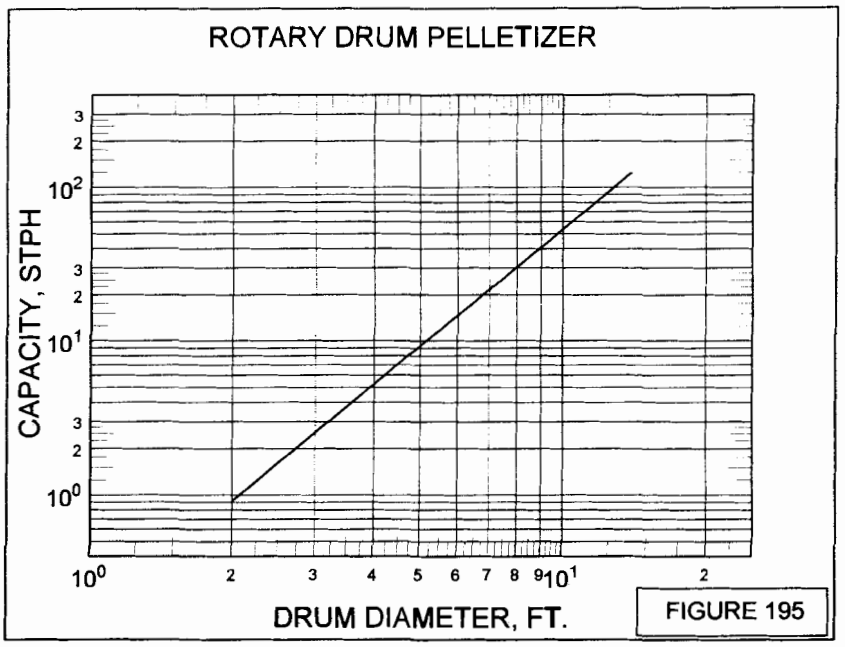


FIGURE 195

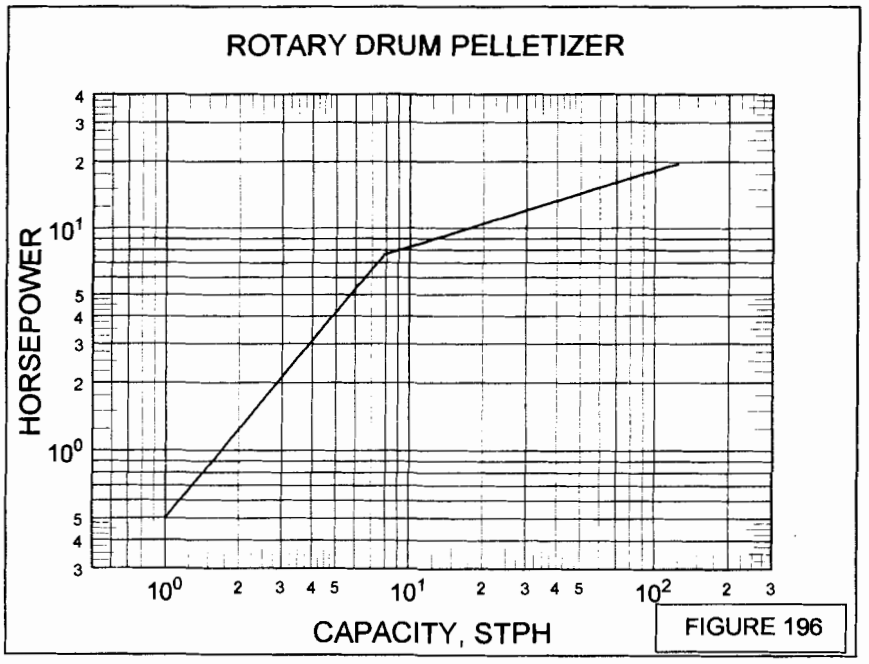


FIGURE 196

ROTATING DISC PELLETIZER

Price = aX^b, US Dollars

where **X is capacity in stph**. Price includes motor.

Range in X, stph: 5 to 125 a = 16210 b = 0.5969

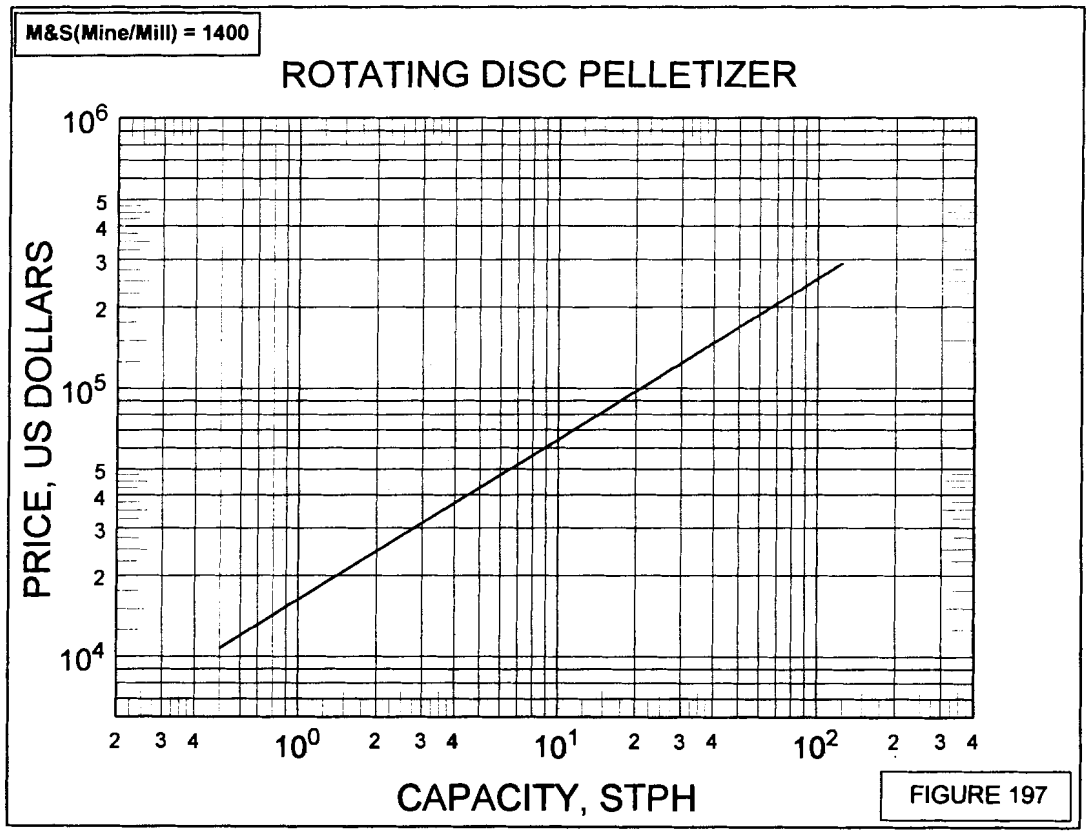


FIGURE 197

ROTATING DISC PELLETIZER (CONTINUED)

Figure 198 shows nominal capacity versus disc diameter, while Figure 199 shows horsepower versus capacity. In both cases, relationships are very rough and depend upon bulk density of feed and feed size.

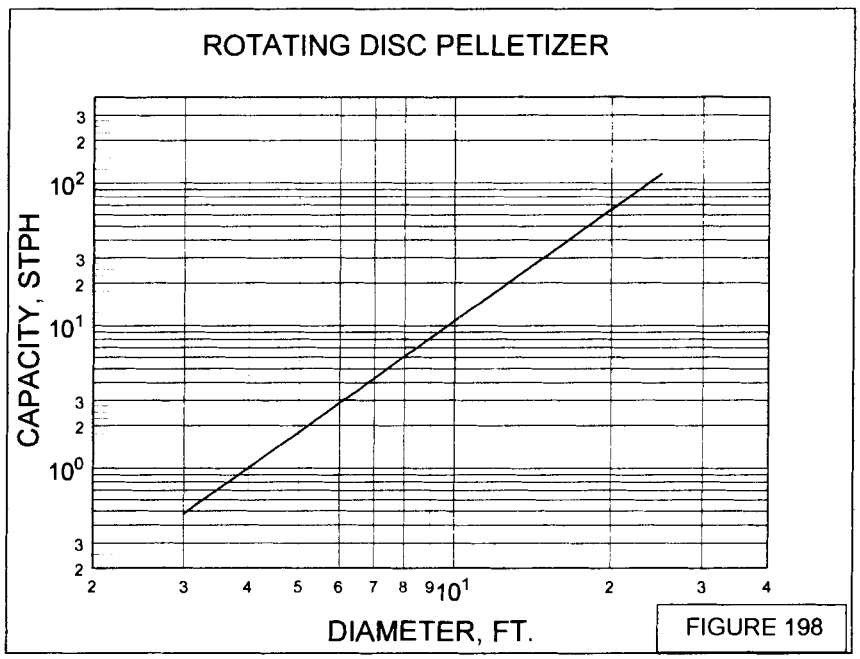


FIGURE 198

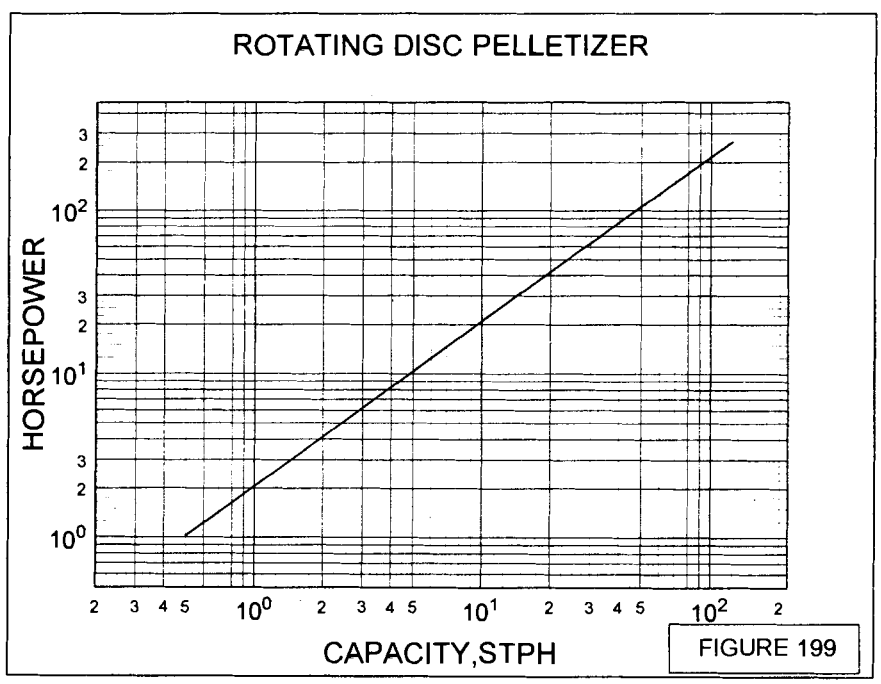


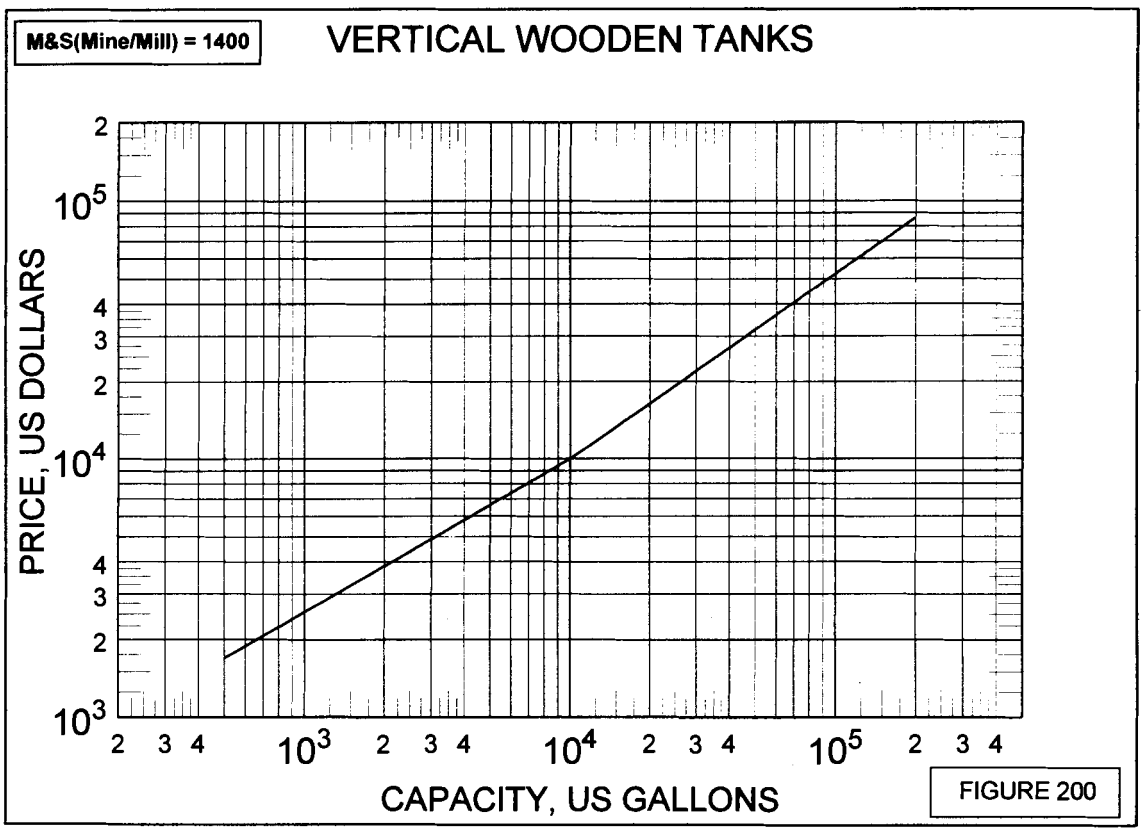
FIGURE 199

VERTICAL WOODEN TANKS

Price = aX^b, US Dollars

where **X is tank capacity in US gallons**. Round, open top, flat bottom to hold material of specific gravity 1.5. Treated Douglas Fir (2 inch and 3 inch). Price includes staves, bottoms, hoops, lugs and chine joists.

Range in X, US gallons:	500 to 10162.4	a = 42.54	b = 0.592
	10162.4 to 200000	a = 12.95	b = 0.7209

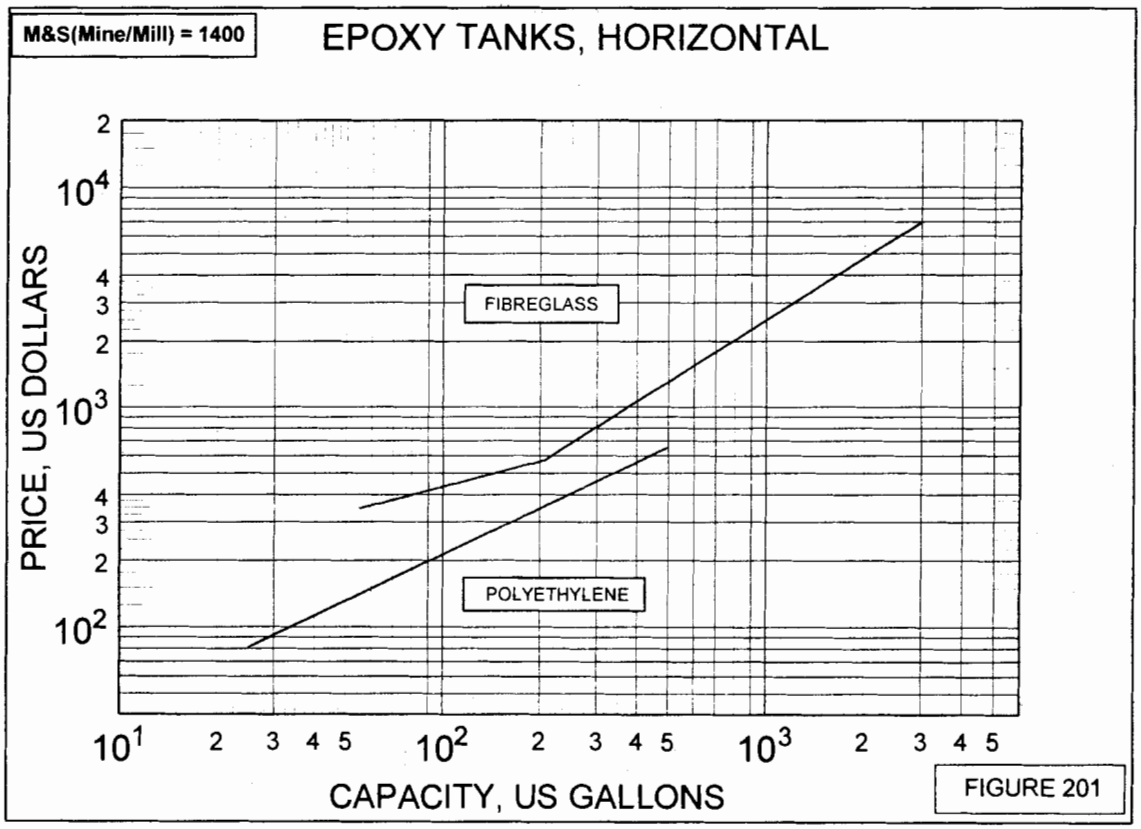


HORIZONTAL EPOXY TANKS

Price = aX^b, US Dollars

where **X is tank capacity in US gallons**. Cylindrical, rounded ends; tanks must be vented.

Fibreglass; Range in X, US Gallons:	55 to 207.3	a = 74.13	b = 0.3826
	207.3 to 3000	a = 3.915	b = 0.934
Polyethylene; Range in X, US Gallons:	25 to 500	a = 8.497	b = 0.6986

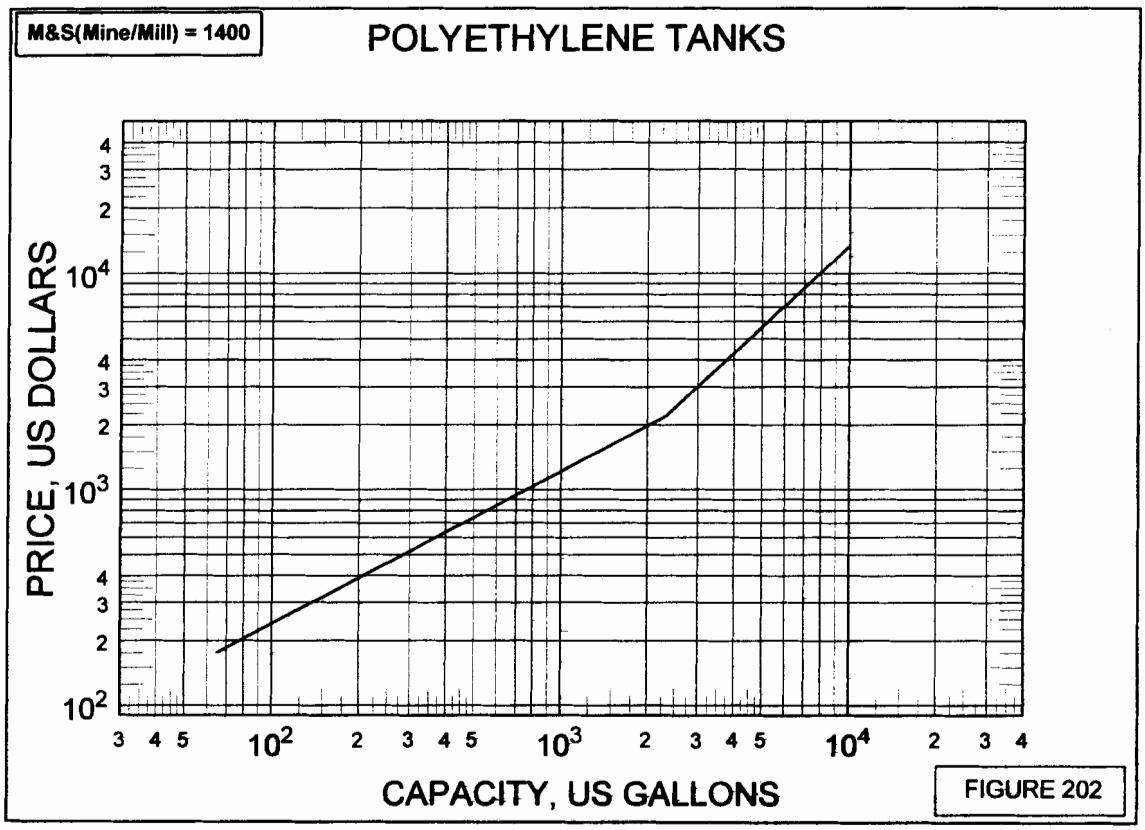


POLYETHYLENE TANKS

Price = aX^b, US Dollars

where **X is tank capacity in US gallons**. Vertical, for corrosive liquids.

Range in X, US gallons:	65 to 2350	a = 9.199	b = 0.7057
	2350 to 10000	a = 0.1518	b = 1.235



FIBREGLASS TANKS

Price = aX^b, US Dollars

where **X is tank capacity in US gallons**. Range in diameter is from 3 ft to 7.5 ft.

Range in X, US gallons:	58 to 870	a = 21.47	b = 0.5923
	870 to 6000	a = 2.588	b = 0.9052

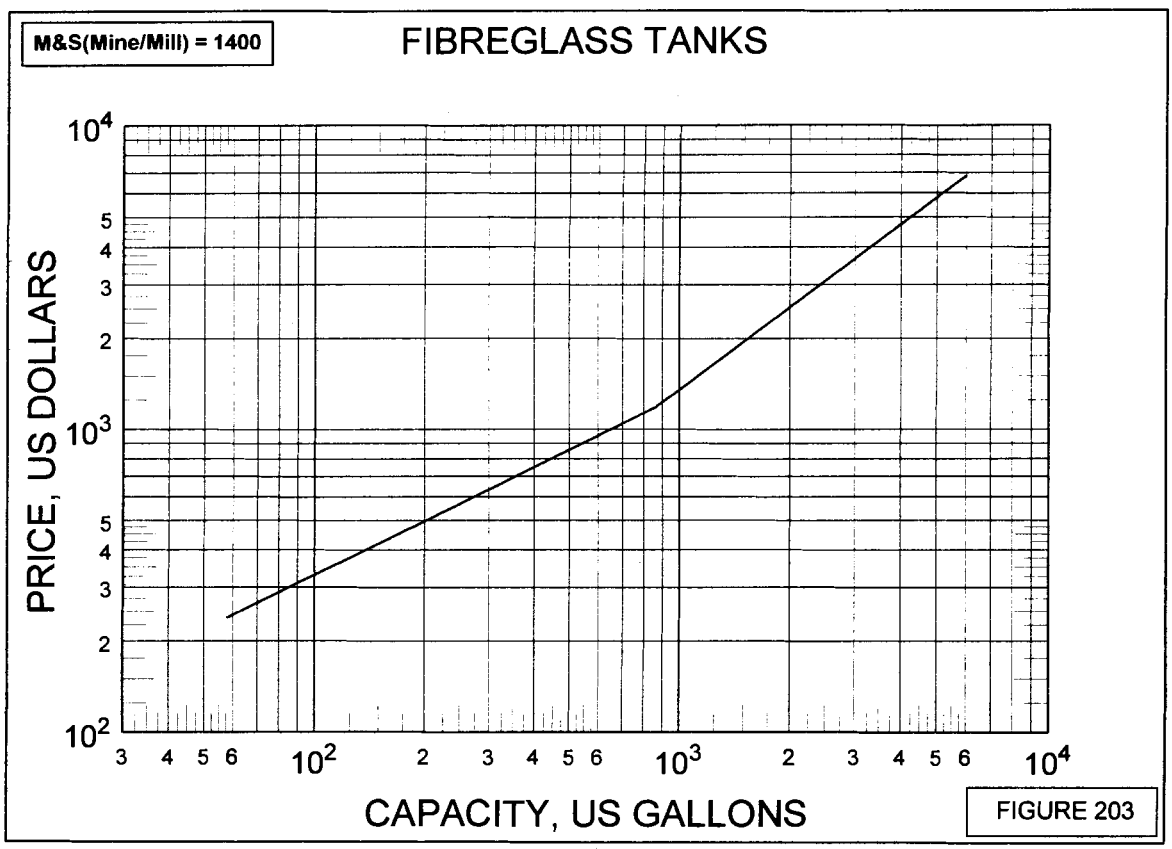


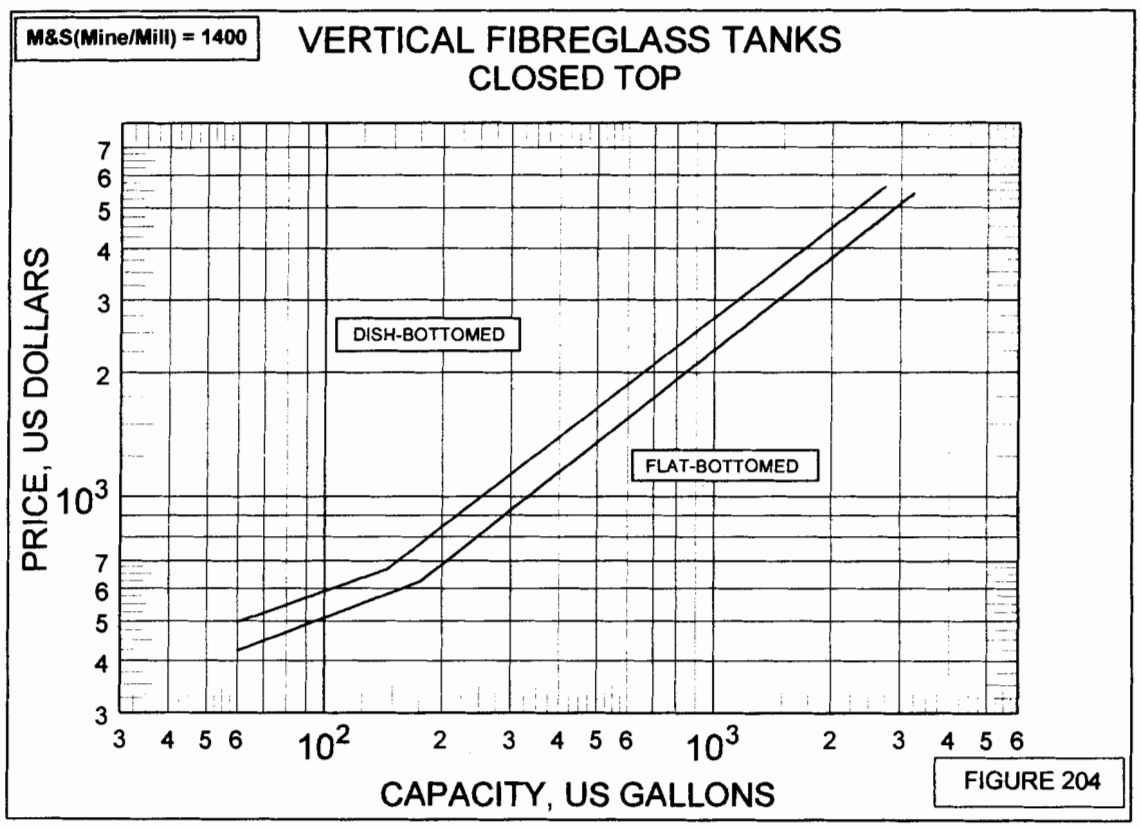
FIGURE 203

VERTICAL FIBREGLASS TANKS
(Closed Top)

Price = aX^b, US Dollars

where **X is tank capacity in US gallons**. Closed, annular top.

Dish-Bottomed; Range in X, US gallons:	60 to 144	a = 125.5	b = 0.3355
	144 to 2750	a = 18.13	b = 0.7243
Flat-Bottomed; Range in X, US gallons:	60 to 176	a = 97.74	b = 0.3577
	176 to 3250	a = 13.37	b = 0.7424



GLASS LINED STEEL TANKS

Price = aX^b, US Dollars

where **X is tank capacity in US gallons**. Vertical tanks with rounded bottoms; top entry, bottom exit.

Range in X, US gallons:	10 to 321	a = 3733	b = 0.2363
	321 to 10000	a = 593.1	b = 0.5551

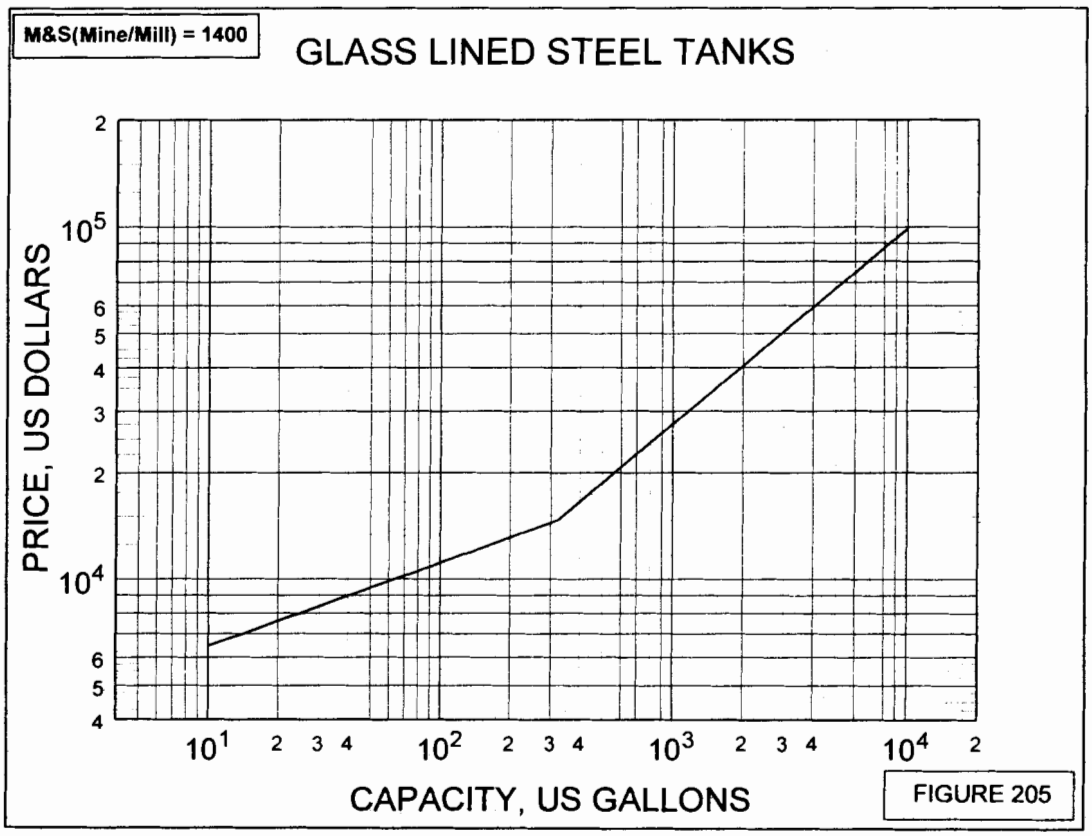


FIGURE 205

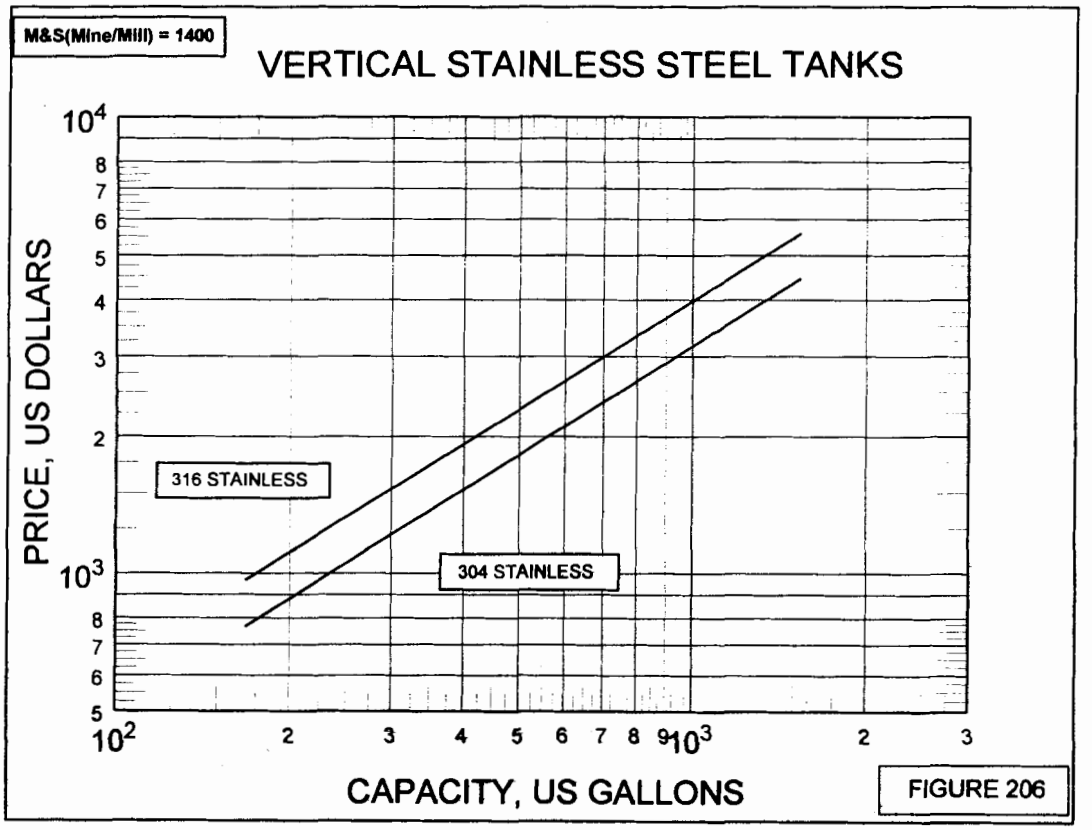
VERTICAL STAINLESS STEEL TANKS
(Open Top)

Price = aX^b, US Dollars

where **X is tank capacity in US gallons**. Round, open top tanks with slope bottoms.

316 Stainless; Range in X, US gallons: 168 to 1540 a = 16.78 b = 0.791

304 Stainless; Range in X, US gallons: 168 to 1540 a = 13.08 b = 0.794



STEEL BULK STORAGE TANKS

Price = aX^b, US Dollars

where **X is tank capacity in cubic feet**. Tanks have hopper bottoms.

Range in X, ft^3:			
1277 to 9000	a = 354	b = 0.5372
9000 to 41937	a = 5.706	b = 0.9901

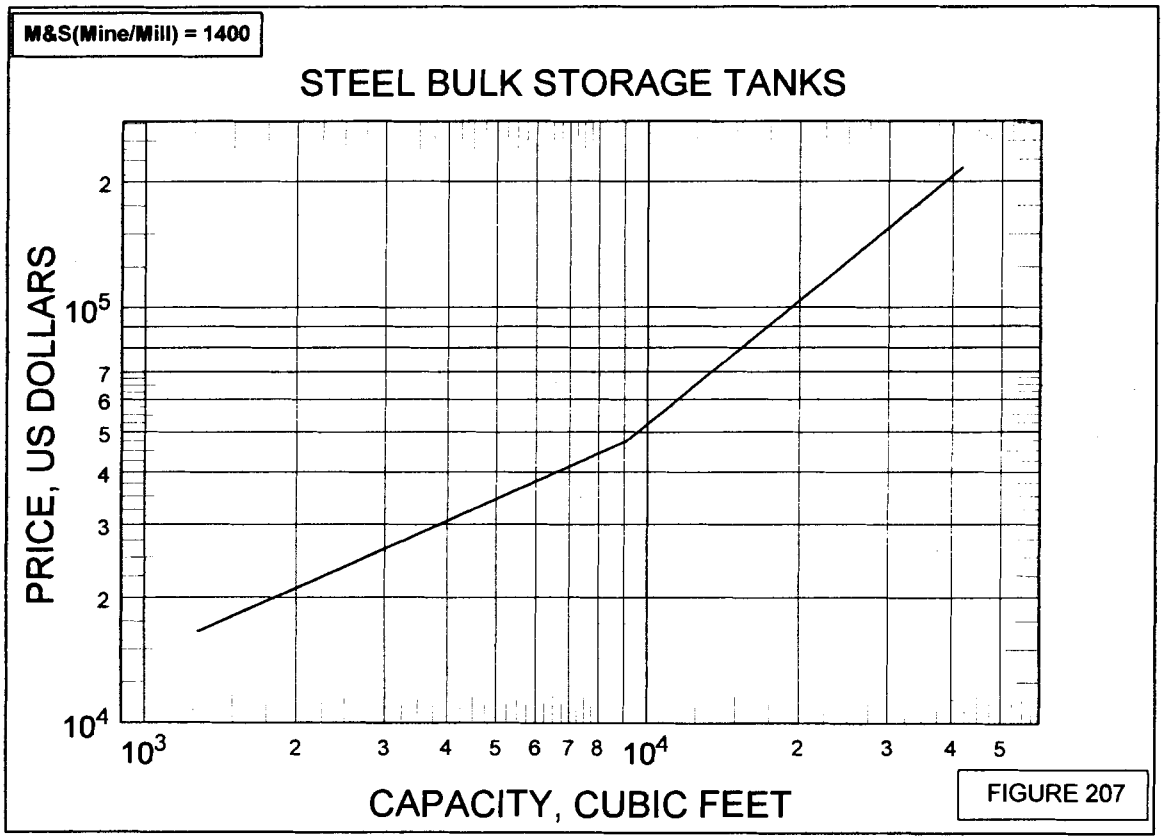


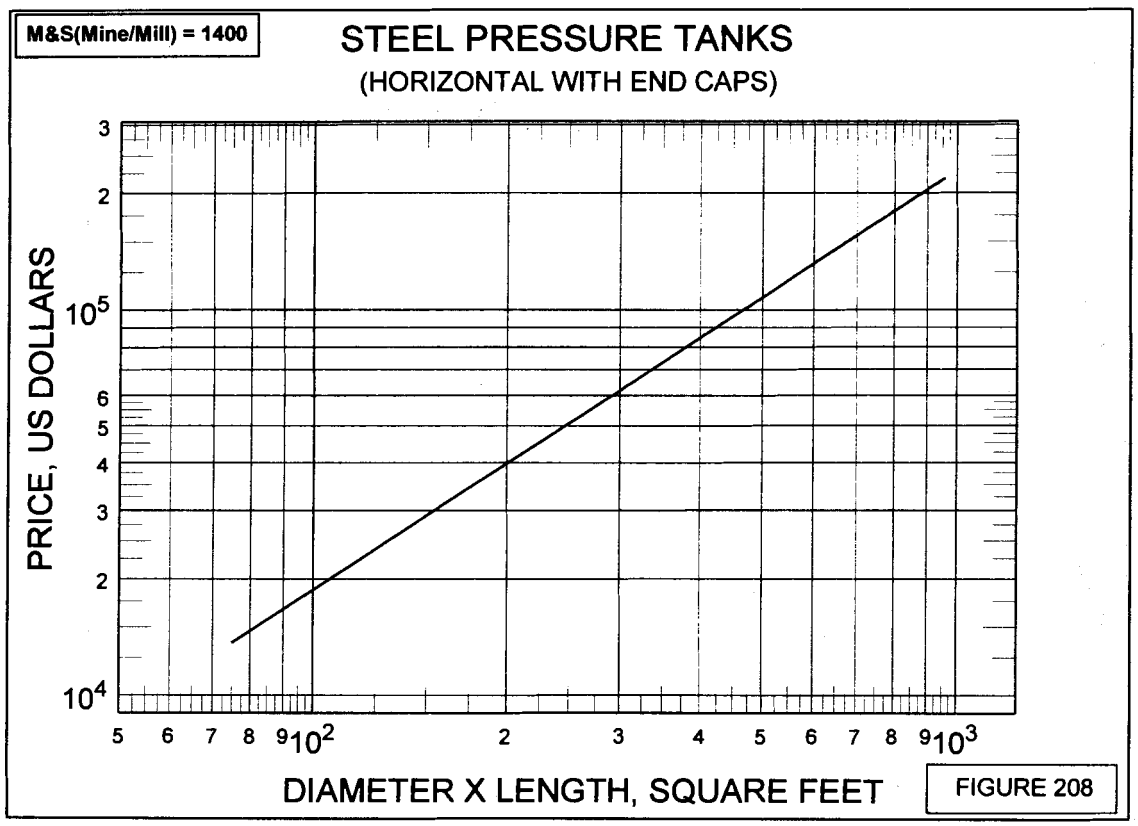
FIGURE 207

STEEL PRESSURE TANKS
(Horizontal With End Caps)

Price = aX^b, US Dollars

where **X is, in square feet, diameter times length**. Shell thickness increases with diameter from 1.5 inch to 2 inch.

Range in X, ft^2: 75 to 960 a = 124.7 b = 1.087



STEEL FUEL TANKS

Price = aX^b, US Dollars

where **X is tank capacity in US gallons**.

Range in X, US gallons:	1000 to 40000	a = 13.97	b = 0.6788
	40000 to 50000	a = .0004283	b = 1.685

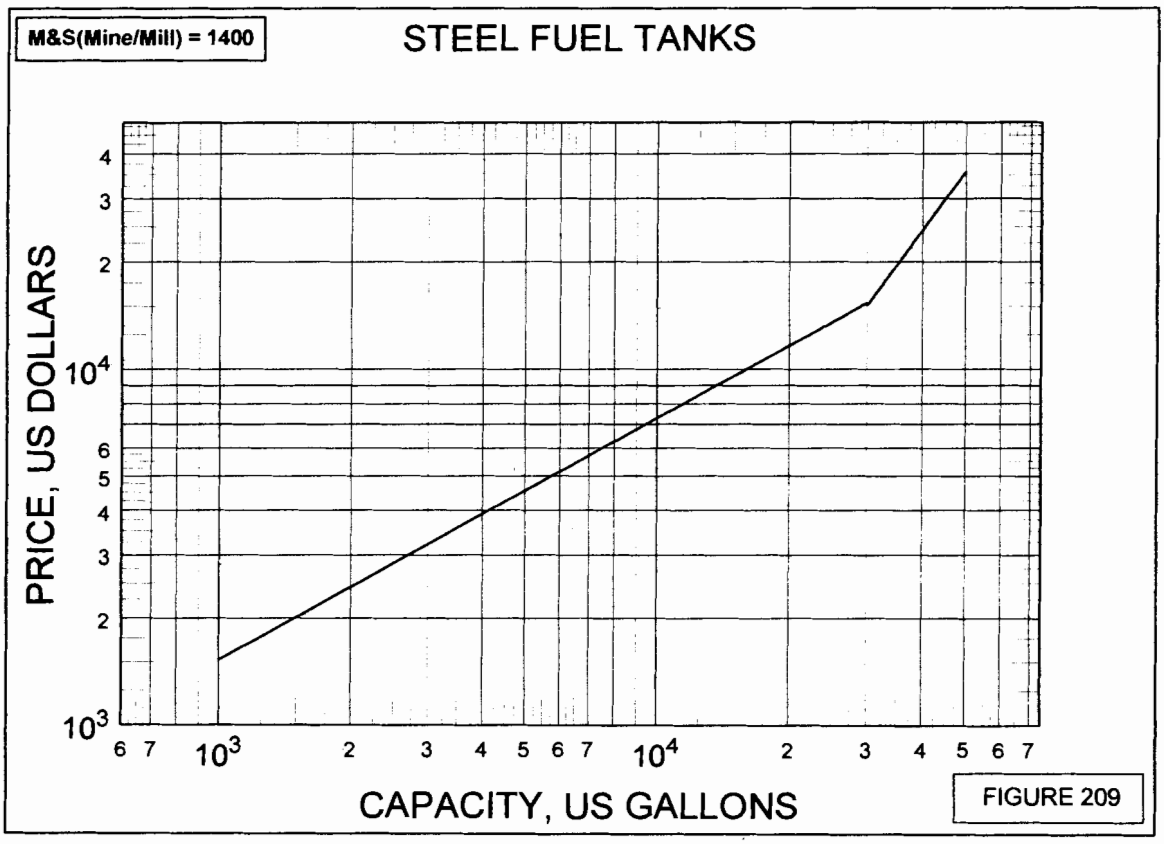


FIGURE 209

XIII. Liquid-Liquid Separation Equipment

Liquid-Liquid separation equipment has been categorized as **Autoclaves**, **Reactors**, **Glass-Lined**, **Carbon Regenerating Kilns**, **Ion Exchange Columns**, **Mixer-Settlers**, **Electrowinning Cells** and **Bullion Furnaces**.

(1) Example Estimates - - - - - - - Page 292

(2) Autoclaves - - - - - - - - Page 293

(3) Reactors, Glass-Lined - - - - - - - Page 294

(4) Carbon Regenerating Kilns - - - - - - Page 295

(5) Ion Exchange Column - - - - - - - Page 296

(6) Mixer-Settlers (Solvent Extraction) - - - - - Page 297

(7) Copper Electrowinning Plants - - - - - Page 298

(8) Electrowinning Cells - - - - - - - Page 299

(9) Bullion Furnaces - - - - - - - - Page 300

EXAMPLE ESTIMATES

In this section, prices are related to size (mainly volume or capacity). Note that some items relevant to gold metallurgy were costed, along with ion exchange columns used in uranium processing. A classical mixer-settler is likewise costed.

Example 1:

An ion exchange column is necessary for the transfer of ions between resin and solution. The volume of liquid to be used for the transfer is 100 cubic feet, and the type of exchanger needed is anionic. Approximately, how much will this exchange column cost?

From Figure 213, page 296, the exchanger must have a volume in US gallons. This amounts to 100 x 7.84 = 784 US gallons. Hence the cost is around US76,000 at an M&S(M/M) Index of 1400. If the current index is at 1095, then this becomes:

$$\left[\frac{1095}{1400}\right]76000 = \text{US\$59,500.}$$

Example 2:

A furnace is necessary to melt bullion, separate any slag and ultimately pour dore' metal into ingot form. What is the estimated cost of a 1000 lb unit.

Figure 217, page 300, shows that the cost is about US$ 26,000 at an M&S(M/M) Index of 1400. Normally, this cost would be updated to a current index.

Example 3:

A high temperature reaction at 2000 PSI must be conducted in a pressure autoclave. The volume of the unit is estimated to be 1000 US gallons. What is the estimated price of the unit?

Figure 210, page 293, shows that this unit will cost about US$427,000 at an M&S(M/M) Index of 1400. If the index is 1095 the current price is estimated as:

$$\left[\frac{1095}{1400}\right]427000 = \text{US\$334,000.}$$

AUTOCLAVES

Price = aX^b, US Dollars

where **X is capacity in US gallons**. Rated pressures at least 2000 PSI. Cost includes motor, stand, heating system and stirrer.

Range in X, US gallons:	.08 to 2.46	a = 10570	b = 0.3404
	2.46 to 1000	a = 8632	b = 0.5649

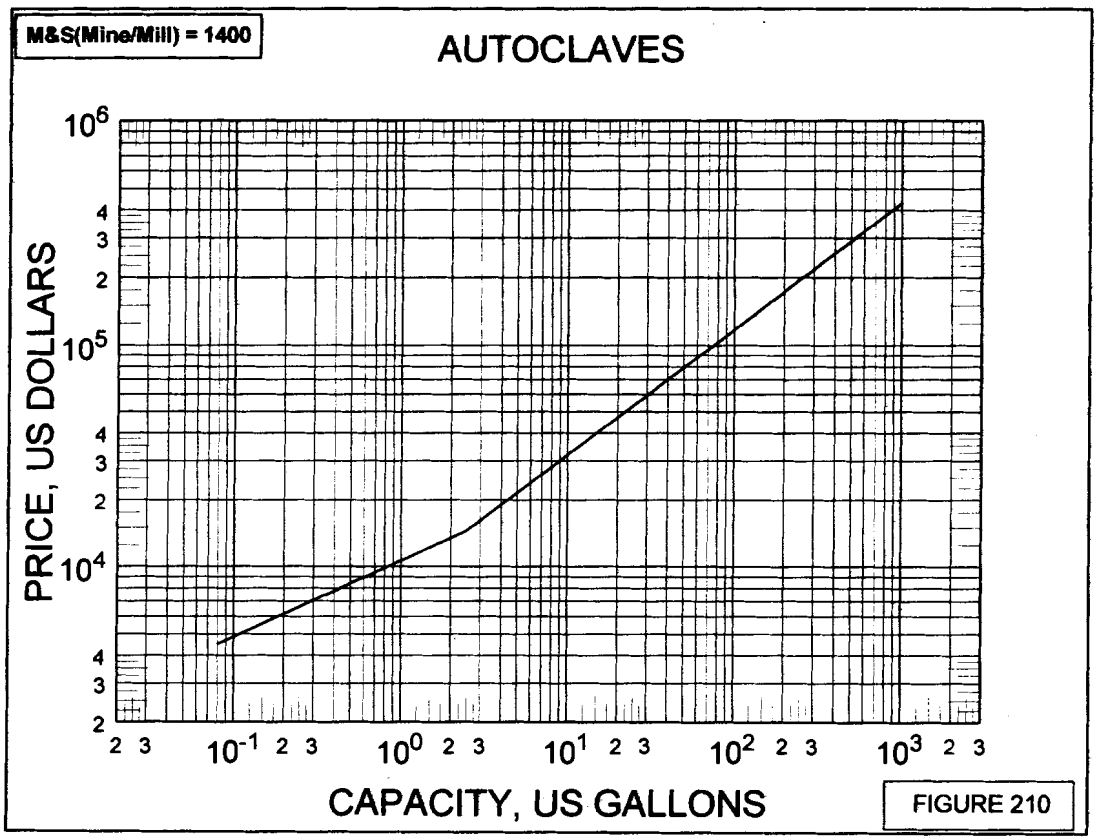


FIGURE 210

GLASS LINED REACTORS

Price = aX^b, US Dollars

where **X is capacity in US gallons**. Price includes drive, motor and mixer.

Range in X, US gallons: 30 to 5000 a = 6751 b = 0.3674

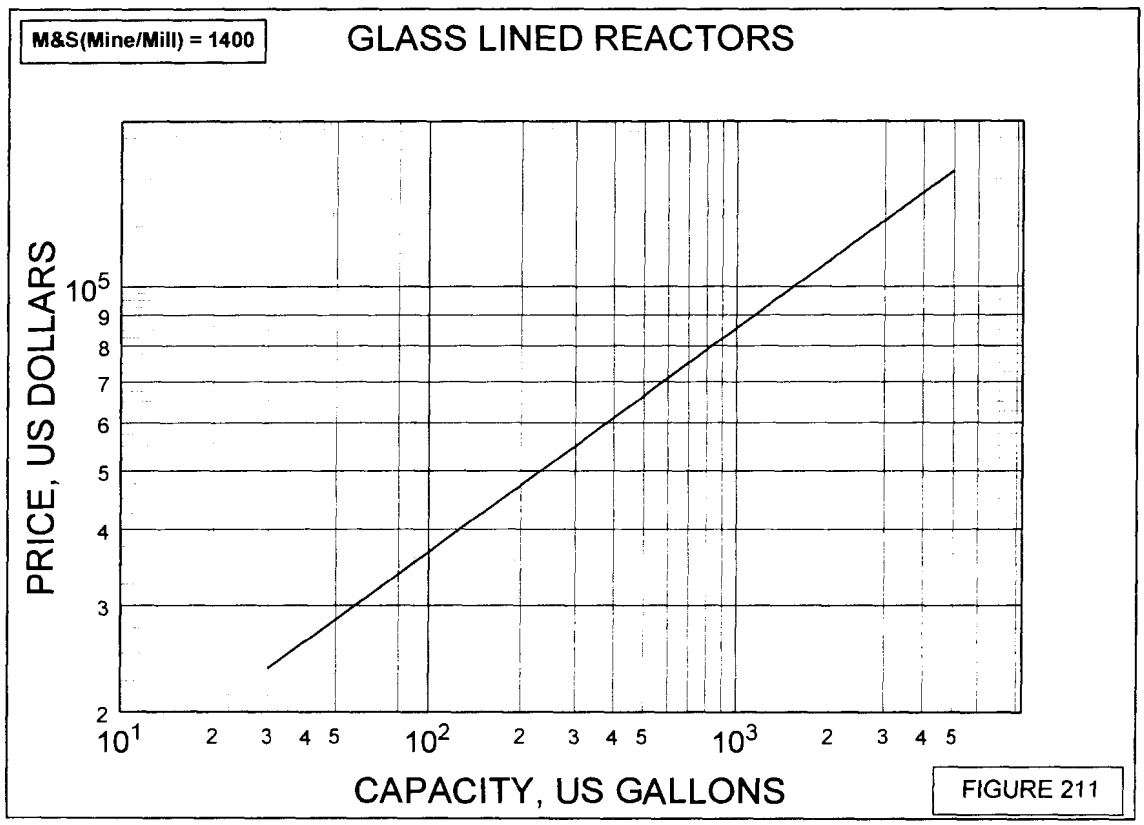


FIGURE 211

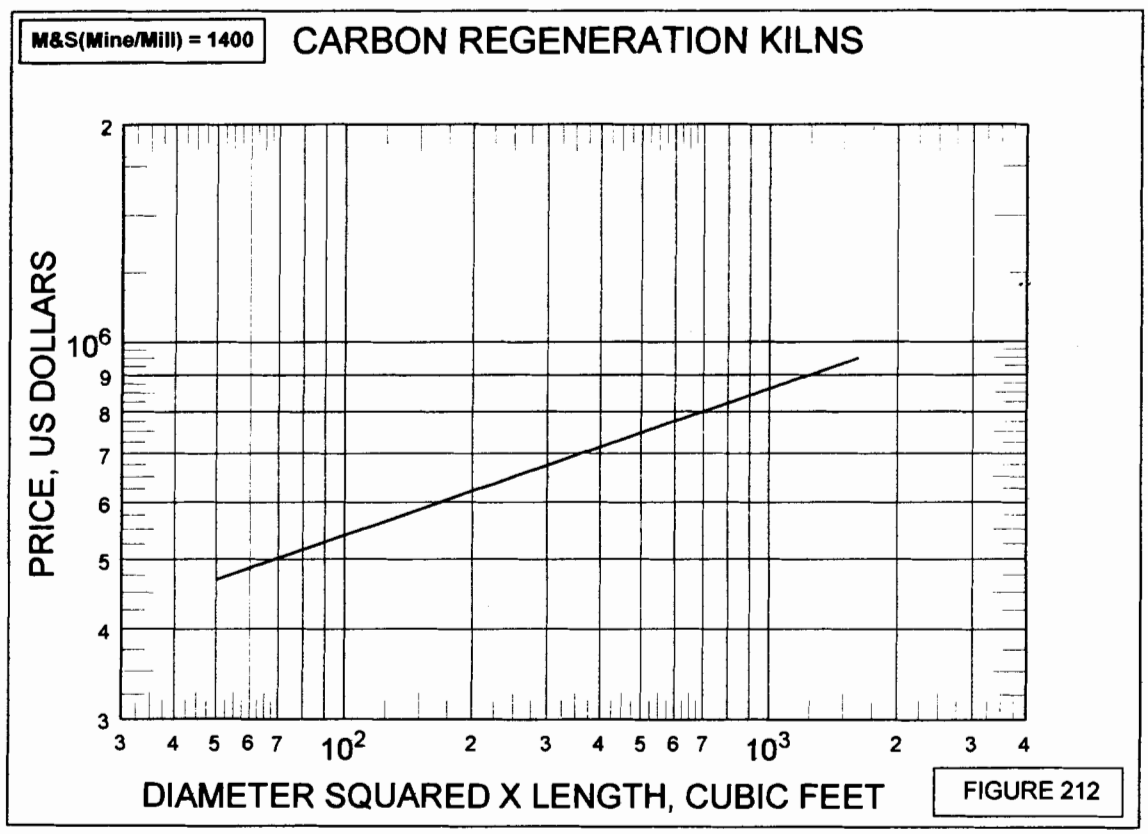
CARBON REGENERATION KILNS

Price = aX^b, US Dollars

where **X is, in cubic feet, the diameter squared times the length**. Rotary kilns varying from 2 ft to 8 ft diameter.

Range in X, ft^3: 50 to 1620 a = 211600 b = 0.2029

Diameter x Length	HP, nominal
3 ft x 20 ft	10
4 ft x 25 ft	15
6 ft x 45 ft	30



ION EXCHANGE COLUMNS

Price = aX^b, US Dollars

where **X is capacity in US gallons**. Price includes fittings and supports; exchange resin suitable for Uranium processing.

Range in X, US gallons: 213 to 17939 a = 2064 b = 0.5397

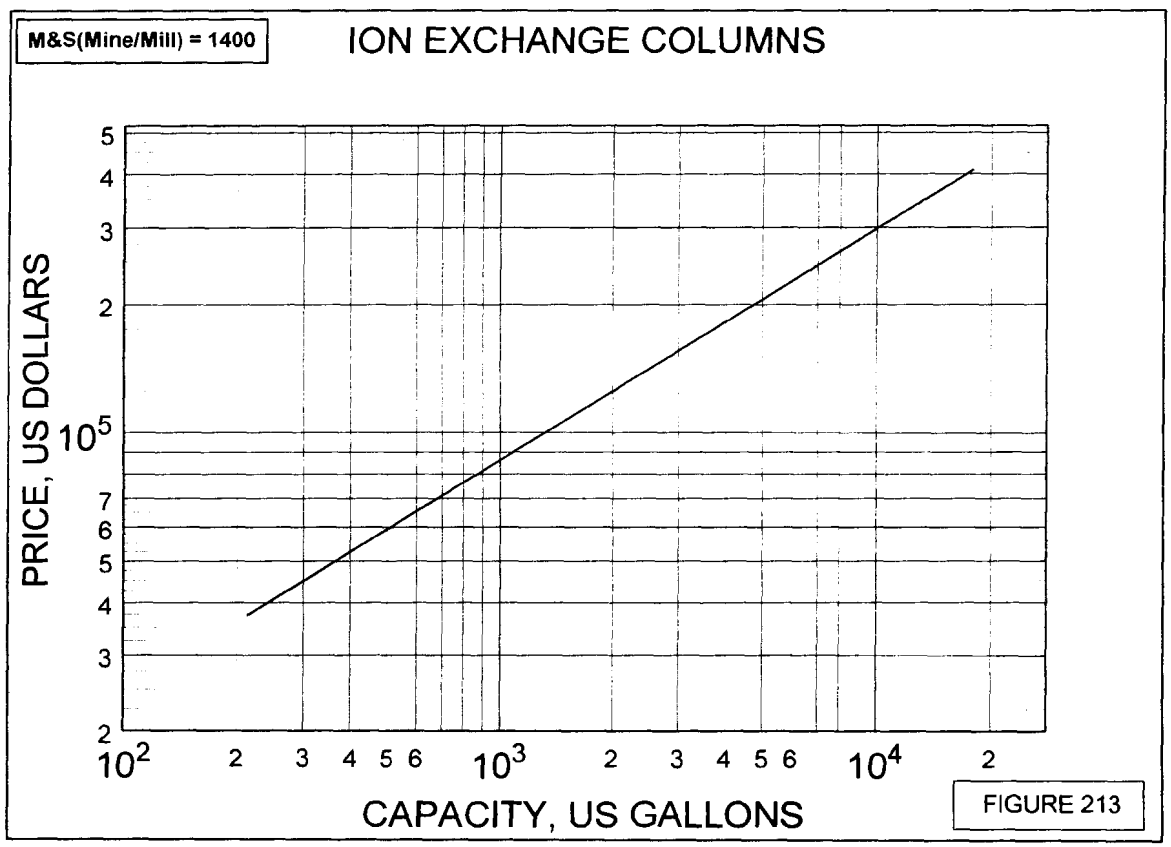


FIGURE 213

MIXER-SETTLERS (SOLVENT EXTRACTION)

Price = aX^b, US Dollars

where **X is mixer volume in US gallons**. At an M&S(M/M) Index of 1400, capital costs for SX plants below **include:** delivered equipment and bulk materials; installation labor/subcontracts at current rates; field construction management/supervision; engineering/project services; other; **exclude:** access roads to site; off-site power/utility installations; mining equipment; permits; financing costs. **Note:** Settler area (ft^2) = $0.527X^{0.9228}$, for X between 2 and 1500.

Range in X, US gallons: 2 to 1500 a = 3854 b = 0.448

Solvent Extraction Plant Capital Costs

Flowrate, gpm	Circuit Definition	Cost, US Dollars	Construction Materials
4000	2+1, 1-Train	13.9×10^6	316 - SS
8100	2+1, 2-Train	17.9×10^6	HDPE/Concrete
8600	2+1, 2-Train	*36.4×10^6	FRP/Concrete
10600	2+1+1, 2-Train	40.0×10^6	FRP/Concrete
12200	2+1, 3-Train	25.8×10^6	HDPE/Concrete
16000	2+1, 4-Train	41.9×10^6	316 SS

*uses more corrosion resistant materials due to high Cl^- concentration in leach solution

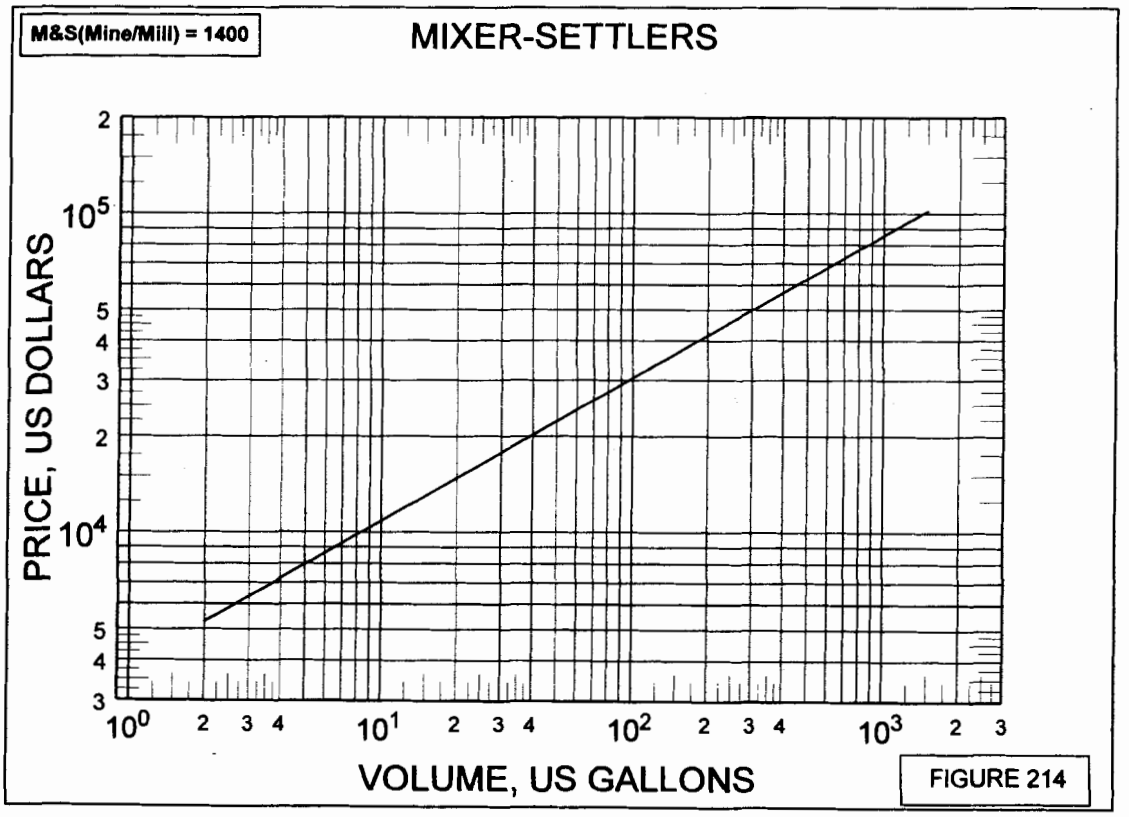


FIGURE 214

COPPER ELECTROWINNING PLANTS

Price = aX^b, US Dollars

where **X is cathode copper tons per day**. At an M&S(M/M) Index of 1400, prices include: delivered equipment and bulk material costs; installation labor/subcontracts at current rates; field construction management/supervision; engineering/project costs; others; and exclude: access roads to site, costs for off-site power, water, other utility; mining equipment; permits; financing costs.

Range in X, tpd: 2 to 227 a = 180400 b = 0.6807

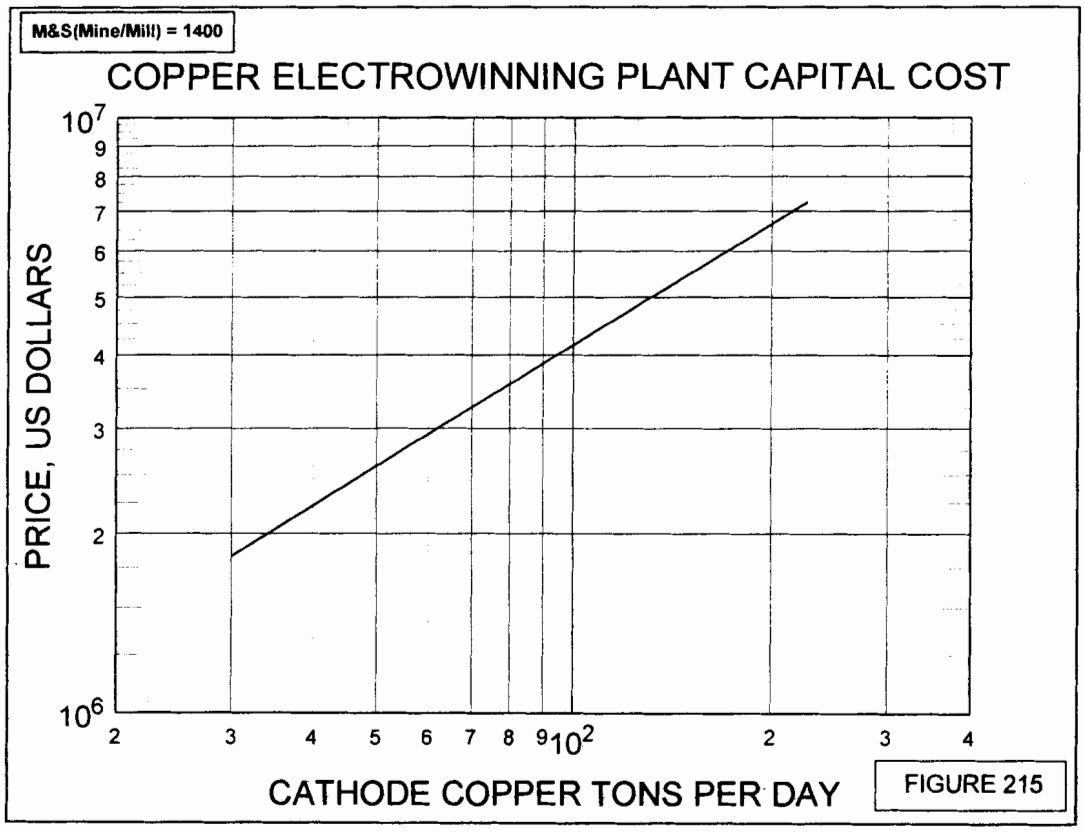


FIGURE 215

ELECTROWINNING CELLS

Price = aX^b, US Dollars

where **X is cell volume in cubic feet**. Polypropylene flux cells are in parallel with 1 lb of steel wool (50 oz Au per lb of steel wool) per 4 ft^3 of cells.

Range in X, ft^3: 4 to 125 a = 3501 b = 0.5051

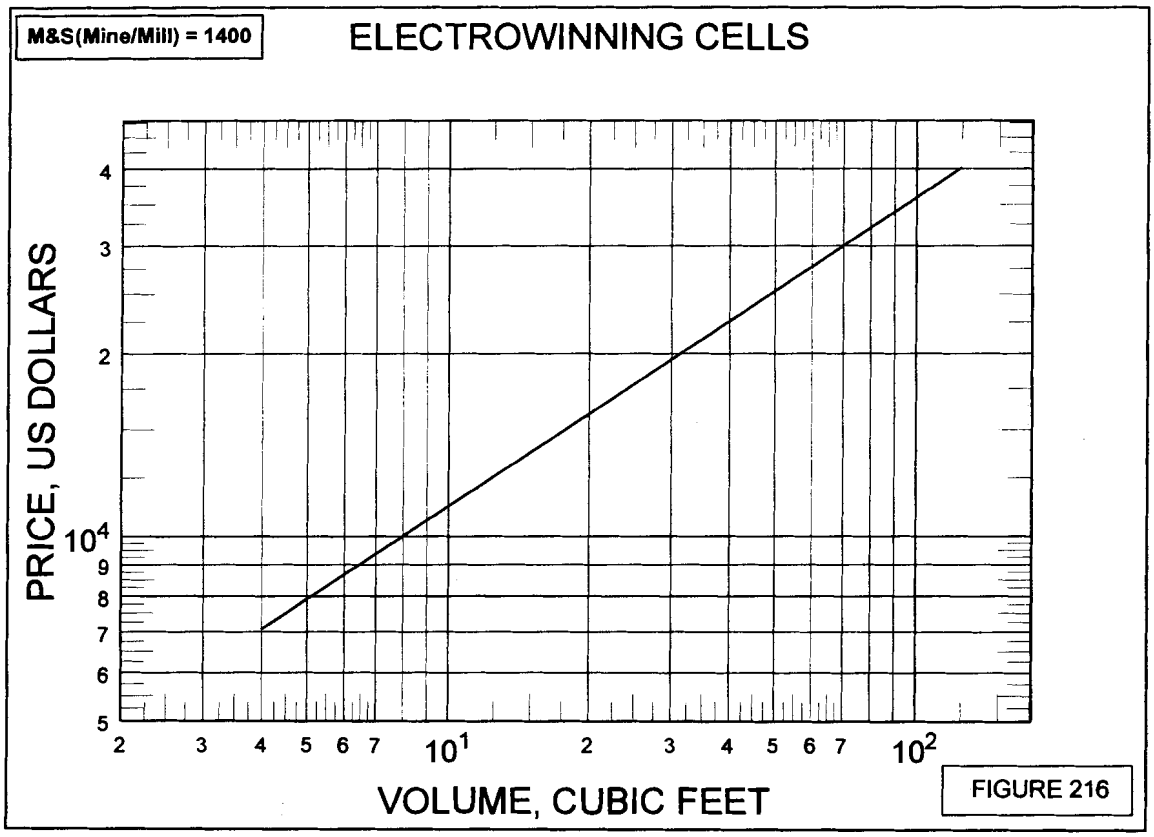


FIGURE 216

BULLION FURNACES

Price = aX^b, US Dollars

where **X is furnace capacity in pounds**.

Range in X, pounds: 280 to 1870 a = 1408 b = 0.4195

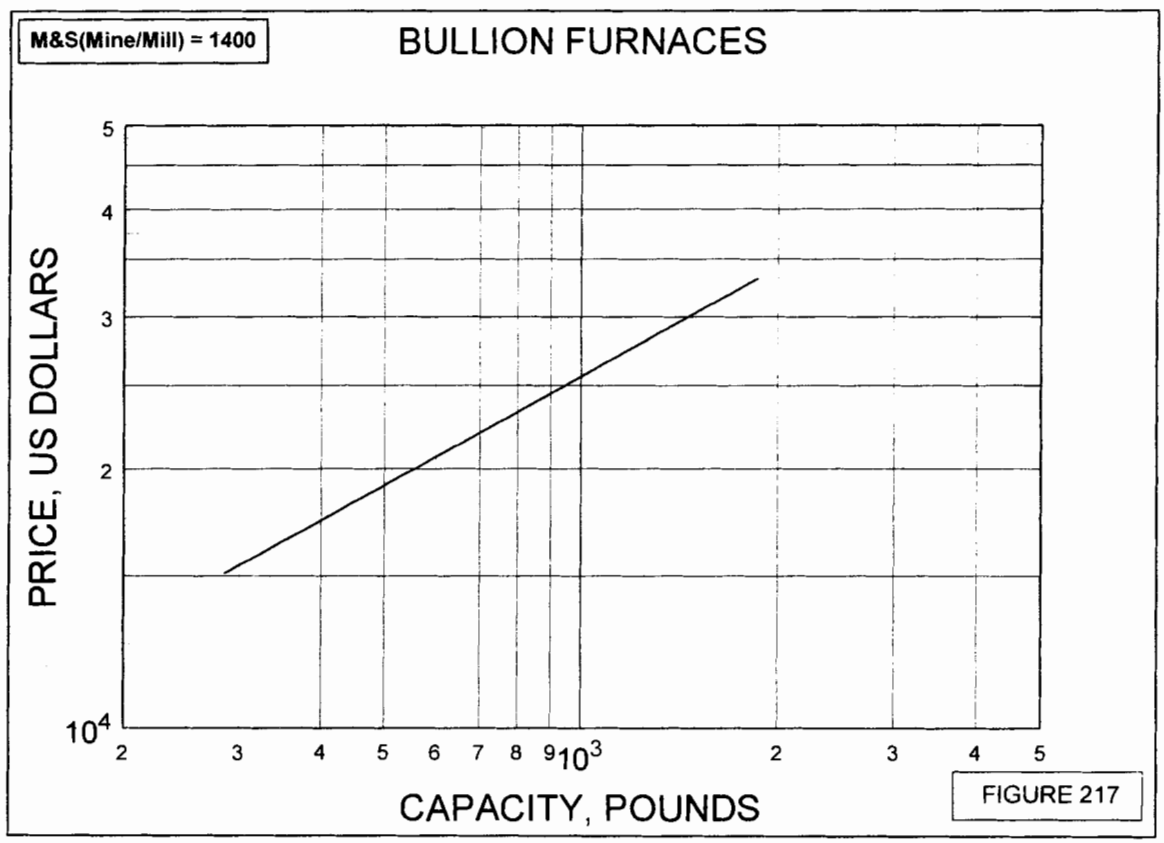


FIGURE 217

RULES OF THUMB

The estimation of the cost of items which do not appear in this handbook depends upon the following categories: (a) the item is not in the handbook, (b) the material of construction differs from that in the handbook and (c) the size of the item is not displayed in the handbook (i.e., the cost parameter is not within the range of interest). Moreover, there are certain features associated with installation costs, quickie plant costs, import duties and the updating of costs that merit further consideration.

Costs Not In Handbook

(1) Item Not Listed

If the item does not appear, then three courses of action are available:

(a) Check the references shown on pages 1, 2 and 305.

(b) If a cost cannot be found anywhere and the material of construction is steel, then a rough estimate is that the cost will be about US$6.00 per pound of equipment at an M&S(M/M) Index of 1400. Note that the weight must be known.

(c) If a single cost and a parameter such as capacity or size or horsepower is available, the 6/10 rule can be used. This rule is:

$$\frac{(Cost)_1}{(Cost)_2} = \left[\frac{(Parameter)_1}{(Parameter)_2}\right]^{0.6}$$

where it is known that Cost is sensitive to Parameter. The exponent 0.6 is the average of hundreds of actual exponents. Sometimes, the rule is sufficiently accurate to allow the cost of an item computed from it to be employed for capital cost estimations of Type II accuracy.

Example of Use:

A siphon sizer of diameter 40 feet was known to cost $80,000 in June of 1988. Table 3, page 7, shows that the M&S(M/M) Index is 870 in 1988, while the current value is 1095. In the absence of further information, estimate the current cost of a 50 foot diameter unit.

Using the six-tenths rule: $$\frac{(Cost)_1}{(80000)} = \left[\frac{50}{40}\right]^{0.6}$$

so that $(Cost)_1$ = $91461.

Updating this value, $$(Cost)_{now} = \left[\frac{1095}{870}\right]91461$$

$$= \$115{,}115.$$

(2) Different Material of Construction

If the item has been costed in this handbook but the material of construction is wrong, then factors can be used which will provide rough estimates. Choosing steel as a basis the factors are:

Carbon Steel	1.00
Aluminum	1.3
304 stainless steel	1.5
316 stainless steel	1.9
Monel	3.6
Inconel	4.4
Nickel	6.3
Fiberglass Polyester(FRP)	1.2
Rubber Lined Carbon Steel	1.6

(3) Cost Parameter in Wrong Range

If the item is costed, but the desired parameter value does not appear, the graph may be extrapolated. However, extrapolation is hazardous. It is likely that the lower end of the graph will tend to become horizontal, while at the upper end the equipment has a parameter value that is not realistic for the equipment. In such cases, multiple units rather than one very large unit may serve as a first approximation to the cost.

Installation/Structural Steel/Concrete/Plant Costs

Installed equipment costs should be based on estimated man hours multiplied by labor rate, with labor accounting for 40 to 50 percent of installed equipment. However, Table 10, page 32, suggests that purchased major equipment costs should be summed and the sum then multiplied by the single factor 1.43 to determine installed equipment costs----clearly a rough approximation. In some situations, a more accurate method is to estimate the installed cost of each individual item, add this estimate to the purchased cost, obtain the installed equipment cost per item, and then sum the individual installed costs for total installed costs. This procedure avoids the following problem: Suppose that a low-carbon steel tank costs $3200, so that the installed cost is about $4576. If the tank is ordered in stainless steel, the cost will be about $6080 and the installed cost is then $8694 (using the 1.43 ratio). But why should it cost $1376 to install a low-carbon steel tank, while a similar stainless steel tank costs $2614 to install? The only difference in tanks is the material of construction.

Since the factor 1.43 is an average, it is clear that neither of these installation costs is correct. One possibility is to choose the lowest cost; another is to average the two. Probably, the average is best when the item is virtually the only process equipment made of stainless steel. However, if most of the major process equipment is of stainless steel construction, the probable installation cost likely will be that based on carbon steel construction.

The average weight of structural steel depends on the type of plant being designed. For processing plants, if the bay line area is known in square feet, roughly 20 lb/ft^2 of heavy steel is

required. Light steel requirements are about 15 lb/ft^2.

The price of structural steel per ton (fabricated and erected) is about US$4410 at an M&S(M/M) Index of 1400.

For the US gulf coast, poured concrete will cost between US$ 645 to US$ 675 per cubic meter. Such costs vary considerably with geographic location (i.e., the Andes mountains east of Santiago, Chile versus Winnemucca, Nevada, USA).

Complete plant costs may be estimated by means of the 0.6 rule. Of interest is the following expression transmitted during a conversation with Dr. K. K. Humphrey of the University of West Virginia in the USA:

$$C = 10TS$$

where C is the cost of a coal preparation plant, T is the capacity in tph, and S is the cost, $/ton, of erected structural steel. For example, if a plant treats 24,000 tpd or 1000 tph and if the cost of the steel is $4410/ton, then C = 10 x 1000 x 4410 = $44,100,000.

Import Duties

The following points are relevant to import duties if applicable:

(1) Check to determine whether the foreign manufacturer has a Canadian outlet. If there is a Canadian outlet and if the item is assembled in Canada, there may be no duty involved.

(2) If the equipment to be imported is used in the processing, smelting or refining of ores, metals or minerals, the equipment may be duty free. This should be checked carefully.

(3) If an import duty must be paid, add approximately 15 percent to the purchase cost to cover duties.

Updating Costs

To update obsolete costs, when cost index values are not available, Figures 218 and 219 on page 304 can be useful. Figure 218 is a linear graph of the M&S(M/M) Index versus year from 1960 to 1981. Distinct regions appear. From 1960 to 1965, the index changes very little. From 1965 to 1973, the index increases by about 3 1/2 percent per year. From 1973 to 1979 the average increase is about 10 1/2 percent per year. This becomes about 14 1/2 percent per year from 1979 to 1982.

Figure 219, page 304, shows how the M&S(M/M) Index varies with year from 1982 to 1996. Here, 3 distinct regions appear. From 1982 to 1987 the average increase in the index is about 1.3% per year; from 1987 to 1989 it is about 4.25%. However, from 1989 to 1996, the average increase per year becomes about 2.32 percent.

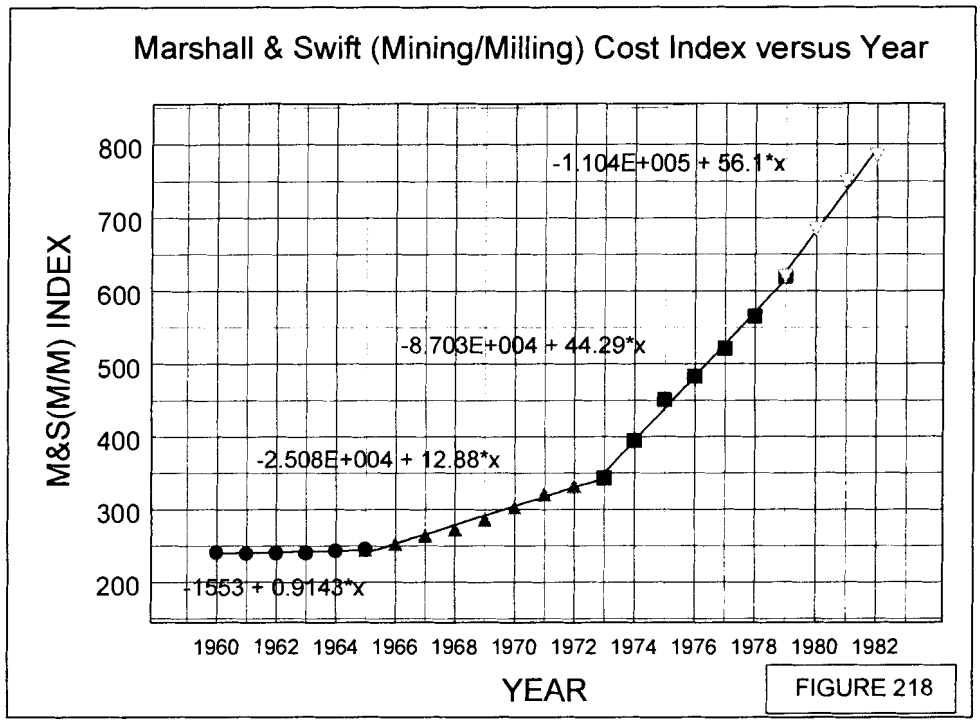


FIGURE 218

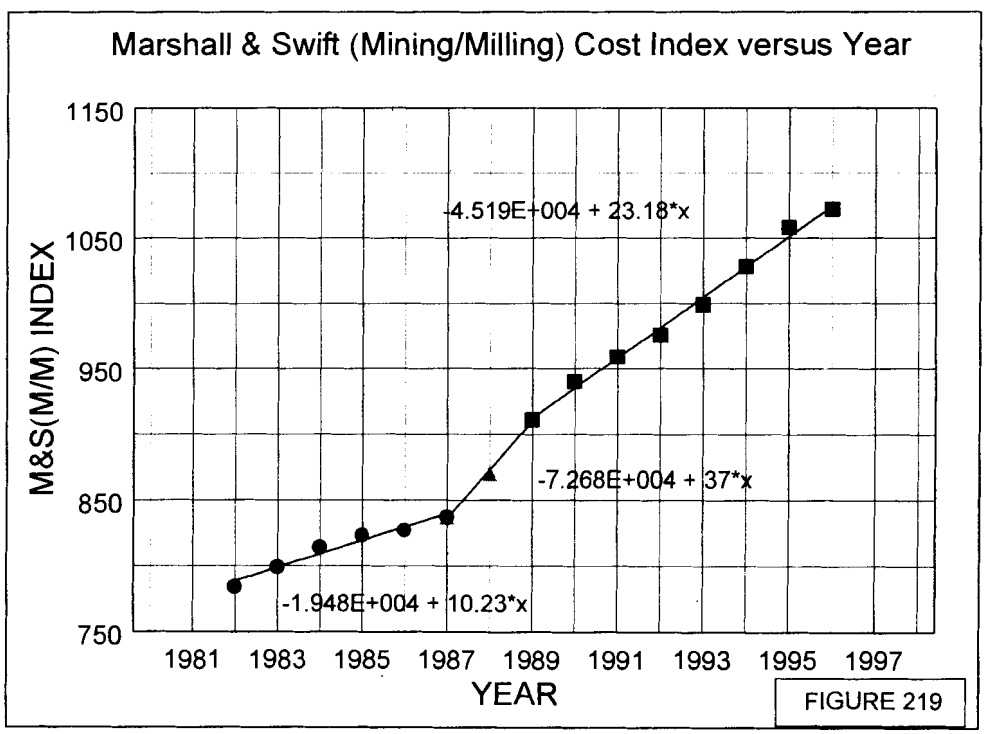


FIGURE 219

ADDITIONAL SOURCES OF COST INFORMATION

Journals

1. Chemical Engineering
2. Chemical Engineering Progress
3. Coal Age
4. Cost Engineering
5. Industrial Engineering
6. Oil and Gas Journal
7. Petroleum Refiner
8. Canadian Mining Journals Buyers Guide (Sourcebook)
9. Engineering and Mining Journal
10. Engineering News Record

Equipment References

For detailed information on sizes, capacities and horsepower requirements, it is recommended that the reader consult bulletins supplied by manufacturers. See page 3 for trade directories.

Technical Papers

1. O'Hara, T. A., "Quick Guides to the Evaluation of Orebodies", **CIM Bulletin,** February, 1980, p. 87-99.

2. Mular, A. L., "The Estimation of Preliminary Capital Costs", Ch. 3 in **Mineral Processing Plant Design,** Mular, Andrew L. and Bhappu, Roshan B., Eds.,, SME-AIME, Littleton, Colorado, 1980, 946 pages.

3. Humphreys, K. K. and Mular, A. L., "Capital and Operating Cost Estimation", Ch. 6 in **Design and Installation of Comminution Circuits** Mular, Andrew L. and Jergensen, Gerald V. II, Eds., SME-AIME, Littleton, Colorado, 1982, 1022 pages.

4. Corneille, E. K., "Design, Capital and Operating Costs of Mineral Processing Plants", **Mineral Processing and Extractive Metallurgy Review, Vol. 2**, Gordon and Breach, Science Publishers Inc., 1987, p. 255-288.

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